ALTERNATING BUNDLES AND THE SPLITTING OF EXTERIOR POWERS

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Abstract. The alternating-map bifunctor over a nontrivially normed field acts analytically on analytic bundles when precomposition is analytic. For normed E and D, precomposition is analytic for all Banach values exactly when one class vanishes: the class of the exact sequence of completed projective tensor and exterior powers of E, pushed along an evaluation map. When $k! = 0$, no splitting of this sequence for $E = K^k$ with the maximum norm commutes with the permutation matrix of a single p-cycle, where $p = \operatorname{char} K$. Source, target and parameter splittings, chiefly ordering splittings, make the class vanish; Ramsey forcing over finite residue fields obstructs it. Over subfields of $\kappa((X))$, κ finite, this decides several classes of pairs, and rigid Banach spaces give vanishing classes that no splitting decomposition explains. The last section treats ambient analyticity as a structure, not a class of maps: the fixed envelope Mult^k gives Bourbaki's formulas on ambient sections in every characteristic, while over every field of characteristic $p > 0$ ambient analyticity is not closed under composition. Seventy-six statements, listed in Section 13, have Lean proofs of corresponding statements.

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The original counterexample and its initial write-up were developed with Anthropic's Claude. Subsequent research and exposition used Claude and OpenAI's Codex, and the organization of the paper was prepared with Claude and Codex. AI assistants contributed to mathematical arguments and independent proof audits. The original Lean formalization used Codex and Autoform. The remaining claims and the per-claim ledger were formalized with Anthropic's Claude Code in October 2026. The splitting theory and the classification of good pairs, that is, the results of Sections 3–12 and Appendices J–O not listed in Section 13, and Proposition 10.4 were developed and written with Claude and are not formalized. Section 14 and Appendix Q were developed and written with Claude and are not formalized. Section 13 records the verification scope.

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1. Introduction

A functor on normed spaces extends fiberwise to C^n vector bundles, with its natural transformations, when C^n families of linear maps induce C^n families (Theorem 2.1). We ask when the bifunctor $(E,F)\mapsto \mathrm{Alt}^k(E;F)$ of alternating k-linear maps over a nontrivially normed field K carries analytic bundles to analytic bundles. For normed spaces E, D and F this concerns $A^k_{E,D;F}\colon \mathcal{L}(E,D)\to\mathcal{L}(\mathrm{Alt}^k(D;F),\mathrm{Alt}^k(E;F))$, $A^k_{E,D;F}(u)(m)=m\circ(u,\dots,u)$. Let $j_{E,F}\colon \mathrm{Alt}^k(E;F)\to\mathrm{Mult}^k(E;F)$ be the isometric inclusion. Composed with the induced isometric inclusion $j_{E,F}\circ -\colon \mathcal{L}(\mathrm{Alt}^k(D;F),\mathrm{Alt}^k(E;F))\to\mathcal{L}(\mathrm{Alt}^k(D;F),\mathrm{Mult}^k(E;F))$, the map $A^k_{E,D;F}$ is the diagonal of a bounded k-linear map (Proposition 2.2 with $g=\mathrm{id}_F$), and Theorem 2.3 decides when such a diagonal descends analytically through $j_{E,F}\circ -$. By Proposition 2.5, $A^k_{E,D;F}$ is analytic at some point if and only if it has a *lift*: a bounded k-linear map $Q\colon \mathcal{L}(E,D)^k\to\mathcal{L}(\mathrm{Alt}^k(D;F),\mathrm{Alt}^k(E;F))$ with $Q(u,\dots,u)=A^k_{E,D;F}(u)$.

The splitting question. Let $\mathcal{N}^k_E$ be the kernel of the quotient map $\widehat{\wedge}^k_E$ from the completed projective tensor power of E onto its completed projective exterior power (Section 3):

$$0\longrightarrow \mathcal{N}^k_E\longrightarrow \widehat{\otimes}^k_\pi E\xrightarrow{\widehat{\wedge}^k_E}\widehat{\Lambda}^k_\pi E\longrightarrow 0. \tag{1.1}$$

If $k!\neq 0$ in K, antisymmetrization divided by $k!$ splits (1.1) naturally in E and solves every lift problem (Propositions 4.1 and 5.1(1)). If $k!=0$ in K, one p-cycle of the coordinates of K^k, $p=\mathrm{char}\,K$, already forbids natural splittings (Theorem 4.2, Corollary 4.4). The question becomes when the class $\mathrm{ob}_k(E,D)$ obtained from (1.1) vanishes. It vanishes exactly when (E,D) is *good for Banach values in degree k*: $A^k_{E,D;F}$ is analytic for every Banach space F; the pair is *good in degree k* if this holds for every normed space F (Definition 3.3, Theorem A).

The class vanishes under a *source splitting* (S^k_E), a bounded section of $\widehat{\wedge}^k_E$; a *target splitting* (R^k_D), a bounded retraction of the antisymmetrization alt^k_D; or a *parameter splitting*, a bounded projection of $\widehat{\otimes}^k_\pi\mathcal{L}(E,D)$ onto the closed span of the powers $u^{\otimes k}$, which makes $\mathcal{L}(E,D)$ *polarizable in degree k*. Relative splittings, decompositions and block assemblies combine them. The class can be nonzero: *forced asymmetry* makes lifts asymmetric (Section 4), and over a finite residue field *Ramsey forcing* turns this asymmetry into a triangular pattern of coefficients that norm estimates forbid (Section 7).

Main results.

Theorem A (descent and one obstruction class)**.** *Let E and D be normed spaces and $k\geq 1$.*

1. *For every normed space F, $A^k_{E,D;F}$ is C^∞ on $\mathcal{L}(E,D)$; it is analytic at one point if and only if it has a lift, and then it is a continuous polynomial.*
2. *The following are equivalent:* (a) *(E,D) is good for Banach values in degree k;* (b) *$u\mapsto\Lambda^k_\pi u$ has a bounded k-linear lift $\mathcal{L}(E,D)^k\to\mathcal{L}(\Lambda^k_\pi E,\widehat{\Lambda}^k_\pi D)$;* (c) *$\mathrm{ob}_k(E,D)=0$, where $\mathrm{ob}_k(E,D)$ is the pushout of the class of (1.1) along the restriction $\mathrm{ev}^k_{E,D}|_{\mathcal{N}^k_E}\colon\mathcal{N}^k_E\to\mathrm{Mult}^k_0(\mathcal{L}(E,D);\widehat{\Lambda}^k_\pi D)$ of the evaluation map $\mathrm{ev}^k_{E,D}(x_1\otimes\cdots\otimes x_k)(u_1,\dots,u_k)=u_1x_1\wedge\cdots\wedge u_kx_k$, with Mult^k_0 the bounded k-linear maps with zero diagonal;* (d) *some bounded k-linear map $\Psi\colon E^k\to\mathrm{Mult}^k_0(\mathcal{L}(E,D);\widehat{\Lambda}^k_\pi D)$ satisfies $\Psi(x_1,\dots,x_k)=\mathrm{ev}^k_{E,D}(x_1\otimes\cdots\otimes x_k)$ whenever two of the x_a coincide.*
3. *Let $(e_t)_{t\in\Sigma}$ be a linearly independent family in E with dense span V, indexed by a totally ordered set Σ; every normed space has one. The* ordering lift *$Q_<(u)$, for $u\in\mathcal{L}(E,D)^k$, is the linear map $\Lambda^k_K V\to\widehat{\Lambda}^k_\pi D$ sending $e_{t_1}\wedge\cdots\wedge e_{t_k}$, $t_1<\cdots<t_k$, to $u_1e_{t_1}\wedge\cdots\wedge u_ke_{t_k}$.*

Composed with the canonical map $\Lambda^k_K V \to \Lambda^k_\pi E$, every lift Q as in (b) *is $Q_< + \Phi$ for a correction Φ, k-linear in u with $\Phi(v,\dots,v) = 0$ for $v \in \mathcal{L}(E,D)$ (Theorem 3.18). The pair (E,D) is good for Banach values in degree k if and only if, for some such Φ and some C, $\|Q_<(u)y + \Phi(u)y\| \le C\|u_1\| \cdots \|u_k\|\|y\|_\pi$ for all u and $y \in \Lambda^k_K V$, where $\|y\|_\pi$ is the projective seminorm.*

Theorem A collects Lemma 2.4, Propositions 2.2 and 2.5, Lemma 3.4, Theorem 3.15, Corollary 3.20 and Theorem 3.18.

Theorem B (canonical splittings exist exactly when $k! \neq 0$)**.** *Let $k \ge 2$.*

1. *If $k! \neq 0$ in K, then $\mathrm{alt}^k_E/k!$, $\wedge^k_D/k!$ and the symmetrizer $(1/k!)\sum_{\sigma\in S_k}\sigma$ on $\widehat{\otimes}^k_\pi P$ split the source, target and parameter sequences naturally in E, D and P. Every pair is good in degree k, and Alt^k is analytic on every hom space.*
2. *Suppose $k! = 0$ in K. For every norm on K^k with bounded coordinate functionals, no section of $\wedge^k_{K^k}$, retraction of $\mathrm{alt}^k_{K^k}$ or projection onto the symmetric tensors commutes with the permutation matrix of one p-cycle of the coordinates. If E has k linearly independent bounded functionals and $\widehat{\Lambda}^k_\pi D \neq 0$, no lift of $u \mapsto \Lambda^k_\pi u$ is symmetric, no section of $\widehat{\wedge}^k_E$ takes sign-isotypic values and no retraction of alt^k_D is sign-equivariant.*
3. *If $k! = 0$ in K, then $A^k_{E,E;F}$ is analytic at no point for some Banach spaces E and F, so Alt^k is analytic on all hom spaces, or on those of Banach spaces, if and only if $k! \neq 0$. If K is moreover complete and not spherically complete, $A^k_{E,D;K}$ is analytic at no point for some nonarchimedean Banach spaces E and D.*
4. *If K is complete and nonarchimedean, $\mathrm{Alt}^k(-;K)$ is analytic on all hom spaces of normed spaces if and only if $k! \neq 0$ in K or K is spherically complete.*
5. *Let a group act on a set I, and on $K^{(I)}$ by permuting the unit vectors. On the algebraic powers of $K^{(I)}$, $\wedge^k$ has an equivariant section and alt^k an equivariant retraction if and only if the stabilizer of each k-subset of I induces on it a group of order nonzero in K; an equivariant projection onto the span of the pure powers exists if and only if the stabilizer of each k-multiset on I induces on its support a group of order nonzero in K. The obstruction to an equivariant section is a family of Tate classes. One element inducing a permutation of order p on some k-subset forbids all three equivariant splittings, also on completed powers of normed spaces with a compatible biorthogonal system of vectors and bounded functionals.*

Theorem B collects Propositions 4.1 and 5.1, Theorem 4.2, Proposition 4.6, Theorem 7.2, Corollary 7.3, Theorems 7.4 and 4.3, Corollary 4.4 and Proposition 4.5. Corollary 5.15 lists the bundle settings in which fiberwise alternating maps form an analytic bifunctor; Proposition 10.1, Corollary 10.2 and Proposition 10.3 give analytic atlases with nonanalytic induced alternating or cotangent atlases, the latter over complete fields of characteristic $p > 0$ that are not spherically complete, in degrees $k \ge p$; over complete fields, ambient analytic forms keep pullback, wedge products and exterior derivatives (Theorem 10.7).

Theorem C (how the class vanishes)**.** *Let E and D be normed spaces and $k \ge 1$.*

1. *The pair (E,D) is good for Banach values in degree k if (S^k_E) or (R^k_D) holds, if $\mathcal{L}(E,D)$ is polarizable in degree k, if (E,D) has a splitting decomposition of degree k, or if (E,D) is an ℓ^1 block assembly of pairs with uniformly bounded lifts in all degrees $j \le k$. For decompositions into $\mathrm{S}^{\le k}$-split and $\mathrm{R}^{\le k}$-split pieces, two pieces suffice.*
2. *If E has a linearly independent family, indexed by a totally ordered set, with dense span V such that the strict triangle projection $\mathcal{P}_<$ is bounded on $V \otimes_\pi V$, for instance one*

with a lower r-estimate for some $r < 2$, then (S^n_E) and (R^n_E) hold and E is polarizable in degree n, for every $n \geq 1$.

(3) *If K is nonarchimedean, E has a bounded linearly independent family indexed by a totally ordered set, with dense span on which E is ℓ^2-dominated, and D has an equivalent r-norm, $r \in (2, \infty)$, or an equivalent nonarchimedean norm, then (E, D) is good for Banach values in every degree.*

(4) *Suppose* $\operatorname{char} K = p > 0$. *The least degree, if any, in which (E, D) is not good, or not good for Banach values, is a power of p. If E^* and D^* are infinite-dimensional, the set of degrees in which (E, D) is good, or good for Banach values, is $\{1, \dots, p^h - 1\}$ for some $h \geq 1$ or $\{1, 2, \dots\}$. For every normed space G, each of (S^p_G), (R^p_G) and polarizability in degree p implies the same condition in every degree.*

(5) *If K is nonarchimedean and E has a bounded linearly independent family $(e_t)_{t\in\Sigma}$, indexed by a totally ordered set, with dense span V and coordinate functionals $x \mapsto x(t)$ uniformly bounded on V, then the class of (1.1) pushes out to zero along every bounded linear map from $\mathcal{N}^k_E$ to a Banach space with an equivalent nonarchimedean norm. In particular* $\mathrm{ob}_k(E, D) = 0$ *if $\widehat{\Lambda}^k_\pi D$ has such a norm.*

Theorem C collects Proposition 3.5, Corollary 3.17, Theorem 6.13, Propositions 6.15 and 6.27, Theorem 5.6, Proposition 5.7, Theorem 5.9, Corollaries 5.10 and 6.3, and Theorems 6.5, 6.6, 6.10 and 3.22. A bounded retraction of $j_{E,F}$ gives analyticity for that value space F only (Proposition 5.1, Theorems 5.2 and 5.4).

Theorem D (forcing)**.** *Assume condition* (L)*: κ is a finite field of characteristic p, K is a subfield of $\kappa((X))$ on which the absolute value $|a| = \varrho^{\operatorname{ord} a}$, $0 < \varrho < 1$, is nontrivial, and $k \geq p$.*

(1) *For $2 \leq r \leq \infty$, the pair $(\ell^r(\mathbb{N}, K), \ell^r(\mathbb{N}, K))$, with $c_0(\mathbb{N}, K)$ for $r = \infty$, is not good for Banach values in degree k.*

(2) *If symmetric sequence spaces E and D form a pair good for Banach values in degree k, then for some C, the* product triangle bound (PTB)

$$\|\mathcal{P}_<\big((ax) \otimes (by)\big)\|_{D\otimes_\pi D} \leq C\|\mu_a\|_{E\to D}\|x\|_E\|\mu_b\|_{E\to D}\|y\|_E$$

holds for all finitely supported a, x, b, y, where μ_a is multiplication by a. If moreover $\|\xi\|_D \leq C'\|\xi\|_r$ for all finitely supported ξ and some $r \in (2, \infty]$, then E is ℓ^2-dominated.

(3) *If E has an equivalent nonarchimedean norm with $\dim_{\widehat{K}} \widehat{E} = \infty$, then the pairs $(E, c_0(\mathbb{N}, \widehat{K}))$ and $(c_0(\mathbb{N}, K), E)$ are not good for Banach values in degree k.*

Theorem D collects Theorems 7.7 and 7.13, Proposition 7.15, and Theorem 7.19 with its proof, all proved in Appendix K; Lemma 7.1 reduces condition (L) to fields dense in a Laurent series field. Over every nontrivially normed field K of characteristic 2, polynomial identities replace Ramsey's theorem: for $E = c_0(\mathbb{N}, K)$, $\ell^\infty(\mathbb{N}, K)$ and $\ell^r(\mathbb{N}, K)$, $2 < r < \infty$, the pair (E, E) is not good for Banach values in any degree $k \geq 2$, and neither (S^k_E) nor (R^k_E) holds (Theorem 7.20). Over Laurent series fields over infinite fields and over densely valued fields, the case $E = c_0(\mathbb{N}, K)$ for odd p and the case $E = \ell^2(\mathbb{N}, K)$ for $p = 2$ are open (Question 12.9). In characteristic 2, Proposition 11.21 removes condition (L) from Theorems 11.14 and 11.19(4), from Theorem 11.17(3) except its last sentence, and from Corollary 11.15(2) for $E = c_0(\mathbb{N}, K)$ and for $E = \ell^{r_2}(\mathbb{N}, K)$, $r_2 > 2$.

Theorem E (good pairs in classes)**.** *Let κ, ϱ and K be as in condition* (L)*, and let good mean good for Banach values.*

(1) *Let E be a symmetric sequence space in which c_{00} is dense. The following are equivalent: (E, E) is good in some degree $k \geq p$; (c_0, E) is good in some degree $k \geq p$; $\mathcal{P}_<$ is bounded*

on $E \otimes_\pi E$; (S^k_E) holds for some $k \geq p$; (R^k_E) holds for some $k \geq p$; E is polarizable in some degree $k \geq p$; (E, D) and (D, E) are good in every degree for every normed space D. If they hold, E is ℓ^2-dominated and $\varphi_E(m)^2 \geq c\, m \ln m$ for some $c > 0$ and all $m \geq 1$, with $\varphi_E(m)$ the norm of the indicator of $\{1, \dots, m\}$.

(2) *Let E be as in (1) and D a symmetric sequence space with an equivalent r-norm, $r \in (2, \infty)$, or an equivalent nonarchimedean norm. Then (E, D) is good in some degree $k \geq p$ if and only if* (PTB) *holds, if and only if E is ℓ^2-dominated, if and only if (E, D) is good in every degree.*

(3) *For $1 \leq r_1, r_2 \leq \infty$, with c_0 in place of ℓ^∞, the pair (ℓ^{r_1}, ℓ^{r_2}) is good in some degree $k \geq p$ if and only if it is good in every degree, if and only if $\min(r_1, r_2) < 2$ or $r_1 = 2 < r_2$.*

(4) *If E has an equivalent nonarchimedean norm and $k \geq p$, then (E, D) is good in degree k for every normed space D if and only if (D, E) is good in degree k for every normed space D, if and only if $\dim_{\widehat{K}} \widehat{E} < \infty$. If K is complete and $k \geq p$, two spaces with equivalent nonarchimedean norms form a pair good in degree k if and only if one of them is finite-dimensional.*

For the pairs in (1)–(3) and in the last sentence of (4), the good degrees form $\{1, \dots, p-1\}$ or $\{1, 2, \dots\}$, and every good pair has a splitting decomposition by absolute or relative ordering splittings or by finite-coordinate splittings.

Theorem E collects Theorems 8.1, 8.4 and 7.19, Corollaries 8.5 and 8.6, and Remark 8.7. Items (1) and (2), the last sentence for the pairs in them, and Theorem D(2) hold for spreading sequence spaces, whose norms need only be invariant under increasing relabelings of the unit vectors (Theorem 8.9); this class contains spaces with no equivalent symmetric norm (Example 8.10).

Theorem F (goodness without splitting decompositions)**.** (1) *Over every separable complete nonarchimedean field K there are nonzero infinite-dimensional Banach spaces $\mathfrak{X}_1, \mathfrak{X}_2, \dots$ with $\mathcal{L}(\mathfrak{X}_\alpha, \mathfrak{X}_\alpha) = K \operatorname{id}$ and $\mathcal{L}(\mathfrak{X}_\alpha, \mathfrak{X}_\beta) = 0$ for $\alpha \neq \beta$.*

(2) *Over $\kappa((X))$ with κ finite, suitable choices in (1) give $\mathfrak{X} = \mathfrak{X}_1$ with $(\mathfrak{X}, \mathfrak{X})$ good in every degree, while for every $k \geq p$ neither $(\mathrm{S}^k_{\mathfrak{X}})$ nor $(\mathrm{R}^k_{\mathfrak{X}})$ holds and $(\mathfrak{X}, \mathfrak{X})$ has no splitting decomposition of degree k.*

(3) *Over $\kappa((X))$ with κ finite and these choices, the completed ℓ^1-sum E of the spaces $\mathfrak{X}_\alpha$ gives an ℓ^1 block assembly (E, E) that is good for Banach values in every degree and has no splitting decomposition of any degree $k \geq p$, even with parameter pieces.*

Theorem F collects Theorems 9.2, 9.3 and 9.4, proved in Appendix M.

Bundles need goodness only for triples $(P; E, D)$, along analytic operator families on open subsets of the base model P (Section 11). Goodness of a triple reduces to homogeneous polynomial families and to familywise classes (Theorems 11.5 and 11.7). Under (L), $(\ell^2(\mathbb{N}, K); E, E)$ with $E = \ell^r(\mathbb{N}, K)$, $2 < r < \infty$, is good in every degree, although neither is $\ell^2(\mathbb{N}, K)$ polarizable nor (E, E) good for Banach values in degree k (Theorems 11.9 and 11.13, Proposition 11.11, Corollary 8.2, Theorem 7.7). For a structure group with an open subgroup that lies in a linear space of operators and contains a ball of it around the identity, goodness is that of the linear space, not that of the closed linear span of the group (Theorem 11.19, Proposition 11.20).

Theorem G (what bundles need)**.** *Let P be a normed space and $k \geq 1$.*

(1) *Let $\mathcal{C}$ be a class of Banach spaces closed under finite direct sums. Fiberwise alternating maps with Banach value fibers preserve the analyticity of bundles with typical fibers in $\mathcal{C}$ over every analytic manifold modeled on P if and only if $x \mapsto \Lambda^k_\pi u(x)$ is analytic for*

every analytic family u of operators between spaces of $\mathcal{C}$ on an open subset of P. For a single Banach fiber, families of invertible operators give the same condition.

(2) *The condition holds for all normed fibers if P is polarizable in every degree $n \le k$, and over every P for fibers E, D with $(E, \widehat{D})$ good for Banach values in degree k. In characteristic p, the condition in all degrees $j \le k$ is decided by the lifting constants of homogeneous polynomial families in the degrees $p^i \le k$. Under* (L)*, it holds for all normed fibers over $P = \ell^r(\mathbb{N}, K)$, $1 \le r < \infty$, if and only if $r < 2$, and it fails for some normed fibers over every P with an equivalent nonarchimedean norm and $\dim_{\widehat{K}} \widehat{P} = \infty$, such as $c_0(\mathbb{N}, K)$.*

(3) *Let K be complete and P a Banach space with $P \cong P^3$. The bundles $\widehat{\Lambda}^k_\pi TM$ are analytic for every analytic manifold M modeled on P if and only if the condition holds for $E = D = P$. Under* (L)*, $c_0(\mathbb{N}, K)$ and $\ell^r(\mathbb{N}, K)$, $4 < r < \infty$, model manifolds M with a nonanalytic bundle $\widehat{\Lambda}^k_\pi TM$.*

Theorem G collects Proposition 11.2, Corollary 11.3, Theorems 11.5, 11.9 and 11.10, Proposition 11.11 and Theorem 11.17.

Section 14 treats ambient analyticity as a structure, not a class of maps. Fixed envelopes come first: in every characteristic the transitions act analytically on Mult^k, preserving the alternating forms, and this gives n° 7.8.1–7.8.4 of [3] on ambient sections of filtered analytic bundles (Theorem 14.4). If the enlargement may depend on the map, the maps so obtained are not a usable class: over every field of characteristic $p > 0$ they are not closed under composition (Theorem 14.17), and the mechanism is that enlarging the target carries maps into the space, not analytic maps out of it (Section 14.4). For K complete and scalar values, $\mathrm{Mult}^k(-; K)$ is initial among the functorial envelopes of $\mathrm{Alt}^k(-; K)$ with a multilinear action (Theorem 14.32). On tame pairs, such as ultrametric spaces or $\ell^1(J, K)$, ambient analyticity has composition and inverse functions, and there it coincides with divided-power analyticity, which takes bounded polynomial laws as Taylor coefficients (Theorems 14.38 and 14.39). A sequel [II] studies divided-power analyticity.

Section 12 collects questions; those on ambient analyticity are at the end of Section 14, and the proofs of that section are in Appendix Q. Every theorem, proposition, lemma and corollary of the paper is proved in the text, except Theorem D.3 and Lemma M.9, which are cited; Section 13 lists the statements that also have Lean proofs of corresponding statements. Sections 2, 3.1, 3.2, 5.1, 5.3, 5.4, 7.2 and 10 with Appendices A–I form the formalized core; besides it, they use only Definition 3.8, the sign $\epsilon(S)$ of Section 6.1 and condition (L) of Section 7.1. Proofs that would interrupt the text are collected in Appendix P.

Related work. The bundle construction follows Lang's functors of class C^n [15, Ch. III, §4] and Bourbaki's vector functors [3, n° 7.6]. Bourbaki defines the alternating bifunctor when K has characteristic 0 or the bundles have finite rank, and states that it is not known whether this restriction can be removed [3, n° 7.8]; the paragraph *Bourbaki's question* in Section 10 shows that it cannot over complete fields of characteristic $p > 0$ that are not spherically complete. Gouëzel's construction [7, 8] covers characteristic zero; the finite-dimensional case in positive characteristic, noted in [7], is Proposition 5.1(2) over complete fields. Closed-range reflection of smoothness is formalized in [18]. The nonarchimedean extension mechanism is a norm-controlled form of spherical completeness [11, 23, 24, 26]. Completed nonarchimedean exterior powers go back to Serre [25] and Gruson [9], who do not study sections of the quotient map. The rigid spaces of Section 9 transfer a construction of Kalton and Roberts [14]. Corollary 3.21 detects the class by the distance of a z-linear map from the linear maps, as for twisted sums [12, 13, 5].

Conventions. $\mathcal{L}(E,F)$ denotes bounded linear maps with the operator norm, $\mathrm{Mult}^k(E;F)$ bounded k-linear maps, and $\mathrm{Alt}^k(E;F)$ those vanishing whenever two arguments coincide; this condition, not antisymmetry, defines alternation in characteristic two. The degree-zero space is F. Products carry the maximum norm. E denotes a source fiber, D a target fiber, F a value space (a *target fiber* in Corollary 5.15 and Appendix C), G an intermediate space, and P a base model or parameter space. *Finite coordinates* means a bounded linear isomorphism with K^d; equivalently, a basis $e_1,\dots,e_d$ with bounded coordinate functionals $e_1^*,\dots,e_d^*$, which we also call finite continuous coordinates (e_i,e_i^*); over a complete field this is finite dimension. Hats denote completions (Section 3). Exponents are written r; ℓ^r, lower r-estimates, r-norms and ℓ^2-domination of a family are defined in Section 5, symmetric sequence spaces in Definition 7.10. A finite field κ has $|\kappa|$ elements. Condition (L) is fixed in Section 7. A norm is *nonarchimedean* if $\|x+y\| \le \max(\|x\|,\|y\|)$. Every normed field of characteristic $p>0$ is nonarchimedean: Frobenius gives $|x+y|^{p^m} \le 2\max(|x|,|y|)^{p^m}$; take roots and let $m\to\infty$.

C^n ($n\le\infty$) means iterated Fréchet differentiability; analyticity means local expansion in bounded multilinear coefficients with a positive norm-controlled radius. On manifolds, C^ω also requires analytic iterated derivatives, as in Mathlib [16]. On open sets the two agree, even for incomplete ranges: the derivative series converges in the completion to the Fréchet derivative, which exists in the operator space since nearby points are points of analyticity.

2. Alternating bundles and analytic descent

Throughout, Vec_K is the category of normed K-spaces, not necessarily complete, and bounded linear maps, with hom spaces $\mathcal{L}(E,D)$. For $n\in\mathbb{N}\cup\{\infty,\omega\}$, C^n is as in the conventions of Section 1, and C^0 means continuous.

2.1. **The bundle construction.** Fix a C^n manifold M modeled on a normed space P. Write $\mathsf{VB}^n_K(M)$ for the category of C^n normed vector bundles over M, with morphisms over id_M whose local *operator-valued* expressions, not merely the maps of total spaces, are C^n. Thus a morphism from a bundle with fiber E to one with fiber F is locally a C^n map $x\mapsto T(x)\in\mathcal{L}(E,F)$. This condition is independent of the trivializations, since composition is bounded bilinear.

Let $\epsilon=(\epsilon_1,\dots,\epsilon_r)$ specify a variance in each variable, and put $\mathbf{C}^\epsilon=\mathsf{C}_1\times\cdots\times\mathsf{C}_r$ with $\mathsf{C}_a=\mathsf{Vec}_K$ or $\mathsf{Vec}_K^{\mathrm{op}}$. A functor $\mathcal{F}\colon\mathbf{C}^\epsilon\to\mathsf{Vec}_K$ is C^n if all its maps on products of hom spaces are jointly C^n. Let $\mathbf{B}^\epsilon_M$ denote the corresponding product of bundle categories, with the same variances. Natural transformations between such functors have bounded linear components, since their codomain is Vec_K.

Theorem 2.1 (lifting linear constructions to bundles)**.** *For a fixed C^n manifold M, fiberwise application defines, up to canonical natural isomorphism, a functor*

$$\mathcal{B}^n_M\colon \mathrm{Fun}^{C^n}(\mathbf{C}^\epsilon,\mathsf{Vec}_K)\longrightarrow \mathrm{Fun}(\mathbf{B}^\epsilon_M,\mathsf{VB}^n_K(M)),$$
$$\mathcal{F}\longmapsto\mathcal{F}_M,\qquad \eta\longmapsto\eta_M.$$

Here the category on the left is the full subcategory of the ordinary functor category on the functors regular on hom spaces. The resulting bundles, bundle morphisms, and natural transformations are given fiberwise by $\mathcal{F}$, its action on maps, and the components of η, respectively.

The same conclusion holds if regularity on whole hom spaces is replaced by preservation of C^n families parametrized by open subsets of the model space P.

Proof. Step 1 (transitions). Over a common trivializing cover of the input bundles, let the C^n operator-valued maps g^a_{ij} convert coordinates of the ath bundle from chart j to chart i, so that $g^a_{ij}g^a_{j\ell}=g^a_{i\ell}$ and $g^a_{ii}=\mathrm{id}$. Let $\mathbf{g}_{ij}(x)$ be the tuple of arrows in $\mathbf{C}^\epsilon$ with entry $g^a_{ij}(x)$ in

a covariant variable and $g^a_{ji}(x)$ in a contravariant one. By functoriality, $\theta_{ij}(x) = \mathcal{F}(\mathbf{g}_{ij}(x))$ satisfies $\theta_{ij}\theta_{j\ell} = \theta_{i\ell}$. The maps θ_{ij} are C^n by composition, so they glue the products with typical fiber $\mathcal{F}(E_1, \dots, E_r)$ into a C^n vector bundle; its coordinate changes are C^n because operator evaluation is bounded bilinear.

Step 2 (morphisms). Applied to the local operator families of a tuple of bundle morphisms, reversed in the contravariant variables, $\mathcal{F}$ gives families compatible with the θ_{ij} by functoriality and C^n by composition; this defines $\mathcal{F}_M$. The same argument for changes of trivialization identifies different choices canonically and preserves identities and composition.

Step 3 (natural transformations). For $\eta\colon \mathcal{F} \Rightarrow \mathcal{G}$ and $\mathbf{E} = (E_1, \dots, E_r)$, let η_M be locally the constant operator $\eta_{\mathbf{E}}$. Naturality gives $\eta_{\mathbf{E}}\,\mathcal{F}(\mathbf{g}_{ij}(x)) = \mathcal{G}(\mathbf{g}_{ij}(x))\,\eta_{\mathbf{E}}$, so η_M is a bundle morphism, C^n because its local expression is constant. The same identity for arbitrary input morphisms proves naturality of η_M, and identities and compositions of transformations are preserved fiberwise.

Step 4 (familywise version). In a base chart, every transition and local morphism family is a C^n family on an open subset of P, so preservation of such families gives the regularity used in Steps 1 and 2; Step 3 uses none. □

The familywise hypothesis, that $\mathcal{F} \circ \gamma$ be C^n for each C^n map γ on an open subset of P into $\prod_a \operatorname{Hom}_{\mathsf{C}_a}(E_a, D_a)$, is weaker than regularity on whole hom spaces. The theorem also holds for restricted categories of fibers containing the bundle morphisms; compare [8, 15] and [3, n°s 7.6.1–7.6.4].

2.2. **Alternating maps inside multilinear maps.** For normed spaces E, F, let $\operatorname{Mult}^k(E; F)$ be the normed space of bounded k-linear maps $E^k \to F$, and $\operatorname{Alt}^k(E; F)$ its subspace of alternating maps, those vanishing whenever two arguments agree. The inclusion

$$j_{E,F}\colon \operatorname{Alt}^k(E; F) \longrightarrow \operatorname{Mult}^k(E; F)$$

is a linear isometry with closed range, since alternation is an intersection of closed conditions under evaluation at fixed tuples. Both constructions are contravariant in E and covariant in F, and the inclusions form a natural transformation. If F is complete, so are both spaces, since operator-norm Cauchy families have bounded pointwise limits satisfying the same identities; so the alternating bifunctor restricts to Banach spaces.

For $f \in \mathcal{L}(E, D)$ and $g \in \mathcal{L}(F, F')$, their joint action is

$$B(f, g)(m)(x_1, \dots, x_k) = g\big(m(fx_1, \dots, fx_k)\big).$$

On alternating maps, write

$$A = A^k_{E,D;F}\colon \mathcal{L}(E, D) \longrightarrow \mathcal{L}(\operatorname{Alt}^k(D; F), \operatorname{Alt}^k(E; F)), \qquad A(f) = B(f, \mathrm{id}_F),$$

for precomposition. Since $\|B(f, g)(m)\| \le \|g\|\,\|m\|\,\|f\|^k$, these actions are bounded operators.

Proposition 2.2 (ambient polynomial action)**.** *The joint action on multilinear maps*

$$B\colon \mathcal{L}(E, D) \times \mathcal{L}(F, F') \longrightarrow \mathcal{L}\big(\operatorname{Mult}^k(D; F), \operatorname{Mult}^k(E; F')\big)$$

is the diagonal of a bounded $(k+1)$-linear map, and hence is C^ω. For the second argument id_F*, the precomposition action $A^k_{E,D;F}$ followed by $j_{E,F} \circ -$ is the diagonal of the bounded k-linear map $(f_1, \dots, f_k) \mapsto \big[m \mapsto m \circ (f_1, \dots, f_k)\big]$, read on alternating m, of norm at most 1. The restriction of B to alternating inputs is analytic as a map into*

$$\mathcal{L}\big(\operatorname{Alt}^k(D; F), \operatorname{Mult}^k(E; F')\big),$$

and all its values belong to the closed subspace of operators taking values in $\operatorname{Alt}^k(E; F')$*.*

Proof. For $v_i = (f_i, g_i)$ in the product of hom spaces, define

$$\Phi(v_0, \dots, v_k)(m)(x_1, \dots, x_k) = g_0\big(m(f_1 x_1, \dots, f_k x_k)\big).$$

This map is multilinear in the v_i, has norm at most 1, and has diagonal B. It depends on v_0 only through g_0, so fixing $g_0 = \mathrm{id}_F$ leaves a k-linear map of norm at most 1 whose diagonal, read on alternating m, is $j_{E,F} \circ A^k_{E,D;F}$. A diagonal of a bounded multilinear map has finite polynomial derivatives of every order, so it is C^ω even with incomplete range. Restricting m to alternating maps is bounded linear and preserves analyticity. On the diagonal, two equal x_i give two equal fx_i, so the output is alternating; the operator subspace is closed because $j_{E,F'}$ has closed range, applied pointwise. For $k = 0$, the formula is postcomposition by g_0. □

For bundles with analytic transition families g_{ij} and h_{ij}, the induced alternating transitions $m \mapsto h_{ij}(x) \circ m \circ (g_{ji}(x), \dots, g_{ji}(x))$ are therefore analytic as maps into the multilinear operator space. Whether their power-series coefficients can be chosen in the alternating operator space is the descent problem of Section 2.3.

2.3. **Analytic descent through closed subspaces.** Throughout this subsection, P, W, Z are normed K-spaces, not necessarily complete, and $j\colon W \to Z$ is a linear isometry with closed range; we identify W with its image when convenient. If $U \subseteq P$ is open and $f\colon U \to W$, an ambient expansion at $x_0 \in U$ means

$$jf(x_0 + h) = \sum_{n \geq 0} b_n(h, \dots, h), \qquad \sup_n \|b_n\| R^n < \infty \tag{2.1}$$

for some $R > 0$, with bounded n-linear $b_n\colon P^n \to Z$ (b_0 a constant) and convergence for $\|h\| < R$. For $j = \mathrm{id}_W$ this is, after shrinking R, the usual local definition of analyticity. For nonarchimedean spaces over a complete nonarchimedean field, this use of multilinear representatives agrees with the polynomial lifting norms of [2, 7.14–7.19].

Theorem 2.3 (the coefficient descent criterion)**.** *Suppose* (2.1) *holds. Then every diagonal $b_n(h, \dots, h)$ belongs to $j(W)$. The map f is analytic at x_0 if and only if there are bounded n-linear maps $q_n\colon P^n \to W$ and $r > 0$ such that*

$$jq_n(h, \dots, h) = b_n(h, \dots, h) \quad (n \geq 0,\ h \in P), \qquad \sup_n \|q_n\| r^n < \infty.$$

Only the diagonals must agree; the original mixed values of b_n need not lie in $j(W)$.

Proof. Step 1 (diagonals in $j(W)$). Fix $h \in P$. On the scalar line $x_0 + th$, the coefficients $b_n(h, \dots, h)$ of (2.1) satisfy a geometric bound, and $b_0 = jf(x_0)$. If the diagonals of degree below n lie in $j(W)$, then for small nonzero t

$$t^{-n}\Big(jf(x_0 + th) - \sum_{m<n} t^m b_m(h, \dots, h)\Big) \in j(W),$$

and this vector tends to $b_n(h, \dots, h)$ as $t \to 0$, by the tail estimate in the proof of Lemma A.1; since $j(W)$ is closed, $b_n(h, \dots, h) \in j(W)$.

Step 2 (the criterion). If f is analytic, apply j to one of its expansions; on every scalar line, Lemma A.1, applied to the difference of the two series, gives the diagonal identities. Conversely, shrink r below R and let $\|h\| < r$. The partial sums of $\sum_n q_n(h, \dots, h)$ in W have images converging to $jf(x_0 + h)$, so they converge to $f(x_0 + h)$ because j is an isometry, and the bound on $\|q_n\| r^n$ gives a positive radius. □

Lemma 2.4 (smooth descent)**.** *For $n \in \mathbb{N} \cup \{\infty\}$, a map $f\colon U \to W$ is C^n if and only if jf is C^n.*

Proof. Continuity reflects through an isometry. If jf has Fréchet derivative T at x, then $Th = \lim_{t\to 0,\, t\neq 0} t^{-1}(jf(x+th) - jf(x)) \in j(W)$, so $T = jS$ for a unique bounded linear $S\colon P \to W$ of the same norm; the isometry transfers the remainder estimate, so $Df(x) = S$. Postcomposition by j is a closed linear isometry $\mathcal{L}(P, W) \to \mathcal{L}(P, Z)$, since its image consists of the operators with values in $j(W)$, a closed condition under each evaluation. The same argument applied to the derivative maps proves every finite order by induction, hence C^∞. □

By Proposition 2.2 and Lemma 2.4 with postcomposition by $j_{E,F'}$ as j, the joint action on alternating maps is C^∞, and so is its case $g = \mathrm{id}_F$, the precomposition action $A^k_{E,D;F}$; this is the argument formalized in [18]. With Theorem 2.1, this constructs alternating bundles and their morphisms at every finite order and at C^∞; analyticity is the additional coefficient problem.

We call a finite sum of diagonals of bounded multilinear maps a *continuous polynomial*.

Proposition 2.5 (one lift, every base point)**.** *Let $k \geq 1$, and suppose $a\colon H \to W$ satisfies $ja(h) = \beta(h, \dots, h)$ for a bounded k-linear map $\beta\colon H^k \to Z$. The following are equivalent:*

1. *a is analytic at some point of H;*
2. *there is a bounded k-linear $Q\colon H^k \to W$ with $Q(h, \dots, h) = a(h)$;*
3. *a has a finite power-series expansion at every point, of infinite radius.*

Consequently each precomposition action $A^k_{E,D;F}$ is either a continuous polynomial on its whole domain or analytic nowhere.

Proof. An expansion at h_0, compared along $h_0 + th$ with the polynomial $\beta(h_0 + th, \dots, h_0 + th)$, has degree-$k$ diagonal $\beta(h, \dots, h)$ by Lemma A.1; its degree-k coefficient is the required Q. Conversely, expanding $Q(h_0 + h, \dots, h_0 + h)$ by multilinearity gives a finite expansion whose degree-i coefficient, the sum over the i occupied slots, has norm at most $\binom{k}{i}\|Q\|\|h_0\|^{k-i}$. The last assertion follows from Proposition 2.2. □

Retractions are splittings. A bounded linear retraction ρ of $j_{E,F}$ turns the representative $\beta(u_1, \dots, u_k)(m) = m \circ (u_1, \dots, u_k)$ of $j_{E,F} \circ A^k_{E,D;F}$ into the lift $\rho \circ \beta$ (Proposition 5.1). Section 3 treats all Banach F at once by one exact sequence and one class (Theorem 3.15).

3. Exterior powers and the obstruction class

3.1. **Projective tensor and exterior powers.** For $k \geq 1$ give the algebraic exterior power $\Lambda^k_K E$ the seminorm

$$\|z\|_\pi = \inf_{z = \sum_j x_{j1} \wedge \cdots \wedge x_{jk}} \sum_j \prod_{a=1}^{k} \|x_{ja}\|.$$

Its quotient by the zero-seminorm subspace is the normed space $\Lambda^k_\pi E$, with universal alternating map $\omega_E(x_1, \dots, x_k) = [x_1 \wedge \cdots \wedge x_k]$. Put $\Lambda^0_\pi E = K$ and $\omega_E() = 1$.

Proposition 3.1 (representation)**.** *There is a natural linear isometry*

$$\mathrm{Alt}^k(E; F) \cong \mathcal{L}(\Lambda^k_\pi E, F)$$

for every normed F. Under it, precomposition by $u\colon E \to D$ is precomposition by $\Lambda^k_\pi u\colon \Lambda^k_\pi E \to \Lambda^k_\pi D$.

Proof. The algebraic linearization $\widetilde{m}$ of a bounded alternating map m satisfies $\|\widetilde{m}(z)\| \leq \|m\|\|z\|_\pi$, by the definition of $\|z\|_\pi$; so it kills the null subspace and descends contractively. Conversely, $\|\omega_E(x_1, \dots, x_k)\| \leq \prod_a \|x_a\|$, so composing a bounded linear map with ω_E is

contractive. The two operations are inverse, hence isometric. Applying u to each wedge factor gives the intertwining identity and $\|\Lambda^k_\pi u\| \le \|u\|^k$ for $k \ge 1$; in degree zero the action is the identity on K. □

If F is complete, the same correspondence holds with the completion $\widehat{\Lambda^k_\pi E}$.

The projective tensor product $X_1 \otimes_\pi \cdots \otimes_\pi X_n$ of normed spaces, and $\otimes^k_\pi E$ with k factors E, are defined in the same way with $\otimes$ in place of $\wedge$. As in Proposition 3.1, bounded n-linear maps m on $X_1 \times \cdots \times X_n$ correspond isometrically to bounded linear maps $m^\otimes$ on $X_1 \otimes_\pi \cdots \otimes_\pi X_n$. For bounded operators u_i, $u_1 \otimes \cdots \otimes u_n$ has norm at most $\prod_i \|u_i\|$ and is multilinear in $(u_1, \dots, u_n)$; $u^{\otimes k} = u \otimes \cdots \otimes u$. Hats denote completions, maps extend to completions by continuity, and every normed space is a subspace of its completion. Let

$$\wedge^k_E \colon \otimes^k_\pi E \to \Lambda^k_\pi E, \qquad \mathrm{alt}^k_E \colon \Lambda^k_\pi E \to \otimes^k_\pi E$$

be the maps with $x_1 \otimes \cdots \otimes x_k \mapsto x_1 \wedge \cdots \wedge x_k$ and $x_1 \wedge \cdots \wedge x_k \mapsto \sum_{\sigma \in S_k} \operatorname{sgn}(\sigma)\, x_{\sigma(1)} \otimes \cdots \otimes x_{\sigma(k)}$. The first is $\omega^\otimes_E$, a contraction. The second is well defined in every characteristic, since for $x_a = x_b$ the permutations σ and $(a\ b) \circ \sigma$ give equal tensors with opposite signs; its norm is at most the real number $k!$, and $\wedge^k_E \circ \mathrm{alt}^k_E = k!$. Both maps are natural: $\wedge^k_D \circ u^{\otimes k} = \Lambda^k_\pi u \circ \wedge^k_E$ and $u^{\otimes k} \circ \mathrm{alt}^k_E = \mathrm{alt}^k_D \circ \Lambda^k_\pi u$ for $u \in \mathcal{L}(E, D)$.

3.2. **Universal and scalar tests.** Write $Y^* = \mathcal{L}(Y, K)$ and $J_Y(y)(\lambda) = \lambda(y)$ for the canonical contraction into the continuous bidual; no point-separation or Hahn–Banach assumption is implicit.

Theorem 3.2 (universal and scalar tests)**.** *Let U be an open parameter domain and $W \colon U \to \mathcal{L}(X, Y)$ a family. Define*

$$R_F(t)(T) = T \circ W(t), \qquad R_F \colon U \to \mathcal{L}(\mathcal{L}(Y, F), \mathcal{L}(X, F)).$$

For power-series analyticity, or for any fixed class C^n, the following statements hold.

1. *W is regular if and only if R_F is regular for every normed F; the single target Y suffices.*
2. *R_K is regular if and only if $J_Y W$ is regular in $\mathcal{L}(X, Y^{**})$, and if and only if R_{G^*} is regular for every normed G. Any one nonzero continuous dual G^* suffices for the last test.*
3. *If J_Y has a bounded linear left inverse, regularity of R_K implies regularity of W. If J_F has such a left inverse, regularity of R_K implies regularity of R_F.*

Proof. All identities below use bounded linear operations, which preserve each regularity class. Precomposition depends bounded-linearly on W, and $W(t) = R_Y(t)(\mathrm{id}_Y)$; this proves (1). Exchanging the arguments defines a linear isometry

$$\Theta \colon \mathcal{L}(Y^*, X^*) \longrightarrow \mathcal{L}(X, Y^{**}), \qquad (\Theta(B)x)(\lambda) = (B\lambda)(x),$$

since both operator norms are the least C bounding the displayed scalar by $C\|x\|\|\lambda\|$; its inverse exchanges the arguments again, and $\Theta(R_K(t)) = J_Y W(t)$. The isometry $\mathcal{L}(X, G^*) \cong \mathcal{L}(G, X^*)$, $T \mapsto [g \mapsto (x \mapsto T(x)(g))]$, identifies $R_{G^*}(t)$ with $S \mapsto R_K(t) \circ S$, which gives regularity for every dual target. Conversely, choose $v \in G^*$ and $g \in G$ with $v(g) = 1$; the injection $\alpha \mapsto v\alpha$ and evaluation at g compress R_{G^*} to R_K. This proves (2). For (3), if $P_Y J_Y = \mathrm{id}_Y$, then $W = P_Y \Theta(R_K)$; if $P_F J_F = \mathrm{id}_F$, then $R_F(t)(T) = P_F \circ R_{F^{**}}(t)(J_F \circ T)$. □

3.3. **Good pairs and one test target.** A *Banach space* is a complete normed K-space.

Definition 3.3. A pair (E, D) of normed spaces is *good in degree k* if $A^k_{E,D;F}$ is analytic for every normed F, and *good for Banach values in degree k* if this holds for every Banach space F.

Lemma 3.4 (one test target)**.** *The pair (E, D) is good in degree k if and only if $u \mapsto \Lambda^k_\pi u$ is analytic at some point of $\mathcal{L}(E, D)$, and if and only if there is a bounded k-linear $Q\colon \mathcal{L}(E, D)^k \to \mathcal{L}(\Lambda^k_\pi E, \Lambda^k_\pi D)$ with $Q(u, \dots, u) = \Lambda^k_\pi u$. The same holds for Banach values, with $\mathcal{L}(\Lambda^k_\pi E, \widehat{\Lambda}^k_\pi D)$ in place of $\mathcal{L}(\Lambda^k_\pi E, \Lambda^k_\pi D)$, also as the space in which $u \mapsto \Lambda^k_\pi u$ is analytic.*

Proof. Apply Theorem 3.2(1) with $X = \Lambda^k_\pi E$, $W(u) = \Lambda^k_\pi u$, and $Y = \Lambda^k_\pi D$, respectively $Y = \widehat{\Lambda}^k_\pi D$. By Proposition 3.1 and its completed form, R_F is $A^k_{E,D;F}$ for every normed F, respectively every Banach F, and the target Y suffices. Proposition 2.5 applies to W, with the closed isometry $T \mapsto T \circ \omega_E$ of $\mathcal{L}(\Lambda^k_\pi E, Y)$ onto $\mathrm{Alt}^k(E; Y) \subseteq \mathrm{Mult}^k(E; Y)$ and the ambient map $(u_1, \dots, u_k) \mapsto \omega_D \circ (u_1, \dots, u_k)$. □

We use two splitting conditions on normed spaces:

- (S^k_E) there is a bounded linear $s\colon \Lambda^k_\pi E \to \widehat{\otimes}^k_\pi E$ with $\widehat{\wedge}^k_E \circ s = \mathrm{id}$;
- (R^k_D) there is a bounded linear $\rho\colon \otimes^k_\pi D \to \widehat{\Lambda}^k_\pi D$ with $\rho \circ \mathrm{alt}^k_D = \mathrm{id}$.

They hold *without completion* if s takes values in $\otimes^k_\pi E$, respectively ρ in $\Lambda^k_\pi D$. Both hold without completion if $k! \neq 0$ in K, with $s = \mathrm{alt}^k_E / k!$ and $\rho = \wedge^k_D / k!$, and if the space has finite continuous coordinates (e_i, e_i^*), with

$$s(x_1 \wedge \cdots \wedge x_k) = \sum_{i_1 < \cdots < i_k} \det\big(e^*_{i_a}(x_b)\big)_{a,b}\, e_{i_1} \otimes \cdots \otimes e_{i_k},$$

$$\rho(y_1 \otimes \cdots \otimes y_k) = \sum_{i_1 < \cdots < i_k} \prod_a e^*_{i_a}(y_a)\, e_{i_1} \wedge \cdots \wedge e_{i_k}.$$

Proposition 3.5 (splittings)**.** *Let E and D be normed spaces.*

1. *If (S^k_E) holds, then (E, D) is good for Banach values in degree k for every normed D: $Q(u_1, \dots, u_k) = \widehat{\wedge}^k_D \circ (u_1 \otimes \cdots \otimes u_k) \circ s$ is a lift as in Lemma 3.4, of norm at most $\|s\|$.*
2. *If (R^k_D) holds, then (E, D) is good for Banach values in degree k for every normed E, with the lift $Q(u_1, \dots, u_k) = \rho \circ (u_1 \otimes \cdots \otimes u_k) \circ \mathrm{alt}^k_E$ of norm at most $k!\|\rho\|$.*
3. *If the condition holds without completion, the pairs in (1), respectively (2), are good in degree k.*
4. *(S^k_E) holds if and only if the inclusions $j_{E,F}\colon \mathrm{Alt}^k(E; F) \to \mathrm{Mult}^k(E; F)$, for all Banach spaces F, have bounded linear retractions that commute with postcomposition by bounded linear maps $F \to F'$. It holds without completion if and only if such retractions exist for all normed F.*

The proof is in Appendix P.

3.4. **Three exact sequences.** A sequence $\mathcal{Y}\colon 0 \to Y_1 \xrightarrow{i} Y_2 \xrightarrow{q} Y_3 \to 0$ of Banach spaces and bounded linear maps is *exact* if i is injective, $i(Y_1) = \ker q$, and for a constant C, $\|y\| \le C\|iy\|$ on Y_1 and every $z \in Y_3$ is qy for some y with $\|y\| \le C\|z\|$. It *splits* if q has a bounded linear right inverse. Exact sequences $\mathcal{Y}, \mathcal{Y}''$ with the same end terms are *equivalent* if a bounded linear $T\colon Y_2 \to Y_2''$ satisfies $Ti = i''$ and $q''T = q$; such a T is bijective with bounded inverse, since q is open and i'' is bounded below, so this is an equivalence relation, and it preserves splitting. $\mathrm{Ext}(Y_3, Y_1)$ is the set of classes $[\mathcal{Y}]$, and 0 is the class of $0 \to Y_1 \to Y_1 \oplus Y_3 \to Y_3 \to 0$, with the inclusion, the projection and the norm $\|y_1\| + \|y_3\|$. Then $[\mathcal{Y}] = 0$ if and only if $\mathcal{Y}$ splits:

a left inverse λ of i (Lemma 3.6(2)) gives the equivalence $y \mapsto (\lambda y, qy)$. For a bounded linear $\alpha\colon Y_1 \to Y'$ into a Banach space, the *pushout* $\alpha_*\mathcal{Y}$ is $0 \to Y' \xrightarrow{i_\alpha} Y_\alpha \xrightarrow{q_\alpha} Y_3 \to 0$, where Y_α is the quotient of $Y' \oplus Y_2$, normed by $\|y'\| + \|y\|$, by $\{(\alpha y, -iy) : y \in Y_1\}$, which is closed because $y \mapsto (\alpha y, -iy)$ is bounded below on the complete space Y_1; $i_\alpha y' = [(y', 0)]$ and $q_\alpha[(y', y)] = qy$. Put $\alpha_*[\mathcal{Y}] = [\alpha_*\mathcal{Y}]$; an equivalence T induces the equivalence $[(y', y)] \mapsto [(y', Ty)]$, so this depends only on $[\mathcal{Y}]$.

Lemma 3.6 (splitting criteria)**.** *Let $\mathcal{Y}$ and $\mathcal{Y}'\colon 0 \to Y_1' \xrightarrow{i'} Y_2' \xrightarrow{q'} Y_3' \to 0$ be exact.*

1. *Let $\alpha_a\colon Y_a \to Y_a'$, $a = 1, 2, 3$, be bounded linear maps with $\alpha_2 i = i'\alpha_1$ and $q'\alpha_2 = \alpha_3 q$. Then $\alpha_1 = \tilde\alpha_1 i$ for a bounded linear $\tilde\alpha_1\colon Y_2 \to Y_1'$ if and only if $\alpha_3 = q'\tilde\alpha_3$ for a bounded linear $\tilde\alpha_3\colon Y_3 \to Y_2'$, and the two can be chosen with $\alpha_2 = i'\tilde\alpha_1 + \tilde\alpha_3 q$.*
2. *$\mathcal{Y}$ splits if and only if i has a bounded linear left inverse.*
3. *For a bounded linear $\alpha\colon Y_1 \to Y'$ into a Banach space, $\alpha_*\mathcal{Y}$ is exact, and $\alpha_*[\mathcal{Y}] = 0$ if and only if α extends to Y_2, that is, $\alpha = \tilde\alpha i$ for a bounded linear $\tilde\alpha\colon Y_2 \to Y'$.*

The proof is in Appendix P.

Lemma 3.7. *For a normed space E and $k \geq 1$, $\wedge_E^k$ is a metric surjection: $\|y\| = \inf\{\|z\| : \wedge_E^k z = y\}$ for $y \in \Lambda_\pi^k E$. Its extension $\widehat\wedge_E^k\colon \widehat\otimes_\pi^k E \to \widehat\Lambda_\pi^k E$ is surjective and again a metric surjection.*

The proof is in Appendix P.

Definition 3.8. Let E, D, P be normed spaces and $k, n \geq 1$.

(1) The *source sequence* of E in degree k is the exact sequence (Lemma 3.7)

$$\mathcal{S}_E^k\colon \quad 0 \to \mathcal{N}_E^k \to \widehat\otimes_\pi^k E \xrightarrow{\widehat\wedge_E^k} \widehat\Lambda_\pi^k E \to 0, \qquad \mathcal{N}_E^k = \ker \widehat\wedge_E^k, \tag{3.1}$$

whose first map is the inclusion.

(2) The *target sequence* of D in degree k is $\mathcal{R}_D^k\colon 0 \to \widehat\Lambda_\pi^k D \xrightarrow{\mathrm{alt}_D^k} \widehat\otimes_\pi^k D \to \widehat\otimes_\pi^k D/\overline{\mathrm{alt}_D^k(\widehat\Lambda_\pi^k D)} \to 0$, with the quotient map last. It is exact if and only if alt_D^k is bounded below on $\widehat\Lambda_\pi^k D$.

(3) Let $\widehat\Gamma_\pi^n P$ be the closed linear span in $\widehat\otimes_\pi^n P$ of the pure powers $h^{\otimes n} = h \otimes \cdots \otimes h$, $h \in P$. The *parameter sequence* of P in degree n is the exact sequence $\mathcal{G}_P^n\colon 0 \to \widehat\Gamma_\pi^n P \to \widehat\otimes_\pi^n P \to \widehat\otimes_\pi^n P/\widehat\Gamma_\pi^n P \to 0$. The space P is *polarizable in degree n*, written (Γ_P^n), if $\mathcal{G}_P^n$ splits, that is (Lemma 3.6(2)), if there is a bounded linear projection R_n of $\widehat\otimes_\pi^n P$ onto $\widehat\Gamma_\pi^n P$. Let $g_n(P)$ be the infimum of the norms of such projections.

(4) For a Banach space Y, $\mathrm{Mult}_0^n(P; Y)$ is the closed subspace of $\mathrm{Mult}^n(P; Y)$ of the maps Φ with zero diagonal, $\Phi(h, \dots, h) = 0$ for $h \in P$, and $\mathrm{Poly}^n(P; Y)$ is the quotient of $\mathrm{Mult}^n(P; Y)$ by $\mathrm{Mult}_0^n(P; Y)$, with the quotient norm; through $\Phi \mapsto (h \mapsto \Phi(h, \dots, h))$ its elements correspond to the diagonals of bounded n-linear maps $P^n \to Y$, the n-homogeneous continuous polynomials of Section 2.3. The resulting exact sequence is

$$\mathcal{M}^n(P; Y)\colon \quad 0 \to \mathrm{Mult}_0^n(P; Y) \to \mathrm{Mult}^n(P; Y) \to \mathrm{Poly}^n(P; Y) \to 0.$$

Lemma 3.9. *Condition (S_E^k) holds if and only if $\mathcal{S}_E^k$ in (3.1) splits, and if and only if $\mathcal{N}_E^k$ is complemented in $\widehat\otimes_\pi^k E$. A section s gives the projection $\mathrm{id} - s\,\widehat\wedge_E^k$ onto $\mathcal{N}_E^k$, of norm at most $1 + \|s\|$; a bounded projection Π onto $\mathcal{N}_E^k$ gives the section $s(\widehat\wedge_E^k z) = (\mathrm{id} - \Pi)z$, of norm at most $\|\mathrm{id} - \Pi\|$.*

Proof. A section on $\Lambda^k_\pi E$ extends by continuity to a right inverse of $\widehat{\wedge}^k_E$, and a right inverse restricts to a section. The map $\mathrm{id} - s\,\widehat{\wedge}^k_E$ is the identity on $\mathcal{N}^k_E$ and is killed by $\widehat{\wedge}^k_E$. Conversely, $\mathrm{id} - \Pi$ vanishes on $\mathcal{N}^k_E$, so the formula defines s, and Lemma 3.7 bounds its norm. □

Lemma 3.10. *For a normed space D and $k \geq 1$ the following are equivalent:* (1) (R^k_D); (2) $\mathcal{R}^k_D$ *is exact and splits;* (3) $\omega_D = \mathcal{A}_D\eta$ *for some* $\eta \in \mathrm{Mult}^k(D; \widehat{\Lambda}^k_\pi D)$, *where* $(\mathcal{A}_D\eta)(y_1, \dots, y_k) = \sum_{\sigma\in S_k} \mathrm{sgn}(\sigma)\, \eta(y_{\sigma(1)}, \dots, y_{\sigma(k)})$ *is unnormalized antisymmetrization.*

The proof is in Appendix P.

Lemma 3.11. *For a normed space P and $n \geq 1$ the following are equivalent.*

(1) *P is polarizable in degree n.*
(2) *The power map $P \to \widehat{\Gamma}^n_\pi P$, $h \mapsto h^{\otimes n}$, is the diagonal of a bounded n-linear map $P^n \to \widehat{\Gamma}^n_\pi P$.*
(3) *For every closed subspace W of a Banach space Z, every bounded n-linear $\Phi\colon P^n \to Z$ with $\Phi(h, \dots, h) \in W$ for all $h \in P$, and every $\delta > 0$, there is a bounded n-linear $Q\colon P^n \to W$ with $Q(h, \dots, h) = \Phi(h, \dots, h)$ for all h and $\|Q\| \leq (g_n(P) + \delta)\|\Phi\|$.*

Under these conditions $\mathcal{M}^n(P; Y)$ splits for every Banach space Y, by right inverses that commute with postcomposition by bounded linear maps $Y \to Y'$.

The proof is in Appendix P.

Definition 3.12. Let $w\colon E \to G$ be a bounded linear map of normed spaces and $k \geq 1$. Then w is S^k*-split* if $\widehat{\wedge}^k_G \circ s = \Lambda^k_\pi w$ for a bounded linear $s\colon \Lambda^k_\pi E \to \widehat{\otimes}^k_\pi G$, and R^k*-split* if $\rho \circ \mathrm{alt}^k_E = \Lambda^k_\pi w$ for a bounded linear $\rho\colon \otimes^k_\pi E \to \widehat{\Lambda}^k_\pi G$; it is split *without completion* if s takes values in $\otimes^k_\pi G$, respectively ρ in $\Lambda^k_\pi G$. It is $\mathrm{S}^{\leq k}$*-split* ($\mathrm{R}^{\leq k}$*-split*) if it is S^j-split (R^j-split) for every $j \leq k$.

Thus (S^k_E) says that id_E is S^k-split, and (R^k_D) that id_D is R^k-split. Every w is S^1- and R^1-split, and S^j- and R^j-split whenever $j! \neq 0$ in K, with $s = w^{\otimes j}\,\mathrm{alt}^j_E\,/j!$ and $\rho = \wedge^j_G w^{\otimes j}/j!$, since $\wedge^j_G w^{\otimes j}\,\mathrm{alt}^j_E = \Lambda^j_\pi w\ \wedge^j_E \mathrm{alt}^j_E = j!\,\Lambda^j_\pi w$.

Proposition 3.13 (split operators)**.** *Let $w\colon E \to G$ be a bounded linear map of normed spaces and $k \geq 1$.*

(1) *The map $w^{\otimes k}$ sends $\mathcal{N}^k_E$ into $\mathcal{N}^k_G$, and the following are equivalent: (a) w is S^k-split; (b) $\widehat{\Lambda}^k_\pi w = \widehat{\wedge}^k_G \circ \tilde{s}$ for a bounded linear $\tilde{s}\colon \widehat{\Lambda}^k_\pi E \to \widehat{\otimes}^k_\pi G$; (c) $w^{\otimes k}|_{\mathcal{N}^k_E}\colon \mathcal{N}^k_E \to \mathcal{N}^k_G$ extends to a bounded linear map $\widehat{\otimes}^k_\pi E \to \mathcal{N}^k_G$; (d) the* splitting class $\mathrm{sp}_k(w) = (w^{\otimes k}|_{\mathcal{N}^k_E})_*[\mathcal{S}^k_E] \in \mathrm{Ext}(\widehat{\Lambda}^k_\pi E, \mathcal{N}^k_G)$ *vanishes. In particular (S^k_E) holds if and only if $\mathrm{sp}_k(\mathrm{id}_E) = 0$, if and only if $[\mathcal{S}^k_E] = 0$ (Lemma 3.9).*
(2) *If w is S^k-split with s, and $v\colon G \to G_1$, $v'\colon G_0 \to E$ are bounded linear, then vwv' is S^k-split with $v^{\otimes k} s\, \Lambda^k_\pi v'$. If w is R^k-split with ρ, then vwv' is R^k-split with $\widehat{\Lambda}^k_\pi v\, \rho\, v'^{\otimes k}$. The same holds without completion.*

The proof is in Appendix P.

3.5. **The obstruction class.** Throughout this subsection E and D are normed spaces and $k \geq 1$; $\mathrm{Mult}^k(\mathcal{L}(E, D); \widehat{\Lambda}^k_\pi D)$ is a Banach space, and $\mathrm{Mult}^k_0(\mathcal{L}(E, D); \widehat{\Lambda}^k_\pi D)$ is its closed subspace of maps with zero diagonal.

Lemma 3.14 (evaluation)**.** *There is a unique bounded linear map* $\mathrm{ev} = \mathrm{ev}^k_{E,D}\colon \widehat{\otimes}^k_\pi E \to \mathrm{Mult}^k(\mathcal{L}(E,D);\widehat{\Lambda}^k_\pi D)$ *with*

$$\mathrm{ev}(x_1\otimes\cdots\otimes x_k)(u_1,\dots,u_k) = u_1x_1\wedge\cdots\wedge u_kx_k, \tag{3.2}$$

and $\|\mathrm{ev}\|\le 1$*. It satisfies* $\mathrm{ev}(z)(u,\dots,u) = \widehat{\Lambda}^k_\pi u(\widehat{\wedge}^k_E z)$ *for* $z\in\widehat{\otimes}^k_\pi E$ *and* $u\in\mathcal{L}(E,D)$*; in particular* ev *maps* $\mathcal{N}^k_E$ *into* $\mathrm{Mult}^k_0(\mathcal{L}(E,D);\widehat{\Lambda}^k_\pi D)$*.*

Proof. For $x\in E^k$ the right side of (3.2) is k-linear in $(u_1,\dots,u_k)$, of norm at most $\prod_a\|x_a\|$, and k-linear in x. Its linearization in x is a contraction from $\otimes^k_\pi E$ into a Banach space and extends to $\widehat{\otimes}^k_\pi E$; pure tensors span a dense subspace, which gives uniqueness. For fixed u, both sides of the identity are bounded linear in z and agree on pure tensors. □

Let q be the quotient map of $\mathcal{M}^k(\mathcal{L}(E,D);\widehat{\Lambda}^k_\pi D)$. Since $q\circ\mathrm{ev}$ vanishes on $\mathcal{N}^k_E$, Lemma 3.7 gives a unique bounded linear $\overline{\mathrm{ev}}\colon \widehat{\Lambda}^k_\pi E\to \mathrm{Poly}^k(\mathcal{L}(E,D);\widehat{\Lambda}^k_\pi D)$ with $q\circ\mathrm{ev} = \overline{\mathrm{ev}}\circ\widehat{\wedge}^k_E$. By Lemma 3.14, $\overline{\mathrm{ev}}(y)$ is the class of every map in $\mathrm{Mult}^k(\mathcal{L}(E,D);\widehat{\Lambda}^k_\pi D)$ with diagonal $u\mapsto\widehat{\Lambda}^k_\pi u(y)$.

The *obstruction class* of the pair (E,D) in degree k is

$$\begin{aligned}\mathrm{ob}_k(E,D) &= (\mathrm{ev}\,|_{\mathcal{N}^k_E})_*[\mathcal{S}^k_E]\\ &\in \mathrm{Ext}\big(\widehat{\Lambda}^k_\pi E, \mathrm{Mult}^k_0(\mathcal{L}(E,D);\widehat{\Lambda}^k_\pi D)\big).\end{aligned} \tag{3.3}$$

Theorem 3.15 (obstruction theorem)**.** *Let* E *and* D *be normed spaces and* $k\ge 1$*. The following are equivalent.*

1. *The pair* (E,D) *is good for Banach values in degree* k*.*
2. *The restriction* $\mathrm{ev}\,|_{\mathcal{N}^k_E}$ *extends to a bounded linear map* $\widehat{\otimes}^k_\pi E\to\mathrm{Mult}^k_0(\mathcal{L}(E,D);\widehat{\Lambda}^k_\pi D)$*.*
3. $\mathrm{ob}_k(E,D)=0$ *in* (3.3)*.*

More precisely, a lift $Q\colon\mathcal{L}(E,D)^k\to\mathcal{L}(\Lambda^k_\pi E,\widehat{\Lambda}^k_\pi D)$ *as in Lemma 3.4 gives an extension in (2) of norm at most* $1+\|Q\|$*, and an extension* ξ *in (2) gives such a lift of norm at most* $1+\|\xi\|$*.*

Proof. By Lemma 3.4, (1) holds if and only if a lift Q exists.

Step 1 (lifts give extensions). Bounded k-linear maps $Q\colon\mathcal{L}(E,D)^k\to\mathcal{L}(\Lambda^k_\pi E,\widehat{\Lambda}^k_\pi D)$ correspond to bounded linear maps $Q^\sharp\colon\widehat{\Lambda}^k_\pi E\to\mathrm{Mult}^k(\mathcal{L}(E,D);\widehat{\Lambda}^k_\pi D)$, by $Q^\sharp(y)(u_1,\dots,u_k) = Q(u_1,\dots,u_k)y$ for $y\in\Lambda^k_\pi E$, with $\|Q^\sharp\| = \|Q\|$. Given a lift Q, put $\xi = \mathrm{ev} - Q^\sharp\widehat{\wedge}^k_E$; it equals ev on $\mathcal{N}^k_E = \ker\widehat{\wedge}^k_E$. By Lemma 3.14, $\xi(z)(u,\dots,u) = \Lambda^k_\pi u(\wedge^k_E z) - Q(u,\dots,u)(\wedge^k_E z) = 0$ for $z\in\otimes^k_\pi E$, hence for all z by continuity. So ξ is an extension as in (2). Since $\|\mathrm{ev}\|\le 1$ by Lemma 3.14 and $\widehat{\wedge}^k_E$ is contractive, $\|\xi\|\le\|\mathrm{ev}\|+\|Q^\sharp\|\le 1+\|Q\|$.

Step 2 (extensions give lifts). Given an extension ξ as in (2), the map $\mathrm{ev}-\xi$ vanishes on $\mathcal{N}^k_E$. Since $\widehat{\wedge}^k_E$ is a metric surjection with kernel $\mathcal{N}^k_E$ (Lemma 3.7), $\mathrm{ev}-\xi = Q^\sharp\widehat{\wedge}^k_E$ for a bounded linear $Q^\sharp$ with $\|Q^\sharp\| = \|\mathrm{ev}-\xi\|\le 1+\|\xi\|$, using $\|\mathrm{ev}\|\le 1$. The corresponding Q satisfies $Q(u,\dots,u)(\wedge^k_E z) = \mathrm{ev}(z)(u,\dots,u) - \xi(z)(u,\dots,u) = \Lambda^k_\pi u(\wedge^k_E z)$ for $z\in\otimes^k_\pi E$, by Lemma 3.14 and since $\xi(z)$ has zero diagonal; so $Q(u,\dots,u) = \Lambda^k_\pi u$, since $\wedge^k_E$ is surjective.

Step 3 (the class). Conditions (2) and (3) are equivalent by Lemma 3.6(3). □

If D is dense in a normed K-space D', the canonical maps $\otimes^k_\pi D\to\otimes^k_\pi D'$ and $\Lambda^k_\pi D\to\Lambda^k_\pi D'$ are isometric with dense range; so, with $\widehat{D}$ regarded as a normed K-space, $\widehat{\otimes}^k_\pi\widehat{D} = \widehat{\otimes}^k_\pi D$ and $\widehat{\Lambda}^k_\pi\widehat{D} = \widehat{\Lambda}^k_\pi D$. The maps are contractive. By continuity ω_D extends to a contractive map $\omega'\colon D'^k\to\widehat{\Lambda}^k_\pi D$, alternating because tuples with two equal entries in D'^k are limits of such tuples in D^k; its linearization (Proposition 3.1) is a contraction $\Lambda^k_\pi D'\to\widehat{\Lambda}^k_\pi D$ restricting to the

inclusion on $\Lambda^k_\pi D$, so the maps are isometric. Density follows from $x_1\wedge\cdots\wedge x_k - y_1\wedge\cdots\wedge y_k = \sum_a y_1\wedge\cdots\wedge y_{a-1}\wedge(x_a-y_a)\wedge x_{a+1}\wedge\cdots\wedge x_k$. The tensor case is the same.

Theorem 3.16 (ways for the class to vanish)**.** *Let E and D be normed spaces and $k\ge 1$.*

(1) (Source.) *If (S^k_E) holds, then $\mathrm{ob}_k(E,D)=0$. More generally, $\mathrm{ob}_k(E,D)=0$ if $\mathrm{ev}\,|_{\mathcal{N}^k_E} = \tau\circ w^{\otimes k}|_{\mathcal{N}^k_E}$ for an S^k-split $w\colon E\to G$ and a bounded linear $\tau\colon \mathcal{N}^k_G\to \mathrm{Mult}^k_0(\mathcal{L}(E,D);\widehat{\Lambda}^k_\pi D)$. This applies when $w\colon E\to G$ is S^k-split and $\varepsilon\colon\mathcal{L}(E,D)\to\mathcal{L}(G,\widehat{D})$ is bounded linear with $\varepsilon(u)\circ w = u$ for every u, with $\tau(\zeta)(u_1,\dots,u_k) = \widehat{\wedge}^k_{\widehat{D}}\big(\varepsilon(u_1)\otimes\cdots\otimes\varepsilon(u_k)\big)\zeta$.*

(2) (Parameter.) *If the sequence $\mathcal{M}^k(\mathcal{L}(E,D);\widehat{\Lambda}^k_\pi D)$ splits, and in particular if $\mathcal{L}(E,D)$ is polarizable in degree k, then $\mathrm{ob}_k(E,D)=0$. More generally, $\mathrm{ob}_k(E,D)=0$ if there are a normed space P with (Γ^k_P), a bounded linear map $\psi\colon\mathcal{L}(E,D)\to P$ and a bounded k-linear map $\Phi\colon P^k\to\mathcal{L}(\widehat{\otimes}^k_\pi E,\widehat{\Lambda}^k_\pi D)$ such that $\Phi(h,\dots,h)$ vanishes on $\mathcal{N}^k_E$ for $h\in P$ and $\Phi(\psi u,\dots,\psi u)=\widehat{\Lambda}^k_\pi u\circ\widehat{\wedge}^k_E$ for $u\in\mathcal{L}(E,D)$.*

(3) (Target.) *Let $\mathrm{alt}_*\colon \mathrm{Mult}^k_0(\mathcal{L}(E,D);\widehat{\Lambda}^k_\pi D)\to\mathrm{Mult}^k_0(\mathcal{L}(E,D);\widehat{\otimes}^k_\pi D)$ be postcomposition with alt^k_D. Then $(\mathrm{alt}_*\circ\mathrm{ev}\,|_{\mathcal{N}^k_E})_*[\mathcal{S}^k_E]=0$. If (R^k_D) holds, then $\mathrm{ob}_k(E,D)=0$.*

(3′) (Target, relative.) *If $w'\colon G\to\widehat{D}$ is R^k-split and $\varepsilon\colon\mathcal{L}(E,D)\to\mathcal{L}(E,G)$ is bounded linear with $w'\circ\varepsilon(u)=u$ for every u, then $\mathrm{ob}_k(E,D)=0$.*

The proof is in Appendix P.

Corollary 3.17. *Each of (S^k_E), (R^k_D) and $(\Gamma^k_{\mathcal{L}(E,D)})$ makes (E,D) good for Banach values in degree k; so does each of the data (G,w,ε) of Theorem 3.16(1) and (G,w',ε) of Theorem 3.16(3′).*

Proof. Theorems 3.16 and 3.15. □

3.6. **The ordering lift.** Throughout this subsection E and D are normed spaces, $k\ge1$, and $(e_t)_{t\in\Sigma}$ is a linearly independent family in E, indexed by a totally ordered set Σ, whose span V is dense in E; a Hamel basis of E in any total order is one. Write $x=\sum_t x(t)e_t$ for $x\in V$, $\mathrm{Mult}^k=\mathrm{Mult}^k(\mathcal{L}(E,D);\widehat{\Lambda}^k_\pi D)$ and $\mathrm{Mult}^k_0=\mathrm{Mult}^k_0(\mathcal{L}(E,D);\widehat{\Lambda}^k_\pi D)$, and let q and $\overline{\mathrm{ev}}$ be the maps defined after Lemma 3.14. By the description of $\overline{\mathrm{ev}}$ given there, in the notation of the proof of Theorem 3.15, a bounded linear $T\colon\widehat{\Lambda}^k_\pi E\to\mathrm{Mult}^k$ is $Q^\sharp$ for a lift Q as in Lemma 3.4 if and only if $q\circ T=\overline{\mathrm{ev}}$. Let $c_k(E,D)\in[0,\infty]$ be the infimum of $\|Q\|$ over these lifts; by Lemma 3.4 it is finite if and only if (E,D) is good for Banach values in degree k.

Let $\Sigma^k_<\subseteq\Sigma^k$ be the set of strictly increasing t, and put $e^\wedge_t=e_{t_1}\wedge\cdots\wedge e_{t_k}$ and $e^\otimes_t=e_{t_1}\otimes\cdots\otimes e_{t_k}$ for $t\in\Sigma^k$; the $e^\wedge_t$ with $t\in\Sigma^k_<$ form a basis of $\Lambda^k_K V$. Let $\wedge_V\colon\otimes^k_K V\to\Lambda^k_K V$ be the quotient map and $j_\Lambda\colon\Lambda^k_K V\to\widehat{\Lambda}^k_\pi E$, $j_\otimes\colon\otimes^k_K V\to\widehat{\otimes}^k_\pi E$ the canonical maps, so that $\widehat{\wedge}^k_E\circ j_\otimes = j_\Lambda\circ\wedge_V$, and put $\|y\|_\pi=\|j_\Lambda y\|$. By the paragraph before Theorem 3.16, $\|y\|_\pi$ is the projective seminorm of y computed from representations with entries in V, and $j_\Lambda(\Lambda^k_K V)$ is dense in $\widehat{\Lambda}^k_\pi E$; the same holds for $j_\otimes$. Norms in $[0,\infty]$ of linear maps on $\Lambda^k_K V$, for $\|\cdot\|_\pi$, and of k-linear maps on V^k are best constants, as are the norms of lifts Q and of the maps $Q^\sharp$.

The *ordering section* is the linear map $s_<\colon\Lambda^k_K V\to\otimes^k_K V$ with $s_<(e^\wedge_t)=e^\otimes_t$ for $t\in\Sigma^k_<$. Then $\wedge_V\circ s_<=\mathrm{id}$, and $s_<=\mathcal{P}^{(k)}\circ\mathrm{alt}^k$ for the map $\mathcal{P}^{(k)}$ of Proposition 5.7 and the antisymmetrization $\mathrm{alt}^k\colon\Lambda^k_K V\to\otimes^k_K V$. The *ordering lift* assigns to $u_1,\dots,u_k\in\mathcal{L}(E,D)$ the linear map $Q_<(u_1,\dots,u_k)\colon\Lambda^k_K V\to\Lambda^k_\pi D$ with

$$Q_<(u_1,\dots,u_k)(e^\wedge_t)=u_1e_{t_1}\wedge\cdots\wedge u_ke_{t_k}\qquad(t\in\Sigma^k_<);\tag{3.4}$$

on $x_1 \wedge \cdots \wedge x_k$ with $x_1, \dots, x_k \in V$ it is the right side of (J.6). Let $Q^\sharp_< \colon \Lambda^k_K V \to \mathrm{Mult}^k$ be the linear map with $Q^\sharp_<(y)(u_1, \dots, u_k) = Q_<(u_1, \dots, u_k)y$; it takes values in Mult^k by (3.4). A *zero-diagonal correction* is a linear map $L \colon \Lambda^k_K V \to \mathrm{Mult}^k_0$. The proofs of Theorems 3.18 and 3.22, Lemma 3.19 and Corollaries 3.20 and 3.21 are in Appendix P.

Theorem 3.18 (ordering representation)**.** (1) $\mathrm{ev} \circ j_\otimes \circ s_< = Q^\sharp_<$ *and* $q \circ Q^\sharp_< = \overline{\mathrm{ev}} \circ j_\Lambda$.

(2) *For every lift* $Q \colon \mathcal{L}(E,D)^k \to \mathcal{L}(\Lambda^k_\pi E, \widehat{\Lambda}^k_\pi D)$ *as in Lemma 3.4,* $L = Q^\sharp \circ j_\Lambda - Q^\sharp_<$ *is a zero-diagonal correction; equivalently,* $Q(u_1, \dots, u_k) \circ j_\Lambda = Q_<(u_1, \dots, u_k) + \Phi(u_1, \dots, u_k)$, *where* $\Phi(u_1, \dots, u_k)y = L(y)(u_1, \dots, u_k)$ *is* k*-linear in* $(u_1, \dots, u_k)$ *and* $\Phi(u, \dots, u) = 0$ *for* $u \in \mathcal{L}(E,D)$.

(3) *For every zero-diagonal correction* L *with* $\|Q^\sharp_< + L\| < \infty$ *there is a unique lift* Q *with* $Q^\sharp \circ j_\Lambda = Q^\sharp_< + L$, *and* $\|Q\| = \|Q^\sharp_< + L\|$. *Hence* $c_k(E,D) = \inf_L \|Q^\sharp_< + L\|$ *over the zero-diagonal corrections* L, *and* (E,D) *is good for Banach values in degree* k *if and only if* $\|Q^\sharp_< + L\| < \infty$ *for some zero-diagonal correction* L.

(4) (Defect form.) *The formula* $\Omega_<(x) = \mathrm{ev}(x_1 \otimes \cdots \otimes x_k) - Q^\sharp_<(x_1 \wedge \cdots \wedge x_k)$, *for* $x \in V^k$, *defines a* k*-linear map* $\Omega_< \colon V^k \to \mathrm{Mult}^k_0$. *For every zero-diagonal correction* L, *with* $(L \circ \wedge)(x) = L(x_1 \wedge \cdots \wedge x_k)$,

$$\|Q^\sharp_< + L\| \le 1 + \|\Omega_< - L \circ \wedge\|, \qquad \|\Omega_< - L \circ \wedge\| \le 1 + \|Q^\sharp_< + L\|.$$

So (E,D) *is good for Banach values in degree* k *if and only if* $\Omega_<$ *is the sum of a bounded* k*-linear map* $V^k \to \mathrm{Mult}^k_0$ *and an alternating* k*-linear map* $V^k \to \mathrm{Mult}^k_0$. *The infima* $c_k(E,D)$ *and* $\inf_L \|\Omega_< - L \circ \wedge\|$ *are finite together, and then they differ by at most* 1.

If $\|\mathcal{P}_<(x \otimes y)\| \le C\|x\|\|y\|$ for $x, y \in V$, as in Proposition 5.7, no correction is needed: the lift Q of Proposition 3.5(1) built from the section $s = \mathcal{P}^{(k)} \circ \mathrm{alt}^k_E$ of Proposition 5.7(2) satisfies $Q(u_1, \dots, u_k) \circ j_\Lambda = Q_<(u_1, \dots, u_k)$, since $s \circ j_\Lambda = j_\otimes \circ s_<$.

Lemma 3.19. *The space* $\mathcal{N}^k_E$ *is the closure in* $\widehat{\otimes}^k_\pi E$ *of the span of the pure tensors* $x_1 \otimes \cdots \otimes x_k$ *with* $x_a = x_{a+1}$ *for some* $a < k$, *and also of the span of the pure tensors with two equal entries. The pure tensors of the first kind with all entries in* V *span a dense subspace of* $\mathcal{N}^k_E$.

Corollary 3.20 (adjacent diagonals)**.** *The pair* (E,D) *is good for Banach values in degree* k *if and only if there is a bounded* k*-linear* $\Psi \colon E^k \to \mathrm{Mult}^k_0$ *with*

$$\Psi(x_1, \dots, x_k)(u_1, \dots, u_k) = u_1 x_1 \wedge \cdots \wedge u_k x_k$$

whenever two of the x_a *coincide, equivalently whenever* $x_a = x_{a+1}$ *for some* $a < k$. *A lift* Q *gives such a* Ψ *with* $\|\Psi\| \le 1 + \|Q\|$, *and such a* Ψ *gives a lift* Q *with* $\|Q\| \le 1 + \|\Psi\|$.

Corollary 3.21 (z-linear form)**.** *Let* $\delta > 0$, *and let* $\nu \colon \widehat{\Lambda}^k_\pi E \to \widehat{\otimes}^k_\pi E$ *be a homogeneous map,* $\nu(cy) = c\,\nu(y)$ *for* $c \in K$, *with* $\widehat{\wedge}^k_E \circ \nu = \mathrm{id}$ *and* $\|\nu(y)\| \le (1+\delta)\|y\|$; *it exists by Lemma 3.7, choosing* $\nu(y)$ *for one point* y *of each orbit* $K^\times y$, $y \ne 0$, *and extending homogeneously. Put* $\Omega^\nu_< = \mathrm{ev} \circ \nu \circ j_\Lambda - Q^\sharp_< \colon \Lambda^k_K V \to \mathrm{Mult}^k$. *Then* $\Omega^\nu_<$ *takes values in* Mult^k_0, *is homogeneous, and is* z-linear*:*

$$\Big\| \Omega^\nu_<\Big(\sum_i y_i\Big) - \sum_i \Omega^\nu_<(y_i) \Big\| \le 2(1+\delta) \sum_i \|y_i\|_\pi$$

for every finite family (y_i) *in* $\Lambda^k_K V$. *Let* $\mathrm{dist}(\Omega^\nu_<)$ *be the infimum, over the linear maps* $L \colon \Lambda^k_K V \to \mathrm{Mult}^k_0$, *of the infimum of the* C *with* $\|\Omega^\nu_<(y) - L(y)\| \le C\|y\|_\pi$ *for all* y. *Then* $\mathrm{dist}(\Omega^\nu_<)$ *and* $c_k(E,D)$ *are finite together, and then they differ by at most* $1 + \delta$. *In particular* $\mathrm{ob}_k(E,D) = 0$ *if and only if* $\Omega^\nu_<$ *lies at finite distance from a linear map* $\Lambda^k_K V \to \mathrm{Mult}^k_0$.

Theorem 3.22 (nonarchimedean values)**.** *Let K be nonarchimedean, and suppose that $C_e = \sup_t \|e_t\| < \infty$ and that $|x(t)| \le C_c\|x\|$ for $x \in V$, $t \in \Sigma$ and a constant C_c. Let Y be a Banach space with a nonarchimedean norm $\|\cdot\|_Y$ and constants $0 < c_- \le c_+$ with $c_-\|\cdot\|_Y \le \|\cdot\| \le c_+\|\cdot\|_Y$, and let $\alpha\colon \mathcal{N}_E^k \to Y$ be bounded linear. Then there is a bounded linear $\tilde\alpha\colon \widehat{\otimes}_\pi^k E \to Y$ of norm at most $2(c_+/c_-)\|\alpha\|(C_cC_e)^k$ with $\tilde\alpha(j_\otimes z) = \alpha\big(j_\otimes(z - s_<(\wedge_V z))\big)$ for $z \in \otimes_K^k V$ and $\tilde\alpha = \alpha$ on $\mathcal{N}_E^k$. Hence $\alpha_*[\mathcal{S}_E^k] = 0$. In particular $\mathrm{ob}_k(E,D) = 0$ whenever $\widehat{\Lambda}_\pi^k D$ has an equivalent nonarchimedean norm.*

4. Canonical splittings and forced asymmetry

Throughout this section K is a nontrivially normed field and $[n] := \{1,\dots,n\}$. For $\sigma \in S_n$ let τ_σ be the isometry of the n-th projective tensor power of a normed space, and of its completion, with $\tau_\sigma(x_1 \otimes \cdots \otimes x_n) = x_{\sigma^{-1}(1)} \otimes \cdots \otimes x_{\sigma^{-1}(n)}$; it commutes with $u^{\otimes n}$ for every bounded operator u, and $\tau_\sigma\tau_{\sigma'} = \tau_{\sigma\sigma'}$. Evaluation on pure tensors gives

$$\mathrm{alt}_E^k \circ \wedge_E^k = \sum_{\sigma\in S_k} \mathrm{sgn}(\sigma)\,\tau_\sigma, \qquad \tau_\sigma \circ \mathrm{alt}_E^k = \mathrm{sgn}(\sigma)\,\mathrm{alt}_E^k, \qquad \wedge_E^k \circ \tau_\sigma = \mathrm{sgn}(\sigma)\,\wedge_E^k, \tag{4.1}$$

and the first identity extends to $\widehat{\otimes}_\pi^k E$ by continuity. For vectors $x_1,\dots,x_n$ and $S \subseteq [n]$ put $v_S := \sum_{i\in S} x_i$. Expanding $v_S^{\otimes n}$ over the maps $f\colon [n] \to S$, and using that $\sum_{S\supseteq f([n])}(-1)^{n-|S|}$ is 1 if f is onto and 0 otherwise, gives in every characteristic

$$\sum_{S\subseteq[n]} (-1)^{n-|S|}\, v_S^{\otimes n} = \sum_{\sigma\in S_n} x_{\sigma(1)} \otimes \cdots \otimes x_{\sigma(n)}. \tag{4.2}$$

Proposition 4.1 (natural splittings)**.** *Let E, D and P be normed spaces.*

1. *If $k! \neq 0$ in K, then $\mathrm{alt}_E^k/k!$ is a section of $\wedge_E^k$ of norm at most $k!/|k!|$ and $\wedge_D^k/k!$ is a retraction of alt_D^k of norm at most $1/|k!|$; both are natural with respect to every $u \in \mathcal{L}(E,D)$ and satisfy the sign rules of (4.1).*
2. *If $n! \neq 0$ in K, then $R_n := \frac{1}{n!}\sum_{\sigma\in S_n}\tau_\sigma$ is a projection of $\widehat{\otimes}_\pi^n P$ onto $\widehat{\Gamma}_\pi^n P$ of norm at most $n!/|n!|$; it commutes with $u^{\otimes n}$ for bounded operators u and satisfies $\tau_\sigma R_n = R_n\tau_\sigma = R_n$.*

Proof. (1) holds by the properties of $\wedge_E^k$ and alt_E^k in Section 3. (2) The norm bound and the commutation rules hold since each τ_σ is an isometry commuting with $u^{\otimes n}$ and $\tau_\sigma\tau_{\sigma'} = \tau_{\sigma\sigma'}$. R_n fixes every pure power, hence $\widehat{\Gamma}_\pi^n P$ pointwise, and by (4.2) and continuity it maps $\widehat{\otimes}_\pi^n P$ into $\widehat{\Gamma}_\pi^n P$. □

So every pair is good in degree k when $k! \neq 0$ (Propositions 3.5(3) and 5.1(1)). A matrix in $GL_k(K)$ is *monomial* if each row and each column has exactly one nonzero entry.

Theorem 4.2 (no equivariant splitting)**.** *Let $k \ge 2$ with $k! = 0$ in K, and let E_0 be K^k with a norm for which the coordinate functionals $e_1^*,\dots,e_k^*$ of the standard basis $e_1,\dots,e_k$ are bounded; by Lemma A.3 every norm qualifies if K is complete. For a monomial g, extend $g^{\otimes k}$ and $\Lambda_\pi^k g$ to the completions.*

1. *No linear $s\colon \Lambda_\pi^k E_0 \to \widehat{\otimes}_\pi^k E_0$ with $\widehat{\wedge}_{E_0}^k \circ s = \mathrm{id}$ satisfies $s \circ \Lambda_\pi^k g = g^{\otimes k} \circ s$ for all monomial g.*
2. *No linear $\rho\colon \otimes_\pi^k E_0 \to \widehat{\Lambda}_\pi^k E_0$ with $\rho \circ \mathrm{alt}_{E_0}^k = \mathrm{id}$ satisfies $\rho \circ g^{\otimes k} = \Lambda_\pi^k g \circ \rho$ for all monomial g.*
3. *No linear projection R of $\widehat{\otimes}_\pi^k E_0$ onto $\widehat{\Gamma}_\pi^k E_0$ satisfies $R \circ g^{\otimes k} = g^{\otimes k} \circ R$ for all monomial g.*

Statements (1)–(3) also hold for the algebraic powers $\otimes_K^k K^k$ *and* $\Lambda_K^k K^k$.

The proof is in Appendix P.

Let a group H act on a set I, and let $V = K^{(I)}$ be the space of finitely supported functions $I \to K$, with unit vectors e_i. Each $h \in \mathsf{H}$ acts on V by $he_i = e_{h(i)}$, on $\otimes_K^k V$ by $h^{\otimes k}$ and on $\Lambda_K^k V$ by $\Lambda_K^k h$; a linear map between two of these spaces is H-*equivariant* if it intertwines the actions of every $h \in \mathsf{H}$. The maps $\wedge^k \colon \otimes_K^k V \to \Lambda_K^k V$, $\mathrm{alt}^k \colon \Lambda_K^k V \to \otimes_K^k V$ and τ_σ are given by the formulas of Section 3.1 and of this section. A k-*subset* is a subset with k elements; for a finite $T \subseteq I$, H_T is its setwise stabilizer and $\bar{\mathsf{H}}_T \subseteq \mathrm{Sym}(T)$ is the group of permutations of T induced by H_T. A k-*multiset* is a finitely supported $\mathbf{m} \colon I \to \mathbb{N}$ with $\sum_i \mathbf{m}(i) = k$; H acts on these by $(h\mathbf{m})(i) = \mathbf{m}(h^{-1}(i))$, $\mathsf{H}_\mathbf{m}$ is the stabilizer of $\mathbf{m}$, and $\bar{\mathsf{H}}_\mathbf{m} \subseteq \mathrm{Sym}(\operatorname{supp}\mathbf{m})$ is the group induced by $\mathsf{H}_\mathbf{m}$. Let $\Gamma_K^k V \subseteq \otimes_K^k V$ be the span of the pure powers $x^{\otimes k}$, $x \in V$; since K is infinite, it is the space of tensors fixed by every τ_σ, by the paragraph at the beginning of Section L.3.

Theorem 4.3 (permutation groups)**.** *Let* H *act on* I*, and let* $k \geq 1$.

(1) *There is an* H*-equivariant linear* $s \colon \Lambda_K^k V \to \otimes_K^k V$ *with* $\wedge^k \circ s = \mathrm{id}$ *if and only if* $|\bar{\mathsf{H}}_T| \neq 0$ *in* K *for every* k*-subset* T *of* I.

(2) *There is an* H*-equivariant linear* $\rho \colon \otimes_K^k V \to \Lambda_K^k V$ *with* $\rho \circ \mathrm{alt}^k = \mathrm{id}$ *if and only if the condition in (1) holds.*

(3) *There is an* H*-equivariant linear projection of* $\otimes_K^k V$ *onto* $\Gamma_K^k V$ *if and only if* $|\bar{\mathsf{H}}_\mathbf{m}| \neq 0$ *in* K *for every* k*-multiset* $\mathbf{m}$ *on* I.

When the condition in (1), (2) or (3) holds, the corresponding map can be chosen to commute also with the operator $e_i \mapsto \lambda_i e_i$ *of* V *for every* $\lambda \in (K^\times)^I$.

(4) *Let* E *be a normed space with vectors* $e_i \in E$ *and bounded linear functionals* e_i^* *on* E*,* $i \in I$*, with* $e_i^*(e_j) = \delta_{ij}$*, and let* H *act on* I *and on* E *by bounded linear maps with*

$$he_i = e_{h(i)}, \qquad e_i^* \circ h = e_{h^{-1}(i)}^* \qquad (h \in \mathsf{H},\ i \in I); \tag{4.3}$$

extend $h^{\otimes k}$ *and* $\Lambda_\pi^k h$ *to the completions. If* $|\bar{\mathsf{H}}_T| = 0$ *in* K *for some* k*-subset* T*, no linear* $s \colon \Lambda_\pi^k E \to \widehat{\otimes}_\pi^k E$ *with* $\widehat{\wedge}_E^k \circ s = \mathrm{id}$ *satisfies* $s \circ \Lambda_\pi^k h = h^{\otimes k} \circ s$ *for all* $h \in \mathsf{H}_T$*, and no linear* $\rho \colon \otimes_\pi^k E \to \widehat{\Lambda}_\pi^k E$ *with* $\rho \circ \mathrm{alt}_E^k = \mathrm{id}$ *satisfies* $\rho \circ h^{\otimes k} = \Lambda_\pi^k h \circ \rho$ *for all* $h \in \mathsf{H}_T$*. If* $|\bar{\mathsf{H}}_\mathbf{m}| = 0$ *in* K *for some* k*-multiset* $\mathbf{m}$*, no linear projection* R *of* $\widehat{\otimes}_\pi^k E$ *onto* $\widehat{\Gamma}_\pi^k E$ *satisfies* $R \circ h^{\otimes k} = h^{\otimes k} \circ R$ *for all* $h \in \mathsf{H}_\mathbf{m}$.

The proof is in Appendix N.

Corollary 4.4 (one element)**.** (1) *In each of (1)–(3) of Theorem 4.3, the following are equivalent: the* H*-equivariant map exists; for every* $h \in \mathsf{H}$*, the map equivariant for the cyclic group generated by* h *exists; no* $h \in \mathsf{H}$ *maps a* k*-subset* T *onto itself, in (1) and (2), or fixes a* k*-multiset* $\mathbf{m}$*, in (3), and induces on* T*, respectively on* $\operatorname{supp}\mathbf{m}$*, a permutation of order* $\operatorname{char} K$*. If* $\operatorname{char} K = 0$*, the three statements hold.*

(2) *Let* $k! = 0$ *in* K*, so that* $p = \operatorname{char} K$ *satisfies* $2 \leq p \leq k$*. Let* E_0 *be as in Theorem 4.2, let* $\nu \in S_k$ *be a cycle of length* p*, and let* g *be the permutation matrix with* $ge_i = e_{\nu(i)}$*. No map* s*,* ρ *or* R *as in (1), (2) or (3) of Theorem 4.2 satisfies the corresponding equivariance identity for this single* g.

The proof is in Appendix P.

For a finite group G acting trivially on K, the zeroth Tate cohomology group is $\widehat{H}^0(\mathsf{G}; K) = K/|\mathsf{G}|K$ [4, Ch. VI]; it is K if $|\mathsf{G}| = 0$ in K and 0 otherwise.

Proposition 4.5 (Tate classes)**.** *Let H act on I, let $k \geq 1$, and let $T \subseteq I$ be a k-subset. For a bijection $f\colon [k] \to T$ put $e_f := e_{f(1)} \otimes \cdots \otimes e_{f(k)}$. Fix a bijection $f_T\colon [k] \to T$, put $\eta_T := e_{f_T(1)} \wedge \cdots \wedge e_{f_T(k)}$, let $U_T \subseteq \otimes^k_K V$ be the span of the tensors e_f over the bijections $f\colon [k] \to T$, put $\mathcal{N}_T := U_T \cap \ker \wedge^k$, and let $\chi_T\colon \mathsf{H}_T \to \{1,-1\}$ be the homomorphism with $\Lambda^k_K h(\eta_T) = \chi_T(h)\eta_T$. Let H_T act on U_T by $h \cdot z := \chi_T(h) h^{\otimes k} z$. Write $H^1(\mathsf{H}_T; \mathcal{N}_T)$ for the space of maps $c\colon \mathsf{H}_T \to \mathcal{N}_T$ with $c(hh') = c(h) + h \cdot c(h')$ for $h, h' \in \mathsf{H}_T$, modulo the maps $h \mapsto h \cdot n - n$ with $n \in \mathcal{N}_T$, and for $a \in K$ let $\delta_T(a) \in H^1(\mathsf{H}_T; \mathcal{N}_T)$ be the class of $h \mapsto a(h \cdot e_{f_T} - e_{f_T})$.*

1. *$\delta_T(1) = 0$ if and only if some $z \in U_T$ satisfies $\wedge^k z = \eta_T$ and $h^{\otimes k} z = \chi_T(h) z$ for every $h \in \mathsf{H}_T$.*
2. *$\ker \delta_T = |\bar{\mathsf{H}}_T| K$. Hence δ_T induces an injective linear map from $\widehat{H}^0(\bar{\mathsf{H}}_T; K) = K/|\bar{\mathsf{H}}_T| K$ into $H^1(\mathsf{H}_T; \mathcal{N}_T)$ that sends the class of 1 to $\delta_T(1)$.*
3. *An H-equivariant linear section of $\wedge^k\colon \otimes^k_K V \to \Lambda^k_K V$ exists if and only if $\delta_T(1) = 0$ for every k-subset T of I.*

The proof is in Appendix N. By (2) and (3), an H-equivariant section exists if and only if the Tate classes $[1] \in \widehat{H}^0(\bar{\mathsf{H}}_T; K)$ vanish for all k-subsets T.

A *lift* is, as in Lemma 3.4, a bounded k-linear map $Q\colon \mathcal{L}(E,D)^k \to \mathcal{L}(\Lambda^k_\pi E, \widehat{\Lambda}^k_\pi D)$ with $Q(u,\dots,u) = \Lambda^k_\pi u$. For $u_1, \dots, u_k \in \mathcal{L}(E,D)$ define $\Lambda_{\mathrm{sym}}(u_1,\dots,u_k) \in \mathcal{L}(\Lambda^k_\pi E, \Lambda^k_\pi D)$ by

$$\Lambda_{\mathrm{sym}}(u_1,\dots,u_k)(x_1 \wedge \cdots \wedge x_k) := \sum_{\sigma \in S_k} u_{\sigma(1)} x_1 \wedge \cdots \wedge u_{\sigma(k)} x_k. \tag{4.4}$$

The right side is alternating in (x_b), since the terms for σ and $\sigma \circ (a\ b)$ cancel when $x_a = x_b$; by Proposition 3.1, Λ_{sym} is a bounded symmetric k-linear map with $\Lambda_{\mathrm{sym}}(u,\dots,u) = k!\,\Lambda^k_\pi u$. If $k! \neq 0$, $\Lambda_{\mathrm{sym}}/k!$ is a symmetric lift, the normalized alternatization of Remark B.15.

Proposition 4.6 (forced asymmetry)**.** *Let E and D be normed spaces.*

1. *Every lift Q satisfies $\sum_{\sigma \in S_k} Q(u_{\sigma(1)}, \dots, u_{\sigma(k)}) = \Lambda_{\mathrm{sym}}(u_1,\dots,u_k)$.*

Suppose now that $k! = 0$ in K.

2. *If E has k linearly independent bounded functionals and $\widehat{\Lambda}^k_\pi D \neq 0$, no lift is symmetric.*
3. *No linear retraction $\rho\colon \otimes^k_\pi D \to \widehat{\Lambda}^k_\pi D$ of alt^k_D satisfies $\rho \circ \tau_\sigma = \mathrm{sgn}(\sigma)\rho$ for all $\sigma \in S_k$, unless $\Lambda^k_\pi D = 0$.*
4. *If E has k linearly independent bounded functionals, no linear section $s\colon \Lambda^k_\pi E \to \widehat{\otimes}^k_\pi E$ of $\widehat{\wedge}^k_E$ takes values in the sign-isotypic tensors $\{y : \tau_\sigma y = \mathrm{sgn}(\sigma) y \text{ for all } \sigma\}$.*

The proof is in Appendix P.

The functional hypothesis in (2) is needed: for $k = p$, $K = \mathbb{F}_p(t)$ and the space $E \subseteq A = \mathbb{F}_p((t))[\epsilon]/(\epsilon^p)$ of Lemma H.5, every bounded functional on E is zero and $\mathcal{L}(E,E) = K \cdot \mathrm{id}$, while $\Lambda^p_\pi E \neq 0$ by Proposition 3.1, since the determinant in the basis $(\epsilon^i)_{0 \leq i < p}$ of A is a bounded alternating map on E^p with value 1 at that basis. So $Q(c_1 \mathrm{id}, \dots, c_p \mathrm{id}) = c_1 \cdots c_p\, \mathrm{id}$ is a symmetric lift.

5. Splittings that exist

5.1. Splittings for one value space.

Proposition 5.1 (splitting the alternating inclusion)**.** *If the inclusion $j_{E,F}\colon \mathrm{Alt}^k(E;F) \hookrightarrow \mathrm{Mult}^k(E;F)$ has a bounded linear retraction ρ, then $A^k_{E,D;F}$ has a bounded k-linear lift of norm at most $\|\rho\|$, for every normed D. Such a lift also exists in each of the following cases:*

1. *$k! \neq 0$ in K;*

(2) *either E or D has a finite algebraic basis with continuous coordinate functionals.*

No completeness hypothesis is required.

Proof. For a retraction, $Q(u_1,\dots,u_k)(m) = \rho\big(m\circ(u_1,\dots,u_k)\big)$ is a lift: the mixed composition has norm at most $\|m\|\prod_i\|u_i\|$, and its diagonal is already alternating.

If $k!\neq 0$, the lift

$$Q(u_1,\dots,u_k)(m)(x) = \frac{1}{k!}\sum_{\sigma\in S_k} m(u_{\sigma(1)}x_1,\dots,u_{\sigma(k)}x_k)$$

vanishes when two x-arguments agree, by pairing the permutations that exchange their positions; on the diagonal all $k!$ summands coincide; and its norm is at most $k!/|k!|$, the numerator counting summands.

For continuous coordinates $(e_i,e_i^*)_{1\le i\le d}$ on E, sum over increasing words $i_1<\cdots<i_k$:

$$Q(u_1,\dots,u_k)(m)(x) = \sum_{i_1<\cdots<i_k}\det\big(e_{i_a}^*(x_b)\big)_{a,b}m(u_1e_{i_1},\dots,u_ke_{i_k}).$$

If the coordinates are on D, replace the summand by $\det(e_{i_a}^*(u_ax_b))_{a,b}\,m(e_{i_1},\dots,e_{i_k})$. Equal columns give alternation, and the exterior-basis expansion gives the diagonal identity. Each is a finite sum of norm at most $k!\sum_{i_1<\cdots<i_k}\prod_a\|e_{i_a}^*\|\|e_{i_a}\|$, empty if $k>d$. □

Over a complete field, finite-dimensional spaces have continuous coordinates (Lemma A.3), so cases (1) and (2) contain Bourbaki's hypotheses, characteristic zero and finite rank [3, n° 7.8]; over incomplete K finite rank does not suffice, since the a-coordinate on $K+Ka\subseteq\widehat{K}$, $a\in\widehat{K}\setminus K$, is discontinuous.

A nonarchimedean normed space is *spherically complete* if every nonempty family of pairwise-intersecting closed balls has a common point. In a nonarchimedean metric, intersecting balls are nested.

Theorem 5.2 (spherically complete targets)**.** *Suppose K is nonarchimedean and F is nonarchimedean and spherically complete. Then, for every normed E, the space $\mathrm{Alt}^k(E;F)$ is nonarchimedean and spherically complete, and its inclusion into $\mathrm{Mult}^k(E;F)$ has a contractive linear retraction. Thus $A^k_{E,D;F}$ is a continuous polynomial for every normed E,D and every k. Neither the field nor the source spaces need be complete.*

The proof is in Appendix P. An equivalent nonarchimedean spherically complete norm on F suffices, since equivalent norms preserve analyticity.

Corollary 5.3 (discretely valued fields)**.** *Suppose K is nonarchimedean with $|K^\times| = r^{\mathbb{Z}}$ for some $0<r<1$. Every Banach target F admitting an equivalent nonarchimedean norm makes $A^k_{E,D;F}$ analytic in every degree and for all normed E,D. The field itself need not be complete.*

Proof. Round the equivalent nonarchimedean norm upwards to $r^{\mathbb{Z}}$, with value zero at zero. The new norm lies between the old one and r^{-1} times it, and satisfies the norm axioms, since rounding is monotone and commutes with multiplication by $r^{\mathbb{Z}}$. It is complete with nonzero distances in $r^{\mathbb{Z}}$, hence spherically complete by Lemma D.1. Apply Theorem 5.2 and transport the resulting lift back through the equivalent norms. □

Changing the target category. Over nonarchimedean K, the maximum in place of the sum in the projective seminorm gives a completed exterior power $\widehat{\Lambda}^k_{\pi,\mathrm{na}}E$ representing $\mathrm{Alt}^k(E;F)$ for complete nonarchimedean F (the universal-property proof of Proposition 3.1; [25, §4]). Over a discretely valued field, $u\mapsto\widehat{\Lambda}^k_{\pi,\mathrm{na}}u$ is analytic: it is $A^k_{E,D;F}$ for $F=\widehat{\Lambda}^k_{\pi,\mathrm{na}}D$, evaluated at the

universal alternating map, so Corollary 5.3 applies. The obstructions of Section 7 concern the sum construction, whose target has no equivalent nonarchimedean norm.

In the next theorem and its proof, K is complete and nonarchimedean, and q_Y, Y^u and $j_Y\colon Y \to Y^u$ are the seminorm, the ultrametric envelope and its canonical map of a normed space Y, constructed in the proof of Proposition I.1.

Theorem 5.4 (ultrametric values)**.** *Let F be a Banach space with a nonarchimedean norm, and suppose that E^u or D^u is a bounded linear retract of $c_0(I,K)$ for some set I: there are bounded linear maps ι into $c_0(I,K)$ and ι' out of it with $\iota'\iota = \mathrm{id}$. Then $A^k_{E,D;F}$ has a bounded lift of norm at most $(\|\iota\|\|\iota'\|)^k$, and is therefore a continuous polynomial. If E or D is of countable type (Appendix I), there are such lifts of norm at most t^{-k} for every $0 < t < 1$.*

The proof is in Appendix P. If $\operatorname{char} K = p > 0$ and $k \geq p$, both hypotheses matter: Theorem 7.2(2) gives a failure with $F = K$ if K is not spherically complete; for $K = \kappa((X))$ and $E = D = \ell^\infty(\mathbb{N}, K) \cong c_0(I,K)$ [25, §1, Proposition 1], Theorem C.1 gives one with the target F_1, which has no equivalent nonarchimedean norm, so this pair is not good for Banach values (Corollary 7.6).

5.2. **Source and target splittings.** By Proposition 3.5, a source splitting (S^k_E) or a target splitting (R^k_D) makes (E,D) good for Banach values in degree k. A Banach space Y is *projective* if, for every bounded linear surjection $\psi\colon Y_1 \to Y_2$ of Banach spaces, every bounded linear $T\colon Y \to Y_2$ equals $\psi \circ T'$ for a bounded linear $T'\colon Y \to Y_1$. For a totally ordered set J, let $J^k_<$ be the set of $i \in J^k$ with $i_1 < \cdots < i_k$, and let $(e_i)_{i\in J}$ and $(\delta_i)_{i\in J^k_<}$ be the unit vectors of $\ell^1(J,K)$ and of $\ell^1(J^k_<, \widehat{K})$.

Proposition 5.5 (projective sources)**.** *Let E be a normed space and $k \geq 1$.*

1. *For every totally ordered set J, the map $\delta_i \mapsto e_{i_1} \wedge \cdots \wedge e_{i_k}$ extends to an isometric isomorphism of $\ell^1(J^k_<, \widehat{K})$ onto $\widehat{\Lambda}^k_\pi \ell^1(J,K)$.*
2. *If $\widehat{\Lambda}^k_\pi E$ is projective, then (S^k_E) holds.*
3. *If E is a bounded linear retract of $\ell^1(J,K)$ for some set J, then (S^n_E) holds for every $n \geq 1$.*

The proof is in Appendix P.

Let Σ be a totally ordered set. An *interval* of Σ is a subset I containing every t with $t_1 < t < t_2$ for some $t_1, t_2 \in I$. For a linearly independent family $(e_t)_{t\in\Sigma}$ in E with span V, write $x(t)$ for the coefficients of $x \in V$ and $\mathcal{P}_I x = \sum_{t\in I} x(t) e_t$.

Theorem 5.6 (lower estimates)**.** *Let K be any nontrivially normed field and E a normed space with a linearly independent family $(e_t)_{t\in\Sigma}$ whose span V is dense. Suppose that for some $0 < r < 2$ and $C \geq 1$,*

$$\Big(\sum_{i=1}^N \|\mathcal{P}_{I_i} x\|^r\Big)^{1/r} \leq C\|x\| \qquad (x \in V,\ I_1, \dots, I_N \text{ pairwise disjoint intervals}). \tag{5.1}$$

Put $C_\sharp = 2C^2/(1 - 2^{1-2/r})$.

1. *The linear map $\mathcal{P}_<$ on $V \otimes V$ with $\mathcal{P}_<(e_t \otimes e_{t'}) = e_t \otimes e_{t'}$ for $t < t'$ and $\mathcal{P}_<(e_t \otimes e_{t'}) = 0$ otherwise satisfies $\|\mathcal{P}_<(x \otimes y)\| \leq C_\sharp \|x\|\|y\|$ for $x, y \in V$. It therefore extends to a bounded linear map $E \otimes_\pi E \to E \widehat{\otimes}_\pi E$ of norm at most $C_\sharp$.*
2. *For every $k \geq 1$, (S^k_E) and (R^k_E) hold, with $\|s\| \leq k!\, C_\sharp^{k-1}$ and $\|\rho\| \leq C_\sharp^{k-1}$. If $E = V$, they hold without completion.*

(3) *For every normed D and every k, the pairs (E,D) and (D,E) are good for Banach values, and they are good if $E=V$.*

The proof is in Appendix J.

A normed space E has a *lower r-estimate* if it has a family $(e_t)_{t\in\Sigma}$ as in Theorem 5.6 satisfying (5.1) for some $C\geq 1$. Examples with $C=1$ are $\ell^r(\Sigma,K)$, $1\leq r<2$, with its unit vectors, its subspace of finitely supported vectors, and, for $r_1,r_2\geq 1$ with $r=\max(r_1,r_2)<2$, the space $\ell^{r_1}(\mathbb{N};\ell^{r_2}(\mathbb{N},K))$ ordered lexicographically, by Minkowski's inequality if $r_1\leq r_2$ and by $\ell^{r_2}\subseteq\ell^{r_1}$ otherwise. Conclusions (2), with norms multiplied by $(\|\iota\|\|\iota'\|)^k$, and (3) pass to every Y with bounded $\iota\colon Y\to E$ and $\iota'\colon E\to Y$, $\iota'\iota=\mathrm{id}_Y$: the maps $\iota'^{\otimes k}\circ s\circ\Lambda^k_\pi\iota$ and $\widehat{\Lambda}^k_\pi\iota'\circ\rho\circ\iota^{\otimes k}$ satisfy (S^k_Y) and (R^k_Y), by naturality.

Proposition 5.7 (ordering splittings)**.** *Let E be a normed space with a linearly independent family $(e_t)_{t\in\Sigma}$ whose span V is dense, and suppose that the map $\mathcal{P}_<$ of Theorem 5.6(1) satisfies $\|\mathcal{P}_<(x\otimes y)\|\leq C\|x\|\|y\|$ for $x,y\in V$. Let $n\geq 1$, and let $\mathcal{P}^{(n)}$ and R_n be the linear maps on $\otimes^n V$ that send $e_{t_1}\otimes\cdots\otimes e_{t_n}$ to itself if $t_1<\cdots<t_n$, respectively to the sum O_t of the tensors $e_{\tilde{t}_1}\otimes\cdots\otimes e_{\tilde{t}_n}$ over the distinct rearrangements $\tilde{t}$ of t if $t_1\leq\cdots\leq t_n$, and to 0 otherwise.*

(1) *$\mathcal{P}^{(n)}$ extends to a bounded linear map $\otimes^n_\pi E\to\widehat{\otimes}^n_\pi E$ of norm at most C^{n-1}.*
(2) *$s=\mathcal{P}^{(n)}\circ\mathrm{alt}^n_E$ and $\rho=\widehat{\wedge}^n_E\circ\mathcal{P}^{(n)}$ satisfy (S^n_E) and (R^n_E), with $\|s\|\leq n!\,C^{n-1}$ and $\|\rho\|\leq C^{n-1}$. If $E=V$, both hold without completion.*
(3) *R_n extends to a projection of $\widehat{\otimes}^n_\pi E$ onto $\widehat{\Gamma}^n_\pi E$, so (Γ^n_E) holds. Its norm is at most the sum of $n!\,(c_1!\cdots c_m!)^{-1}C^{m-1}(1+2C)^{n-m}$ over the compositions $c=(c_1,\dots,c_m)$ of n into positive parts.*

The proof is in Appendix J.

By Theorem 5.6(1), the hypothesis holds with $C=C_\sharp$ for every family satisfying (5.1).

In the rest of this subsection K is nonarchimedean. A linearly independent family $(e_t)_{t\in\Sigma}$ in E whose span V is dense is *ℓ^2-dominated* if $(\sum_t|x(t)|^2)^{1/2}\leq C_2\|x\|$ for $x\in V$ and some constant C_2. For $1\leq r<\infty$, an *r-norm* is a norm ν with $\nu(y+z)^r\leq\nu(y)^r+\nu(z)^r$; a normed space D has an *equivalent r-norm* if it carries an r-norm ν with $a\nu(y)\leq\|y\|\leq b\nu(y)$ for constants $0<a\leq b$.

Proposition 5.8 (r-norms)**.** *Let $1\leq r<\infty$. A normed space D has an equivalent r-norm if and only if there is a constant C with*

$$\Big\|\sum_t \xi_t y_t\Big\|\leq C\Big(\sum_t|\xi_t|^r\Big)^{1/r}\max_t\|y_t\| \tag{5.2}$$

for all finite families (ξ_t) in K and (y_t) in D with a common index set. The proof is in Appendix J.

Theorem 5.9 (relative ordering splittings)**.** *Let E be a normed space with an ℓ^2-dominated linearly independent family $(e_t)_{t\in\Sigma}$ whose span V is dense and with $\sup_t\|e_t\|<\infty$.*

(1) *For every $r\in(2,\infty)$, the coordinate map $V\to\ell^r(\Sigma,\widehat{K})$, $x\mapsto(x(t))_{t\in\Sigma}$, extends to a bounded linear map $w\colon E\to\ell^r(\Sigma,\widehat{K})$, which is S^n-split (Definition 3.12) for every $n\geq 1$.*
(2) *If D has an equivalent r-norm for some $r\in(2,\infty)$, then (E,D) is good for Banach values in every degree, and good in every degree if $E=V$.*

The proof is in Appendix J.

Corollary 5.10. *Let E be as in Theorem 5.9, and let D have an equivalent nonarchimedean norm. Then (E, D) is good for Banach values in every degree, and good in every degree if $E = V$. The extension $E \to c_0(\Sigma, \widehat{K})$ of the coordinate map is S^n-split for every $n \geq 1$.*

The proof is in Appendix P. For example, the sources $\ell^r(\Sigma, K)$, $1 \leq r \leq 2$, are covered, with the targets c_0, ℓ^∞, spaces with an equivalent nonarchimedean norm, and ℓ^r, $r > 2$, which is r-normed by its own norm, since $|\alpha + \beta|^r \leq \max(|\alpha|, |\beta|)^r \leq |\alpha|^r + |\beta|^r$.

5.3. **Parameter splittings.** A *parameter splitting* in degree n, a bounded projection of $\widehat{\otimes}^n_\pi P$ onto $\widehat{\Gamma}^n_\pi P$, exists if and only if P is polarizable in degree n (Definition 3.8).

Theorem 5.11 (finite-coordinate parameters)**.** *Let $j\colon W \to Z$ be a closed linear isometry. If P is boundedly linearly isomorphic to K^d and $U \subseteq P$ is open, then $f\colon U \to W$ is analytic if and only if jf is analytic. For the maximum norm on K^d, ambient coefficients b_n can be replaced by W-valued coefficients q_n with*

$$\|q_n\| \leq d^n \|b_n\| \quad (n \geq 1).$$

If Z is nonarchimedean, the factor d^n can be replaced by 1.

Proof. Theorem 2.3 puts every homogeneous diagonal in W. Expand it in coordinates as $b_n(h, \dots, h) = \sum_{|\alpha|=n} c_\alpha h^\alpha$. Every c_α lies in W, by polynomial uniqueness over the infinite field K in the algebraic quotient Z/W. For each α, let $i_1(\alpha) \leq \cdots \leq i_n(\alpha)$ be the coordinate word with these multiplicities, and set

$$q_n(h_1, \dots, h_n) = \sum_{|\alpha|=n} c_\alpha \prod_{a=1}^{n} (h_a)_{i_a(\alpha)}.$$

Before grouping there are exactly d^n ordered terms, so the triangle inequality gives the bound, and two maximum estimates give the nonarchimedean refinement. The diagonals agree, and the exponential bounds preserve a positive radius in Theorem 2.3. The case $d = 0$ is immediate. □

Over an incomplete field, finite dimension of the base does not suffice: for $K = \mathbb{F}_p(t)$ and $k \geq p$, the end of Appendix H gives a base of dimension $p + 1$ without continuous coordinates and a nonanalytic alternating atlas.

Theorem 5.12 (summable parameter spaces)**.** *Let $j\colon W \to Z$ be a closed linear isometry.*

1. *Suppose K and Z are nonarchimedean and W is complete. For $U \subseteq c_0(I, K)$ open, analyticity of jf implies analyticity of $f\colon U \to W$. Coefficients can be lifted without increasing their norms.*
2. *Suppose Z is complete. Let $a\colon H \to W$ have an ambient bounded homogeneous polynomial representation of fixed degree d, that is, $ja(h) = B(h, \dots, h)$ for a bounded d-linear $B\colon H^d \to Z$ on a normed space H. For every analytic $\gamma\colon U \to H$, with $U \subseteq \ell^1(I, K)$ open, the composite $a\gamma$ is analytic.*

Here c_0 carries the supremum norm and ℓ^1 the ordinary sum norm. The field need not be complete. Each analyticity assertion remains valid for a parameter space P that is a bounded linear retract of the indicated space V, with bounded linear $i\colon P \to V$, $r\colon V \to P$ and $ri = \mathrm{id}_P$; in (1) the lifted coefficients then satisfy $\|q_n\| \leq (\|r\|\|i\|)^n \|b_n\|$.

The proof is in Appendix P. Theorems 5.11 and 5.12 give analyticity at every point of an open set, hence the C^ω conclusion (see Conventions).

Proposition 5.13 (parameter splittings of ℓ^1)**.** *Let $n \geq 1$.*

(1) *For every set J, $\ell^1(J,K)$ is polarizable in degree n, by a projection of norm at most $n!$.*
(2) *Every normed space with finite continuous coordinates is polarizable in degree n.*

The proof is in Appendix P.

Proposition 5.14 (degree two)**.** *For every normed space P, $\mathcal{N}_P^2 = \widehat{\Gamma}_\pi^2 P$. Hence $\mathcal{S}_P^2$ and $\mathcal{G}_P^2$ coincide up to the isometry $\widehat{\Lambda}_\pi^2 P \cong \widehat{\otimes}_\pi^2 P/\mathcal{N}_P^2$, and (S_P^2) holds if and only if P is polarizable in degree 2. A section s gives a projection onto $\widehat{\Gamma}_\pi^2 P$ of norm at most $1+\|s\|$, and a projection R_2 onto $\widehat{\Gamma}_\pi^2 P$ gives a section of norm at most $1+\|R_2\|$.*

The proof is in Appendix P.

5.4. **The bundle theorem.**

Corollary 5.15. *Let M be an analytic manifold over K, with normed model space P. Fiberwise alternating maps give an analytic bifunctor on analytic normed vector bundles and their operator-valued analytic morphisms in each of the following settings:*

Base model	*Hypotheses on the typical fibers*
Arbitrary P	*$k! \neq 0$ in K; or K is nonarchimedean and the target fibers admit equivalent nonarchimedean spherically complete norms; or the source fibers have finite continuous coordinates.*
Finite continuous coordinates	*Arbitrary normed fibers.*
$c_0(I,K)$, or a bounded linear retract	*K nonarchimedean; complete nonarchimedean target fibers.*
$\ell^1(I,K)$, or a bounded linear retract	*Complete target fibers.*

The fiber hypotheses apply to both pairs of bundles at the endpoints of a morphism. Each alternative in the first row specifies a separate setting. At every finite smoothness order and at C^∞, no additional hypothesis on K, P or the fibers is needed.

Proof. The fiber results give a bounded k-linear lift Q of A, and $(u_1,\dots,u_k,g) \mapsto \big(m \mapsto g \circ Q(u_1,\dots,u_k)(m)\big)$ is a bounded $(k+1)$-linear lift of the joint action B, which is therefore analytic. The parameter results apply directly to the ambient polynomial along transition families and local morphism families; the ambient multilinear and operator spaces are complete when the final value-space fiber is complete, and nonarchimedean when it is nonarchimedean. Theorem 2.1 supplies the bundle construction in every case, and Lemma 2.4 the smooth assertion. □

Fields of characteristic zero, spherically complete or not, belong to the first row in every finite degree.

Good pairs of fibers. If (E,D) is good for Banach values in degree k for all E, D in a class $\mathcal{C}$, then Theorem 2.1 applies over every analytic manifold to bundles with source fibers in $\mathcal{C}$ and Banach value fibers: the joint action is polynomial in the precomposition variable (Lemma 3.4, Proposition 2.5) and linear in the other. Such classes are the spaces with a lower r-estimate, $r<2$ (Theorem 5.6), and the symmetric spaces with c_{00} dense and $\mathcal{P}_<$ bounded (Propositions 5.7(2), 3.5(1); under (L), Theorem 8.1). Over complete nonarchimedean K, Theorem 5.4 gives the same for source fibers of countable type and nonarchimedean Banach value fibers.

6. Degrees, decompositions and permanence

6.1. **The degrees in which goodness can fail.** For $a, b \geq 1$ and $n = a + b$, let $\wedge_E^{a,b} \colon \Lambda_\pi^a E \otimes_\pi \Lambda_\pi^b E \to \Lambda_\pi^n E$ be the wedge product, $z \otimes w \mapsto z \wedge w$, and define the shuffle coproduct by

$$\mathrm{sh}_E^{a,b}(x_1 \wedge \cdots \wedge x_n) = \sum_{|S|=a} \epsilon(S)\, x_S \otimes x_{S^c}, \tag{6.1}$$

where S runs over the a-element subsets of $\{1, \dots, n\}$, x_S is the wedge of the x_i, $i \in S$, in increasing order, and $\epsilon(S)$ is the sign of the permutation of $\{1, \dots, n\}$ that lists S in increasing order followed by S^c in increasing order. We write $\binom{n}{a}$ also for its image in K.

Lemma 6.1 (the Hopf identity)**.** *The map $\wedge_E^{a,b}$ is contractive, and $\mathrm{sh}_E^{a,b}$ is a well-defined linear map of norm at most $\binom{n}{a}$, in every characteristic. Both are natural in bounded operators, and*

$$\wedge_E^{a,b} \circ \mathrm{sh}_E^{a,b} = \tbinom{n}{a}\, \mathrm{id}. \tag{6.2}$$

Moreover $\wedge_E^n = \wedge_E^{a,b} \circ (\wedge_E^a \otimes \wedge_E^b)$ and $\mathrm{alt}_E^n = (\mathrm{alt}_E^a \otimes \mathrm{alt}_E^b) \circ \mathrm{sh}_E^{a,b}$, where $\otimes_\pi^n E$ is identified with $\otimes_\pi^a E \otimes_\pi \otimes_\pi^b E$.

The proof is in Appendix P.

Theorem 6.2 (Lucas step)**.** *Let $n = a + b$ with $a, b \geq 1$ and $\binom{n}{a} \neq 0$ in K, let F be normed, and let $G = \mathrm{Alt}^b(D; F)$. If $A_{E,D;G}^a$ and $A_{E,D;F}^b$ are analytic, so is $A_{E,D;F}^n$. In particular, if $n \neq 0$ in K and $A_{E,D;F}^{n-1}$ is analytic, so is $A_{E,D;F}^n$.*

The proof is in Appendix P.

Corollary 6.3 (the good degrees)**.** *Let $\mathrm{char}\, K = p > 0$, and let $\mathfrak{D}$ be one of the following sets of degrees $k \geq 1$: the set $\mathfrak{D}(E, D)$ of those in which (E, D) is good; the set $\mathfrak{D}_B(E, D)$ of those in which it is good for Banach values; or the set $\mathfrak{D}_K(E, D)$ of those for which $A_{E,D;K}^k$ is analytic.*

1. *$\{1, \dots, p-1\} \subseteq \mathfrak{D}$, and $a + b \in \mathfrak{D}$ whenever $a, b \in \mathfrak{D}$ and the addition of a and b in base p has no carry.*
2. *If $\mathfrak{D} \neq \{1, 2, \dots\}$, its least missing element is a power of p. Thus $\mathfrak{D} = \{1, 2, \dots\}$ if and only if $\mathfrak{D}$ contains every power of p.*
3. *Let F be normed, and suppose that $A_{E,D;F}^n$ is not analytic for some n; let n_F be the least such n. Then n_F is divisible by the least degree in which (E, D) is not good.*

The proof is in Appendix P.

Theorem 6.4 (hyperplane step)**.** *Give $E \oplus K$ and $D \oplus K$ the maximum norms, let $k \geq 2$, and let F be normed. If $A_{E\oplus K, D\oplus K;F}^k$ is analytic, then so is $A_{E,D;F}^{k-1}$.*

The proof is in Appendix P.

Theorem 6.5 (no gaps for large duals)**.** *Let $\mathrm{char}\, K = p > 0$, and let E and D be normed spaces whose continuous duals E^* and D^* are infinite-dimensional. Then each of $\mathfrak{D}(E, D)$ and $\mathfrak{D}_B(E, D)$ equals $\{1, \dots, p^h - 1\}$ for a unique $h \geq 1$, or equals $\{1, 2, \dots\}$.*

If $E \cong E \oplus K$ and $D \cong D \oplus K$ by bounded linear isomorphisms, the same holds for $\mathfrak{D}_K(E, D)$: composing with these isomorphisms on both sides shows that $A_{E,D;K}^k$ and $A_{E\oplus K, D\oplus K;K}^k$ are analytic together, so Theorem 6.4 makes $\mathfrak{D}_K(E, D)$ closed under $k \mapsto k - 1$ for $k \geq 2$, and Corollary 6.3(2) identifies its first missing element.

Theorem 6.6 (degree p for one space)**.** *Let* $\operatorname{char} K = p > 0$*, and let* Y *be a normed space. If* (S^p_Y) *holds, then* (S^n_Y) *holds for every* $n \geq 1$*. If* (R^p_Y) *holds, then* (R^n_Y) *holds for every* $n \geq 1$*. Both statements also hold without completion.*

The proofs of Theorems 6.5 and 6.6 are in Appendix L.

Corollary 6.7. *Let* $\operatorname{char} K = p > 0$*, and let* E *and* D *be normed spaces. If* (S^p_E) *holds, or if* (R^p_D) *holds, then* (E, D) *is good for Banach values in every degree. If that condition holds without completion, then* (E, D) *is good in every degree.*

Proof. By Theorem 6.6 the condition holds in every degree, and Proposition 3.5 turns it into goodness in every degree. □

Split operators. For bounded linear $w\colon E \to G$ and $n \geq 1$, a *section for* w *in degree* n is a map s as in Definition 3.12 with $\widehat{\wedge}^n_G \circ s = \Lambda^n_\pi w$, and a *retraction for* w *in degree* n is a map ρ with $\rho \circ \mathrm{alt}^n_E = \Lambda^n_\pi w$. In degree 0 both are id_K, $\otimes^0_\pi E = K$, and $\wedge^{0,b}$, $\wedge^{a,0}$, $\mathrm{sh}^{0,b}$, $\mathrm{sh}^{a,0}$ are the canonical identifications. For bounded linear $w_1, w_2\colon E \to G$ and $x_1, \dots, x_n \in E$, the term of $(w_1 + w_2)x_1 \wedge \cdots \wedge (w_1 + w_2)x_n$ with w_1 in the positions of $S \subseteq \{1, \dots, n\}$ is $\epsilon(S)\,(w_1 x)_S \wedge (w_2 x)_{S^c}$, with $(w_1 x)_S$ formed as x_S from the vectors $w_1 x_i$; so (6.1) gives

$$\Lambda^n_\pi(w_1 + w_2) = \sum_{a=0}^{n} \wedge^{a,n-a}_G \circ \left(\Lambda^a_\pi w_1 \otimes \Lambda^{n-a}_\pi w_2\right) \circ \mathrm{sh}^{a,n-a}_E \,. \tag{6.3}$$

By (6.3) and Lemma 6.1, sections $s^{(l)}_a$ for w_l in all degrees $a \leq n$ give the section $\sum_a (s^{(1)}_a \otimes s^{(2)}_{n-a}) \circ \mathrm{sh}^{a,n-a}_E$ for $w_1 + w_2$ in degree n, and retractions $\rho^{(l)}_a$ give the retraction $\sum_a \wedge^{a,n-a}_G \circ (\rho^{(1)}_a \otimes \rho^{(2)}_{n-a})$. By Lemma 6.1, if $a, b \geq 1$ and $\binom{a+b}{a} \neq 0$ in K, sections s_a, s_b for w in degrees a, b give the section $\binom{a+b}{a}^{-1}(s_a \otimes s_b) \circ \mathrm{sh}^{a,b}_E$ in degree $a + b$, and retractions ρ_a, ρ_b give the retraction $\binom{a+b}{a}^{-1} \wedge^{a,b}_G \circ (\rho_a \otimes \rho_b)$; this is the Lucas step for split operators. Every operator has sections and retractions in the degrees n with $n! \neq 0$ in K (the paragraph after Definition 3.12); so if $\operatorname{char} K = p > 0$, the degrees in which an operator is S-split, or R-split, contain $1, \dots, p-1$ and are closed under carry-free sums. The $\mathrm{S}^{\leq k}$-split maps $E \to G$ form a linear space, as do the $\mathrm{R}^{\leq k}$-split ones, and both are closed under composition with bounded linear maps on either side (Proposition 3.13(2)).

Theorem 6.8 (composition)**.** *Let* $q, m \geq 1$ *and* $n = qm$*, and suppose that the number* $N_{q,m} = (qm)!/\big((q!)^m\, m!\big)$ *of partitions of* $\{1, \dots, n\}$ *into* m *blocks of size* q *is nonzero in* K*; this holds, for instance, when* q *and* m *are powers of* $p = \operatorname{char} K > 0$*. Let* $|N_{q,m}|$ *be the absolute value of its image in* K*; elsewhere* $N_{q,m}$ *and* $m!$ *are real numbers.*

1. *If* $w_1\colon E \to G_1$ *has a section* s_1 *in degree* q *and* $w_2\colon G_1 \to G_2$ *has a section* s_2 *in degree* m*, then* $w_2 w_1$ *has a section* s *in degree* n *with* $\|s\| \leq N_{q,m} \|s_1\|^m \|s_2\|^q / |N_{q,m}|$*.*
2. *If* $w_1\colon E \to G_1$ *has a retraction* ρ_1 *in degree* m *and* $w_2\colon G_1 \to G_2$ *has a retraction* ρ_2 *in degree* q*, then* $w_2 w_1$ *has a retraction* ρ *in degree* n *with* $\|\rho\| \leq (m!)^{q-1} \|\rho_2\|^m \|\rho_1\|^q / |N_{q,m}|$*.*

If the given sections, respectively retractions, are without completion, so is the new one.

Corollary 6.9. *Let* $\operatorname{char} K = p > 0$ *and* $h \geq 1$*.*

1. *A composite of* h *operators that are* S^p*-split is* S^n*-split for every* $n < p^{h+1}$*, and a composite of* h *operators that are* R^p*-split is* R^n*-split for every* $n < p^{h+1}$*.*
2. *An operator that factors through a normed space* Y *with* (S^p_Y) *is* S^n*-split for every* n*, and an operator that factors through a normed space* Y *with* (R^p_Y) *is* R^n*-split for every* n*.*

Theorem 6.10 (degree p for polarizability)**.** *Let* $\operatorname{char} K = p > 0$*. If a normed space* P *is polarizable in degree* p*, then* P *is polarizable in every degree* $n \geq 1$*.*

The proofs of Theorems 6.8 and 6.10 and of Corollary 6.9 are in Appendix L.

6.2. **Relative splittings and decompositions.** Throughout this subsection E and D are normed spaces and $k \geq 1$. Since $\widehat{\Lambda}^n_\pi \widehat{D} = \widehat{\Lambda}^n_\pi D$ (the paragraph before Theorem 3.16), $\Lambda^n_\pi v$ maps $\Lambda^n_\pi E$ into $\widehat{\Lambda}^n_\pi D$ for every $v \in \mathcal{L}(E, \widehat{D})$.

Proposition 6.11 (relative splittings)**.** *Let* G *be a normed space.*

(1) *Let* $w\colon E \to G$ *have a section* s *in degree* k*, and let* $\varepsilon\colon \mathcal{L}(E,D) \to \mathcal{L}(G, \widehat{D})$ *be bounded linear with* $\varepsilon(u) \circ w = u$ *for every* $u \in \mathcal{L}(E,D)$*. Then* (E,D) *is good for Banach values in degree* k*, with the lift*

$$Q(u_1, \dots, u_k) = \widehat{\wedge}^k_{\widehat{D}} \circ \big(\varepsilon(u_1) \otimes \cdots \otimes \varepsilon(u_k)\big) \circ s$$

of norm at most $\|\varepsilon\|^k \|s\|$*.*

(2) *Let* $w'\colon G \to \widehat{D}$ *have a retraction* ρ *in degree* k*, and let* $\varepsilon\colon \mathcal{L}(E,D) \to \mathcal{L}(E,G)$ *be bounded linear with* $w' \circ \varepsilon(u) = u$ *for every* $u \in \mathcal{L}(E,D)$*. Then* (E,D) *is good for Banach values in degree* k*, with the lift*

$$Q(u_1, \dots, u_k) = \rho \circ \big(\varepsilon(u_1) \otimes \cdots \otimes \varepsilon(u_k)\big) \circ \operatorname{alt}^k_E$$

of norm at most $k! \, \|\varepsilon\|^k \|\rho\|$*.*

(3) *If in (1) the section* s *takes values in* $\otimes^k_\pi G$ *and* ε *takes values in* $\mathcal{L}(G,D)$*, or if in (2) the retraction* ρ *takes values in* $\Lambda^k_\pi D$*, then* (E,D) *is good in degree* k*.*

Proof. Both maps Q are bounded and k-linear, with the stated norms. In (1), naturality of $\wedge^k$, extended to completions, gives $Q(u, \dots, u) = \widehat{\Lambda}^k_\pi \varepsilon(u) \circ \widehat{\wedge}^k_G \circ s = \Lambda^k_\pi \varepsilon(u) \circ \Lambda^k_\pi w = \Lambda^k_\pi u$. In (2), naturality of alt^k gives $Q(u, \dots, u) = \rho \circ \operatorname{alt}^k_G \circ \Lambda^k_\pi \varepsilon(u) = \Lambda^k_\pi w' \circ \Lambda^k_\pi \varepsilon(u) = \Lambda^k_\pi u$. Under the hypotheses of (3) both lifts take values in $\mathcal{L}(\Lambda^k_\pi E, \Lambda^k_\pi D)$. Lemma 3.4 concludes. □

Definition 6.12. (1) A *source-side relative splitting* of degree k for (E,D) is a triple (G, w, ε) as in Proposition 6.11(1): $w\colon E \to G$ is S^k-split and $\varepsilon(u) \circ w = u$. A *target-side relative splitting* of degree k is a triple (G, w', ε) as in Proposition 6.11(2): $w'\colon G \to \widehat{D}$ is R^k-split and $w' \circ \varepsilon(u) = u$.

(2) A *piece* for (E,D) is a tuple $(G, G', w, w', \varepsilon)$ of normed spaces G and G', bounded linear maps $w\colon E \to G$ and $w'\colon G' \to \widehat{D}$, and a bounded linear map $\varepsilon\colon \mathcal{L}(E,D) \to \mathcal{L}(G,G')$. It is of *source type* in degree k if w is $\mathrm{S}^{\leq k}$-split, and of *target type* in degree k if w' is $\mathrm{R}^{\leq k}$-split.

(3) A *splitting decomposition* of degree k is a finite family of pieces $(G_i, G_i', w_i, w_i', \varepsilon_i)$, $1 \leq i \leq m$, each of source type or of target type in degree k, such that

$$u = \sum_{i=1}^{m} w_i' \, \varepsilon_i(u) \, w_i \qquad \text{for every } u \in \mathcal{L}(E,D). \tag{6.4}$$

A source-side relative splitting (G, w, ε), respectively a target-side relative splitting (G, w', ε), of degree k also counts as a splitting decomposition of degree k with the single piece $(G, \widehat{D}, w, \mathrm{id}_{\widehat{D}}, \varepsilon)$, respectively $(E, G, \mathrm{id}_E, w', \varepsilon)$.

(4) These structures are *without completion* if their sections take values in the uncompleted tensor powers, their retractions take values in the uncompleted exterior powers $\Lambda^n_\pi D$, the maps $\varepsilon(u)$ of a source-side relative splitting take values in D, and the maps w_i' of a decomposition take values in D.

A *linear family* is a bounded linear map $\theta\colon P \to \mathcal{L}(E, \widehat{D})$ on a normed space P. It is *liftable in degree* $n \geq 1$ if there is a bounded n-linear map $Q\colon P^n \to \mathcal{L}(\Lambda_\pi^n E, \widehat{\Lambda}_\pi^n D)$ with $Q(h, \dots, h) = \Lambda_\pi^n \theta(h)$ for every $h \in P$. For $P = \mathcal{L}(E, D)$, Lemma 3.4 shows that (E, D) is good for Banach values in degree n if and only if the inclusion $\theta_{E,D}\colon \mathcal{L}(E, D) \to \mathcal{L}(E, \widehat{D})$ is liftable in degree n. The proofs of Theorem 6.13, Propositions 6.15 and 6.18 and Lemma 6.16 are in Appendix P.

Theorem 6.13 (decompositions). (1) *For a piece $(G, G', w, w', \varepsilon)$ of source type or of target type in degree k, the map $u \mapsto w'\varepsilon(u)w$ is liftable in every degree $n \leq k$.*
(2) *Suppose that $u = \sum_{i=1}^m \theta_i(u)$ for every $u \in \mathcal{L}(E, D)$, where each $\theta_i\colon \mathcal{L}(E, D) \to \mathcal{L}(E, \widehat{D})$ is bounded linear and liftable in every degree $n \leq k$. Then (E, D) is good for Banach values in degree k. If every $\theta_i(u)$ takes values in D and every lift can be chosen with values in $\mathcal{L}(\Lambda_\pi^n E, \Lambda_\pi^n D)$, then (E, D) is good in degree k.*
(3) *A splitting decomposition of degree k makes (E, D) good for Banach values in degree k, and good in degree k if it is without completion.*

Proposition 6.14 (transfer). *Let $w\colon E \to G$ and $w'\colon G' \to \widehat{D}$ be bounded linear maps of normed spaces, and let $\varepsilon\colon \mathcal{L}(E, D) \to \mathcal{L}(G, G')$ be bounded linear with $u = w'\varepsilon(u)w$ for every u. If (G, G') is good for Banach values in degree k, with a lift Q', then so is (E, D), with the lift*

$$Q(u_1, \dots, u_k) = \widehat{\Lambda}_\pi^k w' \circ Q'\big(\varepsilon(u_1), \dots, \varepsilon(u_k)\big) \circ \Lambda_\pi^k w.$$

Proof. The map Q is bounded and k-linear, with values in $\mathcal{L}(\Lambda_\pi^k E, \widehat{\Lambda}_\pi^k \widehat{D})$ and $\widehat{\Lambda}_\pi^k \widehat{D} = \widehat{\Lambda}_\pi^k D$; its diagonal is $\widehat{\Lambda}_\pi^k w' \circ \Lambda_\pi^k \varepsilon(u) \circ \Lambda_\pi^k w = \Lambda_\pi^k u$. Lemma 3.4 concludes. □

The same formula makes $u \mapsto w'\varepsilon(u)w$ liftable in degree n for every piece $(G, G', w, w', \varepsilon)$ with (G, G') good for Banach values in degree n.

Proposition 6.15 (two pieces suffice). *For a bounded bilinear map $b\colon \mathcal{L}(E, D) \times E \to \widehat{D}$ let $w_b\colon E \to \mathcal{L}(\mathcal{L}(E, D), \widehat{D})$ be the map $w_b(x)(u) = b(u, x)$, and let $w_b'\colon \mathcal{L}(E, D) \otimes_\pi E \to \widehat{D}$ be the linear map with $w_b'(u \otimes x) = b(u, x)$. The pair (E, D) has a splitting decomposition of degree k into pieces of source and target type if and only if there are bounded bilinear maps $b_S, b_R\colon \mathcal{L}(E, D) \times E \to \widehat{D}$ with $b_S(u, x) + b_R(u, x) = ux$, w_{b_S} $\mathrm{S}^{\leq k}$-split and w'_{b_R} $\mathrm{R}^{\leq k}$-split. In that case (E, D) has such a decomposition with at most two pieces.*

Lemma 6.16 (complemented sub-pairs). *Let E_0 and D_0 be normed spaces, $J\colon E_0 \to E$ and $\Pi\colon \widehat{D} \to \widehat{D}_0$ bounded linear maps, and $\Phi\colon \mathcal{L}(E_0, D_0) \to \mathcal{L}(E, D)$ a bounded linear map with $\Pi\, \Phi(v)\, J = v$ for every $v \in \mathcal{L}(E_0, D_0)$. This holds, for instance, if $\Pi' J = \mathrm{id}_{E_0}$ and $\Pi J' = \mathrm{id}_{D_0}$ for bounded linear maps $\Pi'\colon E \to E_0$ and $J'\colon D_0 \to D$, and $\Phi(v) = J' v \Pi'$.*
(1) *If (E, D) has a splitting decomposition of degree k, so does (E_0, D_0).*
(2) *If (E, D) is good for Banach values in degree k, so is (E_0, D_0).*

Definition 6.17. A normed space P is *polarizable in degrees* $\leq k$ if it is polarizable in degree n for every $n \leq k$. A *parameter piece* of degree k for (E, D) is a piece $(G, G', w, w', \varepsilon)$ with $\varepsilon = \varepsilon' \circ \psi$ for bounded linear maps $\psi\colon \mathcal{L}(E, D) \to P$ and $\varepsilon'\colon P \to \mathcal{L}(G, G')$, where P is polarizable in degrees $\leq k$; the maps w and w' are arbitrary. A *generalized decomposition* of degree k is a finite family of pieces satisfying (6.4), each of source type or of target type in degree k, or a parameter piece of degree k. A source-side or target-side relative splitting of degree k also counts as a generalized decomposition of degree k, as in Definition 6.12(3).

Proposition 6.18 (parameter pieces). (1) *For every set I, the space $\ell^1(I, K)$ is polarizable in every degree, and so is every normed space with finite continuous coordinates, for instance every finite-dimensional normed space over a complete field (Lemma A.3).*

(2) *If P_1 and P_2 are polarizable in degrees $\le k$, so is $P_1 \oplus P_2$ with the maximum norm.*
(3) *If P is polarizable in degree n and P_0 is a retract of P, then P_0 is polarizable in degree n.*
(4) *For a parameter piece $(G, G', w, w', \varepsilon)$ of degree k, the map $u \mapsto w'\varepsilon(u)w$ is liftable in every degree $n \le k$. Hence a generalized decomposition of degree k makes (E, D) good for Banach values in degree k.*

Corollary 6.19. *Let* $\operatorname{char} K = p > 0$.

(1) *A parameter piece of degree p is a parameter piece of every degree, and a generalized decomposition of degree p into parameter pieces is one of every degree.*
(2) *If $\mathcal{L}(E, D)$, with the operator norm, is polarizable in degree p, then (E, D) is good for Banach values in every degree.*

Theorem 6.20 (decompositions of degree p)**.** *Let* $\operatorname{char} K = p > 0$. *A splitting decomposition of degree p is a splitting decomposition of degree $p^2 - 1$, with the same pieces. Hence (E, D) is then good for Banach values in every degree $n < p^2$, and good in every degree $n < p^2$ if the decomposition is without completion.*

Proposition 6.21 (target pieces)**.** *Suppose that $E^* \neq 0$ and that (E, D) has a splitting decomposition of degree k all of whose pieces are of target type in degree k. Then (R^k_D) holds, without completion if the decomposition is without completion. If moreover* $\operatorname{char} K = p > 0$ *and $k = p$, then (E, D) is good for Banach values in every degree, and good in every degree if the decomposition is without completion.*

The proofs of Corollary 6.19, Theorem 6.20 and Proposition 6.21 are in Appendix L.

6.3. **The class of an operator family.** Throughout this subsection E, D, F, P and P' are normed spaces and $k \ge 1$; linear families and liftability are as defined before Theorem 6.13. As in Lemma 3.14, with the same proof, a linear family $\theta\colon P \to \mathcal{L}(E, \widehat{D})$ gives a unique bounded linear map $\mathrm{ev}_\theta\colon \widehat{\otimes}^k_\pi E \to \mathrm{Mult}^k(P; \widehat{\Lambda}^k_\pi D)$ with

$$\mathrm{ev}_\theta(x_1 \otimes \cdots \otimes x_k)(h_1, \dots, h_k) = \theta(h_1)x_1 \wedge \cdots \wedge \theta(h_k)x_k \in \widehat{\Lambda}^k_\pi \widehat{D} = \widehat{\Lambda}^k_\pi D;$$

it has norm at most $\|\theta\|^k$, and $\mathrm{ev}_\theta(z)(h, \dots, h) = \widehat{\Lambda}^k_\pi \theta(h)(\widehat{\wedge}^k_E z)$ for $z \in \widehat{\otimes}^k_\pi E$ and $h \in P$, so it maps $\mathcal{N}^k_E$ into $\mathrm{Mult}^k_0(P; \widehat{\Lambda}^k_\pi D)$. Let q be the quotient map of the *coefficient sequence* $\mathcal{M}^k(P; \widehat{\Lambda}^k_\pi D)$ of P (Definition 3.8(4)). Since $q \circ \mathrm{ev}_\theta$ vanishes on $\mathcal{N}^k_E$, Lemma 3.7 gives a bounded linear map $\overline{\mathrm{ev}}_\theta\colon \widehat{\Lambda}^k_\pi E \to \mathrm{Poly}^k(P; \widehat{\Lambda}^k_\pi D)$ with $\overline{\mathrm{ev}}_\theta \circ \widehat{\wedge}^k_E = q \circ \mathrm{ev}_\theta$. The *class* of θ in degree k is

$$\mathrm{ob}_k(\theta) = (\mathrm{ev}_\theta |_{\mathcal{N}^k_E})_*[\mathcal{S}^k_E] \in \mathrm{Ext}\big(\widehat{\Lambda}^k_\pi E, \mathrm{Mult}^k_0(P; \widehat{\Lambda}^k_\pi D)\big), \tag{6.5}$$

and $\mathrm{ob}_k(\theta_{E,D}) = \mathrm{ob}_k(E, D)$, since $\mathrm{ev}_{\theta_{E,D}} = \mathrm{ev}^k_{E,D}$.

For exact sequences $\mathcal{Y}\colon 0 \to Y_1 \xrightarrow{i} Y_2 \xrightarrow{q} Y_3 \to 0$ and $\mathcal{Y}'\colon 0 \to Y'_1 \xrightarrow{i'} Y'_2 \xrightarrow{q'} Y'_3 \to 0$, a *morphism* $\mathcal{Y} \to \mathcal{Y}'$ is a triple of bounded linear maps $\alpha_l\colon Y_l \to Y'_l$, $l = 1, 2, 3$, with $\alpha_2 i = i'\alpha_1$ and $q'\alpha_2 = \alpha_3 q$. For a bounded linear $\gamma\colon Y' \to Y_3$ from a Banach space Y', the *pullback* $\gamma^*\mathcal{Y}\colon 0 \to Y_1 \to Y_\gamma \to Y' \to 0$ has $Y_\gamma = \{(y, y') \in Y_2 \oplus Y' : qy = \gamma y'\}$, normed by $\|y\| + \|y'\|$, with the maps $y_1 \mapsto (iy_1, 0)$ and $(y, y') \mapsto y'$. It is exact, its class $\gamma^*[\mathcal{Y}]$ depends only on $[\mathcal{Y}]$, and $\gamma^*[\mathcal{Y}] = 0$ if and only if $\gamma = q \circ L$ for a bounded linear $L\colon Y' \to Y_2$ (Lemma N.1). Pushouts and pullbacks compose and commute, and a morphism $(\alpha_1, \alpha_2, \alpha_3)\colon \mathcal{Y} \to \mathcal{Y}'$ gives $\alpha_{1*}[\mathcal{Y}] = \alpha_3^*[\mathcal{Y}']$ (Lemma N.2, Corollary N.3).

Theorem 6.22 (the class of a family)**.** *Let $\theta\colon P \to \mathcal{L}(E, \widehat{D})$ be a linear family.*

(1) *The following are equivalent: θ is liftable in degree k; the restriction $\mathrm{ev}_\theta|_{\mathcal{N}_E^k}$ extends to a bounded linear map $\widehat{\otimes}_\pi^k E \to \mathrm{Mult}_0^k(P;\widehat{\Lambda}_\pi^k D)$; $\mathrm{ob}_k(\theta)=0$ in (6.5). A lift Q gives such an extension of norm at most $\|\theta\|^k+\|Q\|$, and such an extension ξ gives a lift of norm at most $\|\theta\|^k+\|\xi\|$.*

(2) *The triple $(\mathrm{ev}_\theta|_{\mathcal{N}_E^k}, \mathrm{ev}_\theta, \overline{\mathrm{ev}}_\theta)$ is a morphism $\mathcal{S}_E^k \to \mathcal{M}^k(P;\widehat{\Lambda}_\pi^k D)$, and $\mathrm{ob}_k(\theta) = \overline{\mathrm{ev}}_\theta^*[\mathcal{M}^k(P;\widehat{\Lambda}_\pi^k D)]$.*

Proof. (1) Steps 1–3 of the proof of Theorem 3.15, with its correspondence $Q \leftrightarrow Q^\sharp$, apply with P, $\theta(h)$, ev_θ and $\mathrm{Mult}^k(P;\widehat{\Lambda}_\pi^k D)$ in place of $\mathcal{L}(E,D)$, u, ev and $\mathrm{Mult}^k(\mathcal{L}(E,D);\widehat{\Lambda}_\pi^k D)$, using $\mathrm{ev}_\theta(z)(h,\dots,h) = \widehat{\Lambda}_\pi^k\theta(h)(\widehat{\wedge}_E^k z)$; the bound $\|\mathrm{ev}_\theta\| \le \|\theta\|^k$ replaces $\|\mathrm{ev}\| \le 1$ in the constants. The proof of (2) is in Appendix N. □

For $\psi \in \mathcal{L}(P',P)$ and a Banach space Y let $\psi^\sharp\colon \mathrm{Mult}^k(P;Y) \to \mathrm{Mult}^k(P';Y)$ be the map $\Phi \mapsto \Phi\circ(\psi\times\cdots\times\psi)$, and for a bounded linear map $\chi\colon Y \to Y'$ of Banach spaces let $\chi_\flat\colon \mathrm{Mult}^k(P;Y) \to \mathrm{Mult}^k(P;Y')$ be the map $\Phi \mapsto \chi\circ\Phi$. Both preserve zero diagonals, and their pushouts are taken along these restrictions.

Proposition 6.23 (naturality)**.** *Let $\theta\colon P \to \mathcal{L}(E,\widehat{D})$ be a linear family.*

(a) *For $\psi \in \mathcal{L}(P',P)$, $\mathrm{ob}_k(\theta\circ\psi) = (\psi^\sharp)_*\,\mathrm{ob}_k(\theta)$.*

(b) *Let E' be a normed space, $w \in \mathcal{L}(E',E)$, and $\theta_w(h) = \theta(h)\circ w$ for $h \in P$. Then $\mathrm{ob}_k(\theta_w) = (\widehat{\Lambda}_\pi^k w)^*\,\mathrm{ob}_k(\theta) = (\mathrm{ev}_\theta|_{\mathcal{N}_E^k})_*\,\mathrm{sp}_k(w)$, with $\mathrm{sp}_k(w)$ as in Proposition 3.13(1).*

(c) *Let D' be a normed space, $w'\colon \widehat{D} \to \widehat{D}'$ bounded linear, and $(w'\theta)(h) = w'\circ\theta(h)$ for $h \in P$. Then $\mathrm{ob}_k(w'\theta) = ((\widehat{\Lambda}_\pi^k w')_\flat)_*\,\mathrm{ob}_k(\theta)$.*

(d) *$((\mathrm{alt}_D^k)_\flat)_*\,\mathrm{ob}_k(\theta) = 0$, with alt_D^k extended to $\widehat{\Lambda}_\pi^k D$. If G is a normed space, $w'\colon G \to \widehat{D}$ is R^k-split and $\theta_0\colon P \to \mathcal{L}(E,G)$ is bounded linear, then the family $h \mapsto w'\circ\theta_0(h)$ is liftable in degree k.*

The proof is in Appendix N.

Corollary 6.24 (transport)**.** (1) *Let G and G' be normed spaces, and let $w\colon E \to G$, $w'\colon G' \to \widehat{D}$ and $\varepsilon\colon \mathcal{L}(E,D) \to \mathcal{L}(G,G')$ be bounded linear maps with $u = w'\varepsilon(u)w$ for every $u \in \mathcal{L}(E,D)$. Then*

$$\mathrm{ob}_k(E,D) = ((\widehat{\Lambda}_\pi^k w')_\flat)_*(\varepsilon^\sharp)_*(\widehat{\Lambda}_\pi^k w)^*\,\mathrm{ob}_k(G,G'),$$

with w' extended to $\widehat{G}'$.

(2) *Let G be a normed space, and let $w\colon E \to G$ and $\varepsilon\colon \mathcal{L}(E,D) \to \mathcal{L}(G,\widehat{D})$ be bounded linear maps with $u = \varepsilon(u)\circ w$ for every $u \in \mathcal{L}(E,D)$. Then $\mathrm{ob}_k(E,D) = (\mathrm{ev}_\varepsilon|_{\mathcal{N}_G^k})_*\,\mathrm{sp}_k(w)$, which vanishes if w is S^k-split.*

The proof is in Appendix P.

Theorem 6.25 (sum formula)**.** *Let P_1 and P_2 be normed spaces, $\theta_1\colon P_1 \to \mathcal{L}(E,\widehat{D})$ and $\theta_2\colon P_2 \to \mathcal{L}(E,\widehat{D})$ linear families, $P = P_1 \oplus P_2$ with the maximum norm, $\theta_1 \boxplus \theta_2\colon P \to \mathcal{L}(E,\widehat{D})$ the family $(h_1,h_2) \mapsto \theta_1(h_1)+\theta_2(h_2)$, and $\iota_1\colon P_1 \to P$, $\iota_2\colon P_2 \to P$ the inclusions. Put $Y = \widehat{\Lambda}_\pi^k D$. For $S \subseteq \{1,\dots,k\}$ and $\Phi \in \mathrm{Mult}^k(P;Y)$ let $\Phi_S(h^1,\dots,h^k)$, where $h^j = (h_1^j,h_2^j) \in P$, be the value of Φ at the tuple whose j-th entry is $(h_1^j,0)$ for $j \in S$ and $(0,h_2^j)$ for $j \notin S$, and let $\Pi^{(a)}\Phi = \sum_{|S|=a}\Phi_S$. The maps $\Pi^{(a)}$, $0 \le a \le k$, are bounded commuting idempotents on $\mathrm{Mult}^k(P;Y)$ with sum* id *that map $\mathrm{Mult}_0^k(P;Y)$ into itself (Lemma N.4). Put $\mathrm{Mult}_0^{(a)} = \Pi^{(a)}\,\mathrm{Mult}_0^k(P;Y)$ and $\mathrm{ob}^{(a)} = (\Pi^{(a)}|_{\mathrm{Mult}_0^k(P;Y)})_*\,\mathrm{ob}_k(\theta_1\boxplus\theta_2) \in \mathrm{Ext}(\widehat{\Lambda}_\pi^k E, \mathrm{Mult}_0^{(a)})$.*

(1) $\mathrm{ob}_k(\theta_1 \boxplus \theta_2) = 0$ *if and only if* $\mathrm{ob}^{(a)} = 0$ *for every* a. *The map* $\iota_1^\sharp$ *restricts to an isomorphism of* $\mathrm{Mult}_0^{(k)}$ *onto* $\mathrm{Mult}_0^k(P_1; Y)$ *with* $(\iota_1^\sharp)_* \mathrm{ob}^{(k)} = \mathrm{ob}_k(\theta_1)$, *and* $\iota_2^\sharp$ *restricts to an isomorphism of* $\mathrm{Mult}_0^{(0)}$ *onto* $\mathrm{Mult}_0^k(P_2; Y)$ *with* $(\iota_2^\sharp)_* \mathrm{ob}^{(0)} = \mathrm{ob}_k(\theta_2)$.

(2) *Let* $0 < a < k$. *Then* $\mathrm{ob}^{(a)} = 0$ *if and only if there is a bounded* k*-linear map* $Q^{(a)} \colon P_1^a \times P_2^{k-a} \to \mathcal{L}(\widehat{\Lambda}_\pi^k E, \widehat{\Lambda}_\pi^k D)$ *with*

$$Q^{(a)}(h_1, \dots, h_1; h_2, \dots, h_2) = \wedge_{\widehat{D}}^{a,k-a} \circ \left(\widehat{\Lambda}_\pi^a \theta_1(h_1) \otimes \widehat{\Lambda}_\pi^{k-a} \theta_2(h_2)\right) \circ \mathrm{sh}_E^{a,k-a}$$

for $h_1 \in P_1$ *and* $h_2 \in P_2$, *the maps on the right extended to completions; the right side is the term of index* a *in* (6.3) *for* $w_1 = \theta_1(h_1)$ *and* $w_2 = \theta_2(h_2)$.

(3) *Let* $0 < a < k$. *If* θ_1 *is liftable in degree* a *and* θ_2 *in degree* $k - a$, *then* $\mathrm{ob}^{(a)} = 0$.

(4) *Suppose that* $P_1 = P_2$, *and let* $\Delta_{P_1} \colon P_1 \to P$ *be the map* $h \mapsto (h, h)$. *If* $\mathrm{ob}^{(a)} = 0$ *for every* a, *then* $\theta_1 + \theta_2 = (\theta_1 \boxplus \theta_2) \circ \Delta_{P_1}$ *is liftable in degree* k. *In particular,* $\theta_1 + \theta_2$ *is liftable in degree* k *if* θ_1 *and* θ_2 *are liftable in every degree at most* k.

(5) *Suppose that* $P_1 = P_2$ *and* $\theta_1 = \theta_2 = \theta$, *let* Δ_{P_1} *be as in (4), and let* $0 < a < k$. *Then* $\Delta_{P_1}^\sharp \circ \Pi^{(a)} \circ \mathrm{ev}_{\theta \boxplus \theta} = \binom{k}{a} \mathrm{ev}_\theta$. *Hence, if* $\binom{k}{a} \neq 0$ *in* K, *then* $\mathrm{ob}_k(\theta) = 0$ *if and only if* $(\Delta_{P_1}^\sharp|_{\mathrm{Mult}_0^{(a)}})_* \mathrm{ob}^{(a)} = 0$, *and* θ *is liftable in degree* k *if it is liftable in degrees* a *and* $k - a$.

The proof is in Appendix N.

Proposition 6.26 (composite families)**.** *Let* $\theta \colon P \to \mathcal{L}(E, \widehat{D})$ *and* $\theta' \colon P' \to \mathcal{L}(D, \widehat{F})$ *be linear families. Extend each* $\theta'(h')$ *to* $\widehat{D}$, *and regard* θ' *also as a family* $P' \to \mathcal{L}(\widehat{D}, \widehat{F})$; *its class is* $\mathrm{ob}_k(\theta')$, *since* $\mathcal{S}_{\widehat{D}}^k = \mathcal{S}_D^k$ *(the paragraph before Theorem 3.16). Let* $\theta' \odot \theta \colon P' \otimes_\pi P \to \mathcal{L}(E, \widehat{F})$ *be the bounded linear map on the uncompleted projective tensor product with* $(\theta' \odot \theta)(h' \otimes h) = \theta'(h') \circ \theta(h)$.

(1) *For* $h \in P$ *let* $\psi_h \colon P' \to P' \otimes_\pi P$ *be the map* $h' \mapsto h' \otimes h$. *Then* $(\psi_h^\sharp)_* \mathrm{ob}_k(\theta' \odot \theta) = (\widehat{\Lambda}_\pi^k \theta(h))^* \mathrm{ob}_k(\theta')$. *For* $h' \in P'$ *let* $\psi'_{h'} \colon P \to P' \otimes_\pi P$ *be the map* $h \mapsto h' \otimes h$. *Then* $((\psi'_{h'})^\sharp)_* \mathrm{ob}_k(\theta' \odot \theta) = ((\widehat{\Lambda}_\pi^k \theta'(h'))_\flat)_* \mathrm{ob}_k(\theta)$.

(2) *Let* $\mathrm{comp} \colon \mathcal{L}(D, F) \otimes_\pi \mathcal{L}(E, D) \to \mathcal{L}(E, F)$ *be the composition map,* $v \otimes u \mapsto vu$. *Then* $\mathrm{ob}_k(\theta_{D,F} \odot \theta_{E,D}) = (\mathrm{comp}^\sharp)_* \mathrm{ob}_k(E, F)$. *In particular, if* (E, F) *is good for Banach values in degree* k, *then* $\theta_{D,F} \odot \theta_{E,D}$ *is liftable in degree* k.

The proof is in Appendix N.

6.4. **Block assemblies and permanence.** The proofs of Propositions 6.27 and 6.28 are in Appendix P.

Proposition 6.27 (block assemblies)**.** *Let* I *be a totally ordered set, and let* E *be the* ℓ^1*-sum, algebraic or completed, of normed spaces* E_i, $i \in I$, *with the norm* $\|x\| = \sum_i \|\Pi_i x\|$, *inclusions* $J_i \colon E_i \to E$ *and block projections* $\Pi_i \colon E \to E_i$. *Let* D_i *be normed spaces and* $J_i' \colon D_i \to D$ *bounded linear maps with* $\|J_i'\| \le 1$. *Let* $\mathcal{U} \subseteq \mathcal{L}(E, D)$ *be a linear subspace with linear maps* $\mathcal{U} \to \mathcal{L}(E_i, D_i)$, $u \mapsto u_i$, *such that* $\|u_i\| \le c\|u\|$ *for a constant* c, *and* $ux = \sum_i J_i' u_i \Pi_i x$ *for every* $x \in E$ *with finitely many nonzero blocks. Suppose that there is a constant* $C \ge 1$ *such that for every* $i \in I$ *and every* $n \le k$ *there is a bounded* n*-linear map* Q_n^i *on* $\mathcal{U}_i = \{u_i : u \in \mathcal{U}\}$ *with values in* $\mathcal{L}(\Lambda_\pi^n E_i, \widehat{\Lambda}_\pi^n D_i)$, *with* $Q_n^i(v, \dots, v) = \Lambda_\pi^n v$ *and* $\|Q_n^i\| \le C$. *Then there is a bounded* k*-linear map* $Q \colon \mathcal{U}^k \to \mathcal{L}(\Lambda_\pi^k E, \widehat{\Lambda}_\pi^k D)$ *with* $Q(u, \dots, u) = \Lambda_\pi^k u$ *and* $\|Q\| \le C^k c^k$. *In particular, if* $\mathcal{U} = \mathcal{L}(E, D)$, *then* (E, D) *is good for Banach values in degree* k.

Proposition 6.28 (permanence)**.** *Let* $k \ge 1$, *and give finite direct sums the maximum norm.*

(1) (Retracts.) *Let Y_0 be a retract of a normed space Y. If Y satisfies (S^k_Y), or (R^k_Y), or is polarizable in degree k, then Y_0 satisfies the same condition. If (E, D) is good for Banach values in degree k and E_0, D_0 are retracts of E, D, then (E_0, D_0) is good for Banach values in degree k.*

(2) (Finite sums.) *For normed spaces Y_1 and Y_2, $(\mathrm{S}^n_{Y_1\oplus Y_2})$ holds for every $n \le k$ if and only if $(\mathrm{S}^n_{Y_1})$ and $(\mathrm{S}^n_{Y_2})$ hold for every $n \le k$; the same holds for (R^n) and for polarizability in degrees $\le k$. If (E_a, D_b) is good for Banach values in every degree $n \le k$ for all $a, b \in \{1, 2\}$, so is $(E_1 \oplus E_2, D_1 \oplus D_2)$.*

(3) (Completion of a space.) *Regard the completion $\widehat{Y}$ of a normed space Y as a normed space over K. Then Y satisfies (S^k_Y), (R^k_Y), or polarizability in degree k, if and only if $\widehat{Y}$ satisfies the same condition.*

(4) (Completion of a pair.) *If D is complete, then (E, D) is good for Banach values in degree k if and only if $(\widehat{E}, D)$ is. If $(E, \widehat{D})$ is good for Banach values in degree k, so is (E, D).*

Every normed space embeds isometrically in a space with all three splittings, and every Banach space is isomorphic to a quotient of such a space.

Theorem 6.29 (a space with all splittings)**.** *Let G be a normed space and $\Omega = \{0,1\}^{\mathbb{N}}$. For $m \ge 0$ and $\xi \in \{0,1\}^m$ let $C_\xi \subseteq \Omega$ be the set of sequences that begin with ξ, a dyadic cell of level m, of Haar measure 2^{-m}. A step function of level m is a function $f = \sum_{\xi\in\{0,1\}^m} 1_{C_\xi} g_\xi$ on Ω with $g_\xi \in G$; its norm $\|f\|_1 = \sum_\xi 2^{-m}\|g_\xi\|$ does not depend on the level used to write it. Let $L^1(\Omega; G)$ be the completion of the normed space of step functions. Then for every $k \ge 1$ the space $Y = L^1(\Omega; G)$ satisfies (S^k_Y) with a section of norm at most 1 and (R^k_Y) with a retraction of norm at most 1, and it is polarizable in degree k with a projection of norm at most $k!$.*

Corollary 6.30 (no hereditary permanence)**.** *Let E and D be Banach spaces, let $\mathrm{Ball}(E)$ be the closed unit ball of E, and let $\ell^1(\mathrm{Ball}(E), \widehat{K}) \to E$ be the bounded linear map that sends each unit vector e_x to x.*

(1) *The map $g \mapsto 1_\Omega g$ is an isometry of E onto a closed subspace of $L^1(\Omega; E)$, and $(L^1(\Omega; E), D)$ is good for Banach values in every degree. Likewise D is isometric to a closed subspace of $L^1(\Omega; D)$, and $(E, L^1(\Omega; D))$ is good for Banach values in every degree.*

(2) *The map $\ell^1(\mathrm{Ball}(E), \widehat{K}) \to E$ is surjective and open, so E is isomorphic to a quotient of $\ell^1(\mathrm{Ball}(E), \widehat{K})$, and $(\ell^1(\mathrm{Ball}(E), \widehat{K}), D)$ is good for Banach values in every degree. Likewise D is isomorphic to a quotient of $\ell^1(\mathrm{Ball}(D), \widehat{K})$, and $(E, \ell^1(\mathrm{Ball}(D), \widehat{K}))$ is good for Banach values in every degree.*

(3) *Assume condition* (L)*. Then $(c_0(\mathbb{N}, \widehat{K}), c_0(\mathbb{N}, \widehat{K}))$ is not good for Banach values in degree k, and $c_0(\mathbb{N}, \widehat{K})$ satisfies neither (S^k) nor (R^k) and is not polarizable in degree k (Theorem 7.7, Corollary 8.3 and Proposition 6.28). Hence goodness for Banach values in degree k passes neither to closed subspaces nor to quotients, in either slot of a pair, and none of (S^k), (R^k) and polarizability in degree k passes to closed subspaces or to quotients.*

The proofs of Theorem 6.29 and Corollary 6.30 are in Appendix L.

Theorem 6.31 (no three-space property in the source slot)**.** *Assume condition* (L)*, and let D be a Banach space with an equivalent nonarchimedean norm and $\dim_{\widehat{K}} D = \infty$, for example $D = c_0(\mathbb{N}, \widehat{K})$. Let $Y_1 = \ell^1(\mathrm{Ball}(D), \widehat{K})$, let $\varpi_D \colon Y_1 \to D$ be the map $e_x \mapsto x$, surjective and open*

by Corollary 6.30(2), and let $\eta\colon Y_1 \to L^1(\Omega;Y_1)$ *be the isometry* $g \mapsto 1_\Omega g$ *of Corollary 6.30(1). Put* $Y_2 = D \oplus L^1(\Omega;Y_1)$ *with the maximum norm and* $Y_3 = L^1(\Omega;Y_1)/\eta(\ker \varpi_D)$*. The maps* $y \mapsto (-\varpi_D y, \eta y)$ *and* $(d,v) \mapsto [v + \eta(y_d)]$*, where* $y_d \in Y_1$ *is any element with* $\varpi_D y_d = d$*, form an exact sequence* $0 \to Y_1 \to Y_2 \to Y_3 \to 0$ *such that:*

(1) (Y_1, D') *is good for Banach values in every degree for every normed space* D'*;*
(2) $\mathcal{L}(Y_3, D) = 0$*, so* (Y_3, D) *is good in every degree;*
(3) (Y_2, D) *is not good for Banach values in degree* k*.*

The proof is in Appendix N.

7. Obstructions

7.1. **The field.** We say that *condition* (L) holds if κ is a finite field of characteristic p, $0 < \varrho < 1$, the Laurent series field $\kappa((X))$ carries the absolute value $|a| = \varrho^{\operatorname{ord} a}$, K is a subfield of $\kappa((X))$ on which this absolute value is nontrivial, and $k \geq p$; for example $K = \kappa((X))$ or $K = \kappa(X)$. Then K is nonarchimedean, and $\widehat{K}$ is the closure of K in the complete field $\kappa((X))$. For $a \in \kappa((X))$ and $n \in \mathbb{Z}$, $\operatorname{coeff}_n(a) \in \kappa$ is the X^n-coefficient of a, applied entrywise to vectors and matrices, and $M_\varrho = \max_{l \geq 0}(l+1)\varrho^l$, which exists since $(l+1)\varrho^l \to 0$. The group $|K^\times|$ is a nontrivial subgroup of $\varrho^{\mathbb{Z}}$, hence $|K^\times| = \varrho^{n_0\mathbb{Z}}$ for an integer $n_0 \geq 1$; under (L) we fix $\vartheta \in K$ with $|\vartheta| = \varrho^{n_0}$, so $|K^\times| = |\vartheta|^{\mathbb{Z}}$.

Lemma 7.1 (normalization of the field)**.** *Assume* (L)*. There are a finite field* κ' *of characteristic* p *and an isometric field isomorphism* σ *from* $\kappa'((X))$*, with the absolute value* $|g| = |\vartheta|^{\operatorname{ord} g}$*, onto* $\widehat{K}$ *with* $\sigma(X) = \vartheta$*. Hence* K *is isometrically isomorphic to a dense subfield of* $\kappa'((X))$*. Consequently, a statement whose hypotheses and conclusion depend only on* K *as a normed field holds for every* K *satisfying* (L) *once it holds for every subfield* K *dense in* $\kappa((X))$*, for every finite field* κ *of characteristic* p *and every* $\varrho \in (0,1)$*.*

The proof is in Appendix P. When K is dense in $\kappa((X))$, we have $\widehat{K} = \kappa((X))$ and $|K^\times| = \varrho^{\mathbb{Z}}$, so $|\vartheta| = \varrho$.

7.2. **Forcing and the two counterexamples.** The factorial hypothesis of Proposition 5.1(1) cannot be removed.

Theorem 7.2 (operator obstructions)**.** *Let* K *be a nontrivially normed field of characteristic* $p > 0$*, and let* $k \geq p$*.*

(1) *There are Banach* K*-spaces* E, F *for which*

$$A^k_{E,E;F}\colon \mathcal{L}(E,E) \longrightarrow \mathcal{L}\big(\operatorname{Alt}^k(E;F), \operatorname{Alt}^k(E;F)\big)$$

is analytic at no point. The target F *admits no equivalent nonarchimedean norm.*

(2) *If* K *is complete and not spherically complete, there are nonarchimedean Banach spaces* E, D *for which* $A^k_{E,D;K}$ *is analytic at no point. They can be chosen as closed subspaces of*

$$\ell^\infty(\Lambda, K^{k+1}) \quad \text{and} \quad \ell^\infty(\Lambda, K^k),$$

respectively, containing the corresponding c_0 *spaces, with* Λ *countable.*

Part (1) is assembled in Appendix E. For κ finite of characteristic p, Theorem C.1 excludes a bounded lift over $\kappa((X))$ for $E_1 = \ell^\infty(\mathbb{N}, \kappa((X)))$ and $F_1 = \widehat{\Lambda}^k_\pi E_1$ (Lemma C.2): through Proposition C.3 a lift gives the map Ψ of Lemma C.4, which Theorem B.9 excludes by Ramsey's theorem, clusters and a staircase matrix. Proposition D.9, which rests on Ingleton's Theorem D.3, and Lemma D.11 pass to every K; by Proposition E.1 the target F has no

equivalent nonarchimedean norm. Part (2) is proved in Appendix F from Lemmas F.4 and F.2 and van der Put's description of the continuous dual of a bounded sequence space [19]. In both parts, Proposition 2.5 turns the absence of a lift into nonanalyticity at every point.

Corollary 7.3 (regularity on whole hom spaces)**.** *For every nontrivially normed field K, the bifunctor $\mathrm{Alt}^k\colon \mathsf{Vec}_K^{\mathrm{op}} \times \mathsf{Vec}_K \to \mathsf{Vec}_K$ is C^∞, and is analytic if and only if $k! \neq 0$ in K. Both assertions remain true on the full subcategory of Banach spaces.*

Proof. Smooth descent applies to the ambient polynomial action. An invertible factorial gives a finite multilinear lift, and bounded bilinear postcomposition supplies the joint bifunctor action. If the factorial vanishes, Theorem 7.2(1) excludes analyticity even on Banach spaces, after fixing the output morphism to be the identity. □

Theorem 7.4 (universal scalar classification)**.** *Let K be a complete nonarchimedean, nontrivially normed field. For any $k \geq 0$, the contravariant functor*

$$\mathrm{Alt}^k(-;K)\colon \mathsf{Vec}_K^{\mathrm{op}} \longrightarrow \mathsf{Vec}_K$$

is analytic on every hom space if and only if $k! \neq 0$ in K or K is spherically complete. The same equivalence holds when the source category is restricted to Banach spaces, or to nonarchimedean Banach spaces.

Proof. An invertible factorial gives a multilinear lift by normalized antisymmetrization. If K is spherically complete, Theorem 5.2 applies with target K. Both arguments permit arbitrary normed sources. If neither condition holds, then K has characteristic $p > 0$ with $k \geq p$, and Theorem 7.2(2) supplies nonarchimedean Banach witnesses against all three assertions. □

By Theorem 3.2(2), the same criterion holds for every nonzero continuous-dual target, and hence for every nonzero finite-dimensional target, which over complete K is isomorphic to a continuous dual (Lemma A.3).

Call a normed space F a *universal target* if $\mathrm{Alt}^k(-;F)$ is analytic on every hom space, that is, if $A^k_{E,D;F}$ is analytic on $\mathcal{L}(E,D)$ for all normed spaces E and D. Over a nonarchimedean field, every nonarchimedean spherically complete F is a universal target, by Theorem 5.2.

Proposition 7.5 (universal targets)**.** *If $\operatorname{char} K = p > 0$, $k \geq p$, and K is complete and not spherically complete, then every universal target has zero continuous dual.*

Proof. Let F be a normed space with a nonzero $\lambda \in F^*$, and choose $v \in F$ with $\lambda(v) = 1$. Let E and D be the spaces of Theorem 7.2(2), and for $a \in \mathrm{Alt}^k(D;K)$ let $va \in \mathrm{Alt}^k(D;F)$ be $y \mapsto a(y)v$. Then $A^k_{E,D;K}(u)(a) = \lambda \circ A^k_{E,D;F}(u)(va)$ for $u \in \mathcal{L}(E,D)$. Composing with the bounded linear maps $a \mapsto va$ and $m \mapsto \lambda \circ m$ is a bounded linear map $\mathcal{L}(\mathrm{Alt}^k(D;F), \mathrm{Alt}^k(E;F)) \to \mathcal{L}(\mathrm{Alt}^k(D;K), \mathrm{Alt}^k(E;K))$. So if $A^k_{E,D;F}$ were analytic at a point, so would be $A^k_{E,D;K}$, which Theorem 7.2(2) excludes. Hence F is not a universal target. □

Corollary 7.6 (bounded sequences)**.** *Let κ be a finite field of characteristic p, $K_1 = \kappa((X))$ with $|X| = \varrho$, $k \geq p$, and $E_1 = \ell^\infty(\mathbb{N}, K_1)$. Then (E_1, E_1) is not good for Banach values in degree k, and neither $(\mathrm{S}^k_{E_1})$ nor $(\mathrm{R}^k_{E_1})$ holds, although E_1 is isometrically isomorphic to $c_0(I, K_1)$ for a set I, and $A^k_{E_1,E_1;F}$ is a continuous polynomial for every Banach space F with an equivalent nonarchimedean norm.*

Proof. The space $F_1 = \widehat{\Lambda}^k_\pi E_1$ is a Banach space (Lemma C.2), and $A^k_{E_1,E_1;F_1}$ is analytic at no point by Theorem C.1; so (E_1, E_1) is not good for Banach values, and by Proposition 3.5(1),(2) neither $(\mathrm{S}^k_{E_1})$ nor $(\mathrm{R}^k_{E_1})$ holds. Since $|K_1^\times| = \varrho^{\mathbb{Z}}$, Corollary 5.3 and Proposition 2.5 give the assertion about nonarchimedean F. The norm of E_1 takes values in $|K_1|$, so E_1 has an orthonormal basis and is isometrically isomorphic to $c_0(I, K_1)$ [25, §1, Prop. 1]. □

7.3. **The ℓ^r threshold.** Throughout this subsection K satisfies (L). For a sequence a, $\mu_a\colon x \mapsto ax$ denotes multiplication by a. The proof of the next theorem follows the proof of Theorem B.9, with norm bounds on the staircase matrix in place of the support bound.

Theorem 7.7 (ℓ^r with $r \geq 2$)**.** *Let $2 \leq r \leq \infty$, and let E be a normed space with $c_{00}(\mathbb{N}, K) \subseteq E \subseteq \ell^r(\mathbb{N}, K)$ and the ℓ^r norm, the supremum norm if $r = \infty$. Suppose $Q\colon c_{00}(\mathbb{N}, K)^k \to \mathcal{L}(\Lambda_\pi^k E, \widehat{\Lambda}_\pi^k E)$ is k-linear, bounded for the supremum norm on $c_{00}(\mathbb{N}, K)$, and satisfies $Q(a, \dots, a) = \Lambda_\pi^k \mu_a$. Then for every $m \geq 1$, with $1/r = 0$ when $r = \infty$,*

$$\begin{aligned} \|Q\| &\geq \frac{(1-\varrho^2)\ln m}{k(k-1)2^k(1+\ln(2|\kappa|))} \quad \text{if } r = 2, \\ m - 1 &\leq M_\varrho k(k-1)\|Q\|(4m)^{2/r}4^{(k-2)/r} \quad \text{if } r > 2. \end{aligned} \tag{7.1}$$

Hence no such Q exists. Consequently (E, E) is not good for Banach values in degree k; $A^k_{E,E;F}$ is analytic at no point for $F = \Lambda_\pi^k E$ and for $F = \widehat{\Lambda}_\pi^k E$; (S^k_E) and (R^k_E) fail; and $\widehat{\Lambda}_\pi^k E$ has no equivalent nonarchimedean norm.

Corollary 7.8 (the ℓ^r threshold)**.** *Let $1 \leq r \leq \infty$, and let $E = \ell^r(\mathbb{N}, K)$, or $E = c_0(\mathbb{N}, K)$ if $r = \infty$. Then (E, E) is good for Banach values in degree k if and only if $r < 2$; in that case (E, D) and (D, E) are good for Banach values for every normed D. The same holds, with good in place of good for Banach values, for the finitely supported sequences with the ℓ^r norm.*

Corollary 7.9 (pairs of ℓ^r spaces)**.** *Let $1 \leq r_1, r_2 \leq \infty$, and let $E = \ell^{r_1}(\mathbb{N}, K)$ and $D = \ell^{r_2}(\mathbb{N}, K)$, with $c_0(\mathbb{N}, K)$ in place of ℓ^∞. The pair (E, D) is good for Banach values in degree k if $r_1 < 2$ or $r_2 < 2$, and it is not if $r_2 = 2 \leq r_1$ or if $r_1, r_2 > 2$.*

The proofs are in Appendix K. The pairs with $r_1 = 2 < r_2$ are good for Banach values in every degree (Corollary 8.5).

7.4. **The product triangle bound.** Throughout this subsection K satisfies (L). For $A \subseteq \mathbb{N}$ let 1_A be its indicator function, for $n \geq 1$ let $[n]$ be the set of the first n elements of $\mathbb{N}$, let e_t ($t \in \mathbb{N}$) be the unit vectors, and for a sequence x let $|x|$ be the sequence of absolute values $|x(t)|$.

Definition 7.10. A *sequence space* is a normed space D with $c_{00}(\mathbb{N}, K) \subseteq D \subseteq \ell^\infty(\mathbb{N}, K)$. It is *symmetric* if there is a norm on $c_{00}(\mathbb{N}, \mathbb{R})$, again written $\|\cdot\|_D$, with $\|f\|_D = \||f|\|_D$, with $\|f\|_D \leq \|g\|_D$ whenever $|f| \leq |g|$ pointwise, invariant under every permutation of $\mathbb{N}$ and equal to 1 at the unit vectors, such that $\|x\|_D = \||x|\|_D$ for $x \in c_{00}(\mathbb{N}, K)$ and $\|x\|_D = \sup_n \|x1_{[n]}\|_D$ for $x \in D$. A sequence space E is *ℓ^2-dominated* if there is C_2 with $\|x\|_2 \leq C_2\|x\|_E$ for all $x \in E$; for symmetric E in which $c_{00}(\mathbb{N}, K)$ is dense, this is the ℓ^2-domination of the unit vectors in the sense of Section 5, since $\|x\|_E = \sup_n \|x1_{[n]}\|_E$. The *fundamental function* of E is $\varphi_E(m) = \|1_{[m]}\|_E$. For a symmetric sequence space E, a sequence space D and $a \in c_{00}(\mathbb{N}, K)$, μ_a is bounded from E to D (Lemma 7.11(2)), and $\|\mu_a\|_{E\to D}$ denotes its norm.

For example, $\ell^r(\mathbb{N}, K)$ with $1 \leq r < \infty$ and $c_0(\mathbb{N}, K)$ are symmetric sequence spaces.

Lemma 7.11 (multiplier norms)**.** *Let E be a symmetric sequence space.*

1. *$\|x\|_\infty \leq \|x\|_E$ for $x \in E$, and $\|bx\|_E \leq \|x\|_E$ for $x \in E$ and every finitely supported b with $|b| \leq 1$.*
2. *For every sequence space D and $a \in c_{00}(\mathbb{N}, K)$, $\mu_a \in \mathcal{L}(E, D)$ and $\|\mu_a\|_{E\to D} = \sup\{\|ax\|_D : x \in c_{00}(\mathbb{N}, K),\ \operatorname{supp} x \subseteq \operatorname{supp} a,\ \|x\|_E \leq 1\}$.*

(3) *If D is symmetric, then $a \mapsto \|\mu_a\|_{E\to D}$ is a norm on $c_{00}(\mathbb{N},K)$ with $\|\mu_a\|_{E\to D} \geq \|a\|_\infty$ and $\|\mu_{e_t}\|_{E\to D} = 1$. It does not change when the coordinates of a are relabeled by an injective map $\mathbb{N}\to\mathbb{N}$ or when a is multiplied by a sequence of absolute value 1, and $\|\mu_a\|_{E\to D} \leq \|\mu_b\|_{E\to D}$ when $|a| \leq |b|$.*

(4) *If $\|y\|_D \leq C\|y\|_E$ for $y \in c_{00}(\mathbb{N},K)$, then $\|\mu_a\|_{E\to D} \leq C\|a\|_\infty$. If D is symmetric, then $\|\mu_a\|_{c_0\to D} = \|a\|_D$ and $\|\mu_a\|_{E\to c_0} = \|a\|_\infty$, where $c_0 = c_0(\mathbb{N},K)$.*

The proof is in Appendix P.

Definition 7.12. Let E be a symmetric sequence space, D a sequence space and $C \geq 0$, and let $\mathcal{P}_<$ be the strict triangle projection of the unit vectors: $\mathcal{P}_<(e_t \otimes e_{t'}) = e_t \otimes e_{t'}$ if $t < t'$ and 0 otherwise. The pair (E,D) satisfies the *product triangle bound* (PTB) with constant C if

$$\|\mathcal{P}_<((ax)\otimes(by))\|_{D\otimes_\pi D} \leq C\|\mu_a\|_{E\to D}\|x\|_E\|\mu_b\|_{E\to D}\|y\|_E \qquad (a,x,b,y \in c_{00}(\mathbb{N},K)). \tag{7.2}$$

We say that $\mathcal{P}_<$ is bounded on $E\otimes_\pi E$ if $\|\mathcal{P}_<(x\otimes y)\|_{E\otimes_\pi E} \leq C\|x\|_E\|y\|_E$ for some C and all $x,y \in c_{00}(\mathbb{N},K)$.

Theorem 7.13 (product triangle bound)**.** *Let E and D be symmetric sequence spaces, and suppose that (E,D) is good for Banach values in degree k, with a lift $Q\colon \mathcal{L}(E,D)^k \to \mathcal{L}(\Lambda^k_\pi E, \widehat{\Lambda}^k_\pi D)$ as in Lemma 3.4. Then (E,D) satisfies* (PTB) *with a constant that depends only on k, K and $\|Q\|$. No inclusion between E and D is assumed, and $c_{00}(\mathbb{N},K)$ need not be dense in either.*

Lemma 7.14 (the triangle of one vector)**.** *Suppose that K is dense in $\kappa((X))$, and let D be a sequence space with $\|y\|_\infty \leq \|y\|_D$ for $y \in D$. Then for every $x \in c_{00}(\mathbb{N},K)$,*

$$\|x\|_2^2 \leq \frac{\|\mathcal{P}_<(x\otimes x)\|_{D\otimes_\pi D}}{1-\varrho} + \frac{\|x\|_\infty^2}{1-\varrho^2}.$$

Proposition 7.15 (ℓ^2-domination)**.** *Let E be a symmetric sequence space and D a sequence space with $\|y\|_\infty \leq \|y\|_D$ for $y \in D$. Suppose that $\|\xi\|_D \leq C'\|\xi\|_r$ for all $\xi \in c_{00}(\mathbb{N},K)$, for some $r \in (2,\infty]$ and $C' \geq 0$, and that (E,D) satisfies* (PTB) *with constant C. Put $\mathfrak{a} = CC'^2/(1-|\vartheta|)$ and $\mathfrak{b} = 1/(1-|\vartheta|^2)$. Then $\|x\|_2 \leq C_2\|x\|_E$ for all $x \in E$, where $C_2^2 = \mathfrak{a}+\mathfrak{b}$ if $r=\infty$ and $C_2^2 = \max\big((2\mathfrak{a})^{r/(r-2)}, 2\mathfrak{b}\big)$ if $r<\infty$. In particular E is ℓ^2-dominated.*

Proposition 7.16 (polarizable symmetric spaces)**.** *Let E be a symmetric sequence space in which $c_{00}(\mathbb{N},K)$ is dense. If (Γ^k_E) holds, then $\mathcal{P}_<$ is bounded on $E\otimes_\pi E$.*

The proofs of Theorem 7.13, Lemma 7.14 and Propositions 7.15 and 7.16 are in Appendix K.

7.5. **Nonarchimedean fibers.** Throughout this subsection K satisfies (L). The proofs, in Appendix K, take K dense in $\kappa((X))$; Lemma 7.1 extends them to every K satisfying (L).

Lemma 7.17 (marking)**.** *Let E be a normed space with an equivalent nonarchimedean norm and $\dim_{\widehat{K}} \widehat{E} = \infty$. There are $e_n \in E$, $n \in \mathbb{N}$, an equivalent nonarchimedean norm $\|\cdot\|'$ on $\widehat{E}$ with $\|\sum_n \xi_n e_n\|' = \max_n|\xi_n|$ for finitely supported (ξ_n) in $\widehat{K}$, and a $\widehat{K}$-linear map $\theta_E\colon \widehat{E} \to c_0(\mathbb{N},\widehat{K})$, contractive for $\|\cdot\|'$, that sends e_n to the nth unit vector. If E is complete, $\widehat{\psi}_E(\xi) = \sum_n \xi_n e_n$ defines an isometry of $c_0(\mathbb{N},\widehat{K})$ into $(E,\|\cdot\|')$ with $\theta_E\widehat{\psi}_E = \mathrm{id}$. If K is complete, θ_E takes values in $c_0(\mathbb{N},K)$.*

Theorem 7.18 (marking obstruction)**.** *Let E, E' be normed spaces, let $\psi\colon c_{00}(\mathbb{N},K) \to E$ and $\theta\colon E' \to \ell^\infty(\mathbb{N},\widehat{K})$ be bounded linear maps, with the supremum norm on $c_{00}(\mathbb{N},K)$, and let $a \mapsto u_a$ be a bounded linear map $c_{00}(\mathbb{N},K) \to \mathcal{L}(E,E')$ with*

$$\theta\big(u_a(\psi x)\big) = ax \qquad (a,x \in c_{00}(\mathbb{N},K)). \tag{7.3}$$

Let $\ell^\infty_{\widehat{K}} = \ell^\infty(\mathbb{N}, \widehat{K})$, regarded over K. Let F be normed, $\nu \in \mathrm{Alt}^k(E'; F)$, and $\theta_F \colon F \to \widehat{\Lambda}^k_\pi \ell^\infty_{\widehat{K}}$ bounded linear with $\theta_F \circ \nu = \omega_{\ell^\infty_{\widehat{K}}} \circ (\theta, \dots, \theta)$. Then no bounded k-linear $Q \colon c_{00}(\mathbb{N}, K)^k \to \mathrm{Alt}^k(E; F)$ satisfies $Q(a, \dots, a) = \nu \circ (u_a, \dots, u_a)$, and $A^k_{E,E';F}$ is analytic at no point. For $F = \Lambda^k_\pi E'$ and $F = \widehat{\Lambda}^k_\pi E'$, with $\nu = \omega_{E'}$ and θ_F induced by θ, the pair (E, E') is therefore not good, respectively not good for Banach values, in degree k.

Theorem 7.19 (nonarchimedean fibers over a local field)**.** *Let E be a normed space with an equivalent nonarchimedean norm.*

1. *The following are equivalent: $\dim_{\widehat{K}} \widehat{E} < \infty$; (E, E') is good for Banach values in degree k for every normed E'; (E', E) is good for Banach values in degree k for every normed E'.*
2. *If E is complete, these conditions are also equivalent to each of: (E, E) is good in degree k; (E, E) is good for Banach values in degree k.*
3. *If K is complete and E' also has an equivalent nonarchimedean norm, then (E, E') is good in degree k if and only if it is good for Banach values, if and only if $\dim E < \infty$ or $\dim E' < \infty$.*

Each failure is shown by an explicit pair whose action with values in the completed exterior power of its second space is analytic at no point; these values admit no equivalent nonarchimedean norm, while complete nonarchimedean values never fail, by Corollary 5.3.

By Theorem 5.6, $(c_0(\mathbb{N}, K), \ell^1(\mathbb{N}, K))$ is good for Banach values; so the hypothesis on E' in (3) cannot be dropped.

7.6. **Characteristic two.** Throughout this subsection condition (L) is not assumed. For a field $\Bbbk$, e_t are the unit vectors of $\Bbbk^n$, $[n] = \{1, \dots, n\}$, $1_S = \sum_{t \in S} e_t$ for $S \subseteq [n]$, also read in $c_{00}(\mathbb{N}, K)$, matrices are written in sans-serif, $\mathcal{P}_< \mathsf{Z}$ keeps the entries of Z above the diagonal and replaces the others by 0, and Ω_w is the alternating matrix of $w \in \Lambda^2 \Bbbk^n$. In characteristic 2, Ω_w is symmetric with zero diagonal and $\Omega_{x \wedge y} = xy^{\mathsf{T}} + yx^{\mathsf{T}}$. For $t \neq t'$, $\pi_{tt'}$ is the coordinate projection of $\Lambda^2 \Bbbk^n$ onto $\Bbbk\, e_t \wedge e_{t'}$, and for $a \in \Bbbk^n$, μ_a is multiplication by a, so that

$$\Lambda^2 \mu_a = \sum_{t<t'} a_t a_{t'}\, \pi_{tt'}. \tag{7.4}$$

For $L \subseteq [n]$ put $L^c = [n] \setminus L$, and for a vector a of indeterminates $a_L = \prod_{t \in L} a_t$; for a polynomial g, $[a_L b_{L^c} x_L y_{L^c}]g$ is the coefficient of the monomial $a_L b_{L^c} x_L y_{L^c}$ in g. The Pfaffian of an alternating matrix of even size over a commutative ring of characteristic 2 is the sum, over the partitions of the index set into pairs, of the products of the entries at the pairs; its square is the determinant, as for the generic alternating matrix over $\mathbb{Z}$. Sequence spaces are as in Definition 7.10, which does not use condition (L), and in this subsection a sequence space E is also assumed to satisfy $\|x\|_\infty \le \|x\|_E$ for $x \in E$; for finitely supported a, multiplication $\mu_a \colon x \mapsto ax$ then lies in $\mathcal{L}(E)$, as $\|ax\|_E \le \sum_t |a_t||x_t| \|e_t\|_E \le \big(\sum_t |a_t| \|e_t\|_E\big) \|x\|_E$. Put $\bar{\varphi}_E(n) = \max_{S \subseteq [n]} \|1_S\|_E$, which is nondecreasing in n. For $n \ge 1$, $c_0(n) = c_0([n], K)$.

Let $\Bbbk$ be a field of characteristic 2 and $n \ge 2$. A *multiplier lift* on $\Bbbk^n$ is a $\Bbbk$-bilinear map $\Psi \colon \Bbbk^n \times \Bbbk^n \to \mathrm{End}_\Bbbk(\Lambda^2 \Bbbk^n)$, $\Psi(a, b) = \sum_{t,t'} a_t b_{t'} \Psi_{tt'}$, with

$$\Psi_{tt'} + \Psi_{t't} = \pi_{tt'} \qquad (t \neq t'). \tag{7.5}$$

Its *output matrix* is $\mathsf{M}_\Psi(a, b, x, y) = \Omega_{\Psi(a,b)(x \wedge y)}$; regarding the entries of a, b, x, y as indeterminates, it is a matrix over $\Bbbk[a, b, x, y]$. The *ordering lift* $\Psi^<$ has $\Psi^<_{tt'} = \pi_{tt'}$ for $t < t'$ and $\Psi^<_{tt'} = 0$ for $t \ge t'$; thus $\Psi^<(a, b)$ is given by formula (3.4) with $k = 2$, $u_1 = \mu_a$, $u_2 = \mu_b$ and the natural order of $[n]$.

If $\Psi(a,a) = \Lambda^2\mu_a$ for all $a \in \Bbbk^n$, then Ψ is a multiplier lift: by (7.4), $\Psi_{tt} = \Lambda^2\mu_{e_t} = 0$ and $\Psi_{tt'} + \Psi_{t't} = \Lambda^2\mu_{e_t+e_{t'}} - \Lambda^2\mu_{e_t} - \Lambda^2\mu_{e_{t'}} = \pi_{tt'}$.

Theorem 7.20 (characteristic two)**.**

(1) (Identities.) *Let $\Bbbk$ be a field of characteristic 2 and $n \geq 2$. If n is even, every multiplier lift Ψ on $\Bbbk^n$ satisfies*

$$\sum_{L\subseteq[n],\ |L|=n/2} \left[a_L\, b_{L^c}\, x_L\, y_{L^c}\right] \operatorname{Pf} \mathsf{M}_\Psi(a,b,x,y) = 1, \tag{7.6}$$

and some $\mathsf{M}_\Psi(a,b,x,y)$ with $a,b,x,y \in \{0,1\}^n$ is invertible. For every $\Bbbk$-linear $\sigma\colon \Lambda^2\Bbbk^n \to \Bbbk^{n\times n}$ with $\wedge \circ \sigma = \mathrm{id}$, where $\wedge\mathsf{Z} = \sum_{t<t'}(\mathsf{Z}_{tt'} + \mathsf{Z}_{t't})\, e_t \wedge e_{t'}$, there are $S,T \subseteq [n]$ with $\operatorname{rank}\sigma(1_S \wedge 1_T) \geq n-1$, while $\sigma^<(x\wedge y) = \mathcal{P}_<(xy^{\mathsf{T}} + yx^{\mathsf{T}})$, $x,y \in \Bbbk^n$, defines such a map with strictly upper triangular values, of rank at most $n-1$. For every $\Bbbk$-linear $\tau\colon \Bbbk^{n\times n} \to \Lambda^2\Bbbk^n$ with $\tau(\Omega_w) = w$ for every w, and even n, there is $S \subseteq [n]$ with $\Omega_{\tau(1_S 1_S^{\mathsf{T}})}$ invertible.

(2) (Multiplier maps.) *Let K be a nontrivially normed field of characteristic 2, $k \geq 2$, and E a sequence space, with $\|e_t\|_E = 1$ for all t if $k \geq 3$. Every k-linear $Q\colon c_{00}(\mathbb{N},K)^k \to \mathcal{L}(\Lambda^k_\pi E, \widehat{\Lambda}^k_\pi E)$ with $\|Q(a_1,\dots,a_k)\| \leq \|Q\| \prod_i \|a_i\|_\infty$ and $Q(a,\dots,a) = \Lambda^k_\pi \mu_a$ satisfies, for every even $n \geq 2$,*

$$n \leq k(k-1)\,\|Q\|\,\bar{\varphi}_E(n)^2. \tag{7.7}$$

(3) (Splittings.) *Let K be as in (2), E a sequence space and $n \geq 2$. Every section s of $\widehat{\wedge}^2_E$ has $\|s\| \geq (n-1)/\bar{\varphi}_E(n)^2$, every retraction ρ of alt^2_E has $\|\rho\| \geq \lfloor n/2\rfloor/\bar{\varphi}_E(n)^2$, and every bounded linear projection R_2 of $\widehat{\otimes}^2_\pi E$ onto $\widehat{\Gamma}^2_\pi E$ has $\|R_2\| \geq n/\bar{\varphi}_E(n)^2$. For the corresponding maps of $c_0(n)$ the bounds are $n-1$, $\lfloor n/2\rfloor$ and n.*

(4) (Consequences.) *Let K be as in (2), and let E be $c_0(\mathbb{N},K)$, $\ell^\infty(\mathbb{N},K)$, $\ell^r(\mathbb{N},K)$ with $2 < r < \infty$, or a symmetric sequence space with $\sup_n n/\varphi_E(n)^2 = \infty$. For every $k \geq 2$, (E,E) is not good for Banach values in degree k, $A^k_{E,E;F}$ is analytic at no point for $F = \Lambda^k_\pi E$ and for $F = \widehat{\Lambda}^k_\pi E$, neither (S^k_E) nor (R^k_E) holds, and $\widehat{\Lambda}^k_\pi E$ has no equivalent nonarchimedean norm; moreover (Γ^2_E) fails. For $2 < r_1, r_2 \leq \infty$, with $c_0(\mathbb{N},K)$ in place of $\ell^\infty(\mathbb{N},K)$, the pair $(\ell^{r_1}(\mathbb{N},K), \ell^{r_2}(\mathbb{N},K))$ is not good for Banach values in any degree $k \geq 2$.*

The proof is in Appendix N. In (7.7), $\bar{\varphi}_E(n) = 1$ for $c_0(\mathbb{N},K)$ and $\ell^\infty(\mathbb{N},K)$, and $\bar{\varphi}_E(n) = n^{1/r}$ for $\ell^r(\mathbb{N},K)$; so $\|Q\| \geq n/(k(k-1))$, respectively $\|Q\| \geq n^{1-2/r}/(k(k-1))$, and for $E = \ell^2(\mathbb{N},K)$ only $\|Q\| \geq 1/(k(k-1))$. The proof uses Q only on vectors supported in $[n+k-2]$, and for $k=2$ the same argument applies to $E = c_0(n)$: for even n, every lift for $(c_0(n), c_0(n))$ in degree 2, for Banach values, has norm at least $n/2$. Over Laurent series fields over infinite fields and over densely valued fields, the case $E = c_0(\mathbb{N},K)$ for odd p is Question 12.9(1), and the case $E = \ell^2(\mathbb{N},K)$ for $p=2$ is Question 12.9(2).

Remark 7.21 (exactness without splitting). Let K be nonarchimedean, of any characteristic, and let $D = c_0(\mathbb{N},K)$. Then

$$\|\operatorname{alt}^2_D w\|_\pi = 2\|w\|_\pi \qquad (w \in \widehat{\Lambda}^2_\pi D), \tag{7.8}$$

so the target sequence $\mathcal{R}^2_D$ of Definition 3.8 is exact. If K has characteristic 2, then $\mathcal{R}^2_D$ does not split: by Lemma 3.10 a splitting would give (R^2_D), which Theorem 7.20(4) excludes. So in characteristic 2 the failure of (R^2_D) is a failure of $\mathcal{R}^2_D$ to split, not of its exactness.

Proof. Let $\mathcal{O}$ be the closed unit ball of K, and let $\|\cdot\|_\infty$ be the supremum norm.

Step 1 (reduction). Since $\operatorname{alt}^2_D(x\wedge y) = x\otimes y - y\otimes x$, the left side of (7.8) is at most the right side. Both sides are continuous in w, and the union of the spaces $\Lambda^2_K K^m$, where $K^m \subseteq D$ is the span of $e_1,\dots,e_m$, is dense in $\widehat{\Lambda}^2_\pi D$. The coordinate projection of D onto K^m is contractive and fixes K^m, so for $w \in \Lambda^2_K K^m$ both norms may be computed in $c_0(m)$. Put $T = \operatorname{alt}^2_D w$, an alternating matrix in $K^{m\times m}$.

Step 2 (elimination). Let $a = T_{ij}$ be an entry of T of maximal absolute value; then $i \neq j$. Let B be the block of T on the rows and columns i,j, and C the part of the columns i,j outside these rows. As $|\det B| = |a|^2$, the matrix CB^{-1} has entries in $\mathcal{O}$. Let g and h be the unit vectors e_i and e_j plus the first and second columns of CB^{-1}, placed in the rows other than i,j; so $\|g\|_\infty = \|h\|_\infty = 1$. The Schur complement gives $T = a(gh^{\mathsf{T}} - hg^{\mathsf{T}}) + T'$, where $T' = \operatorname{alt}^2_D(w - a\, g\wedge h)$ vanishes on the rows and columns i,j. Repeating with T' gives $w = \sum_k a_k\, g_k \wedge h_k$ with $\|g_k\|_\infty = \|h_k\|_\infty = 1$, so $\|w\|_\pi \le \sum_k |a_k|$, and $T = U \operatorname{diag}(a_1 J, a_2 J, \dots) U^{\mathsf{T}}$ with $U \in GL_m(\mathcal{O})$, whose columns are the g_k, the h_k and unit vectors, and $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$.

Step 3 (minors). Let $\mu_1 \ge \mu_2 \ge \dots \ge \mu_{2r}$ be the numbers $|a_k|$, each taken twice, and $\mu_{2r+1} = 0$. By the Cauchy–Binet formula the minors of T and of $\operatorname{diag}(a_k J)$ are $\mathcal{O}$-linear combinations of each other, so the largest absolute value of an $l\times l$ minor of T is $\mu_1\cdots\mu_l$. Let $T = \sum_q x_q y_q^{\mathsf{T}}$ be a finite representation, with costs $c_q = \|x_q\|_\infty \|y_q\|_\infty$ listed in decreasing order. The Cauchy–Binet formula and the ultrametric inequality bound every $l\times l$ minor of T by $c_1\cdots c_l$. So $\mu_1\cdots\mu_l \le c_1\cdots c_l$ for $l \le 2r$, and these c_l are positive. Since $c - \mu \ge \mu\log(c/\mu)$ for $c,\mu > 0$, Abel summation gives

$$\sum_{l\le 2r}(c_l - \mu_l) \ge \sum_{l\le 2r} \mu_l \log\frac{c_l}{\mu_l} = \sum_{l\le 2r}(\mu_l - \mu_{l+1}) \log\frac{c_1\cdots c_l}{\mu_1\cdots\mu_l} \ge 0.$$

Hence $2\|w\|_\pi \le 2\sum_k |a_k| = \sum_l \mu_l \le \sum_q c_q$, and taking the infimum over representations gives $2\|w\|_\pi \le \|\operatorname{alt}^2_D w\|_\pi$. □

8. Which pairs are good

Throughout this section κ, ϱ and K are as in condition (L), and each result states its degrees. Symmetric sequence spaces, their fundamental functions φ_E, ℓ^2-domination and the product triangle bound (PTB) are those of Definitions 7.10 and 7.12. For a sequence space, $\mathcal{P}_<$ is the strict triangle projection of the unit vectors $(e_t)_{t\in\mathbb{N}}$, and c_{00}, c_0 and ℓ^r stand for $c_{00}(\mathbb{N},K)$, $c_0(\mathbb{N},K)$ and $\ell^r(\mathbb{N},K)$.

8.1. **Symmetric spaces.**

Theorem 8.1 (symmetric spaces)**.** *Let E be a symmetric sequence space in which c_{00} is dense. The following are equivalent:*

1. *(E,E) is good for Banach values in some degree $k \ge p$;*
2. *(c_0,E) is good for Banach values in some degree $k \ge p$;*
3. *$\mathcal{P}_<$ is bounded on $E\otimes_\pi E$: there is C with $\|\mathcal{P}_<(x\otimes y)\|_{E\otimes_\pi E} \le C\|x\|_E\|y\|_E$ for all $x,y \in c_{00}$;*
4. *(S^k_E) holds for some $k \ge p$;*
5. *(R^k_E) holds for some $k \ge p$;*
6. *E is polarizable in some degree $k \ge p$;*
7. *for every normed D, the pairs (E,D) and (D,E) are good for Banach values in every degree.*

If these conditions hold, then (S^n_E), (R^n_E) *and* (Γ^n_E) *hold for every* $n \geq 1$, E *is* ℓ^2*-dominated, and there is* $c > 0$ *with* $\varphi_E(m)^2 \geq c\, m \ln m$ *for every* $m \geq 1$.

The proof is in Appendix K.

Corollary 8.2 (small fundamental functions)**.** *Let* E *be* ℓ^r *with* $2 \leq r < \infty$*, or* c_0*, or a symmetric sequence space in which* c_{00} *is dense and* $\varphi_E(m)^2 \leq Cm$ *for some* C *and every* $m \geq 1$*. Then for every* $k \geq p$ *neither* (E, E) *nor* (c_0, E) *is good for Banach values in degree* k*, and* (S^k_E), (R^k_E) *and* (Γ^k_E) *fail.*

Proof. For $E = \ell^r$ we have $\varphi_E(m) = m^{1/r}$, so $\varphi_E(m)^2 \leq m$ when $r \geq 2$, and for $E = c_0$ we have $\varphi_E(m) = 1$; in both cases c_{00} is dense. If one of the pairs were good in a degree $k \geq p$, or one of the splittings held, Theorem 8.1 would give $c\, m \ln m \leq \varphi_E(m)^2 \leq Cm$ for every m, which fails for large m. □

Corollary 8.3. *The space* c_0 *is not polarizable in any degree* $k \geq p$.

Proof. This is Corollary 8.2 for $E = c_0$. □

8.2. **Symmetric sources against r-normed targets.** For the targets of the next theorem, the necessary condition (PTB) of Theorem 7.13 is also sufficient.

Theorem 8.4 (r-normed targets)**.** *Let* E *be a symmetric sequence space in which* c_{00} *is dense, and let* D *be a symmetric sequence space with an equivalent* r*-norm for some* $r \in (2, \infty)$ *or with an equivalent nonarchimedean norm. The following are equivalent:*

(1) (E, D) *is good for Banach values in some degree* $k \geq p$*;*
(2) (E, D) *satisfies* (PTB) *with some constant;*
(3) E *is* ℓ^2*-dominated;*
(4) (E, D) *is good for Banach values in every degree.*

The proof is in Appendix K. Under (3), the coordinate map of E into $\ell^r(\mathbb{N}, \widehat{K})$, with $r = 3$ when D has an equivalent nonarchimedean norm, is a relative ordering splitting: it is S^n-split for every n, and every $u \in \mathcal{L}(E, D)$ factors through it (Theorem 5.9(1) and Step 1 of its proof, Corollary 5.10).

8.3. **The ℓ^r scale.**

Corollary 8.5 (the ℓ^r scale)**.** *Let* $1 \leq r_1, r_2 \leq \infty$*. For the pair* (ℓ^{r_1}, ℓ^{r_2})*, with* c_0 *in place of* ℓ^∞*, the following are equivalent: it is good for Banach values in some degree* $k \geq p$*; it is good for Banach values in every degree;* $\min(r_1, r_2) < 2$*, or* $r_1 = 2 < r_2$*. Hence* $\mathfrak{D}_B(\ell^{r_1}, \ell^{r_2})$ *is given by the following table, in which* $r_i = \infty$ *stands for* c_0*:*

$r_1 \backslash r_2$	$[1, 2)$	2	$(2, \infty)$	∞
$[1, 2)$	$\{1, 2, \dots\}$	$\{1, 2, \dots\}$	$\{1, 2, \dots\}$	$\{1, 2, \dots\}$
2	$\{1, 2, \dots\}$	$\{1, \dots, p-1\}$	$\{1, 2, \dots\}$	$\{1, 2, \dots\}$
$(2, \infty)$	$\{1, 2, \dots\}$	$\{1, \dots, p-1\}$	$\{1, \dots, p-1\}$	$\{1, \dots, p-1\}$
∞	$\{1, 2, \dots\}$	$\{1, \dots, p-1\}$	$\{1, \dots, p-1\}$	$\{1, \dots, p-1\}$

The proof is in Appendix K.

8.4. **Degrees and mechanisms.**

Corollary 8.6 (one critical degree gives all)**.** *For each of the following pairs* (E, D)*, the set* $\mathfrak{D}_B(E, D)$ *is* $\{1, \dots, p-1\}$ *or* $\{1, 2, \dots\}$*:*

(1) (E, E) *and* (c_0, E)*, for* E *as in Theorem 8.1;*
(2) (E, D) *as in Theorem 8.4;*

(3) (ℓ^{r_1}, ℓ^{r_2}) *for* $1 \le r_1, r_2 \le \infty$, *with* c_0 *in place of* ℓ^∞;
(4) (E, D) *for complete* K *and spaces* E, D *with equivalent nonarchimedean norms.*

Proof. By Corollary 6.3(1), $\{1, \dots, p-1\} \subseteq \mathfrak{D}_B(E, D)$. It remains to show that each listed pair that is good for Banach values in one degree $k \ge p$ is good for Banach values in every degree. For (1)–(3) this is the equivalence of conditions (1), (2) and (7) of Theorem 8.1, of (1) and (4) of Theorem 8.4, and Corollary 8.5. For (4), Theorem 7.19(3) shows that goodness in a degree $k \ge p$ forces $\dim E < \infty$ or $\dim D < \infty$; that space has continuous coordinates by Lemma A.3, and Proposition 5.1(2) gives a lift in every degree. □

Remark 8.7 (mechanisms). In every degree, each pair good for Banach values in Theorems 8.1, 8.4 and 7.19 and Corollary 8.5 has a splitting decomposition (Definition 6.12) with one piece of one of three kinds: an ordering splitting of the source or the target (Theorem 5.6, Proposition 5.7); the relative ordering splitting of the source described after Theorem 8.4; or a splitting of the source or the target by finite coordinates on its completion (proof of Theorem 7.19). In each of these results, goodness implies the stated criterion by forcing (Theorems 7.7, 7.13 and 7.18).

8.5. **Spreading spaces.**

Definition 8.8. A sequence space E is *spreading* if it satisfies Definition 7.10 for symmetric sequence spaces with invariance under permutations of $\mathbb{N}$ replaced by the following condition: for $d \ge 1$ and $f \in \mathbb{R}^d$, the norm $\|\sum_{i=1}^d f_i e_{t_i}\|_E$ is the same for all $t_1 < \cdots < t_d$ in $\mathbb{N}$.

Symmetric sequence spaces are spreading. In Banach space terms, the unit vectors of a spreading sequence space are 1-unconditional and 1-spreading, that is, 1-subsymmetric. A map from a subset of $\mathbb{N}$ to $\mathbb{N}$ is *increasing* if it preserves the strict order.

Theorem 8.9 (spreading spaces)**.** *Let* E *and* D *be spreading sequence spaces.*

(1) *Lemma 7.11 holds for* E *and* D, *with relabeling by an injective map* $\mathbb{N} \to \mathbb{N}$ *replaced by relabeling along a map increasing on the support of* a; *so* (PTB) *is defined for* (E, D) *by Definition 7.12.*
(2) *Theorem 7.13, Propositions 7.15 and 7.16, Theorems 8.1 and 8.4, and Corollaries 8.2 and 8.6(1),(2) hold with spreading sequence spaces in place of symmetric ones, and so does Remark 8.7 for the pairs of Theorems 8.1 and 8.4.*
(3) *Over every nontrivially normed field* K *of characteristic* 2, *without condition* (L), *Theorem 7.20(4) holds for every spreading sequence space* E *over* K *with* $\sup_n n/\varphi_E(n)^2 = \infty$.

The proof is in Appendix K. The spreading spaces form a larger class than the symmetric ones, also up to equivalent norms.

Example 8.10 (Garling spaces)**.** In this example we identify $\mathbb{N}$ with $\{1, 2, 3, \dots\}$ through its order. Let $0 < \alpha < 1$, $1 \le q < \infty$ and $\beta = (1-\alpha)/q$. For $f \in c_{00}(\mathbb{N}, \mathbb{R})$ put

$$\|f\|_{\alpha,q} = \sup\Big\{ \Big(\sum_{j=1}^{L} j^{-\alpha} |f(t_j)|^q \Big)^{1/q} : L \ge 1,\ t_1 < \cdots < t_L \Big\},$$

a norm of the kind introduced by Garling [6] and studied in [1], and let $E_{\alpha,q}$ be the closure of c_{00} in the space of all $x \in \ell^\infty(\mathbb{N}, K)$ with $\|x\| = \sup_n \|x 1_{[n]}\|_{\alpha,q} < \infty$.

(1) $E_{\alpha,q}$ is a spreading sequence space in which c_{00} is dense, and $m^{2\beta} \le \varphi_{E_{\alpha,q}}(m)^2 \le (1-\alpha)^{-2/q} m^{2\beta}$ for $m \ge 1$.
(2) No norm on c_{00} invariant under permutations of $\mathbb{N}$ is equivalent to that of $E_{\alpha,q}$, and the unit vectors of $E_{\alpha,q}$, in the order of $\mathbb{N}$, satisfy (5.1) for no $r \in (0, 2)$ and $C \ge 1$.

(3) If $2\beta \le 1$, then for every $k \ge p$ neither $(E_{\alpha,q}, E_{\alpha,q})$ nor $(c_0, E_{\alpha,q})$ is good for Banach values in degree k, and $(\mathrm{S}^k_{E_{\alpha,q}})$, $(\mathrm{R}^k_{E_{\alpha,q}})$ and $(\Gamma^k_{E_{\alpha,q}})$ fail. If $2\beta < 1$, then over every nontrivially normed field K of characteristic 2, with $E_{\alpha,q}$ formed over K in the same way, the pair $(E_{\alpha,q}, E_{\alpha,q})$ is good for Banach values in no degree $k \ge 2$.
(4) If $2\beta > 1$, then $\|x\|_2^2 \le \big(\sum_{m\ge1} m^{-2\beta}\big)\|x\|^2$ for $x \in E_{\alpha,q}$.

The proof is in Appendix K. For $2\beta > 1$ the necessary conditions of Theorem 8.1 hold: $E_{\alpha,q}$ is ℓ^2-dominated by (4), and $\varphi_{E_{\alpha,q}}(m)^2 \ge m^{2\beta}$ grows faster than $m \ln m$; by (2), Theorem 5.6 does not apply to the unit vectors in their natural order (Question 12.16).

9. Goodness without splitting decompositions

The good pairs of Remark 8.7 all have splitting decompositions; this section shows that goodness does not require one.

9.1. One-dimensional operator spaces.

Proposition 9.1. *Let E and D be normed spaces with $\mathcal{L}(E,D) = K\,u_0$ for some nonzero $u_0 \in \mathcal{L}(E,D)$.*

(1) *For every $k \ge 1$ the pair (E,D) is good in degree k, and the map*
$$Q(\lambda_1 u_0, \dots, \lambda_k u_0) = \lambda_1 \cdots \lambda_k \, \Lambda^k_\pi u_0$$
is a lift as in Lemma 3.4.
(2) *If D is complete, then for every $k \ge 1$ the pair (E,D) has a splitting decomposition of degree k if and only if u_0 is S^k-split or R^k-split.*
(3) *If $\dim D \ge 2$, then every bounded linear functional on E is zero.*

The proof is in Appendix P.

9.2. Rigid Banach spaces.

The next theorem transfers the rigid subspace of L_0 of Kalton and Roberts [14] to separable complete nonarchimedean fields.

Theorem 9.2. *Let K be a complete nonarchimedean field with nontrivial absolute value that is separable as a metric space. There are infinite-dimensional Banach spaces $\mathfrak{X}_1, \mathfrak{X}_2, \dots$ over K such that, for all $\alpha, \beta \ge 1$,*
$$\mathcal{L}(\mathfrak{X}_\alpha, \mathfrak{X}_\beta) = \begin{cases} K\,\mathrm{id}_{\mathfrak{X}_\alpha}, & \alpha = \beta, \\ 0, & \alpha \ne \beta. \end{cases}$$

The proof is in Appendix M. By Proposition 9.1(3) with $E = D = \mathfrak{X}_\alpha$ and $u_0 = \mathrm{id}_{\mathfrak{X}_\alpha}$, every bounded linear functional on $\mathfrak{X}_\alpha$ is zero.

Theorem 9.3. *Let κ be a finite field of characteristic p and $K = \kappa((X))$, with the absolute value of condition (L) of Section 7. For suitable choices in the construction of Theorem 9.2, the Banach space $\mathfrak{X} = \mathfrak{X}_1$ has the following properties.*

(1) $\mathcal{L}(\mathfrak{X}, \mathfrak{X}) = K\,\mathrm{id}_{\mathfrak{X}}$.
(2) *The pair $(\mathfrak{X}, \mathfrak{X})$ is good in every degree, hence good for Banach values in every degree.*
(3) *For every $k \ge p$, neither $(\mathrm{S}^k_{\mathfrak{X}})$ nor $(\mathrm{R}^k_{\mathfrak{X}})$ holds.*
(4) *For every $k \ge p$, the pair $(\mathfrak{X}, \mathfrak{X})$ has no splitting decomposition of degree k.*

The proof is in Appendix M. Since $\mathfrak{X}$ has no nonzero bounded functional, item (3) is proved by forcing through L^r-valued readouts (Corollary M.12) instead of the coordinates of Section 7.

Theorem 9.4. *Let κ be a finite field of characteristic p and $K = \kappa((X))$, with the absolute value of condition (L). Let $\mathfrak{X}_1, \mathfrak{X}_2, \dots$ be the spaces of Theorem 9.2, built with the choices of Theorem 9.3, and let E be the completed ℓ^1-sum of the $\mathfrak{X}_\alpha$, with the norm $\|x\| = \sum_\alpha \|x_\alpha\|$ for $x = (x_\alpha)_{\alpha \geq 1}$.*

1. *For a bounded sequence $\lambda = (\lambda_\alpha)_{\alpha \geq 1}$ in K put $u_\lambda x = (\lambda_\alpha x_\alpha)_\alpha$. Every bounded operator on E is some u_λ, and $\|u_\lambda\| = \sup_\alpha |\lambda_\alpha|$.*
2. *The pair (E, E) is good for Banach values in every degree.*
3. *For every $k \geq p$, the pair (E, E) has no generalized decomposition of degree k, that is, no decomposition into finitely many source, target and parameter pieces (Definition 6.17).*

The proof is in Appendix M. In Theorem 9.3 the lift Q of Proposition 9.1(1) is that of one parameter piece through K; in Theorem 9.4 the lift is a block assembly (Proposition 6.27) of infinitely many such lifts, which no finite family of pieces replaces. Question 12.4 asks whether these mechanisms and the operations of Section 6 yield every pair good for Banach values.

10. Bundles, tangent bundles and differential forms

10.1. **From a bad operator family to changes of frame.** Transition functions take values in invertible operators; for a single Banach fiber, invertible families impose the same condition as all operator families (Proposition 11.2(1)).

Proposition 10.1 (shear realization)**.** *Suppose $A = A^k_{E,D;F}$ is not analytic at $u_0 \in \mathcal{L}(E, D)$. Set $H = D \oplus E$, with the maximum norm, and*

$$g(u)(d, e) = (d + ue, e).$$

Both g and its inverse family are affine analytic. Nevertheless, the family of pullback operators $g(u)^*$ on $\mathrm{Alt}^k(H; F)$ is not analytic at u_0.

Proof. The inverse family is $g(-u)$. Let $\pi_D \colon H \to D$ be the projection and $i_E(e) = (0, e)$. Define bounded linear maps $J \colon \mathrm{Alt}^k(D; F) \to \mathrm{Alt}^k(H; F)$ by $Jm = m \circ \pi_D^{\times k}$ and $R \colon \mathrm{Alt}^k(H; F) \to \mathrm{Alt}^k(E; F)$ by restriction along i_E. Since $\pi_D g(u) i_E = u$,

$$R \circ g(u)^* \circ J = A(u). \tag{10.1}$$

Bounded linear compression preserves analyticity, so $u \mapsto g(u)^*$ is not analytic at u_0. □

Corollary 10.2 (shear bundle)**.** *Suppose $A = A^k_{E,D;F}$ is not analytic at $u_0 \in \mathcal{L}(E, D)$, and let H and g be as in Proposition 10.1. Take the base $M = \mathcal{L}(E, D)$ and the trivial bundle $M \times H$. Its product trivialization and the trivialization $(u, z) \mapsto (u, g(u)z)$ form an analytic bundle atlas, and the induced alternating atlas on the trivial bundle with fiber $\mathrm{Alt}^k(H; F)$ is not analytic. Equivalently, the analytic bundle automorphism $(u, z) \mapsto (u, g(u)z)$ does not induce an analytic morphism on the alternating product bundle.*

Proof. The transitions are $g(u)$ and $g(-u)$, affine analytic by Proposition 10.1. The induced alternating transitions are pullback by $g(u)$ and $g(-u)$, and the induced morphism is locally pullback by $g(u)$; by Proposition 10.1, $u \mapsto g(u)^*$ is not analytic at u_0. □

Thus every failure of the hom-space criterion has a bundle realization, and Theorem 7.2 gives fibers for which it occurs at every point of the base. Applied to analytic families $x \mapsto u(x)$, the shear also makes the familywise hypothesis of Theorem 2.1 necessary for all fibers and frames; Section 11 studies it (Corollary 11.3, Theorems 11.9 and 11.10).

10.2. **Tangent bundles and individual forms.**

Proposition 10.3 (tangent and pullback obstructions)**.** *Suppose K is complete, has characteristic $p > 0$, is not spherically complete, and $k \geq p$.*

(1) *There is an analytic Banach manifold with two global analytic charts whose induced transition on scalar alternating k-forms is analytic at no point.*
(2) *There are Banach spaces N, Y, a polynomial analytic map $h\colon N \to Y$, and an analytic map $\omega_0\colon Y \to \mathrm{Alt}^k(Y;K)$ such that $h^*\omega_0$, expressed in the product coordinates on N, is analytic at no point as an $\mathrm{Alt}^k(N;K)$-valued map. After inclusion in $\mathrm{Mult}^k(N;K)$ it is analytic everywhere.*

The proof is in Appendix P.

Assertion (2) separates intrinsic analyticity of individual forms from analyticity of bundle transitions; Section 10.3 builds a calculus on the ambient notion.

The chart ψ of the proof of Proposition 10.3 realizes any nowhere analytic action $A^k_{E,D;F}$; with Theorem 7.2(1) it works over every field of characteristic p.

Proposition 10.4 (tangent obstructions with other values)**.** *Let K be a nontrivially normed field of characteristic $p > 0$, complete or not, and let $k \geq p$.*

(1) *There are a Banach space F and an analytic Banach manifold with two global analytic charts whose induced transition on F-valued alternating k-forms is analytic at no point.*
(2) *There is an analytic Banach manifold with two global analytic charts whose induced transition on tangent-valued alternating k-forms is analytic at no point.*
(3) *If K is complete and not spherically complete, a manifold as in (2) exists with a nonarchimedean Banach model.*

The target F constructed for (1) admits no equivalent nonarchimedean norm; for $K = \mathbb{F}_q((X))$ as in Corollary C.8, the model space can be chosen nonarchimedean. The model space constructed for (2) admits no equivalent nonarchimedean norm.

The proof is in Appendix P.

Bourbaki's question. For $k \geq 1$, Alt^k on Banach spaces is the functor α_k of [3, n° 7.8.1], and $A^k_{E,D;F}(u) = \alpha_k(\mathrm{id}_F, u)$. There K is $\mathbb{R}$, $\mathbb{C}$ or complete nonarchimedean, norms over a nonarchimedean K are nonarchimedean, and class C^r means analytic if $K \neq \mathbb{R}$, in a sense that agrees with ours between Banach spaces. In that setting α_k is a vector functor of class C^r if $k! \neq 0$ in K (Corollary 7.3), and for every k if K is discretely valued (Corollary 5.3 and the proof of Corollary 5.15). It is not one if $k! = 0$ in K and K is not spherically complete, since Theorem 7.2(2) gives nonarchimedean Banach spaces E, D with $A^k_{E,D;K}$ analytic at no point; so the definition in [3, n° 7.8.1] needs a restriction. By Corollary 10.2 and Proposition 10.3, it is needed for the bundles themselves. Source spaces of countable type suffice in every degree, by the following corollary of Theorem 5.4.

Corollary 10.5 (countable type and Bourbaki's α_k)**.** *Let K be complete and nonarchimedean, let E, D be normed spaces, at least one of them of countable type, and let F be a nonarchimedean Banach space. For every normed F', the joint action B of Proposition 2.2 on $\mathcal{L}(E,D)\times\mathcal{L}(F,F')$ has a bounded $(k+1)$-linear lift. In the setting of the paragraph* Bourbaki's question*, α_k is therefore a vector functor of the class considered there on source spaces of countable type, in every degree.*

Proof. If Q lifts $A^k_{E,D;F}$ (Theorem 5.4), then $(u_1, \dots, u_k, g) \mapsto (m \mapsto g \circ Q(u_1, \dots, u_k)(m))$ lifts B, as in the proof of Corollary 5.15. □

Our results do not decide the case of a densely valued, spherically complete field with $k! = 0$ (Section 12). Theorem 14.4 gives the constructions of [3, n° 7.8.1–7.8.4] on ambient sections of filtered analytic bundles, in every characteristic.

A derivative adapted to positive characteristic. Yaël Dillies asked whether $A = A^k_{E,D;F}$ is analytic in the calculus of [2, Definition 4.7 and Theorem 5.4], where a C^k map f has Taylor coefficients $a_n(x,h)$ with $n!\,a_n(x,h) = d^n f(x)(h,\dots,h)$. There A is smooth [2, Remark 4.2 and Lemma 10.1] and $a_k(u_0,u) = A(u)$, while $d^k A(u_0)(u,\dots,u) = k!\,A(u)$ vanishes if $k! = 0$. Analyticity is not restored: it asks for a bounded k-linear lift (Proposition 2.5), and Theorem 7.2 gives spaces with none when $k! = 0$ in K. Section 14 enlarges the target instead; [II] changes the coefficients.

10.3. **Analytic differential forms.** Throughout this subsection K is complete and nontrivially normed, manifold models are normed spaces, and charts and maps are C^ω, so that their derivatives are analytic; on the Banach coefficient spaces this is power-series analyticity.

Definition 10.6. An *ambient analytic scalar k-form* on M is a family $\eta_x \in \mathrm{Alt}^k(T_xM;K)$ whose representative in each chart is analytic after the inclusion

$$j\colon \mathrm{Alt}^k(P;K) \hookrightarrow \mathrm{Mult}^k(P;K).$$

Write $\Omega^k_{\mathrm{amb}}(M)$ for the space of such forms. In degree zero this is the space of analytic scalar functions.

Under a coordinate change $\psi = \phi' \circ \phi^{-1}$ the representatives satisfy $j\eta_\phi(y) = (j\eta_{\phi'}(\psi(y))) \circ (D\psi(y),\dots,D\psi(y))$, which is analytic by bounded multilinear composition; so any analytic atlas may be used.

For an ambient k-form η, an ambient l-form ζ and $S \subseteq \{1,\dots,k+l\}$, let v_S list the arguments v_i, $i \in S$, in increasing order of i, and let $\epsilon(S)$ be the shuffle sign of Section 6.1. The product is the shuffle formula

$$(\eta \wedge \zeta)(v_1,\dots,v_{k+l}) = \sum_{|S|=k} \epsilon(S)\eta(v_S)\zeta(v_{S^c}), \tag{10.2}$$

a bounded bilinear operation on multilinear coefficient spaces, of norm at most $\binom{k+l}{k}$, which is alternating on alternating inputs. On an open subset of a model space P, the exterior derivative of an ambient coefficient map η of degree k is

$$(d\eta)_y(v_0,\dots,v_k) = \sum_{i=0}^{k} (-1)^i D\eta(y)(v_i)(v_0,\dots,\widehat{v_i},\dots,v_k). \tag{10.3}$$

Theorem 10.7 (ambient differential calculus)**.** *Ambient analytic forms define a sheaf of graded algebras on every analytic manifold. Analytic maps induce pullbacks, and there is an exterior derivative*

$$d\colon \Omega^k_{\mathrm{amb}}(M) \longrightarrow \Omega^{k+1}_{\mathrm{amb}}(M).$$

The product is associative and graded commutative, $d^2 = 0$, pullback commutes with both product and exterior derivative, and

$$d(\eta \wedge \zeta) = d\eta \wedge \zeta + (-1)^k \eta \wedge d\zeta \qquad (\eta \in \Omega^k_{\mathrm{amb}}(M)).$$

Thus $\Omega^\bullet_{\mathrm{amb}}$ is a contravariant differential graded algebra construction in every characteristic.

The proof is in Appendix P.

Proposition 10.8 (comparison with alternating-bundle sections)**.** *Every intrinsically analytic alternating-valued coefficient map is ambient analytic. If, for the model space P, the inclusion j has a bounded linear retraction, then the natural alternating cotangent atlas is analytic and its analytic sections are precisely the ambient analytic forms. The same conclusion holds when P has finite continuous coordinates.*

Proof. The first statement follows by composition with j. For a bounded retraction R, the identity $\eta = R(j\eta)$ gives the converse. Moreover, $(u_1, \dots, u_k) \mapsto \big[m \mapsto R\big(m \circ (u_1, \dots, u_k)\big)\big]$ is a bounded multilinear lift of the precomposition action, so the induced alternating cotangent transitions are analytic. If P has finite continuous coordinates $(e_i, e_i^*)_{i \le d}$, the values $m(e_{i_1}, \dots, e_{i_k})$ identify $\mathrm{Mult}^k(P; K)$ with K^{d^k}, with its maximum norm, by a bounded linear isomorphism, so every linear retraction of j is bounded and the previous case applies. □

The retraction exists if $k!$ is invertible, by normalized antisymmetrization, and if K is spherically complete and nonarchimedean, by Theorem 5.2; and if P has finite continuous coordinates, by the proof above.

Exact constant forms in degree p. Here K is any nontrivially normed field of characteristic $p > 0$, Y and F are normed, and $\mathcal{A}_Y \colon \mathrm{Mult}^p(Y; F) \to \mathrm{Alt}^p(Y; F)$ is unnormalized antisymmetrization (Lemma 3.10). Define $(d\eta)_y \in \mathrm{Alt}^p(Y; F)$ by (10.3) for every map η from an open subset of Y to $\mathrm{Alt}^{p-1}(Y; F)$ differentiable at y. Constant forms are closed, and (R^p_Y) is the exactness of one of them.

Proposition 10.9 (exact constant forms)**.** *For $\omega \in \mathrm{Alt}^p(Y; F)$ the following are equivalent:*

1. *there are a map η from an open subset of Y to $\mathrm{Alt}^{p-1}(Y; F)$ and a point y at which η is differentiable with $(d\eta)_y = \omega$;*
2. *$\omega = d\eta$ on Y for a bounded linear map $\eta \colon Y \to \mathrm{Alt}^{p-1}(Y; F)$;*
3. *$\omega = \mathcal{A}_Y \beta$ for some $\beta \in \mathrm{Mult}^p(Y; F)$.*

For $F = \Lambda^p_\pi Y$ and $\omega = \omega_Y$ they hold if and only if alt^p_Y has a bounded linear left inverse, and then (E, Y) is good in degree p for every normed E. For $F = \widehat{\Lambda}^p_\pi Y$ and $\omega = \omega_Y$ they hold if and only if (R^p_Y) holds, and then (E, Y) is good for Banach values in degree p for every normed E.

The proof is in Appendix P.

Under condition (L) of Section 7.1, for $Y = c_0(\mathbb{N}, K)$ the closed constant form ω_Y with values in $\widehat{\Lambda}^p_\pi Y$ has no primitive near any point, by Theorem 7.7.

11. Familywise analyticity and bundles

Throughout, P, E, D are normed spaces and $k \ge 1$; P is the model space of a base, since Theorem 2.1 needs only operator families on open subsets of P.

11.1. **Families of operators.**

Definition 11.1. The triple $(P; E, D)$ is *good in degree k* if for every open $U \subseteq P$ and every analytic $u \colon U \to \mathcal{L}(E, \widehat{D})$ the map $x \mapsto \Lambda^k_\pi u(x) \in \mathcal{L}(\Lambda^k_\pi E, \widehat{\Lambda}^k_\pi D)$ is analytic on U. We write $(P; E)$ for $(P; E, E)$. For a Banach space H and an open $U \subseteq P$, an *invertible family* on U is an analytic $g \colon U \to \mathcal{L}(H)$ whose values are invertible and for which $x \mapsto g(x)^{-1}$ is analytic.

Here $\Lambda^k_\pi u(x)$ maps $\Lambda^k_\pi E$ into $\Lambda^k_\pi \widehat{D} \subseteq \widehat{\Lambda}^k_\pi D$ (paragraph before Theorem 3.16). By Theorem 3.2(1) with $X = \Lambda^k_\pi E$, $Y = \widehat{\Lambda}^k_\pi D$, and the completed form of Proposition 3.1, $x \mapsto \Lambda^k_\pi u(x)$ is analytic at a point, or on U, if and only if $x \mapsto A^k_{E, \widehat{D}; F}(u(x))$ is analytic there for every Banach space F. For a linear family $\theta \colon P \to \mathcal{L}(E, \widehat{D})$ (defined before Theorem 6.13), $h \mapsto \Lambda^k_\pi \theta(h)$

is analytic at some point if and only if θ is liftable in degree k, and then it is a continuous polynomial (Proposition 2.5, as in the proof of Lemma 3.4).

Proposition 11.2 (reductions). (1) *Let H be a Banach space. If $x \mapsto \Lambda^k_\pi g(x)$ is analytic for every invertible family g on every open subset of P, then $(P;H)$ is good in degree k. More precisely, for an analytic $u\colon U \to \mathcal{L}(H)$ and $x_0 \in U$ there are distinct $\lambda_0, \dots, \lambda_k \in K$ and a neighborhood $U_0 \subseteq U$ of x_0 such that every $1+\lambda_i u$, with $1 = \mathrm{id}_H$, is an invertible family on U_0 and $\Lambda^k_\pi u$ is a fixed linear combination of the maps $\Lambda^k_\pi(1+\lambda_i u)$ on U_0.*

(2) *If $(P;E,D)$ is good in degree k, so is every triple of bounded linear retracts of P, E and D, respectively. The triple $(P;E\oplus E',D)$ is good in every degree $j \le k$ if and only if $(P;E,D)$ and $(P;E',D)$ are; the same holds in the target slot.*

(3) *Let E and D be Banach spaces and $H = D \oplus E$ with the maximum norm. Then $x \mapsto \Lambda^j_\pi g(x)$ is analytic for every invertible family g into $\mathcal{L}(H)$ over every open subset of P and every $j \le k$ if and only if the four triples $(P;E,E)$, $(P;E,D)$, $(P;D,E)$, $(P;D,D)$ are good in every degree $j \le k$. If $\mathcal{C}$ is a class of Banach spaces closed under finite direct sums, then, in the single degree k, $x \mapsto \Lambda^k_\pi g(x)$ is analytic for every $G \in \mathcal{C}$ and every invertible family g into $\mathcal{L}(G)$ over an open subset of P if and only if $(P;E,D)$ is good in degree k for all $E, D \in \mathcal{C}$.*

The proof is in Appendix O.

Corollary 11.3 (what bundles need). *Let $\mathcal{C}$ be a class of Banach spaces closed under finite direct sums. The following are equivalent.*

(1) *For every analytic manifold M modeled on P, every analytic vector bundle over M with typical fiber $E \in \mathcal{C}$ and every Banach space F, the atlas induced on the bundle with typical fiber $\mathrm{Alt}^k(E;F)$ is analytic.*

(2) *The triple $(P;E,D)$ is good in degree k for all $E, D \in \mathcal{C}$.*

If they hold, Theorem 2.1 applies to fiberwise alternating maps on analytic bundles with typical fibers in $\mathcal{C}$ and analytic bundles with Banach typical fibers, and to their operator-valued analytic morphisms, over every analytic manifold modeled on P.

The proof is in Appendix O.

11.2. **Homogeneous families.**

Definition 11.4. For $n \ge 0$ and $\beta \in \mathrm{Mult}^n(P;\mathcal{L}(E,\widehat{D}))$ put $u_\beta(h) = \beta(h,\dots,h)$; for $n=0$, β is a constant operator. For $j \ge 1$, a *lift of u_β in degree j* is a bounded jn-linear map $q\colon P^{jn} \to \mathcal{L}(\Lambda^j_\pi E, \widehat{\Lambda}^j_\pi D)$ with $q(h,\dots,h) = \Lambda^j_\pi u_\beta(h)$ for $h \in P$, and $c_j(\beta) \in [0,\infty]$ is the infimum of their norms.

For $n \ge 1$, $c_j(\beta) < \infty$ if and only if $h \mapsto \Lambda^j_\pi u_\beta(h)$ is analytic at some point of P (Lemma O.5). For $j \ge 1$ consider the condition

$$\begin{aligned}(\mathrm{G}_j)\colon\ &\text{for every sequence } (\beta_n)_{n\ge1} \text{ with } \beta_n \in \mathrm{Mult}^n(P;\mathcal{L}(E,\widehat{D})) \text{ and } \|\beta_n\| \le 1,\\ &\text{every } c_j(\beta_n) \text{ is finite and } \limsup_{n\to\infty} c_j(\beta_n)^{1/n} < \infty.\end{aligned} \tag{11.1}$$

Theorem 11.5 (reduction to homogeneous families). (1) *If $(P;E,D)$ is good in degree j, then (G_j) holds.*

(2) *If (G_j) holds for every $j \le k$, then $(P;E,D)$ is good in degree k.*

(3) *Condition (G_j) holds whenever $j! \ne 0$ in K. If $\operatorname{char} K = p > 0$, then $(P;E,D)$ is good in every degree $j \le k$ if and only if (G_{p^i}) holds for every $i \ge 1$ with $p^i \le k$.*

(4) *Let Y_n be the completed projective tensor product of n copies of P and one copy of E, and let $\beta_n^{\mathrm{u}} \in \mathrm{Mult}^n(P;\mathcal{L}(E,Y_n))$ be given by $\beta_n^{\mathrm{u}}(h_1,\dots,h_n)y = h_1\otimes\cdots\otimes h_n\otimes y$. For every normed D, $n\ge 1$ and $\beta\in\mathrm{Mult}^n(P;\mathcal{L}(E,\widehat{D}))$, $c_j(\beta)\le c_j(\beta_n^{\mathrm{u}})\|\beta\|^j$. Consequently $(P;E,D)$ is good in every degree $j\le k$ for every normed D if and only if, for every $j\le k$, every $c_j(\beta_n^{\mathrm{u}})$ is finite and $\limsup_n c_j(\beta_n^{\mathrm{u}})^{1/n}<\infty$; if $\mathrm{char}\,K=p>0$, the degrees $j=p^i\le k$ suffice. Equivalently, $(P;E,D_{\mathrm{u}})$ is good in every degree $j\le k$ for the ℓ^1-sum D_{u} of the spaces Y_n.*

Corollary 11.6 (degrees of triples)**.** *Let $\mathrm{char}\,K=p>0$, and let $\mathfrak{D}(P;E,D)$ be the set of $k\ge 1$ in which $(P;E,D)$ is good.*

(1) *$\{1,\dots,p-1\}\subseteq\mathfrak{D}(P;E,D)$, and $a+b\in\mathfrak{D}(P;E,D)$ whenever $a,b\in\mathfrak{D}(P;E,D)$ and the addition of a and b in base p has no carry. If $\mathfrak{D}(P;E,D)\ne\{1,2,\dots\}$, its least missing element is a power of p.*
(2) *If $E\cong E\oplus K$ and $D\cong D\oplus K$ by bounded linear isomorphisms, then $\mathfrak{D}(P;E,D)$ is $\{1,\dots,p^i-1\}$ for some $i\ge 1$ or $\{1,2,\dots\}$.*
(3) *$\mathfrak{D}(P;E,D)\supseteq\mathfrak{D}_B(E,\widehat{D})$, with equality for $P=\mathcal{L}(E,\widehat{D})$.*
(4) *The set of k for which $(P;E,D)$ is good in degree k for all normed E and D is $\{1,\dots,p^i-1\}$ for some $i\ge 1$, or $\{1,2,\dots\}$.*

The proofs are in Appendix O.

11.3. **The familywise class.** The tensor algebra $\mathcal{T}(P)$ is the Banach space of sequences $(w_n)_{n\ge0}$, $w_n\in\widehat{\otimes}_\pi^n P$, $\widehat{\otimes}_\pi^0 P=\widehat{K}$, with norm $\sum_n\|w_n\|<\infty$. The map $\mathrm{ver}_n\in\mathrm{Mult}^n(P;\mathcal{T}(P))$ places $h_1\otimes\cdots\otimes h_n$ in slot n, with ver_0 the unit of slot 0, and $\mathrm{ver}(h)=\sum_n\mathrm{ver}_n(h,\dots,h)=(h^{\otimes n})_{n\ge0}$ is analytic for $\|h\|<1$ (Lemma O.1). Every analytic germ factors through ver: if u is analytic at x_0 with values in $\mathcal{L}(E,\widehat{D})$, then $u(x_0+\lambda h)=\theta(\mathrm{ver}(h))$ for $\|h\|<1$, for some $\lambda\in K^\times$ and a linear family $\theta\colon\mathcal{T}(P)\to\mathcal{L}(E,\widehat{D})$ (Lemma O.8).

For a Banach space Y and $0<R<1$, $\mathrm{Mult}^\omega_R(P;Y)$ is the Banach space of sequences $(b_N)_{N\ge0}$, $b_N\in\mathrm{Mult}^N(P;Y)$, with norm $\sup_N R^N\|b_N\|<\infty$; $\mathrm{Mult}^\omega_{R,0}(P;Y)$ is its closed subspace where every b_N has zero diagonal, so that $b_0=0$; for $R'\le R$, $\iota_{R'R}$ is the contractive inclusion into $\mathrm{Mult}^\omega_{R'}(P;Y)$. With $I_1,\dots,I_k$ the consecutive blocks of $\{1,\dots,N\}$ of sizes $n_1,\dots,n_k$ and h_I the tuple of the h_i with $i\in I$, the map $\mathrm{ver}^\sharp\colon\mathrm{Mult}^k(\mathcal{T}(P);Y)\to\mathrm{Mult}^\omega_R(P;Y)$,

$$(\mathrm{ver}^\sharp\,\Phi)_N(h_1,\dots,h_N)=\sum_{n_1+\cdots+n_k=N}\Phi\big(\mathrm{ver}_{n_1}(h_{I_1}),\dots,\mathrm{ver}_{n_k}(h_{I_k})\big),$$

is bounded and maps $\mathrm{Mult}^k_0(\mathcal{T}(P);Y)$ into $\mathrm{Mult}^\omega_{R,0}(P;Y)$ (Lemma O.9). For a linear family $\theta\colon\mathcal{T}(P)\to\mathcal{L}(E,\widehat{D})$, with ev_θ and $\mathrm{ob}_k(\theta)$ as in Section 6.3, put

$$\begin{aligned}\mathrm{ob}^\omega_k(\theta;R)&=\big(\mathrm{ver}^\sharp\,\mathrm{ev}_\theta|_{\mathcal{N}_E^k}\big)_*[\mathcal{S}_E^k]=(\mathrm{ver}^\sharp)_*\,\mathrm{ob}_k(\theta)\\&\in\mathrm{Ext}\big(\widehat{\Lambda}^k_\pi E,\mathrm{Mult}^\omega_{R,0}(P;\widehat{\Lambda}^k_\pi D)\big),\end{aligned}\tag{11.2}$$

the second equality by Corollary N.3(b), and $\mathrm{ob}_{k,N}(\theta)=((\mathrm{pr}_N\,\mathrm{ver}^\sharp\,\mathrm{ev}_\theta)|_{\mathcal{N}_E^k})_*[\mathcal{S}_E^k]\in\mathrm{Ext}(\widehat{\Lambda}^k_\pi E,\mathrm{Mult}^N_0(P;\widehat{\Lambda}^k_\pi D))$, with pr_N the Nth term. The class $\mathrm{ob}^\omega_k(\theta)$ *vanishes after shrinking* if $(\iota_{R'R})_*\,\mathrm{ob}^\omega_k(\theta;R)=0$ for some $R'\in(0,R]$; this does not depend on R.

Theorem 11.7 (the familywise class)**.** (1) *For a linear family $\theta\colon\mathcal{T}(P)\to\mathcal{L}(E,\widehat{D})$ the following are equivalent: $x\mapsto\Lambda^k_\pi\theta(\mathrm{ver}(x))$ is analytic at 0; $\mathrm{ob}^\omega_k(\theta)$ vanishes after shrinking; there are $C,R'>0$ such that for every N the map $(\mathrm{pr}_N\,\mathrm{ver}^\sharp\,\mathrm{ev}_\theta)|_{\mathcal{N}_E^k}$ extends to a bounded linear map $\widehat{\otimes}^k_\pi E\to\mathrm{Mult}^N_0(P;\widehat{\Lambda}^k_\pi D)$ of norm at most CR'^{-N}.*

(2) $(P; E, D)$ *is good in degree* k *if and only if* $\mathrm{ob}_k^\omega(\theta)$ *vanishes after shrinking for every linear family* $\theta\colon \mathcal{T}(P) \to \mathcal{L}(E, \widehat{D})$.

(3) *Let* $\theta_{\mathrm{u}}\colon \mathcal{T}(P) \to \mathcal{L}(E, \mathcal{T}(P)\widehat{\otimes}_\pi E)$ *be given by* $\theta_{\mathrm{u}}(w)y = w \otimes y$. *For* θ *as in (1), let* $\tau_\theta\colon \mathcal{T}(P)\widehat{\otimes}_\pi E \to \widehat{D}$ *be the linearization of* $(w, y) \mapsto \theta(w)y$, *and let* $(\widehat{\Lambda}^k_\pi \tau_\theta)_\flat$ *apply* $\widehat{\Lambda}^k_\pi \tau_\theta$ *to every term. Then* $\mathrm{ob}_k^\omega(\theta; R) = ((\widehat{\Lambda}^k_\pi \tau_\theta)_\flat)_* \mathrm{ob}_k^\omega(\theta_{\mathrm{u}}; R)$. *Hence* $(P; E, D)$ *is good in degree* k *for every normed* D *if and only if* $\mathrm{ob}_k^\omega(\theta_{\mathrm{u}})$ *vanishes after shrinking, and if and only if* $x \mapsto \Lambda^k_\pi \mathcal{U}(x)$ *is analytic at* 0 *for the universal germ* $\mathcal{U}(x)y = \mathrm{ver}(x) \otimes y$.

The proof is in Appendix O.

11.4. **Base models good for all fibers.**

Definition 11.8. The space P is *universally good in degree* k, written (U^k_P), if $(P; E, D)$ is good in degree k for all normed E and D. It has *fixed-degree reflection in degree* k, written (FR^k_P), if for all Banach spaces G and Z, every closed subspace $W \subseteq Z$, every bounded k-linear $\Phi\colon G^k \to Z$ with $\Phi(y, \dots, y) \in W$ for $y \in G$, every open $U \subseteq P$ and every analytic $\gamma\colon U \to G$, the map $x \mapsto \Phi(\gamma(x), \dots, \gamma(x))$ is analytic as a W-valued map on U.

Recall $g_n(P)$ from Definition 3.8(3).

Theorem 11.9 (polarizable base models)**.** (1) *The following are equivalent:* (FR^k_P); (FR^k_P) *restricted to affine maps* $\gamma(x) = y_0 + Tx$ *on* $U = P$, *with* T *bounded linear;* P *is polarizable in every degree* $n \le k$.

(2) *If* P *is polarizable in every degree* $n \le k$, *then* (U^k_P) *holds. Quantitatively, let* $\bar{g} \ge \max(1, g_1(P), \dots, g_k(P))$, *with* $\bar{g} > g_n(P)$ *when no projection of norm* $g_n(P)$ *exists. If* $u\colon U \to \mathcal{L}(E, \widehat{D})$ *satisfies* $u(x_0+h) = \sum_n \beta_n(h, \dots, h)$ *for* $\|h\| < R$, *with* $\|\beta_n\| \le CR^{-n}$, *then* $\Lambda^k_\pi u$ *has at* x_0 *an expansion of radius at least* $R/\bar{g}$ *whose* N*th coefficient has norm at most* $k!\, C^k \binom{N+k-1}{k-1} (\bar{g}/R)^N$. *In particular, Theorem 2.1 applies to fiberwise alternating maps on analytic bundles with arbitrary normed typical fibers and analytic bundles with Banach typical fibers over every analytic manifold modeled on* P.

(3) *If* $\mathrm{char}\, K = 0$, *or* $\mathrm{char}\, K = p$ *and* $k < p$, *then* (U^k_P) *holds for every* P. *If* $\mathrm{char}\, K = p$ *and* P *is polarizable in degree* p, *then* (U^n_P) *holds for every* $n \ge 1$. *Every normed space* P *with a lower* r*-estimate for some* $r < 2$ *(Section 5.2) satisfies* (U^n_P) *for every* $n \ge 1$, *over every* K.

(4) *Let* K *be complete and* P *a Banach space. Then* P *has universal analytic reflection (Definition G.2) if and only if* $\sup_n g_n(P)^{1/n} < \infty$, *and* (FR^k_P) *holds for every* k *if and only if* $g_n(P) < \infty$ *for every* n. *If* K *is nonarchimedean and* $P = \ell^1(\mathbb{N}, K)$, *then* (FR^k_P) *and* (U^k_P) *hold for every* k, *but* P *does not have universal analytic reflection.*

Theorem 11.10 (base models under (L))**.** *Assume* (L).

(1) *Let* P *be a symmetric sequence space in which* $c_{00}(\mathbb{N}, K)$ *is dense, let* E_0 *be* $c_{00}(\mathbb{N}, K)$ *with the supremum norm, and let* $\theta_0\colon P \to \mathcal{L}(E_0, P)$ *be given by* $\theta_0(a)y = ay$. *The following are equivalent:* (U^k_P); $a \mapsto \Lambda^k_\pi \theta_0(a)$ *is analytic at some point of* P; $\mathcal{P}_<$ *is bounded on* $P \otimes_\pi P$; P *is polarizable in degree* p; (U^n_P) *holds for every* $n \ge 1$. *They are equivalent to the conditions of Theorem 8.1. In particular, for* $P = \ell^r(\mathbb{N}, K)$ *with* $1 \le r < \infty$, (U^k_P) *holds if and only if* $r < 2$; *and for* $P = \ell^r(\mathbb{N}, K)$ *with* $2 \le r < \infty$, *for* $P = c_0(\mathbb{N}, K)$, *and for every symmetric sequence space* P *in which* $c_{00}(\mathbb{N}, K)$ *is dense and* $\varphi_P(m)^2 \le Cm$ *for some* C, (U^n_P) *holds exactly for* $n \le p - 1$.

(2) *Let* P *be normed, and suppose there are bounded linear maps* $i\colon c_{00}(\mathbb{N}, K) \to P$, *with the supremum norm on* $c_{00}(\mathbb{N}, K)$, *and* $w\colon P \to \ell^\infty(\mathbb{N}, \widehat{K})$ *with* $w \circ i$ *the inclusion.*

Then for the linear family $x \mapsto \mu_{w(x)} \in \mathcal{L}(c_{00}(\mathbb{N}, K), \ell^\infty(\mathbb{N}, \widehat{K}))$, with $\ell^\infty(\mathbb{N}, \widehat{K})$ regarded over K, the map $x \mapsto \Lambda^k_\pi \mu_{w(x)}$ is analytic at no point, so (U^k_P) fails. Such maps exist for $c_0(\Sigma, K)$ and $\ell^\infty(\Sigma, K)$ with Σ infinite, and for every P with an equivalent nonarchimedean norm and infinite-dimensional completion, $\dim_{\widehat{K}} \widehat{P} = \infty$. For P with an equivalent nonarchimedean norm, (U^k_P) holds if and only if P is polarizable in degree p, and if and only if $\dim_{\widehat{K}} \widehat{P} < \infty$.

The proofs are in Appendix O.

11.5. **Fibers and mixed mechanisms.**

Proposition 11.11 (fibers good over every base)**.** (1) *If $(E, \widehat{D})$ is good for Banach values in degree k, with a lift Q as in Lemma 3.4, then $(P; E, D)$ is good in degree k for every normed P, and $c_k(\beta) \le \|Q\|\|\beta\|^k$ for every n and $\beta \in \mathrm{Mult}^n(P; \mathcal{L}(E, \widehat{D}))$.*

(2) *$(\mathcal{L}(E, \widehat{D}); E, D)$ is good in degree k if and only if $(E, \widehat{D})$ is good for Banach values in degree k. Hence $(P; E, D)$ is good in degree k for every normed P if and only if $(E, \widehat{D})$ is good for Banach values in degree k.*

The proof is in Appendix O.

Definition 11.12. A bounded linear map $T\colon P \to P'$ is *Γ^k-split* if there is a bounded k-linear $\Pi\colon P^k \to \widehat{\Gamma}^k_\pi P'$ with $\Pi(a, \dots, a) = (Ta)^{\otimes k}$ for $a \in P$.

By Lemma 3.11, P is polarizable in degree k if and only if id_P is Γ^k-split, and T is Γ^k-split if P or P' is polarizable in degree k: compose $T^{\otimes k}$ with a projection onto $\widehat{\Gamma}^k_\pi P'$, or a projection onto $\widehat{\Gamma}^k_\pi P$ with the extension of $T^{\otimes k}$.

Theorem 11.13 (parameter-relative ordering)**.** *Let $2 < r < \infty$.*

(1) *For every totally ordered set Σ and every $k \ge 1$, the inclusion $\ell^2(\Sigma, \widehat{K}) \to \ell^r(\Sigma, \widehat{K})$, over K, is Γ^k-split.*

(2) *Let K be nonarchimedean, and let P have an ℓ^2-dominated linearly independent family $(e_t)_{t\in\Sigma}$ with dense span V and $\sup_t \|e_t\| < \infty$. Let G be a normed space with an equivalent r-norm, Z a Banach space, $W \subseteq Z$ a closed subspace and $\Phi\colon G^k \to Z$ a bounded k-linear map with $\Phi(y, \dots, y) \in W$ for $y \in G$. For every bounded linear $T\colon P \to G$ there is a bounded k-linear $Q\colon P^k \to W$ with $Q(a, \dots, a) = \Phi(Ta, \dots, Ta)$ for $a \in P$; and for every open $U \subseteq P$ and every analytic $\gamma\colon U \to G$, the map $x \mapsto \Phi(\gamma(x), \dots, \gamma(x))$ is analytic into W.*

(3) *For K and P as in (2), and for D with an equivalent r-norm or an equivalent nonarchimedean norm, $(P; E, D)$ is good in every degree for every normed E.*

Under (L) and for $2 < r < \infty$, (3) makes $(\ell^2(\mathbb{N}, K); \ell^r(\mathbb{N}, K))$ good in every degree, although neither space is polarizable in degree k (Corollary 8.2).

Theorem 11.14 (weighted staircase)**.** *Assume* (L). *Let E be a sequence space with $\|y\|_\infty \le \|y\|_E$ for $y \in E$, and let P be a normed space containing $c_{00}(\mathbb{N}, K)$ as a linear subspace. For $G \in \{E, P\}$ let*

$$\varphi^{\pm}_G(n) = \sup\big\{\|y\|_G : y \in c_{00}(\mathbb{N}, K),\ y(t) \in \{0, 1, -1\} \text{ for all } t,\ |\operatorname{supp} y| \le n\big\}.$$

If $\liminf_{m\to\infty} \varphi^{\pm}_P(2m)^2 \varphi^{\pm}_E(4m)^2/m = 0$, then no k-linear $Q\colon c_{00}(\mathbb{N}, K)^k \to \mathcal{L}(\Lambda^k_\pi E, \widehat{\Lambda}^k_\pi E)$ that is bounded for the norm of P satisfies $Q(a, \dots, a) = \Lambda^k_\pi \mu_a$ for $a \in c_{00}(\mathbb{N}, K)$. Consequently, if $T\colon P \to \mathcal{L}(E, \widehat{E})$ is bounded linear with $T(a) = \mu_a$ for $a \in c_{00}(\mathbb{N}, K)$, then $x \mapsto \Lambda^k_\pi T(x)$ is analytic at no point of P. For a symmetric sequence space G, $\varphi^{\pm}_G = \varphi_G$.

Corollary 11.15 (the table)**.** *Assume* (L)*. Let P be $\ell^{r_1}(\mathbb{N},K)$ with $1\le r_1<\infty$, or $c_0(\mathbb{N},K)$. In (1) and (2), let E be $\ell^{r_2}(\mathbb{N},K)$ with $1\le r_2<\infty$, or $c_0(\mathbb{N},K)$, or, if K is complete, an infinite-dimensional Banach space with an equivalent nonarchimedean norm.*

(1) *$(P;E)$ is good in every degree if $r_1<2$; if $E=\ell^{r_2}(\mathbb{N},K)$ with $r_2<2$; or if $r_1=2$ and E is $\ell^{r_2}(\mathbb{N},K)$ with $r_2>2$, or $c_0(\mathbb{N},K)$, or has an equivalent nonarchimedean norm.*
(2) *There is a linear family $P\to\mathcal{L}(E,\widehat{E})$ along which Λ^k_π is analytic at no point, so $(P;E)$ is not good in degree k, if $P=c_0(\mathbb{N},K)$ and E is not $\ell^{r_2}(\mathbb{N},K)$ with $r_2<2$; or if $2<r_1<\infty$ and E is $c_0(\mathbb{N},K)$, or has an equivalent nonarchimedean norm, or is $\ell^{r_2}(\mathbb{N},K)$ with $1/r_1+1/r_2<1/2$.*
(3) *Let E be a symmetric sequence space in which $c_{00}(\mathbb{N},K)$ is dense. The multiplier family $c_0(\mathbb{N},K)\to\mathcal{L}(E,\widehat{E})$, $a\mapsto\mu_a$, is liftable in some degree $k\ge p$ if and only if (E,E) is good for Banach values in every degree, and if and only if $(c_0(\mathbb{N},K);E)$ is good in every degree.*

The proofs are in Appendix O.

11.6. **Tangent bundles.**

Definition 11.16. A normed space P_0 is *tangent-good in degree k* if for every open $U\subseteq P_0$ and every analytic $f\colon U\to P_0$ whose derivative $Df\colon U\to\mathcal{L}(P_0,P_0)$ is analytic, the map $x\mapsto\Lambda^k_\pi Df(x)\in\mathcal{L}(\Lambda^k_\pi P_0,\widehat{\Lambda}^k_\pi P_0)$ is analytic on U.

Let K be complete and P_0 a Banach space, so that analytic maps between open subsets of P_0 have analytic derivatives [3]. For a manifold M modeled on P_0 with chart transitions f_{ij}, the transitions of $\widehat{\Lambda}^k_\pi TM$ and $\mathrm{Alt}^k(TM;F)$ are $\Lambda^k_\pi Df_{ij}$ and $A^k_{P_0,P_0;F}(Df_{ji})$; so if P_0 is tangent-good in degree k, these bundles are analytic for every Banach space F, by the paragraph after Definition 11.1.

Theorem 11.17 (shear charts)**.** (1) *Let K be complete, let P_0 be a Banach space with a bounded linear isomorphism $J\colon P_1\times D\times E\to P_0$, let $U\subseteq P_1$ be open and let $u\colon U\to\mathcal{L}(E,D)$ be analytic. Then $\psi(x,d,e)=(x,d+u(x)e,e)$ is an analytic diffeomorphism of $U\times D\times E$ onto itself, with inverse $(x,d,e)\mapsto(x,d-u(x)e,e)$, and with $i_E(e)=(0,0,e)$ and $\pi_D(x,d,e)=d$,*

$$\Lambda^k_\pi u(x)=\widehat{\Lambda}^k_\pi\pi_D\circ\Lambda^k_\pi D\psi(x,d,e)\circ\Lambda^k_\pi i_E.$$

If $\Lambda^k_\pi u$ is not analytic at x_0, then neither $(x,d,e)\mapsto\Lambda^k_\pi D\psi(x,d,e)$ nor $(x,d,e)\mapsto\Lambda^k_\pi D(\psi^{-1})(x,d,e)$ is analytic at any point (x_0,d,e), and the manifold $M=J(U\times D\times E)$ with the charts id_M and $J\circ\psi\circ J^{-1}$ has bundles $\widehat{\Lambda}^k_\pi TM$ and $\mathrm{Alt}^k(TM;\widehat{\Lambda}^k_\pi D)$ whose induced atlases are not analytic.

(2) *Let K be complete and P_0 a Banach space with $P_0\cong P_0\times P_0\times P_0$. The following are equivalent: P_0 is tangent-good in degree k; for every analytic manifold M modeled on P_0 the induced atlases of $\widehat{\Lambda}^k_\pi TM$ and of $\mathrm{Alt}^k(TM;F)$, for every Banach space F, are analytic; $(P_0;P_0)$ is good in degree k.*
(3) *Assume* (L)*. For $E=c_0(\mathbb{N},K)$ and for $E=\ell^r(\mathbb{N},K)$ with $4<r<\infty$, the polynomial map $\psi(a,d,e)=(a,d+a\cdot e,e)$ on E^3, with $a\cdot e$ the pointwise product, has the polynomial inverse $(a,d,e)\mapsto(a,d-a\cdot e,e)$, and $x\mapsto\Lambda^k_\pi D\psi(x)$ and $x\mapsto\Lambda^k_\pi D(\psi^{-1})(x)$ are analytic at no point of E^3; since $E\cong E^3$, E is not tangent-good in degree k. If K is complete, no infinite-dimensional Banach space with an equivalent nonarchimedean norm is tangent-good in degree k.*

The proof is in Appendix O.

11.7. **Structure groups.**

Definition 11.18. Let E be a normed space and $\mathrm{GL}(E)$ the group of invertible elements of $\mathcal{L}(E)$. For a subset $\mathsf{S} \subseteq \mathcal{L}(E)$, an S-*family* over P is an analytic map $u\colon U \to \mathcal{L}(E)$, with $U \subseteq P$ open, such that $u(U) \subseteq \mathsf{S}$; and S is *good over P in degree k* if $x \mapsto \Lambda^k_\pi u(x)$ is analytic on U for every S-family u over P. For a linear subspace $\mathsf{V} \subseteq \mathcal{L}(E)$ and $\varepsilon > 0$ put $\mathsf{V}(\varepsilon) = \{v \in \mathsf{V} : \|v\| < \varepsilon\}$ and $1 + \mathsf{V}(\varepsilon) = \{1 + v : v \in \mathsf{V}(\varepsilon)\}$, with $1 = \mathrm{id}_E$.

By Theorem 2.1, a bundle with transitions in a subgroup $\mathsf{H} \subseteq \mathrm{GL}(E)$ needs only H-families.

Theorem 11.19 (structure groups)**.** (1) *Let $\mathsf{H} \subseteq \mathrm{GL}(E)$ be a subgroup, $\mathsf{H}_0 \subseteq \mathsf{H}$ a subgroup that is open in H for the operator norm, and $\mathsf{V}_0 \subseteq \mathcal{L}(E)$ a linear subspace with $\mathsf{H}_0 \subseteq \mathsf{V}_0$ and $1 + \mathsf{V}_0(\varepsilon) \subseteq \mathsf{H}_0$ for some $\varepsilon > 0$. Then H is good over P in degree k if and only if V_0 is. Neither E nor K need be complete. If E is a Banach space, $\mathrm{GL}(E)$ is good over P in degree k if and only if $(P; E)$ is good in degree k.*

(2) *If E is a Banach space with a nonarchimedean norm, the group $\mathrm{Isom}(E)$ of surjective linear isometries of E is good over P in degree k if and only if $(P; E)$ is good in degree k.*

(3) *Let K be nonarchimedean and $E = c_0(\mathbb{N}, K)$ or $E = \ell^r(\mathbb{N}, K)$ with $1 \le r < \infty$. Let $\mathsf{V}_\mathrm{d} = \{\mu_a : a \in \ell^\infty(\mathbb{N}, K)\}$, let $\mathsf{H}_\mathrm{d} = \{\mu_a : a \in \ell^\infty(\mathbb{N}, K),\ \inf_t |a(t)| > 0\}$ be the diagonal group and $\mathsf{H}_\mathrm{m} = \{\sigma_\sharp \mu_a : \mu_a \in \mathsf{H}_\mathrm{d},\ \sigma$ a bijection of $\mathbb{N}\}$ the monomial group, where $\sigma_\sharp e_t = e_{\sigma(t)}$. Then H_m is good over P in degree k if and only if H_d is, and if and only if V_d is.*

(4) *Under* (L)*, the commutative group H_d of $E = c_0(\mathbb{N}, K)$ is not good over $P = c_0(\mathbb{N}, K)$ in degree k.*

Corollary C.7 gives isometric diagonal transitions with the same behavior over Laurent series fields.

Proposition 11.20 (the closed span does not decide)**.** *Assume* (L) *with K complete, and let κ' and ϑ be as in Lemma 7.1, with κ' identified with its image in K, so that the nonzero elements of κ' have absolute value* 1 *and every element of K is a convergent series $\sum_{n \ge n_0} \xi_n \vartheta^n$ with $\xi_n \in \kappa'$. Let $E = D = c_0(\mathbb{N}, K)$, let e_i^* be the ith coordinate functional, let $\mathsf{U} \subseteq \mathcal{L}(E)$ be the additive group of the operators u with $e_i^*(ue_j) \in \kappa'$ for all i, j, and let $\mathsf{H} = \{g(u) : u \in \mathsf{U}\} \subseteq \mathrm{GL}(D \oplus E)$, with g as in Proposition 10.1. Then H is a discrete subgroup, every H-family over every P is locally constant, and H is good over every P in every degree. But the closed linear span of H contains $g(\mu_x)$ for every $x \in c_0(\mathbb{N}, K)$, and $x \mapsto \Lambda^k_\pi g(\mu_x)$ is analytic at no point of $c_0(\mathbb{N}, K)$.*

The proofs are in Appendix O.

11.8. **Characteristic two.**

Proposition 11.21 (families in characteristic two)**.** *Let K be a nontrivially normed field of characteristic* 2*, not necessarily satisfying* (L)*, and let $k \ge 2$.*

(1) *Let E, P, $\varphi_E^\pm$ and $\varphi_P^\pm$ be as in Theorem 11.14. If $\sup_n n/(\varphi_P^\pm(n)^2 \varphi_E^\pm(n)^2) = \infty$, the conclusions of Theorem 11.14 hold.*

(2) *The conclusion of Corollary 11.15(2) holds for $P = c_0(\mathbb{N}, K)$ and E equal to $c_0(\mathbb{N}, K)$ or to $\ell^{r_2}(\mathbb{N}, K)$ with $2 < r_2 < \infty$, and for $P = \ell^{r_1}(\mathbb{N}, K)$ with $2 < r_1 < \infty$ and E equal to $c_0(\mathbb{N}, K)$ or to $\ell^{r_2}(\mathbb{N}, K)$ with $1/r_1 + 1/r_2 < 1/2$.*

(3) *For $E = c_0(\mathbb{N}, K)$ and for $E = \ell^r(\mathbb{N}, K)$ with $4 < r < \infty$, the maps $x \mapsto \Lambda^k_\pi D\psi(x)$ and $x \mapsto \Lambda^k_\pi D(\psi^{-1})(x)$ of Theorem 11.17(3) are analytic at no point of E^3, and E is not tangent-good in degree k.*

(4) *The commutative group* H_{d} *of Theorem 11.19(3), for* $E = c_0(\mathbb{N}, K)$*, is not good over* $P = c_0(\mathbb{N}, K)$ *in degree* k*.*

The proof is in Appendix O. Item (2) does not cover the pair $P = c_0(\mathbb{N}, K)$, $E = \ell^2(\mathbb{N}, K)$ of Corollary 11.15(2). Items (2) and (3) do not cover the spaces with an equivalent nonarchimedean norm in Corollary 11.15(2) and in the last sentence of Theorem 11.17(3), whose proofs use Lemma 7.17, which assumes (L).

12. Questions

The questions below concern $\operatorname{char} K = p > 0$ and $k \geq p$. For $1 < r < \infty$ and $1 \leq b \leq \infty$, the symmetric sequence space $\ell^{r,b}$ (Definition 7.10) is the closure of c_{00} in the space of all $x \in \ell^\infty(\mathbb{N}, K)$ with $\sup_n \|x1_{[n]}\|_{r,b} < \infty$, where on c_{00} $\|x\|_{r,b} = c_{r,b}\big(\sum_{n\geq 1} n^{b/r-1}(x_n^{**})^b\big)^{1/b}$ for $b < \infty$, $\|x\|_{r,\infty} = c_{r,\infty} \sup_{n\geq 1} n^{1/r} x_n^{**}$, x_n^{**} is the average of the n largest values of $|x(t)|$, and $c_{r,b} > 0$ makes $\|e_1\|_{r,b} = 1$.

Conjecture 12.1 (product triangle conjecture)**.** Let κ and K be as in (L), and let E, D be symmetric sequence spaces in which c_{00} is dense. If (E, D) satisfies (PTB) (Definition 7.12), then (E, D) is good for Banach values in every degree.

The converse holds (Theorem 7.13). The conjecture holds when $D = E$ or $E = c_0$, since indicator functions of supports in (7.2) bound $\mathcal{P}_<$ on $D \otimes_\pi D$ (Theorem 8.1), and when D has an equivalent r-norm, $2 < r < \infty$, or nonarchimedean norm (Theorem 8.4). The same holds for spreading sequence spaces (Theorem 8.9).

Conjecture 12.2. Let κ be a finite field of characteristic p, $K = \kappa((X))$ with the absolute value of (L), and $1 < r < 2 \leq b \leq \infty$. Then $\mathcal{P}_<$ is unbounded on $\ell^{r,b} \otimes_\pi \ell^{r,b}$.

By Theorem 8.1, the conjecture for given r, b is equivalent to the failure of $(\mathrm{S}^p_{\ell^{r,b}})$, of $(\mathrm{R}^p_{\ell^{r,b}})$, or of $(\Gamma^p_{\ell^{r,b}})$, and to $(\ell^{r,b}, \ell^{r,b})$ and $(c_0, \ell^{r,b})$ being good for Banach values in no degree $k \geq p$.

Conjecture 12.3 (sequence lattices)**.** Assume (L) with K complete. Let E and D be sequence lattices, that is, each is a Banach space Y with $c_{00}(\mathbb{N}, K) \subseteq Y \subseteq \ell^\infty(\mathbb{N}, K)$ in which c_{00} is dense, with $\|\cdot\|_\infty \leq \|\cdot\|_Y$, $\sup_t \|e_t\|_Y < \infty$, and $\|x\|_Y \leq \|y\|_Y$ whenever $|x(t)| \leq |y(t)|$ for every t. Then (E, D) is good for Banach values in degree k if and only if it has a splitting decomposition of degree k (Definition 6.12).

Question 12.4. Is every pair that is good for Banach values in all degrees $\leq k$ obtained from pairs (E, D) with (S^n_E) for all $n \leq k$, pairs with (R^n_D) for all $n \leq k$, relative splittings through $\mathrm{S}^{\leq k}$-split or $\mathrm{R}^{\leq k}$-split maps, and parameter pieces, by finite decompositions (Theorem 6.13, Proposition 6.18), transfer (Proposition 6.14) and block assemblies (Proposition 6.27)?

Question 12.5. Let κ and K be as in (L). Do (S^k_Y), (R^k_Y) and (Γ^k_Y) coincide for every Banach space Y and every $k \geq p$?

They coincide for spreading, in particular symmetric, sequence spaces in which c_{00} is dense (Theorems 8.1 and 8.9), and (S^2_Y) is equivalent to (Γ^2_Y) for every normed Y (Proposition 5.14).

Question 12.6. Let W be a closed subspace of a Banach space Z. If W and Z/W satisfy (S^k), does Z? The same question for (R^k) and for polarizability in degree k.

Under (L), goodness for Banach values with a fixed target has no three-space property in the source slot, even with an ℓ^1 kernel (Theorem 6.31).

Question 12.7. Let $k \geq p$. If (E, D) is good for Banach values in degree k for every normed D, does (S^k_E) hold?

Under (L) the answer is yes for spreading, in particular symmetric, sequence spaces E in which c_{00} is dense (Theorems 8.1 and 8.9).

Question 12.8. Assume (L). Does every S^k-split operator factor through a normed space G with (S^k_G)? Does the inclusion $\ell^2(\mathbb{N}, K) \to c_0(\mathbb{N}, \widehat{K})$, which is S^n-split for every n (Corollary 5.10), factor through such a space?

Question 12.9. Let K be a field of Laurent series in X over an infinite field of characteristic p, with an X-adic absolute value, or a densely valued field of characteristic p.

(1) If p is odd, is $(c_0(\mathbb{N}, K), c_0(\mathbb{N}, K))$ good for Banach values in degree p?
(2) If $p = 2$, is $(\ell^2(\mathbb{N}, K), \ell^2(\mathbb{N}, K))$ good for Banach values in degree 2?

For $p = 2$, $(c_0(\mathbb{N}, K), c_0(\mathbb{N}, K))$ is good for Banach values in no degree $k \geq 2$, over every nontrivially normed field of characteristic 2 (Theorem 7.20(4)); for $E = \ell^2(\mathbb{N}, K)$, $\bar{\varphi}_E(n)^2 = n$ and (7.7) gives no contradiction. For odd p, a positive answer to (1) needs a zero-diagonal correction making the ordering lift bounded (Theorem 3.18).

Question 12.10. If a pair is good in degree p, is it good in every degree? The same question for goodness for Banach values. Is every S^p-split operator S^n-split for every n, and every R^p-split operator R^n-split for every n?

The least degree in which a pair fails to be good is a power of p (Corollary 6.3). For the conditions (S^p_Y) and (R^p_Y) of one normed space Y, Theorem 6.6 answers the third question.

Question 12.11. Let $\operatorname{char} K = p$.

(1) If a normed space P satisfies (U^p_P) (Definition 11.8), is P polarizable in degree p?
(2) Assume (L). Is $(\ell^{r_1}(\mathbb{N}, K); \ell^{r_2}(\mathbb{N}, K))$ good in degree k when $r_1 = r_2 = 2$, when $2 < r_1 < \infty$ and $r_2 = 2$, or when $2 < r_1, r_2 < \infty$ and $1/r_1 + 1/r_2 \geq 1/2$?
(3) Do Theorems 11.5 and 11.7 hold for one Banach value space F, with $x \mapsto A^k_{E,\widehat{D};F}(u(x))$ in place of $x \mapsto \Lambda^k_\pi u(x)$?

Polarizability in the degrees $n \leq k$ gives (U^k_P) (Theorem 11.9(2)); under (L), (U^k_P) gives polarizability in degree p when P is a symmetric sequence space in which $c_{00}(\mathbb{N}, K)$ is dense or has an equivalent nonarchimedean norm (Theorem 11.10). The cases in (2) are those left open by Corollary 11.15; for $r_1 = r_2 = r$ and K complete they ask whether $\ell^r(\mathbb{N}, K)$ is tangent-good in degree k (Theorem 11.17(2)).

Question 12.12. Let P, E and D be normed spaces. If every linear family $P \to \mathcal{L}(E, \widehat{D})$ is liftable in degree k, is $(P; E, D)$ good in degree k?

The answer is yes for $P = \mathcal{L}(E, \widehat{D})$, whose identity is a linear family (Lemma 3.4, Proposition 11.11(2)); for fixed-degree reflection, affine families suffice (Theorem 11.9(1)).

Question 12.13. If $(P; E, D)$ is good in degree k, is it good in every degree $j < k$?

The answer is yes when $E \cong E \oplus K$ and $D \cong D \oplus K$, and for goodness for all normed E and D (Corollary 11.6(2), (4)).

Question 12.14. Is there a linear family $\theta\colon \mathcal{T}(P) \to \mathcal{L}(E, \widehat{D})$ for which every class $\mathrm{ob}_{k,N}(\theta)$ vanishes but $\mathrm{ob}^\omega_k(\theta)$ does not vanish after shrinking?

By Theorem 11.7(1), $\mathrm{ob}_k^{\omega}(\theta)$ vanishes after shrinking exactly when the classes $\mathrm{ob}_{k,N}(\theta)$ vanish with extension costs growing at most geometrically in N. For general closed subspaces, Taylor coefficients can have lifts without a common radius (Corollary G.5).

Question 12.15. Let E, D and F be normed spaces. If (E, D) and (D, F) are good for Banach values in degree k, is the composite family $\theta_{D,F} \odot \theta_{E,D}$ of Proposition 6.26 liftable in degree k?

The composite family is liftable in degree k when (E, F) is good for Banach values in that degree (Proposition 6.26(2)).

Question 12.16. Let κ and K be as in (L), $1 \le q < 2$ and $0 < \alpha < 1 - q/2$. Is $(E_{\alpha,q}, E_{\alpha,q})$ (Example 8.10) good for Banach values in some degree $k \ge p$?

By Theorem 8.9, this holds if and only if $\mathcal{P}_<$ is bounded on $E_{\alpha,q} \otimes_\pi E_{\alpha,q}$, and then the pair is good for Banach values in every degree. The necessary conditions of Theorem 8.1 hold, and Theorem 5.6 does not apply to the unit vectors in their natural order (Example 8.10(2),(4)).
Targets, bases and domains. An equivalent nonarchimedean, spherically complete norm makes a Banach space F a universal target (Section 7.2, Theorem 5.2), but the given norm of F need not be spherically complete (Remark I.3). Over complete, not spherically complete fields universal targets have zero dual (Proposition 7.5); we do not know whether that suffices, nor whether over densely valued, spherically complete fields every nonarchimedean Banach space is one. We do not know which P satisfy (U_P^k) (Question 12.11(1)); for Banach P over complete fields universal analytic reflection suffices (Theorem G.3) but, over nonarchimedean fields, is not necessary (Theorem 11.9(4)). We do not know whether compatibility with all analytic coordinate changes forces the ambient and intrinsic analytic forms to agree; one product structure does not (Proposition 10.3). Over the t-adic field $\mathbb{F}_p(t)$ a largest full analytic domain need not exist (Theorem H.4); we do not know whether this happens over complete fields. Under (L) with K complete, $\mathrm{Alt}^k(TM; K)$ is analytic on every analytic manifold M (Corollary 5.3); we do not know this over incomplete fields such as $\kappa(X)$, nor for tangent-valued forms on $\ell^r(\mathbb{N}, K)$, $4 < r < \infty$ (Theorem 11.17(3)). Nor do we know whether Bourbaki's α_k is a vector functor of class C^r over densely valued, spherically complete fields with $k! = 0$.
Ambient analyticity. Questions 14.35 and 14.42 state what is open on ambient analyticity: whether it is closed under composition in characteristic 0, or in characteristic p on ultrametric sources with a non-ultrametric final target; whether Proposition 14.18 applies to $c_0(\mathbb{N}, K)$ over densely valued K of odd characteristic; whether a source P with $\mathfrak{k}(P) < \infty$ (Definition 14.21) admits a divided-power analytic germ into a Banach space that is not ambient analytic; whether $c_0(\mathbb{N}, K)$ is tame (Definition 14.36); whether the sections ambient for $\mathrm{Mult}^k(-; K)$ are ambient for every functorial envelope; whether a section of $\mathrm{Alt}^k(TM; K)$ can be ambient analytic in one analytic chart and not in another; and whether $\mathrm{Alt}^k(M_1; M_0)$ has an analytic structure for which every analytic automorphism of (M_1, M_0) induces an analytic morphism. One more question is open. Let K be complete of characteristic p, not spherically complete, and let $\mathfrak{E}$ be a functorial envelope of $\mathrm{Alt}^k(-; K)$ (Definition 14.30) with $\mathrm{Alt}^k(E; K) \subseteq \mathfrak{E}(E) \subseteq \mathrm{Mult}^k(E; K)$, the inclusions being isometric and natural in E. We do not know whether $\mathfrak{E}(E)$ is always the range of a bounded linear projection of $\mathrm{Mult}^k(E; K)$ natural in E.

13. Formal verification

The Lean proofs are in [17]. Its file `challenge.md` is a ledger of 76 claims of this paper. Each claim has a challenge file, stating it in Lean with `sorry` as proof, and a solution file proving the same statement. For a repository identifier `ID` such as `Thm4_4`, `python3 scripts/check_claim.py ID` builds both files and checks that the proof uses only the axioms

`propext`, `Classical.choice` and `Quot.sound` and that the two statements agree when printed with all implicit arguments (`pp.all`). All 76 claims pass with Lean `4.34.0-rc2` and the Mathlib fork `Deicyde/mathlib4` at `2b73d98` (upstream `0383a80e64` plus the commit of [18]). The six statements of the comparator challenge in the same repository pass the comparator `https://github.com/leanprover/comparator` at revision `19e111e`.

Table 1 gives the identifiers of the 24 claims in Sections 2–10. The other 52 claims are the appendix statements A.1–A.3, B.2–B.13, B.15, C.1–C.4, C.6–C.8, D.1–D.9, D.11, E.1, E.2, F.1–F.5, G.1, G.3–G.5, H.3–H.7 and I.1–I.3; the identifier of each is formed from its type and number, for example `LemB_2` for Lemma B.2.

TABLE 1. Repository identifiers of the ledger claims in Sections 2–10

Statement	Identifier	Statement	Identifier
Theorem 2.1	`Thm2_1`	Theorem 5.12(2)	`Thm4_5b`
Proposition 2.2	`Prop2_2`	Corollary 5.15	`Cor4_6`
Theorem 2.3	`Thm3_1`	Theorem 7.2(1)	`Thm6_1a`
Lemma 2.4	`Lem3_2`	Theorem 7.2(2)	`Thm6_1b`
Proposition 2.5	`Prop3_3`	Corollary 7.3	`Cor6_2`
Proposition 3.1	`Prop5_1`	Theorem 7.4	`Thm6_3`
Theorem 3.2	`Thm5_2`	Proposition 7.5	`Sec9_zeroDual`
Proposition 5.1	`Prop4_1`	Proposition 10.1	`Prop6_4`
Theorem 5.2	`Thm4_2`	Corollary 10.2	`Prop6_4_bundle`
Corollary 5.3	`Cor4_3`	Proposition 10.3	`Prop6_5`
Theorem 5.11	`Thm4_4`	Theorem 10.7	`Thm7_2`
Theorem 5.12(1)	`Thm4_5a`	Proposition 10.8	`Prop7_3`

A numbered statement of this paper has a Lean proof of a corresponding statement exactly when it is one of the 24 claims of Table 1 or the 52 claims just listed. Excluded are the definitions, the conjectures and questions of Section 12, Remarks B.14, C.5 and D.10, Proposition 10.4, the coefficient bound on retracts in Theorem 5.12(1), and all other theorems, propositions, lemmas, corollaries and remarks of Sections 3–12 and 14 and Appendices J–O and Q. The excluded results have written proofs only, except Lemma M.9, which is cited from [20], and apart from the Lean declarations named below.

`challenge.md` and the docstrings of the challenge files list where a ledger statement differs from the text. The main differences: Theorem 10.7 and Proposition 10.8 are stated on open subsets of a normed space, since Mathlib has no differential forms on manifolds; the Lean statement of Proposition 2.2 concerns the joint action B and does not include the clause for $g = \mathrm{id}_F$; the Lean statement of Theorem 2.1 groups the variables by variance, uses the canonical trivializations and does not state independence of the trivializing cover; Corollary C.7 is stated for the explicit transition and section maps, and Corollary 5.15 for objects and morphisms without the functor laws.

In [17], `alternatingMapAction` is the joint action B of Section 2, and the alternating bundle construction rests on the family-preservation premise: for $n \in \mathbb{N} \cup \{\infty, \omega\}$, a C^n manifold M, normed spaces E, D, F, F' and $B\colon \mathcal{L}(E,D) \times \mathcal{L}(F,F') \to \mathcal{L}(\mathrm{Alt}^k(D;F), \mathrm{Alt}^k(E;F'))$,

$$U \subseteq M \text{ open}, \quad \gamma\colon M \to \mathcal{L}(E,D) \times \mathcal{L}(F,F'), \quad \gamma|_U \text{ is } C^n \quad \Longrightarrow \quad (B \circ \gamma)|_U \text{ is } C^n. \tag{13.1}$$

The predicate `PreservesAlternatingFamilies` states (13.1) and typechecks against [17]. The declarations

```
contMDiffOn_continuousAlternatingMapCoordChange_of_family
alternatingVectorPrebundle_isContMDiff_of_family
```

`contMDiffVectorBundle_alternating_of_family`
`contMDiffVectorBundle_alternating_of_finiteCoordinates`
`alternatingBundleHom_of_family`

of [17] index alternating maps by `Fin k`. The first three assume (13.1) with $D = E$ and $F' = F$ and show that the alternating coordinate changes, prebundle and bundle are C^n. The fourth shows that over an analytic manifold modeled on a space P with a continuous linear isomorphism $P \to K^d$ the alternating bundle of two analytic bundles is analytic, with the premise from Theorem 5.11. The fifth assumes (13.1) and turns C^n bundle morphisms into a C^n morphism of the induced alternating bundles when these are C^n.

14. Ambient analyticity: a structure, not a class of maps

Ambient analyticity is not a usable class of analytic maps between normed spaces: over every field of characteristic p it is not closed under composition, and for K complete it has no inverse function theorem, depends on the target chart, and gives no transitive compatibility of charts; in characteristic 0 this is open. On tame pairs (Section 14.6), such as ultrametric spaces or $\ell^1(J)$, it has composition and inverse functions, but there it coincides with divided-power analyticity, so this is no separate argument for it. The mechanism behind the failure of Theorem 14.17 is that enlarging the target carries maps into the space, not analytic maps out of it. Within analytic maps on normed spaces, the formulas of n° 7.8.1–7.8.4 of [3] use a structure instead: the fixed envelope Mult^k, on which the transitions act analytically, preserving the alternating forms. In every characteristic this gives those formulas on ambient sections of filtered analytic bundles, and, for K complete and scalar values, $\mathrm{Mult}^k(-;K)$ is initial among the functorial envelopes of $\mathrm{Alt}^k(-;K)$ with a multilinear action. In general it does not give Bourbaki's vector functor in the analytic class, a direct factor, or an analytic induced atlas.

Section 14.1 holds over every nontrivially normed field. From Section 14.2 on, K is nonarchimedean. Norms need not be complete or ultrametric unless this is said.

14.1. **Fixed envelopes.**

Definition 14.1 (filtered analytic vector bundle)**.** A *filtered space* $\mathbb{V} = (\tilde V; V_1, V_2)$ is a normed space $\tilde V$ with closed subspaces $V_2 \subseteq V_1 \subseteq \tilde V$. Its *fiber* is $V = V_1/V_2$ with the quotient norm, and $\mathrm{qt}\colon V_1 \to V$ is the quotient map. For filtered spaces $\mathbb{V}$, $\mathbb{V}'$ put

$$\mathcal{L}(\mathbb{V},\mathbb{V}') = \{T \in \mathcal{L}(\tilde V,\tilde V') :\ TV_1 \subseteq V_1',\ TV_2 \subseteq V_2'\},$$

a closed subspace of $\mathcal{L}(\tilde V,\tilde V')$, and let $\rho\colon \mathcal{L}(\mathbb{V},\mathbb{V}') \to \mathcal{L}(V,V')$ be the induced map on fibers, $\rho(T)\,\mathrm{qt}(v) = \mathrm{qt}'(Tv)$ for $v \in V_1$. It is linear and multiplicative, and $\|\rho(T)\| \le \|T\|$.

A *filtered analytic vector bundle* over an analytic manifold M is an analytic vector bundle $\widetilde{\mathcal{B}}$ with typical fiber $\tilde V$, closed subspaces $V_2 \subseteq V_1 \subseteq \tilde V$, and an analytic atlas of $\widetilde{\mathcal{B}}$ whose transitions Θ_{ab}, which convert coordinates of chart b to coordinates of chart a, take values in $\mathcal{L}(\mathbb{V},\mathbb{V})$. In this section a *chart* of $\widetilde{\mathcal{B}}$ is a chart of this atlas. The inverse $\Theta_{\mathsf{ba}}(x)$ of $\Theta_{\mathsf{ab}}(x)$ also preserves the filtration, so $\Theta_{\mathsf{ab}}(x)$ maps V_1 onto V_1 and V_2 onto V_2. Hence the subsets $\mathcal{B}_1, \mathcal{B}_2 \subseteq \widetilde{\mathcal{B}}$ given in each chart by V_1, V_2 are well defined and fiberwise closed, and $\mathcal{B} = \mathcal{B}_1/\mathcal{B}_2$ is a bundle of vector spaces with fiber V and transitions $\rho(\Theta_{\mathsf{ab}})$. The bundle is of *sub type* if $V_2 = 0$.

Definition 14.2 (ambient sections, operations)**.** A section of $\mathcal{B}$ over an open subset of M is *ambient* if, near each point and in some chart, its local expression is $\mathrm{qt}\circ\varsigma$, with ς analytic into $\tilde V$ and with values in V_1. For sub type these are the analytic sections of $\widetilde{\mathcal{B}}$ lying in $\mathcal{B}_1$.

For filtered spaces $\mathbb{V}^{(1)},\dots,\mathbb{V}^{(d)},\mathbb{V}'$, with quotient maps $\mathrm{qt}^{(i)}$ and qt', let $\mathcal{L}(\mathbb{V}^{(1)},\dots,\mathbb{V}^{(d)};\mathbb{V}')$ be the space of bounded d-linear maps $T\colon \tilde V^{(1)} \times \cdots \times \tilde V^{(d)} \to \tilde V'$ with $T(V_1^{(1)} \times \cdots \times$

$V_1^{(d)}) \subseteq V_1'$ and with $T(v_1,\dots,v_d) \in V_2'$ whenever all $v_i \in V_1^{(i)}$ and some $v_i \in V_2^{(i)}$. Put $\rho_d(T)(\mathrm{qt}^{(1)}\, v_1,\dots,\mathrm{qt}^{(d)}\, v_d) = \mathrm{qt}'\, T(v_1,\dots,v_d)$ for $v_i \in V_1^{(i)}$. A *d-linear operation* from $\mathcal{B}^{(1)},\dots,\mathcal{B}^{(d)}$ to $\mathcal{B}'$ is a map $\mathcal{B}^{(1)} \times_M \cdots \times_M \mathcal{B}^{(d)} \to \mathcal{B}'$ over id_M, d-linear on fibers, whose local expressions are locally $\rho_d(\Theta(x))$, with Θ analytic into the bounded d-linear maps of the $\tilde{V}^{(i)}$ into $\tilde{V}'$ and with values in $\mathcal{L}(\mathbb{V}^{(1)},\dots,\mathbb{V}^{(d)};\mathbb{V}')$.

Proposition 14.3. *Being ambient does not depend on the chart. Ambient sections form a module over the analytic scalar functions, and a d-linear operation carries ambient sections to an ambient section. If the filtered space is split, that is, if* qt *has a bounded linear section and V_1 has a bounded linear retraction in $\tilde{V}$, then the transitions $\rho(\Theta_{\mathsf{ab}})$ are analytic, and the analytic sections of $\mathcal{B}$ are exactly its ambient sections.*

The proof is in Appendix Q.

The main example is this. Let M_1, M_0 be analytic vector bundles over M with normed fibers E, F and transition families $\mathbf{g}^1_{\mathsf{ab}}, \mathbf{g}^0_{\mathsf{ab}}$ over a common trivializing cover, and fix $k \geq 1$. Let $\mathfrak{m}_k$ be the action on multilinear maps, $\mathfrak{m}_k(\mathbf{f}_1, \mathbf{f}_0)\mathsf{m} = \mathbf{f}_0 \circ \mathsf{m} \circ \mathbf{f}_1^{\times k}$, and let α_k be Bourbaki's functor, with values $\mathrm{Alt}^k(E;F)$. Let $\mathcal{L}(M_1,\dots,M_1;M_0)$ be the bundle with fiber $\mathrm{Mult}^k(E;F)$, as in [3, n° 7.8.1], and with transitions

$$\Theta_{\mathsf{ab}}(x) = \mathfrak{m}_k\big(\mathbf{g}^1_{\mathsf{ba}}(x), \mathbf{g}^0_{\mathsf{ab}}(x)\big).$$

They are jointly analytic, as a bounded multilinear evaluation of analytic operator families (Proposition 2.2 and its proof, which needs no retraction), so this is an analytic bundle. They preserve $\mathrm{Alt}^k(E;F)$, and the inverse of $\Theta_{\mathsf{ab}}(x)$ is $\Theta_{\mathsf{ba}}(x)$. Analytic vector charts of M_1, M_0 induce charts of $\mathcal{L}(M_1,\dots,M_1;M_0)$ and of its fiberwise subset $\mathrm{Alt}^k(M_1;M_0)$, with fiber $\mathrm{Alt}^k(E;F)$; these are the *induced charts*. In the same way $\Lambda^k_\pi E$ is the quotient type: $\tilde{V} = V_1 = \otimes^k_\pi E$, with V_2 the closed kernel of $\wedge^k_E$.

Theorem 14.4 (n° 7.8.1–7.8.4 on ambient sections)**.** *Let K be a nontrivially normed field of any characteristic, complete or not. Let $k \geq 1$, and in items* (3)(b),(c) *also $k = 0$, with $\mathrm{Alt}^0(M_1;\mathcal{Q}) = \mathcal{Q}$ for a bundle $\mathcal{Q}$, filtered by $V_1 = \tilde{V}$ and $V_2 = 0$; this k is Bourbaki's d. Let M be an analytic manifold and M_1, M_0 analytic vector bundles over M with normed fibers.*

1. *α_k is a closed subfunctor of the analytic functor $\mathfrak{m}_k$ on normed spaces, whose hom maps are continuous polynomials with a representative of norm at most* 1.
2. *$\mathrm{Alt}^k(M_1;M_0)$ is a filtered analytic vector bundle of sub type inside $\mathcal{L}(M_1,\dots,M_1;M_0)$: it is fiberwise closed and invariant under the analytic transitions. Its atlas is the one induced by all analytic vector charts of M_1, M_0, so the bundle and its ambient sections do not depend on a trivializing cover. Its underlying bundle is the C^∞ bundle of Theorem 2.1.*
3. *On ambient sections, by the formulas of* [3, n° 7.8] *in* (a)–(d)*, and by* [3, n° 7.6.3] *in* (e)*:*
   (a) *$\omega(\mathsf{s}_1,\dots,\mathsf{s}_k)$ is an analytic section of M_0 for ambient ω and analytic sections s_i of M_1 (n° 7.8.1);*
   (b) *for an* analytic pairing *$\varphi\colon \mathcal{Q} \times_M \mathcal{Q}' \to \mathcal{Q}''$ of analytic bundles, that is, a map bilinear on fibers whose local expressions $x \mapsto \varphi_x$ are analytic into the bounded bilinear maps, and for a degree $k' \geq 0$, the exterior product $\wedge_\varphi$ is a bilinear operation from $\mathrm{Alt}^k(M_1;\mathcal{Q})$ and $\mathrm{Alt}^{k'}(M_1;\mathcal{Q}')$ to $\mathrm{Alt}^{k+k'}(M_1;\mathcal{Q}'')$ (n° 7.8.2);*
   (c) *for an analytic bundle A of associative commutative unital algebras, whose multiplication is an analytic pairing, the ambient sections of the bundles $\mathrm{Alt}^k(M_1;\mathsf{A})$,*

$k \geq 0$, form over each open U a graded algebra $\Omega^*(U)$ over the analytic scalars, associative and anticommutative (n° 7.8.3);

(d) *for $k \geq 1$ the interior product i is a bilinear operation from M_1 and $\mathrm{Alt}^k(M_1; \mathsf{A})$ to $\mathrm{Alt}^{k-1}(M_1; \mathsf{A})$ (n° 7.8.4), and Bourbaki's formulas (1)–(7) of n° 7.8.2–7.8.4 hold;*

(e) *for analytic $\gamma\colon M' \to M$, the bundle $\mathrm{Alt}^k(\gamma^* M_1; \gamma^* M_0)$ is the inverse image of the filtered bundle $\mathrm{Alt}^k(M_1; M_0)$, with transitions $\Theta_{\mathsf{ab}} \circ \gamma$, and $\omega \circ \gamma$ is an ambient section of it for ambient ω (n° 7.6.3).*

(4) *For $M_1 = TM$, M_0 the trivial line bundle $M \times K$, and K complete, the ambient sections of $\mathrm{Alt}^k(M_1; M_0)$ are the forms of $\Omega^k_{\mathrm{amb}}(M)$ of Section 10.3.*

The proof is in Appendix Q.

What lifts to Mult^k are the operations: evaluation, the shuffle formula for φ and insertion of a vector act on multilinear maps, with analytic local expressions. Not every identity lifts: $i(\mathsf{s})^2 = 0$ holds only on Alt^k, as an identity in each fiber. The identities of item (3) are claimed only on Alt^k, which is all that item needs. Theorem 14.4 concerns n° 7.8.1–7.8.4 and ambient sections. It does not say that α_k is a vector functor of class C^r in the sense of [3, n° 7.6.1], and n° 7.8.5 is not treated. Inverse images come from n° 7.6.3, since n° 7.8 has no pullback of forms.

Theorem 14.5 (what this does not give in general)**.** *In items (1)–(4) let K have characteristic $p > 0$ and let $k \geq p$.*

(1) *There are M, M_1, M_0 for which the charts of $\mathrm{Alt}^k(M_1; M_0)$ induced by analytic vector charts of M_1, M_0 are not analytically compatible (Corollary 10.2 with Theorem 7.2, and Proposition 10.4; for scalar values Proposition 10.3, K complete and not spherically complete).*

(2) *There are E, F such that $\mathrm{Alt}^k(E; F)$ has no bounded linear retraction in $\mathrm{Mult}^k(E; F)$. So $\mathrm{Alt}^k(M_1; M_0)$ need not be locally a direct factor of $\mathcal{L}(M_1, \dots, M_1; M_0)$.*

(3) *If K is complete and not spherically complete, ambient sections need not be analytic in induced charts (Proposition 10.3(2)).*

(4) *In the situation of Corollary 10.2, no analytic vector bundle structure on $\mathrm{Alt}^k(M \times H; M \times F)$ contains the induced product chart and makes the map induced by the shear automorphism an analytic morphism. By Proposition 2.5, $A^k_{E,D;F}$ is then analytic nowhere, and a structure for which all analytic automorphisms of $M \times H$, over all open subsets of M, induce analytic morphisms contains no chart induced by an analytic vector chart of $M \times H$ and the product chart of $M \times F$.*

Now let K be any nontrivially normed field and $k \geq 1$, with fibers E, F of M_1, M_0.

(5) *If α_k is analytic on the hom spaces $\mathcal{L}(E, E) \times \mathcal{L}(F, F)$ of the fibers (for instance if $k! \neq 0$, Corollary 7.3; or in the setting of Corollary 10.5 with E of countable type), then $\mathrm{Alt}^k(M_1; M_0)$ is an analytic vector bundle and its analytic sections are ambient. If moreover $j_{E,F}$ has a bounded linear retraction (if $k! \neq 0$, or if F is as in Theorem 5.2), then ambient sections are exactly the analytic ones.*

The proof is in Appendix Q.

Remark 14.6. In condition (b) of [3, n° 7.6.1] only analyticity fails: with Fréchet C^∞ in place of analytic the condition holds (Section 2). The inputs M_1, M_0 must be analytic, and the induced charts do not make the output $\mathrm{Alt}^k(M_1; M_0)$ an analytic bundle in general, so the construction is not iterated here. A single induced chart can still be an atlas, as for the trivial bundle of Corollary 10.2. Whether there is an analytic structure for which every analytic automorphism of (M_1, M_0) induces an analytic morphism is asked in Question 14.42.

14.2. **Maps that become analytic after an enlargement.** Section 14.1 used one enlargement, chosen with the fiber, and the transitions act on it. Now the enlargement is allowed to depend on the map. From here on K is nonarchimedean, so K is infinite, and $\varpi \in K$ is fixed with $0 < |\varpi| < 1$.

Notation. A normed space is *ultrametric* if its norm is nonarchimedean. P, P' are source spaces and W, Y are targets, all normed. For $n \geq 0$, $P^{\otimes n}$ is the algebraic tensor power and π is its projective seminorm; for $n = 0$ we read $P^{\otimes 0} = K$, $\pi = |\cdot|$, $h^{\otimes 0} = 1$, and $\|h\|^0 = 1$ also at $h = 0$. $\Gamma^n(P) \subseteq P^{\otimes n}$ is the span of the powers $h^{\otimes n}$, $h \in P$; it is the space of symmetric tensors, by the opening paragraph of Section L.3. $\Gamma^n_\pi(P)$ is $\Gamma^n(P)$ with the restriction of π, and on $\Gamma^n(P)$ we also use the seminorm $\|t\|_\Gamma = \inf\{\sum_i |c_i| \|h_i\|^n : \ t = \sum_i c_i h_i^{\otimes n}\}$, so that $\pi \leq \|\cdot\|_\Gamma$. For $h_1, \dots, h_r \in P$ and integers $\alpha_i \geq 0$ with $|\alpha| = \alpha_1 + \cdots + \alpha_r = n$, the *orbit sum* $O_\alpha(h_1, \dots, h_r)$ is the sum of the tensors $h_{i_1} \otimes \cdots \otimes h_{i_n}$ over the tuples $(i_1, \dots, i_n)$ in which each i occurs α_i times, so that, with $\lambda^\alpha = \prod_i \lambda_i^{\alpha_i}$,

$$\Big(\sum_i \lambda_i h_i\Big)^{\otimes n} = \sum_{|\alpha| = n} \lambda^\alpha O_\alpha(h_1, \dots, h_r) \qquad (\lambda \in K^r), \tag{14.1}$$

and $O_\alpha(h_1, \dots, h_r) \in \Gamma^n(P)$, since each coefficient of a polynomial with values in a subspace lies in that subspace, by interpolation.

Definition 14.7 (some j)**.** Let $U \subseteq P$ be open, $f\colon U \to W$, $x_0 \in U$, $R > 0$ with $B(x_0, R) \subseteq U$, and $\mathfrak{b} \geq 0$. An *ambient expansion* of f at x_0 with radius R and bound $\mathfrak{b}$ is a linear isometry $j\colon W \to Z$ into a normed space and bounded n-linear maps $b_n\colon P^n \to Z$, $n \geq 0$, with $\|b_n\| \leq \mathfrak{b} R^{-n}$, with $b_n(h, \dots, h) \in j(W)$ for all $h \in P$, and with

$$jf(x_0 + h) = \sum_{n \geq 0} b_n(h, \dots, h) \qquad (\|h\| < R). \tag{14.2}$$

The map f is *ambient analytic*, or *AA*, at x_0 with radius R and bound $\mathfrak{b}$ if it has such an expansion. It is AA at x_0 if this holds for some $R, \mathfrak{b}$, and AA on U if it is AA at every point of U. For $Z = W$ and $j = \mathrm{id}_W$ we say *analytic* at x_0 with radius R and bound $\mathfrak{b}$, and analytic with radius R if this holds for some $\mathfrak{b}$.

This is a condition on a map into the one space W, not on a map up to changes of chart in W. The term extends the ambient expansion of Section 2.3, where $j(W)$ is closed by assumption; here it need not be. If $j(W)$ is closed in Z, the condition on diagonals is automatic, by Step 1 of the proof of Theorem 2.3. Replacing Z by its completion keeps all the conditions, so Z may be taken Banach. If W is Banach, then $j(W)$ is complete, hence closed, and the definition is (2.1) with its radius and bound named. For K complete and some incomplete W the condition on diagonals cannot be dropped, see [II].

Definition 14.8 (laws, divided-power germs)**.** Let $n \geq 0$.

(1) A map $a\colon P \to W$ is a *bounded polynomial law of degree* n, or a *law*, if, for all $h_1, \dots, h_r \in P$, the map $\lambda \mapsto a(\lambda_1 h_1 + \cdots + \lambda_r h_r)$ on K^r is a homogeneous polynomial of degree n with coefficients in W, and $\|a\| = \sup_{h \neq 0} \|a(h)\| / \|h\|^n$ is finite; for $P = 0$ read $\|a\| = \|a(0)\|$. For $|\alpha| = n$, $a^{[\alpha]}(h_1, \dots, h_r) \in W$ is the coefficient of λ^α.

(2) A map $a\colon P \to W$ is *ambient of degree* n if $ja(h) = \beta(h, \dots, h)$ for all $h \in P$, for a linear isometry j of W into a normed space Z and a bounded n-linear map $\beta\colon P^n \to Z$.

(3) For $U, x_0, R, \mathfrak{b}$ as in Definition 14.7, a map $f\colon U \to W$ is *divided-power analytic*, or *DP*, at x_0 with radius R and bound $\mathfrak{b}$ if $f(x_0 + h) = \sum_n a_n(h)$ for $\|h\| < R$, with laws $a_n\colon P \to W$ of degree n and $\|a_n\| \leq \mathfrak{b} R^{-n}$. It is DP at x_0 with radius R if this holds for some $\mathfrak{b}$, DP at x_0 if it holds for some $R, \mathfrak{b}$, and DP on U if it is DP at every point of

U. The a_n are the *coefficient laws* of f at x_0. By Lemma A.1 on the lines $\lambda \mapsto x_0 + \lambda h$, they depend only on f.

A sequel [II] studies this class. By Lemma Q.1, a law a of degree n has a unique linearization: a linear map $L_a \colon \Gamma^n(P) \to W$ with $L_a(h^{\otimes n}) = a(h)$. Its *ambient constant* is

$$\operatorname{amb}(a) = \sup\{\|L_a(t)\|/\pi(t) : \ t \in \Gamma^n(P)\} \in [0, \infty],$$

with $c/0 = \infty$ for $c > 0$ and $0/0 = 0$. By the same lemma, a is ambient of degree n if and only if $\operatorname{amb}(a) < \infty$. Also $\|a\| \le \operatorname{amb}(a)$, since $\|L_a(h^{\otimes n})\| \le \operatorname{amb}(a)\pi(h^{\otimes n}) \le \operatorname{amb}(a)\|h\|^n$.

Proposition 14.9 (no choice of enlargement)**.**

(1) *A map $f \colon U \to W$ is AA at x_0 with radius R and bound $\mathfrak{b}$ if and only if it is DP at x_0 with radius R and its coefficient laws satisfy $\operatorname{amb}(a_n) \le \mathfrak{b}R^{-n}$ for all n. It is then DP with bound $\mathfrak{b}$, since $\|a_n\| \le \operatorname{amb}(a_n)$, and $ja_n(h) = b_n(h, \dots, h)$ for every ambient expansion with this radius.*

(2) *A law a of degree n such that $j \circ a$ is analytic at one point, for a linear isometry j of W into a normed space, is ambient of degree n.*

The proof is in Appendix Q.

Remark 14.10. The sections of Definition 14.2 are the case of one fixed j. For a filtered space $\mathbb{V}$ let $\iota_{\mathbb{V}} \colon V \to \tilde{V}/V_2$ be the inclusion, an isometry whose range is closed, as its preimage in $\tilde{V}$ is V_1. The map $S \mapsto \iota_{\mathbb{V}} \circ S \circ \mathrm{qt}$ is a linear isometry J of $\mathcal{L}(V)$ into $\mathcal{L}(V_1, \tilde{V}/V_2)$, since each $x \in V$ has lifts $v \in V_1$ with $\|v\|$ arbitrarily close to $\|x\|$. Its range is closed: it consists of the operators that vanish on V_2 and have values in $\iota_{\mathbb{V}}(V)$. For $T \in \mathcal{L}(\mathbb{V}, \mathbb{V})$, $\mathsf{J}\rho(T)$ is $T|_{V_1}$ followed by the quotient map $\tilde{V} \to \tilde{V}/V_2$, a bounded linear function of $T \in \mathcal{L}(\tilde{V})$. So the transitions $\rho(\Theta_{\mathsf{ab}})$ are AA at every point through the one enlargement J, and each ambient section is AA through the one isometry $\iota_{\mathbb{V}}$; for sub type, ambient sections are exactly the sections that are analytic through $\iota_{\mathbb{V}}$. What Section 14.1 uses is only this case, precomposition with analytic maps, and bounded multilinear operations on the fixed $\tilde{V}$: no step composes two AA maps.

Theorem 14.11 (one enlargement)**.** *Let $(f_i)_{i \in I}$ be a set of germs into one normed space W, with arbitrary sources, each f_i AA at a point x_i with radius R_i and bound $\mathfrak{b}_i$. There is one linear isometry $j \colon W \to Z$ into a Banach space through which every f_i has an ambient expansion with radius R_i and bound $\mathfrak{b}_i$. In particular, a map that is AA on U has one j for all points of U.*

The proof is in Appendix Q.

Proposition 14.12 (what is kept)**.** *Let $U \subseteq P$ be open and $x_0 \in U$.*

Target descent. *Let $f_Y \colon U \to Y$ be DP at x_0, with $j_Y \circ f_Y$ AA at x_0 with radius R' and bound $\mathfrak{b}'$ for a linear isometry $j_Y \colon Y \to Z_Y$. Then f_Y is AA at x_0 with radius R' and bound $\mathfrak{b}'$.*

Now let $f \colon U \to W$ be AA at x_0 with radius R and bound $\mathfrak{b}$.

(1) *The following are AA at the relevant point:*
   (a) (linear postcomposition) *$\Upsilon \circ f$, for bounded linear $\Upsilon \colon W \to Y$;*
   (b) (analytic precomposition) *$f \circ \gamma$ at x_0', for γ from an open subset of P' into U, analytic at x_0', with $\gamma(x_0') = x_0$;*
   (c) (pairing, sums) *$(f, f') \colon U \to W \oplus W'$, and $f + f'$ if $W' = W$, for $f' \colon U \to W'$ AA at x_0;*
   (d) (scalar factors) *ζf, for $\zeta \colon U \to K$ analytic at x_0;*
   (e) (germs) *every map that agrees with f near x_0.*

(2) *Let W be Banach. Then f is AA at every x_1 with $\varrho = \|x_1 - x_0\| < R$, with radius $R - \varrho$ and bound $\mathfrak{b}R/(R-\varrho)$. If f is AA on U, then it is C^∞, and every $D^\ell f\colon U \to \mathrm{Mult}^\ell(P;W)$ is AA on U.*

The proof is in Appendix Q.

Remark 14.13. For a Banach space W, Definition 14.7 says: f is AA with radius R and bound $\mathfrak{b}$ if and only if $j \circ f$ is analytic with this radius and bound for one linear isometry j of W into a Banach space. So AA is reached from analyticity by one target descent, and this is close to the definition. Analytic maps do not have the descent, for K of characteristic $p > 0$, $k \geq p$ and the spaces E, F of Theorem 7.2(1): following $A^k_{E,E;F}$ by $j_{E,F} \circ -$ makes it analytic (Proposition 2.2), and $A^k_{E,E;F}$ is analytic at no point. For K complete of any characteristic, see Theorem 14.33(1).

Remark 14.14. This is the link with the title of the paper. Let W be Banach and f AA at x_0 with radius R, bound $\mathfrak{b} > 0$ and coefficient laws a_n. Let Z_f be the completion of the pushout (Q.7) of f, with its isometry j_f (classes $[v, y]$, $j_f(v) = [v, 0]$, $y = (y_n)_n$, and ι_n the inclusion of the n-th summand, as there). Then f is analytic with radius R if and only if j_f has a bounded linear retraction.

Step 1 (a retraction gives an expansion). In Z_f we have $[L_{a_n}(t), 0] = [0, \iota_n t]$ for $t \in \Gamma^n(P)$, and $\|[0, \iota_n y]\| \leq \mathfrak{b}R^{-n}\pi(y)$. If r is a retraction, the maps $q_n(h_1, \dots, h_n) = \mathsf{r}[0, \iota_n(h_1 \otimes \cdots \otimes h_n)]$ have norm at most $\|\mathsf{r}\|\mathfrak{b}R^{-n}$ and diagonals a_n, so they expand f in W.

Step 2 (an expansion gives a retraction). Let $q_n\colon P^n \to W$ expand f with radius R and $\|q_n\| \leq \mathfrak{b}'R^{-n}$. By Lemma A.1 on lines, $q_n(h, \dots, h) = a_n(h)$, so the linearization $q_n^\otimes$ of q_n on $P^{\otimes n}$ agrees with L_{a_n} on powers, hence on $\Gamma^n(P)$. So $\mathsf{r}[v, y] = v + \sum_n q_n^\otimes(y_n)$ is well defined on the pushout, bounded by $\max(1, \mathfrak{b}'/\mathfrak{b})$, and extends to a retraction of j_f on Z_f.

At fixed radius this is formal: a map out of a subspace extends if and only if its pushout splits. A retraction of $j_{E,F}$ gives this for $A^k_{E,D;F}$ (Proposition 5.1). For one germ at radius R the exact condition is that its own j_f has a retraction. As f is also AA with every radius $R' \leq R$ and bound $\mathfrak{b}$, the germ is analytic at x_0 if and only if this holds at some radius $R' \leq R$.

14.3. **Composition fails.** The failure comes from one constant: the average cost of replacing elementary tensors by symmetric tensors.

Definition 14.15 (averaged polarization constant, bad space)**.** Let $X \neq 0$ be a normed space and $n \geq 1$. A *family* is a finite set I and elementary tensors $y = (y_l)_{l \in I}$ in $X^{\otimes n}$, given with their factors, $y_l = y_l^1 \otimes \cdots \otimes y_l^n$, such that $\|y_l\|_{\mathrm{e}} = \|y_l^1\| \cdots \|y_l^n\|$ is nonzero for some l. Let E_0 be the span of the y_l, and $\mathcal{P}(y)$ the set of linear maps $\Psi\colon E_0 \to \Gamma^n(X)$ that fix $E_0 \cap \Gamma^n(X)$. Put

$$\mathfrak{l}(y) = \inf_{\Psi \in \mathcal{P}(y)} \frac{\sum_l \pi(\Psi y_l)}{\sum_l \|y_l\|_{\mathrm{e}}}, \qquad \mathfrak{l}_n(X) = \sup_y \mathfrak{l}(y) \in [0, \infty]. \tag{14.3}$$

Let $\bar{E} = E_0/(E_0 \cap \Gamma^n(X))$, with $\bar{v}$ the image of $v \in E_0$, and let $\mathcal{N} \subseteq \ell^1(I, K)$ be the space of the vectors $(\eta(\bar{y}_l))_{l \in I}$, for η a linear form on $\bar{E}$. It is closed: it is the annihilator of $\{\lambda \in K^I : \sum_l \lambda_l y_l \in \Gamma^n(X)\}$. The *universal space* of the family is $G(y) = \ell^1(I, K)/\mathcal{N}$ with the quotient norm. Let ε_l be the image of the unit vector at l, and $t(y) = \sum_l y_l \otimes \varepsilon_l$.

For a normed space G, let π also denote the projective seminorm of $X^{\otimes n} \otimes G$, and for $t \in \Gamma^n(X) \otimes G$ let

$$\mu(t) = \inf\Big\{\sum_q \pi(s_q)\|w_q\| : \ t = \sum_q s_q \otimes w_q, \ s_q \in \Gamma^n(X), \ w_q \in G\Big\}, \tag{14.4}$$

the projective seminorm of $\Gamma^n_\pi(X)\otimes G$. Always $\pi\le\mu$.

The ℓ^1-sum of normed spaces B_i is the space of the families $(u_i)_i$, $u_i\in B_i$, with $\sum_i\|u_i\|<\infty$, and this sum as norm. Now let $\mathfrak{l}_n(X)=\infty$ in (14.3). A *bad space* is the ℓ^1-sum G of the $G(y^{(i)})$, $i\ge1$, for families with $\sup_i \mathfrak{l}(y^{(i)})=\infty$. Bounded forms separate the points of any ℓ^1-sum of spaces $G(y)$: all linear forms on K^I are bounded for $\ell^1(I,K)$, those vanishing on $\mathcal{N}$ separate the points of $G(y)$, and we compose them with the coordinate projections. So if π is a norm on $X^{\otimes n}$ and G is such a sum, then μ is a norm: for $t\ne0$ a bounded form η on G has $(1\otimes\eta)t\ne0$, and $\mu(t)\ge\pi(t)\ge\|\eta\|^{-1}\pi((1\otimes\eta)t)>0$. In that case, for G a bad space, put, with maximum norms on direct sums and μ as in (14.4), $H_X=X\oplus G$, $W_X=\Gamma^n_\pi(X)\oplus G$, $Y_X=(\Gamma^n(X)\otimes G,\mu)$, and

$$f_X\colon H_X\to W_X,\quad f_X(x,w)=(x^{\otimes n},w);\qquad g_X\colon W_X\to Y_X,\quad g_X(s,w)=s\otimes w. \tag{14.5}$$

Then g_X is the diagonal of the bounded bilinear map $\beta_X((s,w),(s',w'))=s\otimes w'$.

Definition 14.16 (atomic spaces)**.** Let $\operatorname{char}K=p$. For $m\ge1$, X_m is K^m with the quotient norm of $\ell^1(\mathbb{F}_p^m\setminus\{0\},K)$ under $e_{\mathsf{x}}\mapsto\mathsf{x}$:

$$\|h\|_{X_m}=\inf\Big\{\sum_{\mathsf{x}}|\lambda_{\mathsf{x}}|:\ h=\sum_{\mathsf{x}}\lambda_{\mathsf{x}}\mathsf{x}\Big\}.$$

Vectors of $\mathbb{F}_p^m$ have norm at most 1, and the coordinate forms of K^m are bounded by 1 on X_m. Let X^{at} be the ℓ^1-sum of the X_m, $m\ge1$. The coordinate forms separate the points of $(X^{\mathrm{at}})^{\otimes n}$, so π is a norm there. Let $I_m=(\mathbb{F}_p^m)^p$, and $y^{(m)}$ the family of all $y^1\otimes\cdots\otimes y^p$, $(y^1,\dots,y^p)\in I_m$. Let $G_m=G(y^{(m)})$; it is the same computed in X_m or in X^{at}, as $\Gamma^p(X^{\mathrm{at}})\cap X_m^{\otimes p}=\Gamma^p(X_m)$. Let G^{at} be the ℓ^1-sum of the G_m, and

$$H^{\mathrm{at}}=X^{\mathrm{at}}\oplus G^{\mathrm{at}},\qquad W^{\mathrm{at}}=\Gamma^p_\pi(X^{\mathrm{at}})\oplus G^{\mathrm{at}},\qquad Y^{\mathrm{at}}=(\Gamma^p(X^{\mathrm{at}})\otimes G^{\mathrm{at}},\mu),$$

with maximum norms on direct sums, and with f_X,g_X as in (14.5) for $X=X^{\mathrm{at}}$, $n=p$. Let $\widehat{W}^{\mathrm{at}},\widehat{Y}^{\mathrm{at}}$ be the completions, as normed K-spaces. If K is complete, the X_m,G_m are finite-dimensional, so complete; then $X^{\mathrm{at}},G^{\mathrm{at}}$ are the completed sums and H^{at} is a Banach space.

Theorem 14.17 (failure of composition)**.** *Let K be any nontrivially normed field of characteristic p, complete or not. On H^{at}, $f_X\colon H^{\mathrm{at}}\to W^{\mathrm{at}}$ is ambient analytic at every point, with every radius, the bound depending on the radius and the point; $g_X\colon W^{\mathrm{at}}\to Y^{\mathrm{at}}$ is the diagonal of a bounded bilinear map, so analytic everywhere; $g_X\circ f_X$ is a bounded law of degree $p+1$, divided-power analytic everywhere, and ambient analytic at no point, also as a map into $\widehat{Y}^{\mathrm{at}}$. So ambient analyticity is not closed under composition, not even under* $\mathrm{An}\circ\mathrm{AA}$*, an analytic map after an ambient analytic one. Moreover $\mathfrak{k}_{(p,1)}(H^{\mathrm{at}})=\infty$ and $\mathfrak{a}_{p+1}(H^{\mathrm{at}})=\infty$, for the constants of Definitions 14.21 and 14.36.*

The proof is in Appendix Q.

Over every K the map β_X extends to the completions, so g_X extends to the diagonal $\hat{g}_X\colon\widehat{W}^{\mathrm{at}}\to\widehat{Y}^{\mathrm{at}}$ of a bounded bilinear map. By Proposition 14.12(1)(a), f_X is ambient analytic into $\widehat{W}^{\mathrm{at}}$. So the failure also occurs with complete intermediate and final targets.

Proposition 14.18 (general form)**.** *Let X be normed with π a norm on $X^{\otimes n}$, $n\ge2$, $\mathfrak{l}_n(X)=\infty$, and G a bad space. Then f_X is ambient analytic at every point, with every radius, the bound depending on the radius and the point; g_X is the diagonal of a bounded bilinear map, so analytic everywhere; $g_X\circ f_X$ is a bounded law of degree $n+1$, divided-power analytic everywhere, and ambient analytic at no point, also as a map into the completion of Y_X; and $\mathfrak{k}_{(n,1)}(H_X)=\infty$.*

The proof is in Appendix Q. Theorem 14.17 is the case $X = X^{\mathrm{at}}$, $G = G^{\mathrm{at}}$, $n = p$, by Lemma Q.9; the constant $\mathfrak{a}_{p+1}(H^{\mathrm{at}})$ is infinite since $\mathfrak{k}_{\vec{n}} \le \mathfrak{a}_N$ by Definition 14.21.

Corollary 14.19 (inverse functions, charts, sections)**.** *Let K be complete of characteristic p, and let $H_1 = H^{\mathrm{at}} \oplus \widehat{W}^{\mathrm{at}} \oplus \widehat{Y}^{\mathrm{at}}$ with the maximum norm, a Banach space. For $z_1 \in H^{\mathrm{at}}$, $z_2 \in \widehat{W}^{\mathrm{at}}$, $z_3 \in \widehat{Y}^{\mathrm{at}}$ put*

$$\begin{aligned} \Delta(z_1, z_2, z_3) &= (z_1,\ z_2 - f_X(z_1),\ z_3 - \hat{g}_X(z_2)), \\ \Delta'(z_1, z_2, z_3) &= (z_1,\ z_2 + f_X(z_1),\ z_3 + \hat{g}_X(z_2 + f_X(z_1))). \end{aligned} \tag{14.6}$$

1. *Δ is a bijection of H_1 with inverse Δ'. It is AA at every point, and its derivative is bijective with bounded inverse at every point. Δ' is DP everywhere and AA nowhere. So AA has no inverse function theorem.*
2. *$\Delta''(z_2, z_3) = (z_2, z_3 + \hat{g}_X(z_2))$ is an analytic automorphism of $\widehat{W}^{\mathrm{at}} \oplus \widehat{Y}^{\mathrm{at}}$. The map $(f_X, 0)$ is AA everywhere, and $\Delta'' \circ (f_X, 0)$ is AA nowhere. So being AA depends on the analytic chart of the target.*
3. *On H_1 let $\phi_1(z) = (z_1, z_2 + f_X(z_1), z_3)$ and $\psi(z) = (z_1, z_2, z_3 + \hat{g}_X(z_2))$. Call two global charts AA-compatible if both transitions are AA everywhere. Then* id *and ϕ_1 are AA-compatible, ϕ_1 and $\psi \circ \phi_1$ are AA-compatible, and* id *and $\psi \circ \phi_1$ are not. So AA-compatibility is not transitive.*
4. *Let E^{at} be the completion of $\Gamma^p_\pi(X^{\mathrm{at}})$, direct sum $\widehat{Y}^{\mathrm{at}}$, with the maximum norm. For $w \in G^{\mathrm{at}}$, the map $s \mapsto s \otimes w$ from $\Gamma^p_\pi(X^{\mathrm{at}})$ to Y^{at} has norm at most $\|w\|$, so it extends to the completion of $\Gamma^p_\pi(X^{\mathrm{at}})$, with values in $\widehat{Y}^{\mathrm{at}}$. For $x = (x_X, x_G)$ in H^{at} let $\mathsf{S}(x)(s, z_3) = (s, z_3 + s \otimes x_G)$. Then S and $x \mapsto \mathsf{S}(x)^{-1}$ are affine bounded maps from H^{at} to $\mathcal{L}(\mathsf{E}^{\mathrm{at}})$, hence analytic, so S is a change of trivialization of the trivial analytic vector bundle $H^{\mathrm{at}} \times \mathsf{E}^{\mathrm{at}}$. The section $\mathsf{v}(x) = (x_X^{\otimes p}, 0)$ is AA everywhere, and the section $\mathsf{S}\mathsf{v}\colon x \mapsto \mathsf{S}(x)\mathsf{v}(x)$ is AA nowhere. So being AA depends on the trivialization, even for a trivial bundle with affine transitions.*

The proof is in Appendix Q.

Remark 14.20 (scope). Whether composition fails in characteristic 0 is open (Question 14.42). A failure in which the composite is a bounded law, as here, cannot have an ultrametric final target, by Theorem 14.39(4). The source spaces above are built from quotients of ℓ^1-spaces, and this construction cannot be made ultrametric: the quotient of $c_0(\mathbb{F}_p^m \setminus \{0\}, K)$ under $e_{\mathsf{x}} \mapsto \mathsf{x}$ is K^m with the maximum norm, isometrically, because the vectors x have entries of absolute value at most 1 and the unit vectors are among them. Over K of characteristic p, on $c_0 \oplus G$, with G the ℓ^1-sum of the universal spaces of the families of all tensors $y^1 \otimes \cdots \otimes y^p$ with $y^1, \dots, y^p \in \mathbb{F}_p^m \subseteq c_0$, the same failure holds if K is discretely valued or $p = 2$, see [II]; over densely valued K of odd characteristic this is open.

Composition is decided by the following constants of the source.

Definition 14.21 (composition constants)**.** Let P be a normed space and $\mathcal{C}(N, r)$ the set of compositions $\vec{n} = (n_1, \dots, n_r)$ of N into r positive parts. Since $h^{\otimes N} = h^{\otimes n_1} \otimes \cdots \otimes h^{\otimes n_r}$, we have $\Gamma^N(P) \subseteq \Gamma^{n_1}(P) \otimes \cdots \otimes \Gamma^{n_r}(P)$. For t in the latter space let $\pi_{\vec{n}}(t)$ be its projective seminorm in $\Gamma^{n_1}_\pi(P) \otimes \cdots \otimes \Gamma^{n_r}_\pi(P)$, and put, with $0/0 = 0$ and $c/0 = \infty$ for $c > 0$,

$$\mathfrak{k}_{\vec{n}}(P) = \sup_{t \in \Gamma^N(P)} \frac{\pi_{\vec{n}}(t)}{\pi(t)} \in [0, \infty], \qquad \mathfrak{k}(P) = \sup_{N \ge 1} \max_{1 \le r \le N} \max_{\vec{n} \in \mathcal{C}(N,r)} \mathfrak{k}_{\vec{n}}(P)^{1/N}.$$

Write $\widehat{\pi}_{\vec{n}}, \widehat{\mathfrak{k}}_{\vec{n}}(P), \widehat{\mathfrak{k}}(P)$ for the same quantities with each $\Gamma^{n_i}_\pi(P)$ replaced by its separated completion, with t mapped there. On $\Gamma^N(P)$ we have $\pi \le \pi_{\vec{n}} \le \|\cdot\|_\Gamma$: expand the factors of a

representation for the lower bound, and note that $t = \sum_q \lambda_q h_q^{\otimes N}$ has cost at most $\sum_q |\lambda_q| \|h_q\|^N$ in $\pi_{\vec{n}}$. So $\mathfrak{k}_{\vec{n}}(P) \le \mathfrak{a}_N(P)$, with $\mathfrak{a}_N(P) = \sup_t \|t\|_\Gamma / \pi(t)$ as in Definition 14.36.

Theorem 14.22 (composition criterion)**.** *Let $P \neq 0$ be a normed space. Consider:*

(i) *for all normed W, Y: if f maps an open subset of P into W and is AA at x_0, and g maps an open subset of W into Y and is AA at $f(x_0)$, then $g \circ f$ is AA at x_0;*
(ii) *the same for Banach W, Y and g analytic at $f(x_0)$;*
(iii) $\mathfrak{k}(P) < \infty$;
(iii′) $\widehat{\mathfrak{k}}(P) < \infty$.

Then (iii)⇒(i)⇒(ii)⇒(iii′)*, and* (iii′)⇔(iii) *if K is complete. Over every K,* (iii′)*,* (i) *for Banach W, and* (ii) *are equivalent.*

If $\mathfrak{k}_{\vec{n}}(P) = \infty$ for one $\vec{n} \in \mathcal{C}(N, r)$, let $W_{\vec{n}} = \bigoplus_i \Gamma^{n_i}_\pi(P)$, with the maximum norm, and $Y_{\vec{n}} = (\bigotimes_i \Gamma^{n_i}(P), \pi_{\vec{n}})$, or their separated quotients if these are only seminormed. Then the polynomial $f_{\vec{n}}(h) = (h^{\otimes n_1}, \dots, h^{\otimes n_r})$ is AA at every point of P, the map $g_{\vec{n}}(s_1, \dots, s_r) = s_1 \otimes \cdots \otimes s_r$ is the diagonal of a bounded r-linear map, and $g_{\vec{n}} \circ f_{\vec{n}}$ is AA at no point of P.

The proof is in Appendix Q.

14.4. **Why it fails.** Ambient analyticity means analyticity after moving the target W into a larger space Z. That move carries maps into W along. It does not carry analytic maps out of W: an analytic g on W need not extend to Z. Extension, with $g \circ f$ divided-power analytic, is a sufficient condition for $g \circ f$ to be ambient analytic.

Proposition 14.23 (extension along the enlargement)**.** *Let $f\colon U \to W$ be AA at x_0 with an ambient expansion through $j\colon W \to Z$, and let g be a germ at $f(x_0)$ into Y. Suppose that $j_Y \circ g = \tilde{g} \circ j$ near $f(x_0)$, for a linear isometry $j_Y\colon Y \to Z_Y$ and a map $\tilde{g}$ from an open subset of Z into Z_Y that is AA at $j(f(x_0))$. If $g \circ f$ is DP at x_0, then $g \circ f$ is AA at x_0.*

Proof. Step 1 (the composite after j_Y). Near x_0 we have $j_Y \circ g \circ f = \tilde{g} \circ (j \circ f)$, and $j \circ f$ is analytic at x_0 by (14.2). So $j_Y \circ g \circ f$ is AA at x_0, by Proposition 14.12(1)(b) and (1)(e).

Step 2 (descent). As $g \circ f$ is DP at x_0, the target descent of Proposition 14.12 gives the claim. □

For the maps $(x, w) \mapsto \beta_0(x^{\otimes n}, w)$, of which $g_X \circ f_X$ is one, ambient analyticity is equivalent to the extension of one bilinear map.

Theorem 14.24 (one bilinear map)**.** *Let $X \neq 0, G, Y$ be normed, $n \ge 2$, $\beta_0\colon \Gamma^n_\pi(X) \times G \to Y$ bounded bilinear, and L_{β_0} its linearization on $\Gamma^n(X) \otimes G$. The following are equivalent:*

(a) *$(x, w) \mapsto \beta_0(x^{\otimes n}, w)$ is AA at one point of $X \oplus G$;*
(b) *it is AA at every point of $X \oplus G$;*
(c) *there are a linear isometry $j_Y\colon Y \to Z_Y$ and a bounded bilinear map $\tilde{\beta}$ on $(X^{\otimes n}, \pi) \times G$ that equals $j_Y \circ \beta_0$ on $\Gamma^n(X) \times G$;*
(d) *$\|L_{\beta_0} t\| \le C\pi(t)$ for some C and all $t \in \Gamma^n(X) \otimes G$, with π the projective seminorm of $X^{\otimes n} \otimes G$.*

For fixed X and n, all G, Y, β_0 satisfy these if and only if $\mathfrak{l}_n(X) < \infty$.

The proof is in Appendix Q. The statement on $g_X \circ f_X$ in Proposition 14.18 is the case $\beta_0(s, w) = s \otimes w$, where L_{β_0} is the identity and (d) fails. So the failure is exactly the non-extension of one bilinear map.

Let $\operatorname{char} K = p$. Fixing one Z does not help. Theorem 14.11 gives one enlargement for any set of germs, and $g_X \circ f_X$ is still AA nowhere. The outer map must act on the envelope. For K

complete, the map Δ'' of Corollary 14.19(2) is an analytic automorphism that destroys ambient analyticity. So ambient analyticity sees W as a normed space, not as an analytic manifold. The following language records this.

Definition 14.25 (pairs)**.** A *pair* (Z, W) is a Banach space Z with a closed subspace W. A *morphism germ* $\Phi\colon (Z, W) \to (Z', W')$ at $w_0 \in W$ is the germ at w_0 of an analytic map from an open subset of Z into Z' with $\Phi(W) \subseteq W'$ near w_0. Put $\mathrm{Fg}(Z, W) = W$ and $\mathrm{Fg}\,\Phi = \Phi|_W$.

Proposition 14.26 (ambient analyticity as an image)**.** *Pairs with a base point $w_0 \in W$ and morphism germs preserving base points form a category, and* Fg *is a functor to pointed Banach spaces and germs of maps. Let $j\colon W \to Z$, $j'\colon W' \to Z'$ be the inclusions. For a morphism germ Φ we have $j' \circ \Phi|_W = \Phi \circ j$, so $\mathrm{Fg}\,\Phi$ is AA. Conversely, for Banach P, W, a germ from P to W is AA if and only if, after identifying W with a closed subspace of a Banach space Z, it is $\mathrm{Fg}\,\Phi$ for a morphism germ $\Phi\colon (P, P) \to (Z, W)$.*

Fg is not injective on objects, so its image need not be a subcategory: $\mathrm{Fg}\,\Phi_2 \circ \mathrm{Fg}\,\Phi_1$ is defined when Φ_1 lands in (Z, W) and Φ_2 starts at (W, W), while $\Phi_2 \circ \Phi_1$ is not. Let W be Banach and let $j_1\colon W \to Z_1$, $j_2\colon W \to Z_2$ be linear isometries into Banach spaces. The quotient Z_3 of the ℓ^1-sum of Z_1, Z_2 by $\{(j_1 w, -j_2 w) : w \in W\}$ is Banach, with linear isometries $i_1\colon Z_1 \to Z_3$, $i_2\colon Z_2 \to Z_3$ and $i_1 j_1 = i_2 j_2$. So a morphism germ Φ_1 into $(Z_1, j_1(W))$ gives the morphism germ $i_1 \circ \Phi_1$ into $(Z_3, i_1 j_1(W))$, with the same image under Fg *after identifying W with its images.*

The proof is in Appendix Q. No such construction extends Φ_2. Let K be complete of characteristic p, let $j\colon \widehat{W}^{\mathrm{at}} \to Z$ be a linear isometry into a Banach space through which f_X is analytic, so that $\Phi_1 = j \circ f_X$ is a morphism germ from $(H^{\mathrm{at}}, H^{\mathrm{at}})$ to $(Z, j(\widehat{W}^{\mathrm{at}}))$, and let $\Phi_2 = \hat{g}_X$ on $(\widehat{W}^{\mathrm{at}}, \widehat{W}^{\mathrm{at}})$. An analytic extension of $\hat{g}_X$ along j, after a linear isometry of its target, would make $g_X \circ f_X$ ambient analytic, by Proposition 14.23.

A *subquotient* is a filtered space $(Z; W_1, W_2)$ as in Definition 14.1, with Z Banach, read as a pair with values in W_1/W_2. The pairs of Definition 14.25 are the case $W_1 = W$, $W_2 = 0$, and we use only these. Subquotients are allowed as targets, the source being a pair, since the image stays ambient analytic by bounded linear postcomposition, and as fibers with a linear action over an analytic base, as in Definition 14.1. If the source is a subquotient as well, then for K complete and Banach spaces the image consists of all divided-power analytic germs, after identifying the source, with an equivalent norm, with a quotient of an ℓ^1-space, see [II].

Theorem 14.27 (inverse germs of pairs)**.** *Let $\Phi\colon (Z, W) \to (Z', W')$ be a morphism germ at w_0 that has an analytic inverse germ Φ^{-1} at $\Phi(w_0)$ on Z', and let $\mathsf{T}_1 = D\Phi(w_0)$. Then Φ^{-1} is a morphism germ if and only if $\mathsf{T}_1(W) = W'$.*

Let K be complete of characteristic p, let $z^0 \in H_1$ for H_1 as in Corollary 14.19, and let $j\colon H_1 \to Z$, $j'\colon H_1 \to Z'$ be linear isometries into Banach spaces. Then no analytic germ Φ from Z to Z' at $j(z^0)$ with an analytic inverse germ satisfies $\Phi \circ j = j' \circ \Delta$ near z^0.

The proof is in Appendix Q. Such a Φ would be a morphism germ with $\mathsf{T}_1(j(H_1)) = j'(H_1)$, and its inverse germ would make Δ' ambient analytic.

Fix a pair (Z, W) with inclusion j. Say that the pair *carries* a germ f into W if $j \circ f$ is analytic, and *lifts* a germ on W if the germ is $\mathrm{Fg}\,\Phi_g$ for a morphism germ $\Phi_g\colon (Z, W) \to (Z, W)$. A *local automorphism* is a morphism germ $\Phi\colon (Z, W) \to (Z, W)$ at a point $w_0 \in W$ with an analytic inverse germ and with $D\Phi(w_0)(W) = W$.

Corollary 14.28 (one pair)**.** *The germs of local automorphisms of a fixed pair (Z, W) at points of W form a groupoid. If the pair carries f, then it carries $\Phi|_W \circ f$ for a local automorphism Φ*

at the value of f at the base point. If the pair lifts g, then it lifts $\Phi|_W \circ g \circ (\Phi'|_W)^{-1}$ for local automorphisms Φ at the value of g and Φ' at the base point of g. By Corollary 14.19(2), for K complete of characteristic p, no pair $(Z, \widehat{W}^{\mathrm{at}} \oplus \widehat{Y}^{\mathrm{at}})$ carries $(f_X, 0)$ at a point x_0 and lifts Δ'' at $(f_X, 0)(x_0)$.

The proof is in Appendix Q.

Remark 14.29 (why Section 14.1 survives). Let the model P of M and the value fiber F be Banach. The transitions of $\mathrm{Alt}^k(M_1; M_0)$ in Section 14.1 give the local automorphisms $(x, \mathsf{m}) \mapsto (\psi_{\mathsf{ab}}(x), \Theta_{\mathsf{ab}}(x)\mathsf{m})$ of the pair $(P \times \mathrm{Mult}^k(E;F), P \times \mathrm{Alt}^k(E;F))$, with ψ_{ab} the change of chart of M. They are jointly analytic, as a bounded bilinear evaluation of an analytic operator family, and their inverses are transitions, so each derivative maps $P \times \mathrm{Alt}^k(E;F)$ onto itself, by difference quotients. Each d-linear operation of Theorem 14.4(3) acts on the envelope, and inverse images change the base. Only the operations lift: $i(\mathsf{s})^2 = 0$ holds on Alt^k, fiberwise, and not on Mult^k. One fixed enlargement alone does not suffice, by Theorem 14.11. What Section 14.1 uses is that the transitions act on the envelope.

14.5. **Which envelope.** Section 14.1 used the envelope Mult^k. Here we compare it with the other envelopes. Throughout this subsection K is complete, as in Section 10.3, so $\mathrm{Alt}^k(E;K)$ is closed in every isometric enlargement.

Definition 14.30 (functorial envelope)**.** Fix $k \geq 1$. Let $\mathcal{V}$ be one of two categories, with bounded linear maps as morphisms: (A) all normed spaces; (B) all ultrametric normed spaces. A *functorial envelope* of $\mathrm{Alt}^k(-;K)$ on $\mathcal{V}$ is a contravariant functor $\mathfrak{E}$ from $\mathcal{V}$ to normed spaces, whose hom maps $T \mapsto \mathfrak{E}(T)$ are analytic at 0, with linear isometries $j_E\colon \mathrm{Alt}^k(E;K) \to \mathfrak{E}(E)$ such that $\mathfrak{E}(T) j_D(\omega) = j_E(\omega \circ T^{\times k})$ for $T \in \mathcal{L}(E, D)$.

For a manifold M as in Section 10.3, modeled on E in $\mathcal{V}$, a section ω of $\mathrm{Alt}^k(TM;K)$ is $\mathfrak{E}$-*ambient* if, in each chart, $j_E \circ \omega$ is analytic into the completion of $\mathfrak{E}(E)$. The completion is used because a change of chart re-expands the coefficients as infinite sums. By Lemma Q.10 and the change-of-chart formula of Section 10.3, one atlas suffices. Write $\Omega_{\mathfrak{E}}$ for the class of these sections.

Example. Let $\mathcal{U}$ be a normed space and $u_0 \in \mathcal{U}$ with $\|u_0\| = 1$. Then $\mathrm{Mult}^k(-;\mathcal{U})$ is an envelope, with $\mathfrak{E}(T)\mathsf{m} = \mathsf{m} \circ T^{\times k}$ and $j_E(\omega) = \omega\, u_0$. We define $\Omega_{\mathcal{U}}$, with u_0 understood, as the class of the envelope $\mathrm{Mult}^k(-;\widehat{\mathcal{U}})$. For $\mathcal{U} = K$, $u_0 = 1$ we have $\Omega_K = \Omega^k_{\mathrm{amb}}$ of Section 10.3.

Coefficient space. Let $\mathsf{L}_k = K^k$, with the sum norm in case (A) and the maximum norm in case (B), and unit vectors $e_1, \dots, e_k$. Let det be the determinant on L_k; its norm is 1. For $v \in E^k$ let $T_v\colon \mathsf{L}_k \to E$, $T_v e_i = v_i$. For $\lambda \in K^k$ let $\mathsf{t}_\lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_k)$. By Lemma Q.10, $\mathfrak{E}(\lambda \mathrm{id}_E) = \sum_n \lambda^n \mathsf{q}^E_n$ near $\lambda = 0$, with natural bounded orthogonal idempotents q^E_n; the subfunctor $\mathfrak{E}_k(E) = \mathsf{q}^E_k \mathfrak{E}(E)$ has hom maps that are diagonals of bounded k-linear maps. So $\lambda \mapsto \mathfrak{E}_k(\mathsf{t}_\lambda)$ is a polynomial, homogeneous of degree k. Multiplicativity, and K being infinite, give

$$\mathfrak{E}_k(\mathsf{t}_\lambda) = \sum_{|\alpha|=k} \lambda^\alpha \mathsf{p}_\alpha,$$

with bounded orthogonal idempotents p_α of sum id. Put $\mathsf{p} = \mathsf{p}_{(1,\dots,1)}$. The *coefficient space* and *coefficient vector* of $\mathfrak{E}$ are

$$\mathcal{U}_{\mathfrak{E}} = \mathsf{p}\, \mathfrak{E}_k(\mathsf{L}_k), \qquad u_{\mathfrak{E}} = j_{\mathsf{L}_k}(\det), \qquad \|u_{\mathfrak{E}}\| = 1.$$

Here $u_{\mathfrak{E}} \in \mathcal{U}_{\mathfrak{E}}$, as $\det \circ \mathsf{t}_\lambda^{\times k} = \lambda_1 \cdots \lambda_k \det$. For $\mathrm{Mult}^k(-;\mathcal{U})$ the coefficient space is isomorphic to $\mathcal{U}^{k!}$: it consists of the maps supported on the words that are permutations of $(1, \dots, k)$. The line $K u_{\mathfrak{E}}$ is *split* if a bounded linear form η_0 on $\mathcal{U}_{\mathfrak{E}}$ has $\eta_0(u_{\mathfrak{E}}) = 1$.

Theorem 14.31 (comparison)**.** *Let $(\mathfrak{E}, j)$ be a functorial envelope.*

1. *For $\xi \in \mathfrak{E}_k(E)$ and $v \in E^k$ put $\chi_E(\xi)(v_1,\dots,v_k) = \mathsf{p}\,\mathfrak{E}_k(T_v)\xi$. Then χ_E is a bounded linear map $\mathfrak{E}_k(E) \to \operatorname{Mult}^k(E;\mathcal{U}_{\mathfrak{E}})$, natural in E, and $\chi_E(j_E\omega) = \omega\, u_{\mathfrak{E}}$.*
2. *For every normed $\mathcal{U}'$, each natural family of bounded linear maps $\mathfrak{E}(E) \to \operatorname{Mult}^k(E;\mathcal{U}')$ equals $\mathsf{h}_* \circ \chi_E \circ \mathsf{q}_k^E$ for exactly one $\mathsf{h} \in \mathcal{L}(\mathcal{U}_{\mathfrak{E}},\mathcal{U}')$, and every such h gives a natural family.*
3. *$\Omega_{\mathfrak{E}} \subseteq \Omega_{\mathcal{U}_{\mathfrak{E}}}$, for the vector $u_{\mathfrak{E}}$. If the line $Ku_{\mathfrak{E}}$ is split, $\Omega_{\mathfrak{E}} \subseteq \Omega_K$.*

The proof is in Appendix Q.

A *multilinear action* on an envelope is a family of bounded k-linear maps $\beta^{\mathfrak{E}}_{E,D}\colon \mathcal{L}(E,D)^k \to \mathcal{L}(\mathfrak{E}(D),\mathfrak{E}(E))$ such that, for $T, T_i \in \mathcal{L}(E,D)$ and $T_i' \in \mathcal{L}(E',E)$,

$$\begin{gathered}\beta^{\mathfrak{E}}_{E,E}(\mathrm{id},\dots,\mathrm{id}) = \mathrm{id}, \qquad \beta^{\mathfrak{E}}_{E,D}(T,\dots,T) = \mathfrak{E}(T),\\ \beta^{\mathfrak{E}}_{E',D}(T_1T_1',\dots,T_kT_k') = \beta^{\mathfrak{E}}_{E',E}(T_1',\dots,T_k') \circ \beta^{\mathfrak{E}}_{E,D}(T_1,\dots,T_k).\end{gathered} \tag{14.7}$$

The envelope $\mathfrak{E} = \operatorname{Mult}^k(-;\mathcal{U})$ has one: $\beta^{\mathfrak{E}}_{E,D}(T_1,\dots,T_k)\mathsf{m} = \mathsf{m} \circ (T_1 \times \cdots \times T_k)$. Every multilinear form is obtained from an alternating form by this action (Lemma Q.11).

A *morphism* of envelopes $\mathfrak{E} \to \mathfrak{E}'$ is a family of bounded linear maps $\mathfrak{E}(E) \to \mathfrak{E}'(E)$, natural in E, that commutes with j and j'. If both envelopes have a multilinear action, the morphism is *equivariant* if it also commutes with $\beta^{\mathfrak{E}}$ and $\beta^{\mathfrak{E}'}$.

Theorem 14.32 (Mult^k is initial)**.** (1) *Let $(\mathfrak{E}, j, \beta^{\mathfrak{E}})$ be an envelope with a multilinear action. There is a unique family of bounded linear maps $\mathsf{i}_E\colon \operatorname{Mult}^k(E;K) \to \mathfrak{E}(E)$ that extends j_E and satisfies $\mathsf{i}_E(\mathsf{m}\circ(T_1\times\cdots\times T_k)) = \beta^{\mathfrak{E}}_{E,D}(T_1,\dots,T_k)\,\mathsf{i}_D(\mathsf{m})$. So $\operatorname{Mult}^k(-;K)$ is initial among envelopes with a multilinear action and equivariant morphisms.*

2. *For such an envelope $\Omega_K \subseteq \Omega_{\mathfrak{E}}$, with equality if the line $Ku_{\mathfrak{E}}$ is split.*
3. *For $k \geq 2$, $\operatorname{Mult}^k(-;K)$ is not initial among all envelopes and morphisms: it has more than one morphism to itself. For $k = 1$ it is initial among all envelopes and morphisms.*

The proof is in Appendix Q.

Now the upper bound Ω_K of Theorem 14.31(3) can fail when the line is not split. Let $\mathfrak{V}$ be the space $K \oplus \bigoplus_{N\geq 1} K^N$ with the norm $\|(c,(c_{N,i}))\| = |c| + \sum_N \frac{1}{N}\sum_{i\leq N}|c_{N,i}|$. Let $\mathsf{r}_N = (1, -(1,\dots,1)_N)$, with $(1,\dots,1)_N$ in the summand K^N, and let $\mathfrak{U}$ be the separated quotient of $\mathfrak{V}$ by the span of the r_N. Put $u_0 = [1,0]$ and let $\mathsf{u}_{N,i}$ be the class of the i-th unit vector of K^N. Then

$$u_0 = \sum_{i\leq N} \mathsf{u}_{N,i}, \qquad \|\mathsf{u}_{N,i}\| \leq \frac{1}{N}, \qquad \|u_0\| = 1. \tag{14.8}$$

For a manifold with a chosen atlas, write Ω_{AA} for the class of sections whose expression in each chart of the atlas is ambient analytic into $\operatorname{Alt}^k(E;K)$, with an enlargement chosen per chart.

Theorem 14.33 (more than Ω_K)**.** *Let K be complete, of any characteristic. Choose $\mathsf{e}_n \in K$ with $n^n \leq |\mathsf{e}_n| < |\varpi|^{-1}n^n$. For h in the open unit ball of $\ell^1(\mathbb{N},K)$ put $\mathsf{f}(h) = \sum_{n\geq 1}\mathsf{e}_n h_1\cdots h_n$. On the object $E = \ell^1(\mathbb{N},K)\oplus K^k$ of case (A), with the sum norm, put $\omega_{\mathsf{f}}(x,y) = \mathsf{f}(x)\,\mathrm{d}y_1\wedge\cdots\wedge\mathrm{d}y_k$. Let O be the open subset $\{\|x\| < \mathrm{e}^{-1}\}$ of E, with its identity chart, where e is Euler's number.*

1. *f is ambient analytic at 0 and not analytic at 0.*
2. *$\Omega_{\widehat{\mathfrak{U}}} \supsetneq \Omega_K$ on O: the form ω_{f} lies in $\Omega_{\widehat{\mathfrak{U}}}$ and not in Ω_K. And $\widehat{\mathfrak{U}}$ has no bounded linear form nonzero at u_0. So for the envelope $\operatorname{Mult}^k(-;\widehat{\mathfrak{U}})$ the line $Ku_{\mathfrak{E}}$ is not split: the coordinates of $u_{\mathfrak{E}}$ in $\widehat{\mathfrak{U}}^{k!}$ are $\pm u_0$.*

(3) *A chart-by-chart enlargement gives more:* $\Omega_K \subsetneq \Omega_{\mathrm{AA}}$ *on* O*, again by* ω_{f}*.*
(4) *Let* $\operatorname{char} K = p$*, put* $M = H^{\mathrm{at}}$*, and let* E^{at}*,* S*,* v *be as in Corollary 14.19(4). Put* $M_1 = M \times K^k$*, with the maximum norm on* K^k*, and* $M_0 = M \times \mathsf{E}^{\mathrm{at}}$*. With* det *the determinant on* K^k*, the section* $\det \cdot \mathsf{v}$ *of* $\operatorname{Alt}^k(M_1; M_0)$ *is ambient analytic at every point, for an enlargement chosen per chart. In the vector bundle chart of* M_0 *changed by* S *its expression* $\det \cdot (\mathsf{S}\mathsf{v})$ *is ambient analytic at no point.*

The proof is in Appendix Q.

Corollary 14.34 (where nothing is lost)**.** *Let* K *be complete and* $k \geq 1$*. Consider the following property of an ultrametric Banach space* E*: every map from an open subset of* E *into* $\operatorname{Alt}^k(E; K)$ *that is divided-power analytic at a point is analytic at that point as a map into* $\operatorname{Mult}^k(E; K)$*. If* $\operatorname{char} K = 0$*, every ultrametric Banach space has this property. If* $\operatorname{char} K = p$*, every ultrametric Banach space has it if and only if* K *is spherically complete.*

The proof is in Appendix Q.

Question 14.35 (a lower bound)**.** Let $\operatorname{char} K = p$, let $k \geq p$, and let K be not spherically complete. Is $\Omega_K \subseteq \Omega_{\mathfrak{E}}$ for every functorial envelope? This is open. For all other K and k the inclusion holds: if $k! \neq 0$, or if K is spherically complete (Theorem 5.2), then $\operatorname{Alt}^k(E; K)$ has a bounded linear retraction ϑ_E in $\operatorname{Mult}^k(E; K)$, and $j_E \circ \vartheta_E$ is a bounded linear extension of j_E to $\operatorname{Mult}^k(E; K)$. The inclusion also holds with a multilinear action, by Theorem 14.32. It would follow if every j_E extended to a bounded linear map on $\operatorname{Mult}^k(E; K)$. The converse of this last implication is open.

14.6. **Tame pairs.** The calculus of composition and inverse functions uses only tameness, a condition on a pair of spaces. Tameness is sufficient for composition; for K complete, the exact condition on the source is $\mathfrak{k}(P) < \infty$ (Theorem 14.22). Ultrametric norms are used in four other places, for four reasons.

Finiteness in each degree, from an ultrametric target. Values add without counting the summands, so every bounded law into an ultrametric space is ambient (Theorem 14.39(4)). For a general source a hypothesis on the target is needed: over a field of characteristic p the composite of Theorem 14.17, on the non-ultrametric source H^{at}, is a bounded law that is not ambient. No such hypothesis is needed for $P = \ell^1(J)$ (item (2)), or for ultrametric P in characteristic 0 (item (3)). For $P = c_0$ in characteristic p, see Question 14.42(4).

Removal of the factor n^n*, from an ultrametric source.* This factor is what a non-ultrametric norm on P costs in Theorem 14.39(4). An ultrametric norm on P removes it from the bound on $\mathfrak{a}_n(P, W)$ (item (1)), and so does $P = \ell^1(J)$ (item (2)). Whether it can be removed for every P is Question 14.42(4).

The ambient space. For P, W ultrametric the enlargement can be built with maxima in place of sums, so it can be taken ultrametric (Theorem 14.39(1)).

An extension theorem. In characteristic p, agreement with analytic maps uses the extension of contractions into spherically complete spaces (Proposition 14.40(1)). The ultrametric source is needed for this: the germ of Theorem 14.33(1) has source $\ell^1(\mathbb{N}, K)$ and target K, and K may be spherically complete.

Definition 14.36 (tame)**.** For normed P, W and $n \geq 1$ let $\mathfrak{a}_n(P, W)$ be the supremum of $\operatorname{amb}(a)/\|a\|$ over the nonzero bounded laws $a\colon P \to W$ of degree n, and 0 if there are none. The pair (P, W) is *tame* if $\sup_{n \geq 1} \mathfrak{a}_n(P, W)^{1/n} < \infty$. The space P is *tame* if (P, W) is tame for every normed W. Put $\mathfrak{a}_n(P) = \sup\{\|t\|_\Gamma / \pi(t) : t \in \Gamma^n(P)\}$, with $c/0 = \infty$ for $c > 0$ and $0/0 = 0$.

The bound $\|L_a(t)\| \le \|a\|\|t\|_\Gamma$ gives $\mathfrak{a}_n(P,W) \le \mathfrak{a}_n(P)$. Equality holds for W_n, the separated quotient of $(\Gamma^n(P), \|\cdot\|_\Gamma)$: the law $h \mapsto [h^{\otimes n}]$ has norm at most 1 and linearization $t \mapsto [t]$ (Lemma Q.1(2)), and if this law vanishes then $\mathfrak{a}_n(P) = 0$. Enlarging a target isometrically does not lower $\mathfrak{a}_n$, and the ℓ^1-sum of the W_n contains each W_n. So P is tame if and only if $\sup_n \mathfrak{a}_n(P)^{1/n} < \infty$.

Proposition 14.37 (tame is exact)**.** *Let W be a Banach space. The following are equivalent: (i) (P,W) is tame; (ii) every map from an open subset of P into W that is divided-power analytic at a point is ambient analytic there; (iii) the same at the point 0, for germs given by one series of bounded laws.*

The proof is in Appendix Q.

Theorem 14.38 (calculus on a tame pair)**.** (1) *Let (P,W) be tame, with $\mathfrak{a}_n(P,W) \le C\mathfrak{q}^n$ for $n \ge 1$ and $\mathfrak{q}, C \ge 1$. A germ from P to W that is divided-power analytic at x_0 with radius R and bound $\mathfrak{b}$ is ambient analytic there with radius $R/\mathfrak{q}$ and bound $C\mathfrak{b}$.*

(2) *Let (P,W) be tame and Y any normed space. Let f, from an open subset of P into Y, be divided-power analytic (for instance ambient analytic) at x_0, and g, from an open subset of Y into W, be divided-power analytic (for instance ambient analytic) at $f(x_0)$. Then $g \circ f$ is ambient analytic at x_0.*

(3) *Let P be Banach and (P,P) tame. Let f, from an open subset of P into P, be divided-power analytic at x_0, with coefficient law a_1 bijective with bounded inverse. Then f has a local inverse germ, which is ambient analytic at $f(x_0)$ and at every point of a ball around it.*

The proof is in Appendix Q.

Item (2) uses only the pair (P,W) and item (3) only (P,P). On a tame pair ambient analyticity coincides with divided-power analyticity, by item (1), so this is no separate argument for it.

Now the main examples. Let $\mathfrak{c}_K$ be the generator below 1 of $|K^\times|$ if K is discretely valued, and $\mathfrak{c}_K = 1$ otherwise. On $P^{\otimes n}$ let

$$\pi_{\max}(t) = \inf\Big\{\max_l \prod_{i=1}^n \|x_{l,i}\| : \ t = \sum_l x_{l,1} \otimes \cdots \otimes x_{l,n}\Big\},$$

an ultrametric seminorm with $\pi_{\max} \le \pi$. Put $C_0 = 8\mathrm{e}/|\varpi|^2$. Let $\ell^1(J) = \ell^1(J,K)$ for a set J, and $\ell^1_{00}(J)$ its subspace of finitely supported vectors. If $\operatorname{char} K = 0$, at most one prime p_0 has $|p_0| < 1$ in K; put $\mathfrak{d} = |p_0|^{-1/(p_0-1)}$ if there is one, and $\mathfrak{d} = 1$ otherwise.

Theorem 14.39 (examples)**.** *Let $n \ge 1$.*

(1) *Let P, W be ultrametric. For every bounded law $a\colon P \to W$ of degree n,*

$$\|L_a(t)\| \le (|\varpi|\mathfrak{c}_K)^{-n}\|a\|\,\pi_{\max}(t) \qquad (t \in \Gamma^n(P)). \tag{14.9}$$

So $\mathfrak{a}_n(P,W) \le (|\varpi|\mathfrak{c}_K)^{-n}$. A germ from P to W that is divided-power analytic at x_0 with radius R and bound $\mathfrak{b}$ is ambient analytic there with radius $|\varpi|\mathfrak{c}_K R$ and bound $\mathfrak{b}$, and the ambient space can be taken ultrametric. Hence between ultrametric spaces the divided-power analytic and the ambient analytic germs coincide.

(2) *If $P = \ell^1(J)$ or $P = \ell^1_{00}(J)$, then $\mathfrak{a}_n(P) \le C_0^n$.*

(3) *Let $\operatorname{char} K = 0$ and P be ultrametric. Then $\mathfrak{a}_n(P) \le (8\mathfrak{d}/|\varpi|^2)^n$. For every normed W, every bounded law $a\colon P \to W$ of degree n is the diagonal of an n-linear map into W of norm at most $(8\mathfrak{d}/|\varpi|^2)^n\|a\|$.*

(4) *Let P be any normed space and W ultrametric. For every bounded law $a\colon P \to W$ of degree n,*

$$\|L_a(t)\| \le \Big(\frac{8n}{|\varpi|^2 \mathfrak{c}_K}\Big)^n \|a\|\, \pi_{\max}(t) \qquad (t \in \Gamma^n(P)). \tag{14.10}$$

So every bounded law into W is ambient and $\mathfrak{a}_n(P,W) < \infty$ in each degree. Whether (P,W) is always tame is open.

The proof is in Appendix Q.

Proposition 14.40 (agreement with analytic maps)**.** (1) *Let P, W be ultrametric and W spherically complete. Every bounded law $a\colon P \to W$ of degree $n \ge 0$ is the diagonal of an n-linear map into W of norm at most $(|\varpi|\mathfrak{c}_K)^{-n}\|a\|$. So a germ from P to W that is divided-power analytic at x_0 with radius R and bound $\mathfrak{b}$ is analytic there with radius $|\varpi|\mathfrak{c}_K R$ and bound $\mathfrak{b}$. The same holds for every ultrametric Banach space W over a discretely valued K, with an extra factor $\mathfrak{c}_K^{-1}$ in both bounds.*

(2) *If $\operatorname{char} K = 0$ and P is ultrametric, then for every normed W the divided-power analytic, ambient analytic and analytic germs from P to W coincide (Theorem 14.39(3)).*

(3) *If K is complete, not spherically complete, of characteristic p, there is a function on an ultrametric Banach space that is ambient analytic at every point and analytic at no point.*

The proof is in Appendix Q. For finer agreement statements, see [II].

Remark 14.41 (one enlargement for a class of sources). *Ultrametric sources.* Let K be spherically complete and W ultrametric with $\|W\| \subseteq |K|$. Let $\mathcal{F}_W$ be the set of K-linear maps $\phi\colon W \to K$ with $|\phi(w)| \le \|w\|$, and $\mathrm{ev}_W\colon W \to \ell^\infty(\mathcal{F}_W, K)$, $\mathrm{ev}_W(w) = (\phi(w))_\phi$. This is an isometry. Indeed, for $w \ne 0$ choose $c_w \in K$ with $|c_w| = \|w\|$; the isometry $\lambda w \mapsto \lambda c_w$ on Kw extends to some $\phi \in \mathcal{F}_W$, by Step 1 of the proof of Theorem 5.2. The space $\ell^\infty(\mathcal{F}_W, K)$ is ultrametric and spherically complete: if balls $\overline{B}(z_\theta, r_\theta)$ meet pairwise, then so do their coordinates in K, and common points of the coordinates form a bounded family, which lies in every ball. So if P is ultrametric and f into W is divided-power analytic at x_0 with radius R and bound $\mathfrak{b}$, then $\mathrm{ev}_W \circ f$ is analytic at x_0 with radius $|\varpi|\mathfrak{c}_K R$ and bound $\mathfrak{b}$, by Proposition 14.40(1). One isometry serves all ultrametric sources.

The sources $\ell^1(J)$. Here K need not be spherically complete or complete: the definition of $\mathfrak{U}$ and (14.8) use no completeness. Let W be normed, and $W \mathbin{\widehat{\otimes}_\pi} \mathfrak{U}$ the separated completion of $(W \otimes \mathfrak{U}, \pi)$. The map $w \mapsto w \otimes u_0$ is an isometry. It is contractive, as $\|u_0\| = 1$. For the converse, the bilinear map $(w, (c, (c_{N,i}))) \mapsto (cw, (c_{N,i}w))$ from $W \times \mathfrak{V}$ into $W \oplus \bigoplus_N W^N$, with the norm of $\mathfrak{V}$ and $\|\cdot\|$ of W in place of $|\cdot|$, has norm at most 1. Let lin be its linearization, so $\|\mathrm{lin}(t)\| \le \pi(t)$. For finitely many $w_N \in W$,

$$\Big\|\mathrm{lin}\Big(w \otimes (1,0) + \sum_N w_N \otimes \mathsf{r}_N\Big)\Big\| = \Big\|w + \sum_N w_N\Big\| + \sum_N \|w_N\| \ge \|w\|,$$

and by continuity the same bound holds with $\sum_N w_N \otimes \mathsf{r}_N$ replaced by any tensor in $W \otimes \ker(\mathfrak{V} \to \mathfrak{U})$, as this kernel is the closure of the span of the r_N. Lifting a representation of $w \otimes u_0$ in $W \otimes \mathfrak{U}$ to $\mathfrak{V}$ with arbitrarily small loss gives $\pi(w \otimes u_0) \ge \|w\|$.

Let f, from an open subset of $\ell^1(J)$ into W, be ambient analytic at x_0 with radius R, bound $\mathfrak{b}$ and coefficient laws a_n. Fix $n \ge 1$. For a finitely supported $\alpha \in \mathbb{N}^J$ with $|\alpha| = n$, a word $\mathsf{w} \in J^n$ has *content* α if each $\theta \in J$ occurs α_θ times in it. There are $N_\alpha = n!/\prod_\theta \alpha_\theta!$ such words. The orbit sum $O_\alpha(e)$, with e listing the unit vectors e_θ for θ in the support of α, has $\pi(O_\alpha(e)) \le N_\alpha$. So $w_\alpha = L_{a_n}(O_\alpha(e))$ has $\|w_\alpha\| \le \mathrm{amb}(a_n) N_\alpha$. Number the words of content

α from 1 to N_α, and let $\mathsf{z}_\mathsf{w} = w_\alpha \otimes \mathsf{u}_{N_\alpha,i}$ if w is the i-th. By (14.8), $\|\mathsf{z}_\mathsf{w}\| \le \mathrm{amb}(a_n)$ and the z_w of content α sum to $w_\alpha \otimes u_0$. Then

$$b_n(h^1, \dots, h^n) = \sum_{\mathsf{w} \in J^n} h^1_{\mathsf{w}_1} \cdots h^n_{\mathsf{w}_n}\, \mathsf{z}_\mathsf{w}$$

converges absolutely, $\|b_n\| \le \mathrm{amb}(a_n) \le \mathfrak{b} R^{-n}$, and for finitely supported h, by (14.1),

$$b_n(h, \dots, h) = \sum_\alpha h^\alpha w_\alpha \otimes u_0 = L_{a_n}(h^{\otimes n}) \otimes u_0 = a_n(h) \otimes u_0.$$

Both ends are continuous in h, so this holds for all h. With $b_0 = f(x_0) \otimes u_0$, the map f becomes analytic at x_0 with the same radius and bound, after one isometry that depends only on W.

14.7. **Conclusion and questions.** Between ultrametric spaces, ambient analyticity coincides with divided-power analyticity; it is closed under composition and, on Banach spaces, has the inverse function theorem. For general norms over a field of characteristic p it is not closed under composition, and, for K complete, it has no inverse function theorem and its charts and sections depend on choices; in characteristic 0 this is open. Over a field of characteristic p, repairing composition alone does not single out a class: there are composition-closed classes of germs with Banach values strictly between ambient and divided-power analyticity, two of them with an inverse function theorem between Banach spaces [II]. Paper [II] studies divided-power analyticity, the smallest class of germs with Banach values that contains the analytic germs and has target and source descent.

Question 14.42. Each of the following is open.

(1) Let $\mathrm{char}\, K = 0$. Is $\mathfrak{k}(P) < \infty$ for every normed P? Is every divided-power analytic germ from a normed space to a Banach space ambient analytic?
(2) Let $\mathrm{char}\, K = p$. Is ambient analyticity closed under composition on every ultrametric source, with a non-ultrametric final target? Let moreover K be densely valued and p odd. Is $\mathfrak{l}_p(c_0) = \infty$, so that Proposition 14.18 applies with $X = c_0$ and $n = p$?
(3) Is there a normed P with $\mathfrak{k}(P) < \infty$ and a divided-power analytic germ from P into a Banach space that is not ambient analytic?
(4) Is (P, W) tame for every normed P and every ultrametric W? Let $\mathrm{char}\, K = p$ and $c_0 = c_0(\mathbb{N}, K)$. Is c_0 tame? Is every bounded law from c_0 into a normed space ambient?
(5) Let $\mathrm{char}\, K = p$ and $k \ge p$. Is there, for all analytic vector bundles M_1, M_0 over an analytic manifold, an analytic vector bundle structure on $\mathrm{Alt}^k(M_1; M_0)$, compatible with its C^∞ structure, for which every analytic automorphism of (M_1, M_0) induces an analytic morphism?
(6) Let K be complete and $k \ge 1$. Is there an analytic manifold M, modeled on a normed space, and a section of $\mathrm{Alt}^k(TM; K)$ that is ambient analytic in one analytic chart, through some enlargement, and not in another?

Appendix A. Preliminaries for the obstruction proofs

The constructions below use three elementary facts about valued fields and power series. We record their proofs to make explicit the scope of the coefficient argument, including incomplete base fields.

Lemma A.1 (one-variable uniqueness)**.** *Let K be nontrivially normed and Z a normed K-space, not necessarily complete. Let $z_0, z_1, \dots \in Z$, $\rho > 0$ and $M \ge 0$ with $\|z_n\| \rho^n \le M$ for all n. If $\sum_n t^n z_n$ converges to 0 for every $t \in K$ with $|t| < \rho$, then $z_n = 0$ for all n.*

Proof. Otherwise let N be least with $z_N \neq 0$. For $0 < |t| < \rho/2$ the series $\sum_{n>N} t^{n-N} z_n$ converges, being t^{-N} times the convergent tail $\sum_{n>N} t^n z_n$, and $t^N\big(z_N + \sum_{n>N} t^{n-N} z_n\big) = 0$. Hence

$$\|z_N\| = \Big\| \sum_{n>N} t^{n-N} z_n \Big\| \leq \sum_{n>N} M\rho^{-n}|t|^{n-N} = M\rho^{-N}\frac{|t|/\rho}{1-|t|/\rho},$$

where the norm of a convergent series is bounded by the sum of the norms by continuity of the norm on partial sums. Since K is nontrivially normed, there are $t \neq 0$ with $|t|$ arbitrarily small, so $\|z_N\| = 0$, a contradiction. □

Lemma A.2. *A normed field of characteristic $p > 0$ is nonarchimedean.*

Proof. Frobenius and the triangle inequality give, for every $m \geq 0$,

$$|x+y|^{p^m} = |x^{p^m} + y^{p^m}| \leq 2\max(|x|,|y|)^{p^m}.$$

Taking p^m-th roots and letting $m \to \infty$ proves the assertion. □

Lemma A.3. *Let $\mathbb{K}$ be a complete nontrivially normed field and U a finite-dimensional normed $\mathbb{K}$-space. Then every linear functional on U is bounded, and U is complete.*

Proof. We prove both assertions by induction on $n = \dim U$, the case $n = 0$ being immediate. Suppose $n \geq 1$ and let $g \neq 0$ be a linear functional on U. Its kernel H has dimension $n-1$, so the induction hypothesis makes H complete and therefore closed in U. Choose v with $g(v) = 1$. Since $v \notin H$, $\delta := \inf_{h\in H}\|v-h\| > 0$. Write $x = g(x)v + h$ with $h \in H$. If $g(x) \neq 0$, then

$$\|x\| = |g(x)|\,\|v + g(x)^{-1}h\| \geq |g(x)|\delta.$$

Thus $|g(x)| \leq \delta^{-1}\|x\|$ for every x, proving boundedness.

Choose a basis of U. Its coordinate functionals are now bounded, so the coordinate map $U \to \mathbb{K}^n$, with the max norm on $\mathbb{K}^n$, is bounded. The inverse map $(c_i) \mapsto \sum_i c_i e_i$ is bounded by $\sum_i \|e_i\|$. Hence U is bi-Lipschitz equivalent to the complete space $\mathbb{K}^n$, and is itself complete. □

When the scalar field needs to be displayed, $A^{k,L}$ denotes the degree-k precomposition action formed over L. For the precomposition action A, a *bounded lift* means a bounded k-linear map on its operator domain whose diagonal is A, as in Proposition 2.5. That proposition applies at every base point; Lemma A.1 provides the one-variable coefficient comparison used in its proof.

Appendix B. The multiplier theorem over a finite field

This appendix proves Theorem B.9: over a finite field with $k! = 0$, coordinatewise multiplication has no lift satisfying (Pol1) with uniformly bounded support dimension. Ramsey's theorem reduces the coefficients to cluster values (Lemmas B.10, B.11), (Pol1) makes them nonconstant (Lemma B.13), and the support bound forbids the resulting staircase (Lemma B.12).

B.1. **Exterior powers, supports and determinant arrays.** Let L be a field, V an L-vector space and $k \geq 1$. We write $\Lambda^k V = \Lambda^k_L V$ for the algebraic exterior power, in which $x_1 \wedge \cdots \wedge x_k = 0$ whenever two x_i coincide. We use the standard facts: for a subspace $W \subseteq V$ the map $\Lambda^k W \to \Lambda^k V$ is injective, and we regard $\Lambda^k W$ as a subspace of $\Lambda^k V$; if $u_1, \ldots, u_d$ is a basis of a finite-dimensional space U, the wedges $u_I := u_{i_1} \wedge \cdots \wedge u_{i_k}$, $I = \{i_1 < \cdots < i_k\}$, form a basis of $\Lambda^k U$; and every $\omega \in \Lambda^k V$ lies in $\Lambda^k U$ for some finite-dimensional $U \subseteq V$.

Definition B.1. The *support dimension* of $\omega \in \Lambda^k V$ is

$$\operatorname{sdim}(\omega) := \min\{\dim W : \ W \subseteq V \text{ finite-dimensional}, \ \omega \in \Lambda^k W\}.$$

The definition gives $\operatorname{sdim}(0) = 0$, $\operatorname{sdim}(c\omega) = \operatorname{sdim}(\omega)$ for $c \neq 0$, $\operatorname{sdim}(\omega + \omega') \leq \operatorname{sdim}(\omega) + \operatorname{sdim}(\omega')$ (use $W + W'$), and $\operatorname{sdim}(\omega) \leq \dim \operatorname{span} S$ if ω is a sum of wedges of vectors from a finite set S. If $V \subseteq V'$, then sdim computed in V' is at most sdim computed in V.

For spaces of functions, exterior vectors can be detected by determinants of coordinate evaluations. These arrays also turn bounds on support dimension into ordinary matrix-rank bounds. Let S be a set and $V \subseteq L^S$ a space of functions on S. The *determinant array* is the linear map

$$\Omega\colon \Lambda^k V \to L^{S^k}, \qquad \Omega_{y_1 \wedge \cdots \wedge y_k}(c_1, \ldots, c_k) := \det\big(y_b(c_a)\big)_{a,b=1}^{k}.$$

It is well defined because the right side is multilinear and alternating in $(y_1, \ldots, y_k)$ (these are the columns). For each ω, the function Ω_ω is alternating in $(c_1, \ldots, c_k)$ (the rows); in particular $\Omega_\omega(c_{\sigma(1)}, \ldots, c_{\sigma(k)}) = \operatorname{sgn}(\sigma)\, \Omega_\omega(c_1, \ldots, c_k)$.

Lemma B.2. *The determinant array $\Omega\colon \Lambda^k V \to L^{S^k}$ is injective.*

Proof. Let $\Omega_\omega = 0$ and choose a finite-dimensional $U \subseteq V$ with $\omega \in \Lambda^k U$. The evaluations $\operatorname{ev}_s|_U$, $s \in S$, separate the points of U (a function in U vanishing at every s is 0), so they span the dual space U^*. Choose $s_1, \ldots, s_d \in S$ such that the $\operatorname{ev}_{s_i}|_U$ form a basis of U^*, and let $u_1, \ldots, u_d$ be the dual basis of U, so that $u_j(s_i) = \delta_{ij}$. For increasing k-subsets I, J of $\{1, \ldots, d\}$ we get $\Omega_{u_I}(s_{j_1}, \ldots, s_{j_k}) = \det(u_{i_b}(s_{j_a}))_{a,b} = \delta_{IJ}$. Writing $\omega = \sum_I c_I u_I$ therefore gives $0 = \Omega_\omega(s_{j_1}, \ldots, s_{j_k}) = c_J$ for every J, so $\omega = 0$. □

Lemma B.3 (support bounds flattening rank)**.** *Let $V \subseteq L^S$, $\omega \in \Lambda^k V$ and $\alpha \in \{1, \ldots, k\}$. Let $R \subseteq S$ and $R' \subseteq S^{\{1,\ldots,k\}\setminus\{\alpha\}}$ be finite. The $R \times R'$ matrix M with entries $M(c_\alpha, c') := \Omega_\omega(c)$, where c has c_α in slot α and the entries of c' in the other slots, has rank at most* $\operatorname{sdim}(\omega)$.

Proof. Write $\omega = \sum_j y_{j,1} \wedge \cdots \wedge y_{j,k}$ with all $y_{j,b}$ in a subspace W of dimension $\operatorname{sdim}(\omega)$. Expanding each determinant along row α, $\det(y_{j,b}(c_a))_{a,b} = \sum_b \pm y_{j,b}(c_\alpha) \cdot m_{j,b}(c')$, where the minor $m_{j,b}(c')$ does not involve c_α. So for fixed c' the column $c_\alpha \mapsto M(c_\alpha, c')$ is a linear combination of the restrictions $y_{j,b}|_R$. All columns lie in the restriction of W to R, a space of dimension at most $\operatorname{sdim}(\omega)$. □

A second way to obtain lower bounds on support is to contract all but one exterior factor. The resulting vectors must lie in every subspace supporting the original exterior vector.

Lemma B.4 (contraction)**.** *Let $k \geq 2$ and $\varphi = (\varphi_1, \ldots, \varphi_{k-1}) \in (V^*)^{k-1}$. There is a linear map $c_\varphi\colon \Lambda^k V \to V$ with*

$$c_\varphi(x_1 \wedge \cdots \wedge x_k) = \sum_{i=1}^{k} (-1)^{k+i} \det\big(\varphi_a(x_b)\big)_{1 \leq a \leq k-1,\ b \neq i}\, x_i,$$

and $c_\varphi(\Lambda^k W) \subseteq W$ for every subspace W. Consequently $\operatorname{sdim}(\omega) \geq \dim \operatorname{span}\{c_\varphi(\omega) : \varphi \in (V^*)^{k-1}\}$.

Proof. The right side is multilinear in (x_b). For $\psi \in V^*$, applying ψ to it gives, by Laplace expansion along the last row, the determinant of the $k \times k$ matrix with entries $\varphi_a(x_b)$ in the rows $a < k$ and $\psi(x_b)$ in the last row; it vanishes when two columns coincide. Since V^* separates the points of V, the right side vanishes whenever $x_a = x_b$ for some $a \neq b$, so it factors through $\Lambda^k V$. Each term is a multiple of some x_i, which gives $c_\varphi(\Lambda^k W) \subseteq W$. If $\omega \in \Lambda^k W$ with $\dim W = \operatorname{sdim}(\omega)$, all $c_\varphi(\omega)$ lie in W. □

Proposition B.5 (canonical exterior support)**.** *Let L be any field, V an L-vector space, $k \geq 2$, and $\omega \in \Lambda^k V$. Then*

$$S(\omega) := \operatorname{span}\{c_\varphi(\omega) : \varphi \in (V^*)^{k-1}\}$$

is the unique smallest subspace supporting ω. In particular, $\omega \in \Lambda^k S(\omega)$ and $\operatorname{sdim}(\omega) = \dim S(\omega)$.

Proof. Choose a finite-dimensional supporting subspace $U \subseteq V$. Lemma B.4 gives $S(\omega) \subseteq U$, so $S(\omega)$ is finite-dimensional. Choose a basis $e_1, \dots, e_d$ of U whose first r vectors form a basis of $S(\omega)$, and extend its coordinate functionals algebraically to V. Write $\omega = \sum_{|I|=k} a_I e_I$ in the exterior basis of U. If $j \in I$ and $j > r$, contract against the $k-1$ coordinate functionals indexed by $I \setminus \{j\}$. The e_j-coordinate of this contraction is $\pm a_I$: no other basis wedge contributes to that coordinate. The contraction belongs to $S(\omega)$, so this coordinate is zero. Hence $a_I = 0$ whenever I meets $\{r+1, \dots, d\}$, proving $\omega \in \Lambda^k S(\omega)$. Every supporting subspace contains all the contractions by Lemma B.4, and therefore contains $S(\omega)$. This proves minimality and the dimension formula. Only unit signs occur, so the argument is valid in every characteristic. □

The same conclusion holds in degree one: under $\Lambda^1 V \simeq V$, the empty-tuple contraction is the identity and $S(\omega) = L\omega$. In degree zero, every element of $\Lambda^0 V = L$ is supported by the zero subspace, so its support dimension is zero; no contraction to V is needed.

For matrix calculations, choose any finite-dimensional supporting subspace U and a basis of it. Form the finite matrix whose columns are the coordinates of the contractions against $(k-1)$-tuples of coordinate functionals, extended to V. Arbitrary functionals restrict to linear combinations of these coordinates on U; multilinearity in the covectors therefore shows that the column span is exactly $S(\omega)$. The matrix consequently has rank $\operatorname{sdim}(\omega)$. This uses ordinary matrix rank, and entails no choice of a rank convention for an infinite matrix.

B.2. **Lifts and three diagonal identities.** Fix a field L, an integer $k \geq 1$, L-vector spaces A and V, and a linear map $\mu\colon A \to \operatorname{End}_L(V)$, $a \mapsto \mu_a$. A *lift of μ in degree k* is a $2k$-linear map $\Psi\colon A^k \times V^k \to \Lambda^k V$ over L that is alternating in the V-slots; its diagonal identity will be specified separately. For every $n \geq 1$ write $[n] := \{1, \dots, n\}$, and for $\alpha \in \mathbb{N}^m$ put $|\alpha| := \sum_i \alpha_i$. Given $m \geq 1$, a map $f\colon [k] \to [m]$, a family $b = (b_1, \dots, b_m) \in A^m$ and $\lambda \in L^m$, put

$$b_f := (b_{f(1)}, \dots, b_{f(k)}), \quad \operatorname{type} f := \big(|f^{-1}(1)|, \dots, |f^{-1}(m)|\big) \in \mathbb{N}^m, \quad \lambda_f := \prod_{j=1}^{k} \lambda_{f(j)} = \lambda^{\operatorname{type} f}.$$

The *pointwise identity* is

$$\Psi(a, \dots, a; x_1, \dots, x_k) = \mu_a x_1 \wedge \cdots \wedge \mu_a x_k \qquad \text{for all } a \in A,\ x \in V^k. \tag{Pw}$$

The *polarized identity* is: for all $m \geq 1$, $b \in A^m$, $\alpha \in \mathbb{N}^m$ with $|\alpha| = k$, and $x \in V^k$,

$$\sum_{\operatorname{type} f = \alpha} \Psi(b_f; x) = \sum_{\operatorname{type} f = \alpha} \mu_{b_{f(1)}} x_1 \wedge \cdots \wedge \mu_{b_{f(k)}} x_k. \tag{Pol}$$

Its *multilinear instance* is: for all $b_1, \dots, b_k \in A$ and $x \in V^k$,

$$\sum_{\sigma \in S_k} \Psi(b_{\sigma(1)}, \dots, b_{\sigma(k)}; x) = \sum_{\sigma \in S_k} \mu_{b_{\sigma(1)}} x_1 \wedge \cdots \wedge \mu_{b_{\sigma(k)}} x_k. \tag{Pol1}$$

Here $\sum_{\operatorname{type} f = \alpha}$ runs over the maps $f\colon [k] \to [m]$ of type α. (Pol1) is the instance $m = k$, $\alpha = (1, \dots, 1)$ of (Pol), since the maps of type $(1, \dots, 1)$ are the permutations. For $b \in A^m$ and

$x \in V^k$ put $a_\lambda := \sum_i \lambda_i b_i$ and

$$\Delta_{b,x}(\lambda) := \Psi(a_\lambda, \dots, a_\lambda; x) - \mu_{a_\lambda} x_1 \wedge \cdots \wedge \mu_{a_\lambda} x_k = \sum_{|\alpha|=k} \lambda^\alpha \, \delta_\alpha(b, x),$$

$$\delta_\alpha(b, x) := \sum_{\text{type } f=\alpha} \Big[\Psi(b_f; x) - \mu_{b_{f(1)}} x_1 \wedge \cdots \wedge \mu_{b_{f(k)}} x_k \Big].$$

Multilinearity of Ψ in the A-slots, linearity of μ and multilinearity of the wedge give the displayed expansion. Thus $\Delta_{b,x}$ is a homogeneous polynomial of degree k with coefficients in $\Lambda^k V$. Identity (Pol) says that every coefficient $\delta_\alpha(b, x)$ is zero. Identity (Pw) says instead that $\Delta_{b,x}$ vanishes at every point of L^m; conversely, this pointwise vanishing gives (Pw) by taking $m = 1$ and $\lambda = 1$. The following two identity principles identify the cases in which pointwise vanishing forces all coefficients to vanish.

Lemma B.6 (identity principle)**.** *Let $\mathbb{K}$ be an infinite field, Y a $\mathbb{K}$-vector space and $g(\lambda) = \sum_\alpha \lambda^\alpha y_\alpha$ a polynomial in $\lambda \in \mathbb{K}^m$ with finitely many nonzero coefficients $y_\alpha \in Y$. If $g(\lambda) = 0$ for all $\lambda \in \mathbb{K}^m$, then every $y_\alpha = 0$.*

Proof. The finitely many coefficients y_α span a finite-dimensional subspace of Y. Applying its coordinate functionals reduces the claim to scalar polynomials. For these, induction on m reduces to the fact that a nonzero polynomial in one variable has only finitely many roots. □

Lemma B.7 (homogeneous forms over a finite field)**.** *Let q be a prime power, $L' \supseteq \mathbb{F}_q$ a field, $1 \le e \le q$, and $g \in L'[\lambda_1, \dots, \lambda_m]$ homogeneous of degree e with $g(\lambda) = 0$ for every $\lambda \in \mathbb{F}_q^m$. Then $g = 0$.*

Proof. Call a monomial λ^β *reduced* if every $\beta_i \le q-1$. The indicator function of a point $a \in \mathbb{F}_q^m$ is $\prod_i \big(1 - (\lambda_i - a_i)^{q-1}\big)$, which expands into reduced monomials. So the q^m reduced monomials span the q^m-dimensional L'-space of functions $\mathbb{F}_q^m \to L'$, and they are linearly independent as functions. A monomial of degree $e \le q$ that is not reduced is λ_i^q (and then $e = q \ge 2$); as a function on $\mathbb{F}_q^m$ it agrees with the reduced monomial λ_i, of degree $1 < e$. If $e = q$, replace each λ_i^q in g by λ_i. These degree-one images are distinct, and homogeneity separates them from all the remaining monomials, which still have degree q. The replacement therefore sends the monomials of g injectively to reduced monomials without changing the associated function. Linear independence of the reduced monomials forces every coefficient to vanish. If $e < q$, no replacement is needed. □

Proposition B.8 (pointwise versus polarized)**.** *Let Ψ be a lift of μ in degree k over a field L.*

1. (Pol)⇒(Pw)⇒(Pol1)*, over every field.*
2. *If L is infinite, or if $L = \mathbb{F}_q$ and $k \le q$, then* (Pw)⇒(Pol)*.*
3. *If $L = \mathbb{F}_q$ and $k \ge q + 1$, then* (Pw) *does not imply* (Pol) *in general.*
4. *If $k! = 0$ in L, then* (Pol1) *does not imply* (Pw) *in general.*

Proof. (1) Inclusion–exclusion. The implication (Pol)⇒(Pw) is the case $m = 1$ and $\alpha = (k)$. To prove (Pw)⇒(Pol1), fix $b_1, \dots, b_k \in A$ and $x \in V^k$, and put $b_S := \sum_{i \in S} b_i$ for $S \subseteq [k]$. Multilinearity gives

$$\sum_{S \subseteq [k]} (-1)^{k-|S|} \Psi(b_S, \dots, b_S; x) = \sum_S (-1)^{k-|S|} \sum_{f\colon [k] \to S} \Psi(b_f; x)$$

$$= \sum_{f\colon [k] \to [k]} \Psi(b_f; x) \sum_{S \supseteq f([k])} (-1)^{k-|S|} = \sum_{f \text{ bijective}} \Psi(b_f; x),$$

because $\sum_{S\supseteq T}(-1)^{k-|S|} = (1-1)^{k-|T|}$ is 1 if $T=[k]$ and 0 otherwise. The same computation applies to the multilinear map $(a_1,\dots,a_k)\mapsto \mu_{a_1}x_1\wedge\cdots\wedge\mu_{a_k}x_k$, whose diagonal is the right side of (Pw). Evaluating (Pw) at the 2^k subset sums b_S therefore gives (Pol1). The computation is valid in every characteristic and involves no division by $k!$.

(2) The identity principles. The coefficients of $\Delta_{b,x}$ span a finite-dimensional subspace of $\Lambda^k V$; in a basis of it each coordinate of $\Delta_{b,x}$ is a scalar form of degree k vanishing on L^m. It is zero by Lemma B.6 if L is infinite, and by Lemma B.7 if $L=\mathbb{F}_q$ and $k\le q$.

(3) Pointwise vanishing without full polarization. Let $\dim A\ge 3$, with $b_1,b_2,b_3\in A$ and functionals g_1,g_2,g_3 on A such that $g_i(b_j)=\delta_{ij}$, and let $y_1,\dots,y_k\in V$ be linearly independent. Put

$$\theta(a_1,\dots,a_k) := \big[g_1(a_1)\cdots g_1(a_q)\,g_2(a_{q+1}) - g_1(a_1)\,g_2(a_2)\cdots g_2(a_{q+1})\big]\cdot g_3(a_{q+2})\cdots g_3(a_k)$$

(for $k=q+1$ the g_3-factor is empty and $\dim A\ge 2$ suffices), and $\Theta(a;x) := \theta(a)\,x_1\wedge\cdots\wedge x_k$. Then Θ is a lift of $\mu=0$. It satisfies (Pw), because $\theta(a,\dots,a) = (g_1(a)^q g_2(a) - g_1(a)g_2(a)^q)\,g_3(a)^{k-q-1} = 0$ since $c^q=c$ for $c\in\mathbb{F}_q$. It violates (Pol) for $\alpha=(q,1,k-q-1)$ and $b=(b_1,b_2,b_3)$: the first product contributes to $\sum_{\text{type } f=\alpha}\theta(b_f)$ exactly for $f=(1,\dots,1,2,3,\dots,3)$, with value 1; the second product would need $f(2)=\cdots=f(q+1)=2$, that is, at least $q\ge 2$ values equal to 2, which is impossible for type α. So the coefficient $\delta_\alpha(b,y)$ of Θ is $y_1\wedge\cdots\wedge y_k\neq 0$. When $k=q+1$, the third label and its functional can be omitted throughout, with $\alpha=(q,1)$.

(4) The multilinear identity without the pointwise identity. Take $\bar a\in A$ and a functional g_1 with $g_1(\bar a)=1$, and $y_1,\dots,y_k$ linearly independent. The lift $\Theta'(a;x) := g_1(a_1)\cdots g_1(a_k)\,x_1\wedge\cdots\wedge x_k$ of $\mu=0$ satisfies (Pol1), since $\sum_\sigma \Theta'(b_\sigma;x) = k!\,g_1(b_1)\cdots g_1(b_k)\,x_1\wedge\cdots\wedge x_k = 0$ for all $b_1,\dots,b_k\in A$ and $x\in V^k$, but $\Theta'(\bar a,\dots,\bar a;y) = y_1\wedge\cdots\wedge y_k\neq 0$. □

The finite-field obstruction below requires only (Pol1), which follows from either (Pw) or (Pol). The distinction between the latter two remains important: over $\mathbb{F}_q$ they need not agree when $k>q$. The lift constructed in Appendix C nevertheless satisfies the full identity (Pol) in every degree: the identity is polarized over an infinite Laurent-series field before constant coefficients are taken (Lemma C.4).

B.3. **Statement and proof strategy.** For the proof of the theorem, assume that L is a *finite* field with $k!=0$ in L, that is, $\operatorname{char} L = p\le k$; in particular $k\ge 2$. Put $V := L^{\mathbb{N}}$ and let $V_{\mathrm{fin}} := L^{(\mathbb{N})}\subseteq V$ be the subspace of finitely supported sequences. For $u,x\in V$ we write ux for the coordinatewise product; so $\mu_u x = ux$ is the multiplier family.

Theorem B.9 (multipliers over a finite field)**.** *Let L be a finite field and $k \geq 1$ with $k! = 0$ in L.*

(1) *There is no $2k$-linear map $\Psi\colon V_{\mathrm{fin}}^k \times V_{\mathrm{fin}}^k \to \Lambda_L^k V$ over L such that*
   (a) *Ψ is antisymmetric in the last k slots;*
   (b) *Ψ satisfies (Pol1) for $\mu_u x = ux$: for all $u_1, \dots, u_k, x_1, \dots, x_k \in V_{\mathrm{fin}}$,*

$$\sum_{\sigma \in S_k} \Psi(u_{\sigma(1)}, \dots, u_{\sigma(k)}; x_1, \dots, x_k) = \sum_{\sigma \in S_k} (u_{\sigma(1)} x_1) \wedge \cdots \wedge (u_{\sigma(k)} x_k);$$

   (c) *for some $d \in \mathbb{N}$, $\operatorname{sdim}(\Psi(u; x)) \leq d$ for all $u, x \in V_{\mathrm{fin}}^k$ with entries in $\{0, \pm 1\}$.*

(2) *Let $E_0 = V$ or $E_0 = V_{\mathrm{fin}}$. There is no $2k$-linear map $\Psi\colon E_0^k \times E_0^k \to \Lambda_L^k E_0$ over L that is antisymmetric in the last k slots, satisfies (Pol1) for all finitely supported u, x, and has $\operatorname{sdim}(\Psi(u; x)) \leq d$ for all $u, x \in E_0^k$, for some d.*

Reduction to (1). Given a map as in (2), restrict its inputs to $V_{\mathrm{fin}}^k \times V_{\mathrm{fin}}^k$ and compose its output with the injective map $\Lambda^k E_0 \to \Lambda^k V$. This preserves antisymmetry and (Pol1). Support dimension can only decrease under the inclusion of the ambient space (§B.1), so the resulting map satisfies (a)–(c).

We therefore prove (1). Suppose that Ψ satisfies its hypotheses, and fix the bound d in (c). Write $e_i \in V_{\mathrm{fin}}$ for the unit vectors, $e_a := (e_{a_1}, \dots, e_{a_k})$ for $a \in \mathbb{N}^k$, and Ω for the determinant array of §B.1 with $S = \mathbb{N}$.

The proof has three steps. Ramsey's theorem first makes the coefficients of $\Omega\Psi$ depend only on the relative order and equalities among their indices. Special weights on four-point clusters then cancel every contribution from an occupied cluster containing no output index. This cancellation produces a two-valued staircase matrix from one exterior vector. Its bounded support forces the two values to agree, and hence makes all cluster order patterns have a common value. Finally, antisymmetry and (Pol1) say that the sum of these $k!$ equal values is 1, contradicting $k! = 0$ in L.

B.4. **Order-homogeneity from Ramsey's theorem.** We use Ramsey's theorem [20] in its infinite form: *for every $n \geq 1$ and every coloring of the n-element subsets of $\mathbb{N}$ with finitely many colors there is an infinite $H \subseteq \mathbb{N}$ all of whose n-element subsets have the same color.*

For a tuple $z \in \mathbb{N}^N$ with distinct values $y_1 < \cdots < y_r$, its *pattern* is the map $i \mapsto j$ determined by $z_i = y_j$. In particular, repeated entries are retained in the pattern. Two tuples z and z' have the same pattern if and only if $z_i < z_j \Leftrightarrow z'_i < z'_j$ and $z_i = z_j \Leftrightarrow z'_i = z'_j$ for all i, j.

Lemma B.10 (pattern homogeneity)**.** *For every $N \geq 1$ and every function $T\colon \mathbb{N}^N \to L$ there is an infinite $H \subseteq \mathbb{N}$ such that $T(z) = T(z')$ whenever $z, z' \in H^N$ have the same pattern.*

Proof. Color an N-element set $s = \{s_1 < \cdots < s_N\} \subseteq \mathbb{N}$ by the vector

$$\big(T(s_{\beta(1)}, \dots, s_{\beta(N)})\big)_{\beta\colon [N] \to [N]} \in L^{N^N}.$$

The coloring records all maps β, including those that repeat indices. It has finitely many values because L is finite; this is the only use of the finiteness of L in the obstruction proof. Ramsey's theorem gives an infinite H on which the color is constant, say equal to $(t_\beta)_\beta$. Let $z \in H^N$ have value set $\{y_1 < \cdots < y_r\}$. Since H is unbounded, add $N - r$ elements of H above y_r to obtain an N-element set $s \subseteq H$. Padding above the largest value leaves the ranks unchanged: $s_j = y_j$ for $j \leq r$, and $z = (s_{\beta(1)}, \dots, s_{\beta(N)})$, where $\beta(i)$ is the rank of z_i in $\{y_1, \dots, y_r\}$. The map β depends only on the pattern of z, and $T(z) = t_\beta$. □

Coefficients. For $(a;y;c)\in\mathbb{N}^k\times\mathbb{N}^k\times\mathbb{N}^k$ put

$$T(a;y;c):=\Omega_{\Psi(e_a;e_y)}(c_1,\dots,c_k)\in L.$$

Lemma B.10 with $N=3k$ gives an infinite set $H=\{h_0<h_1<h_2<\cdots\}$ on which T depends only on the pattern. For $u_j,v_j\in V_{\mathrm{fin}}$ supported in H, multilinearity of Ψ and linearity of Ω give

$$\Omega_{\Psi(u;v)}(c)=\sum_{a,y\in H^k}\prod_{j=1}^{k}u_j(a_j)\prod_{j=1}^{k}v_j(y_j)\cdot T(a;y;c). \tag{B.1}$$

Although the sum is indexed by $H^k\times H^k$, only finitely many summands are nonzero, because all the input vectors have finite support. Every output tuple c used below also lies in H^k, so pattern homogeneity applies to all $3k$ indices in (B.1).

B.5. **Clusters and the cluster value.** A *cluster* is a set $C=\{h_n,h_{n+1},h_{n+2},h_{n+3}\}$ of four *consecutive* elements of H; we write $C(i):=h_{n+i-1}$ for $i=1,\dots,4$. Since a cluster is an interval of H, two disjoint clusters are *comparable*: every element of one is smaller than every element of the other; we write $C<C'$. Distinct clusters need not be adjacent in H.

We choose two weight vectors, one for each kind of input slot:

$$s=(s_1,s_2,s_3,s_4):=(1,0,-1,0) \qquad \text{on the operator slots,}$$
$$w=(w_1,w_2,w_3,w_4):=(1,-1,1,-1) \qquad \text{on the vector slots.}$$

Over $\mathbb{Z}$, hence over every field,

$$\sum_i s_i=\sum_i w_i=\sum_i s_iw_i=\sum_{i<i'}s_iw_{i'}=\sum_{i>i'}s_iw_{i'}=0,\qquad s_1w_1=1. \tag{W}$$

Indeed $\sum_i s_i=1-1$, $\sum_i w_i=1-1+1-1$, $s_1w_1=1$, and

$$\sum_i s_iw_i=s_1w_1+s_3w_3=1-1=0,$$
$$\sum_{i<i'}s_iw_{i'}=s_1(w_2+w_3+w_4)+s_3w_4=-1+1=0,$$
$$\sum_{i>i'}s_iw_{i'}=s_3(w_1+w_2)=-(1-1)=0.$$

The first two zero sums will cancel a cluster occupied by only one input index. The other three will cancel a cluster occupied by an operator index and a vector index, according to their relative order. For a cluster C put

$$s_C:=\sum_{i=1}^{4}s_ie_{C(i)},\qquad w_C:=\sum_{i=1}^{4}w_ie_{C(i)}.$$

These are vectors of V_{fin} with entries in $\{0,\pm1\}$, and $s_Cw_C=e_{C(1)}-e_{C(3)}$.

The cluster value. For pairwise disjoint clusters $C_1,\dots,C_k$, one for each slot, define

$$\chi(C_1,\dots,C_k):=\Omega_{\Psi(s_{C_1},\dots,s_{C_k};\,w_{C_1},\dots,w_{C_k})}\big(C_1(1),\dots,C_k(1)\big)$$
$$=\sum_{i,i'\in[4]^k}\prod_{j=1}^{k}s_{i_j}w_{i'_j}\cdot T\big(C_1(i_1),\dots,C_k(i_k);\,C_1(i'_1),\dots,C_k(i'_k);\,C_1(1),\dots,C_k(1)\big).$$

Let $\tau\in S_k$ be the *order pattern* of $(C_1,\dots,C_k)$: $\tau(j)$ is the position of C_j among $C_1,\dots,C_k$. The pattern of every $3k$-tuple in the sum is determined by (i,i') and τ: entries in different clusters compare as the clusters do, and the entries $C_j(i_j)$, $C_j(i'_j)$, $C_j(1)$ in the same cluster

compare as their indices do, including equalities. Pattern homogeneity therefore shows that the displayed sum depends only on τ. Every permutation occurs as the order pattern of some disjoint clusters in H; we denote the resulting scalar by $\chi(\tau)$.

Lemma B.11 (cancellation of output-free clusters)**.** *Let $\mathcal{C}_1, \dots, \mathcal{C}_k$ be finite families of clusters that are pairwise disjoint as sets of clusters, such that all clusters in $\bigcup_j \mathcal{C}_j$ are pairwise disjoint. Put $u_j := \sum_{C \in \mathcal{C}_j} s_C$ and $v_j := \sum_{C \in \mathcal{C}_j} w_C$. Let $C_l \in \mathcal{C}_l$ for $l = 1, \dots, k$ and $c := (C_1(1), \dots, C_k(1))$. Then $\Omega_{\Psi(u;v)}(c) = \chi(C_1, \dots, C_k)$.*

Proof. We show that every cluster assignment except the one matching the output clusters has total contribution zero. Expand by (B.1). Since the clusters are disjoint, a nonzero term has, for each slot j, an operator index $a_j = C_j^{\mathrm{op}}(i_j)$ and a vector index $y_j = C_j^{\mathrm{vec}}(i'_j)$ with $C_j^{\mathrm{op}}, C_j^{\mathrm{vec}} \in \mathcal{C}_j$, and weight $\prod_j s_{i_j} w_{i'_j}$. Group the terms according to the *cluster assignment* $(C_j^{\mathrm{op}}, C_j^{\mathrm{vec}})_j$.

Call an occupied cluster *output-free* if it is none of $C_1, \dots, C_k$. Suppose a group has such a cluster Γ. Disjointness of the families assigns Γ to exactly one slot j, so the only tuple entries it can contain are a_j and y_j. Disjointness of the clusters excludes every output index from Γ.

Fix the positions of all indices outside Γ and sum over the positions inside it. Since Γ consists of consecutive elements of H, every outside entry lies either below the whole cluster or above it. Moving an index within Γ cannot change any comparison with an outside entry. By pattern homogeneity, the coefficient T therefore depends on the internal positions only as follows:

(i) if only $a_j \in \Gamma$, not at all, and the partial sum is $(\sum_i s_i) \cdot T = 0$;
(ii) if only $y_j \in \Gamma$, not at all, and the partial sum is $(\sum_i w_i) \cdot T = 0$;
(iii) if both lie in Γ, only through whether $i_j < i'_j$, $i_j = i'_j$ or $i_j > i'_j$; with the corresponding values $T_<, T_=, T_>$ of T the partial sum is

$$T_< \sum_{i<i'} s_i w_{i'} + T_= \sum_i s_i w_i + T_> \sum_{i>i'} s_i w_{i'} = 0.$$

In each case the weights of the other slots are a common factor, and the displayed partial sum vanishes by (W). Thus every assignment with an output-free cluster contributes zero. In any surviving assignment, every occupied cluster must be an output cluster. The only output cluster belonging to $\mathcal{C}_j$ is C_j, which forces $C_j^{\mathrm{op}} = C_j^{\mathrm{vec}} = C_j$ for every j. The contribution of this one assignment is exactly $\chi(C_1, \dots, C_k)$. □

B.6. **The staircase and the diagonal.** We now use the support bound to compare two order patterns that differ by an adjacent exchange. The key is to realize arbitrarily large staircase matrices as flattenings of a single value of Ψ.

Lemma B.12 (staircase)**.** *Let $\tau \in S_k$, let $\alpha \neq \beta$ be slots with $\tau(\alpha) = \ell$ and $\tau(\beta) = \ell + 1$, and let τ' be τ with these two positions exchanged, so $\tau'(\alpha) = \ell + 1$, $\tau'(\beta) = \ell$ and $\tau'(\gamma) = \tau(\gamma)$ otherwise. Then $\chi(\tau) = \chi(\tau')$.*

Proof. Let $m \geq 1$. Choose pairwise disjoint clusters, in increasing order in H: first a cluster C_γ for each slot γ with $\tau(\gamma) < \ell$, ordered as τ prescribes; then $2m$ clusters $A_1 < B_1 < A_2 < B_2 < \cdots < A_m < B_m$; then a cluster C_γ for each slot γ with $\tau(\gamma) > \ell + 1$, ordered as τ prescribes. Put $\mathcal{C}_\alpha := \{A_1, \dots, A_m\}$, $\mathcal{C}_\beta := \{B_1, \dots, B_m\}$ and $\mathcal{C}_\gamma := \{C_\gamma\}$ for the other slots, and let u, v be as in Lemma B.11 and $\omega := \Psi(u; v)$. Disjointness of the clusters ensures that the entries of u and v still lie in $\{0, \pm 1\}$, so $\mathrm{sdim}(\omega) \leq d$ by hypothesis (c) of Theorem B.9(1). For this fixed m, keep u, v and hence ω fixed while varying only the output indices.

Consider the $m \times m$ matrix M with rows indexed by $c_\alpha = A_n(1)$ and columns by $c_\beta = B_{n'}(1)$ $(n, n' = 1, \dots, m)$, all other output indices being fixed at $c_\gamma = C_\gamma(1)$, and entries $M_{nn'} := \Omega_\omega(c)$.

By Lemma B.11, applied with the output clusters $A_n \in \mathcal{C}_\alpha$, $B_{n'} \in \mathcal{C}_\beta$ and $C_\gamma \in \mathcal{C}_\gamma$, $M_{nn'}$ is the cluster value of the tuple with A_n in slot α, $B_{n'}$ in slot β, and C_γ in every other slot. The interlacing order gives $A_n < B_{n'}$ exactly when $n \le n'$. The tuple therefore has pattern τ on and above the matrix diagonal, and pattern τ' below it.

Put $\delta := \chi(\tau) - \chi(\tau')$, let $\mathbf{1} \in L^m$ be the all-ones column, and let U be the upper unitriangular matrix with $U_{nn'} = 1$ for $n \le n'$ and $U_{nn'} = 0$ otherwise. Then

$$M = \begin{pmatrix} \chi(\tau) & \chi(\tau) & \cdots & \chi(\tau) \\ \chi(\tau') & \chi(\tau) & \cdots & \chi(\tau) \\ \vdots & \ddots & \ddots & \vdots \\ \chi(\tau') & \cdots & \chi(\tau') & \chi(\tau) \end{pmatrix} = \chi(\tau')\,\mathbf{1}\mathbf{1}^{\mathsf{T}} + \delta\, U,$$

The rows vary only slot α; each column fixes all the other slots, including the chosen value in slot β. Thus M is a submatrix of the flattening of this one array Ω_ω along slot α. Lemma B.3 gives $\operatorname{rank} M \le \operatorname{sdim}(\omega) \le d$.

If $\delta \neq 0$, the matrix δU is invertible, whereas the constant matrix $\chi(\tau')\mathbf{1}\mathbf{1}^{\mathsf{T}}$ has rank at most one. Applying rank subadditivity to $\delta U = M - \chi(\tau')\mathbf{1}\mathbf{1}^{\mathsf{T}}$ gives

$$m = \operatorname{rank}(\delta U) \le \operatorname{rank} M + 1 \le d + 1.$$

The construction is available for every $m \ge 1$; choosing $m = d + 2$ is a contradiction. Hence $\delta = 0$, as required. □

The support bound has now forced equality between adjacent order patterns. The remaining identity comes from (Pol1); it does not use the support bound.

Lemma B.13 (the diagonal)**.** $\sum_{\tau \in S_k} \chi(\tau) = 1$.

Proof. Take clusters $C_1 < \cdots < C_k$, and put $u_j := s_{C_j}$, $v_j := w_{C_j}$ and $c := (C_1(1), \dots, C_k(1))$. Apply $\omega \mapsto \Omega_\omega(c)$ to both sides of (Pol1), with the operator vectors $u_1, \dots, u_k$ and the vector arguments $v_1, \dots, v_k$.

Right side. If $\sigma \neq \mathrm{id}$, some slot j has $\sigma(j) \neq j$. Disjoint supports then give $u_{\sigma(j)} v_j = 0$, so the corresponding wedge vanishes. Only $\sigma = \mathrm{id}$ survives. It gives $\Omega_{(u_1 v_1) \wedge \cdots \wedge (u_k v_k)}(c) = \det\big((u_b v_b)(C_a(1))\big)_{a,b}$. Since $u_b v_b = e_{C_b(1)} - e_{C_b(3)}$ and $C_a(1) \neq C_b(3)$ for all a, b, this is the determinant of the identity matrix, so the right side is 1.

Left side. Fix $\sigma \in S_k$ and write $u_\sigma := (u_{\sigma(1)}, \dots, u_{\sigma(k)})$ and $v_\sigma := (v_{\sigma(1)}, \dots, v_{\sigma(k)})$. To identify the summand with a cluster value, we must permute both the vector arguments and the output coordinates to match the operator arguments. Each permutation contributes the same sign. More precisely, antisymmetry of Ψ in the vector slots gives $\Psi(u_\sigma; v) = \operatorname{sgn}(\sigma)\Psi(u_\sigma; v_\sigma)$. For every $\omega \in \Lambda^k V$ the function Ω_ω is alternating in its arguments, so $\Omega_\omega(c) = \operatorname{sgn}(\sigma)\,\Omega_\omega(c_{\sigma(1)}, \dots, c_{\sigma(k)})$; we apply this with $\omega = \Psi(u_\sigma; v_\sigma)$. Since $\operatorname{sgn}(\sigma)^2 = 1$,

$$\begin{aligned} \Omega_{\Psi(u_\sigma; v)}(c) &= \operatorname{sgn}(\sigma)\,\Omega_{\Psi(u_\sigma; v_\sigma)}(c) = \operatorname{sgn}(\sigma)^2\,\Omega_{\Psi(u_\sigma; v_\sigma)}(c_{\sigma(1)}, \dots, c_{\sigma(k)}) \\ &= \Omega_{\Psi(u_\sigma; v_\sigma)}(c_{\sigma(1)}, \dots, c_{\sigma(k)}). \end{aligned}$$

In the last expression slot j has operator $s_{C_{\sigma(j)}}$, vector $w_{C_{\sigma(j)}}$ and output $C_{\sigma(j)}(1)$, so it equals $\chi(C_{\sigma(1)}, \dots, C_{\sigma(k)})$ by definition. Its order pattern is σ, since $C_{\sigma(j)}$ is the $\sigma(j)$-th cluster in increasing order. Thus the convention for patterns gives σ, not σ^{-1}. No cluster cancellation is needed in this calculation.

As σ runs over S_k, each order pattern occurs exactly once. Comparing the two sides gives

$$\sum_{\sigma \in S_k} \Omega_{\Psi(u_\sigma; v)}(c) = \sum_{\tau \in S_k} \chi(\tau) = 1.$$ □

Proof of Theorem B.9. Exchanging adjacent ranks acts on an order pattern by $\tau \mapsto (\ell\ \ell{+}1) \circ \tau$, $1 \le \ell < k$. These transpositions generate S_k, and Lemma B.12 makes χ invariant under each exchange. Hence χ is constant; write its value as χ_0. Lemma B.13 now gives $k!\,\chi_0 = 1$ in L, contradicting $k! = 0$. This proves (1), and the preceding reduction gives (2). □

Remark B.14 (where the hypotheses enter). Ramsey homogeneity is the only step in the obstruction proof that uses the finiteness of L. Multilinearity then gives the cluster expansion and cancellation. The support bound is used in Lemma B.12 only at the finitely many test inputs obtained by taking $m = d + 2$; all have entries in $\{0, \pm 1\}$. Antisymmetry and (Pol1) enter in Lemma B.13, which does not use that bound. Finally, $k! = 0$ is used only to turn the diagonal identity into a contradiction.

The inclusion–exclusion argument in Proposition B.8(1) uses only multilinearity in the operator slots. The theorem therefore remains true with the pointwise identity (Pw) at L-points in place of (Pol1), for this family of multipliers. In characteristic 2 antisymmetry is the same as symmetry, and the two signs in Lemma B.13 are trivial.

Remark B.15 (when $k!$ is invertible). The vanishing of $k!$ is essential. If instead $k! \neq 0$ in L, the *normalized alternatization*

$$\Psi_{\mathrm{alt}}(u; x) := \frac{1}{k!} \sum_{\sigma \in S_k} (u_{\sigma(1)} x_1) \wedge \cdots \wedge (u_{\sigma(k)} x_k)$$

is alternating in x and satisfies (Pol1). Every summand is supported in the span of the k^2 vectors $u_i x_j$, so $\operatorname{sdim}(\Psi_{\mathrm{alt}}(u; x)) \le k^2$. The preceding homogeneity, cancellation, staircase and diagonal arguments apply, but now yield $\chi \equiv 1/k!$: for pairwise disjoint clusters only $\sigma = \mathrm{id}$ survives in $\Psi_{\mathrm{alt}}(s_{C_1}, \dots, s_{C_k}; w_{C_1}, \dots, w_{C_k})$, and the determinant at $(C_1(1), \dots, C_k(1))$ is that of the identity. The final equation $k! \cdot (1/k!) = 1$ is then consistent. So the obstruction is exactly the vanishing of $k!$, not positive characteristic as such. For example, at $(p, k) = (5, 3)$ every step is consistent, since $3! = 6 = 1$ in $\mathbb{F}_5$ and $\chi_0 = 1$.

Appendix C. The Banach exterior target over a Laurent series field

We now realize the finite-field obstruction of Theorem B.9 in a Banach-space setting. The space F_1 of Theorem C.1 is $\widehat{\Lambda}^k_\pi E_1$, the test target of Lemma 3.4 for the pair (E_1, E_1), so Theorem C.1 says that (E_1, E_1) is not good for Banach values in degree k (Corollary 7.6).

Theorem C.1. *Let κ be a finite field of characteristic p, let $k \ge p$ and $\varrho \in (0, 1)$, and let $K_1 = \kappa((X))$ with $|X| = \varrho$. Let $E_1 = \ell^\infty(\mathbb{N}, K_1)$, and let F_1 be the completion of $\Lambda^k_{K_1} E_1$ under the projective exterior norm. Then $A^{k,K_1}_{E_1,E_1;F_1}$ admits no bounded k-linear lift over K_1. Consequently it is analytic at no point of $\mathcal{L}(E_1, E_1)$.*

C.1. **Setting.** Throughout this section $k \ge 1$, κ is a field, $\varrho \in (0, 1)$, and $K_1 := \kappa((X))$ is the Laurent series field with absolute value $|a| := \varrho^{\operatorname{ord} a}$ ($|0| = 0$). This field is complete, nontrivially normed and nonarchimedean, and $|c| = 1$ for every $c \in \kappa^\times$. Write $\operatorname{coeff}_n(a) \in \kappa$ for the X^n-coefficient of $a \in K_1$. For a set S and $y \in \ell^\infty(S, K_1)$ we take the constant coefficient coordinatewise:

$$\operatorname{coeff}_0(y) := \big(\operatorname{coeff}_0(y(s))\big)_{s \in S} \in \kappa^S.$$

The resulting *constant-coefficient map* coeff_0 is κ-linear but not K_1-linear. A nonzero Laurent series of absolute value less than one has order at least one. Consequently,

$$\text{(C.1)}\quad \|y\|_\infty < 1 \implies \operatorname{coeff}_0(y) = 0, \qquad \text{hence} \qquad \|y - y'\|_\infty < 1 \implies \operatorname{coeff}_0(y) = \operatorname{coeff}_0(y').$$

Put $E_1 := \ell^\infty(\mathbb{N}, K_1)$ with the sup norm, a nonarchimedean K_1-Banach space, and let $E_0 := \kappa^{\mathbb{N}} \subseteq E_1$ be the κ-subspace of sequences with entries in κ. Every nonzero $z \in E_0$ has $\|z\|_\infty = 1$. For $x \in E_1$ and $n \in \mathbb{Z}$ let $x_{[n]} := (\operatorname{coeff}_n(x(i)))_{i\in\mathbb{N}} \in E_0$ be the n-th Laurent component. For $x \neq 0$ the set $\{\operatorname{ord} x(i) : x(i) \neq 0\}$ is bounded below, because $\varrho^{\operatorname{ord} x(i)} \leq \|x\|_\infty$, so it has a minimum $\nu(x)$; then $\|x\|_\infty = \varrho^{\nu(x)}$ and $x_{[n]} = 0$ for $n < \nu(x)$. Finally put

$$M_\varrho := \max_{l\in\mathbb{N}} (l+1)\varrho^l.$$

The maximum exists because $(l+1)\varrho^l \to 0$. This constant will control the number of Laurent components that can contribute to a constant coefficient. For $\varrho \leq \frac{1}{2}$ one has $(l+1)\varrho^l \leq (l+1)2^{-l} \leq 1$, so $M_\varrho = 1$.

C.2. **The projective exterior norm.** On $\Lambda := \Lambda^k_{K_1} E_1$ put

$$\|\omega\|_\pi := \inf\Big\{\sum_j \prod_{i=1}^k \|x_{j,i}\|_\infty \;:\; \omega = \sum_j x_{j,1} \wedge \cdots \wedge x_{j,k}\Big\},$$

where the infimum is over finite decompositions and the cost is an ordinary sum. Let $\Omega^{K_1} \colon \Lambda \to \ell^\infty(\mathbb{N}^k, K_1)$ be the determinant array of §B.1 with $L = K_1$, $S = \mathbb{N}$ and $V = E_1$, so that $\Omega^{K_1}(x_1 \wedge \cdots \wedge x_k)(i_1, \ldots, i_k) = \det(x_b(i_a))_{a,b}$; its values are bounded by the next lemma.

Lemma C.2. *$\|\cdot\|_\pi$ is a K_1-norm on Λ with $\|x_1 \wedge \cdots \wedge x_k\|_\pi \leq \prod_i \|x_i\|_\infty$, and Ω^{K_1} is K_1-linear and injective with $\|\Omega^{K_1}(\omega)\|_\infty \leq \|\omega\|_\pi$. Let F_1 be the completion of $(\Lambda, \|\cdot\|_\pi)$, a K_1-Banach space, $J \colon F_1 \to \ell^\infty(\mathbb{N}^k, K_1)$ the continuous extension of Ω^{K_1} (so $\|J\| \leq 1$), and $\omega_{E_1} \colon E_1^k \to F_1$, $\omega_{E_1}(x) := x_1 \wedge \cdots \wedge x_k$. Then $\omega_{E_1} \in \operatorname{Alt}^k_{K_1}(E_1; F_1)$, $\|\omega_{E_1}\| \leq 1$, and $J(\omega_{E_1}(x)) = \Omega^{K_1}(x_1 \wedge \cdots \wedge x_k)$.*

Proof. For a pure wedge, each entry of the determinant array is a signed sum of products $x_1(i_{\sigma(1)}) \cdots x_k(i_{\sigma(k)})$. Each product has absolute value at most $\prod_b \|x_b\|_\infty$, so the nonarchimedean inequality in K_1 gives the same bound for the determinant. For any finite decomposition of ω, it follows that

$$\|\Omega^{K_1}(\omega)\|_\infty \leq \max_j \prod_i \|x_{j,i}\|_\infty \leq \sum_j \prod_i \|x_{j,i}\|_\infty.$$

Thus the array is bounded, and taking the infimum gives $\|\Omega^{K_1}(\omega)\|_\infty \leq \|\omega\|_\pi$. The map Ω^{K_1} is linear and injective by Lemma B.2, with $L = K_1$, $S = \mathbb{N}$ and $V = E_1$. In particular, $\|\omega\|_\pi > 0$ whenever $\omega \neq 0$.

The remaining norm laws follow directly from decompositions. Concatenation gives the triangle inequality. Since $k \geq 1$, a scalar can be absorbed into the first factor of each wedge; for $\lambda \neq 0$ this gives $\|\lambda\omega\|_\pi \leq |\lambda|\|\omega\|_\pi$, and applying the same inequality to λ^{-1} gives equality. The pure-wedge bound follows by using a single summand.

The array space $\ell^\infty(\mathbb{N}^k, K_1)$ is complete, so Ω^{K_1} extends to the contraction J on F_1. The wedge map into F_1 is K_1-multilinear and vanishes whenever two arguments coincide. The pure-wedge bound makes it continuous with norm at most one, and the asserted identity with J holds on its values by construction. □

For $k \geq 2$, F_1 is not nonarchimedean (Proposition E.1).

C.3. **The support estimate.** We next show that the constant coefficient of every completed determinant array is represented by a finite algebraic exterior vector. The quantitative bound comes from counting the Laurent components needed as wedge *factors*; the number of wedge summands may be much larger.

Let $\Omega^\kappa\colon \Lambda^k_\kappa E_0 \to \kappa^{\mathbb{N}^k}$ be the determinant array over κ; it is injective by Lemma B.2 with $L=\kappa$ and $V=E_0$. For $y_1,\dots,y_k\in E_0$ we have $\Omega^{K_1}(y_1\wedge\cdots\wedge y_k)=\Omega^\kappa(y_1\wedge\cdots\wedge y_k)$: the determinants are the same, with entries in $\kappa\subseteq K_1$. Support dimensions of elements of $\Lambda^k_\kappa E_0$ are computed in E_0 over κ.

Proposition C.3 (support estimate)**.** *For every $b\in F_1$ there is a unique $\eta(b)\in\Lambda^k_\kappa E_0$ with $\Omega^\kappa(\eta(b))=\operatorname{coeff}_0(Jb)$. The map η is κ-linear, and*

$$\operatorname{sdim}\big(\eta(b)\big)\le kM_\varrho\,\|b\|_{F_1}\qquad \text{for all } b\in F_1. \tag{C.2}$$

Proof. Step 1: one decomposable. Let $x_1,\dots,x_k\in E_1$. If some $x_b=0$, the wedge is zero and we take $\eta=0$. Otherwise put $\nu_b:=\nu(x_b)$ and $l:=-\sum_b\nu_b$, so that $\prod_b\|x_b\|_\infty=\varrho^{-l}$. The possible exponent tuples contributing to a constant coefficient form the set

$$\mathcal{N}:=\Big\{n\in\mathbb{Z}^k:\ n_b\ge\nu_b \text{ for all } b,\ \sum_b n_b=0\Big\}.$$

For such a tuple, the nonnegative integers $n_b-\nu_b$ sum to l. Hence $\mathcal{N}$ is empty if $l<0$; if $l\ge0$, every coordinate satisfies $\nu_b\le n_b\le\nu_b+l$, so $\mathcal{N}$ is finite.

Fix $(i_1,\dots,i_k)\in\mathbb{N}^k$ and expand each entry as $x_b(i_a)=\sum_{n\ge\nu_b}\operatorname{coeff}_n(x_b(i_a))X^n$. In the determinant expansion

$$\det(x_b(i_a))_{a,b}=\sum_\sigma\operatorname{sgn}(\sigma)\prod_b x_b(i_{\sigma(b)}),$$

the constant coefficient of each product is its finite coefficient convolution over $\mathcal{N}$. Exchanging this finite sum with the permutation sum gives

$$\operatorname{coeff}_0\det\big(x_b(i_a)\big)_{a,b}=\sum_{n\in\mathcal{N}}\det\big(x_{b,[n_b]}(i_a)\big)_{a,b}.$$

The same finite set $\mathcal{N}$ works for every array coordinate. Thus $\operatorname{coeff}_0\big(\Omega^{K_1}(x_1\wedge\cdots\wedge x_k)\big)=\Omega^\kappa(\eta)$, where

$$\eta:=\sum_{n\in\mathcal{N}}x_{1,[n_1]}\wedge\cdots\wedge x_{k,[n_k]}\in\Lambda^k_\kappa E_0.$$

This is a finite sum in the algebraic exterior power. When $l<0$ it is zero. When $l\ge0$, all its wedge factors belong to the set $\{x_{b,[n]}:1\le b\le k,\ \nu_b\le n\le\nu_b+l\}$, which has at most $k(l+1)$ elements. Taking their span gives

$$\operatorname{sdim}(\eta)\le k(l+1)=k\,(l+1)\varrho^l\cdot\varrho^{-l}\le kM_\varrho\prod_b\|x_b\|_\infty.$$

The final bound therefore holds in both cases.

Step 2: an element of Λ. Write $\omega=\sum_j w_j$, where $w_j=x_{j,1}\wedge\cdots\wedge x_{j,k}$, and take η_j from Step 1. Additivity gives

$$\operatorname{coeff}_0(\Omega^{K_1}(\omega))=\sum_j\operatorname{coeff}_0(\Omega^{K_1}(w_j))=\Omega^\kappa\Big(\sum_j\eta_j\Big).$$

Injectivity of Ω^κ shows that $\eta(\omega):=\sum_j\eta_j$ is independent of the chosen decomposition. Subadditivity of support dimension now yields

$$\operatorname{sdim}(\eta(\omega))\le\sum_j\operatorname{sdim}(\eta_j)\le kM_\varrho\sum_j\prod_i\|x_{j,i}\|_\infty.$$

The left side is fixed while the decomposition varies. Taking the infimum therefore gives $\operatorname{sdim}(\eta(\omega)) \le kM_\varrho\|\omega\|_\pi$, without requiring a minimizing decomposition.

Step 3: completion. Choose $\omega_n \in \Lambda$ with $\|b-\omega_n\|_{F_1} \to 0$. Since J is a contraction,

$$\|Jb - \Omega^{K_1}(\omega_n)\|_\infty \le \|b-\omega_n\|_{F_1} < 1 \qquad (n \ge n_0).$$

Equation (C.1) therefore gives the *exact* identities

$$\operatorname{coeff}_0(Jb) = \operatorname{coeff}_0(\Omega^{K_1}(\omega_n)) = \Omega^\kappa(\eta(\omega_n)) \qquad (n \ge n_0).$$

By injectivity of Ω^κ, the algebraic vectors $\eta(\omega_n)$ are eventually equal. Define $\eta(b)$ to be this common value. Its array is $\operatorname{coeff}_0(Jb)$, so the same injectivity also proves independence of the approximating sequence and uniqueness. In particular, no limit in $\Lambda^k_\kappa E_0$ is involved.

For every $n \ge n_0$ the fixed vector $\eta(b)$ satisfies $\operatorname{sdim}(\eta(b)) \le kM_\varrho\|\omega_n\|_\pi$. Since $\|\omega_n\|_\pi \to \|b\|_{F_1}$, this gives (C.2). Finally, the array characterization proves κ-linearity: J is K_1-linear, coeff_0 is κ-linear, and Ω^κ is linear and injective. □

C.4. **The coefficient lift.** The support estimate converts norm control into an algebraic obstruction. We apply it to a hypothetical lift, evaluated on coordinatewise multiplication operators and on the universal wedge map ω_{E_1}.

Suppose that P is a bounded k-linear lift of $A^{k,K_1}_{E_1,E_1;F_1}$ over K_1. For $a \in E_1$ let $\mu_a \in \mathcal{L}(E_1,E_1)$ be coordinatewise multiplication, $\mu_a x = ax$. The map $a \mapsto \mu_a$ is K_1-linear and $\|\mu_a\| \le \|a\|_\infty$. Define

$$\Phi(a_1,\dots,a_k;x_1,\dots,x_k) := P(\mu_{a_1},\dots,\mu_{a_k})(\omega_{E_1})(x_1,\dots,x_k) \in F_1,$$
$$\Psi := \eta\circ\Phi|_{E_0^{2k}} \colon E_0^k \times E_0^k \to \Lambda^k_\kappa E_0.$$

Since $\Omega^\kappa \circ \eta = \operatorname{coeff}_0 \circ J$, the map Ψ is $\operatorname{coeff}_0 \circ J \circ \Phi|_{E_0^{2k}}$, read in $\Lambda^k_\kappa E_0$ through the injective map Ω^κ.

Lemma C.4. *The map Ψ has the following properties.*

- (Ψ1) *Ψ is $2k$-linear over κ.*
- (Ψ2) *Ψ is alternating in the last k slots.*
- (Ψ3) $\operatorname{sdim}(\Psi(z)) \le d := \lfloor kM_\varrho\|P\|\rfloor$ *for* all $z \in E_0^{2k}$.
- (Ψ4) *Ψ satisfies* (Pol) *for the multiplier family $\mu^\kappa \colon E_0 \to \operatorname{End}_\kappa(E_0)$, $\mu^\kappa_a x = ax$: for all $m \ge 1$, $b_1,\dots,b_m \in E_0$, $\alpha \in \mathbb{N}^m$ with $|\alpha| = k$, and $x \in E_0^k$,*

$$\sum_{\text{type } f=\alpha} \Psi(b_f;x) = \sum_{\text{type } f=\alpha} (b_{f(1)}x_1)\wedge\cdots\wedge(b_{f(k)}x_k) \qquad \text{in } \Lambda^k_\kappa E_0.$$

In particular Ψ satisfies (Pol1).

Proof. The map Φ is K_1-multilinear, so its restriction to E_0^{2k} is κ-multilinear. Composing with the κ-linear map η proves (Ψ1). Since $P(\mu_{a_1},\dots,\mu_{a_k})(\omega_{E_1})$ belongs to $\operatorname{Alt}^k_{K_1}(E_1;F_1)$, it vanishes when two of the last k arguments coincide; the same holds after applying η, proving (Ψ2).

For (Ψ3), all inputs from E_0 have norm at most one. Thus

$$\|\Phi(a;x)\| \le \|P\| \prod_i \|\mu_{a_i}\| \, \|\omega_{E_1}\| \prod_i \|x_i\|_\infty \le \|P\|.$$

Apply (C.2) and use that support dimension is an integer to obtain the stated bound by d.

To prove (Ψ4), we first compare coefficients over the infinite field K_1, while the values still lie in F_1. Fix $b_1, \dots, b_m \in E_0$ and $x \in E_0^k$. For $\lambda \in K_1^m$ put $a_\lambda := \sum_i \lambda_i b_i \in E_1$. The diagonal identity for the lift gives

$$\Phi(a_\lambda, \dots, a_\lambda; x) = P(\mu_{a_\lambda}, \dots, \mu_{a_\lambda})(\omega_{E_1})(x) = A(\mu_{a_\lambda})(\omega_{E_1})(x) = \omega_{E_1}(a_\lambda x_1, \dots, a_\lambda x_k).$$

Expanding both sides by multilinearity gives $\sum_\alpha \lambda^\alpha (L_\alpha - R_\alpha) = 0$ for all $\lambda \in K_1^m$, where

$$L_\alpha := \sum_{\text{type } f = \alpha} \Phi(b_f; x) \in F_1, \qquad R_\alpha := \sum_{\text{type } f = \alpha} \omega_{E_1}(b_{f(1)} x_1, \dots, b_{f(k)} x_k) \in F_1.$$

The field K_1 is infinite even when κ is finite. Lemma B.6, applied in the K_1-vector space F_1, therefore yields $L_\alpha = R_\alpha$ for every α.

We can now apply η. Additivity gives $\eta(L_\alpha) = \sum_{\text{type } f=\alpha} \Psi(b_f; x)$. To identify the other side, take $y_1, \dots, y_k \in E_0$. By Lemma C.2 and the determinant-array agreement preceding Proposition C.3,

$$J(\omega_{E_1}(y)) = \Omega^{K_1}(y_1 \wedge \cdots \wedge y_k) = \Omega^{\kappa}(y_1 \wedge \cdots \wedge y_k).$$

This array has entries in κ, so taking its constant coefficient leaves it unchanged. Uniqueness in Proposition C.3 then gives $\eta(\omega_{E_1}(y)) = y_1 \wedge \cdots \wedge y_k \in \Lambda_\kappa^k E_0$. Using $y_j = b_{f(j)} x_j$ in each summand of R_α proves

$$\eta(R_\alpha) = \sum_{\text{type } f = \alpha} (b_{f(1)} x_1) \wedge \cdots \wedge (b_{f(k)} x_k),$$

which is the required identity. □

The order of these operations is essential for the full identity (Ψ4). Evaluating the diagonal only at κ-points would give (Pw) there, which for $\kappa = \mathbb{F}_q$ and $k > q$ is strictly weaker than (Pol) (Proposition B.8(3)). The proof below uses only the consequence (Pol1).

C.5. **No bounded lift over a Laurent series field.** We now combine the support estimate (Proposition C.3) and the coefficient lift (Lemma C.4) with the finite-field theorem of Appendix B (Theorem B.9) to prove Theorem C.1.

Proof of Theorem C.1. Suppose P exists, and form Ψ as in §C.4. By Lemma C.4, $\Psi\colon E_0^k \times E_0^k \to \Lambda_\kappa^k E_0$ with $E_0 = \kappa^{\mathbb{N}}$ is $2k$-linear over κ, alternating and hence antisymmetric in the last k slots, satisfies (Pol1), and has $\operatorname{sdim}(\Psi(z)) \le d$ for all inputs z. Moreover $k! = 0$ in κ, since $k \ge p = \operatorname{char}\kappa$. This contradicts Theorem B.9(2) with $L = \kappa$ and $E_0 = \kappa^{\mathbb{N}}$. The last sentence follows from Proposition 2.5. □

Remark C.5. Lemmas C.2 and C.4 and Proposition C.3 hold for every field κ, and nothing in them uses a normalization of ϱ: the constant M_ϱ makes the argument uniform in $\varrho \in (0,1)$. Only the final appeal to Theorem B.9 needs κ finite. We do not need, and do not discuss, the analogue of Theorem C.1 for infinite κ with this particular pair (E_1, F_1). For an infinite κ of characteristic p the field $\kappa((X))$ is covered by Theorem 7.2(1), with the pair obtained by ascent from its subfield $\mathbb{F}_p((X))$.

C.6. **Multiplier families and c_0 bases.** The proof of Theorem C.1 uses the hypothetical lift only on coordinatewise multiplication operators and only at the single form ω_{E_1}. Moreover, Theorem B.9(1) needs only finitely supported test vectors with entries in $\{0, \pm 1\}$. Both observations give a much smaller bad parameter space. Throughout this subsection κ, k, ϱ, K_1, E_1, F_1 and ω_{E_1} are as in Theorem C.1, $c_0(\mathbb{N}, K_1) \subseteq E_1$ carries the supremum norm, and $\mu_a \in \mathcal{L}(E_1, E_1)$ denotes multiplication by a.

Proposition C.6 (multipliers on c_0)**.** *The homogeneous map*

$$\sigma\colon c_0(\mathbb{N},K_1)\to \mathrm{Alt}^k_{K_1}(E_1;F_1),\qquad \sigma(a)(x_1,\dots,x_k)=\omega_{E_1}(ax_1,\dots,ax_k)=(ax_1)\wedge\cdots\wedge(ax_k),$$

has no bounded k-linear lift with values in $\mathrm{Alt}^k_{K_1}(E_1;F_1)$. Consequently, for every $u_0\in\mathcal{L}(E_1,E_1)$, the map $a\mapsto A^{k,K_1}_{E_1,E_1;F_1}(u_0+\mu_a)(\omega_{E_1})$ is analytic at no point of $c_0(\mathbb{N},K_1)$. In particular $a\mapsto A^{k,K_1}_{E_1,E_1;F_1}(u_0+\mu_a)$ is analytic at no point.

Proof of Proposition C.6. Suppose $p\colon c_0(\mathbb{N},K_1)^k\to\mathrm{Alt}^k_{K_1}(E_1;F_1)$ is a bounded k-linear lift of σ. Let $V_{\mathrm{fin}}=\kappa^{(\mathbb{N})}$ be the finitely supported sequences with entries in κ; it lies in both $c_0(\mathbb{N},K_1)$ and E_0. Define $\Psi(a;x)=\eta\big(p(a_1,\dots,a_k)(x_1,\dots,x_k)\big)$ for $a,x\in V_{\mathrm{fin}}^k$. The proof of Lemma C.4 applies verbatim. The map Ψ is $2k$-linear over κ and alternating in the last k slots. Its inputs have supremum norm at most one, so (C.2) gives $\operatorname{sdim}\Psi(a;x)\le\lfloor kM_\varrho\|p\|\rfloor$. The polarization points $a_\lambda=\sum_i\lambda_i b_i$, with $b_i\in V_{\mathrm{fin}}$, are finitely supported and hence lie in $c_0(\mathbb{N},K_1)$, so the identity (Pol), and in particular (Pol1), follows from the diagonal identity as before. This contradicts Theorem B.9(1), since $k!=0$ in κ.

For the second assertion, suppose that the map is analytic at a_0, with expansion $\sum_n p_n$ there. For $h\in c_0(\mathbb{N},K_1)$ and $t\in K_1$, multilinearity gives, in $\mathrm{Mult}^k_{K_1}(E_1;F_1)$,

$$\omega_{E_1}\circ(u_0+\mu_{a_0}+t\mu_h,\dots,u_0+\mu_{a_0}+t\mu_h)=\sum_{r=0}^{k}t^rC_r(h),\qquad C_k(h)=\sigma(h).$$

Comparing this with the given expansion on scalar lines, Lemma A.1 shows, exactly as in Proposition 2.5, that p_k is a bounded lift of σ. This is impossible. The operator-valued statement follows by evaluation at ω_{E_1}. □

Corollary C.7 (diagonal transitions over a c_0 base)**.** *Let U be the open unit ball of $c_0(\mathbb{N},K_1)$, an analytic manifold with one chart. Let $\mathcal{V}\to U$ be the trivial bundle with fiber E_1 and the two global trivializations $\tau_0(a,z)=(a,z)$ and $\tau_1(a,z)=(a,\mu_{1+a}z)$, and let $\mathcal{B}=U\times F_1$. Then $\mathcal{V}$ is an analytic normed vector bundle whose transitions are isometric diagonal automorphisms of E_1. The naturally induced atlas on $\mathrm{Alt}^k(\mathcal{V};\mathcal{B})$ is not analytic: the section that is constantly ω_{E_1} in the chart induced by τ_1 is, in the chart induced by τ_0, the map $a\mapsto\omega_{E_1}\circ(\mu_{1+a},\dots,\mu_{1+a})$, which is analytic at no point of U.*

Proof. For $a\in U$ every coordinate satisfies $|1+a_i|=1$, so μ_{1+a} is an isometry of E_1. Its inverse $\mu_{(1+a)^{-1}}=\sum_{n\ge0}(-\mu_a)^n$ is a power series of radius one in a, since $\mu_a^n=\mu_{a^n}$ has norm at most $\|a\|^n$. Hence both transitions are analytic. If $\theta_a\in\mathrm{Alt}^k(\mathcal{V}_a;F_1)$ has coordinates $m_0(a)$ and $m_1(a)$ in the two induced charts, then $m_0(a)=m_1(a)\circ(\mu_{1+a},\dots,\mu_{1+a})$. With $m_1\equiv\omega_{E_1}$, this is the map of Proposition C.6 with $u_0=\mathrm{id}$, which is analytic at no point. □

The obstruction concerns the atlas obtained from the two analytic frames. The underlying pointwise bundle is trivial and has an analytic product structure; the failure is compatibility of that structure with the natural construction from all analytic frames. The complete target F_1 admits no equivalent nonarchimedean norm by Proposition E.1, so this example does not contradict Corollary 5.3.

Corollary C.8 (finite dimension is sharp for nonarchimedean bases)**.** *Let $K=\mathbb{F}_q((X))$, where q is a power of p, with an X-adic absolute value, and let $k\ge p$. Let P be a K-Banach space admitting an equivalent nonarchimedean norm. The alternating construction preserves analytic bundles and their operator-valued analytic morphisms over every analytic manifold modeled on P, with arbitrary normed fibers, if and only if P is finite-dimensional.*

Proof. If P is finite-dimensional, Lemma A.3 supplies continuous coordinates and Corollary 5.15 applies. Conversely, suppose that P is infinite-dimensional. Round an equivalent nonarchimedean norm upwards to the discrete value group $|K^\times|$; the result is still an equivalent nonarchimedean K-norm, and its nonzero values lie in $|K^\times|$. Serre's orthonormal-basis theorem [25, §1, Proposition 1] then identifies P topologically with $c_0(I, K)$ for an infinite set I. Choosing a countable subset of I gives bounded linear maps

$$\iota\colon c_0(\mathbb{N}, K) \longrightarrow P, \qquad \pi\colon P \longrightarrow c_0(\mathbb{N}, K), \qquad \pi\iota = \mathrm{id}.$$

Apply the Laurent construction with $\kappa = \mathbb{F}_q$ and $K_1 = K$. On the open set $V = \{x \in P : \|\pi x\| < 1\}$, take the trivial bundle with fiber E_1 and two analytic frames whose transition is $\mu_{1+\pi(x)}$, as in Corollary C.7, and the trivial target bundle with fiber F_1. If the induced alternating transition were analytic at $x_0 \in V$, restriction along $b \mapsto x_0 + \iota b$ would make

$$b \longmapsto A^k_{E_1,E_1;F_1}\left(\mu_{1+\pi(x_0)} + \mu_b\right)$$

analytic at zero. This contradicts Proposition C.6 with $u_0 = \mu_{1+\pi(x_0)}$. □

The dichotomy permits arbitrary normed target fibers. In the nonarchimedean Banach target category, Corollary 5.3 gives analyticity over every base for this discretely valued field.

Appendix D. Ascent to an arbitrary base field

Proposition D.9, applied in Appendix E with E_1 and F_1 as in Theorem C.1, carries analyticity at no point from a spherically complete Laurent subfield K_1 to the completion K' of the prescribed field, since a lift after scalar extension descends to K_1. Lemma D.11 then passes from K' to the field itself.

D.1. **The Laurent subfield.** We use spherical completeness in the sense used in Theorem 5.2: every nonempty family of pairwise-meeting closed balls has a common point. For the field case of the following lemma, see Schneider [24, Lemmas 1.3 and 1.6].

Lemma D.1 (discrete distances)**.** *Let Y be a complete nonarchimedean space whose distances lie in $r^{\mathbb{Z}} \cup \{0\}$ for some $r \in (0, 1)$. Then Y is spherically complete.*

Proof. Write $\mathcal{D}(a, \rho) = \{y \in Y : \mathrm{d}(y, a) \le \rho\}$. In a nonarchimedean space, every point of a ball is a center. Consequently, whenever two closed balls meet, the one with the smaller radius lies in the other. The discrete distance assumption also gives

$$\mathcal{D}(a, \rho) = \mathcal{D}(a, \rho^\flat), \qquad \rho^\flat := \max\{s \in r^{\mathbb{Z}} \cup \{0\} : s \le \rho\}.$$

We may therefore replace every radius by a value in $r^{\mathbb{Z}} \cup \{0\}$.

Let $(\mathcal{D}(a_i, \rho_i))_{i\in I}$ be a nonempty family of pairwise-meeting closed balls. If its radii have a least element ρ_{i_0}, then $\mathcal{D}(a_{i_0}, \rho_{i_0})$ lies in every ball of the family, so a_{i_0} is a common point. Otherwise, discreteness implies that all the radii are positive and that their infimum is zero. Choose $i_1, i_2, \dots$ so that ρ_{i_n} decreases strictly to zero. The corresponding balls are nested, and

$$\mathrm{d}(a_{i_{n'}}, a_{i_n}) \le \rho_{i_n} \qquad (n' \ge n).$$

Their centers form a Cauchy sequence. By completeness it has a limit a, and closedness puts a in every ball of the sequence. For any $i \in I$, choose n with $\rho_{i_n} \le \rho_i$. The n-th ball then lies in $\mathcal{D}(a_i, \rho_i)$, so a belongs to every ball of the original family. □

Lemma D.2 (Laurent subfield)**.** *Let K' be a complete nontrivially normed field of characteristic p, and let $t \in K'$ with $0 < |t| = r < 1$.*

(1) *K' is nonarchimedean, and $|c| = 1$ for every $c \in \mathbb{F}_p^\times$.*

(2) *For* $a_m, \dots, a_M \in \mathbb{F}_p$, *not all zero,* $\left|\sum_{i=m}^{M} a_i t^i\right| = r^{\min\{i : a_i \neq 0\}}$.
(3) *Give* $\mathbb{F}_p((X))$ *the absolute value with* $|X| = r$. *There is a continuous isometric ring homomorphism* $\mathrm{ev}_t \colon \mathbb{F}_p((X)) \to K'$ *such that* $\mathrm{ev}_t\bigl(\sum_{i \geq m} a_i X^i\bigr) = \sum_{i \geq m} a_i t^i$. *Its image* K_1 *is a closed subfield of* K'*; it is the closure of* $\mathbb{F}_p(t)$*, and it is isometrically isomorphic to* $\mathbb{F}_p((X))$.
(4) K_1 *is spherically complete.*

Proof. (1) Lemma A.2 makes K' nonarchimedean. For $c \in \mathbb{F}_p^{\times}$, the identity $c^{p-1} = 1$ gives $|c| = 1$.

(2) The nonzero terms $a_i t^i$ have pairwise distinct absolute values r^i. In a nonarchimedean field, $|x + y| = |x|$ whenever $|y| < |x|$. Repeatedly applying this identity shows that the sum has the absolute value of its largest term, namely $r^{\min\{i : a_i \neq 0\}}$.

(3) Evaluation on Laurent polynomials defines a ring homomorphism $\mathrm{ev}_t \colon \mathbb{F}_p[X, X^{-1}] \to K'$, and (2) shows that $|\mathrm{ev}_t(g)| = r^{\operatorname{ord} g} = |g|$. Laurent polynomials are dense in $\mathbb{F}_p((X))$, since truncations converge. Completeness of K' therefore gives a unique continuous extension to $\mathbb{F}_p((X))$. Passing to limits preserves the isometry and the ring operations. It also gives the stated evaluation formula, because the partial sums of $\sum_{i \geq m} a_i X^i$ map to the partial sums of $\sum_{i \geq m} a_i t^i$.

The image is complete as the isometric image of a complete space, hence closed in K'. It is a subfield: for $g \neq 0$, the identity $\mathrm{ev}_t(g)\mathrm{ev}_t(g^{-1}) = 1$ provides the inverse. This subfield contains $\mathbb{F}_p(t)$, while the Laurent polynomials $\mathbb{F}_p[t, t^{-1}] \subseteq \mathbb{F}_p(t)$ are dense in it. The image is consequently exactly the closure of $\mathbb{F}_p(t)$.

(4) By (3), K_1 is complete and has value group $|K_1^{\times}| = r^{\mathbb{Z}}$. It is nonarchimedean by (1), so its distances lie in $r^{\mathbb{Z}} \cup \{0\}$. Lemma D.1 gives spherical completeness. □

D.2. **Ingleton's theorem and its consequences.** Throughout the rest of this section we fix fields $K_1 \subseteq K'$ with the following properties.

(H1) K_1, with the restricted absolute value, is nontrivially normed, complete and spherically complete.
(H2) K' is nonarchimedean.

By multiplication, K' is a nonarchimedean normed K_1-space. In our application $K' = \widehat{K}$ and K_1 is the field of Lemma D.2.

Theorem D.3 (Ingleton [11]; see also [22])**.** *Let* K_1 *be a spherically complete nonarchimedean normed field,* V *a* K_1*-vector space with a nonarchimedean norm,* $V_0 \subseteq V$ *a subspace,* $C \geq 0$*, and* $T_0 \colon V_0 \to K_1$ *linear with* $|T_0 v| \leq C\|v\|$ *for* $v \in V_0$*. Then* T_0 *extends to a linear map* $T \colon V \to K_1$ *with* $|Tx| \leq C\|x\|$ *for all* $x \in V$.

Schneider [24, Prop. 9.2] gives the more general formulation with a dominating nonarchimedean seminorm.

We will use Theorem D.3 on the nonarchimedean K_1-space K' in two ways: to obtain the scalar retraction below, and to extend finite-dimensional coordinate functionals in the proof of positivity of the projective tensor norm. Both applications have codomain K_1. The spaces E_1 and F_1 introduced later need not be nonarchimedean.

Corollary D.4 (Ingleton projection)**.** *There is a* K_1*-linear map* $\varpi \colon K' \to K_1$ *with* $\varpi|_{K_1} = \mathrm{id}$ *and* $|\varpi(\lambda)| \leq |\lambda|$ *for all* $\lambda \in K'$.

Proof. Theorem D.3 with $V = K'$, $V_0 = K_1$, $T_0 = \mathrm{id}$ and $C = 1$; its hypotheses hold by (H1) and (H2). □

Lemma D.5 (extension)**.** *Let $\mathbb{K} \in \{K_1, K'\}$ and let $C \geq 0$.*

(1) *Let V_0 be a dense $\mathbb{K}$-subspace of a normed $\mathbb{K}$-space V, let Y be a $\mathbb{K}$-Banach space and $T_0 \colon V_0 \to Y$ a $\mathbb{K}$-linear map with $\|T_0 x\| \leq C\|x\|$. Then T_0 extends uniquely to a continuous map $T \colon V \to Y$, which is $\mathbb{K}$-linear with $\|Tx\| \leq C\|x\|$.*

(2) *Let $V_1, \dots, V_k$ be normed K'-spaces with dense subspaces V_i^0, Y a K'-Banach space, and $M \colon \prod_i V_i^0 \to Y$ a K'-multilinear map with $\|M(d)\| \leq C \prod_i \|d_i\|$. Then M extends uniquely to a continuous K'-multilinear map $\overline{M} \colon \prod_i V_i \to Y$ with $\|\overline{M}\| \leq C$.*

Proof. (1) The bound makes T_0 Lipschitz, hence uniformly continuous. Density of V_0 and completeness of Y give a unique continuous extension. Linearity and the norm bound pass to limits by continuity of addition, scalar multiplication and the norm.

(2) The case $k = 0$ is immediate. For $k \geq 1$, multilinearity gives the telescoping estimate

$$\|M(d) - M(d')\| \leq C\rho^{k-1} \sum_i \|d_i - d_i'\|$$

whenever $\|d_i\|, \|d_i'\| \leq \rho$ for every i, with $\rho > 0$. For any $v \in \prod_i V_i$, choose tuples $d^{(n)} \in \prod_i V_i^0$ converging to v. Their coordinates are uniformly bounded, so the estimate makes $M(d^{(n)})$ Cauchy in Y. Define $\overline{M}(v)$ to be its limit. The same estimate, applied to two approximating sequences, proves that this definition is independent of the choices and is continuous on bounded sets. Multilinearity and the product norm bound pass to limits. Finally, density of $\prod_i V_i^0$ proves uniqueness. □

D.3. **Projective base change.** The next construction extends scalars while retaining the original norm and a contractive retraction whenever the original space is complete. The scalar retraction from Corollary D.4 is the essential ingredient; the construction applies to arbitrary normed spaces.

For a normed K_1-space V let $V_{K'} := V \otimes_{K_1} K'$, a K'-vector space through the second factor, and put

$$\|u\|_\pi := \inf\Big\{ \sum_j \|x_j\| \, |\lambda_j| \;:\; u = \sum_j x_j \otimes \lambda_j \Big\},$$

the infimum over finite decompositions, with an ordinary sum. For a bounded K_1-linear $\psi \colon K' \to K_1$ let $T_\psi \colon V_{K'} \to V$ be the K_1-linear map with $T_\psi(x \otimes \lambda) = \psi(\lambda)x$; it exists because $(x, \lambda) \mapsto \psi(\lambda)x$ is K_1-bilinear. Let $\iota_V \colon V \to V_{K'}$, $x \mapsto x \otimes 1$; it is K_1-linear, since $cx \otimes 1 = x \otimes c = c \cdot (x \otimes 1)$ for $c \in K_1$.

Lemma D.6 (projective base change)**.** *Let V be a normed K_1-space; no nonarchimedean hypothesis is made on V.*

(1) *$\|T_\psi u\| \leq \|\psi\| \, \|u\|_\pi$ for every bounded K_1-linear $\psi \colon K' \to K_1$.*

(2) *$\|\cdot\|_\pi$ is a K'-norm on $V_{K'}$.*

(3) *ι_V is isometric.*

(4) *Let $V \widehat{\otimes}_\pi K'$ be the completion of $(V_{K'}, \|\cdot\|_\pi)$. If V is complete, then T_ϖ, with ϖ as in Corollary D.4, extends to a K_1-linear map $\Pi_V \colon V \widehat{\otimes}_\pi K' \to V$ with $\|\Pi_V\| \leq 1$ and $\Pi_V \circ \iota_V = \mathrm{id}_V$.*

Proof. (1) For every decomposition $u = \sum_j x_j \otimes \lambda_j$,

$$\|T_\psi u\| = \Big\| \sum_j \psi(\lambda_j) x_j \Big\| \leq \sum_j \|\psi\| |\lambda_j| \|x_j\|.$$

Taking the infimum over decompositions proves the contraction estimate.

(2) Concatenating decompositions gives the triangle inequality. Multiplying their second factors by $\mu \in K'$ gives $\|\mu u\|_\pi \le |\mu|\|u\|_\pi$; for $\mu \neq 0$, applying the same inequality with μ^{-1} gives equality. The zero-scalar case is immediate.

It remains to prove positivity. Let $u \neq 0$. Choose a finite tensor representation and rewrite its second factors in a basis of their K_1-span. This gives

$$u = \sum_{j=1}^{n} x_j \otimes \lambda_j$$

with $\lambda_1, \dots, \lambda_n$ linearly independent over K_1. Some x_{j_0} is nonzero, since otherwise $u = 0$. On the finite-dimensional subspace $U := \operatorname{span}_{K_1}\{\lambda_1, \dots, \lambda_n\} \subseteq K'$, take the coordinate functional $\psi_0(\lambda_j) = \delta_{jj_0}$. It is bounded by some $C \ge 0$ by Lemma A.3, using completeness of K_1 in (H1). Apply Theorem D.3 to the nonarchimedean K_1-space K' to extend it to $\psi\colon K' \to K_1$ with $\|\psi\| \le C$. Then $T_\psi u = x_{j_0}$, so (1) gives

$$0 < \|x_{j_0}\| \le C\|u\|_\pi.$$

Thus $\|u\|_\pi > 0$. This argument detects a nonzero vector coefficient by a functional on the *scalar* factor K'; it requires no separating continuous dual on V.

(3) The elementary tensor representation gives $\|x \otimes 1\|_\pi \le \|x\|$. For the reverse inequality, use $\varpi(1) = 1$ and $\|\varpi\| \le 1$ in (1):

$$\|x\| = \|T_\varpi(x \otimes 1)\| \le \|x \otimes 1\|_\pi.$$

(4) When V is complete, the contraction T_ϖ on the dense tensor subspace extends by Lemma D.5(1), with $\mathbb{K} = K_1$, to a K_1-linear contraction Π_V with values in V. On the embedded copy of V it satisfies $\Pi_V \iota_V x = T_\varpi(x \otimes 1) = \varpi(1)x = x$. □

We now fix the spaces to which the descent argument will be applied. Let E_1 be a normed K_1-space and F_1 a K_1-Banach space, and put

$$E := E_1 \widehat{\otimes}_\pi K', \qquad F := F_1 \widehat{\otimes}_\pi K',$$

the completions of $((E_1)_{K'}, \|\cdot\|_\pi)$ and $((F_1)_{K'}, \|\cdot\|_\pi)$. They are K'-Banach spaces; $\iota_E\colon E_1 \to E$ and $\iota_F\colon F_1 \to F$ are K_1-linear isometries, and $\Pi_F\colon F \to F_1$ is K_1-linear with $\|\Pi_F\| \le 1$ and $\Pi_F \circ \iota_F = \mathrm{id}_{F_1}$ (Lemma D.6).

Lemma D.7 (operators)**.** *For $f \in \mathcal{L}_{K_1}(E_1, E_1)$ the map $f \otimes \mathrm{id}$ on $(E_1)_{K'}$ is K'-linear with $\|(f \otimes \mathrm{id})u\|_\pi \le \|f\|\|u\|_\pi$. Its extension $f_{K'} \in \mathcal{L}_{K'}(E, E)$ satisfies $\|f_{K'}\| \le \|f\|$ and $f_{K'} \circ \iota_E = \iota_E \circ f$, and $f \mapsto f_{K'}$ is K_1-linear.*

Proof. For any decomposition $u = \sum_j x_j \otimes \lambda_j$, the image under $f \otimes \mathrm{id}$ has the decomposition $\sum_j f(x_j) \otimes \lambda_j$, whose cost is at most $\|f\| \sum_j \|x_j\||\lambda_j|$. Taking the infimum proves the norm bound on the algebraic tensor product. Since E is complete, Lemma D.5(1), with $\mathbb{K} = K'$, gives the continuous extension $f_{K'}$ with the same bound. Its action on $x \otimes 1$ proves the embedding identity. Additivity and K_1-homogeneity in f hold on elementary tensors, hence on their span, and then on E by continuity. □

Lemma D.8 (forms)**.** *For $m \in \mathrm{Alt}^k_{K_1}(E_1; F_1)$ there is $m_{K'} \in \mathrm{Alt}^k_{K'}(E; F)$ with $\|m_{K'}\| \le \|m\|$ and*

$$m_{K'}(\iota_E x_1, \dots, \iota_E x_k) = \iota_F\big(m(x_1, \dots, x_k)\big) \qquad (x_i \in E_1),$$

and $m \mapsto m_{K'}$ is K_1-linear.

Proof. We first define the form on algebraic tensors, then establish a bound independent of their decompositions, and finally pass to the completions. Fix a K_1-basis $(\lambda_\beta)_{\beta\in\mathfrak{B}}$ of K'. Every $u \in (E_1)_{K'}$ has a unique finite expansion $u = \sum_\beta u^\beta \otimes \lambda_\beta$, with $u^\beta \in E_1$, and each coordinate map $u \mapsto u^\beta$ is K_1-linear. Define

$$m^{\mathrm{alg}}(u_1,\dots,u_k) := \sum_{\beta_1,\dots,\beta_k} m\big(u_1^{\beta_1},\dots,u_k^{\beta_k}\big) \otimes \lambda_{\beta_1}\cdots\lambda_{\beta_k} \in (F_1)_{K'}.$$

The sum is finite, since each input has finitely many nonzero coordinates.

(i) Linearity of the coordinate maps and multilinearity of m make m^{alg} K_1-multilinear.

(ii) On elementary tensors, the definition gives

$$m^{\mathrm{alg}}(x_1\otimes\mu_1,\dots,x_k\otimes\mu_k) = m(x_1,\dots,x_k)\otimes\mu_1\cdots\mu_k \qquad (x_i\in E_1,\ \mu_i\in K').$$

Indeed, writing $\mu_i = \sum_\beta c_{i\beta}\lambda_\beta$ with $c_{i\beta}\in K_1$ gives $x_i\otimes\mu_i = \sum_\beta (c_{i\beta}x_i)\otimes\lambda_\beta$. Substitution in the defining sum yields

$$\sum_{\beta_1,\dots,\beta_k}\Big(\prod_i c_{i\beta_i}\Big)m(x)\otimes\prod_i\lambda_{\beta_i} = m(x)\otimes\prod_i\Big(\sum_\beta c_{i\beta}\lambda_\beta\Big),$$

which is the asserted formula.

(iii) The map m^{alg} is K'-multilinear. By additivity in each slot, it suffices to check K'-homogeneity when every argument is an elementary tensor; there it follows from (ii). The same formula determines the map on all tuples of algebraic tensors, so the initial basis choice does not affect it.

(iv) For arbitrary decompositions $u_i = \sum_j x_{ij}\otimes\mu_{ij}$, multilinearity and (ii) give

$$m^{\mathrm{alg}}(u) = \sum_{j_1,\dots,j_k} m(x_{1j_1},\dots,x_{kj_k})\otimes\prod_i\mu_{ij_i}.$$

The cost of this tensor decomposition is at most $\|m\|\prod_i\big(\sum_j\|x_{ij}\|\|\mu_{ij}\|\big)$. Taking the infimum separately over the decompositions in each slot proves

$$\|m^{\mathrm{alg}}(u_1,\dots,u_k)\|_\pi \le \|m\|\prod_i\|u_i\|_\pi.$$

(v) To prove strong alternation, suppose two distinct slots a,b both contain $u=\sum_j x_j\otimes\mu_j$. Use this *same* decomposition in both slots and expand all arguments into elementary tensors. Terms using the same index j in slots a and b vanish because m has a repeated argument. For $j\neq j'$, the term using indices (j,j') in these slots pairs with the term using (j',j). Their scalar factors agree, while the values of m are negatives of one another, so the pair sums to zero. Thus m^{alg} vanishes whenever any two arguments coincide. Pair cancellation remains valid in characteristic 2; merely checking $m^{\mathrm{alg}}(u,\dots,u)=0$ would not establish strong alternation for $k\ge 3$.

(vi) Regard m^{alg} as taking values in the completion F. By (iii), (iv) and Lemma D.5(2), it extends to a continuous K'-multilinear map $m_{K'}\colon E^k\to F$ with $\|m_{K'}\|\le\|m\|$. Strong alternation also passes to the limit: if $v_a=v_b$, approximate this common vector by one sequence in $(E_1)_{K'}$, and approximate the other entries separately. Every approximating tuple has equal entries in slots a,b, so (v) and continuity give $m_{K'}(v)=0$.

(vii) Taking every scalar factor equal to one in (ii) gives

$$m_{K'}(\iota_E x_1,\dots,\iota_E x_k) = m^{\mathrm{alg}}(x_1\otimes 1,\dots,x_k\otimes 1) = m(x)\otimes 1 = \iota_F(m(x)).$$

The defining formula is K_1-linear in m. These linear identities pass to the continuous extensions by uniqueness, proving K_1-linearity of $m\mapsto m_{K'}$. □

Proposition D.9 (descent of a lift)**.** *Assume* (H1) *and* (H2)*. Let E_1 be a normed K_1-space, F_1 a K_1-Banach space, and E, F as above. If $A^{k,K'}_{E,E;F}$ admits a bounded k-linear lift P over K', then $A^{k,K_1}_{E_1,E_1;F_1}$ admits a bounded k-linear lift P_1 over K_1 with $\|P_1\| \le \|P\|$. Equivalently, by Proposition 2.5 over K_1 and over K': if $A^{k,K_1}_{E_1,E_1;F_1}$ is analytic at no point, then $A^{k,K'}_{E,E;F}$ is analytic at no point either.*

Proof. Extend the operator and form inputs to K', apply the given lift, and retract the value to F_1. Explicitly, for $f_1, \dots, f_k \in \mathcal{L}_{K_1}(E_1, E_1)$, $m \in \mathrm{Alt}^k_{K_1}(E_1; F_1)$ and $x \in E_1^k$, set

$$P_1(f_1, \dots, f_k)(m)(x_1, \dots, x_k) := \Pi_F\Big(P\big((f_1)_{K'}, \dots, (f_k)_{K'}\big)(m_{K'})(\iota_E x_1, \dots, \iota_E x_k)\Big).$$

We check the norm bound and then the diagonal identity.

For fixed f and m, the map $P((f_i)_{K'})(m_{K'})$ is a continuous alternating K'-multilinear form on E. Precomposing each slot with the K_1-linear isometry ι_E and postcomposing with Π_F gives a K_1-multilinear map on E_1, still vanishing whenever two arguments coincide. Lemmas D.6–D.8 give

$$\|P_1(f)(m)\| \le \|\Pi_F\|\|P\| \prod_i \|(f_i)_{K'}\| \|m_{K'}\| \le \|P\| \prod_i \|f_i\| \|m\|.$$

In particular, $P_1(f)(m) \in \mathrm{Alt}^k_{K_1}(E_1; F_1)$. The form extension is K_1-linear in m, the operator extension is K_1-linear in each f_i, and P is K'-multilinear with values in K'-linear operators. Consequently P_1 is K_1-linear in m and K_1-multilinear in $(f_1, \dots, f_k)$. The displayed estimate proves boundedness at both levels and yields $\|P_1\| \le \|P\|$.

On the diagonal, use $P(g, \dots, g) = A^{k,K'}(g)$ with $g = f_{K'}$, followed by the embedding identities in Lemmas D.7 and D.8 and the retraction identity $\Pi_F \iota_F = \mathrm{id}$:

$$\begin{aligned} P_1(f, \dots, f)(m)(x) &= \Pi_F\big(m_{K'}(f_{K'}\iota_E x_1, \dots, f_{K'}\iota_E x_k)\big) \\ &= \Pi_F\big(m_{K'}(\iota_E f x_1, \dots, \iota_E f x_k)\big) \\ &= \Pi_F\big(\iota_F(m(f x_1, \dots, f x_k))\big) \\ &= m(f x_1, \dots, f x_k) = A^{k,K_1}(f)(m)(x). \end{aligned}$$

□

Remark D.10. Proposition D.9 is characteristic-free and uses no hypothesis on k. It transports negative results upward only, from K_1 to K'. Ingleton's theorem is used only on K', with no nonarchimedean norm hypothesis on F_1. The completeness of F_1 is used essentially: it is what makes the retraction Π_F land in F_1.

D.4. **Incomplete base fields.** The preceding construction takes place over the completion $\widehat{K}$ of the prescribed field. To return to K, we restrict scalars on the resulting Banach spaces. Density of K in $\widehat{K}$ forces every continuous K-multilinear map between these spaces to be $\widehat{K}$-multilinear, so a lift cannot appear under this restriction.

Lemma D.11. *Let K be a nontrivially normed field with completion $\widehat{K}$, and let E, E', F be $\widehat{K}$-Banach spaces, regarded as K-Banach spaces by restriction of scalars (with the same norms). Then $\mathcal{L}_K(E, E') = \mathcal{L}_{\widehat{K}}(E, E')$, $\mathrm{Alt}^k_K(E; F) = \mathrm{Alt}^k_{\widehat{K}}(E; F)$ and $\mathrm{Alt}^k_K(E'; F) = \mathrm{Alt}^k_{\widehat{K}}(E'; F)$ with the same norms, $A^{k,K}_{E,E';F} = A^{k,\widehat{K}}_{E,E';F}$, and every bounded k-linear lift of $A^{k,K}_{E,E';F}$ over K is a bounded k-linear lift of $A^{k,\widehat{K}}_{E,E';F}$ over $\widehat{K}$.*

Proof. Let g be a continuous K-linear map between normed spaces that already carry compatible $\widehat{K}$-scalar actions. For $\lambda \in \widehat{K}$, choose $\lambda_n \in K$ with $\lambda_n \to \lambda$. Since $\|\lambda_n x - \lambda x\| = |\lambda_n - \lambda|\|x\|$,

continuity gives

$$g(\lambda x) = \lim_n g(\lambda_n x) = \lim_n \lambda_n g(x) = \lambda g(x).$$

Thus g is $\widehat{K}$-linear. Applying the same argument with all but one slot fixed shows that every continuous K-multilinear map is $\widehat{K}$-multilinear. Alternation is the same equal-input vanishing condition over either field.

The spaces of bounded maps therefore coincide as sets. Their operator norms agree as well, since the norms are defined by the same pointwise bounds on the same vectors. They are $\widehat{K}$-Banach spaces, and $A^{k,K}_{E,E';F}$ and $A^{k,\widehat{K}}_{E,E';F}$ agree by their defining precomposition formula. Finally, apply the slotwise density argument once more to a bounded k-linear lift P between these map spaces. It becomes $\widehat{K}$-multilinear with the same norm and the same diagonal identity, and hence is a lift over $\widehat{K}$. □

Appendix E. Proof of the operator obstruction theorem

We show that the target spaces F_1 and F admit no equivalent nonarchimedean norm (Proposition E.1) and prove Theorem 7.2(1), which with Proposition 5.1(1) gives Corollary 7.3.

E.1. **The target has no equivalent nonarchimedean norm.**

Proposition E.1. *Let κ be any field, $k \geq 2$, $\varrho \in (0,1)$, and let $K_1 = \kappa((X))$, E_1, E_0 and F_1 be as in Appendix C. Let $e_i \in E_0$ be the unit vectors and $\omega_j := e_{kj+1} \wedge \cdots \wedge e_{kj+k} \in F_1$ $(j \geq 0)$, wedges on pairwise disjoint blocks of indices. Then $\|\omega_j\|_{F_1} = 1$ and*

$$\frac{N}{M_\varrho} \;\leq\; \Big\| \sum_{j<N} \omega_j \Big\|_{F_1} \;\leq\; N \qquad (N \geq 1).$$

Hence the norm of F_1 is not nonarchimedean, and F_1 admits no equivalent nonarchimedean norm. The same holds for $F = F_1 \widehat{\otimes}_\pi K'$ as in Appendix D, for any $K' \supseteq K_1$ satisfying (H1) *and* (H2)*, regarded as a normed space over K' or over any subfield of K'.*

Proof. Each ω_j is a wedge of unit vectors, so $\|\omega_j\|_{F_1} \leq 1$ by the defining projective norm. Evaluation of its determinant array at the block indices gives $\det I = 1$, hence $\|\omega_j\|_{F_1} \geq \|J\omega_j\|_\infty \geq 1$. The triangle inequality now gives the upper bound N.

For the lower bound, write ξ_N for the sum of the same block wedges in $\Lambda^k_\kappa E_0$, and put $\omega := \sum_{j<N} \omega_j \in F_1$. Its array $J\omega$ is the constant Laurent array $\Omega^\kappa(\xi_N)$. Thus $\mathrm{coeff}_0(J\omega) = \Omega^\kappa(\xi_N)$, and uniqueness in Proposition C.3 gives $\eta(\omega) = \xi_N$.

We apply Lemma B.4 to ξ_N. Fix one of the first N blocks and an index i_0 in it. Take for φ the $k-1$ coordinate functionals corresponding to the other indices of that block. On every other block all these functionals vanish, so its contraction is zero; here $k \geq 2$ ensures that the minors have positive size. On the chosen block, the coefficient of e_{i_0} is ± 1, while every other cofactor contains the zero column obtained by evaluating the functionals at e_{i_0}. Consequently $c_\varphi(\xi_N) = \pm e_{i_0}$. These contractions span all kN unit vectors in the first N blocks, so

$$kN \leq \mathrm{sdim}(\xi_N) = \mathrm{sdim}(\eta(\omega)) \leq kM_\varrho \|\omega\|_{F_1}$$

by (C.2). This proves the lower bound.

If $\|\cdot\|_u$ were an equivalent nonarchimedean norm, choose positive constants c, c' with $\|\cdot\|_{F_1} \leq c\|\cdot\|_u \leq c'\|\cdot\|_{F_1}$. Then

$$\frac{N}{M_\varrho} \leq \Big\| \sum_{j<N} \omega_j \Big\|_{F_1} \leq c \max_{j<N} \|\omega_j\|_u \leq c'$$

for every $N \geq 1$, a contradiction. In particular, the original norm is not nonarchimedean. This argument uses only the nonarchimedean inequality for $\|\cdot\|_u$, not its scalar homogeneity.

Finally, $\iota_F \colon F_1 \to F$ is a K_1-linear isometry by Lemma D.6(3). The images $\iota_F \omega_j$ satisfy the same unit-norm and sum-growth estimates, so the argument applies to F as well. Restriction to a subfield leaves these estimates unchanged. □

If $\varrho \leq \frac{1}{2}$, then $M_\varrho = 1$ and $\|\sum_{j<N} \omega_j\|_{F_1} = N$ exactly.

E.2. **Proof of Theorem 7.2(1).**

Proof. Let K be a nontrivially normed field of characteristic $p > 0$, and let $k \geq p$.

(1) *Choose a Laurent subfield.* By Lemma A.2, K is nonarchimedean. Its completion $\widehat{K}$ is a complete, nontrivially normed, nonarchimedean field of characteristic p. Choose $t \in K$ with $0 < |t| = \varrho < 1$, and let K_1 be the closure of $\mathbb{F}_p(t)$ in $\widehat{K}$. Lemma D.2 identifies K_1 isometrically with $\mathbb{F}_p((X))$, with $|X| = \varrho$, by ev_t. In particular, K_1 is complete and spherically complete.

(2) *Obtain the obstruction over K_1.* Put $E_1 := \ell^\infty(\mathbb{N}, K_1)$, and let F_1 be the completion of $\Lambda^k_{K_1} E_1$ under the projective exterior norm. Transport of scalars along ev_t preserves the norms, exterior powers and completions in this construction. It also identifies the operator and alternating-map spaces, their precomposition maps, and bounded lifts. Theorem C.1, with $\kappa = \mathbb{F}_p$, therefore shows that $A^{k,K_1}_{E_1,E_1;F_1}$ has no bounded k-linear lift.

(3) *Ascend to $\widehat{K}$.* The inclusion $K_1 \subseteq \widehat{K}$ satisfies (H1) and (H2), and F_1 is complete. Proposition D.9, with $K' = \widehat{K}$, gives Banach spaces

$$E := E_1 \widehat{\otimes}_\pi \widehat{K}, \qquad F := F_1 \widehat{\otimes}_\pi \widehat{K}$$

for which $A^{k,\widehat{K}}_{E,E;F}$ has no bounded k-linear lift.

(4) *Descend to K.* Regard the same E and F as K-Banach spaces. By Lemma D.11, a bounded lift of $A^{k,K}_{E,E;F}$ would extend to a bounded lift of $A^{k,\widehat{K}}_{E,E;F}$, which is impossible. Proposition 2.5 applies over the possibly incomplete field K: $A^{k,K}_{E,E;F}$ is analytic at no point. Reindexing the arguments by a bijection gives the same conclusion, with the same spaces E, F, for every finite index set with k elements.

(5) *Exclude an equivalent nonarchimedean target norm.* Since $k \geq p \geq 2$, Proposition E.1 applies to F_1 and F, after the scalar identification in step (2). Its conclusion persists when F is regarded over K. □

Remark E.2. If $K_1 = \widehat{K}$, as when $K = \mathbb{F}_p((X))$ and $t = X$, the completed tensor products identify isometrically with E_1 and F_1: use Lemma D.6(3) and $x \otimes \lambda = \lambda x \otimes 1$. Thus E is nonarchimedean in this case. For $K = \mathbb{F}_q((X))$, $q = p^n$, with its X-adic absolute value, one can instead apply Theorem C.1 directly with $\kappa = \mathbb{F}_q$, obtaining $E = \ell^\infty(\mathbb{N}, K)$ and $F = F_1$. The proof over a general base field does not require, or assert, that E is nonarchimedean.

Appendix F. A counterexample with scalar values

We prove Theorem 7.2(2) in the form below: Lemma F.4 rules out k-linear lifts of the determinant vanishing on prescribed subspace pairs, and the chain spaces of (F.6) carry these conditions to Banach spaces. Its spaces E, E' are the pair (E, D) of Theorem 7.2(2).

Theorem F.1 (scalar counterexample)**.** *Let K be a complete, nontrivially normed field that is not spherically complete, and let $k \geq 1$ satisfy $k! = 0$ in K. There are nonarchimedean Banach spaces E, E' for which*

$$A^k_{E,E';K} \colon \mathcal{L}(E, E') \longrightarrow \mathcal{L}\big(\mathrm{Alt}^k(E'; K), \mathrm{Alt}^k(E; K)\big)$$

is analytic at no point. The spaces may be taken to be closed subspaces of $\ell^\infty(\Lambda; K^{k+1})$ and $\ell^\infty(\Lambda; K^k)$, respectively, containing the corresponding c_0 spaces, where Λ is countable. Both spaces are nonseparable.

The factorial hypothesis says precisely that K has characteristic $p > 0$ and $k \geq p$. In particular $k \geq 2$, and K is nonarchimedean by Lemma A.2. We first establish the three ingredients of the construction.

F.1. **The countable multilinear tail property.** For a set I, a family $(a_i)_{i \in I}$ is *null* if, for every $\varepsilon > 0$, only finitely many indices satisfy $|a_i| \geq \varepsilon$. Over a complete nonarchimedean field, every null scalar family is unconditionally summable: its finite partial sums form a Cauchy net, since the norm of a difference is at most the largest omitted term. Its sum has norm at most $\sup_i |a_i|$, and such sums may be regrouped.

We derive the multilinear tail property from van der Put's description of the dual of countable ℓ^∞. The induction is independent of the characteristic.

Lemma F.2 (multilinear tails)**.** *Let K be complete, nonarchimedean and not spherically complete, let I be countably infinite, and let $d \geq 1$. Every bounded d-linear map $\lambda \colon \ell^\infty(I, K)^d \to K$ has an expansion*

$$\lambda(x^1, \ldots, x^d) = \sum_{(i_1, \ldots, i_d) \in I^d} a_{i_1, \ldots, i_d} x^1_{i_1} \cdots x^d_{i_d}, \qquad a_{i_1, \ldots, i_d} = \lambda(e_{i_1}, \ldots, e_{i_d}), \tag{F.1}$$

where $a \in c_0(I^d, K)$. Consequently, for every $\varepsilon > 0$, there is a finite $S \subseteq I$ such that $|\lambda(x^1, \ldots, x^d)| \leq \varepsilon$ whenever all $\|x^r\| \leq 1$ and at least one x^r vanishes on S. In particular $\lambda(e_i, \ldots, e_i) \to 0$ outside finite subsets of I.

Proof. After reindexing, take $I = \mathbb{N}$. Van der Put's theorem [19, Théorème 4.1], applied to countably many copies of K (a countable index set satisfies its nonmeasurability hypothesis), gives

$$\varphi(x) = \sum_i x_i \varphi(e_i), \qquad (\varphi(e_i))_i \in c_0(\mathbb{N}, K) \quad \big(\varphi \in \mathcal{L}(\ell^\infty, K)\big).$$

This is the case $d = 1$. Suppose the assertion about the array and expansion holds in smaller degrees. Fixing a coordinate vector in any slot of λ gives a form of degree $d - 1$. Hence, for every $\varepsilon > 0$, each slice of the set

$$W = \{(i_1, \ldots, i_d) : |a_{i_1, \ldots, i_d}| \geq \varepsilon\}$$

obtained by fixing one coordinate is finite.

If W were infinite, we could choose $w^{(0)}, w^{(1)}, \ldots \in W$ with the coordinates $w^{(n)}_r$ distinct as n varies, for each r: a finite list of forbidden coordinates excludes only finitely many elements of W. Define an isometric embedding $T_r \colon \ell^\infty(\mathbb{N}, K) \to \ell^\infty(\mathbb{N}, K)$ by putting $(T_r x)_{w^{(n)}_r} = x_n$ and zero elsewhere. The pulled-back form $\mu = \lambda \circ (T_1, \ldots, T_d)$ has array b with $|b_{n, \ldots, n}| \geq \varepsilon$ for all n. Its slices are null by induction. Thus

$$F_n = \{v \in \mathbb{N}^d : |b_v| \geq \varepsilon,\ v_r = n \text{ for some } r\}$$

is finite. Choose $n_1 < n_2 < \cdots$ so that n_s exceeds every coordinate occurring in $F_{n_1} \cup \cdots \cup F_{n_{s-1}}$. Every nonconstant tuple in $\{n_s : s \geq 1\}^d$ then has coefficient of norm $< \varepsilon$: its least coordinate

puts a large tuple in the corresponding F_{n_r}, whereas its greatest coordinate violates the choice of the subsequence.

Let y be the indicator of $\{n_s : s \geq 1\}$ and set $\psi(x) = \mu(x, y, \dots, y)$. The expansion of each fixed-coordinate slice gives

$$\psi(e_{n_s}) = b_{n_s,\dots,n_s} + \sum_{v' \in \{n_r : r \geq 1\}^{d-1},\ v' \neq (n_s,\dots,n_s)} b_{n_s,v'}.$$

The summands in the last sum form a null family and all have norm $< \varepsilon$. Their supremum is also $< \varepsilon$: outside a finite set they have norm $< \varepsilon/2$, and the remaining finite maximum is $< \varepsilon$. The nonarchimedean inequality therefore gives $|\psi(e_{n_s})| \geq \varepsilon$ for every s, contradicting the displayed dual formula. Thus a is null. Applying that formula in the first slot and the inductive expansion in the remaining slots gives (F.1); the null full array permits regrouping the iterated sums.

Finally choose a finite S containing all coordinates occurring in tuples with $|a_{i_1,\dots,i_d}| > \varepsilon$. If one input vanishes on S, all these terms vanish. Every remaining term has norm at most ε on the product of unit balls, so the same holds for the sum. Apply this conclusion to coordinate vectors outside S to obtain the last assertion. □

F.2. **Chains and a finite-dimensional incompatibility.** Let L be a nonempty finite set and $\Lambda = L^{<\omega}$ the set of finite words in L, including the empty word. Write $s \preceq t$ when t extends s, and write sj for the word obtained by appending $j \in L$. A *j-chain* is a set $T = \{s_0, s_1, \dots\}$ with $s_r j \preceq s_{r+1}$ for every r. Let Γ_j denote all such chains.

Lemma F.3 (chain gap)**.** *Chains with different labels have at most one common word. If sets $a_j \subseteq \Lambda$, $j \in L$, contain all but finitely many words of every chain in Γ_j, then $\bigcap_{j \in L} a_j \neq \varnothing$.*

Proof. If distinct words s, t belong to both a j-chain and an h-chain, take s shorter than t. Then t extends both sj and sh, forcing $j = h$.

For each word r and label j, there is an extension $s \succeq r$ such that every extension of sj belongs to a_j. Otherwise start with $s_0 = r$ and choose successively an extension $s_{m+1} \succeq s_m j$ outside a_j. This is a j-chain with infinitely many words outside a_j, a contradiction. Enumerate $L = \{j_1, \dots, j_q\}$ and start at the empty word r_0. Choose $s_m \succeq r_{m-1}$ with the asserted property for j_m, and put $r_m = s_m j_m$. The word r_q extends every $s_m j_m$ and hence belongs to every a_j. □

The next lemma is purely algebraic; the alternating maps in its statement need no topology. Over a field κ, put $V = \kappa^{k+1}$ and $V' = \kappa^k$. Write $e_1, \dots, e_{k+1}$ and $e'_1, \dots, e'_k$ for their bases, ε^a for the coordinate forms on V, and δ' for the determinant on V'. Set

$$N = \{\varepsilon^a : 1 \leq a \leq k+1\} \cup \{\varepsilon^a + \varepsilon^b : 1 \leq a < b \leq k+1\},$$
$$\Sigma_\nu = \ker \nu, \qquad C'_s = \{z : z_s = 0\}, \qquad H'_{cd} = \{z : z_c = z_d\},$$

and consider the finite family of pairs

$$\mathcal{F} = \{(V, H'_{cd}) : c < d\} \cup \{(\Sigma_\nu, C'_s) : \nu \in N,\ 1 \leq s \leq k\}. \tag{F.2}$$

Let $g_0 \colon V \to V'$ be the coordinate projection onto the first k coordinates.

Lemma F.4 (finite fiber obstruction)**.** *Suppose $k! = 0$ in κ. There is no k-linear map*

$$\tau \colon \operatorname{Hom}_\kappa(V, V')^k \longrightarrow \operatorname{Alt}^k_\kappa(V; \kappa)$$

satisfying both

(1) $\tau(g_0, \dots, g_0) = \delta' \circ (g_0, \dots, g_0)$;

(2) *for every* $(\Sigma, \Sigma') \in \mathcal{F}$*, the restriction* $\tau(g^1, \dots, g^k)|_{\Sigma^k}$ *is zero whenever* $g^r(\Sigma) \subseteq \Sigma'$ *for every* r*.*

Proof. Let δ be the determinant on V. Every alternating k-form α has a unique expression $\alpha = \delta(w_\alpha, \cdot, \dots, \cdot)$: its evaluations on the basis tuples omitting one vector give

$$(w_\alpha)_a = (-1)^{a-1}\alpha(e_1, \dots, \widehat{e}_a, \dots, e_{k+1}).$$

For a nonzero ν, this description also gives

$$\alpha|_{(\ker \nu)^k} = 0 \quad \Longleftrightarrow \quad \nu(w_\alpha) = 0. \tag{F.3}$$

Indeed choose ζ with $\nu(\zeta) = 1$ and a basis $y_1, \dots, y_k$ of $\ker \nu$. Writing $w_\alpha = \nu(w_\alpha)\zeta + z$ with $z \in \ker \nu$ gives $\delta(w_\alpha, y_1, \dots, y_k) = \nu(w_\alpha)\delta(\zeta, y_1, \dots, y_k)$, whose last factor is nonzero.

Suppose τ exists. For a function $a\colon \{1, \dots, k\} \to \{1, \dots, k\}$ define

$$B_a(\varphi_1, \dots, \varphi_k) = \tau(\varphi_1 \otimes e'_{a(1)}, \dots, \varphi_k \otimes e'_{a(k)}), \qquad (\varphi \otimes e'_c)(x) = \varphi(x)e'_c.$$

If a omits s, every input map takes V into C'_s. Applying condition (2) to $(\ker \varepsilon^b, C'_s)$ for each b and using (F.3) makes every coordinate of $w_{B_a(\varphi)}$ zero. Hence $B_a = 0$ unless a is a permutation.

For a permutation a and distinct c, d, replace the target vector in slots $a^{-1}(c)$ and $a^{-1}(d)$ by $e'_c + e'_d$. All input maps then take V into H'_{cd}, so their image under τ is zero. Expansion in these two slots gives $B_a + B_{(c\ d)\circ a}$ and two terms with a missing target coordinate. The latter vanish by the preceding paragraph. Thus, with $B = B_{\mathrm{id}}$,

$$B_a = \operatorname{sgn}(a)B \qquad (a \in S_k). \tag{F.4}$$

Define the $(k+1)$-linear form

$$\beta(\varphi_1, \dots, \varphi_k; \chi) = \chi\big(w_{B(\varphi_1, \dots, \varphi_k)}\big).$$

If the sth argument is $\nu \in N$, its rank-one input vanishes on Σ_ν, and every other rank-one input takes values in C'_s. Conditions (2) and (F.3) give $\beta(\varphi_1, \dots, \nu, \dots, \varphi_k; \nu) = 0$. Use this identity successively for ε^a, ε^b and $\varepsilon^a + \varepsilon^b$ to obtain

$$\beta(\dots, \varepsilon^a, \dots; \varepsilon^b) = -\beta(\dots, \varepsilon^b, \dots; \varepsilon^a).$$

Consequently $t(\sigma) = \beta(\varepsilon^{\sigma(1)}, \dots, \varepsilon^{\sigma(k)}; \varepsilon^{\sigma(k+1)})$ changes sign under every transposition of a slot with the last slot. These transpositions generate S_{k+1}, so $t(\sigma) = \operatorname{sgn}(\sigma)t(\mathrm{id})$.

Expand $g_0 = \sum_{c=1}^k \varepsilon^c \otimes e'_c$ in condition (1). The missing-coordinate terms vanish, and (F.4) gives

$$\varepsilon^1 \wedge \dots \wedge \varepsilon^k = \sum_{a \in S_k} \operatorname{sgn}(a) B(\varepsilon^{a(1)}, \dots, \varepsilon^{a(k)}).$$

Apply $\alpha \mapsto \varepsilon^{k+1}(w_\alpha)$. Its value on the left is $(-1)^k$; on the right it is $\sum_{a \in S_k} \operatorname{sgn}(a)^2 t(\mathrm{id}) = k!t(\mathrm{id}) = 0$. This is a contradiction. All identities remain valid in characteristic 2: repeated arguments vanish by alternation, and no step divides by 2 or by a factorial. □

We will need a quantitative consequence of this finite incompatibility. Give K^{k+1} and K^k their maximum norms and use the same family (F.2) over K.

Lemma F.5 (finite test certificate)**.** *Assume* $k! = 0$ *in* K*, as in Theorem F.1. There are finite sets of tests* $\mathcal{T}_j$*, one for each* $j = (\Sigma^j, \Sigma'^j) \in \mathcal{F}$*, consisting of tuples* $(g^1, \dots, g^k; \xi_1, \dots, \xi_k)$ *with* $g^r\Sigma^j \subseteq \Sigma'^j$ *and* $\xi_r \in \Sigma^j$*, such that every* K*-multilinear* τ *satisfying condition (1) of Lemma F.4 obeys*

$$\max_{j \in \mathcal{F}} \max_{(g;\xi) \in \mathcal{T}_j} |\tau(g^1, \dots, g^k)(\xi_1, \dots, \xi_k)| \geq 1. \tag{F.5}$$

Proof. Write $\operatorname{char} K = p$. All spaces in (F.2) are defined over $\mathbb{F}_p$. For each j, choose an $\mathbb{F}_p$-basis of the subspace Σ^j defined over $\mathbb{F}_p$, and one of

$$\{g \in \operatorname{Hom}_{\mathbb{F}_p}(\mathbb{F}_p^{k+1}, \mathbb{F}_p^k) : g\Sigma^j \subseteq \Sigma'^j\}.$$

Their scalar extensions are bases of the corresponding K-spaces: row reduction of the defining prime-field matrices has the same pivot positions over K. Take all k-tuples from each basis to form $\mathcal{T}_j$. Multilinearity in the maps and the vectors makes these tests equivalent to condition (2).

Choose coordinates θ for the finite-dimensional space of possible τ. Condition (1) is a system $A\theta = b$, and the tests for condition (2) form a system $C\theta = 0$, with all matrices and constants over $\mathbb{F}_p$. Lemma F.4 over $\mathbb{F}_p$ makes the combined system inconsistent. Row reduction of its augmented matrix therefore gives row vectors λ, μ over $\mathbb{F}_p$ with

$$\lambda A + \mu C = 0, \qquad \lambda b = 1.$$

For every K-valued solution of $A\theta = b$ it follows that $\mu C\theta = -1$. Every nonzero coefficient of μ has absolute value 1, since $\mathbb{F}_p^\times$ is finite. The nonarchimedean inequality now gives (F.5). □

F.3. **The Banach spaces defined by chain limits.** Set $L = \mathcal{F}$ and $\Lambda = L^{<\omega}$. Write $\Sigma^j \subseteq V = K^{k+1}$ and $\Sigma'^j \subseteq V' = K^k$ for the pair indexed by $j \in L$. Define

$$\begin{aligned} E &= \{x \in \ell^\infty(\Lambda; V) : \operatorname{dist}(x_i, \Sigma^j) \longrightarrow 0 \text{ along every } T \in \Gamma_j,\ j \in L\}, \\ E' &= \{z \in \ell^\infty(\Lambda; V') : \operatorname{dist}(z_i, \Sigma'^j) \longrightarrow 0 \text{ along every } T \in \Gamma_j,\ j \in L\}. \end{aligned} \tag{F.6}$$

Limits along a chain are taken outside its finite subsets. We verify the properties of these full solution spaces that will enter the proof.

Each subspace in the definition admits a projection R with $\|R\| \le 1$ and $\|1 - R\| \le 1$. For $\ker \varepsilon^a$, delete coordinate a; for $\ker(\varepsilon^a + \varepsilon^b)$, replace coordinate a by the negative of coordinate b; for V, use the identity. On V', delete coordinate s for C'_s, and replace coordinate c by coordinate d for H'_{cd}. These formulas are projections onto the stated spaces. Their norm bounds follow directly from the maximum norm and the nonarchimedean inequality. Moreover

$$\operatorname{dist}(v, \Sigma) = \|(1 - R)v\|. \tag{F.7}$$

Indeed $Rv \in \Sigma$ gives one inequality, and $(1 - R)(v - y) = (1 - R)v$ for every $y \in \Sigma$ gives the other.

For fixed j and $T \in \Gamma_j$, the corresponding condition in (F.6) is thus the inverse image of $c_0(T; V)$ under the bounded linear map $x \mapsto ((1 - R_j)x_i)_{i \in T}$, and similarly for V'. The space c_0 is closed in the sup norm: a uniform approximation within $\varepsilon/2$ by a null family leaves only finitely many coordinates of norm at least ε. Each condition is therefore closed and linear. Their intersection is a closed subspace of the Banach space $\ell^\infty(\Lambda; V)$, respectively $\ell^\infty(\Lambda; V')$, so E and E' are nonarchimedean Banach spaces.

Both contain their corresponding c_0 spaces. They are also modules over $\ell^\infty(\Lambda, K)$ under coordinatewise multiplication, with $\|tx\| \le \|t\|\|x\|$: each residual family in (F.7) is multiplied by the same bounded scalar family and remains null. Finally, if $T \in \Gamma_j$, every bounded Σ^j-valued family supported on T belongs to E. For chains with label j its residual is zero; for a chain with another label, its support meets that chain in at most one word by Lemma F.3. The identical assertion holds for E' and Σ'^j.

Since $k \ge 2$, the label $j = (V, H'_{12})$ occurs and both its spaces contain a vector of norm 1. Fix a j-chain T. Multiplying either such vector by arbitrary bounded scalar families on T gives an isometric embedding of $\ell^\infty(\mathbb{N}, K)$ into E, respectively E'. The binary-valued sequences form an uncountable set at pairwise distance 1, so both spaces are nonseparable.

F.4. **The global contradiction.**

Proof of Theorem F.1. Use the spaces (F.6). Suppose that $A = A^k_{E,E';K}$ has a bounded k-linear lift P. For $i \in \Lambda$, let $\xi_{[i]} \in E$ denote the family equal to $\xi \in V$ at i and zero elsewhere. Single-coordinate families belong to $c_0 \subseteq E$, and likewise in E'. For $g \in \mathrm{Hom}_K(V, V')$ let $g_{[i]} \in \mathcal{L}(E, E')$ send x to the family equal to $g(x_i)$ at i and zero elsewhere. Its norm is at most $\|g\|$. Set

$$\delta'_i(z_1, \dots, z_k) = \delta'(z_{1,i}, \dots, z_{k,i}) \in \mathrm{Alt}^k(E'; K).$$

The determinant estimate for the maximum norm gives $\|\delta'_i\| \le 1$. Define the fiber map

$$\pi_i(g^1, \dots, g^k)(\xi_1, \dots, \xi_k) = P(g^1_{[i]}, \dots, g^k_{[i]})(\delta'_i)(\xi_{1[i]}, \dots, \xi_{k[i]}). \tag{F.8}$$

It is k-linear in the maps and alternating in the vectors, and the lift identity gives

$$\pi_i(g, \dots, g) = \delta' \circ (g, \dots, g) \qquad \text{for every } g. \tag{F.9}$$

In particular every π_i satisfies condition (1) of Lemma F.4.

Fix $j = (\Sigma, \Sigma') \in L$, a chain $T \in \Gamma_j$, maps $g^1, \dots, g^k$ with $g^r\Sigma \subseteq \Sigma'$, and vectors $\xi_1, \dots, \xi_k \in \Sigma$. We claim that

$$\pi_i(g^1, \dots, g^k)(\xi_1, \dots, \xi_k) \longrightarrow 0 \qquad (i \in T). \tag{F.10}$$

For $t \in \ell^\infty(\Lambda, K)$ define u^r_t by

$$(u^r_t x)_i = \begin{cases} t_i g^r(x_i), & i \in T, \\ 0, & i \notin T. \end{cases}$$

This is a bounded map from E to E'. To see that its values lie in E', let R be the projection onto Σ above and decompose, on T,

$$t_i g^r(x_i) = t_i g^r(Rx_i) + t_i g^r((1 - R)x_i).$$

The first term is a bounded Σ'-valued family supported on T, hence belongs to E'. The second is null on T by (F.7), hence its zero extension belongs to $c_0 \subseteq E'$. The coordinate formula gives $\|u^r_t\| \le \|t\|\|g^r\|$, so $t \mapsto u^r_t$ is bounded linear with values in $\mathcal{L}(E, E')$.

For $c \in \ell^\infty(\Lambda, K)$ put

$$m_c(z_1, \dots, z_k) = \sum_{i \in T} c_i \delta'(z_{1,i}, \dots, z_{k,i}). \tag{F.11}$$

This sum converges. In fact, if R' projects onto Σ', write $z_{r,i} = a_{r,i} + b_{r,i}$ with $a_{r,i} = R'z_{r,i}$ and $b_{r,i} = (1 - R')z_{r,i}$. Both families are bounded by $\|z_r\|$, and $b_{r,i} \to 0$ along T. Expanding the determinant, the term with only a's is zero because $\dim \Sigma' = k - 1$. Every remaining term contains an error factor, giving

$$|\delta'(z_{1,i}, \dots, z_{k,i})| \le \max_{1 \le r \le k} \left(\|b_{r,i}\| \prod_{q \ne r} \|z_q\| \right) \longrightarrow 0.$$

Multiplication by bounded c preserves this limit. Completeness of K now gives the unconditional scalar sum in (F.11). Linearity in each argument and in c follows by taking limits of finite partial sums; repeated arguments make every summand zero. Moreover the determinant estimate gives

$$|m_c(z_1, \dots, z_k)| \le \|c\| \prod_{r=1}^{k} \|z_r\|.$$

Thus $c \mapsto m_c$ is a bounded linear map into $\mathrm{Alt}^k(E'; K)$, of norm at most 1.

Let x^r be the family equal to ξ_r on T and zero elsewhere; it belongs to E by the chain-generator inclusion already proved. Coordinatewise multiplication defines a bounded linear map $\rho \mapsto \rho x^r$ from $\ell^\infty(\Lambda, K)$ to E. Consequently

$$\Phi(t^1, \dots, t^k, c, r^1, \dots, r^k) = P(u^1_{t^1}, \dots, u^k_{t^k})(m_c)(r^1 x^1, \dots, r^k x^k)$$

is a bounded $(2k+1)$-linear scalar form on $\ell^\infty(\Lambda, K)$, with bound

$$\|P\| \prod_{s=1}^{k} \big(\|g^s\| \|\xi_s\|\big) \prod_{s=1}^{k} \|t^s\| \, \|c\| \prod_{s=1}^{k} \|r^s\|.$$

The set Λ is countably infinite, so Lemma F.2 gives $\Phi(e_i, \dots, e_i) \to 0$ outside finite sets. For $i \in T$ we have $u^s_{e_i} = g^s_{[i]}$, $m_{e_i} = \delta'_i$ and $e_i x^s = \xi_{s[i]}$. Equation (F.8) therefore identifies this diagonal value with the expression in (F.10), proving the claim.

Choose the finite tests of Lemma F.5, and define

$$G_j(i) = \max_{(g;\xi)\in\mathcal{T}_j} |\pi_i(g^1, \dots, g^k)(\xi_1, \dots, \xi_k)|, \qquad a_j = \{i \in \Lambda : G_j(i) < 1\}.$$

For fixed j and $T \in \Gamma_j$, equation (F.10) applies to every test in the finite set $\mathcal{T}_j$. Hence $G_j(i) \to 0$ along T, and $T \setminus a_j$ is finite. Lemma F.3 supplies an i belonging to every a_j. But (F.9) and Lemma F.5 force $G_j(i) \geq 1$ for some j, a contradiction.

There is therefore no bounded k-linear lift of A. Proposition 2.5 applies in this degree without dividing by $k!$: analyticity at any point would make the degree-k Taylor coefficient such a lift. Thus A is analytic at no point, as claimed. □

Appendix G. Parameter splittings and universal analytic reflection

A normed space P is polarizable in degree n (condition (Γ^n_P), Definition 3.8) if a bounded projection R_n of $\widehat{\otimes}^n_\pi P$ onto $\widehat{\Gamma}^n_\pi P$ exists; for Banach P over complete K this is reflection in degree n (Proposition G.1). Analytic descent (Section 2.3) of every Banach-valued analytic map through every closed inclusion is stronger; we characterize it and separate the two by an example.

Throughout this appendix, K is complete and nontrivially normed, and P is Banach. For $n \geq 1$, let $\widehat{\otimes}^n_\pi P$ be the completed projective tensor power as in Section 3.1; it represents bounded n-linear maps from P^n to Banach spaces. Recall from Definition 3.8 the closed subspace

$$\widehat{\Gamma}^n_\pi P = \overline{\operatorname{span}\{x^{\otimes n} : x \in P\}} \subseteq \widehat{\otimes}^n_\pi P.$$

Proposition G.1 (homogeneous reflection)**.** *For a fixed $n \geq 1$, the following conditions are equivalent.*

1. *For every Banach space Z, closed subspace $W \subseteq Z$, and bounded n-linear map $B\colon P^n \to Z$ with diagonal values in W, there is a bounded W-valued n-linear map with the same diagonal.*
2. *The subspace $\widehat{\Gamma}^n_\pi P$ is the range of a bounded projection in $\widehat{\otimes}^n_\pi P$.*

Proof. Suppose first that a projection R_n exists. Linearize B to a bounded map $\widetilde{B}\colon \widehat{\otimes}^n_\pi P \to Z$. The diagonal hypothesis and closedness of W imply $\widetilde{B}(\widehat{\Gamma}^n_\pi P) \subseteq W$. Composing $\widetilde{B}|_{\widehat{\Gamma}^n_\pi P}$ with R_n and the canonical tensor map gives the required multilinear representative, of norm at most $\|B\|\|R_n\|$. Conversely, take $Z = \widehat{\otimes}^n_\pi P$, $W = \widehat{\Gamma}^n_\pi P$, and B equal to the canonical tensor map. Linearizing the representative supplied by the first condition gives a bounded map into $\widehat{\Gamma}^n_\pi P$ fixing every pure power. It fixes their linear span and, by continuity, their closed span, so it is a projection. □

Definition G.2. The Banach parameter space P has *universal analytic reflection* if, for every Banach space Z, closed subspace $W \subseteq Z$, every open $U \subseteq P$, and every map $f\colon U \to W$, analyticity of the composite $U \to W \hookrightarrow Z$ implies analyticity of f.

Theorem G.3 (analytic tensor criterion)**.** *A Banach space P has universal analytic reflection if and only if there are projections $R_n\colon \widehat{\otimes}_\pi^n P \to \widehat{\Gamma}_\pi^n P$ for all $n \geq 1$ such that*

$$\limsup_{n\to\infty} \|R_n\|^{1/n} < \infty. \tag{G.1}$$

Proof. Assume first that the projections exist. Expand an ambient analytic map with values in W at a point. The scalar-line uniqueness argument in Theorem 2.3 puts every homogeneous diagonal in W. The construction in Proposition G.1 replaces its coefficient p_n by a W-valued coefficient of norm at most $\|p_n\|\|R_n\|$ and with the same diagonal. The exponential bound preserves a positive radius of convergence. The degree-zero coefficient is already in W, and the resulting expansion represents the given map.

Conversely, form the Banach spaces with supremum norms

$$Z = c_0\big(\{\widehat{\otimes}_\pi^n P\}_{n\geq 1}\big), \qquad W = c_0\big(\{\widehat{\Gamma}_\pi^n P\}_{n\geq 1}\big) \subseteq Z.$$

Here c_0 denotes families tending to zero in their respective Banach coordinate spaces; W is closed. On the open unit ball of P, define

$$f(x) = (x^{\otimes n})_{n\geq 1}.$$

Since $\|x^{\otimes n}\| \leq \|x\|^n$, this vector lies in W. As a Z-valued map it is analytic: its degree-n coefficient is the canonical tensor map placed in coordinate n, and has norm at most one. This gives an expansion at zero. To re-expand at a with $\|a\| < 1$, choose $r > 0$ with $\|a\| + r < 1$. Expanding each tensor power in $a + h$ gives translated coefficients, and the sum of all term bounds is at most $\sum_{n\geq 1}(\|a\| + r)^n < \infty$. Completeness of Z supplies the coefficients and the rearranged sum, proving analyticity throughout the unit ball. Universal reflection gives a W-valued expansion $\sum_m q_m(x, \dots, x)$ near zero with $\|q_m\| \leq C\rho^{-m}$ for some $C, \rho > 0$. Compose this expansion with the projection $\pi_n\colon W \to \widehat{\Gamma}_\pi^n P$ and compare on scalar lines with $x \mapsto x^{\otimes n}$. Uniqueness of scalar power-series coefficients gives

$$\pi_n q_n(x, \dots, x) = x^{\otimes n} \qquad (x \in P).$$

Linearizing $\pi_n q_n$ therefore gives a projection R_n onto $\widehat{\Gamma}_\pi^n P$ with $\|R_n\| \leq \|q_n\| \leq C\rho^{-n}$, proving (G.1). □

G.1. **Projection bounds and the special role of fixed degree.** For $P = K^d$ with the maximum norm, the coordinate construction in Theorem 5.11, applied to the canonical tensor map, gives $\|R_n\| \leq d^n$ for $d \geq 1$. A bounded linear change of coordinates introduces only a fixed distortion factor to the n-th power, so it does not change the exponential-growth condition.

For $P = \ell^1(I, K)$, the proof of Proposition 5.13(1) identifies $\widehat{\otimes}_\pi^n P$ isometrically with $\ell^1(I^n, K)$, with unit vectors e_w for $w \in I^n$, puts the sum of the e_w over each permutation orbit in I^n into $\widehat{\Gamma}_\pi^n P$, and gives a projection onto $\widehat{\Gamma}_\pi^n P$ of norm at most $n!$. In these coordinates every element of $\widehat{\Gamma}_\pi^n P$ is constant on each permutation orbit, since every pure power is. Over a nonarchimedean field and for infinite I, the bound $n!$ is optimal even among projections that do not preserve the individual orbit blocks.

Proposition G.4. *If K is nonarchimedean and complete, and I is infinite, then*

$$\inf\{\|R\| : R\colon \widehat{\otimes}_\pi^n \ell^1(I, K) \to \widehat{\Gamma}_\pi^n \ell^1(I, K) \text{ is a bounded projection}\} = n!.$$

Consequently $\ell^1(I, K)$ does not have universal analytic reflection.

Proof. Proposition 5.13(1) gives the upper bound. For the lower bound, take an orbit $\mathcal{O}$ of a tuple with n distinct indices; it has $n!$ elements. For each $w \in \mathcal{O}$, let λ_w be the common coefficient on $\mathcal{O}$ of $R(e_w)$. The orbit sum $\sum_{w\in\mathcal{O}} e_w$ belongs to $\widehat{\Gamma}^n_\pi P$ and is fixed by R, hence

$$\sum_{w\in\mathcal{O}} \lambda_w = 1.$$

The nonarchimedean inequality implies that some $|\lambda_w| \geq 1$. For this w, the ordinary sum norm satisfies $\|R(e_w)\|_1 \geq n!|\lambda_w| \geq n!$. Thus $\|R\| \geq n!$. Finally, $(n!)^{1/n}$ is unbounded, so Theorem G.3 excludes universal analytic reflection. □

Corollary G.5 (entire polynomial approximation of a nonanalytic smooth map)**.** *Let K be a complete nonarchimedean nontrivially normed field and put $P = \ell^1(\mathbb{N}, K)$, with the ordinary sum norm. There are Banach spaces $W \subseteq Z$, with W closed, and a map f on the open unit ball of P such that f is analytic into Z and C^∞ into W, but is not analytic into W at zero. Moreover, f is a uniform limit on every smaller ball of entire W-valued polynomials.*

Proof. Take

$$Z = c_0\big(\{\widehat{\otimes}^n_\pi P\}_{n\geq 1}\big), \qquad W = c_0\big(\{\widehat{\Gamma}^n_\pi P\}_{n\geq 1}\big), \qquad f(x) = (x^{\otimes n})_{n\geq 1}.$$

The spaces are Banach and W is closed in Z. As in the proof of Theorem G.3, the ambient expansion has coefficient norms at most one and defines an analytic map on the open unit ball. Smooth reflection through the closed isometric inclusion makes f C^∞ as a map into W.

Suppose a W-valued expansion at zero has coefficients q_n. Uniqueness on scalar lines gives

$$\pi_n q_n(x, \dots, x) = x^{\otimes n},$$

where $\pi_n \colon W \to \widehat{\Gamma}^n_\pi P$ is the coordinate projection. Linearizing $\pi_n q_n$ gives a projection $\widehat{\otimes}^n_\pi P \to \widehat{\Gamma}^n_\pi P$ of norm at most $\|q_n\|$. Proposition G.4 therefore gives $\|q_n\| \geq n!$. Since $(n!)^{1/n}$ is unbounded, these coefficients give a formal multilinear series of radius zero, contradicting the assumed analytic expansion.

For each n, choose the sorted-orbit projection $R_n \colon \widehat{\otimes}^n_\pi P \to \widehat{\Gamma}^n_\pi P$. Its composition with the canonical tensor map is a bounded multilinear representative of $x \mapsto x^{\otimes n}$ with values in $\widehat{\Gamma}^n_\pi P$. Consequently the truncation

$$f_N(x) = (x, x^{\otimes 2}, \dots, x^{\otimes N}, 0, \dots)$$

is an entire W-valued polynomial. For $0 < r < 1$ and $\|x\| \leq r$,

$$\|f(x) - f_N(x)\|_W \leq r^{N+1},$$

which proves the asserted uniform convergence. □

There is no conflict with Theorem 5.12(2): the ambient representation there has one fixed degree d, so factoring a general analytic parameter map through a free ℓ^1 space incurs a single factorial bound $d!$, rather than a sequence of factorial bounds across Taylor degrees. Over $\mathbb{R}$ and $\mathbb{C}$ every Banach space has universal analytic reflection, by Theorem G.3: the symmetrization $\frac{1}{n!}\sum_{\sigma\in S_n} \sigma$ has norm at most one on $\widehat{\otimes}^n_\pi P$ and, by polarization, is a projection onto $\widehat{\Gamma}^n_\pi P$.

Appendix H. Why a largest full analytic domain need not exist

In this appendix a pair (E, F) consists of a source space and a value space. It is split in degree k (Definition H.2) when $j_{E,F}$ has a bounded linear retraction, a splitting for the single value space F. Split pairs form an analytic domain (Definition H.1), since a splitting at the destination of a morphism suffices, with no compatibility between pairs (Lemma H.3).

There need not be a largest analytic domain: the endomorphism actions at two objects can each be analytic while their cross-action is not. As in Proposition 2.5, a bounded d-linear *lift* of a homogeneous map f is a bounded multilinear map B with $B(x,\dots,x)=f(x)$.

Definition H.1. A full subcategory $\mathcal{D}\subseteq \mathsf{Vec}_K^{\mathrm{op}}\times\mathsf{Vec}_K$ is an *analytic domain* for Alt^k if the restriction of Alt^k to $\mathcal{D}$ is analytic on every hom space.

Definition H.2. A pair (E,F) is *split in degree k* if the isometric inclusion

$$j_{E,F}\colon\ \mathrm{Alt}^k(E;F)\longrightarrow \mathrm{Mult}^k(E;F)$$

has a bounded linear retraction.

Lemma H.3. *Every morphism-space action whose destination is a split pair is analytic. In particular, the full subcategory of split pairs is an analytic domain.*

Proof. Let (E,F) be split, with retraction R, and let the other object be (D,G). For $u\colon E\to D$ and $v\colon G\to F$, the action is

$$m\longmapsto R\big(v\circ m\circ(u,\dots,u)\big).$$

Before applying R, replace the k copies of u by independent variables. This gives a bounded $(k+1)$-linear lift of the joint action, with values in $\mathcal{L}(\mathrm{Alt}^k(D;G),\mathrm{Mult}^k(E;F))$. Postcomposition with R gives a lift with the required alternating codomain. No compatibility between the retractions chosen for different objects is needed. □

Theorem H.4 (no largest full analytic domain)**.** *Let p be prime, let $k\geq p$, and equip $K=\mathbb{F}_p(t)$ with its t-adic absolute value. There is no largest full analytic domain for*

$$\mathrm{Alt}^k\colon \mathsf{Vec}_K^{\mathrm{op}}\times\mathsf{Vec}_K\longrightarrow\mathsf{Vec}_K.$$

The assertion remains true if the domains must be closed under isomorphisms, and if their objects are restricted to pairs of finite-dimensional, nonarchimedean normed K-spaces. In the latter case, isomorphism closure is understood inside that restricted ambient category.

More precisely, there are objects $X_k=(E_k,G)$ and $Y_k=(D_k,G)$ whose singleton full subcategories are analytic, such that Y_k is split and the action on morphisms from Y_k to X_k is not analytic.

Here finite-dimensional means finite *algebraic* dimension over the incomplete field K. These spaces need not admit continuous coordinates over K. The theorem does not give a counterexample over a complete field, or a counterexample with Banach objects.

We give the entire construction. Put $L=\mathbb{F}_p((t))$, the completion of K. We first work in degree p, then extend to every $k\geq p$ by adjoining coordinate summands.

H.1. **The rigid source and the split pair.** We use the following elementary completion principle. Let K be dense in a complete valued extension L, let (E_i) be a finite family of dense K-subspaces of Banach L-spaces H_i, and let Z be Banach over L. A bounded K-multilinear map $\prod_i E_i\to Z$ extends uniquely to a bounded L-multilinear map $\prod_i H_i\to Z$, with the same norm. Indeed, approximate each input by vectors in E_i. A telescoping expansion and the multilinear bound make the resulting values Cauchy, independently of the approximations. The bound persists in the limit, and density of K in L gives L-linearity in each slot. In the applications below the extended codomain is the complete space L or A, while membership of values on the original inputs in the smaller spaces is retained.

Put $L=\mathbb{F}_p((t))$, the completion of K, and consider the normed L-algebra

$$A=L[\epsilon]/(\epsilon^p),\qquad e_i=\epsilon^i\quad(0\leq i<p),$$

with the maximum norm in this basis. Multiplication satisfies $\|xy\| \le \|x\|\,\|y\|$. Choose elements $a_0, \dots, a_{p-1} \in L$, together with the finitely many auxiliary scalars τ_w introduced below, algebraically independent over K. Such a choice is possible because K is countable and L is uncountable: the elements of L algebraic over any finitely generated extension of K form a countable set. Set

$$a = \sum_{i=0}^{p-1} a_i e_i, \qquad E = K^p + Ka \subseteq A.$$

Every subspace used below carries the norm induced by A or L. In particular, E has K-dimension $p+1$ and completion A.

Lemma H.5. *Every bounded endomorphism of E is multiplication by a scalar in K, and every bounded linear map $E \to K$ is zero.*

Proof. A bounded K-linear map extends to completions, and the extension is L-linear by density of K in L. If it preserves E, its columns, the images of the e_i, give a matrix $M + ac^{\mathsf{T}}$, with $M \in M_p(K)$ and $c \in K^p$. Its value at a lies in $K^p + Ka$. Comparison of homogeneous degrees in the algebraically independent a_i shows first that $a(c^{\mathsf{T}}a) = 0$, hence $c = 0$, and then that $M = sI$ for some $s \in K$.

Similarly, the extension of a bounded map $E \to K$ is an L-linear functional on A. Its coefficients belong to K, since its values on the e_i do. Its value on a also lies in K, so algebraic independence makes all its coefficients zero. □

Multiplication by a has lower triangular Toeplitz matrix

$$(M_a)_{ij} = \begin{cases} a_{i-j}, & i \ge j, \\ 0, & i < j. \end{cases}$$

Define

$$D_0 = \operatorname{span}_K\{e_i, \epsilon^i a : 0 \le i < p\} + Ka^2, \qquad \mathcal{W} = \{e_i : 0 \le i < p\} \cup \{e_i + e_j : 0 \le i < j < p\},$$

and put

$$D = D_0 + \sum_{w \in \mathcal{W}} K\tau_w w, \qquad G = \operatorname{span}_K\{\det(d_1, \dots, d_p) : d_i \in D\} \subseteq L.$$

The determinant is taken in the basis $(e_0, \dots, e_{p-1})$. These are finite-dimensional K-spaces, with completions A and L: D contains K^p, and G contains 1. By multilinearity, G is spanned by determinants of the displayed finite generating set for D. We may therefore identify G with a finite-dimensional subspace of $K[a_0, \dots, a_{p-1}, (\tau_w)_w]$. Every element has degree at most one in each individual τ_w, since a nonzero determinant cannot repeat the generator $\tau_w w$. Algebraic independence gives

$$\tau_w^2 G \cap G = \{0\} \qquad (w \in \mathcal{W}). \tag{H.1}$$

Lemma H.6. *Every bounded p-linear map $D^p \to G$ is alternating. Thus (D, G) is split in degree p.*

Proof. Compose such a map m with $G \hookrightarrow L$ and extend to a bounded L-multilinear map on A^p. Fix two argument positions. Put the same $w \in \mathcal{W}$ in those positions and arbitrary standard basis vectors in the others, and call the value $c \in G$. Replacing both copies of w by $\tau_w w \in D$ shows that $\tau_w^2 c \in G$, so (H.1) gives $c = 0$. Taking $w = e_i$ gives vanishing on repeated basis vectors. Taking $w = e_i + e_j$ then gives the signed interchange identity in the two positions. Expanding two equal arbitrary arguments in the basis proves that the completed

map, and hence m, is alternating, including when $p = 2$. The inclusion of alternating maps into multilinear maps is therefore an isometric bijection, whose inverse is the desired retraction. □

H.2. **A missing coefficient.** The coefficient space of alternating maps on E is

$$C = \{\lambda \in G : a_i\lambda \in G \text{ for every } i\}.$$

Indeed, an alternating map extends to $\lambda \det$ on the p-dimensional completion A. Determinants of tuples from E span $\operatorname{span}_K\{1, a_0, \dots, a_{p-1}\}$: a determinant can contain at most one copy of a, and replacing one basis vector by a realizes each a_i up to sign. Consequently

$$\operatorname{Alt}^p(E; G) = C \det . \tag{H.2}$$

This identification is isometric, since the determinant bound gives one inequality and evaluation on the standard basis gives the reverse inequality.

Algebraically set every τ_w to zero. The image of G is

$$G_0 = \operatorname{span}_K\{\det(d_1, \dots, d_p) : d_i \in D_0\}.$$

Put $C_0 = \{\lambda \in G_0 : a_i\lambda \in G_0 \text{ for every } i\}$. Specialization sends C into C_0; no continuity of specialization is asserted. The displayed generators of D_0 are homogeneous polynomial vectors of degrees zero, one, and two. Their determinants are homogeneous, so G_0 is graded, and the defining conditions show that C_0 is graded as well.

Lemma H.7. *The homogeneous degree-one part of C_0 is zero.*

Proof. The degree-two part of G_0 is spanned by the 2×2 minors of M_a and the coordinates of a^2. These correspond respectively to choosing two degree-one columns, or one degree-two column, in a determinant of generators. On homogeneous quadratic polynomials define the K-linear map $\Phi\colon K[a_0, \dots, a_{p-1}]_2 \to K[T]$ by

$$\Phi(a_i a_j) = \begin{cases} T^{i+j}, & i + j \geq p - 1, \\ 0, & i + j < p - 1. \end{cases}$$

Every 2×2 minor of M_a is killed by Φ. When both product terms are nonzero, their index sums coincide. If only one is nonzero, write the rows as $r < s$ and columns as $i < j$. The surviving term has index sum $r + s - i - j$, and the missing term implies $r < j$. Hence $r + s - i - j \leq s - i - 1 \leq p - 2$, so its image is again zero.

The coefficient of ϵ^r in a^2 is $\sum_{i+j=r} a_i a_j$. Its image is zero for $r < p - 1$; for $r = p - 1$ there are exactly p summands and its image is $pT^{p-1} = 0$. Thus Φ kills the entire degree-two part of G_0. If $\ell = \sum_j c_j a_j$ lies in the degree-one part of C_0, then $a_{p-1}\ell \in G_0$, whereas

$$\Phi(a_{p-1}\ell) = \sum_j c_j T^{p-1+j}.$$

The powers are distinct, so all c_j vanish. □

H.3. **The nonanalytic cross-action.** Set $X = (E, G)$ and $Y = (D, G)$. Lemma H.5 identifies $\mathcal{L}(E, E)$ isometrically with K, and the self-action of X is

$$(s, v) \longmapsto s^p v_*.$$

This is a bounded polynomial, so the singleton full subcategory on X is analytic. The singleton on Y is analytic by Lemmas H.6 and H.3.

For $x \in E$, multiplication by x in A restricts to a bounded map $M_x \colon E \to D$. Indeed, writing $x = b + ha$ with $b \in K^p$ and $h \in K$, its values on the generators e_j, a lie in

$$K^p + \sum_i K\epsilon^i a + Ka^2 \subseteq D.$$

Submultiplicativity gives a bounded linear family $x \mapsto M_x$ from E to $\mathcal{L}(E, D)$. Its matrix is lower triangular with diagonal entry x_0, the e_0-coordinate of x in A, so $\det M_x = x_0^p$. In the cross-action from Y to X, fix the target morphism to id_G, evaluate at $\det \in \mathrm{Alt}^p(D; G)$, and use (H.2). The resulting map is

$$f \colon E \longrightarrow C, \qquad f(x) = x_0^p. \tag{H.3}$$

Its values lie in C because it was obtained by pulling back an alternating G-valued map.

Suppose that f were analytic at zero. Since $f(sx) = s^p f(x)$, uniqueness of power-series coefficients on scalar lines gives a bounded p-linear map $B \colon E^p \to C$ with diagonal f. This is the homogeneous coefficient argument used in Proposition 2.5; it requires no factorial division or completeness. Compose B with $C \hookrightarrow L$ and extend to an L-multilinear map on A^p. Put

$$b_i = B(e_0, \dots, e_0, e_i) \in C.$$

There are $p - 1$ initial copies of e_0, and $b_0 = f(e_0) = 1$. Evaluation at a in the last position gives $\sum_i a_i b_i \in C$. Specializing all τ_w to zero yields elements $d_i \in C_0$ with

$$d_0 = 1, \qquad \sum_i a_i d_i \in C_0.$$

The degree-one part of this last expression is $\sum_i a_i (d_i)_0$, where $(d_i)_0 \in K$ is the constant term. Its a_0-coefficient equals one. This contradicts Lemma H.7. Thus (H.3), and therefore the cross-action from Y to X, is not analytic.

H.4. **Higher degrees and the categorical conclusion.** For $k = p + r$, let

$$E_k = E \oplus K^r, \qquad D_k = D \oplus K^r$$

with maximum norms. Their completions have dimension k over L. Determinants of tuples from E_k still span $\mathrm{span}_K\{1, a_0, \dots, a_{p-1}\}$, so extraction of the top-degree coefficient identifies

$$\mathrm{Alt}^k(E_k; G) = C \det_k \tag{H.4}$$

isometrically. In particular, the coefficient takes values in C, a fact needed to retain the preceding obstruction after increasing the degree.

By Lemma H.5, the extension of every bounded endomorphism of E_k to its L-completion $A \oplus L^r$ has block form

$$\begin{pmatrix} sI_p & B \\ 0 & T \end{pmatrix}, \qquad s \in K, \quad T \in M_r(K).$$

The scalar s and entries of T are bounded linear functions of the endomorphism, as is seen by evaluation on the standard basis. Its determinant is $s^p \det T$, so its action on (H.4) is multiplication by this bounded polynomial. Including postcomposition shows that the singleton full subcategory on $X_k = (E_k, G)$ is analytic.

The pair $Y_k = (D_k, G)$ is split. Given $m \in \mathrm{Mult}^k(D_k; G)$, restrict the first p arguments to D and fix the last r arguments at the standard auxiliary basis $e'_1, \dots, e'_r$ of K^r. The resulting

p-linear map α_m is alternating by Lemma H.6. Write $\pi_D\colon D_k \to D$ and $\pi_r\colon D_k \to K^r$ for the two projections. Explicitly, set

$$(\mathcal{R}m)(v_1,\dots,v_k) = \sum_{\substack{S\subseteq\{1,\dots,k\}\\ |S|=p}} \varepsilon(S)\,\alpha_m(\pi_D v_{s_1},\dots,\pi_D v_{s_p})\det(\pi_r v_{t_1},\dots,\pi_r v_{t_r}),$$

where $s_1<\cdots<s_p$ list S, $t_1<\cdots<t_r$ list its complement, and $\varepsilon(S)$ is the sign of the permutation $(s_1,\dots,s_p,t_1,\dots,t_r)$. The determinant and the sign are scalars multiplying the G-valued factor. In degree $r=0$ the determinant is 1. This shuffle expression is alternating by the same repeated-argument cancellation as the wedge product, involves no factorial division, and gives a bounded linear map $\mathcal{R}\colon \mathrm{Mult}^k(D_k;G)\to \mathrm{Alt}^k(D_k;G)$ with norm at most $\binom{k}{p}$. It fixes every alternating map: after extension to the k-dimensional completion, both maps are determined by their value on the standard basis, where the shuffle formula has exactly one nonzero term. This is the required retraction. For $r>0$, we do not need or assert that every k-linear map on D_k is alternating.

Finally, restrict the cross-action along the bounded affine family

$$E \longrightarrow \mathcal{L}(E_k,D_k), \qquad x\longmapsto M_x\oplus I_r.$$

Evaluation at $\det_k$, which is G-valued on D_k by Laplace expansion along the first p rows, and coefficient extraction through (H.4) give exactly the nonanalytic map (H.3). Thus the cross-action from Y_k to X_k is not analytic.

A largest analytic full subcategory would have to contain both individually analytic singleton subcategories, and fullness would then require this cross-action to be analytic. This is impossible. If closure under isomorphisms is required, replace each singleton by its full isomorphism closure. Every action between two copies is transported from the self-action by bounded linear changes of coordinates, so both closures remain analytic. The same contradiction proves Theorem H.4.

Finite-dimensional bases. The same spaces show that Theorem 5.11 needs continuous coordinates on the base, not only finite algebraic dimension. Regard E as an analytic manifold with one chart; by Lemma H.5 it has no continuous coordinates. Over it take the trivial bundle with fiber $D_k\oplus E_k$ and the two frames id and $S_x(d,e)=\big(d+(M_x\oplus I_r)e,e\big)$, together with the trivial bundle with fiber G; S_x and its inverse $(d,e)\mapsto\big(d-(M_x\oplus I_r)e,e\big)$ are affine in x, so these bundles are analytic. By (10.1), evaluation at $\det_k$ and (H.4), analyticity of the induced alternating atlas at $x=0$ would make (H.3) analytic at zero, which was excluded above.

Appendix I. Further coordinate methods

The proof of Proposition I.1 gives a splitting for scalar values: it writes down an explicit section of unnormalized antisymmetrization.

I.1. **Countable-type scalar sources.** The scalar obstruction of Theorem 7.2(2) requires large source spaces in a stronger sense than infinite dimension. A normed space is *of countable type* if a countable subset has dense linear span.

Proposition I.1 (countable-type scalar sources)**.** *Let K be complete and nonarchimedean. If either E or D is of countable type, then $A^k_{E,D;K}$ is analytic in every degree. Neither completeness nor a nonarchimedean norm is required of E,D. In particular, the two spaces in Theorem 7.2(2) must both fail to be of countable type.*

Proof. The case $k=0$ is constant. For a normed space V, set

$$q_V(x)=\inf_{x=\sum_i x_i}\max_i\|x_i\|,$$

where the decompositions use nonempty finite families, allowing a zero term. Scaling and concatenating them show that q_V is a nonarchimedean seminorm. Let V^u be the completion of its separated quotient; the canonical map $j_V\colon V \to V^u$ is contractive and has dense range. Expanding a bounded scalar multilinear form m along decompositions of each argument gives

$$|m(x_1,\dots,x_k)| \le \|m\| \prod_a q_V(x_a).$$

Thus scalar multilinear forms extend uniquely and isometrically to V^u, as do alternating forms. Extension preserves alternation by density of tuples with repeated entries, and commutes with permutations. Every bounded $u\colon V \to W$ induces $u^u\colon V^u \to W^u$ with $\|u^u\| \le \|u\|$. Uniqueness on the dense image makes $u \mapsto u^u$ bounded linear, and extension of forms intertwines precomposition by u and u^u.

If V is of countable type, so is V^u. By van der Put's basis theorem [19, Proposition 1.4], it has a t-orthogonal basis (b_i) for any fixed $0 < t < 1$. Writing f_i for its coordinate functionals, define

$$S(m)(x_1,\dots,x_k) = \sum_{i_1<\cdots<i_k} m(b_{i_1},\dots,b_{i_k}) \prod_{a=1}^{k} f_{i_a}(x_a).$$

The weighted coordinates $|f_i(x)|\|b_i\|$ form a null family bounded by $t^{-1}\|x\|$. Products of finitely many bounded null families are null, so the displayed sum converges unconditionally in K and $\|S(m)\| \le t^{-k}\|m\|$. Let $\mathcal{A}_V$ be unnormalized antisymmetrization, the signed sum over permutations of the arguments. It is bounded and alternating-valued in every characteristic. On finite-coordinate vectors, sorting distinct indices shows $\mathcal{A}_V S(m) = m$ for alternating m; repeated indices contribute zero. Density proves the same identity everywhere. Hence $R = \mathcal{A}_V S$ retracts all multilinear forms onto alternating forms, while $S|_{\mathrm{Alt}}$ is a bounded right inverse of $\mathcal{A}_V$. Transport through j_V gives both operators on the original V.

If E is of countable type, a lift of A is $R_E\big(m \circ (u_1,\dots,u_k)\big)$. If D is of countable type, use instead $\mathcal{A}_E\big(S_D(m) \circ (u_1,\dots,u_k)\big)$. Both are bounded multilinear in the u_i, linear in m, and have diagonal $A(u)(m)$; the second identity uses that antisymmetrization commutes with simultaneous precomposition. Proposition 2.5 concludes the proof. □

I.2. **Continuous algebraic polynomials on c_0.** Theorem 5.12(1) starts with bounded multilinear coefficients in an ambient space. For one homogeneous polynomial, interpolation supplies a lift even when only the diagonal is known to be continuous. The resulting bound is degreewise; it gives no assertion about a common radius across an infinite sequence of degrees.

Proposition I.2 (continuous algebraic polynomials on c_0)**.** *Let K be nontrivially normed and nonarchimedean, let Z be complete and nonarchimedean, and let $n \ge 1$. Suppose $p\colon c_0(I,K) \to Z$ is continuous and is an algebraic homogeneous polynomial: there exists a K-multilinear map $b\colon c_0(I,K)^n \to Z$, not assumed continuous, with $p(x) = b(x,\dots,x)$. Then p has a bounded n-linear lift q. More precisely, there is $C_{K,n} < \infty$, independent of I, Z, p, such that*

$$\|q\| \le C_{K,n}\|p\|_{\mathrm{diag}}, \qquad \|p\|_{\mathrm{diag}} := \inf\{D \ge 0 : \|p(x)\| \le D\|x\|^n \text{ for all } x\}.$$

No completeness assumption on K is needed.

Proof. Continuity at zero and $p(ax) = a^n p(x)$ imply $\|p\|_{\mathrm{diag}} < \infty$. Explicitly, choose $t \in K$ with $0 < |t| < 1$ and $\delta > 0$ such that $\|p(x)\| \le 1$ when $\|x\| < \delta$. For $x \ne 0$, choose an integer m with $|t|\delta \le |t^m|\|x\| < \delta$. Homogeneity gives $\|p(x)\| \le (|t|\delta)^{-n}\|x\|^n$.

Fix a total order on I. On each finite coordinate subspace, the algebraic formula for p is an ordinary homogeneous polynomial of degree n. Polynomial uniqueness over the infinite field K makes its coefficients compatible as the finite coordinate set varies. Denote the coefficient

of x^α by c_α, where $\alpha\colon I \to \mathbb{N}$ is finitely supported and $\sum_i \alpha_i = n$. Its support has at most n elements.

Choose distinct $a_0, \dots, a_n$ in the valuation ring of K, and let $L_n \geq 1$ bound the absolute values of all entries of the inverse Vandermonde matrix $(a_i^j)_{0\leq i,j\leq n}$. If the support of α has $d \leq n$ elements, interpolate the restriction of p to those coordinates on the grid $\{a_0, \dots, a_n\}^d$. Each grid vector has norm at most one. Applying the inverse Vandermonde matrix in each coordinate and using the nonarchimedean inequality in Z yields

$$\|c_\alpha\| \leq L_n^d \|p\|_{\mathrm{diag}} \leq L_n^n \|p\|_{\mathrm{diag}}.$$

For a nondecreasing word $i_1 \leq \cdots \leq i_n$, write $c_{i_1,\dots,i_n}$ for the coefficient with those multiplicities. Define

$$q(x_1, \dots, x_n) = \sum_{i_1 \leq \cdots \leq i_n} c_{i_1,\dots,i_n} (x_1)_{i_1} \cdots (x_n)_{i_n}.$$

The coefficients are uniformly bounded and each input is a null family, so the summands tend to zero outside finite sets of words. Completeness and the nonarchimedean inequality give convergence, multilinearity, and the bound $\|q\| \leq L_n^n \|p\|_{\mathrm{diag}}$. On finite supports its diagonal is p; density and continuity extend the identity to all of $c_0(I, K)$. Thus $C_{K,n} = L_n^n$ suffices. □

This bound is uniform in the number of coordinates, but the interpolation nodes and L_n depend on the degree. No exponential growth bound for $C_{K,n}$ is asserted. Consequently this proposition does not justify replacing lifting norms by diagonal norms in an arbitrary infinite analytic expansion. Theorem 5.12(1) obtains its stronger, radius-preserving conclusion from the already bounded ambient coefficients.

I.3. **A nonspherical target norm with analytic precomposition.**

Remark I.3. Spherical completeness of the given target norm is not necessary for analyticity, even in degrees at least the characteristic. Let $K = \mathbb{F}_p((t))$ with $|t| = 1/2$ and let $F = c_0(\mathbb{N}, K)$, the space of sequences tending to zero. Define

$$w_j = 1 + \frac{1}{j+2}, \qquad \|x\|_w = \sup_{j\geq 0} w_j |x_j|.$$

This is a nonarchimedean norm satisfying

$$\|x\|_\infty \leq \|x\|_w \leq \frac{3}{2} \|x\|_\infty.$$

Thus F is Banach for either norm. The usual supremum norm has nonzero values in $2^{\mathbb{Z}}$, since every nonzero null sequence attains its maximum absolute value. The discrete-distance argument in the proof of Corollary 5.3 therefore makes the supremum norm spherically complete. Theorem 5.2 gives analytic precomposition in every finite degree for that norm, for all normed E, E'. Equivalent norms preserve the bounded maps and their analytic power-series expansions: the norms of coefficients change by a fixed factor and convergence is unchanged. Thus precomposition remains analytic for $\|\cdot\|_w$, also directly by Corollary 5.3.

Nevertheless, the weighted norm is not spherically complete. Write e_j for the coordinate vectors, set $a_n = \sum_{j<n} e_j$, and consider the closed balls

$$B_n = \overline{B}_w(a_n, w_n).$$

They are nested because $w_{n+1} < w_n$ and $\|a_{n+1} - a_n\|_w = w_n$; the nonarchimedean inequality gives $B_{n+1} \subseteq B_n$. If x belonged to all of them, then, for each j and any $n > j$,

$$|x_j - 1| \leq \frac{w_n}{w_j} < 1.$$

Hence $|x_j| = 1$ for every j, contradicting $x_j \to 0$. Their intersection is empty. This example separates spherical completeness of a particular norm from analyticity, which is invariant under equivalent norms. It does not exclude a characterization using an equivalent spherically complete norm: this target has exactly such a norm.

Appendix J. Ordering splittings

Throughout, K is a nontrivially normed field, Σ is a totally ordered set, and $\vartheta \in K$ is the fixed element with $0 < |\vartheta| < 1$.

J.1. **Lower estimates.**

Proof of Theorem 5.6. (1) *Step 1 (the weight ν).* Let $x, y \in V$ be nonzero and $\Sigma_0 \subseteq \Sigma$ a finite set containing their supports. In this proof an interval is a set of consecutive elements of Σ_0; disjoint ones lie in disjoint intervals of Σ, so (5.1) applies to them. For an interval I let $\nu_x(I)$ be the supremum of $\sum_i \|\mathcal{P}_{I_i} x\|^r$ over the families of pairwise disjoint intervals $I_i \subseteq I$. Then ν_x is monotone and superadditive on disjoint intervals, $\|\mathcal{P}_I x\|^r \le \nu_x(I)$ and $\nu_x(\Sigma_0) \le C^r \|x\|^r$. Put $a_I = \nu_x(I)/\nu_x(\Sigma_0)$, $b_I = \nu_y(I)/\nu_y(\Sigma_0)$ and $\nu(I) = a_I + b_I$, so that $\nu(\Sigma_0) = 2$.

Step 2 (the binary tree). Build a binary tree of intervals with root Σ_0 of depth 0. A node I is a leaf if it has at most one element or $\nu(I) = 0$. Otherwise let t_I be the least $t \in I$ with $\nu(\{t' \in I : t' \le t\}) \ge \nu(I)/2$; the children of I, of depth one more, are $I_< = \{t \in I : t < t_I\}$ and $I_> = \{t \in I : t > t_I\}$. Then $\nu(I_<) < \nu(I)/2$ by minimality of t_I and $\nu(I_>) \le \nu(I)/2$ by superadditivity, so the nodes of depth d are pairwise disjoint and satisfy $\nu \le 2^{1-d}$.

Step 3 (the sum over the nodes). For an interval I let z_I be the sum of $x(t)y(t')\, e_t \otimes e_{t'}$ over $t < t'$ in I, so that $\mathcal{P}_<(x \otimes y) = z_{\Sigma_0}$. At an internal node

$$z_I = z_{I_<} + z_{I_>} + \mathcal{P}_{I_<} x \otimes \mathcal{P}_{I \setminus I_<} y + \mathcal{P}_{\{t_I\}} x \otimes \mathcal{P}_{I_>} y,$$

and $z_I = 0$ at a leaf I: it has at most one element, or $\nu(I) = 0$ and x vanishes on I. So $\mathcal{P}_<(x \otimes y)$ is the sum of the last two terms over the internal nodes. At a node I they have projective norm at most

$$2\nu_x(I)^{1/r} \nu_y(I)^{1/r} \le 2C^2 (a_I b_I)^{1/r} \|x\| \|y\|.$$

At depth d, $a_I b_I \le (\nu(I)/2)^2 \le 4^{-d}$, and $\sum_I (a_I b_I)^{1/2} \le 1$ by the Cauchy–Schwarz inequality, as $\sum_I a_I \le 1$ and $\sum_I b_I \le 1$; so $\sum_I (a_I b_I)^{1/r} \le 2^{-d(2/r-1)}$. Summing over d gives $\|\mathcal{P}_<(x \otimes y)\| \le C_\sharp \|x\| \|y\|$. The bounded bilinear map $(x, y) \mapsto \mathcal{P}_<(x \otimes y)$ extends from $V \times V$ to $E \times E$ with values in $E \widehat{\otimes}_\pi E$; its linearization is the asserted extension.

(2) For $1 \le a < k$ let $\mathcal{P}^{(a)}_< = \mathrm{id}^{\otimes(a-1)} \otimes \mathcal{P}_< \otimes \mathrm{id}^{\otimes(k-a-1)}$, a map $\otimes^k_\pi E \to \widehat{\otimes}^k_\pi E$ of norm at most $C_\sharp$, extended to the completion, and $\mathcal{P}^{(k)} = \mathcal{P}^{(k-1)}_< \circ \cdots \circ \mathcal{P}^{(1)}_<$. Then $\|\mathcal{P}^{(k)}\| \le C_\sharp^{k-1}$, and $\mathcal{P}^{(k)}$ fixes $e_{t_1} \otimes \cdots \otimes e_{t_k}$ if $t_1 < \cdots < t_k$ and maps it to 0 otherwise. Put $s = \mathcal{P}^{(k)} \circ \mathrm{alt}^k_E$ and $\rho = \widehat{\wedge}^k_E \circ \mathcal{P}^{(k)}$. For $x_1, \dots, x_k \in V$,

$$s(x_1 \wedge \cdots \wedge x_k) = \sum_{t_1 < \cdots < t_k} \det\big(x_b(t_a)\big)_{a,b}\, e_{t_1} \otimes \cdots \otimes e_{t_k},$$

whose image under $\widehat{\wedge}^k_E$ is $x_1 \wedge \cdots \wedge x_k$. Such wedges span a dense subspace of $\Lambda^k_\pi E$, as V is dense and $\|x_1 \wedge \cdots \wedge x_k\| \le \prod_i \|x_i\|$. Hence $\widehat{\wedge}^k_E \circ s = \mathrm{id}$ and $\rho \circ \mathrm{alt}^k_E = \widehat{\wedge}^k_E \circ s = \mathrm{id}$. If $E = V$, then $\mathcal{P}_<$ preserves $V \otimes V$, so s and ρ take values in the uncompleted spaces.

(3) follows from (2) and Proposition 3.5. □

J.2. **Ordering splittings in every degree.**

Proof of Proposition 5.7. Step 1 (source and target). As at the end of part (1) of the proof of Theorem 5.6, $\mathcal{P}_<$ extends to a map $E \otimes_\pi E \to E\widehat{\otimes}_\pi E$ of norm at most C, and part (2) of that proof, with C in place of $C_\sharp$ and n in place of k, proves (1) and (2).

Step 2 (diagonal operators). For $n = 1$ take $R_1 = \mathrm{id}$; let $n \geq 2$. Let $(1\,2)$ be the isometry exchanging the factors of $E \otimes_\pi E$, $\mathcal{P}_> = (1\,2) \circ \mathcal{P}_< \circ (1\,2)$ and $\Delta = \mathrm{id} - \mathcal{P}_< - \mathcal{P}_>$, so that $\|\Delta\| \leq 1 + 2C$. For $T \in \{\mathcal{P}_<, \Delta\}$ and $1 \leq a < n$ let $T^{(a)} = \mathrm{id}^{\otimes(a-1)} \otimes T \otimes \mathrm{id}^{\otimes(n-a-1)}$, extended to $\widehat{\otimes}^n_\pi E$, of norm at most $\|T\|$. For $\mathbf{t} \in \Sigma^n$ and $e_\mathbf{t} = e_{t_1} \otimes \cdots \otimes e_{t_n}$, the vector $\mathcal{P}^{(a)}_< e_\mathbf{t}$, respectively $\Delta^{(a)} e_\mathbf{t}$, is $e_\mathbf{t}$ if $t_a < t_{a+1}$, respectively $t_a = t_{a+1}$, and 0 otherwise. So these operators preserve $\otimes^n V$ and commute on it, hence on $\widehat{\otimes}^n_\pi E$.

Step 3 (the projection). For a composition $c = (c_1, \ldots, c_m)$ of n let $J(c)$ be the set of partial sums $c_1 + \cdots + c_i$, $i < m$, and $\Pi_c = \prod_{a \in J(c)} \mathcal{P}^{(a)}_< \prod_{a \notin J(c)} \Delta^{(a)}$, with a ranging over $\{1, \ldots, n-1\}$. Then $\Pi_c e_\mathbf{t} = e_\mathbf{t}$ if $\mathbf{t}$ is nondecreasing with constant blocks of lengths $c_1, \ldots, c_m$, and $\Pi_c e_\mathbf{t} = 0$ otherwise. The stabilizer of such a $\mathbf{t}$ in S_n, acting by isometric permutations of the factors, is $S_{c_1} \times \cdots \times S_{c_m}$; if Ω_c is a set of representatives of its left cosets, then $O_\mathbf{t} = \sum_{\sigma \in \Omega_c} \sigma e_\mathbf{t}$. Hence $R_n = \sum_c \sum_{\sigma \in \Omega_c} \sigma \circ \Pi_c$, the first sum over all compositions of n, satisfies $R_n e_\mathbf{t} = O_\mathbf{t}$ for nondecreasing $\mathbf{t}$ and $R_n e_\mathbf{t} = 0$ otherwise, and $\|R_n\| \leq \sum_c n!\,(c_1! \cdots c_m!)^{-1} C^{m-1}(1 + 2C)^{n-m}$.

Step 4 (the image). Let Γ_V be the span of the $O_\mathbf{t}$ with $\mathbf{t}$ nondecreasing. Then $R_n(\otimes^n V) = \Gamma_V$ and R_n fixes Γ_V, so by continuity R_n is a projection of $\widehat{\otimes}^n_\pi E$ onto $\overline{\Gamma_V}$. For $h \in V$, $h^{\otimes n} = \sum_{\mathbf{t}\ \text{nondecreasing}} h(t_1) \cdots h(t_n) O_\mathbf{t} \in \Gamma_V$; as $h \mapsto h^{\otimes n}$ is continuous and V is dense, $\widehat{\Gamma}^n_\pi E \subseteq \overline{\Gamma_V}$. Conversely, by the argument of Step 1 of the proof of Proposition 5.13, with the vectors e_t in place of the unit vectors, $O_\mathbf{t}$ is a linear combination of pure powers. Hence $\overline{\Gamma_V} = \widehat{\Gamma}^n_\pi E$, and (Γ^n_E) holds. □

J.3. **Boxes in chain sets.** Throughout this subsection and the next, $2 < r < \infty$ is fixed. For finite $T_1, \ldots, T_k \subseteq \Sigma$ the *chain set* is $\Xi(T_1, \ldots, T_k) = \{t \in T_1 \times \cdots \times T_k : t_1 < \cdots < t_k\}$, with $\Xi() = \{()\}$. A *box* is a product $J_1 \times \cdots \times J_k$ with $J_a \subseteq T_a$, of *cost* $\prod_a |J_a|^{1/r}$; $\chi(T_1, \ldots, T_k)$ is the least total cost of a partition of $\Xi(T_1, \ldots, T_k)$ into boxes contained in it. So $\chi = 0$ if the chain set is empty, $\chi() = 1$, and $\chi(T_1) = |T_1|^{1/r}$, as $y \mapsto y^{1/r}$ is subadditive.

For $y, y' > 0$ let $\psi(y, y') = \min(y, y')^{1-1/r} \max(y, y')^{1/r}$. A *matching* of a finite set S is a set $\mathfrak{m}$ of pairwise disjoint two-element subsets of S, *near-perfect* if it has $\lfloor |S|/2 \rfloor$ pairs. Let $\mathrm{Mat}(S)$ be the set of near-perfect matchings of S and $\mathrm{Mat}_q = \mathrm{Mat}(\{1, \ldots, q\})$. For positive reals $y = (y_a)_{a \in S}$ put

$$\langle y \rangle = \max_\mathfrak{m} \prod_{\{a,b\} \in \mathfrak{m}} \min(y_a, y_b), \qquad \Psi(y) = \prod_{a \in S} y_a^{1/r}\, \langle y \rangle^{1-2/r},$$

the maximum over all matchings of S, empty products being 1. For a matching $\mathfrak{m}$ let $\Psi_\mathfrak{m}(y)$ be the product of $\psi(y_a, y_b)$ over the pairs $\{a, b\} \in \mathfrak{m}$ and of $y_a^{1/r}$ over the unmatched a; as $\psi(y, y') = (yy')^{1/r} \min(y, y')^{1-2/r}$, it is $\Psi(y)$ with $\langle y \rangle$ replaced by the product of the minima over $\mathfrak{m}$. Write $y_{S'} = (y_a)_{a \in S'}$ for $S' \subseteq S$. Let $\lambda_1 = 1 + (1 - 2^{-1/r})^{-1}$, $\lambda_2 = 1 + 2^{1-2/r}(2^{1-2/r} - 1)^{-1}$ and $\lambda = \max(\lambda_1, \lambda_2)$. Four elementary facts are used.

(F1) $\psi(y, y') \leq y^{1-1/r} y'^{1/r}$ and $\psi(y, y') \leq y^{1/r} y'^{1-1/r}$, with equality in the first if $y \leq y'$: one bound equals $\psi(y, y')$, the other is $(\max / \min)^{1-2/r}$ times larger.

(F2) $\Psi_\mathfrak{m}(y) \leq \Psi(y)$ for every matching $\mathfrak{m}$ of S, as $1 - 2/r > 0$. If all $y_a \geq 1$, then $\Psi(y) = \Psi_\mathfrak{m}(y)$ for some $\mathfrak{m} \in \mathrm{Mat}(S)$, as every matching extends to a near-perfect one and the added minima are ≥ 1; so $\Psi(y) \leq \sum_{\mathfrak{m} \in \mathrm{Mat}(S)} \Psi_\mathfrak{m}(y)$.

(F3) Ψ is nondecreasing in each variable.

(F4) Let $N \ge 1$, $y_{b,i} \ge 0$ for b in a finite set S and $1 \le i \le N$, $\alpha_b \ge 0$ and $\Theta = \sum_b \alpha_b$. Then, with $0^0 = 1$, $\sum_{i=1}^N \prod_b y_{b,i}^{\alpha_b} \le N^{(1-\Theta)_+} \prod_b (\sum_{i=1}^N y_{b,i})^{\alpha_b}$. For $\Theta \le 1$ this is Hölder's inequality, with the constant function 1 if $\Theta < 1$; for $\Theta > 1$, apply the case $\Theta = 1$ to the exponents α_b/Θ and use $\sum_i z_i^\Theta \le (\sum_i z_i)^\Theta$ for $z_i = \prod_b y_{b,i}^{\alpha_b/\Theta}$.

The *hull* of a finite nonempty $Y \subseteq \Sigma$ is $\operatorname{hull}(Y) = \{t \in \Sigma : \min Y \le t \le \max Y\}$.

Lemma J.1 (pair tree)**.** *Let $T', T'' \subseteq \Sigma$ be finite and nonempty, $m = \min(|T'|, |T''|)$ and $h = \lceil \log_2 m \rceil$; call T' the* tree side *if $|T'| \le |T''|$, and T'' otherwise. There are a finite set $\mathcal{T}$ with a partition $\mathcal{T} = \mathcal{T}_0 \sqcup \cdots \sqcup \mathcal{T}_{h-1} \sqcup \mathcal{T}_*$ into* levels*, and nonempty sets $T_i' \subseteq T'$, $T_i'' \subseteq T''$ with $\max T_i' < \min T_i''$ for $i \in \mathcal{T}$, such that*

$$\{(t', t'') \in T' \times T'' : t' < t''\} = \bigsqcup_{i \in \mathcal{T}} T_i' \times T_i'' \tag{J.1}$$

and, if Y_i denotes the one of T_i', T_i'' that lies in the tree side: (T1) *$|\mathcal{T}_d| \le 2^d$ for $d < h$, and $|\mathcal{T}_*| \le m$;* (T2) *$|Y_i| \le m2^{-d}$ for $i \in \mathcal{T}_d$, and $|Y_i| = 1$ for $i \in \mathcal{T}_*$;* (T3) *within each level, the sets T_i' have pairwise disjoint hulls, and so have the sets T_i''.*

Proof. Step 1 (the tree). Assume first $|T'| \le |T''|$. The nodes are pairs (U, I) of a nonempty set U of consecutive elements of T' and an interval I of Σ; the root (T', Σ) has depth 0. If $|U| \ge 2$, let U_L consist of the first $\lceil |U|/2 \rceil$ elements of U and U_R of the others, $t_U = \max U_L$, $I_L = \{t \in I : t \le t_U\}$ and $I_R = \{t \in I : t > t_U\}$; the node has the children (U_L, I_L) and (U_R, I_R), of depth one more, and contributes the pair $(U_L, T'' \cap I_R)$ to $\mathcal{T}_d$, d its depth, if $T'' \cap I_R \ne \emptyset$. A node $(\{t_0\}, I)$ is a leaf and contributes $(\{t_0\}, \{t \in T'' \cap I : t > t_0\})$ to $\mathcal{T}_*$ if this set is nonempty. Each pair satisfies $\max T_i' < \min T_i''$.

Step 2 (partition). Put $\Upsilon(U, I) = \{(t', t'') \in U \times (T'' \cap I) : t' < t''\}$. If $|U| \ge 2$, then $\Upsilon(U, I) = U_L \times (T'' \cap I_R) \sqcup \Upsilon(U_L, I_L) \sqcup \Upsilon(U_R, I_R)$: for $t' \in U_L$ the cases $t'' > t_U$ and $t'' \le t_U$ give the first two pieces, and every $t' \in U_R$ exceeds t_U. At a leaf, Υ is the contributed pair or empty. Unwinding from $\Upsilon(T', \Sigma)$ gives (J.1).

Step 3 (counting and hulls). By induction a node of depth d has $|U| \le \lceil m/2^d \rceil$, as children have at most $\lceil |U|/2 \rceil$ elements and $\lceil \lceil y \rceil / 2 \rceil = \lceil y/2 \rceil$; there are at most 2^d such nodes. If $|U| \ge 2$, then $m > 2^d$, so $d < h$, and $|U_L| \le \lceil m/2^{d+1} \rceil \le m/2^d$, as $\lceil y \rceil \le 2y$ for $y > 1/2$. The leaves have distinct singletons $U \subseteq T'$. This proves (T1) and (T2). Children split the block U and the interval I of their parent into disjoint parts, so two nodes neither of which descends from the other, such as two nodes of one depth or two leaves, have disjoint blocks U, one before the other, and disjoint intervals I. A pair contributed by a node has its first member in U and its second in I; this proves (T3).

Step 4 (the other case). If $|T''| < |T'|$, apply Steps 1–3 to T'', T' and the reversed order of Σ, which changes no hull. □

Lemma J.2 (level summation)**.** *Let $m \ge 1$, $h = \lceil \log_2 m \rceil$, and let $\mathcal{T} = \mathcal{T}_0 \sqcup \cdots \sqcup \mathcal{T}_{h-1} \sqcup \mathcal{T}_*$ be a finite set with $|\mathcal{T}_d| \le 2^d$ and $|\mathcal{T}_*| \le m$. Let $g \colon \mathcal{T} \to [0, \infty)$ satisfy $g \le m2^{-d}$ on $\mathcal{T}_d$ and $g \le 1$ on $\mathcal{T}_*$. For b in a finite set S let $y_b \colon \mathcal{T} \to [0, \infty)$ have sum at most Y_b over each $\mathcal{T}_d$ and over $\mathcal{T}_*$. Let $\alpha_0 \ge 1/r$, $\alpha_b \ge 0$, $\Theta = \sum_{b \in S} \alpha_b$, $Y = \prod_b Y_b^{\alpha_b}$ and $\Phi = \sum_{i \in \mathcal{T}} g(i)^{\alpha_0} \prod_b y_b(i)^{\alpha_b}$.* (1) *If $\Theta + \alpha_0 \ge 1 + 1/r$, then $\Phi \le \lambda_1 m^{\alpha_0} Y$.* (2) *If $\alpha_0 = \Theta = 1/r$, then $\Phi \le \lambda_2 m^{1-1/r} Y$.*

Proof. Put $\zeta = (1 - \Theta)_+ - \alpha_0$. By (F4), the part of Φ over $\mathcal{T}_d$ is at most $(m2^{-d})^{\alpha_0} 2^{d(1-\Theta)_+} Y = m^{\alpha_0} 2^{d\zeta} Y$, and the part over $\mathcal{T}_*$ at most $m^{(1-\Theta)_+} Y$. (1) Here $\zeta \le -1/r$ and $(1 - \Theta)_+ \le \alpha_0$, as $(1 - \Theta)_+ = 0$ if $\Theta \ge 1$ and $1 - \Theta \le \alpha_0 - 1/r$ if $\Theta < 1$; so $\Phi \le m^{\alpha_0} (\sum_{d \ge 0} 2^{-d/r} + 1) Y = \lambda_1 m^{\alpha_0} Y$.

(2) Here $(1-\Theta)_+ = 1-1/r$ and $\zeta = 1-2/r > 0$. For $m \geq 2$, $2^h < 2m$ gives $\sum_{d<h} 2^{d\zeta} \leq 2^\zeta m^\zeta/(2^\zeta - 1)$, and $1/r + \zeta = 1 - 1/r$; for $m = 1$ there are no levels $\mathcal{T}_d$. Adding the two parts gives $\Phi \leq \lambda_2 m^{1-1/r} Y$. □

Theorem J.3 (weighted boxes)**.** *Let $\eta_0 = \eta_1 = 1$ and $\eta_k = \lambda \sum_{l=2}^k |\mathrm{Mat}_{l-1}|\, \eta_{l-1}\eta_{k-l}$ for $k \geq 2$. For every $k \geq 1$ and all finite nonempty $T_1, \dots, T_k \subseteq \Sigma$ with $m_a = |T_a|$ elements,*

$$\chi(T_1, \dots, T_k) \leq \eta_k \Psi(m_1, \dots, m_k) = \eta_k \prod_{a=1}^k m_a^{1/r} \langle m \rangle^{1-2/r}.$$

Proof. We induct on k; the cases $k \leq 1$ are the values of $\chi()$ and $\chi(T_1)$ computed after the definition of χ. Let $k \geq 2$ and $m = (m_1, \dots, m_k)$.

Step 1 (decomposition). Apply Lemma J.1 to $T' = T_1$ and $T'' = T_2$. For $i \in \mathcal{T}$ and $T \subseteq \Sigma$ let $I_i = \mathrm{hull}(T_i'')$ and $T^{>i} = \{t \in T : t > \max T_i''\}$, and for $2 \leq l \leq k$ let

$$\Xi_{i,l} = T_i' \times \Xi(T_i'', T_3 \cap I_i, \dots, T_l \cap I_i) \times \Xi(T_{l+1}^{>i}, \dots, T_k^{>i}).$$

These sets partition $\Xi(T_1, \dots, T_k)$: for a chain t, (t_1, t_2) lies in exactly one $T_i' \times T_i''$ by (J.1), and if l is the largest a with $t_a \leq \max T_i''$, then $t_a \in T_a \cap I_i$ for $3 \leq a \leq l$ and $t_a \in T_a^{>i}$ for $a > l$; conversely, elements of $\Xi_{i,l}$ are chains, as $\max T_i' < \min T_i''$ and $t_l \leq \max T_i'' < t_{l+1}$. Products of box partitions of the three factors are box partitions of $\Xi_{i,l}$ with the product cost, so

$$\chi(T_1, \dots, T_k) \leq \sum_{l=2}^k \sum_{i \in \mathcal{T}} |T_i'|^{1/r}\, \chi(T_i'', T_3 \cap I_i, \dots, T_l \cap I_i)\, \chi(T_{l+1}^{>i}, \dots, T_k^{>i}). \tag{J.2}$$

Step 2 (exponents). Fix l, and let $S_l = \{3, \dots, l\}$ and $S_l^+ = \{l+1, \dots, k\}$. By induction and (F3) the last factor in (J.2) is at most $\eta_{k-l}\Psi(m_{S_l^+})$. Put $m_{2,i} = |T_i''|$ and $m_{a,i} = |T_a \cap I_i|$ for $a \in S_l$. By induction and (F2), the middle factor is at most $\eta_{l-1} \sum_{\mathfrak{m}} \Psi_{\mathfrak{m}}(m_{2,i}, (m_{a,i})_{a \in S_l})$, the sum over $\mathfrak{m} \in \mathrm{Mat}(\{2\} \cup S_l)$, or zero if some $m_{a,i} = 0$. Fix $\mathfrak{m}$. For a pair $\{a, b\} \in \mathfrak{m}$ with $m_a \leq m_b$, ties broken by the index, let $\alpha_a = 1 - 1/r$ and $\alpha_b = 1/r$; let $\alpha_a = 1/r$ for an unmatched a. By (F1),

$$\Psi_{\mathfrak{m}}\big(m_{2,i}, (m_{a,i})_{a \in S_l}\big) \leq \prod_{a \in \{2\} \cup S_l} m_{a,i}^{\alpha_a}, \qquad \prod_{a \in \{2\} \cup S_l} m_a^{\alpha_a} = \Psi_{\mathfrak{m}}(m_{\{2\} \cup S_l}). \tag{J.3}$$

The sum $\Theta_{\mathfrak{m}}$ of the α_a is at least 1 if $S_l \neq \emptyset$, since $\mathfrak{m}$ then has a pair, and is $1/r$ if $S_l = \emptyset$.

Step 3 (level sums). Let $\Phi_{\mathfrak{m}} = \sum_{i \in \mathcal{T}} |T_i'|^{1/r} \prod_{a \in \{2\} \cup S_l} m_{a,i}^{\alpha_a}$. By (T3) the sums of $|T_i'|$, $m_{2,i}$ and $m_{a,i}$ over each level are at most m_1, m_2 and m_a. Apply Lemma J.2 with $\min(m_1, m_2)$ in the role of m, with $g(i) = |Y_i|$ for the set Y_i of Lemma J.1 and α_0 its exponent in $\Phi_{\mathfrak{m}}$, which is $1/r$ or $\alpha_2 \geq 1/r$, and with the other factors of $\Phi_{\mathfrak{m}}$ as the functions y_b; (T2) gives the bounds on g, and the exponents add up to $1/r + \Theta_{\mathfrak{m}}$. If $S_l \neq \emptyset$, part (1) and (J.3) give $\Phi_{\mathfrak{m}} \leq \lambda m_1^{1/r} \Psi_{\mathfrak{m}}(m_{\{2\} \cup S_l})$; if $S_l = \emptyset$, part (2) gives $\Phi_{\mathfrak{m}} \leq \lambda \psi(m_1, m_2)$.

Step 4 (assembly). By (F2), $\Psi(m_{S_l^+}) = \Psi_{\mathfrak{m}^+}(m_{S_l^+})$ for some $\mathfrak{m}^+ \in \mathrm{Mat}(S_l^+)$. Then $m_1^{1/r} \Psi_{\mathfrak{m}}(m_{\{2\} \cup S_l}) \Psi_{\mathfrak{m}^+}(m_{S_l^+})$ is $\Psi_{\mathfrak{m} \cup \mathfrak{m}^+}(m)$, and $\psi(m_1, m_2) \Psi_{\mathfrak{m}^+}(m_{S_l^+})$ is $\Psi_{\{\{1,2\}\} \cup \mathfrak{m}^+}(m)$; both are at most $\Psi(m)$ by (F2). As $\{2\} \cup S_l$ has $l-1$ elements, the l-th term of (J.2) is at most $\lambda |\mathrm{Mat}_{l-1}| \eta_{l-1}\eta_{k-l} \Psi(m)$; summing over l proves the claim. □

J.4. **Triangular truncation from ℓ^2 into ℓ^r.** Let G_r be $c_{00}(\Sigma, K)$ with the norm $\|\cdot\|_r$; its completion is $\ell^r(\Sigma, \widehat{K})$. Let δ_t be the unit vectors, and for $g_1, \dots, g_k \in c_{00}(\Sigma, K)$ let $\mathcal{P}^{(k)}(g_1 \otimes \cdots \otimes g_k) = \sum_{t_1 < \cdots < t_k} g_1(t_1) \cdots g_k(t_k)\, \delta_{t_1} \otimes \cdots \otimes \delta_{t_k}$, the k-fold triangular truncation for the family (δ_t).

Lemma J.4 (flat vectors)**.** *If $g_1,\dots,g_k \in c_{00}(\Sigma,K)$ satisfy $\|g_a\|_\infty \le 1$ and have nonempty supports T_a with m_a elements, then $\|\mathcal{P}^{(k)}(g_1\otimes\cdots\otimes g_k)\|_{\otimes^k_\pi G_r} \le \chi(T_1,\dots,T_k) \le \eta_k\Psi(m_1,\dots,m_k)$.*

Proof. If boxes $J^1_1\times\cdots\times J^i_k$ partition $\Xi(T_1,\dots,T_k)$, then $\mathcal{P}^{(k)}(g_1\otimes\cdots\otimes g_k) = \sum_i (g_1 1_{J^i_1})\otimes\cdots\otimes(g_k 1_{J^i_k})$, as both sides have the coefficient $g_1(t_1)\cdots g_k(t_k)$ at each chain t and no other terms, and $\|g_a 1_J\|_r \le |J|^{1/r}$. Theorem J.3 gives the second inequality. □

Theorem J.5 (triangular truncation)**.** *Let $k\ge 1$, $\tau = |\vartheta|$, $\gamma = 1/2 - 1/r$ and*

$$C_{k,r} = \eta_k |\mathrm{Mat}_k| \Big(\frac{2\tau^{-2}}{1-\tau^\gamma}\Big)^{\lfloor k/2\rfloor} \Big(\frac{\tau^{-2/r}}{1-\tau^{2\gamma}}\Big)^{k-2\lfloor k/2\rfloor}.$$

For $x_1,\dots,x_k \in c_{00}(\Sigma,K)$,

$$\|\mathcal{P}^{(k)}(x_1\otimes\cdots\otimes x_k)\|_{\otimes^k_\pi G_r} \le C_{k,r}\tau^{-k}\prod_{a=1}^k \|x_a\|_2. \tag{J.4}$$

Consequently $\mathcal{P}^{(k)}$ extends to a bounded linear map from $\otimes^k_\pi \ell^2(\Sigma,K)$ into $\widehat{\otimes}^k_\pi \ell^r(\Sigma,\widehat{K})$ of norm at most $C_{k,r}\tau^{-k}$.

Proof. Step 1 (bands). For $x\in c_{00}(\Sigma,K)$ and $i\in\mathbb{Z}$ let $\Sigma_i(x) = \{t : \tau^{i+1} < |x(t)| \le \tau^i\}$, $m_i(x) = |\Sigma_i(x)|$ and $x^{(i)} = x1_{\Sigma_i(x)}$. Then $x = \sum_i x^{(i)}$ is a finite sum, $x^{(i)} = \vartheta^i g$ with $\|g\|_\infty \le 1$, and $\sum_i \tau^{2i} m_i(x) \le \tau^{-2}\|x\|_2^2$. Expanding each x_a in bands and applying Lemma J.4 and (F2), the left side of (J.4) is at most

$$\eta_k \sum_{\mathfrak{m}\in\mathrm{Mat}_k} \prod_{\{a,b\}\in\mathfrak{m}} \Phi_2(x_a,x_b) \prod_{a \text{ unmatched}} \Phi_1(x_a),$$

where $\Phi_1(x) = \sum_i \tau^i m_i(x)^{1/r}$ and $\Phi_2(x,y) = \sum_{i,l}\tau^{i+l}\psi(m_i(x),m_l(y))$, summed over the indices with $m_i(x), m_l(y)\ge 1$; for fixed $\mathfrak{m}$ the sum over the bands factors, as each slot occurs in one factor of $\Psi_\mathfrak{m}$.

Step 2 (single sums). If some $x_a = 0$ there is nothing to prove. Replacing x_a by $\vartheta^c x_a$, $c\in\mathbb{Z}$, multiplies both sides of (J.4) by τ^c, so we may assume $\tau < \|x_a\|_2 \le 1$ for every a. If $\|x\|_2 \le 1$, then $m_i(x)\ge 1$ forces $\tau^{i+1} < 1$, so $i\ge 0$, and $\tau^{2i}m_i(x) \le \tau^{-2}$; hence $\tau^i m_i(x)^{1/r} = (\tau^{2i}m_i(x))^{1/r}\tau^{2\gamma i} \le \tau^{-2/r}\tau^{2\gamma i}$ and $\Phi_1(x) \le \tau^{-2/r}/(1-\tau^{2\gamma})$.

Step 3 (pair sums). Let $\|x\|_2,\|y\|_2 \le 1$, $\alpha_i = \tau^i m_i(x)^{1/2}$ and $\alpha'_l = \tau^l m_l(y)^{1/2}$, so that $\sum_i \alpha_i^2$ and $\sum_l \alpha_l'^2$ are at most τ^{-2}. For $b',b''>0$, $\psi(b',b'') = (b'b'')^{1/2}\min(b'/b'', b''/b')^\gamma \le (b'b'')^{1/2}\min(b'/b'',b''/b')^{\gamma/2}$, as the minimum is at most 1. With $m_i(x)/m_l(y) = (\alpha_i/\alpha'_l)^2\tau^{2(l-i)}$, the term of $\Phi_2(x,y)$ at (i,l) is at most $\alpha_i^{1+\gamma}\alpha_l'^{1-\gamma}\tau^{\gamma(l-i)}$ if $l\ge i$ and at most $\alpha_i^{1-\gamma}\alpha_l'^{1+\gamma}\tau^{\gamma(i-l)}$ if $l<i$. For fixed $l-i$, Hölder's inequality with exponents $2/(1\pm\gamma)$ bounds the sum of these terms by $\tau^{-2}\tau^{\gamma|l-i|}$, so $\Phi_2(x,y) \le \tau^{-2}(1+\tau^\gamma)/(1-\tau^\gamma) \le 2\tau^{-2}/(1-\tau^\gamma)$.

Step 4 (conclusion). Each $\mathfrak{m}\in\mathrm{Mat}_k$ has $\lfloor k/2\rfloor$ pairs and $k-2\lfloor k/2\rfloor$ unmatched slots, so the left side of (J.4) is at most $C_{k,r} < C_{k,r}\tau^{-k}\prod_a\|x_a\|_2$. As $c_{00}(\Sigma,K)$ is dense in $\ell^2(\Sigma,K)$, the bounded k-linear map $(x_1,\dots,x_k)\mapsto\mathcal{P}^{(k)}(x_1\otimes\cdots\otimes x_k)$ extends to $\ell^2(\Sigma,K)^k$ with values in $\widehat{\otimes}^k_\pi G_r$; its linearization, followed by the contraction induced by the isometry $G_r\to\ell^r(\Sigma,\widehat{K})$, is the asserted extension. □

J.5. **Relative ordering splittings.**

Proof of Proposition 5.8. Let ν be an r-norm on D with $a\nu \le \|\cdot\| \le b\nu$, $0<a\le b$. As $\nu(\sum_t z_t)^r \le \sum_t \nu(z_t)^r$ for finite families, the left side of (5.2) is at most $b(\sum_t|\xi_t|^r\nu(y_t)^r)^{1/r} \le$

$(b/a)(\sum_t|\xi_t|^r)^{1/r}\max_t\|y_t\|$. Conversely, assume (5.2) with constant C, and let $\nu(y)$ be the infimum of $(\sum_i\|y_i\|^r)^{1/r}$ over the finite decompositions $y=\sum_i y_i$. Then $\nu\le\|\cdot\|$; $\nu(\xi y)=|\xi|\nu(y)$ for $\xi\in K^\times$, as $y_i\mapsto\xi y_i$ matches the decompositions of y and ξy; and $\nu(y+z)^r\le\nu(y)^r+\nu(z)^r$, by concatenating decompositions. Let $y=\sum_i y_i$ with all $y_i\neq 0$, and $\xi_i=\vartheta^{c_i}$ with c_i the largest integer such that $|\vartheta|^{c_i}\ge\|y_i\|$. Then $\|\xi_i^{-1}y_i\|\le 1$ and $|\xi_i|<\|y_i\|/|\vartheta|$, so (5.2) gives $\|y\|\le C(\sum_i|\xi_i|^r)^{1/r}\le(C/|\vartheta|)(\sum_i\|y_i\|^r)^{1/r}$, hence $\|y\|\le(C/|\vartheta|)\nu(y)$. So ν vanishes only at 0; it is a norm, since $r\ge 1$, and an equivalent r-norm. □

Proof of Theorem 5.9. Let $C_e=\sup_t\|e_t\|$, let C_2 be the constant of ℓ^2-domination, fix a degree $k\ge 1$ and $r\in(2,\infty)$, and let G_r, δ_t, τ and $C_{k,r}$ be as in Subsection J.4. In Steps 1 and 3 we assume, as in (2), that D has an r-norm ν with $c\nu\le\|\cdot\|\le c'\nu$.

Step 1 (factorization). For $u\in\mathcal{L}(E,D)$ let $\tilde u\colon G_r\to D$, $\tilde u(\xi)=\sum_t\xi(t)\,ue_t$. Then $\nu(\tilde u\xi)^r\le\sum_t|\xi(t)|^r\nu(ue_t)^r\le(C_e\|u\|/c)^r\|\xi\|_r^r$, so $\|\tilde u\|\le(c'/c)C_e\|u\|$; $\tilde u$ is linear in u, and $\tilde u\delta_t=ue_t$. For $x\in V$, $\tilde u(wx)=\sum_t x(t)\,ue_t=ux$, so the extension $\tilde u\colon\ell^r(\Sigma,\widehat K)\to\widehat D$ satisfies $\tilde u\circ w=u$, as V is dense; with Step 2, $(\ell^r(\Sigma,\widehat K),w,u\mapsto\tilde u)$ is a source-side relative splitting of degree k (Definition 6.12).

Step 2 (relative section). For $x\in V$ the coordinate vector $wx=(x(t))_t$ satisfies $\|wx\|_r\le\|wx\|_2\le C_2\|x\|$, so w extends to a bounded map $w\colon E\to\ell^r(\Sigma,\widehat K)$. By Theorem J.5 the k-linear map $(x_1,\dots,x_k)\mapsto\mathcal{P}^{(k)}(wx_1\otimes\cdots\otimes wx_k)$ from V^k to $\otimes_\pi^k G_r$ has norm at most $C_{k,r}\tau^{-k}C_2^k$, so it extends to E^k with values in $\widehat\otimes_\pi^k G_r$. Let L be its linearization and $s=L\circ\mathrm{alt}_E^k$; then $\|s\|\le k!\,C_{k,r}\tau^{-k}C_2^k$ and, for $x_1,\dots,x_k\in V$,

$$s(x_1\wedge\cdots\wedge x_k)=\sum_{t_1<\cdots<t_k}\det\big(x_b(t_a)\big)_{a,b}\,\delta_{t_1}\otimes\cdots\otimes\delta_{t_k}. \tag{J.5}$$

Expanding $wx_b=\sum_t x_b(t)\delta_t$, dropping repeated indices and sorting shows that $\widehat\wedge^k$ maps the right side to $wx_1\wedge\cdots\wedge wx_k$. The wedges $x_1\wedge\cdots\wedge x_k$ with $x_a\in V$ span a dense subspace of $\Lambda_\pi^k E$, so $\widehat\wedge^k\circ s=\Lambda_\pi^k w$, with s read in $\widehat\otimes_\pi^k\ell^r(\Sigma,\widehat K)$. Hence w is S^k-split, which proves (1). If $E=V$, s takes values in $\otimes_\pi^k G_r$.

Step 3 (lift). Put $Q(u_1,\dots,u_k)=\widehat\wedge_D^k\circ(\tilde u_1\otimes\cdots\otimes\tilde u_k)\circ s$, with $\tilde u_1\otimes\cdots\otimes\tilde u_k$ extended to $\widehat\otimes_\pi^k G_r$. Then Q is k-linear, $\|Q(u_1,\dots,u_k)\|\le\|s\|\,((c'/c)C_e)^k\prod_a\|u_a\|$, and by (J.5), for $x_1,\dots,x_k\in V$, Q is given by the *canonical lift formula*

$$Q(u_1,\dots,u_k)(x_1\wedge\cdots\wedge x_k)=\sum_{t_1<\cdots<t_k}\det\big(x_b(t_a)\big)_{a,b}\,u_1e_{t_1}\wedge\cdots\wedge u_ke_{t_k}. \tag{J.6}$$

For $u_1=\cdots=u_k=u$ the right side of (J.6) is $ux_1\wedge\cdots\wedge ux_k$, by expanding $ux_b=\sum_t x_b(t)\,ue_t$; by density, $Q(u,\dots,u)=\Lambda_\pi^k u$ as maps into $\widehat\Lambda_\pi^k D$. So (E,D) is good for Banach values in degree k by Lemma 3.4. If $E=V$, then Q takes values in $\mathcal{L}(\Lambda_\pi^k E,\Lambda_\pi^k D)$ by Step 2, and (E,D) is good in degree k by the same lemma. □

Appendix K. Forcing over Laurent series fields

Throughout this appendix K satisfies (L); from Section K.3 on, K is dense in $\kappa((X))$, so $\widehat K=\kappa((X))$ and $|\vartheta|=\varrho$. The results of Sections 7 and 8 proved from Section K.3 on, except Lemma 7.14, which assumes this density, depend only on K as a normed field; Lemma 7.1 extends them to every K satisfying (L). Theorem 8.9(3) and the last assertion of Example 8.10(3) are proved in Section K.8 over every nontrivially normed field of characteristic 2. Lemma K.2 holds, with the same proof, for every field κ of characteristic p in place of the finite field of (L); [II] uses it in that generality.

K.1. **Coefficients and costs.** For a normed space E with $c_{00}(\mathbb{N},K)\subseteq E\subseteq \ell^\infty(\mathbb{N},K)$ and $\|x\|_\infty\le\|x\|$, the determinant array Ω of §B.1, with $S=\mathbb{N}$, satisfies $|\Omega_z(c)|\le\|z\|$, as in the proof of Lemma C.2, so it extends to $\widehat{\Lambda}^k_\pi E$ with values in $\ell^\infty(\mathbb{N}^k,\kappa((X)))$. Likewise $z(t,t')$ denotes the continuous extension to $E\widehat{\otimes}_\pi E$ of the entry $x(t)y(t')$ of $x\otimes y$.

For a matrix Z over κ with rows indexed by a finite set R and columns by a finite set R', $\operatorname{cost}(\mathsf{Z})$ is the least value of $\sum_i|\operatorname{supp} g_i|^{1/2}|\operatorname{supp} h_i|^{1/2}$ over the representations $\mathsf{Z}=\sum_i g_i h_i^{\mathsf{T}}$ with $g_i\in\kappa^R$ and $h_i\in\kappa^{R'}$; it is subadditive.

Lemma K.1 (tournament lemma)**.** *Let Z be an $m\times m$ matrix over κ such that every off-diagonal entry of $\mathsf{Z}-\mathsf{Z}^{\mathsf{T}}$ is nonzero. Then $\operatorname{cost}(\mathsf{Z})\ge m\ln m/(1+\ln(2|\kappa|))$.*

Proof. Step 1 (parts). Let $\mathsf{Z}=\sum_i g_i h_i^{\mathsf{T}}$ with all g_i,h_i nonzero, let $\mathcal{V}_i$ be the union of the supports of g_i and h_i, and for $t\in\mathcal{V}_i$ let the part $\mathcal{V}_i(t)$ be the set of points of $\mathcal{V}_i$ at which (g_i,h_i) is a multiple of $(g_i(t),h_i(t))$. The (t,t') entry $g_i(t)h_i(t')-h_i(t)g_i(t')$ of $g_ih_i^{\mathsf{T}}-h_ig_i^{\mathsf{T}}$ is nonzero exactly when $t,t'\in\mathcal{V}_i$ lie in different parts; so every pair $t\ne t'$ lies in some $\mathcal{V}_i$, in different parts.

Step 2 (deletion). Following Hansel's deletion argument [10] with weighted choices, choose independently for each i one part of $\mathcal{V}_i$, with probability proportional to its size, and delete the points of $\mathcal{V}_i$ outside it. At most one point survives, since two survivors would lie in different parts of some $\mathcal{V}_i$. The point t survives with probability $\exp(-L_t)$, where $L_t=\sum_{i:t\in\mathcal{V}_i}\ln(|\mathcal{V}_i|/|\mathcal{V}_i(t)|)$; so $\sum_t\exp(-L_t)\le 1$, and convexity of the exponential gives $\sum_t L_t\ge m\ln m$. Moreover $\sum_t L_t=\sum_i \operatorname{ent}_i$ with $\operatorname{ent}_i=\sum_{t\in\mathcal{V}_i}\ln(|\mathcal{V}_i|/|\mathcal{V}_i(t)|)$.

Step 3 (one term). Exchanging g_i and h_i, which does not change the parts, we may assume that the sizes N_1 and N_2 of their supports satisfy $N_1\le N_2$, so $N=|\mathcal{V}_i|\le 2N_2$. It remains to show $\operatorname{ent}_i\le(1+\ln(2|\kappa|))\sqrt{N_1N_2}$. The points of $\mathcal{V}_i$ outside the support of g_i form one part, which contributes $(N-N_1)\ln(N/(N-N_1))\le N_1$. The other parts divide the support of g_i by the value of h_i/g_i into at most $|\kappa|$ sets; since a distribution on at most $|\kappa|$ points has entropy at most $\ln|\kappa|$, they contribute at most $N_1\ln(N/N_1)+N_1\ln|\kappa|$. Hence

$$\operatorname{ent}_i\le N_1\Big(1+\ln\frac{2|\kappa|N_2}{N_1}\Big)\le(1+\ln(2|\kappa|))\sqrt{N_1N_2},$$

the second inequality with $x=N_2/N_1$, since $x^{-1/2}(1+\ln(2|\kappa|x))$ is decreasing for $x\ge1$, as $\ln(2|\kappa|)>1$. □

Lemma K.2 (constant coefficients)**.** *Let R,R' be finite sets and $r\in\{2,\infty\}$, let z belong to the completion of $\ell^r(R,K)\otimes_\pi\ell^r(R',K)$, and let $\operatorname{coeff}_0 z\in\kappa^{R\times R'}$ be the matrix of constant coefficients of its entries.*

(1) *If $r=2$, then $\operatorname{cost}(\operatorname{coeff}_0 z)\le\|z\|/(1-\varrho^2)$.*
(2) *If $r=\infty$, then $\operatorname{rank}(\operatorname{coeff}_0 z)\le M_\varrho\|z\|$.*

Proof. For $x\in\ell^r(R,K)$ and $n\in\mathbb{Z}$ let $x_{[n]}\in\kappa^R$ have entries $\operatorname{coeff}_n(x(t))$. Then $\operatorname{coeff}_0(x\otimes y)=\sum_n x_{[n]}y_{[-n]}^{\mathsf{T}}$, a finite sum. (1) The vector $x_{[n]}$ is supported in $O_n(x)=\{t:x(t)\ne0,\ \operatorname{ord}x(t)\le n\}$, and $\sum_n|O_n(x)|\varrho^{2n}=\|x\|^2/(1-\varrho^2)$. By the Cauchy–Schwarz inequality, $\operatorname{cost}\operatorname{coeff}_0(x\otimes y)\le\sum_n|O_n(x)|^{1/2}|O_{-n}(y)|^{1/2}\le\|x\|\|y\|/(1-\varrho^2)$. (2) If $\|x\|=\varrho^\nu$ and $\|y\|=\varrho^{\nu'}$, only the $l+1$ values $\nu\le n\le-\nu'$ contribute, where $l=-\nu-\nu'$, and none if $l<0$. Since $l+1\le M_\varrho\varrho^{-l}$, this gives $\operatorname{rank}\operatorname{coeff}_0(x\otimes y)\le M_\varrho\|x\|\|y\|$.

For general z, write $z=\sum_j x_j\otimes y_j$ as an absolutely convergent series with $\sum_j\|x_j\|\|y_j\|$ as close to $\|z\|$ as we like, by approximating z quickly with algebraic tensors and decomposing the first approximation and the successive differences nearly optimally. The terms with $\|x_j\|\|y_j\|<1$

have entries of absolute value at most ϱ, since $|x(t)| \le \|x\|$ and absolute values lie in $\varrho^{\mathbb{Z}}$; by the nonarchimedean inequality so does their sum, which therefore does not change coeff_0. The finitely many remaining terms give the bounds with any number larger than $\|z\|$ in place of $\|z\|$, by subadditivity of cost and rank, and hence with $\|z\|$ itself. □

Lemma K.3 (two-slot slice)**.** *Let E be a normed space with $c_{00}(\mathbb{N}, K) \subseteq E \subseteq \ell^\infty(\mathbb{N}, K)$ and $\|x\|_\infty \le \|x\|$, let $\gamma_1 \ne \gamma_2$ be slots in $\{1, \dots, k\}$, and fix $c_\gamma \in \mathbb{N}$ for the remaining slots γ. There is a bounded linear $\Upsilon\colon \widehat{\Lambda}^k_\pi E \to E \widehat{\otimes}_\pi E$ of norm at most $k(k-1)$ such that $(\Upsilon z)(t, t') = \Omega_z(c)$, where c has t in slot γ_1, t' in slot γ_2 and c_γ in each slot γ.*

Proof. For $x_1, \dots, x_k \in E$, expansion along the rows γ_1, γ_2 gives

$$\det(x_b(c_a))_{a,b} = \sum_{b \ne b'} x_b(c_{\gamma_1}) x_{b'}(c_{\gamma_2}) \det_{bb'}(x),$$

where $\det_{bb'}(x)$, the sum of $\mathrm{sgn}(\sigma) \prod_{a \ne \gamma_1, \gamma_2} x_{\sigma(a)}(c_a)$ over the permutations with $\sigma(\gamma_1) = b$ and $\sigma(\gamma_2) = b'$, does not involve $c_{\gamma_1}, c_{\gamma_2}$, and $|\det_{bb'}(x)| \le \prod_{b'' \ne b, b'} \|x_{b''}\|$ by the nonarchimedean inequality. So the k-linear map $x \mapsto \sum_{b \ne b'} \det_{bb'}(x)\, x_b \otimes x_{b'}$ into $E \otimes_\pi E$ has norm at most $k(k-1)$, and its values have the stated determinants as entries. It is alternating, since these entries vanish when two x_b coincide and a tensor in $E \otimes E$ with all entries zero is zero. Proposition 3.1 gives Υ on $\Lambda^k_\pi E$, and continuity extends it and the entry formula to the completion. □

K.2. The ℓ^r threshold.

Proof of Theorem 7.7. Step 1 (a coefficient form). For finitely supported $a_1, \dots, a_k, x_1, \dots, x_k$ with entries in $\mathbb{F}_p$ and $c \in \mathbb{N}^k$ put

$$\Xi(a; x)(c) = \mathrm{coeff}_0\big(\Omega_{Q(a_1, \dots, a_k)(x_1 \wedge \cdots \wedge x_k)}(c)\big) \in \kappa,$$

which is $\mathbb{F}_p$-multilinear in (a, x) and alternating in x and in c. For such $b_1, \dots, b_k, x_1, \dots, x_k$ and $\lambda \in K^k$, with $a_\lambda = \sum_i \lambda_i b_i$, the identity $Q(a_\lambda, \dots, a_\lambda)(x_1 \wedge \cdots \wedge x_k) = (a_\lambda x_1) \wedge \cdots \wedge (a_\lambda x_k)$ is an equality of polynomials in λ with coefficients in $\widehat{\Lambda}^k_\pi E$. Comparing the coefficients of $\lambda_1 \cdots \lambda_k$ (Lemma B.6) and applying $\mathrm{coeff}_0 \circ \Omega$ gives

$$\sum_{\sigma \in S_k} \Xi(b_{\sigma(1)}, \dots, b_{\sigma(k)}; x)(c) = \sum_{\sigma \in S_k} \det\big((b_{\sigma(j)} x_j)(c_i)\big)_{i,j}, \tag{K.1}$$

since these determinants lie in $\mathbb{F}_p$. The proofs of Lemmas B.10, B.11 and B.13 use only multilinearity, alternation and (K.1) of the map Ψ of Appendix B, and the expansion (B.1). For Ξ the coefficient function $T(d; y; c) = \Xi(e_{d_1}, \dots, e_{d_k}; e_{y_1}, \dots, e_{y_k})(c)$ takes values in the finite field κ and (B.1) follows from $\mathbb{F}_p$-multilinearity, the inputs there having entries in $\{0, \pm 1\}$; so those three lemmas hold for Ξ. Since $k! = 0$ in κ, Lemma B.13 shows that the cluster value χ is not constant; as adjacent rank exchanges generate S_k, there are τ and ℓ with $\delta = \chi(\tau) - \chi(\tau') \ne 0$, where $\tau' = (\ell\ \ell{+}1) \circ \tau$ as in Lemma B.12. Put $\gamma_1 = \tau^{-1}(\ell)$ and $\gamma_2 = \tau^{-1}(\ell + 1)$.

Step 2 (the quantitative bound). Fix $m \ge 1$, and let $a = (a_1, \dots, a_k)$ and $x = (x_1, \dots, x_k)$ be the inputs in the proof of Lemma B.12, with γ_1, γ_2 as the slots α, β there. By Lemma B.11, the $m \times m$ matrix Z of the values $\Xi(a; x)(c)$ with $A_n(1)$ in slot γ_1, $B_{n'}(1)$ in slot γ_2 and $C_\gamma(1)$ in each other slot γ is $\mathsf{Z} = \chi(\tau')\mathbf{1}\mathbf{1}^{\mathsf{T}} + \delta U$. All entries of a and x lie in $\{0, \pm 1\}$, and x_{γ_1}, x_{γ_2} have $4m$ nonzero entries and the other x_γ four, so $\zeta = Q(a_1, \dots, a_k)(x_1 \wedge \cdots \wedge x_k)$ has $\|\zeta\| \le \|Q\|(4m)^{2/r} 4^{(k-2)/r}$. The slice Υ of Lemma K.3 for the slots γ_1, γ_2 and $c_\gamma = C_\gamma(1)$, followed by restriction of both tensor factors to the coordinates $\{A_n(1)\}$ and $\{B_{n'}(1)\}$, gives z with $\|z\| \le$

$k(k-1)\|\zeta\|$ and $\operatorname{coeff}_0 z = \mathsf{Z}$. If $r = 2$, Lemma K.2(1) gives $\operatorname{cost}(\mathsf{Z}) \le k(k-1)2^k m\|Q\|/(1-\varrho^2)$, while $\mathsf{Z} - \mathsf{Z}^{\mathsf{T}} = \delta(U - U^{\mathsf{T}})$ and Lemma K.1 give $\operatorname{cost}(\mathsf{Z}) \ge m \ln m/(1+\ln(2|\kappa|))$. If $r > 2$, the ℓ^r norms dominate the supremum norms, and Lemma K.2(2) gives $m-1 \le \operatorname{rank}\mathsf{Z} \le M_\varrho k(k-1)\|\zeta\|$, the first inequality since δU is invertible. This proves (7.1), which fails for large m.

Step 3 (lifts and splittings). A lift Q_0 as in Lemma 3.4, for Banach values, would give such a Q, namely $Q(a_1, \dots, a_k) = Q_0(\mu_{a_1}, \dots, \mu_{a_k})$, since $\|\mu_a\| \le \|a\|_\infty$. Lemma 3.4, Theorem 3.2(1) and the dichotomy of Proposition 2.5 give the assertion about $A^k_{E,E;F}$, Proposition 3.5 the failure of the splittings, and Corollary 5.3 the last assertion, since an equivalent nonarchimedean norm on $\widehat\Lambda^k_\pi E$ would make $A^k_{E,E;\widehat\Lambda^k_\pi E}$ analytic. □

Proof of Corollary 7.8. Theorem 5.6, with the examples following it, and Theorem 7.7. □

Proof of Corollary 7.9. The positive cases are Theorem 5.6. In the negative cases, repeat the proof of Theorem 7.7 with $Q(a_1, \dots, a_k) = Q_0(\mu_{a_1}, \dots, \mu_{a_k})$ for a lift Q_0 on $\mathcal{L}(E, D)$ as in Lemma 3.4, with $\widehat\Lambda^k_\pi D$ in place of $\widehat\Lambda^k_\pi E$ and the slice formed in $D\widehat\otimes_\pi D$. If $r_1 \le r_2$, then $\|\mu_a\| \le \|a\|_\infty$ and $\|\zeta\| \le \|Q_0\|(4m)^{2/r_1}4^{(k-2)/r_1}$. If $r_1 > r_2$, Hölder's inequality gives $\|\mu_a\| \le \|a\|_{r_0}$ with $1/r_0 = 1/r_2 - 1/r_1$; as a_{γ_1}, a_{γ_2} have $2m$ nonzero entries and the other a_γ two, $\|\zeta\| \le \|Q_0\|(2m)^{2/r_0}2^{(k-2)/r_0}(4m)^{2/r_1}4^{(k-2)/r_1}$. In both cases $\|\zeta\| \le 4^k\|Q_0\|m^{2/\min(r_1,r_2)}$. If $r_2 = 2 \le r_1$, Lemmas K.2(1) and K.1 give $m\ln m/(1+\ln(2|\kappa|)) \le k(k-1)4^k m\|Q_0\|/(1-\varrho^2)$; if $r_1, r_2 > 2$, Lemma K.2(2) gives $m - 1 \le M_\varrho k(k-1)4^k\|Q_0\|m^{2/\min(r_1,r_2)}$. Both fail for large m. □

K.3. **A trace norm over $\kappa((X))$.** Let $\mathcal{O} = \kappa[[X]]$, the closed unit ball of $\kappa((X))$. For finite sets R, R' we identify $z \in \kappa((X))^{R\times R'}$ with $\sum_{t,t'} z(t,t')\, e_t \otimes e_{t'}$; its projective norm $\|z\|_\pi$ over $\kappa((X))$ for the supremum norms on $\kappa((X))^R$ and $\kappa((X))^{R'}$ is the infimum of $\sum_a \|g_a\|_\infty\|h_a\|_\infty$ over the finite representations $z = \sum_a g_a h_a^{\mathsf{T}}$. Fix enumerations $t_1, t_2, \dots$ of R and $t'_1, t'_2, \dots$ of R'. Since $\mathcal{O}$ is a principal ideal domain whose nonzero elements are units times powers of X, the elementary divisor theorem, applied to $X^c z \in \mathcal{O}^{R\times R'}$ for a large integer c, writes every z as

$$z = U_1 \Delta U_2, \qquad U_1 \in GL_R(\mathcal{O}), \quad U_2 \in GL_{R'}(\mathcal{O}), \tag{K.2}$$

where Δ has the entries $X^{d_1}, \dots, X^{d_{\operatorname{rank} z}}$, with integers $d_1 \le \dots \le d_{\operatorname{rank} z}$, at the positions $(t_1, t'_1), \dots, (t_{\operatorname{rank} z}, t'_{\operatorname{rank} z})$ and zeros elsewhere. For $l \in \mathbb{Z}$ let $\operatorname{rank}_l(z)$ be the number of indices i with $d_i \le l$.

Lemma K.4 (a trace norm)**.** *Let R, R' be finite sets and $z \in \kappa((X))^{R\times R'}$.*

(1) *The numbers $\operatorname{rank}_l(z)$ do not depend on the factorization (K.2).*
(2) (Level formula.) $\|z\|_\pi = \sum_i \varrho^{d_i} = (1-\varrho)\sum_{l\in\mathbb{Z}} \varrho^l \operatorname{rank}_l(z)$.
(3) (Submatrices.) *If z_0 is the submatrix of z with rows in $S \subseteq R$ and columns in $S' \subseteq R'$, then $\operatorname{rank}_l(z_0) \le \operatorname{rank}_l(z)$ for every $l \in \mathbb{Z}$.*
(4) (Rank at a level.) *If $|R| = |R'|$, $e \in \mathbb{Z}$, and $z = X^e w$ for a matrix $w \in \mathcal{O}^{R\times R'}$ whose reduction modulo X is invertible over κ, then $\operatorname{rank}_e(z) = |R|$.*

Proof. For $l \ge 0$ let $\mathcal{O}_l = \mathcal{O}/X^{l+1}\mathcal{O}$, a local ring with residue field κ, and let $\operatorname{gen}(Y)$ be the least number of generators of a finitely generated $\mathcal{O}_l$-module Y. (a) If an $\mathcal{O}_l$-module is generated by N elements, so is each submodule: the module is a quotient of $\mathcal{O}^N$, and the preimage of the submodule is free of rank at most N, since $\mathcal{O}$ is a principal ideal domain. (b) $\operatorname{gen}(Y) = \dim_\kappa Y/XY$ by Nakayama's lemma; so $\bigoplus_{i=1}^N \mathcal{O}_l/X^{c_i}\mathcal{O}_l$ needs exactly N generators when every $c_i \ge 1$.

Step 1 (generators). Let $w \in \mathcal{O}^{R\times R'}$, $l \ge 0$, and let $\operatorname{Col}_l(w) \subseteq \mathcal{O}_l^R$ be the submodule generated by the columns of w reduced modulo X^{l+1}. If $w = U_1\Delta U_2$ is a factorization (K.2),

then $\Delta = U_1^{-1} w U_2^{-1}$ has entries in $\mathcal{O}$, so all $d_i \geq 0$; we claim $\mathrm{gen}(\mathrm{Col}_l(w)) = \mathrm{rank}_l(w)$. The columns of w and of $U_1\Delta$ generate the same submodule, since U_2 is invertible over $\mathcal{O}$; multiplication by U_1 is an automorphism of $\mathcal{O}_l^R$; and the columns of Δ generate a module isomorphic to $\bigoplus_{d_i \leq l} \mathcal{O}_l / X^{l+1-d_i}\mathcal{O}_l$, which needs exactly $\mathrm{rank}_l(w)$ generators by (b).

Step 2 (item (1)). Choose $c \in \mathbb{Z}$ with $X^c z \in \mathcal{O}^{R\times R'}$. Multiplying (K.2) by X^c gives a factorization of $X^c z$ with the exponents $d_i + c \geq 0$. So $\mathrm{rank}_l(z)$ is 0 if $l + c < 0$ and $\mathrm{gen}(\mathrm{Col}_{l+c}(X^c z))$ by Step 1 otherwise; neither depends on the factorization.

Step 3 (item (2)). The two right sides agree, since $(1-\varrho)\sum_{l\geq d} \varrho^l = \varrho^d$. Writing z as the sum over i of X^{d_i} times the product of the column t_i of U_1 and the row t_i' of U_2, vectors over $\mathcal{O}$, gives a representation of cost at most $\sum_i \varrho^{d_i}$. Conversely, let $z \neq 0$ and $z = \sum_{a=1}^N g_a h_a^{\mathsf{T}}$ with all g_a, h_a nonzero. Since $|\kappa((X))^\times| = \varrho^{\mathbb{Z}}$, $g_a = X^{\nu_a} g_a'$ and $h_a = X^{\nu_a'} h_a'$ with vectors g_a', h_a' over $\mathcal{O}$ of supremum norm 1; with $m_a = \nu_a + \nu_a'$ the cost is $\sum_a \varrho^{m_a}$. Multiplying z and every g_a by X^c multiplies the cost and $\sum_i \varrho^{d_i}$ by ϱ^c, so we may assume every $m_a \geq 0$; then $z \in \mathcal{O}^{R\times R'}$. For $l \geq 0$, $z \equiv \sum_{m_a \leq l} X^{m_a} g_a' h_a'^{\mathsf{T}}$ modulo X^{l+1}, so $\mathrm{Col}_l(z)$ lies in the submodule generated by the reductions of the g_a' with $m_a \leq l$, and (a) with Step 1 gives $\mathrm{rank}_l(z) \leq N_l = |\{a : m_a \leq l\}|$; for $l < 0$ both are 0. Hence $\sum_a \varrho^{m_a} = (1-\varrho)\sum_l \varrho^l N_l \geq (1-\varrho)\sum_l \varrho^l \mathrm{rank}_l(z)$, and the infimum over representations gives the lower bound.

Step 4 (item (3)). Multiplying z and z_0 by the same X^c replaces l by $l + c$ on both sides, so we may assume $z \in \mathcal{O}^{R\times R'}$ and $l \geq 0$, both sides vanishing for $l < 0$. Deleting columns replaces $\mathrm{Col}_l(z)$ by a submodule, which needs no more generators by (a); deleting rows replaces it by its image under the coordinate projection $\mathcal{O}_l^R \to \mathcal{O}_l^S$, which is generated by the images of generators. Step 1 gives $\mathrm{rank}_l(z_0) \leq \mathrm{rank}_l(z)$.

Step 5 (item (4)). Let $\Pi_0 \in \mathcal{O}^{R\times R'}$ have the entry 1 at each position (t_i, t_i') and 0 elsewhere. The determinant of $\Pi_0^{-1} w$ reduces to a nonzero element of κ, so it is a unit and $\Pi_0^{-1} w \in GL_{R'}(\mathcal{O})$. Then $z = I \cdot (X^e \Pi_0) \cdot \Pi_0^{-1} w$ is a factorization (K.2) with all $|R|$ exponents equal to e, so $\mathrm{rank}_e(z) = |R|$. □

K.4. **Coefficient functions of multiplier lifts.** Throughout this subsection G and D are sequence spaces (Definition 7.10) with $\|e_t\|_G \leq 1$ and $\|y\|_\infty \leq \|y\|_D$, Ω is the determinant array on $\widehat{\Lambda}_\pi^k D$ of Section K.1, and $\widetilde{Q}\colon c_{00}(\mathbb{N}, K)^k \to \mathcal{L}(\Lambda_\pi^k G, \widehat{\Lambda}_\pi^k D)$ is a *multiplier lift*: a k-linear map with

$$\widetilde{Q}(b,\dots,b)(x_1 \wedge \cdots \wedge x_k) = (bx_1) \wedge \cdots \wedge (bx_k) \qquad (b, x_1, \dots, x_k \in c_{00}(\mathbb{N}, K)) \tag{K.3}$$

and $\|\widetilde{Q}(e_{a_1}, \dots, e_{a_k})\| \leq \varrho^{-l_0}$ for all $a \in \mathbb{N}^k$ and some integer $l_0 \geq 0$. For $l \in \mathbb{Z}$ and $a, y, c \in \mathbb{N}^k$ put

$$\Omega_{a;y}(c) = \Omega_{\widetilde{Q}(e_{a_1},\dots,e_{a_k})(e_{y_1}\wedge\cdots\wedge e_{y_k})}(c), \qquad T_l(a;y;c) = \mathrm{coeff}_l\, \Omega_{a;y}(c) \in \kappa.$$

Since $|\Omega_z(c)| \leq \|z\|$, $\|e_{y_1} \wedge \cdots \wedge e_{y_k}\| \leq 1$ and the nonzero absolute values of $\kappa((X))$ lie in $\varrho^{\mathbb{Z}}$, a nonzero $\Omega_{a;y}(c)$ has order at least $-l_0$, so $T_l = 0$ for $l < -l_0$. For $g_1, \dots, g_k, h_1, \dots, h_k \in c_{00}(\mathbb{N}, K)$ and $c \in \mathbb{N}^k$, multilinearity gives the finite expansion

$$\Omega_{\widetilde{Q}(g_1,\dots,g_k)(h_1\wedge\cdots\wedge h_k)}(c) = \sum_{a,y\in\mathbb{N}^k} \prod_{\gamma=1}^{k} g_\gamma(a_\gamma) h_\gamma(y_\gamma) \cdot \Omega_{a;y}(c). \tag{K.4}$$

Lemma K.5 (homogeneity)**.** *For every integer $l_1 \geq 0$ there is an infinite set $I \subseteq \mathbb{N}$ such that for every $l \leq l_1$, $T_l(\mathbf{z}) = T_l(\mathbf{z}')$ whenever $\mathbf{z}, \mathbf{z}' \in I^{3k}$ have the same pattern in the sense of Appendix B.*

Proof. The map $\mathbf{z} \mapsto (T_l(\mathbf{z}))_{-l_0 \le l \le l_1}$ takes values in $\kappa^{l_0+l_1+1}$, a finite set whose size is a power of p. Identifying it with a finite field of that size, Lemma B.10 with $N = 3k$ gives I; for $l < -l_0$, $T_l = 0$. □

For the rest of this subsection fix an integer $l_1 \ge 0$ and I as in Lemma K.5. As in Appendix B, clusters are the sets $C = \{C(1) < C(2) < C(3) < C(4)\}$ of four consecutive elements of I, disjoint clusters are compared by $C < C'$, and the operator and vector weights $o = (1, 0, -1, 0)$ and $w = (1, -1, 1, -1)$ satisfy (W), with o in place of s. Put $o_C = \sum_i o_i e_{C(i)}$ and $w_C = \sum_i w_i e_{C(i)}$, so that $o_C w_C = e_{C(1)} - e_{C(3)}$. For $l \le l_1$ and pairwise disjoint clusters $C_1, \dots, C_k$ put

$$\begin{aligned}\chi_l(C_1, \dots, C_k) &= \operatorname{coeff}_l \Omega_{\widetilde{Q}(o_{C_1}, \dots, o_{C_k})(w_{C_1} \wedge \cdots \wedge w_{C_k})}\big(C_1(1), \dots, C_k(1)\big) \\ &= \sum_{i, i' \in [4]^k} \prod_{\gamma=1}^{k} o_{i_\gamma} w_{i'_\gamma} \cdot T_l\big(C_1(i_1), \dots, C_k(i_k); C_1(i'_1), \dots, C_k(i'_k); C_1(1), \dots, C_k(1)\big),\end{aligned}$$

by (K.4), the weights being constants in $\{0, \pm 1\}$. As in Appendix B, the pattern of each $3k$-tuple in the sum is determined by (i, i') and the order pattern $\tau \in S_k$ of $(C_1, \dots, C_k)$, so by Lemma K.5 the value depends only on τ; we write it $\chi_l(\tau)$. Also $\chi_l = 0$ for $l < -l_0$.

Lemma K.6 (two-level cancellation)**.** *Let $\mathcal{C}_1, \dots, \mathcal{C}_k$ be finite families of clusters that are pairwise disjoint as sets of clusters, such that all clusters in $\bigcup_\gamma \mathcal{C}_\gamma$ are pairwise disjoint, and let $\operatorname{lev}^o_\gamma, \operatorname{lev}^w_\gamma : \mathcal{C}_\gamma \to \mathbb{Z}_{\ge 0}$ be functions. Put*

$$g_\gamma = \sum_{C \in \mathcal{C}_\gamma} \vartheta^{\operatorname{lev}^o_\gamma(C)} o_C, \qquad h_\gamma = \sum_{C \in \mathcal{C}_\gamma} \vartheta^{\operatorname{lev}^w_\gamma(C)} w_C \qquad (\gamma = 1, \dots, k).$$

Let $C_\gamma \in \mathcal{C}_\gamma$ for $\gamma = 1, \dots, k$, let $c = (C_1(1), \dots, C_k(1))$, and let $\lambda = \sum_\gamma \big(\operatorname{lev}^o_\gamma(C_\gamma) + \operatorname{lev}^w_\gamma(C_\gamma)\big)$. Then for every integer $l \le l_1$,

$$\operatorname{coeff}_l \Omega_{\widetilde{Q}(g_1, \dots, g_k)(h_1 \wedge \cdots \wedge h_k)}(c) = \sum_{l' \ge 0} \operatorname{coeff}_{l'}(\vartheta^\lambda)\, \chi_{l-l'}(C_1, \dots, C_k),$$

a finite sum.

Proof. Step 1 (expansion). Expand by (K.4). As the clusters are disjoint and g_γ, h_γ are supported in the union of the clusters of $\mathcal{C}_\gamma$, a nonzero term has, in each slot γ, an operator index $a_\gamma = C^{\mathrm{op}}_\gamma(i_\gamma)$ and a vector index $y_\gamma = C^{\mathrm{vec}}_\gamma(i'_\gamma)$ with $C^{\mathrm{op}}_\gamma, C^{\mathrm{vec}}_\gamma \in \mathcal{C}_\gamma$, and the coefficient $\prod_\gamma o_{i_\gamma} w_{i'_\gamma} \cdot \vartheta^{\lambda'}$, where $\lambda' = \sum_\gamma \big(\operatorname{lev}^o_\gamma(C^{\mathrm{op}}_\gamma) + \operatorname{lev}^w_\gamma(C^{\mathrm{vec}}_\gamma)\big) \ge 0$ depends only on the *cluster assignment* $(C^{\mathrm{op}}_\gamma, C^{\mathrm{vec}}_\gamma)_\gamma$. Since $\vartheta^{\lambda'}$ has nonnegative order and $T_{l''} = 0$ for $l'' < -l_0$, $\operatorname{coeff}_l(\vartheta^{\lambda'} \Omega_{a;y}(c)) = \sum_{l' \ge 0} \operatorname{coeff}_{l'}(\vartheta^{\lambda'})\, T_{l-l'}(a; y; c)$, a finite sum.

Step 2 (output-free clusters). Group the terms by cluster assignment and by l'. Suppose an assignment has an occupied cluster C' that is none of $C_1, \dots, C_k$. Then C' belongs to exactly one family $\mathcal{C}_\gamma$, so only a_γ and y_γ can lie in C', and no output index does. Fix l' and all positions outside C', and sum over the positions inside C'. The factor $\operatorname{coeff}_{l'}(\vartheta^{\lambda'})$ and the weights of the other slots are common factors, and $T_{l-l'}$ is pattern-homogeneous on I since $l - l' \le l_1$. As C' consists of consecutive elements of I, cases (i)–(iii) of the proof of Lemma B.11 apply with $T_{l-l'}$ in place of T, and (W) makes the partial sum vanish.

Step 3 (the surviving assignment). In a surviving assignment every occupied cluster is an output cluster, and the only output cluster in $\mathcal{C}_\gamma$ is C_γ; so $C^{\mathrm{op}}_\gamma = C^{\mathrm{vec}}_\gamma = C_\gamma$ for every γ, $\lambda' = \lambda$, and the contribution is $\sum_{l' \ge 0} \operatorname{coeff}_{l'}(\vartheta^\lambda) \chi_{l-l'}(C_1, \dots, C_k)$. □

Lemma K.7 (the diagonal)**.** *For every integer $l \le l_1$, $\sum_{\tau\in S_k}\chi_l(\tau)$ equals 1 if $l=0$ and 0 otherwise.*

Proof. Step 1 (polarization). Let $b_1,\dots,b_k,x_1,\dots,x_k \in c_{00}(\mathbb{N},K)$ and $a_\alpha=\sum_i \alpha_i b_i$ for $\alpha\in K^k$. Both sides of (K.3) with $b=a_\alpha$ are polynomials in α, and comparing the coefficients of $\alpha_1\cdots\alpha_k$ by Lemma B.6 gives

$$\sum_{\sigma\in S_k}\widetilde{Q}(b_{\sigma(1)},\dots,b_{\sigma(k)})(x_1\wedge\cdots\wedge x_k)=\sum_{\sigma\in S_k}(b_{\sigma(1)}x_1)\wedge\cdots\wedge(b_{\sigma(k)}x_k). \tag{K.5}$$

Step 2 (right side). Take clusters $C_1<\cdots<C_k$, $b_\gamma=o_{C_\gamma}$, $x_\gamma=w_{C_\gamma}$ and $c=(C_1(1),\dots,C_k(1))$, and apply $\Omega_{\cdot}(c)$ to (K.5). On the right, a permutation $\sigma\neq\mathrm{id}$ has a slot γ with $\sigma(\gamma)\neq\gamma$, and $o_{C_{\sigma(\gamma)}}w_{C_\gamma}=0$ by disjointness; the identity contributes $\det\big((o_{C_b}w_{C_b})(C_a(1))\big)_{a,b}=1$, since $o_C w_C=e_{C(1)}-e_{C(3)}$.

Step 3 (left side). On the left, the sign manipulation in the proof of Lemma B.13, which uses only that the wedge and Ω_z are alternating, identifies the summand of σ with

$$\Omega_{\widetilde{Q}(o_{C_{\sigma(1)}},\dots,o_{C_{\sigma(k)}})(w_{C_{\sigma(1)}}\wedge\cdots\wedge w_{C_{\sigma(k)}})}\big(C_{\sigma(1)}(1),\dots,C_{\sigma(k)}(1)\big),$$

whose lth coefficient is $\chi_l(\sigma)$, since $(C_{\sigma(1)},\dots,C_{\sigma(k)})$ has order pattern σ. Taking coeff_l gives $\sum_\sigma\chi_l(\sigma)=\mathrm{coeff}_l(1)$. □

Corollary K.8 (an adjacent transposition)**.** *There are $\tau\in S_k$ and $\ell\in\{1,\dots,k-1\}$ with $\chi_0(\tau)\neq\chi_0(\tau')$, where $\tau'=(\ell\ \ell{+}1)\circ\tau$. For such τ,ℓ put $\gamma_1=\tau^{-1}(\ell)$, $\gamma_2=\tau^{-1}(\ell+1)$, $\delta_l=\chi_l(\tau)-\chi_l(\tau')$ for $l\le l_1$, and let l_* be the least $l\le l_1$ with $\delta_l\neq0$. Then $-l_0\le l_*\le 0$.*

Proof. Otherwise χ_0 is invariant under the adjacent transpositions, which generate S_k, hence constant, and Lemma K.7 with $l=0$ gives $k!\,\chi_0=1$, impossible since $k!=0$ in κ. As $\delta_0\neq0$ and $\delta_l=0$ for $l<-l_0$, $-l_0\le l_*\le 0$. □

K.5. **The product triangle bound.** For a symmetric sequence space D and finite $J\subseteq\mathbb{N}$, D_J is the span of the e_t, $t\in J$; we use three facts.

(P1) For finite $J,J'\subseteq\mathbb{N}$, a tensor in $D_J\otimes D_{J'}$ has the same projective norm in $D_J\otimes_\pi D_{J'}$ as in $D\otimes_\pi D$, since multiplication by 1_J is a contractive projection of D onto D_J (Lemma 7.11(1)).

(P2) Relabeling the rows and the columns of finitely supported tensors by injective maps $\mathbb{N}\to\mathbb{N}$, and multiplying rows and columns by sequences in K of absolute value 1, are isometries for the norm of $D\otimes_\pi D$: by symmetry of the norm they map each D_J isometrically onto some $D_{J''}$, and (P1) applies.

(P3) For finite $J,J'\subseteq\mathbb{N}$, the completion of $D_J\otimes_\pi D_{J'}$ is $\widehat{K}^{J\times J'}$, with a norm between the maximum and the sum of the absolute values of the entries, since $\|\cdot\|_\infty\le\|\cdot\|_D$ and $\|e_t\|_D=1$; as K is dense in $\widehat{K}$ and this norm is continuous for the maximum norm, $\|\lambda z\|=|\lambda|\|z\|$ for $\lambda\in\widehat{K}$. Multiplication by $1_J\otimes 1_{J'}$ extends to a contraction of $D\widehat{\otimes}_\pi D$ onto this completion whose value at z is the restriction of z to $J\times J'$ (Section K.1).

Proof of Theorem 7.13. Let $C_Q=\|Q\|$, $l_0=\max(0,\lceil\log_{1/\varrho}C_Q\rceil)$ and $C_\star=64\cdot 8^{k-2}k(k-1)\,C_Q+\varrho^{-l_0}+1$. We show (PTB) with constant $C_\star$. For general K satisfying (L), Lemma 7.1 then gives (PTB) with $|\vartheta|$ in place of ϱ in $C_\star$ and l_0.

Step 0 (a multiplier lift). For $a_1,\dots,a_k \in c_{00}(\mathbb{N},K)$ put $\widetilde{Q}(a_1,\dots,a_k) = Q(\mu_{a_1},\dots,\mu_{a_k})$, a k-linear map with

$$\|\widetilde{Q}(a_1,\dots,a_k)\| \le C_Q \prod_{\gamma=1}^{k} \|\mu_{a_\gamma}\|_{E\to D}. \tag{K.6}$$

Since $Q(u,\dots,u) = \Lambda_\pi^k u$, $\widetilde{Q}$ satisfies (K.3); since $\|e_t\|_E = 1$, $\|y\|_\infty \le \|y\|_D$ and $\|\mu_{e_t}\|_{E\to D} = 1$ (Lemma 7.11(1),(3)), it is a multiplier lift with $G = E$, as $\|\widetilde{Q}(e_{a_1},\dots,e_{a_k})\| \le C_Q \le \varrho^{-l_0}$. The rest of the proof uses Q only through these properties of $\widetilde{Q}$ and (K.6).

Step 1 (configuration). Assume first:

(A) $\operatorname{supp} a = \operatorname{supp} x = J_1$ and $\operatorname{supp} b = \operatorname{supp} y = J_2$ with $J_1 \cap J_2 = \emptyset$, and every nonzero entry of a, x, b and y is a nonnegative power of ϑ.

If one of the four vectors is 0, both sides of (7.2) vanish; so let them be nonzero and put $N_0 = \|\mu_a\|_{E\to D}\|x\|_E\|\mu_b\|_{E\to D}\|y\|_E > 0$. List $J_1 \cup J_2$ as $f_1 < \dots < f_m$, and let $l_1 \ge 0$ be the least integer with $m^2\varrho^{l_1+1} \le N_0$. Take I as in Lemma K.5 for this l_1, and τ, ℓ, γ_1, γ_2, l_* as in Corollary K.8. Choose pairwise disjoint clusters in increasing order: a cluster C_γ for each slot γ with $\tau(\gamma) < \ell$, ordered as τ prescribes; then clusters $Y_1 < \dots < Y_m$; then a cluster C_γ for each slot γ with $\tau(\gamma) > \ell+1$, ordered as τ prescribes. Let

$$\begin{aligned}
&\mathcal{C}_{\gamma_1} = \{Y_i : f_i \in J_1\}, && \operatorname{lev}^o_{\gamma_1}(Y_i) = \operatorname{ord} a(f_i), && \operatorname{lev}^w_{\gamma_1}(Y_i) = \operatorname{ord} x(f_i),\\
&\mathcal{C}_{\gamma_2} = \{Y_i : f_i \in J_2\}, && \operatorname{lev}^o_{\gamma_2}(Y_i) = \operatorname{ord} b(f_i), && \operatorname{lev}^w_{\gamma_2}(Y_i) = \operatorname{ord} y(f_i),
\end{aligned}$$

and $\mathcal{C}_\gamma = \{C_\gamma\}$ with both levels 0 for $\gamma \ne \gamma_1,\gamma_2$; these families are pairwise disjoint since $J_1 \cap J_2 = \emptyset$. Let g_γ and h_γ be as in Lemma K.6, and $\zeta = \widetilde{Q}(g_1,\dots,g_k)(h_1\wedge\dots\wedge h_k) \in \widehat{\Lambda}^k_\pi D$.

Step 2 (a bounded slice). As ϑ has order 1, (A) gives $g_{\gamma_1} = g' - g''$ with $g' = \sum_{f_i\in J_1} a(f_i)e_{Y_i(1)}$ and g'' the same sum with $Y_i(3)$ in place of $Y_i(1)$, two relabelings of a; so $\|\mu_{g_{\gamma_1}}\|_{E\to D} \le 2\|\mu_a\|_{E\to D}$ by Lemma 7.11(3). Likewise $\|\mu_{g_{\gamma_2}}\|_{E\to D} \le 2\|\mu_b\|_{E\to D}$ and $\|\mu_{g_\gamma}\|_{E\to D} \le 2$ for the other slots, and since h_γ is a combination, with the coefficients $w_1,\dots,w_4$, of four relabelings of x, of y or of a unit vector, $\|h_{\gamma_1}\|_E \le 4\|x\|_E$, $\|h_{\gamma_2}\|_E \le 4\|y\|_E$ and $\|h_\gamma\|_E \le 4$. So (K.6) gives $\|\zeta\| \le 64\cdot 8^{k-2}C_Q N_0$. Let $\Upsilon\colon \widehat{\Lambda}^k_\pi D \to D\widehat{\otimes}_\pi D$ be the slice of Lemma K.3 for the slots γ_1,γ_2 and $c_\gamma = C_\gamma(1)$, let $R = \{Y_i(1) : f_i \in J_1\}$ and $R' = \{Y_i(1) : f_i \in J_2\}$, and let $z \in \widehat{K}^{R\times R'}$ be the restriction of $\Upsilon\zeta$ to $R\times R'$ (fact (P3)). Then, in the completion of $D_R \otimes_\pi D_{R'}$,

$$\|z\| \le 64\cdot 8^{k-2}k(k-1)\,C_Q N_0. \tag{K.7}$$

Step 3 (reading all coefficients). Let $t = Y_i(1) \in R$ and $t' = Y_{i'}(1) \in R'$. By Lemma K.3, $z(t,t') = \Omega_\zeta(c)$, where c has $Y_i(1)$ in slot γ_1, $Y_{i'}(1)$ in slot γ_2 and $C_\gamma(1)$ in every other slot γ. The clusters C_γ lie below or above all of $Y_1,\dots,Y_m$ as τ prescribes, so the output clusters have order pattern $\sigma = \tau$ if $f_i < f_{i'}$ and $\sigma = \tau'$ if $f_i > f_{i'}$. The level sum of Lemma K.6 is $\lambda = \operatorname{ord} a(f_i) + \operatorname{ord} x(f_i) + \operatorname{ord} b(f_{i'}) + \operatorname{ord} y(f_{i'})$, and $\vartheta^\lambda = (ax)(f_i)\,(by)(f_{i'})$ by (A). For $\sigma' \in S_k$ put $\hat\chi(\sigma') = \sum_{l=-l_0}^{l_1}\chi_l(\sigma')X^l \in \kappa((X))$. For $l \le l_1$, $\operatorname{coeff}_l(\vartheta^\lambda\hat\chi(\sigma)) = \sum_{l'\ge 0}\operatorname{coeff}_{l'}(\vartheta^\lambda)\chi_{l-l'}(\sigma)$, since ϑ^λ has nonnegative order and $\chi_{l''} = 0$ for $l'' < -l_0$. So by Lemma K.6, $z(t,t') - \vartheta^\lambda\hat\chi(\sigma)$ has no nonzero coefficient of index at most l_1, and

$$|z(t,t') - (ax)(f_i)\,(by)(f_{i'})\,\hat\chi(\sigma)| \le \varrho^{l_1+1}.$$

The relabeling $Y_i(1) \mapsto f_i$ of rows and columns is increasing on $R\cup R'$, and by (P2) and continuity it is an isometry onto the completion of $D_{J_1}\otimes_\pi D_{J_2}$; let $z^\sharp$ be the image of z. With

$\tilde a = ax$ and $\tilde b = by$, as matrices indexed by $J_1 \times J_2$,

$$z^\sharp = \hat\chi(\tau')\,(\tilde a \otimes \tilde b) + \hat\delta \cdot \mathcal{P}_<(\tilde a \otimes \tilde b) + z_1, \qquad \hat\delta = \hat\chi(\tau) - \hat\chi(\tau') = \sum_{l=-l_0}^{l_1} \delta_l X^l, \tag{K.8}$$

where $z_1 \in \widehat{K}^{J_1\times J_2}$ has entries of absolute value at most ϱ^{l_1+1}. The lowest nonzero coefficient of $\hat\delta$ is δ_{l_*} with $l_* \le 0$, so $|\hat\delta| = \varrho^{l_*} \ge 1$, and $|\hat\chi(\tau')| \le \varrho^{-l_0}$.

Step 4 (conclusion under (A)). In the completion of $D_{J_1} \otimes_\pi D_{J_2}$, a $\widehat{K}$-normed space by (P3), $\|\tilde a \otimes \tilde b\| \le \|ax\|_D \|by\|_D \le N_0$ by Lemma 7.11(2), $\|z_1\| \le m^2 \varrho^{l_1+1} \le N_0$ by (P3) and the choice of l_1, and $\|z^\sharp\| = \|z\|$ is bounded by (K.7). By (K.8),

$$\|\mathcal{P}_<(\tilde a \otimes \tilde b)\| = |\hat\delta|^{-1}\|z^\sharp - \hat\chi(\tau')(\tilde a \otimes \tilde b) - z_1\| \le \big(64 \cdot 8^{k-2} k(k-1)\, C_Q + \varrho^{-l_0} + 1\big) N_0 = C_\star N_0.$$

By (P1), the left side is $\|\mathcal{P}_<((ax)\otimes(by))\|_{D\otimes_\pi D}$. The constant does not depend on I, τ, ℓ or l_1, which were chosen after a, x, b, y.

Step 5 (removing (A)). Both sides of (7.2) are homogeneous of degree one in each of a, x, b, y. The changes (a) and (b) below do not increase $\|\mu_a\|_{E\to D}$, $\|x\|_E$, $\|\mu_b\|_{E\to D}$ or $\|y\|_E$ (Lemma 7.11(1),(3) and symmetry of the norms), and they keep $\mathcal{P}_<((ax)\otimes(by))$ or change it by an isometry of (P2). (a) *Supports.* Replace (a, x) by $(a1_J, x1_J)$ with $J = \operatorname{supp}(ax)$, and likewise (b, y); then $\operatorname{supp} a = \operatorname{supp} x = J_1$ and $\operatorname{supp} b = \operatorname{supp} y = J_2$. If $J_1 \cap J_2 \ne \emptyset$, list $J_1 \cup J_2$ as $f_1 < \cdots < f_m$ and move the entries of a and x at $f_i \in J_1$ to the position $2i$ and those of b and y at $f_i \in J_2$ to the position $2i-1$; since $2i < 2i' - 1$ exactly when $i < i'$, this relabels the rows and columns of $\mathcal{P}_<((ax)\otimes(by))$, and the new supports are disjoint. (b) *Units.* Since $|K^\times| = |\vartheta|^{\mathbb{Z}}$, each nonzero entry of a is $a(t) = \theta_a(t)\vartheta^{\operatorname{ord} a(t)}$ with $|\theta_a(t)| = 1$. Replace a by the vector a' with the entries $\vartheta^{\operatorname{ord} a(t)}$ on $\operatorname{supp} a$ and 0 elsewhere, and likewise x, b, y; this multiplies the rows of $\mathcal{P}_<((ax)\otimes(by))$ by $(\theta_a\theta_x)^{-1}$ and its columns by $(\theta_b\theta_y)^{-1}$. (c) *Levels.* By homogeneity we may multiply a, x, b, y by powers of ϑ that make all exponents nonnegative. After (a)–(c), (A) holds, and Steps 1–4 give (7.2) with constant $C_\star$. □

Proof of Lemma 7.14. We may assume $x \ne 0$. Let $R = \operatorname{supp} x$, and let $z \in K^{R\times R}$ have the entries $z(t,t') = x(t)x(t')$ for $t < t'$ and 0 otherwise; $\|\cdot\|_\pi$ is the norm of Section K.3.

Step 1 (reduction to matrices). Restricting to R the vectors of a representation $\mathcal{P}_<(x\otimes x) = \sum_a g_a \otimes h_a$ with $g_a, h_a \in D$ gives a representation $z = \sum_a (g_a|_R)(h_a|_R)^{\mathsf T}$ with $\|g_a|_R\|_\infty \le \|g_a\|_D$ and $\|h_a|_R\|_\infty \le \|h_a\|_D$. Hence $\|z\|_\pi \le \|\mathcal{P}_<(x\otimes x)\|_{D\otimes_\pi D}$.

Step 2 (normalization). Write $\|x\|_\infty = \varrho^c$ with $c \in \mathbb{Z}$ and put $x' = \vartheta^{-c}x$; the matrix $z' = \vartheta^{-2c}z$ is formed from x' in the same way, and $\|z\|_\pi = \varrho^{2c}\|z'\|_\pi$. Since $|K^\times| = |\vartheta|^{\mathbb{Z}}$, each nonzero entry is $x'(t) = \theta_t\vartheta^{\lambda_t}$ with an integer $\lambda_t \ge 0$ and $|\theta_t| = 1$, so $z' \in \mathcal{O}^{R\times R}$. For $i \ge 0$ let m_i be the number of $t \in R$ with $\lambda_t = i$.

Step 3 (one level). Fix i with $m_i \ge 2$, and let $t_1 < \cdots < t_{m_i}$ be the $t \in R$ with $\lambda_t = i$. The submatrix z_i' of z' with rows $t_1, \dots, t_{m_i-1}$ and columns $t_2, \dots, t_{m_i}$ has the entry $\theta_{t_a}\theta_{t_{b+1}}\vartheta^{2i}$ in position (a, b) if $a \le b$, and 0 otherwise. As $\vartheta = X\eta_0$ with a unit η_0 of $\mathcal{O}$, $z_i' = X^{2i}\eta_0^{2i}\Theta_1 U \Theta_2$, where Θ_1 and Θ_2 are diagonal with the units θ_{t_a} and $\theta_{t_{b+1}}$ and U is the upper unitriangular $(m_i-1)\times(m_i-1)$ matrix of ones; the reduction of $\eta_0^{2i}\Theta_1 U\Theta_2$ modulo X is invertible. Lemma K.4(4),(3) gives $\operatorname{rank}_{2i}(z') \ge \operatorname{rank}_{2i}(z_i') = m_i - 1$.

Step 4 (conclusion). The indices $2i$ are distinct, so the level formula of Lemma K.4(2) gives $\|z'\|_\pi \ge (1-\varrho)\sum_{i:m_i\ge1} \varrho^{2i}(m_i - 1)$, and

$$\|x'\|_2^2 = \sum_{i:m_i\ge1} (m_i - 1)\varrho^{2i} + \sum_{i:m_i\ge1} \varrho^{2i} \le \frac{\|z'\|_\pi}{1-\varrho} + \frac{1}{1-\varrho^2}.$$

Multiplying by $\varrho^{2c} = \|x\|_\infty^2$ and using Step 1 gives the lemma. □

Proof of Proposition 7.15. For $m \geq 1$ and $1 \leq r' \leq \infty$ put

$$\psi_{r'}(m) = \sup\{\|\xi\|_{r'}/\|\xi\|_E : \xi \in c_{00}(\mathbb{N}, K) \setminus \{0\},\ |\operatorname{supp}\xi| \leq m\},$$

which is finite, since $\|\xi\|_{r'} \leq m^{1/r'}\|\xi\|_\infty \leq m^{1/r'}\|\xi\|_E$ by Lemma 7.11(1); and $\psi_\infty(m) \leq 1$.

Step 1 (multipliers of indicators). For finite $J \subseteq \mathbb{N}$ and $x \in E$, $\mu_{1_J}x = x1_J$ and $\|x1_J\|_D \leq C'\|x1_J\|_r \leq C'\psi_r(|J|)\|x1_J\|_E \leq C'\psi_r(|J|)\|x\|_E$, so $\|\mu_{1_J}\|_{E\to D} \leq C'\psi_r(|J|)$.

Step 2 (one inequality for each support size). Let $x \in c_{00}(\mathbb{N}, K) \setminus \{0\}$ with $J = \operatorname{supp} x$ of size at most m. (PTB) with $a = b = 1_J$ and $y = x$, and Step 1, give $\|\mathcal{P}_<(x \otimes x)\|_{D\otimes_\pi D} \leq CC'^2\psi_r(m)^2\|x\|_E^2$; Lemma 7.14 and $\|x\|_\infty \leq \|x\|_E$ give $\|x\|_2^2 \leq (\mathfrak{a}\psi_r(m)^2 + \mathfrak{b})\|x\|_E^2$. So

$$\psi_2(m)^2 \leq \mathfrak{a}\,\psi_r(m)^2 + \mathfrak{b} \qquad (m \geq 1). \tag{K.9}$$

Step 3 (interpolation). If $r = \infty$, then $\psi_\infty \leq 1$ and (K.9) give $\psi_2(m)^2 \leq \mathfrak{a} + \mathfrak{b} = C_2^2$. Let $r < \infty$. For ξ with $|\operatorname{supp}\xi| \leq m$, $\|\xi\|_r^r \leq \|\xi\|_\infty^{r-2}\|\xi\|_2^2 \leq \psi_2(m)^2\|\xi\|_E^r$, so $\psi_r(m)^2 \leq \psi_2(m)^{4/r}$ and (K.9) gives $\psi_2(m)^2 \leq \mathfrak{a}\,\psi_2(m)^{4/r} + \mathfrak{b}$. If $\psi_2(m)^2 > 2\mathfrak{b}$, then $\psi_2(m)^2 < 2\mathfrak{a}\,\psi_2(m)^{4/r}$, so $\psi_2(m)^2 < (2\mathfrak{a})^{r/(r-2)}$. In both cases $\psi_2(m)^2 \leq C_2^2$.

Step 4 (all of E). So $\|x\|_2 \leq C_2\|x\|_E$ for $x \in c_{00}(\mathbb{N}, K)$, and for $x \in E$, $\|x\|_2 = \sup_n\|x1_{[n]}\|_2 \leq C_2 \sup_n\|x1_{[n]}\|_E = C_2\|x\|_E$ by Definition 7.10. □

Proof of Proposition 7.16. Write $c_0 = c_0(\mathbb{N}, K)$, a symmetric sequence space, and let $g_k(E)$, finite by (Γ_E^k), be the infimum of the norms of projections of $\widehat{\otimes}_\pi^k E$ onto $\widehat{\Gamma}_\pi^k E$ (Definition 3.8).

Step 1 (a multiplier lift for (c_0, E)). Let $Z = \mathcal{L}(\widehat{\otimes}_\pi^k c_0, \widehat{\Lambda}_\pi^k E)$ and let $W \subseteq Z$ be the closed subspace of operators vanishing on $\mathcal{N}_{c_0}^k = \ker \widehat{\wedge}_{c_0}^k$. Since $\|\mu_a\|_{c_0\to E} = \|a\|_E$ (Lemma 7.11(4)), $c_{00}(\mathbb{N}, K)$ is dense in E and Z is a Banach space, the map $\Phi(a_1, \dots, a_k) = \widehat{\wedge}_E^k \circ (\mu_{a_1} \otimes \cdots \otimes \mu_{a_k})$ extends from $c_{00}(\mathbb{N}, K)^k$ to a k-linear $\Phi\colon E^k \to Z$ of norm at most 1. Its diagonal lies in W, since $\Phi(a, \dots, a) = \widehat{\Lambda}_\pi^k\mu_a \circ \widehat{\wedge}_{c_0}^k$ for $a \in c_{00}(\mathbb{N}, K)$ by naturality of the wedge and W is closed. Lemma 3.11(3), with $P = E$, $n = k$ and $\delta = 1$, gives a bounded k-linear $\Phi'\colon E^k \to W$ with the diagonal of Φ and $\|\Phi'\| \leq g_k(E) + 1$. By Lemma 3.7, $T \mapsto T \circ \widehat{\wedge}_{c_0}^k$ is a linear isometry of $\mathcal{L}(\widehat{\Lambda}_\pi^k c_0, \widehat{\Lambda}_\pi^k E)$ onto W. For $a_1, \dots, a_k \in c_{00}(\mathbb{N}, K)$ let $\widetilde{Q}(a_1, \dots, a_k)$ be the restriction to $\Lambda_\pi^k c_0$ of the preimage of $\Phi'(a_1, \dots, a_k)$. Then $\widetilde{Q}$ is k-linear, $\widetilde{Q}(a, \dots, a) = \Lambda_\pi^k\mu_a$, and

$$\|\widetilde{Q}(a_1, \dots, a_k)\| \leq (g_k(E) + 1)\prod_{\gamma=1}^k \|\mu_{a_\gamma}\|_{c_0\to E}.$$

Step 2 (the triangle bound). By Lemma 7.11(1),(3), $\widetilde{Q}$ is a multiplier lift with $G = c_0$, target E and $l_0 = \max(0, \lceil \log_{1/\varrho}(g_k(E) + 1)\rceil)$, and it satisfies (K.6) for (c_0, E) with $C_Q = g_k(E) + 1$. So Steps 1–5 of the proof of Theorem 7.13 give (PTB) for (c_0, E), and the choice for $G = c_0$ in Step 1 of the proof of Theorem 8.1 bounds $\mathcal{P}_<$ on $E \otimes_\pi E$. □

K.6. **Nonarchimedean fibers.**

Proof of Lemma 7.17. Step 1 (a normalized basis). Extend an equivalent nonarchimedean norm of E to $\widehat{E}$ and round it up to $\varrho^{\mathbb{Z}} \cup \{0\}$, as in the proof of Corollary 5.3; this gives $\|\cdot\|'$. We construct $f_1, f_2, \dots \in \widehat{E}$ with $\|\sum_n \xi_n f_n\|' = \max_n|\xi_n|$. Given $f_1, \dots, f_n$, their span V_n is closed, being finite-dimensional over the complete field $\widehat{K}$ (Lemma A.3). For $v \in \widehat{E} \setminus V_n$ the distances $\|v - w\|'$, $w \in V_n$, lie in $\varrho^{\mathbb{Z}}$ and are at least $\operatorname{dist}(v, V_n) > 0$, so some w_0 attains their minimum,

and the nonarchimedean inequality gives $\|\lambda(v-w_0)+w\|' = \max(|\lambda|\|v-w_0\|', \|w\|')$ for $\lambda \in \widehat{K}$ and $w \in V_n$. Let f_{n+1} be a multiple of $v-w_0$ of norm one.

Step 2 (the projection). The closed span V_∞ of the f_n is isometric to $c_0(\mathbb{N}, \widehat{K})$, complete with distances in $\varrho^{\mathbb{Z}} \cup \{0\}$, hence spherically complete (Lemma D.1); the extension argument in the proof of Theorem 5.2, over $\widehat{K}$, gives a contractive projection Π of $\widehat{E}$ onto V_∞. Choose $e_n \in E$ with $\|e_n - f_n\|' \le \varrho$. Then $\|\sum_n \xi_n(e_n-f_n)\|' < \max_n|\xi_n|$ for $\xi \neq 0$, which gives the norm formula for the e_n. The operator Π' on V_∞ with $\Pi'(\sum_n \xi_n f_n) = \Pi(\sum_n \xi_n e_n)$ satisfies $\|\Pi' - \mathrm{id}_{V_\infty}\| \le \varrho < 1$, so it is invertible with $\|\Pi'^{-1}\| \le 1$. Let $\theta_E(x)$ be the coordinate sequence of $\Pi'^{-1}\Pi x$ in the basis (f_n); then $\theta_E(e_n)$ is the nth unit vector. If E is complete, then $E = \widehat{E}$ is a $\widehat{K}$-Banach space and the series defining $\widehat{\psi}_E$ converge; if K is complete, $K = \widehat{K}$. □

Proof of Theorem 7.18. Step 1 (the form Φ). Let F_1 and η be as in Appendix C for $K_1 = \widehat{K}$, so that $E_1 = \ell^\infty_{\widehat{K}}$, and let η' be η composed with the map $\widehat{\Lambda}^k_\pi \ell^\infty_{\widehat{K}} \to F_1$ induced by the identity of $\ell^\infty_{\widehat{K}}$, a contraction since a decomposition over K is one over $\widehat{K}$ with the same sum $\sum_j \prod_i \|x_{ji}\|$. Suppose Q exists, and for $a, x \in c_{00}(\mathbb{N}, K)^k$ put

$$\Phi(a;x) = \theta_F\big(Q(a_1,\dots,a_k)(\psi x_1,\dots,\psi x_k)\big) \in \widehat{\Lambda}^k_\pi \ell^\infty_{\widehat{K}}.$$

It is k-linear in a and in x, alternating in x, of norm at most $\|\theta_F\|\|Q\|\|\psi\|^k$, and $\Phi(a,\dots,a;x) = \omega_{\ell^\infty_{\widehat{K}}}(ax_1,\dots,ax_k)$ by (7.3).

Step 2 (reduction to κ). For a, x with entries in $\mathbb{F}_p$, $\Psi(a;x) = \eta'(\Phi(a;x)) \in \Lambda^k_\kappa(\kappa^{\mathbb{N}})$ is $\mathbb{F}_p$-multilinear and alternating in x, and by Proposition C.3, $\operatorname{sdim}\Psi(a;x) \le \lfloor kM_\varrho\|\theta_F\|\|Q\|\|\psi\|^k \rfloor$ when all entries lie in $\{0, \pm 1\}$. As in Step 1 of the proof of Theorem 7.7, polarizing the diagonal identity over the infinite field K and applying η', which sends $\omega_{\ell^\infty_{\widehat{K}}}(y_1,\dots,y_k)$ to $y_1 \wedge \cdots \wedge y_k$ when every y_j has entries in κ (proof of Lemma C.4), gives (Pol1) for Ψ on such inputs. Let Ψ^κ be the κ-multilinear map on $(\kappa^{(\mathbb{N})})^{2k}$ that agrees with Ψ on unit vectors. Antisymmetry in the last k slots and (Pol1) are identities between κ-multilinear expressions that hold on unit vectors, so they hold for Ψ^κ, and $\Psi^\kappa = \Psi$ on inputs with entries in $\{0, \pm 1\}$. Since $k! = 0$ in κ, this contradicts Theorem B.9(1).

Step 3 (consequences). If $A^k_{E,E';F}$ were analytic at a point, Proposition 2.5 would give a bounded lift Q_0, and $Q(a_1,\dots,a_k) = Q_0(u_{a_1},\dots,u_{a_k})(\nu)$ would be excluded. The last assertion follows from Lemma 3.4. □

Proof of Theorem 7.19. Step 1 (finite dimension). If $\dim_{\widehat{K}} \widehat{E} < \infty$, then $\widehat{E}$ has a $\widehat{K}$-basis (e_i) with continuous coordinates (ε_i) (Lemma A.3). For $u_1,\dots,u_k \in \mathcal{L}(E,E')$, with extensions $\widehat{u}_i$ to the completions, put

$$Q(u_1,\dots,u_k)(x_1\wedge\cdots\wedge x_k) = \sum_{i_1<\cdots<i_k} \det\big(\varepsilon_{i_a}(x_b)\big)_{a,b}\, \widehat{\omega}_{E'}(\widehat{u}_1 e_{i_1},\dots,\widehat{u}_k e_{i_k}),$$

where the continuous extension $\widehat{\omega}_{E'}$ of $\omega_{E'}$ is $\widehat{K}$-multilinear by density. Expanding $x_b = \sum_i \varepsilon_i(x_b)e_i$ shows, as in Proposition 5.1(2), that Q is a bounded lift of $u \mapsto \Lambda^k_\pi u$ with values in $\mathcal{L}(\Lambda^k_\pi E, \widehat{\Lambda}^k_\pi E')$. For $u_i \in \mathcal{L}(E',E)$ use the terms $\det(\varepsilon_{i_a}(u_a x_b))_{a,b}\,\widehat{\omega}_E(e_{i_1},\dots,e_{i_k})$ instead. If E is complete, then $e_i \in E$, and moving the scalar into the first argument shows that these lifts take values in the uncompleted spaces. This proves the positive directions; in (3) with $\dim E' < \infty$, exchange the roles of E and E' in the second formula.

Step 2 (infinite dimension). Let $\dim_{\widehat{K}} \widehat{E} = \infty$. Take e_n, θ_E and, for complete E, $\widehat{\psi}_E$ from Lemma 7.17, and put $\psi_E(\xi) = \sum_n \xi_n e_n$ on $c_{00}(\mathbb{N}, K)$. Theorem 7.18 applies to the following data, which satisfy (7.3):

(a) the pair $(E, c_0(\mathbb{N}, \widehat{K}))$, with $\psi = \psi_E$, θ the inclusion, and $u_a(x) = a\,\theta_E(x)$;
(b) the pair $(c_0(\mathbb{N}, K), E)$, with ψ the inclusion, $\theta = \theta_E$, and $u_a(x) = \psi_E(ax)$;
(c) for complete E, the pair (E, E), with $\psi = \psi_E$, $\theta = \theta_E$, and $u_a(x) = \widehat{\psi}_E(a\,\theta_E(x))$;
(d) for complete K and infinite-dimensional E, E', the pair (E, E'), with $\psi = \psi_E$, $\theta = \theta_{E'}$, and $u_a(x) = \psi_{E'}(a\,\theta_E(x))$, since θ_E then takes values in $c_0(\mathbb{N}, K)$.

Over complete K, $\dim_{\widehat{K}} \widehat{E} = \dim E$, since finite-dimensional spaces are complete. An equivalent nonarchimedean norm on the values would make the action analytic by Corollary 5.3, since $|K^\times|$ is discrete. □

K.7. **Proofs of the classification results.** As in Section 8, c_{00}, c_0 and ℓ^r stand for $c_{00}(\mathbb{N}, K)$, $c_0(\mathbb{N}, K)$ and $\ell^r(\mathbb{N}, K)$.

Proof of Theorem 8.1. Step 1 ((1)⇒(3) and (2)⇒(3)). Let $G \in \{E, c_0\}$ with (G, E) good for Banach values in a degree $k \geq p$; Theorem 7.13 gives (PTB) for (G, E) with a constant C. For $x, y \in c_{00}$, (7.2) with $a = 1_{\operatorname{supp} x}$ and $b = 1_{\operatorname{supp} y}$ if $G = E$, and with $(x, 1_{\operatorname{supp} x}, y, 1_{\operatorname{supp} y})$ in place of (a, x, b, y) if $G = c_0$, gives $\|\mathcal{P}_<(x \otimes y)\|_{E\otimes_\pi E} \leq C\|x\|_E\|y\|_E$, since $\|\mu_{1_S}\|_{E\to E} \leq 1$, $\|1_S\|_{c_0} = 1$ and $\|\mu_x\|_{c_0\to E} = \|x\|_E$ (Lemma 7.11(1),(4)).

Step 2 ((3)⇒(4), (5), (6), (7)). Condition (3) is the hypothesis of Proposition 5.7 for the unit vectors, a linearly independent family indexed by $\mathbb{N}$ with dense span c_{00}. So (S^n_E), (R^n_E) and (Γ^n_E) hold for every $n \geq 1$, which gives (4), (5) and (6) with $n = p$, and Proposition 3.5(1),(2) gives (7).

Step 3 ((4)⇒(1), (5)⇒(1), (6)⇒(3), (7)⇒(1), (2)). Proposition 3.5(1), respectively (2), with $D = E$ gives the first two implications, and Proposition 7.16 the third; for the last take $D = E$, respectively $D = c_0$, and the degree p.

Step 4 (ℓ^2-domination). Let C be the constant in (3). Lemma 7.14 with $D = E$ and $\|x\|_\infty \leq \|x\|_E$ (Lemma 7.11(1)) gives $\|x\|_2 \leq C_2\|x\|_E$ for $x \in c_{00}$, with $C_2^2 = C/(1-\varrho) + 1/(1-\varrho^2)$, hence for $x \in E$ by taking suprema over the $x1_{[n]}$ (Definition 7.10).

Step 5 (the fundamental function). We may assume $C > 0$. Let $m \geq 1$ and $z_m = \mathcal{P}_<(1_{[m]} \otimes 1_{[m]}) = \sum_{t<t'} e_t \otimes e_{t'}$, the sum over $t, t' \in [m]$. By (3) and fact (P1) of Section K.5, z_m has norm at most $C\varphi_E(m)^2$ in $E_{[m]} \otimes_\pi E_{[m]}$; the identity of $E_{[m]}$ onto $\ell^2([m], K)$ has norm at most C_2 by Step 4, so z_m has norm at most $C_2^2 C\varphi_E(m)^2$ in $\ell^2([m], K) \otimes_\pi \ell^2([m], K)$. The matrix $\mathsf{Z} = \operatorname{coeff}_0 z_m$ is the strictly upper triangular matrix of ones, so every off-diagonal entry of $\mathsf{Z} - \mathsf{Z}^\mathsf{T}$ is ± 1, and Lemmas K.1 and K.2(1) give

$$\frac{m \ln m}{1 + \ln(2|\kappa|)} \leq \operatorname{cost}(\mathsf{Z}) \leq \frac{C_2^2 C\varphi_E(m)^2}{1-\varrho^2},$$

so $\varphi_E(m)^2 \geq c\,m \ln m$ with $c > 0$ independent of m. □

Proof of Theorem 8.4. (1)⇒(2) is Theorem 7.13, and (4)⇒(1) follows with $k = p$.

(2)⇒(3). Let ν be an equivalent r-norm on D, or an equivalent nonarchimedean norm, in which case put $r = \infty$, scaled so that $\nu(y) \leq \|y\|_D \leq C''\nu(y)$ for $y \in D$. Since $\nu(e_t) \leq 1$, iterating $\nu(y+y')^r \leq \nu(y)^r + \nu(y')^r$, respectively $\nu(y+y') \leq \max(\nu(y), \nu(y'))$, gives $\|\xi\|_D \leq C''\nu(\xi) \leq C''\|\xi\|_r$ for $\xi \in c_{00}$. As $\|y\|_\infty \leq \|y\|_D$ (Lemma 7.11(1)), Proposition 7.15 with $C' = C''$ gives (3).

(3)⇒(4). The unit vectors of E are a linearly independent family indexed by $\mathbb{N}$, of norm one with dense span c_{00}, and ℓ^2-dominated by (3) (Definition 7.10). As K is nonarchimedean, Theorem 5.9(2) for $r < \infty$, and Corollary 5.10 for a nonarchimedean norm, give (4). □

Proof of Corollary 8.5. Positive cases. If $r_i < 2$ for some i, then ℓ^{r_i} has a lower r_i-estimate with its unit vectors and $C = 1$ (the examples after Theorem 5.6), and Theorem 5.6(3) makes

the pair good for Banach values in every degree. If $r_1 = 2 < r_2$, the unit vectors of ℓ^2 are ℓ^2-dominated, of norm one with dense span; ℓ^{r_2} is r_2-normed by its own norm if $r_2 < \infty$, as noted after Corollary 5.10, and c_0 has a nonarchimedean norm, so Theorem 5.9(2), respectively Corollary 5.10, makes the pair good for Banach values in every degree.

Negative cases. For the remaining pairs, $r_2 = 2 \le r_1$ or $r_1, r_2 > 2$, Corollary 7.9 excludes every degree $k \ge p$ and Corollary 6.3(1) gives every degree $k < p$. So $\mathfrak{D}_B(\ell^{r_1}, \ell^{r_2})$ is $\{1, 2, \dots\}$ in the positive cases and $\{1, \dots, p-1\}$ otherwise, which gives the equivalence and the table. □

K.8. **Spreading spaces.** For a finite set $S \subseteq \mathbb{N}$, a map $\psi\colon S \to \mathbb{N}$ that is increasing and $x \in c_{00}(\mathbb{N}, K)$ supported in S, put $\psi_* x = \sum_{t\in S} x(t) e_{\psi(t)}$. If E is spreading, then $|\psi_* x| = \sum_{t\in S} |x(t)| e_{\psi(t)}$ and Definition 8.8 give $\|\psi_* x\|_E = \|x\|_E$; in particular $\|1_S\|_E = \varphi_E(|S|)$.

Proof of Theorem 8.9. Step 1 (item (1)). In the proof of Lemma 7.11, invariance under permutations is used only in (3), to see that a vector y supported in $\operatorname{supp} a$ and its relabeling y^γ have the same norm. For γ increasing on $\operatorname{supp} a$ this is the identity $\|\psi_* y\| = \|y\|$ above, with ψ the restriction of γ to $\operatorname{supp} a$. The rest of the proof uses only monotonicity, $\|e_t\| = 1$ and $\|x\| = \sup_n \|x 1_{[n]}\|$.

Step 2 (the facts (P1)–(P3)). They hold for spreading D, with (P2) for relabelings of the rows along a map increasing on the row support and of the columns along a map increasing on the column support: such relabelings map D_J and $D_{J'}$ isometrically onto $D_{\psi(J)}$ and $D_{\psi'(J')}$, so their tensor product is an isometry for the projective norms, and (P1) applies.

Step 3 (Theorem 7.13). Lemmas K.5, K.6 and K.7 and Corollary K.8 concern sequence spaces G and D with $\|e_t\|_G \le 1$ and $\|\cdot\|_\infty \le \|\cdot\|_D$, and these properties hold for spreading spaces by Step 1. In the proof of Theorem 7.13, the norms of E and D enter through Lemma 7.11, the facts (P1)–(P3) and relabelings, and each relabeling is increasing on the support it is applied to: in Step 2, $f_i \mapsto Y_i(c)$ on J_1 and on J_2, $c = 1, \dots, 4$, since $Y_1 < \cdots < Y_m$; in Step 3, $Y_i(1) \mapsto f_i$ on R and on R'; in Step 5(a), $f_i \mapsto 2i$ on J_1 and $f_i \mapsto 2i - 1$ on J_2. Steps 5(b) and (c) use only $\|x\| = \||x|\|$ and homogeneity. So Steps 1 and 2 give the theorem for spreading E and D.

Step 4 (the other results of item (2)). The proofs of Propositions 7.15 and 7.16 and of Theorems 8.1 and 8.4 use the symmetry of the spaces only through Lemma 7.11, Theorem 7.13 and fact (P1), together with $\|e_t\|_D = 1$ in Theorem 8.4; Lemma 7.14 concerns every sequence space D with $\|\cdot\|_\infty \le \|\cdot\|_D$, and c_0 is spreading. Corollaries 8.2 and 8.6(1),(2) follow from these results as before. Remark 8.7 rests, for these pairs, on Proposition 5.7 (Step 2 of the proof of Theorem 8.1) and on the relative ordering splitting described after Theorem 8.4 (Theorem 5.9(1) and Corollary 5.10), which concern arbitrary normed spaces.

Step 5 (item (3)). Here K is any nontrivially normed field of characteristic 2; condition (L) is not used. Steps 7 and 8 of the proof of Theorem 7.20 use of E only that it is a sequence space with $\|e_t\|_E = 1$, $\|x\|_\infty \le \|x\|_E$, $\|\mu_a\|_{E\to E} \le \|a\|_\infty$ and $\sup_n n/\bar\varphi_E(n)^2 = \infty$. For spreading E the first three follow from monotonicity, as there, and $\bar\varphi_E = \varphi_E$, since $\|1_S\|_E = \varphi_E(|S|)$ and φ_E is nondecreasing. □

Proof of the assertions of Example 8.10. Write $\|\cdot\|$ for $\|\cdot\|_{\alpha,q}$ on $c_{00}(\mathbb{N}, \mathbb{R})$, φ for $\varphi_{E_{\alpha,q}}$, and put $\mathfrak{h}_n = \sum_{j\le n} 1/j$.

Step 1 (spreading). Minkowski's inequality along each increasing sequence makes $\|\cdot\|$ a norm. It depends only on $|f|$, is monotone in $|f|$ and dominates $\|f\|_\infty$, and $\|e_t\| = 1$. Let $f = \sum_{i\le d} f_i e_{t_i}$ with $t_1 < \cdots < t_d$. An increasing sequence (u_j) contributes only through its terms in $\{t_1, \dots, t_d\}$, say $u_{j_s} = t_{a_s}$ with $j_1 < j_2 < \cdots$ and $a_1 < a_2 < \cdots$; as $j_s \ge s$, its sum is at most $\sum_s s^{-\alpha} |f_{a_s}|^q$, the sum of the sequence $(t_{a_s})_s$. So $\|f\|^q$ is the largest value of $\sum_s s^{-\alpha} |f_{a_s}|^q$

over the increasing sequences (a_s) in $\{1,\dots,d\}$, which does not depend on $t_1,\dots,t_d$. The other conditions of Definition 8.8 hold by the definition of $E_{\alpha,q}$.

Step 2 (decreasing profiles; item (1)). If $|g(1)|\ge|g(2)|\ge\cdots$, then $|g(u_j)|\le|g(j)|$ for every increasing sequence (u_j), so $\|g\|^q=\sum_j j^{-\alpha}|g(j)|^q$. In particular $\varphi(m)^q=\sum_{j\le m}j^{-\alpha}$, which lies between $m^{1-\alpha}$ and $1+\int_1^m\tau^{-\alpha}\,d\tau\le m^{1-\alpha}/(1-\alpha)$.

Step 3 (blocks). For $n,M\ge1$ let $I_i=\{(i-1)n+1,\dots,in\}$ and $x=\sum_{i\le M}\sum_{j\le n}j^{-\beta}e_{(i-1)n+j}$, a real vector. By Steps 1 and 2 each $x1_{I_i}$ has norm $\mathfrak{h}_n^{1/q}$, since $j^{-\alpha}j^{-\beta q}=j^{-1}$. We show $\|x\|^q\le1+\ln(Mn)+(M-1)/\alpha$. It suffices to consider increasing sequences in the support of x, since other terms only raise the indices of the later ones. Fix one, let m_i be the number of its terms in I_i, and $\lambda_i=m_1+\cdots+m_{i-1}$. The terms in I_i have indices λ_i+s, $s\le m_i$, at positions $(i-1)n+j_s$ with $j_s\ge s$, so they contribute at most $G_i=\sum_{s\le m_i}(\lambda_i+s)^{-\alpha}s^{\alpha-1}$. If $\lambda_i=0$, then $G_i=\mathfrak{h}_{m_i}\le1+\ln m_i$. If $\lambda_i\ge1$, the function $\tau\mapsto(\lambda_i+\tau)^{-\alpha}\tau^{\alpha-1}$ decreases on $(0,\infty)$, so the substitution $\tau=\lambda_i v$ gives

$$G_i\le\int_0^{m_i/\lambda_i}(1+v)^{-\alpha}v^{\alpha-1}\,dv\le\frac{1}{\alpha}+\ln\frac{\lambda_i+m_i}{\lambda_i},$$

using $(1+v)^{-\alpha}\le1$ on $(0,1]$ and $(1+v)^{-\alpha}v^{\alpha-1}\le v^{-1}$ on $[1,\infty)$. Let i_0 be the first i with $m_i>0$. At most $M-1$ later blocks are met, and their logarithms add up to the logarithm of the number of terms divided by m_{i_0}, at most $\ln(Mn/m_{i_0})$. So the sum of the sequence is at most $1+\ln m_{i_0}+(M-1)/\alpha+\ln(Mn/m_{i_0})$.

Step 4 (item (2): symmetric norms). Let $x^*=x\circ\sigma$ be the decreasing rearrangement of x, for a permutation σ of $[Mn]$; it takes the value $j^{-\beta}$ at the M positions $(j-1)M+1,\dots,jM$. By Step 2,

$$\|x^*\|^q=\sum_{j\le n}j^{\alpha-1}\sum_{l\le M}((j-1)M+l)^{-\alpha}\ge M^{1-\alpha}\mathfrak{h}_n,$$

so by Step 3 the ratio $\|x\circ\sigma\|^q/\|x\|^q$ is at least $M^{1-\alpha}\mathfrak{h}_n/(1+\ln(Mn)+(M-1)/\alpha)$, which is unbounded as n and then M grow. For real $f\ge0$ in $c_{00}(\mathbb{N},\mathbb{R})$ let $f'\in c_{00}(\mathbb{N},K)$ have $f'(t)=0$ if $f(t)=0$ and $f'(t)=\vartheta^{c(t)}$ otherwise, with $c(t)$ the largest integer such that $|\vartheta|^{c(t)}\ge f(t)$. Then $f\le|f'|<|\vartheta|^{-1}f$, so $\|f\|\le\|f'\|\le|\vartheta|^{-1}\|f\|$, and $(f\circ\sigma)'=f'\circ\sigma$. Hence $\|x'\circ\sigma\|/\|x'\|\ge|\vartheta|\|x\circ\sigma\|/\|x\|$ is unbounded, which a norm on c_{00} invariant under permutations and equivalent to that of $E_{\alpha,q}$ would exclude.

Step 5 (item (2): lower estimates). Let $r\in(0,2)$ and $C\ge1$, and let x' be as in Step 4. The I_i are intervals, $\|x'1_{I_i}\|\ge\|x1_{I_i}\|=\mathfrak{h}_n^{1/q}$ and $\|x'\|\le|\vartheta|^{-1}\|x\|$, so (5.1) would give $M^{1/r}\mathfrak{h}_n^{1/q}\le C|\vartheta|^{-1}(1+\ln(Mn)+(M-1)/\alpha)^{1/q}$. For $M=\lceil\ln n\rceil$ and large n the right side is at most $C|\vartheta|^{-1}((2+1/\alpha)\ln n)^{1/q}$ and the left side at least $(\ln n)^{1/r+1/q}$, so $(\ln n)^{1/r}\le C|\vartheta|^{-1}(2+1/\alpha)^{1/q}$, which fails for large n.

Step 6 (items (3) and (4)). If $2\beta\le1$, then $\varphi(m)^2\le(1-\alpha)^{-2/q}m$ by (1), and Corollary 8.2, which holds for spreading spaces by Theorem 8.9(2), gives the first assertion of (3); if $2\beta<1$, then $\sup_m m/\varphi(m)^2=\infty$, and Theorem 8.9(3) gives the second, Steps 1 and 2 holding over every nontrivially normed field. For (4), let $y\in c_{00}$ and let y_m^* be the mth largest value of $|y|$. The positions of the m largest values, in increasing order, give $\|y\|^q\ge\sum_{s\le m}s^{-\alpha}(y_m^*)^q\ge m^{1-\alpha}(y_m^*)^q$, so $y_m^*\le m^{-\beta}\|y\|$ and $\|y\|_2^2\le(\sum_m m^{-2\beta})\|y\|^2$. For $y\in E_{\alpha,q}$ take the supremum over the $y1_{[n]}$. □

Appendix L. Degrees of splittings

This appendix proves the results of Section 6 on degrees, composition, polarizability and permanence. Since K is nontrivially normed, it is infinite, so the coefficients of a polynomial map from K^d to a vector space are linear combinations of its values. Degree 0 follows the paragraph on split operators in Section 6.1, with $\Lambda^0_\pi u = \mathrm{id}_K$.

L.1. **Gaps in the degree sets.**

Lemma L.1 (retracts and lines)**.** *Let E and D be normed spaces and $k \geq 1$.*

1. *Let E_0 and D_0 be normed spaces, and let $\iota_1 \colon E_0 \to E$, $\iota_1' \colon E \to E_0$, $\iota_2 \colon D_0 \to D$ and $\iota_2' \colon D \to D_0$ be bounded linear maps with $\iota_1'\iota_1 = \mathrm{id}_{E_0}$ and $\iota_2'\iota_2 = \mathrm{id}_{D_0}$. If (E, D) is good, or good for Banach values, in degree k, then so is (E_0, D_0). In particular, isomorphic pairs have the same degree sets.*
2. *If $f \colon E \to K^d$ has as coordinates linearly independent bounded linear functionals $f_1, \dots, f_d$, then E is isomorphic to $\ker f \oplus K^d$ with the maximum norm.*
3. *Give $E \oplus K$ and $D \oplus K$ the maximum norms. If (E, D) is good, or good for Banach values, in every degree $n \leq k$, then so is $(E \oplus K, D \oplus K)$ in degree k.*

Proof. (1) For Banach values this is Proposition 6.28(1). For goodness, if Q is a lift for (E, D) as in Lemma 3.4, then $Q'(u_1, \dots, u_k) = \Lambda^k_\pi \iota_2' \circ Q(\iota_2 u_1 \iota_1', \dots, \iota_2 u_k \iota_1') \circ \Lambda^k_\pi \iota_1$ is a lift for (E_0, D_0), since $Q'(u, \dots, u) = \Lambda^k_\pi(\iota_2'\iota_2 u \iota_1'\iota_1) = \Lambda^k_\pi u$ by Proposition 3.1. Isomorphic pairs are retracts of each other in this sense.

(2) The map f is onto, since a nonzero functional c on K^d vanishing on its image would give $\sum_a c_a f_a = 0$. For $x_1, \dots, x_d \in E$ with $f_a(x_b) = \delta_{ab}$, the maps $x \mapsto (x - \sum_a f_a(x) x_a, f(x))$ and $(y, \lambda) \mapsto y + \sum_a \lambda_a x_a$ are mutually inverse bounded linear maps between E and $\ker f \oplus K^d$.

(3) Let J_E, Π_E, J_K, Π_K be the inclusions and projections of $E \oplus K$, and J_D, Π_D, J_K', Π_K' those of $D \oplus K$. Since $J_E\Pi_E + J_K\Pi_K$ and $J_D\Pi_D + J_K'\Pi_K'$ are identities, every $u \in \mathcal{L}(E \oplus K, D \oplus K)$ is $\psi_1(u) + \psi_2(u) + \psi_3(u)$ with $\psi_1(u) = J_D\Pi_D u J_E \Pi_E$, $\psi_2(u) = J_K'\Pi_K' u J_E \Pi_E$ and $\psi_3(u) = u J_K \Pi_K$. For $n \leq k$ and a lift Q_n for (E, D) in degree n, the map $\Lambda^n_\pi J_D \circ Q_n(\Pi_D u_1 J_E, \dots, \Pi_D u_n J_E) \circ \Lambda^n_\pi \Pi_E$, with $\widehat{\Lambda}^n_\pi J_D$ for Banach values, lifts ψ_1 in degree n. The operators $\psi_2(u)$ and $\psi_3(u)$ have rank at most one, so their exterior powers of degree $n \geq 2$ vanish, and ψ_2 and ψ_3 are liftable in every degree, with the lift 0 in degrees $n \geq 2$. Theorem 6.13(2) concludes. □

Theorem L.2 (no gap after the first failure)**.** *Let $\mathrm{char}\, K = p > 0$, let E and D each have d linearly independent bounded linear functionals, and let $\mathfrak{D}$ be $\mathfrak{D}(E, D)$ or $\mathfrak{D}_B(E, D)$. If $\mathfrak{D} \neq \{1, 2, \dots\}$ and n is its least missing element, then none of $n, n+1, \dots, n+d$ lies in $\mathfrak{D}$.*

Proof. Step 1 (splitting off lines). Let $f \colon E \to K^d$ and $g \colon D \to K^d$ have the functionals as coordinates. For $0 \leq l \leq d$ let $E^{(l)} = \ker f \oplus K^{d-l}$ and $D^{(l)} = \ker g \oplus K^{d-l}$ with the maximum norms, so that $E^{(l)} = E^{(l+1)} \oplus K$ and $D^{(l)} = D^{(l+1)} \oplus K$. Let $\mathfrak{D}^{(l)}$ be the set of the same kind for $(E^{(l)}, D^{(l)})$, and $n^{(l)}$ its least missing element, or ∞ if there is none. By Lemma L.1(1),(2), $\mathfrak{D}^{(0)} = \mathfrak{D}$ and $n^{(0)} = n$.

Step 2 (the first failure does not move). The pair $(E^{(l+1)}, D^{(l+1)})$ is a retract of $(E^{(l)}, D^{(l)})$, so $\mathfrak{D}^{(l)} \subseteq \mathfrak{D}^{(l+1)}$ and $n^{(l)} \leq n^{(l+1)}$ by Lemma L.1(1). If $i < n^{(l+1)}$, then $1, \dots, i \in \mathfrak{D}^{(l+1)}$, so $i \in \mathfrak{D}^{(l)}$ by Lemma L.1(3); hence $n^{(l)} \geq n^{(l+1)}$. By induction $n^{(l)} = n$ for every $l \leq d$.

Step 3 (descent). Suppose that $k \in \mathfrak{D}$ with $n < k \leq n + d$, and put $i = k - n$. By Corollary 6.3(1), $n \geq p \geq 2$, so $k, k-1, \dots, n+1 \geq 2$. Theorem 6.4, applied with every normed, respectively Banach, space F, turns a degree $k' \geq 2$ in $\mathfrak{D}^{(l)}$ into the degree $k' - 1$ in $\mathfrak{D}^{(l+1)}$. Applied i times, it gives $n \in \mathfrak{D}^{(i)}$, contradicting $n^{(i)} = n$. □

Proof of Theorem 6.5. Since E^* and D^* are infinite-dimensional, Theorem L.2 applies for every d. So if $\mathfrak{D}$, which is $\mathfrak{D}(E,D)$ or $\mathfrak{D}_B(E,D)$, has a least missing element n, then no degree $k \geq n$ lies in $\mathfrak{D}$, and $\mathfrak{D} = \{1,\dots,n-1\}$. By Corollary 6.3(2), $n = p^h$ for a unique $h \geq 1$. □

L.2. **Column constructions.** Throughout this subsection $q, m \geq 1$ and $n = qm$. For vectors $x_1,\dots,x_n$ and $C \subseteq \{1,\dots,n\}$, x_C is the wedge of the x_c, $c \in C$, in increasing order. For disjoint sets $C_1,\dots,C_l$, $\epsilon(C_1,\dots,C_l)$ is the sign of the listing of C_1, then C_2, and so on, each increasingly, relative to the increasing listing of their union; so $x_{C_1} \wedge \cdots \wedge x_{C_l} = \epsilon(C_1,\dots,C_l)\, x_{C_1 \cup \cdots \cup C_l}$. A listing $\lambda = (\lambda_1,\dots,\lambda_n)$ of $\{1,\dots,n\}$ has the sign $\operatorname{sgn}(\lambda)$ as a permutation, and $N_{q,m}$ is as in Theorem 6.8.

For a normed space G let $\wedge_G^{[q,m]} \colon (\Lambda_\pi^m G)^{\otimes_\pi q} \to \Lambda_\pi^n G$ be the iterated wedge product $\xi_1 \otimes \cdots \otimes \xi_q \mapsto \xi_1 \wedge \cdots \wedge \xi_q$, contractive by Lemma 6.1, and $\operatorname{sh}_G^{[q,m]} = (\mathrm{id} \otimes \cdots \otimes \mathrm{id} \otimes \operatorname{sh}_G^{m,m}) \circ \cdots \circ (\mathrm{id} \otimes \operatorname{sh}_G^{m,(q-2)m}) \circ \operatorname{sh}_G^{m,(q-1)m}$ from $\Lambda_\pi^n G$ to $(\Lambda_\pi^m G)^{\otimes_\pi q}$, so that

$$\operatorname{sh}_G^{[q,m]}(x_1 \wedge \cdots \wedge x_n) = \sum_{(J_1,\dots,J_q)} \epsilon(J_1,\dots,J_q)\, x_{J_1} \otimes \cdots \otimes x_{J_q},$$

the sum over the ordered partitions of $\{1,\dots,n\}$ into q blocks of size m. Both maps are natural in bounded operators, and iterating Lemma 6.1 gives, on $\otimes_\pi^n G = (\otimes_\pi^m G)^{\otimes_\pi q}$,

$$\wedge_G^n = \wedge_G^{[q,m]} \circ (\wedge_G^m)^{\otimes q}, \qquad \operatorname{alt}_G^n = (\operatorname{alt}_G^m)^{\otimes q} \circ \operatorname{sh}_G^{[q,m]}. \tag{L.1}$$

Their extensions to completions keep their names, with a hat on $\wedge_G^{[q,m]}$. In $\otimes_\pi^n G = (\otimes_\pi^q G)^{\otimes_\pi m}$, the tth block of q factors is column t, and the ith factors of the columns form row i; the isometry τ_ζ of Section 4, where $\zeta \in S_n$ sends the position $(t-1)q+i$ to $(i-1)m+t$, passes to the row-major order, in which row i is the ith block of m factors.

Lemma L.3. *If q and m are powers of a prime p, then p does not divide $N_{q,m}$. Hence, if $\operatorname{char} K = p$, the image of $N_{q,m}$ in K is a nonzero element of the prime field, and its absolute value is 1.*

Proof. By Legendre's formula, the exponent of p in $(p^c)!$ is $p^{c-1} + \cdots + 1 = (p^c-1)/(p-1)$. For $q = p^a$ and $m = p^b$, the exponent of p in $N_{q,m} = (qm)!/((q!)^m\, m!)$ is therefore $\big((p^{a+b} - 1) - p^b(p^a-1) - (p^b-1)\big)/(p-1) = 0$. Every nonzero element of $\mathbb{F}_p$ is a root of unity, so its absolute value is 1. □

Proposition L.4 (source column map)**.** *Let $w \colon E \to G$ have a section s in degree q. There is a bounded linear map col_s from $\Lambda_\pi^n E$ to the completion of $(\Lambda_\pi^m G)^{\otimes_\pi q}$, of norm at most $N_{q,m}\|s\|^m$ with $N_{q,m}$ read as a real number, such that*

$$\widehat{\wedge}_G^{[q,m]} \circ \operatorname{col}_s = \eta\, N_{q,m}\, \Lambda_\pi^n w,$$

where $\eta = \pm 1$ depends only on q and m. If s takes values in $\otimes_\pi^q G$, then col_s takes values in $(\Lambda_\pi^m G)^{\otimes_\pi q}$.

Proof. Step 1 (regrouping). Let $\Theta = (\wedge_G^m)^{\otimes q} \circ \tau_\zeta$, a contraction from $\otimes_\pi^n G$ to $(\Lambda_\pi^m G)^{\otimes_\pi q}$. It sends the tensor with columns $g_{1t} \otimes \cdots \otimes g_{qt}$ to $(g_{11} \wedge \cdots \wedge g_{1m}) \otimes \cdots \otimes (g_{q1} \wedge \cdots \wedge g_{qm})$. If $\gamma \in S_m$ and τ'_γ moves column t to column $\gamma(t)$, then $\Theta \circ \tau'_\gamma = \operatorname{sgn}(\gamma)^q\, \Theta$, since γ permutes the factors of each of the q row wedges. By (L.1) and Section 4, $\wedge_G^{[q,m]} \circ \Theta = \wedge_G^n \circ \tau_\zeta = \eta \wedge_G^n$ with $\eta = \operatorname{sgn}(\zeta)$.

Step 2 (the multilinear map). For $x_1,\dots,x_n \in E$ put

$$\Xi_s(x_1,\dots,x_n) = \sum_{\{C_1,\dots,C_m\}} \epsilon(C_1,\dots,C_m)\, \Theta\big(s(x_{C_1}) \otimes \cdots \otimes s(x_{C_m})\big),$$

the sum over the partitions of $\{1,\dots,n\}$ into m blocks of size q, each with some order of its blocks. Reordering the blocks by $\gamma \in S_m$ multiplies $\epsilon(C_1,\dots,C_m)$ by $\operatorname{sgn}(\gamma)^{q^2} = \operatorname{sgn}(\gamma)^q$ and, by Step 1, the Θ-term by $\operatorname{sgn}(\gamma)^q$; so each term depends only on the partition. The map Ξ_s is n-linear of norm at most $N_{q,m}\|s\|^m$.

Step 3 (Ξ_s is alternating). For a listing λ of $\{1,\dots,n\}$, cut into m consecutive groups of q entries, let $\Phi(\lambda) = \operatorname{sgn}(\lambda)\,\Theta\big(s(x_{\lambda_1}\wedge\cdots\wedge x_{\lambda_q})\otimes s(x_{\lambda_{q+1}}\wedge\cdots\wedge x_{\lambda_{2q}})\otimes\dots\big)$. Permuting entries inside a group changes $\operatorname{sgn}(\lambda)$ and that group's wedge by the same sign, so $\Phi(\lambda)$ is unchanged; and the term of a partition is $\Phi(\lambda)$ for λ listing C_1, then C_2, and so on. Let $x_a = x_b$ with $a \neq b$. The terms with a and b in one block vanish. Exchanging a and b is an involution without fixed points on the partitions that separate them, and it replaces λ by the listing λ' with the entries a and b exchanged; since $\operatorname{sgn}(\lambda') = -\operatorname{sgn}(\lambda)$ and $x_a = x_b$, $\Phi(\lambda') = -\Phi(\lambda)$, so these terms cancel in pairs. Proposition 3.1 gives col_s with $\operatorname{col}_s(x_1\wedge\cdots\wedge x_n) = \Xi_s(x)$.

Step 4 (the identity). By Step 1, (L.1) for m blocks of size q, and $\widehat{\wedge}^q_G s = \Lambda^q_\pi w$,

$$\begin{aligned}\widehat{\wedge}^{[q,m]}_G\Theta\big(s(x_{C_1})\otimes\cdots\otimes s(x_{C_m})\big) &= \eta\,\Lambda^q_\pi w(x_{C_1})\wedge\cdots\wedge\Lambda^q_\pi w(x_{C_m})\\ &= \eta\,\epsilon(C_1,\dots,C_m)\,\Lambda^n_\pi w(x_1\wedge\cdots\wedge x_n),\end{aligned}$$

so each of the $N_{q,m}$ terms of Ξ_s contributes $\eta\Lambda^n_\pi w(x_1\wedge\cdots\wedge x_n)$. If s takes values in $\otimes^q_\pi G$, every term lies in $(\Lambda^m_\pi G)^{\otimes_\pi q}$. □

Proposition L.5 (target column map)**.** *Let $w\colon G\to D$ have a retraction ρ in degree q. There is a bounded linear map col^ρ from the completion of $(\widehat\Lambda^m_\pi G)^{\otimes_\pi q}$ to $\widehat\Lambda^n_\pi D$, of norm at most $(m!)^{q-1}\|\rho\|^m$, such that*

$$\operatorname{col}^\rho\circ\operatorname{sh}^{[q,m]}_G = \eta'\,N_{q,m}\,\Lambda^n_\pi w,$$

where $\eta' = \pm1$ depends only on q and m. If ρ takes values in $\Lambda^q_\pi D$, then col^ρ maps $(\Lambda^m_\pi G)^{\otimes_\pi q}$ into $\Lambda^n_\pi D$.

Proof. Step 1 (the map). For vectors $y_{i,t}\in G$ in rows i and columns t, put

$$\Psi_\rho(y) = \sum_{g_2,\dots,g_q\in S_m}\operatorname{sgn}(g_2)\cdots\operatorname{sgn}(g_q)\bigwedge_{t=1}^m\rho\big(y_{1,t}\otimes y_{2,g_2(t)}\otimes\cdots\otimes y_{q,g_q(t)}\big)\in\widehat\Lambda^n_\pi D,$$

the wedge taken in the order $t=1,\dots,m$. It is multilinear of norm at most $(m!)^{q-1}\|\rho\|^m$.

Step 2 (alternation in each row). Let $y_{i,t} = y_{i,t'}$ with $t\neq t'$. If $i\geq 2$, replacing g_i by $(t\ t')\circ g_i$ keeps every factor and changes the sign. If $i=1$, replacing every g_l by $g_l\circ(t\ t')$ exchanges the wedge factors in the positions t and t', which gives $(-1)^{q^2} = (-1)^q$, and multiplies the coefficient by $(-1)^{q-1}$; so the sign changes. Both replacements are involutions without fixed points, so the terms cancel in pairs. For $q=1$, Ψ_ρ is alternating since $\rho = w$. By Proposition 3.1 in each row, Ψ_ρ induces a bounded linear map col^ρ on $(\Lambda^m_\pi G)^{\otimes_\pi q}$, which extends to the completion.

Step 3 (arrays). Let $x_1,\dots,x_n\in G$. For an ordered partition $(J_1,\dots,J_q)$ with $J_i = \{b_{i,1}<\cdots<b_{i,m}\}$, the corresponding term of $\operatorname{col}^\rho\operatorname{sh}^{[q,m]}_G(x_1\wedge\cdots\wedge x_n)$ is $\epsilon(J_1,\dots,J_q)\Psi_\rho(y)$ with $y_{i,t} = x_{b_{i,t}}$. An *array* is a bijection $\alpha\colon\{1,\dots,q\}\times\{1,\dots,m\}\to\{1,\dots,n\}$; $\operatorname{sgn}_{\mathrm{row}}(\alpha)$ and $\operatorname{sgn}_{\mathrm{col}}(\alpha)$ are the signs of its readings row by row and column by column, and $\operatorname{sgn}_{\mathrm{row}} = \eta'\operatorname{sgn}_{\mathrm{col}}$ for a sign η' depending only on q and m. Through $\alpha(1,t) = b_{1,t}$ and $\alpha(i,t) = b_{i,g_i(t)}$ for $i\geq 2$, the pairs $((J_i),(g_i))$ correspond bijectively to the arrays with increasing first row, and $\epsilon(J_1,\dots,J_q)\prod_i\operatorname{sgn}(g_i) = \operatorname{sgn}_{\mathrm{row}}(\alpha)$. Hence

$$\operatorname{col}^\rho\operatorname{sh}^{[q,m]}_G(x_1\wedge\cdots\wedge x_n) = \eta'\sum_\alpha\Phi(\alpha),\qquad \Phi(\alpha) = \operatorname{sgn}_{\mathrm{col}}(\alpha)\bigwedge_{t=1}^m\rho\big(x_{\alpha(1,t)}\otimes\cdots\otimes x_{\alpha(q,t)}\big),$$

the sum over the arrays with increasing first row.

Step 4 (columns). Permuting the columns of an array by $\gamma \in S_m$ leaves Φ unchanged, since $\mathrm{sgn}_{\mathrm{col}}$ and the wedge of m elements of degree q both change by $\mathrm{sgn}(\gamma)^{q^2}$. Group the arrays by the partition $\{C_1, \dots, C_m\}$ formed by their column sets. For a fixed partition, the arrays with increasing first row correspond bijectively to the choices of an ordering o_t of each C_t, since the first entries then fix the order of the columns. By the column invariance of Φ, the partition contributes

$$\epsilon(C_1, \dots, C_m) \sum_{o_1, \dots, o_m} \prod_t \mathrm{sgn}(o_t) \bigwedge_t \rho(x_{o_t}) = \epsilon(C_1, \dots, C_m) \bigwedge_t \rho\big(\mathrm{alt}^q_G(x_{C_t})\big),$$

where x_{o_t} is the tensor of the x_c, $c \in C_t$, in the order o_t, and $\mathrm{sgn}(o_t)$ is relative to the increasing order. Since $\rho\,\mathrm{alt}^q_G = \Lambda^q_\pi w$, this is $\epsilon(C_1, \dots, C_m)\Lambda^q_\pi w(x_{C_1}) \wedge \cdots \wedge \Lambda^q_\pi w(x_{C_m}) = \Lambda^n_\pi w(x_1 \wedge \cdots \wedge x_n)$. Summing over the $N_{q,m}$ partitions gives the identity on pure wedges, hence everywhere. □

Proof of Theorem 6.8. Put $N = N_{q,m}$, invertible in K. (1) Let $s = (\eta N)^{-1}(s_2)^{\otimes q} \circ \mathrm{col}_{s_1}$, with col_{s_1} the map of Proposition L.4 for w_1 and $(s_2)^{\otimes q}$ from the completion of $(\Lambda^m_\pi G_1)^{\otimes_\pi q}$ to $\widehat{\otimes}^n_\pi G_2$, rows in the order $1, \dots, q$. By (L.1) and naturality of $\wedge^{[q,m]}$, $\widehat{\wedge}^n_{G_2}(s_2)^{\otimes q} = \widehat{\wedge}^{[q,m]}_{G_2}(\Lambda^m_\pi w_2)^{\otimes q} = \widehat{\Lambda}^n_\pi w_2 \circ \widehat{\wedge}^{[q,m]}_{G_1}$, so Proposition L.4 gives $\widehat{\wedge}^n_{G_2} s = \Lambda^n_\pi(w_2 w_1)$, and $\|s\| \le N_{q,m}\|s_1\|^m \|s_2\|^q / |N|$.

(2) Let $\rho = (\eta' N)^{-1}\,\mathrm{col}^{\rho_2} \circ (\rho_1)^{\otimes q}$ on $\otimes^n_\pi E = (\otimes^m_\pi E)^{\otimes_\pi q}$, with col^{ρ_2} the map of Proposition L.5 for w_2. By (L.1), $\rho_1\,\mathrm{alt}^m_E = \Lambda^m_\pi w_1$ and naturality of $\mathrm{sh}^{[q,m]}$, the map ρ satisfies $\rho\,\mathrm{alt}^n_E = (\eta' N)^{-1}\,\mathrm{col}^{\rho_2}\,\mathrm{sh}^{[q,m]}_{G_1}\,\Lambda^n_\pi w_1 = \Lambda^n_\pi(w_2 w_1)$, and $\|\rho\| \le (m!)^{q-1}\|\rho_2\|^m \|\rho_1\|^q / |N|$. Uncompleted data give uncompleted maps, by the last assertions of Propositions L.4 and L.5. □

Proof of Theorem 6.6. Let s_p be a section for id_Y in degree p. If s_{p^j} is one in degree p^j, Theorem 6.8(1) with $w_1 = w_2 = \mathrm{id}_Y$, $q = p$ and $m = p^j$, which applies by Lemma L.3, gives one in degree p^{j+1}. So $(\mathrm{S}^{p^j}_Y)$ holds for every $j \ge 0$. By its base-p expansion, every $n \ge 1$ is a carry-free sum of powers of p, so (S^n_Y) holds by the Lucas step for split operators in Section 6.1. For retractions use Theorem 6.8(2) with $w_1 = w_2 = \mathrm{id}_Y$, $m = p^j$ and $q = p$. Uncompleted data stay uncompleted at each step. □

Proof of Corollary 6.9. (1) Let $w = w_h \cdots w_1$ with each w_l S^p-split, and put $\gamma_l = w_l \cdots w_1$. Then γ_l is S^{p^l}-split for $l \le h$: for $l = 1$ by hypothesis, and for $l \ge 2$ by Theorem 6.8(1) applied to γ_{l-1} and w_l with $q = p^{l-1}$ and $m = p$, by Lemma L.3. So $w = (w_h \cdots w_{l+1})\gamma_l$ is S^{p^l}-split for $l \le h$, by Proposition 3.13(2). Every $n < p^{h+1}$ is a carry-free sum of powers p^l with $l \le h$, so w is S^n-split by the Lucas step for split operators in Section 6.1. For retractions use Theorem 6.8(2) with the inner map γ_{l-1}, $m = p^{l-1}$, and the outer map w_l, $q = p$. (2) If $w = v_2 v_1$ with $v_1 \colon E \to Y$, $v_2 \colon Y \to G$ and (S^p_Y), then id_Y is S^n-split for every n by Theorem 6.6, hence so is $w = v_2\,\mathrm{id}_Y\,v_1$ by Proposition 3.13(2). The same argument applies to (R^p_Y). □

L.3. **Polarizability in every degree.** A bounded linear map $v \colon P \to P'$ of normed spaces is *polarizable in degree n* if there is a bounded n-linear map $\Pi \colon P^n \to \widehat{\Gamma}^n_\pi P'$ with $\Pi(h, \dots, h) = (vh)^{\otimes n}$ for $h \in P$; by Lemma 3.11, id_P is polarizable in degree n if and only if P is. Each isometry τ_σ of Section 4 fixes every pure power, hence every point of $\widehat{\Gamma}^n_\pi P'$. Every tensor y in the algebraic tensor power $\otimes^n_K P'$ with $\tau_\sigma y = y$ for all $\sigma \in S_n$ is a linear combination of pure powers: if $e_1, \dots, e_d$ is a basis of a finite-dimensional subspace containing the factors of y, then y is a combination of the sums of the tensors $e_{i_1} \otimes \cdots \otimes e_{i_n}$ over the rearrangements of a fixed index tuple, and each such sum is a coefficient of the polynomial $\lambda \mapsto (\sum_i \lambda_i e_i)^{\otimes n}$ on K^d. For $a, b \ge 0$, $y \in \widehat{\otimes}^a_\pi P'$ and $z \in \widehat{\otimes}^b_\pi P'$, the *shuffle product* $y * z \in \widehat{\otimes}^{a+b}_\pi P'$ is the sum of the $\binom{a+b}{a}$

tensors obtained from $y \otimes z$ by distributing the factors of y and of z over the $a+b$ slots in their orders; it is bilinear with $\|y * z\| \le \binom{a+b}{a}\|y\|\|z\|$.

Lemma L.6 (polarizable maps)**.** *Let $v\colon P \to P'$ be a bounded linear map of normed spaces.*

(1) *The shuffle product maps $\widehat{\Gamma}^a_\pi P' \times \widehat{\Gamma}^b_\pi P'$ into $\widehat{\Gamma}^{a+b}_\pi P'$. If v is polarizable in degrees $a, b \ge 1$, with Π_a and Π_b, and $\binom{a+b}{a} \neq 0$ in K, then v is polarizable in degree $a+b$, with $(h_1, \dots, h_{a+b}) \mapsto \binom{a+b}{a}^{-1} \Pi_a(h_1, \dots, h_a) * \Pi_b(h_{a+1}, \dots, h_{a+b})$.*

(2) *If $n! \neq 0$ in K, then v is polarizable in degree n. Hence, if $\operatorname{char} K = p > 0$, the degrees in which v is polarizable contain $1, \dots, p-1$ and are closed under carry-free sums.*

Proof. (1) For $h, h' \in P'$, $h^{\otimes a} * h'^{\otimes b}$ is the coefficient of λ^a in $(\lambda h + h')^{\otimes(a+b)}$, hence a combination of pure powers; by bilinearity and continuity the shuffle product maps $\widehat{\Gamma}^a_\pi P' \times \widehat{\Gamma}^b_\pi P'$ into $\widehat{\Gamma}^{a+b}_\pi P'$. The stated map has diagonal $\binom{a+b}{a}^{-1}(vh)^{\otimes a} * (vh)^{\otimes b} = (vh)^{\otimes(a+b)}$. (2) By Proposition 4.1(2), $R_n = \frac{1}{n!}\sum_\sigma \tau_\sigma$ is a projection onto $\widehat{\Gamma}^n_\pi P'$, and $\Pi(h_1, \dots, h_n) = R_n(vh_1 \otimes \cdots \otimes vh_n)$ has diagonal $(vh)^{\otimes n}$. By Lucas's theorem, $\binom{a+b}{a} \neq 0$ in K when a and b add without carry in base p, and (1) gives the last assertion. □

Lemma L.7 (composition of polarizable maps)**.** *Let $q, m \ge 1$ and $n = qm$, and suppose that the number $N_{m,q}$ of partitions of $\{1, \dots, n\}$ into q blocks of size m is nonzero in K. If $v_1\colon P \to P'$ is polarizable in degree q, with Π_1, and $v_2\colon P' \to P''$ is polarizable in degree m, with Π_2, then $v_2 v_1$ is polarizable in degree n, with a map of norm at most $N_{m,q}\|\Pi_1\|^m\|\Pi_2\|^q/|N_{m,q}|$.*

Proof. Step 1 (the map). Use the rows, columns and τ_ζ of Subsection L.2, with P' and P'' in place of G. The stabilizer $S_{\mathrm{row}} \subseteq S_n$ of the partition into rows consists of the permutations inside rows composed with permutations of whole rows, and it has $N_{m,q}$ left cosets. For representatives $\sigma_1, \dots, \sigma_{N_{m,q}}$ put $T = \sum_l \tau_{\sigma_l}$ on $\widehat{\otimes}^n_\pi P''$, of norm at most $N_{m,q}$. If $\tau_\sigma Y = Y$ for all $\sigma \in S_{\mathrm{row}}$, then TY does not depend on the representatives, and $\tau_\sigma TY = TY$ for all $\sigma \in S_n$. For an array $(h_{i,t})$ in P let $c_t = \Pi_1(h_{1,t}, \dots, h_{q,t}) \in \widehat{\Gamma}^q_\pi P'$ and

$$Y(h) = (\Pi_2^{\otimes})^{\otimes q}\, \tau_\zeta(c_1 \otimes \cdots \otimes c_m), \qquad \Pi(h) = N_{m,q}^{-1}\, T\, Y(h),$$

where the linearization $\Pi_2^{\otimes}\colon \widehat{\otimes}^m_\pi P' \to \widehat{\Gamma}^m_\pi P''$ of Π_2 acts on each row. The map Π is n-linear with the stated bound.

Step 2 (invariance). A permutation of whole rows permutes the q factors of every column block in the same way, which fixes each c_t, so it fixes $\tau_\zeta(c_1 \otimes \cdots \otimes c_m)$. The operator $(\Pi_2^{\otimes})^{\otimes q}$ commutes with these permutations, and its values are fixed by permutations inside rows, since $\Pi_2^{\otimes}$ takes values in $\widehat{\Gamma}^m_\pi P''$. Hence $\tau_\sigma Y(h) = Y(h)$ for $\sigma \in S_{\mathrm{row}}$.

Step 3 (the diagonal). If every $h_{i,t}$ is h, then $c_t = (v_1 h)^{\otimes q}$, $\tau_\zeta(c_1 \otimes \cdots \otimes c_m) = (v_1 h)^{\otimes n}$, and $\Pi_2^{\otimes}$ sends each row $(v_1 h)^{\otimes m}$ to $(v_2 v_1 h)^{\otimes m}$. So $Y(h) = (v_2 v_1 h)^{\otimes n}$, $TY(h) = N_{m,q}(v_2 v_1 h)^{\otimes n}$ and $\Pi(h, \dots, h) = (v_2 v_1 h)^{\otimes n}$.

Step 4 (values). Fix the array h, and choose $c_t^{(j)} \to c_t$ in the linear span of the pure powers in $\otimes^q_\pi P'$. Expanding, $\tau_\zeta(c_1^{(j)} \otimes \cdots \otimes c_m^{(j)})$ is a linear combination of tensors $\tau_\zeta(h_1'^{\otimes q} \otimes \cdots \otimes h_m'^{\otimes q}) = y^{\otimes q}$ with $y = h_1' \otimes \cdots \otimes h_m'$, and $(\Pi_2^{\otimes})^{\otimes q}(y^{\otimes q}) = z^{\otimes q}$ with $z = \Pi_2^{\otimes}(y) \in \widehat{\Gamma}^m_\pi P''$. For $z_l \to z$ in the linear span of the pure powers, $z_l^{\otimes q}$ lies in the algebraic tensor power and is fixed by τ_σ for $\sigma \in S_{\mathrm{row}}$, so $T z_l^{\otimes q}$ is fixed by every τ_σ and is a linear combination of pure powers; hence $T z^{\otimes q} = \lim_l T z_l^{\otimes q}$ lies in $\widehat{\Gamma}^n_\pi P''$. By linearity and continuity, so does $TY(h)$. □

Proof of Theorem 6.10. If P is polarizable in degrees p and p^i, then Lemma L.7 with $v_1 = v_2 = \mathrm{id}_P$, $q = p$ and $m = p^i$, which applies by Lemma L.3, makes P polarizable in degree p^{i+1}. By

induction P is polarizable in every degree p^i, and Lemma L.6 gives every degree, since every $n \geq 1$ is a carry-free sum of powers of p. □

Proof of Corollary 6.19. (1) The parameter space of a parameter piece of degree p is polarizable in degree p, hence in every degree by Theorem 6.10; so the piece, and every piece of a generalized decomposition of degree p into parameter pieces, is a parameter piece of every degree. (2) By Theorem 6.10, $\mathcal{L}(E, D)$ is polarizable in every degree, and Corollary 3.17 applies in each degree. □

L.4. **Decompositions of degree p.**

Proof of Theorem 6.20. Let w be $\mathrm{S}^{\leq p}$-split, as every S^p-split map is, since every map is S^n-split for $n < p$. Every $n < p^2$ is $c_1 p + c_0$ with base-p digits $c_0, c_1 < p$, a carry-free sum of copies of p and of 1, so w is S^n-split by the Lucas step for split operators in Section 6.1. Thus w is $\mathrm{S}^{\leq p^2-1}$-split, and the same holds for R. Hence every piece of the decomposition, including the single piece of a relative splitting, is of the same type in every degree $n < p^2$. Theorem 6.13(3) gives the goodness; the Lucas step keeps uncompleted data uncompleted. □

Proof of Proposition 6.21. Choose $f \in E^*$ and $x_0 \in E$ with $f(x_0) = 1$, and let the pieces be $(G_i, G_i', w_i, w_i', \varepsilon_i)$. For $y \in D$, the operator $u_y = f(\cdot)\, y \in \mathcal{L}(E, D)$ has $\|u_y\| \leq \|f\|\|y\|$, and (6.4) gives $y = u_y x_0 = \sum_i w_i' \chi_i(y)$ with the bounded linear maps $\chi_i \colon D \to G_i'$, $\chi_i(y) = \varepsilon_i(u_y) w_i x_0$. So the inclusion ι_D of D in $\widehat{D}$ is $\sum_i w_i' \chi_i$, which is $\mathrm{R}^{\leq k}$-split, since every w_i' is, by Proposition 3.13(2) and the paragraph on split operators in Section 6.1. A retraction for ι_D in degree k is a bounded linear $\rho \colon \otimes_\pi^k D \to \widehat{\Lambda}_\pi^k \widehat{D} = \widehat{\Lambda}_\pi^k D$ such that $\rho \operatorname{alt}_D^k$ is the inclusion of $\Lambda_\pi^k D$ in its completion; this is (R_D^k). Without completion, the maps w_i' take values in D and their retractions in the uncompleted exterior powers, and so does ρ. For $k = p$, Corollary 6.7 gives the last assertion. □

L.5. **Permanence and its failure.**

Proof of Theorem 6.29. Step 0 (notation). Order the words of each length m lexicographically. For a normed space U, a step function of level m on Ω^k is a function $z = \sum_\xi 1_{C_{\xi_1} \times \cdots \times C_{\xi_k}} u_\xi$, the sum over k-tuples ξ of words of length m, with $u_\xi \in U$; put $\|z\|_1 = \sum_\xi 2^{-mk} \|u_\xi\|$ and $\|z\|_\infty = \max_\xi \|u_\xi\|$, and let $L^1(\Omega^k; U)$ be the completion of the step functions, in which the U_0-valued ones are dense if U_0 is dense in U. Let $\Delta_m \subseteq \Omega^k$ be the union of the cells $C_{\xi_1} \times \cdots \times C_{\xi_k}$ of level m with $\xi_1 < \cdots < \xi_k$, and T_m the union of those with two equal words; T_m has measure at most $\binom{k}{2} 2^{-m}$, so $\|1_{T_m} z\|_1 \leq \binom{k}{2} 2^{-m} \|z\|_\infty$ for step functions z. Every extension of a word ξ precedes every extension of a later word of the same length, so $\Delta_m \subseteq \Delta_{m'}$ and $\Delta_{m'} \setminus \Delta_m \subseteq T_m$ for $m \leq m'$.

Step 1 (tensor powers). For a finite set I with weights $c_t > 0$, the map $\sum_t e_t \otimes u_t \mapsto (u_t)_t$ is an isometry of $\ell^1(I, c) \otimes_\pi U$ onto U^I with the norm $\sum_t c_t \|u_t\|$, since this decomposition costs $\sum_t c_t \|u_t\|$ and $(\eta, u) \mapsto (\eta_t u)_t$ is contractive. With associativity and commutativity of $\otimes_\pi$, the contractive k-linear map $(f_1, \ldots, f_k) \mapsto [\zeta \mapsto f_1(\zeta_1) \otimes \cdots \otimes f_k(\zeta_k)]$ therefore linearizes to an isometry of the kth projective tensor power of the step functions of level m onto the $\otimes_\pi^k G$-valued step functions of level m on Ω^k; since the tensor power at level m maps contractively into that of all step functions, it is an isometry of the latter onto all $\otimes_\pi^k G$-valued step functions. Passing to completions, $\widehat{\otimes}_\pi^k Y = L^1(\Omega^k; \widehat{\otimes}_\pi^k G)$; then $\widehat{\wedge}_Y^k$ sends $1_{C_1 \times \cdots \times C_k}(g_1 \otimes \cdots \otimes g_k)$ to $(1_{C_1} g_1) \wedge \cdots \wedge (1_{C_k} g_k)$, and alt_Y^k sends $f_1 \wedge \cdots \wedge f_k$ to $\zeta \mapsto \sum_\sigma \operatorname{sgn}(\sigma) f_{\sigma(1)}(\zeta_1) \otimes \cdots \otimes f_{\sigma(k)}(\zeta_k)$.

Step 2 (ordered antisymmetrization). For step functions $f_a = \sum_\xi 1_{C_\xi} g_{a,\xi}$ of level at most m, let $\Phi_m(f)$ be the step function equal to $\sum_\sigma \operatorname{sgn}(\sigma) f_{\sigma(1)}(\zeta_1) \otimes \cdots \otimes f_{\sigma(k)}(\zeta_k)$ on Δ_m and to 0 elsewhere.

Since $(\xi,\sigma)\mapsto v$ with $v_{\sigma(a)}=\xi_a$ is a bijection from the increasing k-tuples of words times S_k onto the k-tuples of distinct words, $\|\Phi_m(f)\|_1\le\sum_v 2^{-mk}\prod_b\|g_{b,v_b}\|\le\prod_b\|f_b\|_1$. For $m'\ge m$, $\Phi_{m'}(f)$ and $\Phi_m(f)$ agree outside T_m by Step 0, so $\|\Phi_{m'}(f)-\Phi_m(f)\|_1\le k!\binom{k}{2}2^{-m}\prod_a\|f_a\|_\infty$. Hence $\Phi(f)=\lim_m\Phi_m(f)$ exists; it is k-linear, alternating, since the terms σ and $\sigma\circ(a\ b)$ cancel when $f_a=f_b$, and bounded by $\prod_a\|f_a\|_1$, so it extends to a contractive alternating map on Y^k.

Step 3 (sections). For the f_a of Step 2, $f_1\wedge\cdots\wedge f_k$ is the sum over k-tuples of words of $\bigwedge_a 1_{C_{\xi_a}}g_{a,\xi_a}$. A tuple of distinct words is $(\xi_{\sigma^{-1}(1)},\dots,\xi_{\sigma^{-1}(k)})$ for a unique increasing tuple ξ and a unique $\sigma\in S_k$, and its wedge is $\operatorname{sgn}(\sigma)\bigwedge_b 1_{C_{\xi_b}}g_{\sigma(b),\xi_b}$; by Step 1 these terms add up to $\widehat{\bigwedge}^k_Y\Phi_m(f)$. Since $\|1_{C_\xi}g\|_1=2^{-m}\|g\|$, the tuples with $\xi_a=\xi_b$ for some $a<b$ contribute at most $\sum_{a<b}2^{-m}\|f_a\|_\infty\|f_b\|_1\prod_{c\ne a,b}\|f_c\|_1$ in norm. Letting $m\to\infty$ gives $f_1\wedge\cdots\wedge f_k=\widehat{\bigwedge}^k_Y\Phi(f)$. By Proposition 3.1, Φ induces $s\colon\Lambda^k_\pi Y\to\widehat{\bigotimes}^k_\pi Y$ with $\|s\|\le1$, and $\widehat{\bigwedge}^k_Y s=\mathrm{id}$ on wedges of step functions, hence everywhere, since these span a dense subspace (the paragraph before Theorem 3.16).

Step 4 (retractions). For a step function z of level at most m and $m'\ge m$, $\|1_{\Delta_{m'}}z-1_{\Delta_m}z\|_1\le\binom{k}{2}2^{-m}\|z\|_\infty$ by Step 0. So $\mathcal{P}_\Delta z=\lim_m 1_{\Delta_m}z$ exists, with $\|\mathcal{P}_\Delta z\|_1\le\|z\|_1$, and $\mathcal{P}_\Delta$ extends to a contraction of $L^1(\Omega^k;\widehat{\bigotimes}^k_\pi G)=\widehat{\bigotimes}^k_\pi Y$. Let $\rho=\widehat{\bigwedge}^k_Y\circ\mathcal{P}_\Delta$ on $\bigotimes^k_\pi Y$, so $\|\rho\|\le1$. For step functions f_a, Step 1 gives $1_{\Delta_m}\operatorname{alt}^k_Y(f_1\wedge\cdots\wedge f_k)=\Phi_m(f)$, so $\rho\operatorname{alt}^k_Y(f_1\wedge\cdots\wedge f_k)=\widehat{\bigwedge}^k_Y\Phi(f)=f_1\wedge\cdots\wedge f_k$ by Step 3; by density $\rho\operatorname{alt}^k_Y=\mathrm{id}$.

Step 5 (polarization). On step functions $\tau_\sigma z(\zeta)=\tau_\sigma\big(z(\zeta_{\sigma(1)},\dots,\zeta_{\sigma(k)})\big)$, so $\tau_\sigma(1_S z)=1_{\sigma S}\tau_\sigma z$ with $\sigma S=\{\zeta:(\zeta_{\sigma(1)},\dots,\zeta_{\sigma(k)})\in S\}$; the sets $\sigma\Delta_m$, $\sigma\in S_k$, are disjoint with union $\Omega^k\setminus T_m$. Put $R_k=\sum_\sigma\tau_\sigma\mathcal{P}_\Delta$, of norm at most $k!$. For a step function h, $h^{\otimes k}$ is fixed by every τ_σ, so $R_k h^{\otimes k}=\lim_m 1_{\Omega^k\setminus T_m}h^{\otimes k}=h^{\otimes k}$; by continuity R_k fixes every point of $\widehat{\Gamma}^k_\pi Y$. For cells $C_1<\cdots<C_k$ of level m and $g_a\in G$, $\sum_\sigma\tau_\sigma(1_{C_1}g_1\otimes\cdots\otimes1_{C_k}g_k)$ is the coefficient of $\lambda_1\cdots\lambda_k$ in $(\sum_a\lambda_a 1_{C_a}g_a)^{\otimes k}$, hence a linear combination of pure powers. For a $\bigotimes^k_\pi G$-valued step function z of level at most m, $1_{\Delta_m}z$ is a finite sum of such tensors, so $R_k z=\lim_m\sum_\sigma\tau_\sigma 1_{\Delta_m}z$ lies in $\widehat{\Gamma}^k_\pi Y$; by continuity R_k maps $\widehat{\bigotimes}^k_\pi Y$ onto $\widehat{\Gamma}^k_\pi Y$. □

Proof of Corollary 6.30. (1) The isometry $g\mapsto 1_\Omega g$ has complete, hence closed, image. By Theorem 6.29, $L^1(\Omega;E)$ and $L^1(\Omega;D)$ satisfy (S^n) and (R^n) for every n, and Proposition 3.5(1),(2) gives the goodness statements.

(2) The map $q_E\colon\ell^1(\mathrm{Ball}(E),\widehat{K})\to E$, $e_x\mapsto x$, is contractive. Fix $c\in K$ with $0<|c|<1$. For $x\ne0$ in E choose $j\in\mathbb{Z}$ with $|c|^{j+1}<\|x\|\le|c|^j$; then $x/c^j\in\mathrm{Ball}(E)$, $q_E(c^j e_{x/c^j})=x$ and $\|c^j e_{x/c^j}\|_1=|c|^j<|c|^{-1}\|x\|$. So q_E maps each open ball of radius δ onto a set containing the open ball of radius $|c|\delta$; it is surjective and open, and E is isomorphic to $\ell^1(\mathrm{Ball}(E),\widehat{K})/\ker q_E$. The unit vectors of $\ell^1(\mathrm{Ball}(E),\widehat{K})$ are linearly independent over K with dense K-span, on which $\|\mathcal{P}_<(x\otimes y)\|\le\|x\|_1\|y\|_1$ for every total order of $\mathrm{Ball}(E)$. So Proposition 5.7 gives (S^n), (R^n) and polarizability in degree n for every n, and Proposition 3.5(1),(2) gives the goodness statements. The same holds for D.

(3) By Theorem 7.7 with $r=\infty$ and Corollary 8.3, $(c_0(\mathbb{N},K),c_0(\mathbb{N},K))$ is not good for Banach values in degree k, and $c_0(\mathbb{N},K)$ satisfies neither (S^k) nor (R^k) and is not polarizable in degree k. Proposition 6.28(3) transfers the last three statements to the completion $Y_0=c_0(\mathbb{N},\widehat{K})$. If (Y_0,Y_0) were good for Banach values in degree k, then Proposition 6.28(4), applied to the source with the complete target Y_0 and then to the target, would make $(c_0(\mathbb{N},K),c_0(\mathbb{N},K))$ good for Banach values in degree k. By (1) and (2) with $E=D=Y_0$, Y_0 is a closed subspace

of $L^1(\Omega;Y_0)$ and isomorphic to a quotient of $\ell^1(\mathrm{Ball}(Y_0),\widehat{K})$; these two spaces satisfy the three conditions in degree k, and replacing either entry of (Y_0,Y_0) by one of them gives a pair that is good for Banach values in degree k. Since the three conditions and goodness for Banach values are invariant under isomorphisms (Proposition 6.28(1)), the last assertion follows. □

Appendix M. Rigid Banach spaces

This appendix proves Theorems 9.2, 9.3 and 9.4. It transfers the rigid subspace of L_0 of Kalton and Roberts [14, §3]: their quasi-norms become r-norms, their exponents increasing to 1 become exponents $r_i \uparrow \infty$, and forcing goes through L^r readouts.

M.1. **Blocks and the Nakano sum.** Throughout Subsections M.1–M.4, K is a complete nonarchimedean field with nontrivial absolute value that is separable as a metric space, and $\vartheta \in K$ is fixed with $0 < |\vartheta| < 1$.

For a finite-dimensional normed space Y and $1 \le r < \infty$, $L^r((0,1);Y)$ is the completion of the step functions $f\colon (0,1) \to Y$ modulo null functions for $\|f\|_r = (\int_0^1 \|f(t)\|^r\,dt)^{1/r}$, and $1 \otimes y$ is the constant function with value y. By the Riesz–Fischer argument, Y being complete (Lemma A.3), each element is a measurable function, unique up to null sets, with the same formula for the norm. If the norm of Y is nonarchimedean, $\|\cdot\|_r$ is an r-norm (Section 5), since $\|f(t)+g(t)\|^r \le \max(\|f(t)\|,\|g(t)\|)^r \le \|f(t)\|^r + \|g(t)\|^r$. An r-norm is an r'-norm for $1 \le r' \le r$, since $(c^r + c'^r)^{1/r} \le (c^{r'} + c'^{r'})^{1/r'}$ for reals $c, c' \ge 0$. The space K^{1+N} carries the maximum norm, with unit vectors $e_1, \dots, e_{1+N}$.

Let $1 \le r_0 \le r_1 \le \cdots$ be real numbers. For a sequence $a = (a_i)_{i\ge 0}$ of reals $a_i \ge 0$ put

$$\nu(a) = \inf\Big\{\varsigma > 0 : \sum_{i\ge 0} (a_i/\varsigma)^{r_i} \le 1\Big\} \in [0,\infty].$$

Convexity of $\varsigma \mapsto \varsigma^{r_i}$, with the weights $\varsigma/(\varsigma+\varsigma')$ and $\varsigma'/(\varsigma+\varsigma')$, gives $\nu(a+a') \le \nu(a) + \nu(a')$. Moreover ν is positively homogeneous and monotone, $\nu(a) = 0$ only for $a = 0$, $\nu(a) = a_i$ if a_i is the only nonzero entry, and $\nu(a) = \sup_h \nu(a^{\le h})$, where $a^{\le h}$ replaces the entries with index above h by zero. For finitely supported a the sum is continuous and decreasing in ς, so

$$\nu(a) \le \varsigma \iff \textstyle\sum_i (a_i/\varsigma)^{r_i} \le 1 \qquad (\varsigma > 0). \tag{M.1}$$

Given Banach spaces L_i, $i \ge 0$, the *Nakano sum* $\mathcal{Z}$ is the normed space of the sequences $f = (f_i)_{i\ge 0}$, $f_i \in L_i$, with $\|f\| := \nu((\|f_i\|)_{i\ge 0}) < \infty$. It is complete: a Cauchy sequence converges in each L_i, since $\|f_i\| \le \|f\|$, and to its coordinatewise limit in $\mathcal{Z}$, since $\nu(a) = \sup_h \nu(a^{\le h})$. We identify L_i isometrically with the i-th block of $\mathcal{Z}$. The projections Π_i, $\Pi_{<i}$, $\Pi_{\le i}$ and $\Pi_{>i}$ keep the coordinates with index equal to, less than, at most and greater than i and replace the others by zero; they are contractive, since ν is monotone, and $\mathrm{id} = \Pi_{<i} + \Pi_i + \Pi_{>i}$. Let $\mathcal{Z}_{<i}$, $\mathcal{Z}_{\le i}$ and $\mathcal{Z}_{>i}$ be the ranges of the last three.

Lemma M.1. *If the norm of every L_m is an r_m-norm and $i \ge 0$, then $\|f^{(1)} + \cdots + f^{(d)}\|^{r_{i+1}} \le \|f^{(1)}\|^{r_{i+1}} + \cdots + \|f^{(d)}\|^{r_{i+1}}$ for $f^{(1)}, \dots, f^{(d)} \in \mathcal{Z}_{>i}$.*

Proof. By induction it suffices to take $d = 2$, $f = f^{(1)}$ and $g = f^{(2)}$. Put $r = r_{i+1}$ and, for $m > i$, $a_m = \|f_m\|$, $a'_m = \|g_m\|$ and $a''_m = \|f_m + g_m\|$; then $a''^r_m \le a^r_m + a'^r_m$, since the norm of L_m is an r_m-norm and $r_m \ge r$. Let ν' be defined like ν on sequences indexed by $m > i$, with the exponents $r_m/r \ge 1$; like ν, it is monotone and subadditive. With $a^r = (a^r_m)_{m>i}$, the substitution $\varsigma = \varsigma'^r$ gives $\nu'(a^r) = \inf\{\varsigma'^r : \sum_{m>i}(a_m/\varsigma')^{r_m} \le 1\} = \nu(a)^r$. Hence

$$\|f+g\|^r = \nu'(a''^r) \le \nu'(a^r + a'^r) \le \nu'(a^r) + \nu'(a'^r) = \|f\|^r + \|g\|^r.$$

□

M.2. **Operators between blocks.**

Lemma M.2 (compare [14, Lemma 2.2])**.** *Let $1 \le r \le r'$, let Y be a finite-dimensional normed space, $N' \ge 0$ an integer and $h \in L^r((0,1);Y)$ a step function. There is a bounded linear map $\Phi\colon L^{r'}((0,1);K^{1+N'}) \to L^r((0,1);Y)$ with $\Phi(1\otimes e_1) = h$ and $\|\Phi g\|_r \le \|h\|_r \|g\|_{r'}$ for every g.*

Proof. If $h = 0$, take $\Phi = 0$. Otherwise let $e_1^*\colon K^{1+N'} \to K$ be the first coordinate, so $e_1^*(e_1) = 1$ and $|e_1^*(y)| \le \|y\|$. The function $\xi(t) = \|h\|_r^{-r} \int_0^t \|h(t')\|^r\,dt'$ maps $[0,1]$ onto $[0,1]$ and is continuous, nondecreasing and piecewise affine, with positive slope exactly where $h \ne 0$. For a step function g the step function $(\Phi g)(t) = e_1^*(g(\xi(t)))\,h(t)$ depends only on the class of g, since $\Phi g = 0$ where $h = 0$ and preimages of null sets under ξ are null where $h \ne 0$. On the intervals where $h \ne 0$, which ξ maps onto $(0,1)$ up to finitely many points, the substitution $t'' = \xi(t)$ has $\|h(t)\|^r\,dt = \|h\|_r^r\,dt''$. Hence

$$\|\Phi g\|_r^r = \|h\|_r^r \int_0^1 |e_1^*(g(t''))|^r\,dt'' \le \|h\|_r^r \|g\|_r^r \le \|h\|_r^r \|g\|_{r'}^r,$$

the last step by Jensen's inequality on the probability space $(0,1)$, since $r \le r'$. Extend Φ by continuity; then $\Phi(1\otimes e_1) = e_1^*(e_1)\,h = h$. □

From here to the end of Subsection M.4, $N_i \ge i$ are integers, $L_i = L^{r_i}((0,1);K^{1+N_i})$ and $\mathbf{e}_i = 1\otimes e_1 \in L_i$ for $i \ge 0$, and $\mathcal{Z}$ is the Nakano sum of the L_i.

Lemma M.3 (compare [14, Lemma 3.2])**.** *Let $i \ge 1$, and let $f \in \mathcal{Z}_{<i}$ have step functions as coordinates and $\|f\| \le 1$. There is a bounded linear map $\Phi\colon L_i \to \mathcal{Z}_{<i}$ with $\Phi\mathbf{e}_i = f$ and $\|\Phi\| \le 1$.*

Proof. By (M.1), $\sum_{m<i} \|\Pi_m f\|^{r_m} \le 1$. For $m < i$, since $r_m \le r_i$, Lemma M.2 gives $\Phi_m\colon L_i \to L_m$ with $\Phi_m \mathbf{e}_i = \Pi_m f$ and $\|\Phi_m g\| \le \|\Pi_m f\|\|g\|$. Put $\Phi g = \sum_{m<i} \Phi_m g$, with $\Phi_m g$ in block m. Then $\Phi\mathbf{e}_i = f$, and for $g \ne 0$, $\sum_{m<i} (\|\Phi_m g\|/\|g\|)^{r_m} \le \sum_{m<i} \|\Pi_m f\|^{r_m} \le 1$, so $\|\Phi g\| \le \|g\|$ by (M.1). □

M.3. **The inductive choices.** Fix an integer $h_0 \ge 0$ with $\sum_{h\ge 0} |\vartheta|^{h+h_0} \le 1/4$ and put $\zeta_h = \vartheta^{h+h_0}$ for $h \ge 0$. Given reals $\eta_i > 0$ and integers $d_i \ge 1$, let $\mathbf{e}_{i,l} \in L_i$, $1 \le l \le d_i$, equal e_1 on $[(l-1)/d_i, l/d_i)$ and 0 elsewhere; then $\sum_l \mathbf{e}_{i,l} = \mathbf{e}_i$ and $\|\mathbf{e}_{i,l}\| = d_i^{-1/r_i}$. Let $Y_i = \operatorname{span}(e_2, \dots, e_{1+N_i}) \subseteq K^{1+N_i}$ and

$$\mathcal{U}_i = \operatorname{span}\{\mathbf{e}_{i,1}, \dots, \mathbf{e}_{i,d_i}\} + \{1\otimes y : y \in Y_i\} \subseteq L_i, \qquad \mathcal{U}_{<i} = \mathcal{U}_0 + \cdots + \mathcal{U}_{i-1} \subseteq \mathcal{Z}_{<i},$$

finite-dimensional spaces of step functions. For $i \ge 1$ let $C_i \ge 1$ satisfy $\|\sum_l x_l\| \le C_i \max_l \|x_l\|$ for every finite family (x_l) in $\mathcal{U}_{<i}$; it exists because the norm of $\mathcal{U}_{<i}$ is equivalent to the maximum of the coordinates in a basis (Lemma A.3).

Lemma M.4 (compare [14, Lemma 3.1])**.** *Let real numbers $\underline{r}_i \ge 1$, $i \ge 0$, be given. One can choose r_i, η_i and d_i inductively so that for every $i \ge 0$:*

(1) *$r_i \ge \max(r_{i-1}, \underline{r}_i, i)$, where $r_{-1} = 1$;*
(2) *$d_i^{-1/r_i} < \eta_i$, that is, $\|\mathbf{e}_{i,l}\| < \eta_i$ for every l;*
(3) *$(\sum_{l=1}^{d_i} \|\mathbf{e}_{i,l}\|^{r_{i+1}})^{1/r_{i+1}} < \eta_i$;*
(4) *$iC_i\eta_i < |\zeta_i|$ if $i \ge 1$.*

Proof. At stage i the data with indices below i are fixed. If $i \ge 1$, then $(\sum_l \|\mathbf{e}_{i-1,l}\|^r)^{1/r}$, a finite sum, tends to $\max_l \|\mathbf{e}_{i-1,l}\| < \eta_{i-1}$ as $r \to \infty$, by (2) for $i-1$; choose r_i satisfying (1) and so large that (3) holds for $i-1$. This determines $\mathcal{U}_{<i}$ and its norm, which involves only $r_0, \dots, r_{i-1}$, hence C_i. Choose $\eta_i > 0$ satisfying (4), and then d_i satisfying (2). □

M.4. **The spaces $\mathfrak{X}_\alpha$.** Throughout this subsection the data of Subsection M.3 are chosen by Lemma M.4 for some lower bounds $\underline{r}_i$. The norm of each L_m is an r_m-norm, since K^{1+N_m} is nonarchimedean, so Lemma M.1 applies.

Index sets. Partition $\{0,1,2,\dots\} = \bigsqcup_{\alpha\ge1}\mathcal{I}_\alpha$ into infinite sets, enumerate $\mathcal{I}_\alpha$ as $i_\alpha(0) < i_\alpha(1) < \cdots$, and put $\mathcal{J}_{i_\alpha(a)} = \{i_\alpha(2^a(2a'+1)) : a' \ge 0\}$ for $a \ge 0$. Since every positive integer is $2^a(2a'+1)$ for exactly one pair (a,a') and $2^a > a$, the sets $\mathcal{J}_i$, $i \in \mathcal{I}_\alpha$, are pairwise disjoint infinite subsets of $\mathcal{I}_\alpha$ with union $\mathcal{I}_\alpha \setminus \{\min\mathcal{I}_\alpha\}$, and $\min\mathcal{J}_i > i$.

Dense sequences. For $i \in \mathcal{I}_\alpha$, the set $\mathcal{F}_i$ of the $f \in \sum_{m\in\mathcal{I}_\alpha,\, m\le i}\mathcal{U}_m$ with $|\vartheta| < \|f\| \le 1$ contains $\mathbf{e}_i$ and lies in a finite-dimensional space over the separable field K; let $\mathcal{F}_i^0 \ni \mathbf{e}_i$ be a countable dense subset. Through a bijection of $\mathcal{J}_i$ onto $\mathcal{F}_i^0 \times \mathbb{N}$, choose $\gamma_h \in \mathcal{F}_i^0$, $h \in \mathcal{J}_i$, such that every element of $\mathcal{F}_i^0$ equals γ_h for infinitely many h. Since $\gamma_h \in \mathcal{Z}_{\le i} \subseteq \mathcal{Z}_{<h}$ has step functions as coordinates and norm at most 1, Lemma M.3 gives $\Phi_h\colon L_h \to \mathcal{Z}_{<h}$ with $\Phi_h\mathbf{e}_h = \gamma_h$ and $\|\Phi_h\| \le 1$. For $h = \min\mathcal{I}_\alpha$ put $\Phi_h = 0$.

The construction. For $\alpha \ge 1$ put $\Delta_\alpha = \sum_{h\in\mathcal{I}_\alpha}\zeta_h\Phi_h\Pi_h$; let $\mathcal{E}_\alpha$ and $\mathcal{Y}_\alpha$ be the closed spans of $\{\mathbf{e}_i : i \in \mathcal{I}_\alpha\}$ and of $\bigcup_{i\in\mathcal{I}_\alpha}\mathcal{U}_i$; put $\mathcal{Q}_\alpha = (\mathrm{id}-\Delta_\alpha)(\mathcal{E}_\alpha)$ and $\mathfrak{X}_\alpha = \mathcal{Y}_\alpha/\mathcal{Q}_\alpha$; and let $[f]_\alpha$ be the class of $f \in \mathcal{Z}$ in $\mathcal{Z}/\mathcal{Q}_\alpha$. The series Δ_α converges absolutely, with norm at most $\sum_h|\zeta_h| \le 1/4$, so the Neumann series inverts $\mathrm{id}-\Delta_\alpha$, with $\|(\mathrm{id}-\Delta_\alpha)^{-1}\| \le 4/3$. Since Φ_h maps into $\mathcal{Z}_{<h}$, Δ_α maps $\mathcal{Z}_{\le i}$ into $\mathcal{Z}_{<i}$, so $\Pi_i\Delta_\alpha\Pi_{\le i} = \Pi_{>i}\Delta_\alpha\Pi_{\le i} = 0$, for every i. For $h \in \mathcal{I}_\alpha$ the vector $(\mathrm{id}-\Delta_\alpha)\mathbf{e}_h$ is $\mathbf{e}_h - \zeta_h\gamma_h$, or $\mathbf{e}_h$ if $h = \min\mathcal{I}_\alpha$, and lies in $\mathcal{Y}_\alpha$. So $\mathcal{Q}_\alpha$, closed as the image of a closed subspace under an isomorphism, lies in $\mathcal{Y}_\alpha$, and the Banach space $\mathfrak{X}_\alpha$ is the closed subspace of $\mathcal{Z}/\mathcal{Q}_\alpha$ of the classes $[f]_\alpha$, $f \in \mathcal{Y}_\alpha$. For $i \in \mathcal{I}_\alpha$ and $h \in \mathcal{J}_i$,

$$[\mathbf{e}_h]_\alpha = \zeta_h[\gamma_h]_\alpha, \tag{M.2}$$

and $\Pi_i\mathcal{E}_\alpha$ lies in the line $K\mathbf{e}_i$, which is closed by Lemma A.3.

Lemma M.5 (compare [14, Lemma 3.3])**.** *If $i \in \mathcal{I}_\alpha$ and $f \in \mathcal{Z}_{\le i}$, then $\inf_{\lambda\in K}\|\Pi_i f - \lambda\mathbf{e}_i\| \le \frac{4}{3}\|[f]_\alpha\|$.*

Proof. Let $t > \|[f]_\alpha\|$, choose $g \in \mathcal{E}_\alpha$ with $\|f - (\mathrm{id}-\Delta_\alpha)g\| < t$, and let $\Pi_i g = \lambda\mathbf{e}_i$. Since $\Pi_i\Delta_\alpha\Pi_{\le i} = 0$, the vector $\Pi_i f - \lambda\mathbf{e}_i = \Pi_i(f - (\mathrm{id}-\Delta_\alpha)g) - \Pi_i\Delta_\alpha\Pi_{>i}g$ has norm less than $t + \frac14\|\Pi_{>i}g\|$. Since $\Pi_{>i}f = 0$ and $\Pi_{>i}\Delta_\alpha\Pi_{\le i} = 0$, the vector $\Pi_{>i}((\mathrm{id}-\Delta_\alpha)g - f) = \Pi_{>i}g - \Pi_{>i}\Delta_\alpha\Pi_{>i}g$ has norm less than t, so $\|\Pi_{>i}g\| < t + \frac14\|\Pi_{>i}g\|$, that is, $\|\Pi_{>i}g\| < \frac43 t$. Hence $\|\Pi_i f - \lambda\mathbf{e}_i\| < \frac43 t$; let t decrease to $\|[f]_\alpha\|$. □

Lemma M.6 (compare [14, Lemma 3.5])**.** *The set $\bigcup_{i\in\mathcal{I}_\alpha}K[\mathbf{e}_i]_\alpha$ is dense in $\mathfrak{X}_\alpha$.*

Proof. Let $\mathcal{C}$ be the closure of this set; it is closed under scalar multiplication. Let $i \in \mathcal{I}_\alpha$. Each $g \in \mathcal{F}_i^0$ equals γ_h for some $h \in \mathcal{J}_i$, so $[g]_\alpha = \zeta_h^{-1}[\mathbf{e}_h]_\alpha \in \mathcal{C}$ by (M.2); by density, $[f]_\alpha \in \mathcal{C}$ for every $f \in \mathcal{F}_i$. A nonzero f in the span of $\bigcup_{m\in\mathcal{I}_\alpha}\mathcal{U}_m$, which is dense in $\mathcal{Y}_\alpha$, lies in $\sum_{m\in\mathcal{I}_\alpha,\, m\le i}\mathcal{U}_m$ for some $i \in \mathcal{I}_\alpha$, and a power of ϑ times f lies in $\mathcal{F}_i$; so $[f]_\alpha \in \mathcal{C}$. □

Lemma M.7. *Let $i \in \mathcal{I}_\alpha$, and let $\iota_i\colon Y_i \to L^{r_i}((0,1);Y_i)$, $y \mapsto 1\otimes y$, be the constant embedding, which is isometric. There are bounded linear maps $J_i\colon Y_i \to \mathfrak{X}_\alpha$ and $\Xi_i\colon \mathfrak{X}_\alpha \to L^{r_i}((0,1);Y_i)$ with $\|J_i\| \le 1$, $\|\Xi_i\| \le 4/3$ and $\Xi_i J_i = \iota_i$.*

Proof. Put $J_i y = [1\otimes y]_\alpha$, where $1\otimes y \in \mathcal{U}_i \subseteq \mathcal{Y}_\alpha$ has norm $\|y\|$. The pointwise coordinate projection $\mathrm{pr}_{Y_i}\colon L_i \to L^{r_i}((0,1);Y_i)$ of K^{1+N_i} onto Y_i along Ke_1 is contractive and kills $\mathbf{e}_i$. So the map $z \mapsto \mathrm{pr}_{Y_i}(\Pi_i(\mathrm{id}-\Delta_\alpha)^{-1}z)$ on $\mathcal{Z}$, of norm at most $4/3$, vanishes on $\mathcal{Q}_\alpha$, since $(\mathrm{id}-\Delta_\alpha)^{-1}\mathcal{Q}_\alpha = \mathcal{E}_\alpha$ and $\Pi_i\mathcal{E}_\alpha \subseteq K\mathbf{e}_i$; let Ξ_i be the restriction to $\mathfrak{X}_\alpha$ of the induced map on $\mathcal{Z}/\mathcal{Q}_\alpha$. For $y \in Y_i$, $(\mathrm{id}-\Delta_\alpha)^{-1}(1\otimes y) - 1\otimes y = \sum_{c\ge1}\Delta_\alpha^c(1\otimes y)$ lies in $\mathcal{Z}_{<i}$, since Δ_α maps $\mathcal{Z}_{\le i}$ into $\mathcal{Z}_{<i}$. Hence $\Xi_i J_i y = \mathrm{pr}_{Y_i}(1\otimes y) = 1\otimes y$. □

Lemma M.8. *Each $\mathfrak{X}_\alpha$ is nonzero and infinite-dimensional.*

Proof. For $i \in \mathcal{I}_\alpha$ and $y \in Y_i$, Lemma M.7 gives $\|y\| = \|\Xi_i J_i y\| \le \frac{4}{3}\|J_i y\|$. So $\mathfrak{X}_\alpha$ contains the subspaces $J_i(Y_i)$ of dimension $N_i \ge i$, for the infinitely many $i \in \mathcal{I}_\alpha$. □

Proof of Theorem 9.2. The argument follows [14, Theorem 3.6]. Make the construction with any integers $N_i \ge i$ and any lower bounds $\underline{r}_i$; the Banach spaces $\mathfrak{X}_\alpha$ are infinite-dimensional by Lemma M.8. Let $u \in \mathcal{L}(\mathfrak{X}_\alpha, \mathfrak{X}_\beta)$; multiplying u by a power of ϑ, we may assume $\|u\| < 1$.

Step 1 (approximation). Fix $i \in \mathcal{I}_\alpha$; the set $\mathcal{J}_i' = \{h \in \mathcal{J}_i : \gamma_h = \mathbf{e}_i\}$ is infinite. Let $h \in \mathcal{J}_i'$; then $h > i \ge 0$ and $[\mathbf{e}_h]_\alpha = \zeta_h[\mathbf{e}_i]_\alpha$ by (M.2). Since $\|u[\mathbf{e}_{h,l}]_\alpha\| \le \|u\|\|\mathbf{e}_{h,l}\| < \|\mathbf{e}_{h,l}\|$, there are $g_{h,l} \in \mathcal{Y}_\beta$, $1 \le l \le d_h$, with $[g_{h,l}]_\beta = u[\mathbf{e}_{h,l}]_\alpha$ and $\|g_{h,l}\| < \|\mathbf{e}_{h,l}\| < \eta_h$ (Lemma M.4(2)); put $g_h = \sum_l g_{h,l}$, so that $[g_h]_\beta = u[\mathbf{e}_h]_\alpha$. Since Π_m maps $\mathcal{Y}_\beta$ into the closed space $\mathcal{U}_m$ if $m \in \mathcal{I}_\beta$ and to zero otherwise, $\Pi_{<h} g_{h,l} \in \mathcal{U}_{<h}$ and $\|\Pi_{<h} g_h\| \le C_h \max_l \|\Pi_{<h} g_{h,l}\| < C_h \eta_h$. Lemmas M.1 and M.4(3) give $\|\Pi_{>h} g_h\| \le (\sum_l \|\mathbf{e}_{h,l}\|^{r_{h+1}})^{1/r_{h+1}} < \eta_h$. So $\|g_h - \Pi_h g_h\| < (C_h+1)\eta_h \le 2C_h\eta_h < 2|\zeta_h|/h$ by Lemma M.4(4), and

$$\big\|u[\mathbf{e}_i]_\alpha - \zeta_h^{-1}[\Pi_h g_h]_\beta\big\| = |\zeta_h|^{-1}\|[g_h - \Pi_h g_h]_\beta\| < 2/h \qquad (h \in \mathcal{J}_i'). \tag{M.3}$$

Step 2 (the case $\alpha \ne \beta$). Then $h \notin \mathcal{I}_\beta$, so $\Pi_h g_h = 0$ and $\|u[\mathbf{e}_i]_\alpha\| < 2/h$ for infinitely many h. Hence $u[\mathbf{e}_i]_\alpha = 0$ for every $i \in \mathcal{I}_\alpha$, and $u = 0$ by Lemma M.6.

Step 3 (the case $\alpha = \beta$: eigenvectors). Let $h' < h$ be elements of $\mathcal{J}_i'$. The vector $f = \zeta_h^{-1}\Pi_h g_h - \zeta_{h'}^{-1}\Pi_{h'} g_{h'} \in \mathcal{Z}_{\le h}$ has $\Pi_h f = \zeta_h^{-1}\Pi_h g_h$ and, by (M.3) for h and h', $\|[f]_\alpha\| < 2/h + 2/h'$. Lemma M.5 gives $\xi \in K$ with $\|\zeta_h^{-1}\Pi_h g_h - \xi \mathbf{e}_h\| < \frac{4}{3}(2/h + 2/h')$. Passing to classes and using $[\mathbf{e}_h]_\alpha = \zeta_h[\mathbf{e}_i]_\alpha$ and (M.3), we get $\operatorname{dist}(u[\mathbf{e}_i]_\alpha, K[\mathbf{e}_i]_\alpha) < 2/h + \frac{4}{3}(2/h + 2/h')$. Letting $h \to \infty$ and then $h' \to \infty$ gives $u[\mathbf{e}_i]_\alpha = \lambda_i[\mathbf{e}_i]_\alpha$ for some $\lambda_i \in K$, since $K[\mathbf{e}_i]_\alpha$ is closed (Lemma A.3). If $[\mathbf{e}_i]_\alpha \ne 0$, then λ_i is unique and $|\lambda_i| \le \|u\|$.

Step 4 (one eigenvalue). Let $[\mathbf{e}_i]_\alpha$ and $[\mathbf{e}_{i'}]_\alpha$ be linearly independent and $x = [\mathbf{e}_i]_\alpha + [\mathbf{e}_{i'}]_\alpha$. By Lemma M.6, nonzero $y_\ell \in K[\mathbf{e}_{i_\ell}]_\alpha$, $i_\ell \in \mathcal{I}_\alpha$, converge to x. Since $uy_\ell = \lambda_{i_\ell} y_\ell$ and $|\lambda_{i_\ell}| \le \|u\|$, $\|\lambda_{i_\ell} x - ux\| \le |\lambda_{i_\ell}|\|x - y_\ell\| + \|u\|\|y_\ell - x\| \to 0$. So ux lies in the closed line Kx, say $ux = \lambda x$, and comparing coefficients gives $\lambda_i = \lambda = \lambda_{i'}$. Nonzero dependent $[\mathbf{e}_i]_\alpha$, $[\mathbf{e}_{i'}]_\alpha$ have equal eigenvalues directly. So u is a scalar multiple of the identity on $\bigcup_i K[\mathbf{e}_i]_\alpha$, hence on $\mathfrak{X}_\alpha$ by Lemma M.6. □

M.5. **Forcing through an L^r readout.** Throughout this subsection condition (L) holds with $K = \kappa((X))$; so K is complete, and separable since the Laurent polynomials over κ are dense. Let $c_0(N) = K^N$, $N \ge 1$, carry the maximum norm, with unit vectors $e_1, \dots, e_N$, and let $\mu_a x = ax$ be the coordinatewise product of $a \in K^N$ and $x \in K^N$; then $\|\mu_a\| = \|a\|_\infty$. For $r \ge k$ put $G_{N,r} = L^r((0,1); c_0(N))$, with the constant embedding $\iota_{N,r} \colon c_0(N) \to G_{N,r}$, $x \mapsto 1 \otimes x$. A *readout lift* for $\iota_{N,r}$ is a k-linear map $Q \colon (K^N)^k \to \mathcal{L}(\Lambda_\pi^k c_0(N), \widehat{\Lambda}_\pi^k G_{N,r})$ with $Q(a, \dots, a) = \Lambda_\pi^k(\iota_{N,r}\mu_a)$ for $a \in K^N$; its norm is $\|Q\| = \sup\{\|Q(a_1, \dots, a_k)\| : \|a_j\|_\infty \le 1\}$.

Lemma M.9 (Ramsey [20, Theorem B])**.** *For integers $g, h, C \ge 1$ there is an integer $\operatorname{Ram}(g, h, C)$ such that for every coloring of the g-element subsets of $\{1, \dots, \operatorname{Ram}(g, h, C)\}$ with C colors there is an h-element set all of whose g-element subsets have the same color.*

Lemma M.10. *Let Y be a finite-dimensional normed space and $r \ge k$. The map sending $(g_1, \dots, g_k)$ to the function $t \mapsto g_1(t) \wedge \cdots \wedge g_k(t)$ is a k-linear alternating map from $L^r((0,1); Y)^k$ to $L^{r/k}((0,1); \Lambda_\pi^k Y)$ of norm at most 1. It induces a contraction*

$$\operatorname{pw} \colon \widehat{\Lambda}_\pi^k L^r((0,1); Y) \to L^{r/k}((0,1); \Lambda_\pi^k Y)$$

with $\operatorname{pw}(\Lambda_\pi^k \iota(\xi)) = 1 \otimes \xi$ for $\xi \in \Lambda_\pi^k Y$, where ι is the constant embedding of Y into $L^r((0,1); Y)$.

Proof. For step functions on a common partition the value is a step function with $\|g_1(t)\wedge\cdots\wedge g_k(t)\|\le\prod_j\|g_j(t)\|$, so its $L^{r/k}$-norm is at most $\prod_j\|g_j\|_r$, by Hölder's inequality with k exponents equal to k; it vanishes if two of the g_j coincide. Since step functions are dense and $r/k\ge1$, the map extends to a k-linear alternating map of norm at most 1 into the Banach space $L^{r/k}((0,1);\Lambda^k_\pi Y)$, and Proposition 3.1, in its form for complete targets, gives pw. On constant functions $g_j=1\otimes x_j$ the value is $1\otimes(x_1\wedge\cdots\wedge x_k)$. □

Lemma M.11 (test lemma)**.** *For every $m\ge1$ there are an integer $N(m)\ge m$ and a real number $r^*(m)\ge k$, depending only on m, k and $|\kappa|$, such that for all $N\ge N(m)$ and $r\ge r^*(m)$ every readout lift Q for $\iota_{N,r}$ satisfies $\|Q\|\ge m/(2M_\varrho\,k(k-1))$.*

Proof. Step 0 (parameters and reduction). Put $h=4(2m+2+k)+3k$ and $N(m)=\mathrm{Ram}(3k,h,|\kappa|^{(3k)^{3k}})\ge h\ge m$, and choose $r^*(m)\ge k$ with $\binom{N(m)}{h}\,k!\,(k-1)\,2^{-r^*(m)/k}<1$. Let $N\ge N(m)$, $r\ge r^*(m)$, and Q a readout lift for $\iota_{N,r}$. The extension by zero $\mathrm{ext}\colon c_0(N(m))\to c_0(N)$ and the pointwise restriction $\mathrm{res}\colon G_{N,r}\to G_{N(m),r}$ to the first $N(m)$ coordinates are contractive, and $\mathrm{res}\,\iota_{N,r}\,\mu_{\mathrm{ext}\,a}\,\mathrm{ext}=\iota_{N(m),r}\,\mu_a$. So $(a_1,\dots,a_k)\mapsto\widehat\Lambda^k_\pi\,\mathrm{res}\circ Q(\mathrm{ext}\,a_1,\dots,\mathrm{ext}\,a_k)\circ\Lambda^k_\pi\,\mathrm{ext}$ is a readout lift for $\iota_{N(m),r}$ of norm at most $\|Q\|$, and we may assume $N=N(m)$.

Step 1 (readout). For $a\in(K^N)^k$ and $\xi\in\Lambda^k_\pi c_0(N)$, Lemma M.10 gives an element $\widetilde Q(a;\xi)=\mathrm{pw}(Q(a)(\xi))$ of $L^{r/k}((0,1);\Lambda^k_\pi c_0(N))$, k-linear in a, linear in ξ, with $\|\widetilde Q(a;\xi)\|_{r/k}\le\|Q\|\prod_j\|a_j\|_\infty\|\xi\|$. The k-linear maps $a\mapsto Q(a)(x_1\wedge\cdots\wedge x_k)$ and $a\mapsto\iota_{N,r}(a_1x_1)\wedge\cdots\wedge\iota_{N,r}(a_kx_k)$ agree on the diagonal, so the inclusion–exclusion computation in the proof of Proposition B.8(1), followed by pw, gives for all $a_1,\dots,a_k,x_1,\dots,x_k\in K^N$

$$\sum_{\sigma\in S_k}\widetilde Q(a_{\sigma(1)},\dots,a_{\sigma(k)};x_1\wedge\cdots\wedge x_k)=1\otimes\sum_{\sigma\in S_k}(a_{\sigma(1)}x_1)\wedge\cdots\wedge(a_{\sigma(k)}x_k).\tag{M.4}$$

Step 2 (representatives). For $i\in\{1,\dots,N\}^k$ write $e_i=(e_{i_1},\dots,e_{i_k})$. For each i and each strictly increasing $y\in\{1,\dots,N\}^k$ fix an everywhere defined representative $f_{i,y}\colon(0,1)\to\Lambda^k_\pi c_0(N)$ of $\widetilde Q(e_i;e_{y_1}\wedge\cdots\wedge e_{y_k})$ (Subsection M.1); put $f_{i,y}=\mathrm{sgn}(\sigma)f_{i,y\circ\sigma}$ if y has distinct entries and $y\circ\sigma$ is increasing, and $f_{i,y}=0$ if y has a repeated entry. Then $f_{i,y}$ represents $\widetilde Q(e_i;e_{y_1}\wedge\cdots\wedge e_{y_k})$ for every y. For $t\in(0,1)$ and $a,x\in(\mathbb F_p^N)^k$ put

$$\Psi_t(a;x)=\sum_{i,y\in\{1,\dots,N\}^k}\prod_j a_j(i_j)\prod_j x_j(y_j)\,f_{i,y}(t)\in\Lambda^k_\pi c_0(N).$$

Then $t\mapsto\Psi_t(a;x)$ represents $\widetilde Q(a;x_1\wedge\cdots\wedge x_k)$, and each Ψ_t is $\mathbb F_p$-multilinear and alternating in x: if $x_\gamma=x_{\gamma'}$ with $\gamma\ne\gamma'$, the terms of y and of $y\circ(\gamma\ \gamma')$ have equal weights and $f_{i,y\circ(\gamma\,\gamma')}=-f_{i,y}$, and the terms with $y_\gamma=y_{\gamma'}$ vanish.

Step 3 (test inputs and the point t_0). For an h-element set $\mathcal H\subseteq\{1,\dots,N\}$ let $\mathcal H'$ consist of its $h-3k$ smallest elements; clusters of $\mathcal H'$ are sets of four consecutive elements of $\mathcal H'$, as in Appendix B. Let $(a^{\mathcal H};x^{\mathcal H})$ be the inputs of the proof of Lemma B.13 for the first k clusters of $\mathcal H'$; by (M.4) and Step 2, the set $O_{\mathcal H}$ of the t with $\sum_\sigma\Psi_t(a^{\mathcal H}_{\sigma(1)},\dots,a^{\mathcal H}_{\sigma(k)};x^{\mathcal H})\ne\sum_\sigma(a^{\mathcal H}_{\sigma(1)}x^{\mathcal H}_1)\wedge\cdots\wedge(a^{\mathcal H}_{\sigma(k)}x^{\mathcal H}_k)$ is null. For $\tau\in S_k$ and $1\le\ell<k$, let $(a^{\mathcal H,\tau,\ell};x^{\mathcal H,\tau,\ell})$, with entries in $\{0,\pm1\}$, be the inputs of the proof of Lemma B.12 for τ and ℓ, with $m+1$ pairs of clusters, all clusters taken consecutively from the smallest element of $\mathcal H'$; there are enough, since $4(2m+k)\le h-3k$. For these inputs $(a;x)$, the set $O_{\mathcal H,\tau,\ell}$ of the t with $\|\Psi_t(a;x)\|>2\|\widetilde Q(a;x_1\wedge\cdots\wedge x_k)\|_{r/k}$ has measure at most $2^{-r/k}$, by Chebyshev's inequality for $t\mapsto\|\Psi_t(a;x)\|^{r/k}$, and is null if $\widetilde Q(a;x_1\wedge\cdots\wedge x_k)=0$. There are at most $\binom{N}{h}\,k!\,(k-1)$ sets $O_{\mathcal H,\tau,\ell}$, so the union of all of them

and of the $O_{\mathcal{H}}$ has measure at most $\binom{N}{h}\, k!\,(k-1)\,2^{-r/k} < 1$. Fix $t_0 \in (0,1)$ outside this union, and write $\Psi = \Psi_{t_0}$.

Step 4 (Ramsey at t_0). Let Ω be the determinant array of §B.1 on $\Lambda^k_\pi c_0(N)$, with $S = \{1,\dots,N\}$, and put $T(i;y;c) = \operatorname{coeff}_0 \Omega_{\Psi(e_i;e_y)}(c) \in \kappa$ for $i,y,c \in \{1,\dots,N\}^k$. Color each $3k$-element set $\{d_1 < \dots < d_{3k}\} \subseteq \{1,\dots,N\}$ by the family $(T(d_{\eta(1)},\dots,d_{\eta(3k)}))_\eta$ over all maps $\eta\colon \{1,\dots,3k\} \to \{1,\dots,3k\}$, with at most $|\kappa|^{(3k)^{3k}}$ colors. Lemma M.9 gives a homogeneous h-element set $\mathcal{H}$. Padding with the $3k$ elements of $\mathcal{H} \setminus \mathcal{H}'$, which lie above $\mathcal{H}'$, as in the proof of Lemma B.10, shows that $T(i;y;c)$ depends only on the pattern of (i,y,c) when all its entries lie in $\mathcal{H}'$.

Step 5 (the argument of Appendix B at t_0). For inputs supported in $\mathcal{H}'$ with entries in $\{0,\pm1\}$, $\operatorname{coeff}_0 \Omega_{\Psi(a;x)}(c) = \sum_{i,y \in \mathcal{H}'^k} \prod_j a_j(i_j) \prod_j x_j(y_j)\, T(i;y;c)$, as in (B.1), since Ψ is $\mathbb{F}_p$-multilinear, Ω is linear and coeff_0 is $\mathbb{F}_p$-linear. Define the cluster values $\chi(\tau) \in \kappa$ for clusters of $\mathcal{H}'$ as in Appendix B, with $\operatorname{coeff}_0 \Omega_{\Psi(\cdot)}$ in place of $\Omega_{\Psi(\cdot)}$. The proofs of Lemmas B.11 and B.13 use only this expansion, pattern homogeneity of T, that clusters are intervals of the set on which T is homogeneous, (W), alternation of Ψ in its vector slots and of Ω_z in its arguments, and (Pol1) at $(a^{\mathcal{H}}; x^{\mathcal{H}})$; the last holds since $t_0 \notin O_{\mathcal{H}}$ and the determinants on the right of (Pol1) lie in $\mathbb{F}_p$. So both lemmas hold, and $\sum_\tau \chi(\tau) = 1$ in κ. Since $k! = 0$ in κ, χ is not constant; since exchanges of adjacent ranks generate S_k, some τ and ℓ have $\delta = \chi(\tau) - \chi(\tau') \neq 0$, where $\tau' = (\ell\ \ell{+}1) \circ \tau$.

Let $(a;x) = (a^{\mathcal{H},\tau,\ell}; x^{\mathcal{H},\tau,\ell})$, let $\gamma_1 = \tau^{-1}(\ell)$ and $\gamma_2 = \tau^{-1}(\ell+1)$ be the exchanged slots, let I_1 and I_2 be the sets of first elements of the $m+1$ clusters given to γ_1 and γ_2 in the proof of Lemma B.12, and let c_γ be the first element of the cluster given to any other slot γ. By Lemma B.11, as in that proof, the $(m+1)\times(m+1)$ matrix St of the values $\operatorname{coeff}_0 \Omega_{\Psi(a;x)}(c)$, with an element of I_1 in slot γ_1, an element of I_2 in slot γ_2 and c_γ in each other slot γ of c, is $\chi(\tau')\mathbf{1}\mathbf{1}^{\mathsf{T}} + \delta U$ with U upper unitriangular, so $\operatorname{rank} \mathrm{St} \geq m$. The extension by zero $\operatorname{ext}_\infty\colon c_0(N) \to c_0(\mathbb{N},K)$ makes $\Lambda^k_\pi \operatorname{ext}_\infty$ a contraction that keeps the determinant array on $\{1,\dots,N\}^k$. Let Υ be the slice of Lemma K.3 for $c_0(\mathbb{N},K)$, the slots γ_1,γ_2 and the entries c_γ, and obtain z from $\Upsilon \Lambda^k_\pi \operatorname{ext}_\infty(\Psi(a;x))$ by restricting its two tensor factors to the coordinates in I_1 and in I_2. Then $\operatorname{coeff}_0 z = \mathrm{St}$ and $\|z\| \leq k(k-1)\|\Psi(a;x)\|$, so Lemma K.2(2) gives $m \leq \operatorname{rank} \mathrm{St} \leq M_\varrho k(k-1)\|\Psi(a;x)\|$. Since $t_0 \notin O_{\mathcal{H},\tau,\ell}$ and the entries of a and x lie in $\{0,\pm1\}$, $\|\Psi(a;x)\| \leq 2\|\widetilde{Q}(a;x_1 \wedge \dots \wedge x_k)\|_{r/k} \leq 2\|Q\|$, and $m \leq 2M_\varrho k(k-1)\|Q\|$. □

Corollary M.12. *Let $m \geq 1$, $N \geq N(m)$ and $r \geq r^*(m)$. Let E be a normed space, and $J\colon c_0(N) \to E$ and $\Xi\colon E \to G_{N,r}$ bounded linear maps with $\Xi J = \iota_{N,r}$. Then every s as in (S^k_E) satisfies $\|s\| \geq m/(2M_\varrho k(k-1)(\|\Xi\|\|J\|)^k)$, and every ρ as in (R^k_E) satisfies $\|\rho\| \geq m/(2M_\varrho k(k-1)k!\,(\|\Xi\|\|J\|)^k)$.*

Proof. Let v_a be pointwise multiplication by $a \in K^N$ on $G_{N,r}$; then $\|v_a\| \leq \|a\|_\infty$ and $v_a \iota_{N,r} = \iota_{N,r}\mu_a$. Given s, the k-linear map $Q(a_1,\dots,a_k) = \widehat{\wedge}^k_{G_{N,r}} \circ ((v_{a_1}\Xi) \otimes \dots \otimes (v_{a_k}\Xi)) \circ s \circ \Lambda^k_\pi J$ has norm at most $(\|\Xi\|\|J\|)^k\|s\|$ and, by naturality of $\wedge^k$, diagonal $\widehat{\Lambda}^k_\pi(v_a\Xi) \circ \widehat{\wedge}^k_E \circ s \circ \Lambda^k_\pi J = \Lambda^k_\pi(v_a \Xi J) = \Lambda^k_\pi(\iota_{N,r}\mu_a)$. Given ρ, the k-linear map $Q(a_1,\dots,a_k) = \widehat{\Lambda}^k_\pi \Xi \circ \rho \circ ((J\mu_{a_1}) \otimes \dots \otimes (J\mu_{a_k})) \circ \operatorname{alt}^k_{c_0(N)}$ has norm at most $k!\,(\|\Xi\|\|J\|)^k\|\rho\|$, since $\|\operatorname{alt}^k_{c_0(N)}\| \leq k!$, and, by naturality of alt^k, diagonal $\widehat{\Lambda}^k_\pi \Xi \circ \rho \circ \operatorname{alt}^k_E \circ \Lambda^k_\pi(J\mu_a) = \Lambda^k_\pi(\Xi J \mu_a) = \Lambda^k_\pi(\iota_{N,r}\mu_a)$. Lemma M.11 applies to both. □

M.6. **Proofs of the goodness theorems.** Throughout this subsection κ, K and ϱ are as in Subsection M.5. For $k' \geq p$ and $m \geq 1$ let $N_{k'}(m)$ and $r^*_{k'}(m)$ be the numbers of Lemma M.11 in degree k', and make the construction of Subsections M.3 and M.4 with

$$N_i = \max_{p \leq k' \leq p+i} N_{k'}(i+1), \qquad \underline{r}_i = \max_{p \leq k' \leq p+i} r^*_{k'}(i+1) \qquad (i \geq 0); \tag{M.5}$$

then $N_i \geq i+1$, as Subsection M.2 requires.

Proof of Theorem 9.3. Step 1 (no splitting). Let $\alpha \geq 1$, $k \geq p$, and $i \in \mathcal{I}_\alpha$ with $i \geq k-p$. Identifying Y_i with $c_0(N_i)$ through the basis $e_2, \dots, e_{1+N_i}$ turns the map ι_i of Lemma M.7 into ι_{N_i,r_i}. By (M.5) and Lemma M.4(1), $N_i \geq N_k(i+1)$ and $r_i \geq r_k^*(i+1)$. Corollary M.12, with $m = i+1$, $J = J_i$ and $\Xi = \Xi_i$, shows that every s as in $(\mathrm{S}^k_{\mathfrak{X}_\alpha})$ and every ρ as in $(\mathrm{R}^k_{\mathfrak{X}_\alpha})$ have norm at least $(3/4)^k(i+1)/(2M_\varrho k(k-1)k!)$. Since $\mathcal{I}_\alpha$ is infinite, neither $(\mathrm{S}^k_{\mathfrak{X}_\alpha})$ nor $(\mathrm{R}^k_{\mathfrak{X}_\alpha})$ holds.

Step 2 (the four items). Let $\mathfrak{X} = \mathfrak{X}_1$. Item (1) is Theorem 9.2. Since $\mathrm{id}_{\mathfrak{X}} \neq 0$ (Lemma M.8), Proposition 9.1(1) with $E = D = \mathfrak{X}$ and $u_0 = \mathrm{id}_{\mathfrak{X}}$ makes $(\mathfrak{X}, \mathfrak{X})$ good in every degree, hence good for Banach values by Lemma 3.4; this is (2). Item (3) is Step 1 for $\alpha = 1$. Since $\mathfrak{X}$ is complete, by Proposition 9.1(2) a splitting decomposition of degree k would make $\mathrm{id}_{\mathfrak{X}}$ S^k-split or R^k-split, that is, give $(\mathrm{S}^k_{\mathfrak{X}})$ or $(\mathrm{R}^k_{\mathfrak{X}})$, contrary to (3). This proves (4). □

Proof of Theorem 9.4. The inclusion $\mathrm{in}_\alpha \colon \mathfrak{X}_\alpha \to E$ of the α-th summand and the coordinate projection $\mathrm{pr}_\alpha \colon E \to \mathfrak{X}_\alpha$ are contractive, and $x = \sum_\alpha \mathrm{in}_\alpha \mathrm{pr}_\alpha x$ for $x \in E$.

Step 1 (item (1)). Let $u \in \mathcal{L}(E,E)$. By Theorem 9.2, $\mathrm{pr}_\beta u \,\mathrm{in}_\alpha = 0$ for $\beta \neq \alpha$ and $\mathrm{pr}_\alpha u \,\mathrm{in}_\alpha = \lambda_\alpha \mathrm{id}_{\mathfrak{X}_\alpha}$ for some $\lambda_\alpha \in K$, with $|\lambda_\alpha| \leq \|u\|$ since $\mathfrak{X}_\alpha \neq 0$ (Lemma M.8). Hence $u \,\mathrm{in}_\alpha = \sum_\beta \mathrm{in}_\beta \mathrm{pr}_\beta u \,\mathrm{in}_\alpha = \lambda_\alpha \mathrm{in}_\alpha$ and $ux = \sum_\alpha \lambda_\alpha \mathrm{in}_\alpha \mathrm{pr}_\alpha x = u_\lambda x$. Conversely $\|u_\lambda x\| = \sum_\alpha |\lambda_\alpha| \|\mathrm{pr}_\alpha x\| \leq \sup_\alpha |\lambda_\alpha| \|x\|$, with $\|u_\lambda x\| = |\lambda_\alpha| \|x\|$ on the summand $\mathfrak{X}_\alpha \neq 0$. So $\lambda \mapsto u_\lambda$ is an isometric isomorphism onto $\mathcal{L}(E,E)$ from the space ℓ^∞ of bounded sequences $(\lambda_\alpha)_{\alpha \geq 1}$ in K.

Step 2 (item (2)). Let $k \geq 1$. Proposition 6.27 applies with $I = \{1, 2, \dots\}$ in its natural order, $E_\alpha = D_\alpha = \mathfrak{X}_\alpha$, $D = E$, inclusions in_α, block projections pr_α, $J'_\alpha = \mathrm{in}_\alpha$, $\mathcal{U} = \mathcal{L}(E,E)$, $u_\alpha = \mathrm{pr}_\alpha u \,\mathrm{in}_\alpha$ and $c = C = 1$: $\|u_\alpha\| \leq \|u\|$ and $ux = \sum_\alpha \mathrm{in}_\alpha u_\alpha \mathrm{pr}_\alpha x$ by Step 1, and for $n \leq k$, since $\mathcal{L}(\mathfrak{X}_\alpha, \mathfrak{X}_\alpha) = K\mathrm{id}_{\mathfrak{X}_\alpha}$ with $\mathrm{id}_{\mathfrak{X}_\alpha} \neq 0$, Proposition 9.1(1) with $u_0 = \mathrm{id}_{\mathfrak{X}_\alpha}$ gives the lift $Q^\alpha_n(\theta_1 \mathrm{id}_{\mathfrak{X}_\alpha}, \dots, \theta_n \mathrm{id}_{\mathfrak{X}_\alpha}) = \theta_1 \cdots \theta_n \Lambda^n_\pi \mathrm{id}_{\mathfrak{X}_\alpha}$, $\theta_j \in K$, which satisfies $Q^\alpha_n(v, \dots, v) = \Lambda^n_\pi v$ and $\|Q^\alpha_n\| \leq 1$. So (E,E) is good for Banach values in degree k.

Step 3 (item (3)). Suppose that (E,E) has a generalized decomposition of degree $k \geq p$ (Definition 6.17) with pieces $(G_l, G'_l, w_l, w'_l, \varepsilon_l)$, so that $u = \sum_l w'_l \varepsilon_l(u) w_l$ for $u \in \mathcal{L}(E,E)$; the maps w'_l take values in the complete space E. Fix α and u; the compression $\mathrm{pr}_\alpha w'_l \varepsilon_l(u) w_l \mathrm{in}_\alpha$ lies in $\mathcal{L}(\mathfrak{X}_\alpha, \mathfrak{X}_\alpha) = K\mathrm{id}_{\mathfrak{X}_\alpha}$. For a piece of source or target type, and for the single piece of a relative splitting, w_l is S^k-split or w'_l is R^k-split, so the compression is S^k-split or R^k-split by the ideal property (Proposition 3.13(2)). If it were a nonzero multiple of $\mathrm{id}_{\mathfrak{X}_\alpha}$, the ideal property would make $\mathrm{id}_{\mathfrak{X}_\alpha}$ S^k-split or R^k-split, contrary to Step 1 of the proof of Theorem 9.3; so it is zero.

For a parameter piece, $\varepsilon_l = \varepsilon'_l \circ \psi_l$ with bounded linear $\psi_l \colon \mathcal{L}(E,E) \to P_l$ and $\varepsilon'_l \colon P_l \to \mathcal{L}(G_l, G'_l)$, where P_l is polarizable in degrees $\leq k$. The linear maps $g_{l,\alpha} \colon P_l \to K$ given by $\mathrm{pr}_\alpha w'_l \varepsilon'_l(y) w_l \mathrm{in}_\alpha = g_{l,\alpha}(y) \mathrm{id}_{\mathfrak{X}_\alpha}$ satisfy $|g_{l,\alpha}(y)| \leq \|w'_l\| \|\varepsilon'_l\| \|w_l\| \|y\|$ for every α, since $\mathfrak{X}_\alpha \neq 0$. Compressing $u_\lambda = \sum_l w'_l \varepsilon_l(u_\lambda) w_l$ to the summand α gives $\lambda_\alpha = \sum_l g_{l,\alpha}(\psi_l(u_\lambda))$ for every $\lambda \in \ell^\infty$ and every α, the sum over the parameter pieces. Let P be the direct sum of their spaces P_l with the maximum norm. By Step 1 and the uniform bound, $\mathcal{A} \colon \ell^\infty \to P$, $\mathcal{A}\lambda = (\psi_l(u_\lambda))_l$, and $\mathcal{A}' \colon P \to \ell^\infty$, $\mathcal{A}'(y_l)_l = (\sum_l g_{l,\alpha}(y_l))_{\alpha \geq 1}$, are bounded linear, and $\mathcal{A}'\mathcal{A} = \mathrm{id}_{\ell^\infty}$.

The subspace $c_0 \subseteq \ell^\infty$ of null sequences, a copy of $c_0(\mathbb{N}, K)$, is complete with distances in $\varrho^{\mathbb{Z}} \cup \{0\}$, hence spherically complete by Lemma D.1. The extension argument behind Ingleton's Theorem D.3, in the form for spherically complete values given in the proof of Theorem 5.2, extends id_{c_0} to a contractive linear map $\mathcal{B} \colon \ell^\infty \to c_0$ on the nonarchimedean space ℓ^∞. Then $\mathcal{B}\mathcal{A}' \circ \mathcal{A}|_{c_0} = \mathrm{id}_{c_0}$, so c_0 is a retract of P, which is polarizable in degrees $\leq k$ by Proposition 6.18(2). By Proposition 6.18(3), c_0 is polarizable in degree k, contrary to Corollary 8.3, since K satisfies (L) and $k \geq p$. □

Appendix N. Properties of the obstruction class

This appendix contains the proofs to which Sections 4, 6 and 7 refer. Throughout, K is a nontrivially normed field, hence infinite.

N.1. **Pullbacks and morphisms.** Exact sequences, equivalences, classes and pushouts are as in Section 3.4, morphisms and pullbacks as in Section 6.3. Throughout this subsection $\mathcal{Y}\colon 0 \to Y_1 \xrightarrow{i} Y_2 \xrightarrow{q} Y_3 \to 0$ and $\mathcal{Y}'\colon 0 \to Y_1' \xrightarrow{i'} Y_2' \xrightarrow{q'} Y_3' \to 0$ are exact, with a constant C for $\mathcal{Y}$ as in Section 3.4.

Lemma N.1. *For a bounded linear map $\gamma\colon Y' \to Y_3$ from a Banach space Y', $\gamma^*\mathcal{Y}$ is exact, its class depends only on $[\mathcal{Y}]$, and $\gamma^*[\mathcal{Y}] = 0$ if and only if $\gamma = q \circ L$ for a bounded linear $L\colon Y' \to Y_2$.*

Proof. The space Y_γ is the kernel of $(y, y') \mapsto qy - \gamma y'$, hence complete. The first map has image $\{(y, 0) : qy = 0\}$, the kernel of the second, and $\|(iy_1, 0)\| \geq \|y_1\|/C$. For $y' \in Y'$ some y with $qy = \gamma y'$ has $\|(y, y')\| \leq (1 + C\|\gamma\|)\|y'\|$. An equivalence T of $\mathcal{Y}$ with an exact sequence $\mathcal{Y}''$ induces the equivalence $(y, y') \mapsto (Ty, y')$ of the pullbacks. The splittings of $\gamma^*\mathcal{Y}$, the bounded linear right inverses of $(y, y') \mapsto y'$, are the maps $y' \mapsto (Ly', y')$ with L bounded linear and $qL = \gamma$. □

Lemma N.2. *For a morphism $(\alpha_1, \alpha_2, \alpha_3)\colon \mathcal{Y} \to \mathcal{Y}'$, $\alpha_{1*}[\mathcal{Y}] = \alpha_3^*[\mathcal{Y}']$ in $\operatorname{Ext}(Y_3, Y_1')$.*

Proof. The map $T[(y_1', y)] = (i'y_1' + \alpha_2 y, qy)$ from the middle space of $\alpha_{1*}\mathcal{Y}$, the quotient of $Y_1' \oplus Y_2$ by the vectors $(\alpha_1 y_1, -iy_1)$, to that of $\alpha_3^*\mathcal{Y}'$ is well defined, since $i'\alpha_1 = \alpha_2 i$ and $qi = 0$, bounded, and takes values in the pullback, since $q'(i'y_1' + \alpha_2 y) = \alpha_3 qy$. It sends $[(y_1', 0)]$ to $(i'y_1', 0)$ and has second coordinate qy, so it is an equivalence (Lemmas 3.6(3) and N.1). □

Corollary N.3. (a) *If $(\alpha_1, \alpha_2, \mathrm{id})\colon \mathcal{Y} \to \mathcal{Y}'$ is a morphism, then $[\mathcal{Y}'] = \alpha_{1*}[\mathcal{Y}]$. If $\mathcal{Y}''\colon 0 \to Y_1 \xrightarrow{i''} Y_2'' \to Y_3'' \to 0$ is exact and $(\mathrm{id}, \alpha_2, \gamma)\colon \mathcal{Y}'' \to \mathcal{Y}$ is a morphism, then $[\mathcal{Y}''] = \gamma^*[\mathcal{Y}]$.*

(b) *For bounded linear maps $\alpha, \alpha', \gamma, \gamma'$ of Banach spaces for which the expressions are defined, $(\alpha'\alpha)_* = \alpha'_*\alpha_*$, $(\gamma\gamma')^* = \gamma'^*\gamma^*$, $\mathrm{id}_* = \mathrm{id}^* = \mathrm{id}$ and $\alpha_*\gamma^* = \gamma^*\alpha_*$.*

(c) *For a bounded linear $\alpha\colon Y_1 \to Y'$ into a Banach space, $\alpha_* 0 = 0$, and $(c\alpha)_*[\mathcal{Y}] = 0$ if and only if $\alpha_*[\mathcal{Y}] = 0$, for $c \in K$, $c \neq 0$.*

Proof. (a) Lemma N.2 gives $\alpha_{1*}[\mathcal{Y}] = \mathrm{id}^*[\mathcal{Y}']$ and $\mathrm{id}_*[\mathcal{Y}''] = \gamma^*[\mathcal{Y}]$, and $(y, y_3) \mapsto y$ and $[(y_1, y)] \mapsto i''y_1 + y$ are equivalences of $\mathrm{id}^*\mathcal{Y}'$ with $\mathcal{Y}'$ and of $\mathrm{id}_*\mathcal{Y}''$ with $\mathcal{Y}''$. (b) The triples $(\alpha, y \mapsto [(0, y)], \mathrm{id})\colon \mathcal{Y} \to \alpha_*\mathcal{Y}$ and $(\mathrm{id}, (y, y') \mapsto y, \gamma)\colon \gamma^*\mathcal{Y} \to \mathcal{Y}$ are morphisms, since $[(0, iy_1)] = [(\alpha y_1, 0)]$. Composites of such morphisms, the identity morphism and (a) give the first three identities; the composite $\gamma^*\mathcal{Y} \to \mathcal{Y} \to \alpha_*\mathcal{Y}$ and Lemma N.2 give the last. (c) By Lemma 3.6(3), $\alpha_*[\mathcal{Y}] = 0$ if and only if $\alpha = \tilde{\alpha} i$ for a bounded linear $\tilde{\alpha}$, that is, $c\alpha = (c\tilde{\alpha})i$. If $[\mathcal{Y}] = 0$, then i has a bounded linear left inverse λ (Lemma 3.6(2)), and $\alpha = (\alpha\lambda)i$ gives $\alpha_* 0 = 0$. □

N.2. **Permutation groups.** The notation is that of Section 4, before Theorem 4.3 and in Proposition 4.5.

Proof of Theorem 4.3. Step 1 (notation). For a *tuple* $t\colon [k] \to I$ put $e_t = e_{t(1)} \otimes \cdots \otimes e_{t(k)}$; these form a basis of $\otimes_K^k V$, and $h^{\otimes k} e_t = e_{h \circ t}$. The *type* of t is the k-multiset $i \mapsto |t^{-1}(i)|$; if t has type $\mathbf{m}$, then $h \circ t$ has type $h\mathbf{m}$. For a k-subset T let $\mathrm{Bij}(T)$ be the set of bijections $[k] \to T$. Then $\wedge^k e_f = \operatorname{sgn}(f_T^{-1} f)\eta_T$ for $f \in \mathrm{Bij}(T)$, $\wedge^k e_t = 0$ for t not injective, the η_T form a

basis of $\Lambda^k_K V$, and $\chi_T(h) = \operatorname{sgn}(f_T^{-1} h f_T)$ depends only on the permutation that h induces on T, so χ_T is also a homomorphism on $\bar{\mathsf{H}}_T$. Since $f_T^{-1} h f = (f_T^{-1} h f_T)(f_T^{-1} f)$ in S_k,

$$\operatorname{sgn}(f_T^{-1} h f) = \chi_T(h) \operatorname{sgn}(f_T^{-1} f) \qquad (h \in \mathsf{H}_T,\ f \in \operatorname{Bij}(T)). \tag{N.1}$$

The groups $\bar{\mathsf{H}}_T$ and $\bar{\mathsf{H}}_{\mathbf{m}}$ act freely, by $f \mapsto h \circ f$, on $\operatorname{Bij}(T)$ and on the tuples of type $\mathbf{m}$. This gives the *orbit count*: if $c_f \in \widehat{K}$ satisfy $c_{h\circ f} = \chi_T(h) c_f$ for $h \in \mathsf{H}_T$, then by (N.1) each orbit contributes $|\bar{\mathsf{H}}_T| c_f \operatorname{sgn}(f_T^{-1} f)$ to $\sum_{f \in \operatorname{Bij}(T)} c_f \operatorname{sgn}(f_T^{-1} f)$, which therefore lies in $|\bar{\mathsf{H}}_T| \widehat{K}$; if $c_t \in \widehat{K}$ satisfy $c_{h\circ t} = c_t$ for the tuples t of type $\mathbf{m}$ and $h \in \mathsf{H}_{\mathbf{m}}$, then $\sum_t c_t \in |\bar{\mathsf{H}}_{\mathbf{m}}| \widehat{K}$. Let $O_{\mathbf{m}}$ be the sum of the e_t over the tuples of type $\mathbf{m}$, as in Proposition 5.7. Then $h^{\otimes k} O_{\mathbf{m}} = O_{h\mathbf{m}}$, and the $O_{\mathbf{m}}$ form a basis of $\Gamma^k_K V$, the space of tensors fixed by every τ_σ, since $\tau_\sigma e_t = e_{t\circ\sigma^{-1}}$.

Step 2 (necessity). Assume the hypotheses of (4). For a tuple t let e_t^* be the bounded functional on $\widehat{\otimes}^k_\pi E$ linearizing $(x_j) \mapsto \prod_j e^*_{t(j)}(x_j)$; then $e_t^*(e_{t'}) = \delta_{tt'}$, and $e_t^* \circ h^{\otimes k} = e^*_{h^{-1}\circ t}$ by (4.3). Let ℓ_T be the bounded functional on $\widehat{\Lambda}^k_\pi E$ linearizing $(x_{j'}) \mapsto \det(e^*_{f_T(j)}(x_{j'}))_{j,j'}$ (Proposition 3.1). Then $\ell_T(\eta_T) = 1$; $\ell_T \circ \widehat{\Lambda}^k_\pi h = \chi_T(h) \ell_T$ for $h \in \mathsf{H}_T$, since (4.3) permutes the rows of the determinant by $f_T^{-1} h^{-1} f_T$; and $\ell_T \circ \widehat{\wedge}^k_E = \sum_{f \in \operatorname{Bij}(T)} \operatorname{sgn}(f_T^{-1} f) e_f^*$ by the Leibniz formula.

Let $|\bar{\mathsf{H}}_T| = 0$ in K. For s as in (4), $z = s(\eta_T)$ satisfies $z = \chi_T(h) h^{\otimes k} z$ for $h \in \mathsf{H}_T$, so $c_f = e_f^*(z)$ satisfies $c_{h\circ f} = \chi_T(h) c_f$, and $1 = \ell_T(\widehat{\wedge}^k_E z) = \sum_f c_f \operatorname{sgn}(f_T^{-1} f) = 0$ by the orbit count. For ρ as in (4), $a_f = \ell_T(\rho(e_f))$ satisfies $a_{h\circ f} = \chi_T(h) a_f$, and $\operatorname{alt}^k_E \eta_T = \sum_f \operatorname{sgn}(f_T^{-1} f) e_f$ gives $1 = \ell_T(\rho \operatorname{alt}^k_E \eta_T) = \sum_f \operatorname{sgn}(f_T^{-1} f) a_f = 0$.

Let $|\bar{\mathsf{H}}_{\mathbf{m}}| = 0$ in K, R be as in (4), and $t_{\mathbf{m}}$ a tuple of type $\mathbf{m}$. As $e_t^*(x^{\otimes k}) = \prod_j e^*_{t(j)}(x)$ depends only on the type of t, every e_t^* with t of type $\mathbf{m}$ agrees on $\widehat{\Gamma}^k_\pi E$ with $\varphi_{\mathbf{m}} = e^*_{t_{\mathbf{m}}}$. So for $h \in \mathsf{H}_{\mathbf{m}}$, $\varphi_{\mathbf{m}} \circ h^{\otimes k} = \varphi_{\mathbf{m}}$ on $\widehat{\Gamma}^k_\pi E$, and $c_t = \varphi_{\mathbf{m}}(R e_t)$ satisfies $c_{h\circ t} = c_t$. By Step 1, $O_{\mathbf{m}}$ is a combination of pure powers of vectors in the span of the e_i, so $O_{\mathbf{m}} \in \widehat{\Gamma}^k_\pi E$, and $1 = \varphi_{\mathbf{m}}(R O_{\mathbf{m}}) = \sum_t c_t = 0$ by the orbit count. With the algebraic powers of V, its coordinate functionals and the determinant functional on $\Lambda^k_K V$, the same argument gives the necessity in (1)–(3).

Step 3 (sufficiency in (1)). Let $|\bar{\mathsf{H}}_T| \neq 0$ in K for all T, and let $\mathcal{T}$ be a set of representatives of the H-orbits of k-subsets. For $T \in \mathcal{T}$ put $z_T = |\bar{\mathsf{H}}_T|^{-1} \sum_{g \in \bar{\mathsf{H}}_T} \chi_T(g) e_{g\circ f_T} \in U_T$; then $\wedge^k z_T = \eta_T$ by (N.1), and $h'^{\otimes k} z_T = \chi_T(h') z_T$ for $h' \in \mathsf{H}_T$ by reindexing. The vectors $\Lambda^k_K g\, \eta_T = \pm \eta_{gT}$, one for each k-subset gT, form a basis of $\Lambda^k_K V$; put $s(\Lambda^k_K g\, \eta_T) = g^{\otimes k} z_T$. If $gT = g'T$, then $h' = g^{-1} g' \in \mathsf{H}_T$, and $\Lambda^k_K g' \eta_T$ and $g'^{\otimes k} z_T$ are $\chi_T(h')$ times $\Lambda^k_K g\, \eta_T$ and $g^{\otimes k} z_T$; so s is well defined. It is H-equivariant by construction and a section, since $\wedge^k g^{\otimes k} z_T = \Lambda^k_K g \wedge^k z_T$. For $\lambda \in (K^\times)^I$ let $d_\lambda e_i = \lambda_i e_i$. Then $d_\lambda^{\otimes k}$ and $\Lambda^k_K d_\lambda$ act on $U_{T'}$ and on $K\eta_{T'}$ by $\prod_{i \in T'} \lambda_i$, and $s(K\eta_{T'}) \subseteq U_{T'}$, so s commutes with d_λ.

Step 4 (sufficiency in (2) and (3)). For $T \in \mathcal{T}$ let φ_T be the functional on U_T with $\varphi_T(e_{g\circ f_T}) = |\bar{\mathsf{H}}_T|^{-1} \chi_T(g)$ for $g \in \bar{\mathsf{H}}_T$ that vanishes on the other e_f. Then $\varphi_T \circ h'^{\otimes k} = \chi_T(h') \varphi_T$ on U_T for $h' \in \mathsf{H}_T$, and $\varphi_T(\operatorname{alt}^k \eta_T) = 1$ by (N.1). Put $\rho(e_t) = 0$ for t not injective and $\rho(e_{g\circ f}) = \varphi_T(e_f) \Lambda^k_K g\, \eta_T$ for $g \in \mathsf{H}$, $T \in \mathcal{T}$ and $f \in \operatorname{Bij}(T)$; this is well defined, since $g\circ f = g' \circ f'$ implies $h' = g^{-1} g' \in \mathsf{H}_T$ and $f = h' \circ f'$. The map ρ is equivariant, $\rho(\operatorname{alt}^k \Lambda^k_K g\, \eta_T) = \rho(g^{\otimes k} \operatorname{alt}^k \eta_T) = \Lambda^k_K g\, \eta_T$, and $\rho(U_{T'}) \subseteq K\eta_{T'}$. For (3), choose representatives $\mathbf{m}$ of the H-orbits of k-multisets and tuples $t_{\mathbf{m}}$ of type $\mathbf{m}$. Let $\psi_{\mathbf{m}}(e_t)$ be $|\bar{\mathsf{H}}_{\mathbf{m}}|^{-1}$ for $t \in \bar{\mathsf{H}}_{\mathbf{m}} \circ t_{\mathbf{m}}$ and 0 for the other t of type $\mathbf{m}$; then $\psi_{\mathbf{m}}$ is $\mathsf{H}_{\mathbf{m}}$-invariant and $\psi_{\mathbf{m}}(O_{\mathbf{m}}) = 1$. Put $R(e_{g\circ t}) = \psi_{\mathbf{m}}(e_t) O_{g\mathbf{m}}$ for $g \in \mathsf{H}$ and t of type $\mathbf{m}$. As for ρ, R is well defined and equivariant; it maps into $\Gamma^k_K V$, and

$R(O_{g\mathbf{m}}) = \psi_{\mathbf{m}}(O_{\mathbf{m}})O_{g\mathbf{m}} = O_{g\mathbf{m}}$, so it is a projection onto $\Gamma^k_K V$. It maps e_t into $KO_{\mathbf{m}'}$, $\mathbf{m}'$ the type of t, on which $d_\lambda^{\otimes k}$ acts by the same factor as on e_t. □

Proof of Proposition 4.5. Step 1 (augmentation). Let $\mathrm{aug}_T\colon U_T \to K$ be linear with $\mathrm{aug}_T(e_f) = \mathrm{sgn}(f_T^{-1}f)$. Then $\wedge^k z = \mathrm{aug}_T(z)\eta_T$ for $z \in U_T$, so $\mathcal{N}_T = \ker \mathrm{aug}_T$; and $\mathrm{aug}_T(h\cdot z) = \mathrm{aug}_T(z)$ by (N.1), so $h\cdot e_{f_T} - e_{f_T} \in \mathcal{N}_T$. As $h \mapsto h\cdot y - y$ satisfies the cocycle identity for every $y \in U_T$, δ_T is a well-defined linear map. Call $z \in U_T$ *invariant* if $h\cdot z = z$ for all $h \in \mathsf{H}_T$, that is, $h^{\otimes k} z = \chi_T(h)z$.

Step 2 (item (1)). For $a \in K$, $\delta_T(a) = 0$ means that $z = ae_{f_T} - n$ is invariant for some $n \in \mathcal{N}_T$, and then $\mathrm{aug}_T(z) = a$; conversely, for an invariant z with $\mathrm{aug}_T(z) = a$, the vector $n = ae_{f_T} - z$ lies in $\mathcal{N}_T$ and gives $\delta_T(a) = 0$. For $a = 1$ this is (1).

Step 3 (items (2) and (3)). By Step 2, $\ker\delta_T$ is the image under aug_T of the invariant vectors. A vector $\sum_f c_f e_f$ is invariant if and only if $c_{h\circ f} = \chi_T(h)c_f$, so by the orbit count this image lies in $|\bar{\mathsf{H}}_T|K$; it contains $|\bar{\mathsf{H}}_T|$, the image of $\sum_{g\in\bar{\mathsf{H}}_T}\chi_T(g)e_{g\circ f_T}$. So $\ker\delta_T = |\bar{\mathsf{H}}_T|K$, which gives (2), since δ_T is linear. By (2), $\delta_T(1) = 0$ if and only if $|\bar{\mathsf{H}}_T| \neq 0$ in K, and Theorem 4.3(1) gives (3). □

N.3. **Operator families.** The notation is that of Section 6.3.

Proof of Theorem 6.22(2). The squares commute, since $\mathrm{ev}_\theta(\mathcal{N}_E^k) \subseteq \mathrm{Mult}_0^k(P;\widehat{\Lambda}_\pi^k D)$ and $\overline{\mathrm{ev}}_\theta \circ \widehat{\wedge}_E^k = q\circ \mathrm{ev}_\theta$ by the definition of $\overline{\mathrm{ev}}_\theta$ before Theorem 6.22. Lemma N.2 gives the formula. □

Proof of Proposition 6.23. Step 1 (items (a)–(c)). On pure tensors, hence everywhere, $\mathrm{ev}_{\theta\circ\psi} = \psi^\sharp\circ\mathrm{ev}_\theta$, $\mathrm{ev}_{\theta_w} = \mathrm{ev}_\theta \circ w^{\otimes k}$ and $\mathrm{ev}_{w'\theta} = (\widehat{\Lambda}_\pi^k w')_\flat \circ \mathrm{ev}_\theta$. Corollary N.3(b) gives (a) and (c). For (b), $(w^{\otimes k}|_{\mathcal{N}^k_{E'}}, w^{\otimes k}, \widehat{\Lambda}_\pi^k w)$ is a morphism $\mathcal{S}^k_{E'} \to \mathcal{S}^k_E$ (proof of Proposition 3.13), so Corollary N.3(b) and Lemma N.2 give

$$\begin{aligned}\mathrm{ob}_k(\theta_w) &= (\mathrm{ev}_\theta|_{\mathcal{N}^k_E})_*(w^{\otimes k}|_{\mathcal{N}^k_{E'}})_*[\mathcal{S}^k_{E'}] = (\mathrm{ev}_\theta|_{\mathcal{N}^k_E})_*\,\mathrm{sp}_k(w)\\ &= (\mathrm{ev}_\theta|_{\mathcal{N}^k_E})_*(\widehat{\Lambda}_\pi^k w)^*[\mathcal{S}^k_E] = (\widehat{\Lambda}_\pi^k w)^*\,\mathrm{ob}_k(\theta).\end{aligned}$$

Step 2 (item (d)). The map $(\mathrm{alt}_D^k)_\flat$ preserves zero diagonals, so it induces a bounded linear $\overline{\mathrm{alt}}\colon \mathrm{Poly}^k(P;\widehat{\Lambda}_\pi^k D) \to \mathrm{Poly}^k(P;\widehat{\otimes}_\pi^k D)$ and a morphism $\mathcal{M}^k(P;\widehat{\Lambda}_\pi^k D) \to \mathcal{M}^k(P;\widehat{\otimes}_\pi^k D)$. By Theorem 6.22(2), Lemma N.2 and Corollary N.3(b), the class $((\mathrm{alt}_D^k)_\flat)_*\,\mathrm{ob}_k(\theta)$ equals $(\overline{\mathrm{alt}}\circ\overline{\mathrm{ev}}_\theta)^*[\mathcal{M}^k(P;\widehat{\otimes}_\pi^k D)]$. The bounded linear map $L\colon \widehat{\Lambda}_\pi^k E \to \mathrm{Mult}^k(P;\widehat{\otimes}_\pi^k D)$, $L(y)(h_1,\dots,h_k) = (\theta(h_1)\otimes\cdots\otimes\theta(h_k))\,\mathrm{alt}_E^k y$, has the diagonal $\theta(h)^{\otimes k}\,\mathrm{alt}_E^k y = \mathrm{alt}_D^k\,\widehat{\Lambda}_\pi^k\theta(h)y$, by naturality of alt^k; so its composite with the quotient map of $\mathcal{M}^k(P;\widehat{\otimes}_\pi^k D)$ is $\overline{\mathrm{alt}}\circ\overline{\mathrm{ev}}_\theta$, and Lemma N.1 gives the first assertion. For the second, extend w' to $\widehat{G}$, regard θ_0 as a family $P \to \mathcal{L}(E,\widehat{G})$, and let $\rho\colon \otimes_\pi^k G \to \widehat{\Lambda}_\pi^k D$ satisfy $\rho\circ\mathrm{alt}_G^k = \Lambda_\pi^k w'$, so that $\widehat{\Lambda}_\pi^k w' = \rho\circ\mathrm{alt}_G^k$ on the completions. By (c), Corollary N.3(b),(c) and the first assertion for θ_0, $\mathrm{ob}_k(w'\theta_0) = (\rho_\flat)_*((\mathrm{alt}_G^k)_\flat)_*\,\mathrm{ob}_k(\theta_0) = 0$, and Theorem 6.22(1) concludes. □

Lemma N.4. *Let P_1 and P_2 be normed spaces, $P = P_1 \oplus P_2$ with the maximum norm, Y a Banach space, and Φ_S, $\Pi^{(a)}$ as in Theorem 6.25. For $\Phi \in \mathrm{Mult}^k(P;Y)$, $\Phi = \sum_{S\subseteq[k]}\Phi_S$, $(\Phi_S)_{S'}$ is Φ_S if $S' = S$ and 0 otherwise, and $\|\Phi_S\| \le \|\Phi\|$. Hence the $\Pi^{(a)}$ are bounded commuting idempotents with sum* id. *Each maps $\mathrm{Mult}_0^k(P;Y)$ into itself.*

Proof. Expanding $h^j = (h_1^j, 0) + (0, h_2^j)$ multilinearly gives $\Phi = \sum_S \Phi_S$. The map $(\Phi_S)_{S'}$ evaluates Φ at a tuple whose jth entry is 0 for j in exactly one of S and S', and equals Φ_S if

$S' = S$. Since $(h_1^j, 0)$ and $(0, h_2^j)$ have norm at most $\|h^j\|$, $\|\Phi_S\| \le \|\Phi\|$. For $\Phi \in \mathrm{Mult}_0^k(P;Y)$, $h = (h_1, h_2) \in P$ and $\lambda \in K$,

$$0 = \Phi((\lambda h_1, h_2), \dots, (\lambda h_1, h_2)) = \sum_S \lambda^{|S|} \Phi_S(h, \dots, h) = \sum_{a=0}^{k} \lambda^a (\Pi^{(a)}\Phi)(h, \dots, h).$$

Since K is infinite, the Vandermonde determinant for $k+1$ distinct values of λ shows that $(\Pi^{(a)}\Phi)(h, \dots, h) = 0$ for every a. □

Proof of Theorem 6.25. Put $\mathrm{Mult}_0^k = \mathrm{Mult}_0^k(P;Y)$, $\mathrm{Mult}^{(a)} = \Pi^{(a)}\,\mathrm{Mult}^k(P;Y)$ and $\mathrm{ev} = \mathrm{ev}_{\theta_1 \boxplus \theta_2}$. By Lemma N.4, $\mathrm{Mult}^{(a)}$ and $\mathrm{Mult}_0^{(a)}$ are closed and $\mathrm{Mult}^{(a)} \cap \mathrm{Mult}_0^k = \mathrm{Mult}_0^{(a)}$; by Corollary N.3(b), $\mathrm{ob}^{(a)} = (\Pi^{(a)} \circ \mathrm{ev}\,|_{\mathcal{N}_E^k})_*[\mathcal{S}_E^k]$.

Step 1 (item (1)). By Lemma 3.6(3), $\mathrm{ob}_k(\theta_1 \boxplus \theta_2) = 0$ if and only if $\mathrm{ev}\,|_{\mathcal{N}_E^k}$ extends to a bounded linear $\xi\colon \widehat{\otimes}_\pi^k E \to \mathrm{Mult}_0^k$, and $\mathrm{ob}^{(a)} = 0$ if and only if $\Pi^{(a)}\,\mathrm{ev}\,|_{\mathcal{N}_E^k}$ extends to a bounded linear $\xi_a\colon \widehat{\otimes}_\pi^k E \to \mathrm{Mult}_0^{(a)}$. An extension ξ gives the extensions $\Pi^{(a)}\xi$, and extensions ξ_a give the extension $\sum_a \xi_a$. For $S \ne [k]$, $\Phi_S \circ \iota_1^{\times k}$ evaluates Φ at tuples with a zero entry, so $\iota_1^\sharp = \iota_1^\sharp \Pi^{(k)}$. With the projections $\mathrm{pr}_l\colon P \to P_l$, $l = 1, 2$, $\Phi_{[k]} = (\iota_1^\sharp \Phi) \circ \mathrm{pr}_1^{\times k}$, so $\iota_1^\sharp$ is an isometry of $\mathrm{Mult}^{(k)}$ onto $\mathrm{Mult}^k(P_1;Y)$ with inverse $\mathrm{pr}_1^\sharp$, and both preserve zero diagonals. By Corollary N.3(b) and Proposition 6.23(a), $(\iota_1^\sharp)_*\,\mathrm{ob}^{(k)} = (\iota_1^\sharp)_*\,\mathrm{ob}_k(\theta_1 \boxplus \theta_2) = \mathrm{ob}_k((\theta_1 \boxplus \theta_2) \circ \iota_1) = \mathrm{ob}_k(\theta_1)$. The case of ι_2, with $S = \emptyset$ and pr_2, is the same.

Step 2 (item (2)). Let $\Theta_a(h_1, h_2)$ be the right side of the identity in (2). By Lemma 3.7, the relation $\Pi^{(a)}\,\mathrm{ev} - \xi_a = \Psi \circ \widehat{\wedge}_E^k$ matches the extensions ξ_a of Step 1 with the bounded linear $\Psi\colon \widehat{\Lambda}_\pi^k E \to \mathrm{Mult}^{(a)}$ such that $\Psi(\widehat{\wedge}_E^k z) - \Pi^{(a)}\,\mathrm{ev}(z) \in \mathrm{Mult}_0^k$ for all z. For $z = x_1 \otimes \cdots \otimes x_k$ and $h = (h_1, h_2)$, $\mathrm{ev}(z)_S(h, \dots, h)$ is the wedge, in the order of j, of the vectors $\theta_1(h_1)x_j$, $j \in S$, and $\theta_2(h_2)x_j$, $j \notin S$; by the expansion before (6.3), the sum over $|S| = a$ is $\Theta_a(h_1, h_2)(x_1 \wedge \cdots \wedge x_k)$. By continuity, the condition on Ψ says $\Psi(y)(h, \dots, h) = \Theta_a(h_1, h_2)y$. Given $Q^{(a)}$, the map $\Psi(y)(h^1, \dots, h^k) = Q^{(a)}(h_1^1, \dots, h_1^a; h_2^{a+1}, \dots, h_2^k)y$ satisfies it and lies in $\mathrm{Mult}^{(a)}$, since $\Psi(y) = \Psi(y)_{[a]}$ (Lemma N.4). Conversely, given Ψ, for $g \in P_1^a$, $g' \in P_2^{k-a}$ and $|S| = a$ let $(g; g')_S \in P^k$ have the entries $(g_1, 0), \dots, (g_a, 0)$ in the slots of S and $(0, g'_1), \dots, (0, g'_{k-a})$ in the others, in increasing order, and put $Q^{(a)}(g; g')y = \sum_{|S|=a} \Psi(y)((g; g')_S)$; its diagonal is $(\Pi^{(a)}\Psi(y))(h, \dots, h) = \Theta_a(h_1, h_2)y$.

Step 3 (items (3)–(5)). (3) For lifts Q_a of θ_1 and Q'_{k-a} of θ_2, the map $\wedge_{\widehat{D}}^{a,k-a} \circ (Q_a(g) \otimes Q'_{k-a}(g')) \circ \mathrm{sh}_E^{a,k-a}$, extended to completions, is bounded and k-linear by Lemma 6.1, with diagonal $\Theta_a(h_1, h_2)$; apply Step 2. (4) If all $\mathrm{ob}^{(a)}$ vanish, then $\mathrm{ob}_k(\theta_1 \boxplus \theta_2) = 0$ by (1), so $\mathrm{ob}_k(\theta_1 + \theta_2) = (\Delta_{P_1}^\sharp)_*\,\mathrm{ob}_k(\theta_1 \boxplus \theta_2) = 0$ by Proposition 6.23(a) and Corollary N.3(c), and Theorem 6.22(1) applies. If θ_1 and θ_2 are liftable in every degree at most k, then $\mathrm{ob}_k(\theta_1) = \mathrm{ob}_k(\theta_2) = 0$ by Theorem 6.22(1); applying $(\mathrm{pr}_1^\sharp)_*$ and $(\mathrm{pr}_2^\sharp)_*$ to the identities of (1) gives $\mathrm{ob}^{(k)} = \mathrm{ob}^{(0)} = 0$ by Corollary N.3(b),(c), and the other $\mathrm{ob}^{(a)}$ vanish by (3). (5) For $|S| = a$, $(\mathrm{ev}_{\theta \boxplus \theta}(z))_S \circ \Delta_{P_1}^{\times k} = \mathrm{ev}_\theta(z)$, on pure tensors since $(\theta \boxplus \theta)(h, 0) = (\theta \boxplus \theta)(0, h) = \theta(h)$, hence everywhere; summing over the $\binom{k}{a}$ sets S gives the identity. By Corollary N.3(b), $(\Delta_{P_1}^\sharp|_{\mathrm{Mult}_0^{(a)}})_*\,\mathrm{ob}^{(a)} = (\binom{k}{a}\,\mathrm{ev}_\theta\,|_{\mathcal{N}_E^k})_*[\mathcal{S}_E^k]$, which by Corollary N.3(c) vanishes if and only if $\mathrm{ob}_k(\theta) = 0$. If θ is liftable in degrees a and $k-a$, then $\mathrm{ob}^{(a)} = 0$ by (3), so $\mathrm{ob}_k(\theta) = 0$, and Theorem 6.22(1) applies. □

Proof of Proposition 6.26. (1) As $(\theta' \odot \theta)(\psi_h h') = \theta'(h') \circ \theta(h)$, the family $(\theta' \odot \theta) \circ \psi_h$ is the family θ'_w of Proposition 6.23(b) for θ' on $\widehat{D}$ and $w = \theta(h) \in \mathcal{L}(E, \widehat{D})$, with $\widehat{\Lambda}^k_\pi w$ mapping into $\widehat{\Lambda}^k_\pi \widehat{D} = \widehat{\Lambda}^k_\pi D$; items (a) and (b) give the first identity. Likewise $(\theta' \odot \theta) \circ \psi'_{h'}$ is the family $w'\theta$ of Proposition 6.23(c) for $w' = \theta'(h') \colon \widehat{D} \to \widehat{F}$, and (a) and (c) give the second. (2) On pure tensors $(\theta_{D,F} \odot \theta_{E,D})(v \otimes u) = vu$, so $\theta_{D,F} \odot \theta_{E,D} = \theta_{E,F} \circ \mathrm{comp}$, and Proposition 6.23(a) gives the identity. If (E, F) is good for Banach values in degree k, then $\mathrm{ob}_k(E, F) = 0$ by Theorem 3.15, so the class of $\theta_{D,F} \odot \theta_{E,D}$ vanishes by Corollary N.3(c), and Theorem 6.22(1) applies. □

N.4. **A three-space counterexample.** The Cantor set Ω, its dyadic cells and the spaces $L^1(\Omega; U)$ are as in Theorem 6.29.

Lemma N.5. *Let K be nonarchimedean, U a Banach space, U_0 a closed subspace of $L^1(\Omega; U)$, and Y a normed space with an equivalent nonarchimedean norm. Then every bounded linear map $L^1(\Omega; U)/U_0 \to Y$ is zero.*

Proof. Let ψ be such a map, ψ_0 its composite with the quotient map, and ν a nonarchimedean norm on Y with $\nu \le c_\nu \|\cdot\|$. For a dyadic cell C_ξ of level m, $u \in U$ and $j \ge 0$, $1_{C_\xi} u$ is the sum of the 2^j functions $1_{C_{\xi'}} u$, $C_{\xi'} \subseteq C_\xi$ of level $m + j$, each of norm $2^{-m-j}\|u\|$. So $\nu(\psi_0(1_{C_\xi} u)) \le \max_{\xi'} \nu(\psi_0(1_{C_{\xi'}} u)) \le c_\nu \|\psi_0\| 2^{-m-j} \|u\|$, and $j \to \infty$ gives $\psi_0(1_{C_\xi} u) = 0$. Step functions are dense, so $\psi_0 = 0$, and $\psi = 0$. □

Lemma N.6. *Let Y, Y', Y'' be Banach spaces, $T \colon Y \to Y'$ bounded linear such that every $y' \in Y'$ equals Ty for some y with $\|y\| \le C_T \|y'\|$, $R = \ker T$, and $\eta \colon Y \to Y''$ bounded linear and bounded below. Give $Y' \oplus Y''$ the maximum norm. Then $y \mapsto (-Ty, \eta y)$ and $(y', y'') \mapsto [y'' + \eta(y_{y'})]$, where $Ty_{y'} = y'$, form an exact sequence $0 \to Y \to Y' \oplus Y'' \to Y''/\eta(R) \to 0$.*

Proof. The subspace $\eta(R)$ is complete, hence closed. Two choices of $y_{y'}$ differ by an element of R, so the second map is well defined and linear; it is bounded for the choice $\|y_{y'}\| \le C_T \|y'\|$, and it maps $(0, y'')$ to $[y'']$, where y'' can be chosen with $\|y''\| \le 2\|[y'']\|$. If $y'' + \eta y_{y'} = \eta r$ with $r \in R$, then $(y', y'') = (-Tw, \eta w)$ for $w = r - y_{y'}$; conversely $(-Tw, \eta w)$ maps to $[\eta w - \eta w] = 0$, with $y_{-Tw} = -w$. The first map is bounded below, as η is. □

Proof of Theorem 6.31. Step 1 (exactness). As ϖ_D is open (Corollary 6.30(2)), for some $\delta > 0$ the image of the open unit ball contains the open ball of radius δ. For $d \ne 0$ and $j \in \mathbb{Z}$ with $|\vartheta|^{j+1}\delta \le \|d\| < |\vartheta|^j \delta$, ϑ as in Section 7.1, some y with $\|y\| < 1$ has $\varpi_D y = \vartheta^{-j} d$, and $y_d = \vartheta^j y$ has $\|y_d\| < \|d\|/(|\vartheta|\delta)$. As η is isometric, Lemma N.6 applies with $T = \varpi_D$.

Step 2 (items (1) and (2)). By Proposition 5.5(3), $\ell^1(\mathrm{Ball}(D), K)$ satisfies (S^n) for every n, and it is dense in Y_1; so Proposition 6.28(3) gives $(\mathrm{S}^n_{Y_1})$, and Proposition 3.5(1) gives (1). Under (L), K is nonarchimedean, so Lemma N.5 with $U = Y_1$, the closed subspace $U_0 = \eta(\ker \varpi_D)$ and $Y = D$ gives $\mathcal{L}(Y_3, D) = 0$; the zero map on $\mathcal{L}(Y_3, D)^n$ is then a lift as in Lemma 3.4 for every $n \ge 1$.

Step 3 (item (3)). Let $J \colon D \to Y_2$ be $d \mapsto (d, 0)$ and $\Pi' \colon Y_2 \to D$ the first coordinate. If (Y_2, D) were good for Banach values in degree k, Lemma 6.16(2), with J, Π' and $J' = \Pi = \mathrm{id}_D$, would make (D, D) good for Banach values in degree k, contrary to Theorem 7.19(2), since D is complete, has an equivalent nonarchimedean norm and $\dim_{\widehat{K}} D = \infty$. □

N.5. **Characteristic two.** Throughout this subsection the notation is that of Section 7.6, $\Bbbk$ is a field of characteristic 2, and K is a nontrivially normed field of characteristic 2. By Lemma A.2, K is nonarchimedean, and so is its completion $\widehat{K}$; K is infinite, as a finite field

has only the trivial absolute value. For z in $\Lambda^k K^n$ or in $K^n \otimes K^n$, with coordinates z_c,

$$\max_c |z_c| \le \|z\|_\pi \le \sum_c |z_c|, \tag{N.2}$$

since each coordinate of $x_1 \wedge \cdots \wedge x_k$ is a $k \times k$ minor, of absolute value at most $\prod_a \|x_a\|$ by the ultrametric inequality. So the completions of $\Lambda^k_\pi c_0(n)$ and $c_0(n) \otimes_\pi c_0(n)$ are coordinate spaces over $\widehat{K}$ with norms equivalent to the coordinate supremum norm, and their elements are identified with their coordinate vectors.

Definition N.7. Let $J \subseteq [n]$, let $x|_J$ be the restriction of $x \in \Bbbk^n$ to J, and let $\mathrm{Sym}_J(\Bbbk)$ be the space of symmetric $J \times J$ matrices. A *J-polarization* is a $\Bbbk$-bilinear map $\mathsf{N} \colon \Bbbk^n \times \Bbbk^n \to \mathrm{Sym}_J(\Bbbk)$ with $\mathsf{N}(x,x) = x|_J (x|_J)^\mathsf{T}$. Regarding the entries of x and y also as indeterminates, $\mathsf{N}(x,y) = \sum_{t,t'} x_t y_{t'} \mathsf{N}(e_t, e_{t'})$ is a matrix over $\Bbbk[x,y]$. For $L, L' \subseteq [n]$ put $x_L y_{L'} = \prod_{t \in L} x_t \prod_{t' \in L'} y_{t'}$, and for $g \in \Bbbk[x,y]$ let $c_g(L,L')$ be the coefficient of $x_L y_{L'}$ in g.

Lemma N.8. *Let J be finite, M a symmetric $J \times J$ matrix over a commutative ring of characteristic 2, λ a ring element, and $\mathsf{M}[J']$ the principal submatrix on $J' \subseteq J$, with $\det \mathsf{M}[\emptyset] = 1$. For $a \ne a'$ in J,*

$$\det(\mathsf{M} + \lambda(e_a e_{a'}^\mathsf{T} + e_{a'} e_a^\mathsf{T})) = \det \mathsf{M} + \lambda^2 \det \mathsf{M}[J \setminus \{a, a'\}],$$
$$\det(\mathsf{M} + \lambda e_a e_a^\mathsf{T}) = \det \mathsf{M} + \lambda \det \mathsf{M}[J \setminus \{a\}].$$

Proof. The second identity is the expansion along row a. For the first, expand multilinearly in the rows a and a', which change by $\lambda e_{a'}^\mathsf{T}$ and λe_a^T. The two first-order terms are λ times the cofactors of M at (a,a') and (a',a), which are equal as M is symmetric, so they cancel; the second-order term is $\pm\lambda^2 \det \mathsf{M}[J \setminus \{a,a'\}]$, and signs do not matter in characteristic 2. □

Lemma N.9. *Let $J \subseteq [n]$, and let N and N' be J-polarizations. Then $c_{\det \mathsf{N}}(L,L') = c_{\det \mathsf{N}}(L',L)$ and $c_{\det \mathsf{N}}(L,L) = c_{\det \mathsf{N}'}(L,L)$ for all $L, L' \subseteq [n]$.*

Proof. Step 1 (swap symmetry). Expanding $\mathsf{N}(x+y, x+y)$ at unit vectors gives $\mathsf{N}(y,x) = \mathsf{N}(x,y) + x|_J (y|_J)^\mathsf{T} + y|_J (x|_J)^\mathsf{T}$, and $x|_J (y|_J)^\mathsf{T} + y|_J (x|_J)^\mathsf{T}$, whose diagonal vanishes, is the sum of the matrices $(x_a y_{a'} + x_{a'} y_a)(e_a e_{a'}^\mathsf{T} + e_{a'} e_a^\mathsf{T})$ over $a < a'$ in J. Adding them one at a time, Lemma N.8 changes the determinant by terms $(x_a^2 y_{a'}^2 + x_{a'}^2 y_a^2) g$, whose monomials contain a squared variable. So the coefficient of $x_L y_{L'}$ in $\det \mathsf{N}(x,y)$ equals that in $\det \mathsf{N}(y,x)$, which is $c_{\det \mathsf{N}}(L',L)$.

Step 2 (invariance). The map $\mathsf{A} = \mathsf{N}' - \mathsf{N}$ is bilinear with symmetric values and $\mathsf{A}(x,x) = 0$, so $\mathsf{A}(e_t, e_t) = 0$, $\mathsf{A}(e_{t'}, e_t) = \mathsf{A}(e_t, e_{t'})$ and $\mathsf{A}(x,y) = \sum_{t<t'} (x_t y_{t'} + x_{t'} y_t) \mathsf{A}(e_t, e_{t'})$. Writing each $\mathsf{A}(e_t, e_{t'})$ as a combination of the matrices $e_a e_a^\mathsf{T}$ and $e_a e_{a'}^\mathsf{T} + e_{a'} e_a^\mathsf{T}$, we pass from N to N' through J-polarizations, each obtained from the previous one, N_1, by adding $\lambda(x_t y_{t'} + x_{t'} y_t)\mathsf{E}$ with $\lambda \in \Bbbk$ and E one of these matrices. For $\mathsf{E} = e_a e_{a'}^\mathsf{T} + e_{a'} e_a^\mathsf{T}$, Lemma N.8 changes the determinant by $\lambda^2 (x_t^2 y_{t'}^2 + x_{t'}^2 y_t^2) g$, which has no term $x_L y_L$. For $\mathsf{E} = e_a e_a^\mathsf{T}$, the change is $\lambda(x_t y_{t'} + x_{t'} y_t) g$ with $g = \det \mathsf{N}_1(x,y)[J \setminus \{a\}]$; its coefficient of $x_L y_L$ is 0 unless $t, t' \in L$, when it is $\lambda(c_g(L \setminus \{t\}, L \setminus \{t'\}) + c_g(L \setminus \{t'\}, L \setminus \{t\})) = 0$ by Step 1 for the $(J \setminus \{a\})$-polarization $\mathsf{N}_1[J \setminus \{a\}]$. □

Lemma N.10. *The map with $\mathsf{N}_<(x,y)_{tt'} = x_{\max(t,t')} y_{\min(t,t')}$ is an $[n]$-polarization,*

$$\det \mathsf{N}_<(x,y) = y_1 x_n \prod_{t=1}^{n-1} (x_t y_{t+1} + x_{t+1} y_t),$$

and $c_{\det \mathsf{N}_<}([n],[n]) = 1$.

Proof. The map is bilinear with symmetric values, and $\mathsf{N}_<(x,x)_{tt'} = x_t x_{t'}$. For $n = 1$ the determinant is $x_1 y_1$. For $n \geq 2$, over the field of rational functions, subtracting x_n/x_{n-1} times row $n-1$ from row n leaves the last row $(0, \dots, 0, (x_n/x_{n-1})(x_{n-1}y_n + x_n y_{n-1}))$ below the matrix $\mathsf{N}_<$ for $n-1$. By induction $\det \mathsf{N}_< = x_1 y_1 \prod_{t=2}^{n} (x_t/x_{t-1})(x_{t-1}y_t + x_t y_{t-1})$, which telescopes to the formula. A term of the product equal to $x_{[n]}y_{[n]}$ takes $x_{n-1}y_n$ from the factor of index $n-1$, since x_n is used, then $x_{n-2}y_{n-1}$ from the factor of index $n-2$, and so on; so the coefficient is 1. □

Lemma N.11. *Let g be a polynomial in x and y over a commutative ring of characteristic 2, of degree at most n in x and at most n in y. Then $\sum_{S,T\subseteq[n]} g(1_S, 1_T)$ is the coefficient of $x_{[n]}y_{[n]}$ in g. Likewise, if g is a polynomial in x of degree at most n, then $\sum_{S\subseteq[n]} g(1_S)$ is the coefficient of $x_1 \cdots x_n$.*

Proof. A monomial $x^\alpha y^\beta$ takes the value 1 at $(1_S, 1_T)$ if $\operatorname{supp}\alpha \subseteq S$ and $\operatorname{supp}\beta \subseteq T$, and 0 otherwise, so its sum over S, T is $2^{2n-|\operatorname{supp}\alpha|-|\operatorname{supp}\beta|}$. This vanishes unless both supports are $[n]$, which under the degree bounds happens only for $x_{[n]}y_{[n]}$. The second statement is proved in the same way. □

Proposition N.12. *Let $n \geq 2$ and $\sigma\colon \Lambda^2\Bbbk^n \to \Bbbk^{n\times n}$ be $\Bbbk$-linear with $\wedge \circ \sigma = \mathrm{id}$. Then $\mathsf{N}(x,y) = \sigma(x\wedge y) + xy^{\mathsf{T}}$ is an $[n]$-polarization, $\sum_{S,T\subseteq[n]} \det \mathsf{N}(1_S, 1_T) = 1$, and $\operatorname{rank}\sigma(1_S \wedge 1_T) \geq n-1$ for some $S, T \subseteq [n]$.*

Proof. The map $\Phi(x,y) = \sigma(x\wedge y)$ is bilinear, $\Phi(x,x) = 0$, and $\Phi(x,y) + \Phi(x,y)^{\mathsf{T}} = \Omega_{x\wedge y} = xy^{\mathsf{T}} + yx^{\mathsf{T}}$; so N is bilinear, $\mathsf{N}(x,y)^{\mathsf{T}} = \mathsf{N}(x,y)$ and $\mathsf{N}(x,x) = xx^{\mathsf{T}}$. By Lemmas N.9 and N.10, $c_{\det \mathsf{N}}([n],[n]) = c_{\det \mathsf{N}_<}([n],[n]) = 1$, and $\det \mathsf{N}(x,y)$ has degree n in x and in y, so Lemma N.11 gives the identity. Hence some $\mathsf{N}(1_S, 1_T)$ is invertible, and $\sigma(1_S \wedge 1_T) = \mathsf{N}(1_S, 1_T) + 1_S 1_T^{\mathsf{T}}$ has rank at least $n-1$. □

Proposition N.13. *Let $n \geq 2$ and $\tau\colon \Bbbk^{n\times n} \to \Lambda^2\Bbbk^n$ be $\Bbbk$-linear with $\tau(\Omega_w) = w$ for $w \in \Lambda^2\Bbbk^n$. If n is even, $\sum_{S\subseteq[n]} \operatorname{Pf}\Omega_{\tau(1_S 1_S^{\mathsf{T}})} = 1$, and some $\Omega_{\tau(1_S 1_S^{\mathsf{T}})}$ is invertible. For every n, some $S \subseteq [n]$ has $\operatorname{rank}\Omega_{\tau(1_S 1_S^{\mathsf{T}})} \geq 2\lfloor n/2 \rfloor$.*

Proof. Step 1 (even n). Let n be even and $w_t = \tau(e_t e_t^{\mathsf{T}})$. Since $\tau(e_t e_{t'}^{\mathsf{T}}) + \tau(e_{t'} e_t^{\mathsf{T}}) = \tau(\Omega_{e_t \wedge e_{t'}}) = e_t \wedge e_{t'}$, $q(x) = \tau(xx^{\mathsf{T}})$ equals $\sum_t x_t^2 w_t + \sum_{t<t'} x_t x_{t'} e_t \wedge e_{t'}$, and $\Omega_{q(x)}$ has the off-diagonal entries $x_t x_{t'} + \sum_{t''} x_{t''}^2 (\Omega_{w_{t''}})_{tt'}$. In a product over a partition into pairs, a factor $x_{t''}^2$ gives a squared variable, so the coefficient of $x_1 \cdots x_n$ in $\operatorname{Pf}\Omega_{q(x)}$, a polynomial of degree n, is the number $(n-1)(n-3)\cdots 1$ of these partitions, which is odd. Lemma N.11 gives the identity, so some $\operatorname{Pf}\Omega_{q(1_S)}$ is nonzero, and $\Omega_{q(1_S)}$ is invertible.

Step 2 (odd n). For odd n let $\iota'\colon \Bbbk^{n-1} \to \Bbbk^n$ be the inclusion of the first $n-1$ coordinates and $\tau'(\mathsf{Z}) = \Lambda^2 \iota'^{\mathsf{T}} \tau(\iota' \mathsf{Z} \iota'^{\mathsf{T}})$. As $\iota' \Omega_w \iota'^{\mathsf{T}} = \Omega_{\Lambda^2 \iota' w}$ and $\Omega_{\Lambda^2 \iota'^{\mathsf{T}} w} = \iota'^{\mathsf{T}} \Omega_w \iota'$, the map τ' satisfies the hypothesis for $n-1$, and $\operatorname{rank}\Omega_{\tau'(\mathsf{Z})} \leq \operatorname{rank}\Omega_{\tau(\iota' \mathsf{Z} \iota'^{\mathsf{T}})}$; the even case for τ' and $\iota' 1_S 1_S^{\mathsf{T}} \iota'^{\mathsf{T}} = 1_S 1_S^{\mathsf{T}}$ give the last assertion. □

Proposition N.14 (Pfaffian identity)**.** *Let $n \geq 2$ be even and Ψ a multiplier lift on $\Bbbk^n$. Then (7.6) holds, and*

$$\sum_{L} \sum_{a,x} \sum_{b,y} \operatorname{Pf} \mathsf{M}_\Psi(a,b,x,y) = 1, \tag{N.3}$$

the sums over the $L \subseteq [n]$ with $|L| = n/2$, over the $a, x \in \{0,1\}^n$ supported in L, and over the $b, y \in \{0,1\}^n$ supported in L^c.

Proof. Step 1 (decomposition). For $t \neq t'$ put $\Psi^{\mathrm{cor}}_{tt'} = \Psi_{tt'} - \Psi^{<}_{tt'}$. By (7.5), $\Psi^{\mathrm{cor}}_{tt'} + \Psi^{\mathrm{cor}}_{t't} = \pi_{tt'} - \pi_{tt'} = 0$, so $\Psi^{\mathrm{cor}}_{tt'} = \Psi^{\mathrm{cor}}_{t't}$; write $\Psi^{\mathrm{cor}}_{\{t,t'\}}$ for it. For a 2-subset $\varepsilon = \{t, t'\}$ of $[n]$ with $t < t'$ put $e_\varepsilon = e_t \wedge e_{t'}$, and write $\Psi^{\mathrm{cor}}_{\{t,t'\}}[\varepsilon, \varepsilon']$ for the entry of $\Psi^{\mathrm{cor}}_{\{t,t'\}}$ in the row of e_ε and the column of $e_{\varepsilon'}$. Fix L with $|L| = n/2$. In the monomial $a_L b_{L^c} x_L y_{L^c}$ every a-variable has its index in L and every b-variable in L^c, so only the $\Psi_{tt'}$ with $t \in L$ and $t' \in L^c$, in particular $t \neq t'$, contribute. As the coordinate of $x \wedge y$ at $e_{\{u,u'\}}$ is $x_u y_{u'} + x_{u'} y_u$, for $t \neq t'$ and $u \neq u'$ the coefficient of $a_t b_{t'} x_u y_{u'}$ in the entry of M_Ψ at a 2-subset ε is

$$O_\varepsilon(t, t', u, u') + \Psi^{\mathrm{cor}}_{\{t,t'\}}[\varepsilon, \{u, u'\}], \tag{N.4}$$

where $O_\varepsilon(t, t', u, u') = 1$ if $t < t'$ and $\{t, t'\} = \{u, u'\} = \varepsilon$, and $O_\varepsilon(t, t', u, u') = 0$ otherwise.

Step 2 (configurations). Expanding the Pfaffian over the perfect matchings $\mathfrak{m}$ of $[n]$, whose elements we call edges, and each entry by (N.4), the coefficient of $a_L b_{L^c} x_L y_{L^c}$ in $\operatorname{Pf} \mathsf{M}_\Psi$ is the sum of the values of the *configurations* for L. A configuration consists of a perfect matching $\mathfrak{m}$, an assignment $\varepsilon \mapsto (t_\varepsilon, t'_\varepsilon, u_\varepsilon, u'_\varepsilon)$ on its edges such that each element of L occurs once among the t_ε and once among the u_ε, and each element of L^c once among the t'_ε and once among the u'_ε, and a choice at each edge of one of the two summands of (N.4), where the first summand may be chosen only if it equals 1. Its value is the product of the chosen summands. An edge is an *ordering edge* or a *correction edge* according to the chosen summand.

Step 3 (ordering edges). An ordering edge ε has $\{t_\varepsilon, t'_\varepsilon\} = \{u_\varepsilon, u'_\varepsilon\} = \varepsilon$ and $t_\varepsilon < t'_\varepsilon$. As $t_\varepsilon, u_\varepsilon \in L$ and $t'_\varepsilon, u'_\varepsilon \notin L$, this forces $t_\varepsilon = u_\varepsilon = \min \varepsilon \in L$ and $t'_\varepsilon = u'_\varepsilon = \max \varepsilon \notin L$, and its summand is 1.

Step 4 (correction data). Let $\varepsilon_1, \dots, \varepsilon_m$ be the correction edges of a configuration, and put $J_i = \{t_{\varepsilon_i}, t'_{\varepsilon_i}\}$ and $J'_i = \{u_{\varepsilon_i}, u'_{\varepsilon_i}\}$. The value of the configuration is

$$\prod_{i=1}^{m} \Psi^{\mathrm{cor}}_{J_i}[\varepsilon_i, J'_i], \tag{N.5}$$

and the set $\Delta = \{(\varepsilon_i, J_i, J'_i) : i \leq m\}$ is its *correction data*. Each element of $[n]$ occurs exactly once among the $t_\varepsilon, t'_\varepsilon$, once among the $u_\varepsilon, u'_\varepsilon$ and once among the edges of $\mathfrak{m}$, and the ordering edges use the same two elements in all three roles (Step 3). So $J_1, \dots, J_m$, the sets $J'_1, \dots, J'_m$ and the edges $\varepsilon_1, \dots, \varepsilon_m$ are three partitions of one set $U \subseteq [n]$, and the ordering edges form a perfect matching $\mathfrak{m}'$ of $[n] \setminus U$. Moreover $L \cap U$ meets each J_i and each J'_i in exactly one element.

Step 5 (counting). Fix a set $\Delta = \{(\varepsilon_i, J_i, J'_i) : i \leq m\}$, $m \geq 0$, of triples of 2-subsets in which the ε_i, the J_i and the J'_i partition the same set U, and call a subset of U meeting each J_i and each J'_i in exactly one element a *transversal* of Δ. The configurations, for all L together, with correction data Δ correspond bijectively to the pairs of a transversal L_U and a perfect matching $\mathfrak{m}'$ of $[n] \setminus U$: given the pair, put $L = L_U \cup \{\min \varepsilon : \varepsilon \in \mathfrak{m}'\}$ and $\mathfrak{m} = \{\varepsilon_1, \dots, \varepsilon_m\} \cup \mathfrak{m}'$, let t_{ε_i} and u_{ε_i} be the elements of J_i and J'_i in L_U and t'_{ε_i}, u'_{ε_i} the other ones, and use Step 3 on the edges of $\mathfrak{m}'$; by Steps 3 and 4 every configuration arises from exactly one pair. All of them have the value (N.5), and $[n] \setminus U$ has $(n - 2m - 1)(n - 2m - 3) \cdots 1$ perfect matchings (one if $U = [n]$), an odd number. The multigraph on U with the edges $J_1, \dots, J_m, J'_1, \dots, J'_m$ has degree 2 at every vertex, with one edge of each kind, so it is a disjoint union of γ cycles of even length alternating between the kinds ($J_i = J'_j$ gives a cycle of length 2). A transversal contains exactly one endpoint of each of these edges, so on each cycle it is one of the two colour classes of its proper 2-colouring; there are 2^γ transversals, and $\gamma \geq 1$ if $m \geq 1$.

Step 6 (summation). Summing over L sums the values of all configurations. Grouped by their correction data, those with $m \geq 1$ contribute 2^γ times an odd number times (N.5), which

is 0, and those with $m = 0$ are the pairs $(\emptyset, \mathfrak{m}')$, of value 1, contributing $(n-1)(n-3)\cdots 1 = 1$. This is (7.6).

Step 7 (0/1 inputs). Fix L and restrict $\operatorname{Pf} \mathsf{M}_\Psi$ to a, x supported in L and b, y supported in L^c. The result is a polynomial of degree at most $2n$ in the $2n$ variables a_t, x_t $(t \in L)$ and $b_{t'}, y_{t'}$ $(t' \in L^c)$. As in the proof of Lemma N.11, the sum of the values of such a polynomial over $\{0,1\}^{2n}$ is the coefficient of the product of all its variables, here of $a_L b_{L^c} x_L y_{L^c}$. Summing over L, (7.6) gives (N.3). □

Lemma N.15 (Pfaffian and determinant bounds)**.**

(1) *Let $n \geq 2$ be even. For z in $\Lambda^2_\pi c_0(n)$ or in its completion, $\|z\|_\pi \geq \frac{n}{2} \, |\operatorname{Pf} \Omega_z|^{2/n}$.*

(2) *Let $n \geq 1$. For Z in $c_0(n) \otimes_\pi c_0(n)$ or in its completion, $\|\mathsf{Z}\|_\pi \geq n \, |\det \mathsf{Z}|^{1/n}$.*

Proof. Step 1 (expansion). Let $z = \sum_{i \in I_0} x_i \wedge y_i$ with $x_i, y_i \in K^n$ and I_0 a finite ordered set. Then

$$\operatorname{Pf} \Omega_z = \sum_{I \subseteq I_0, \ |I| = n/2} \det(x_{i_1}, y_{i_1}, \dots, x_{i_{n/2}}, y_{i_{n/2}}), \qquad I = \{i_1 < \cdots < i_{n/2}\}. \tag{N.6}$$

Indeed, read with the Pfaffian of the alternating matrix of z over $\mathbb{Z}$, (N.6) is a polynomial identity with integer coefficients in the entries of the x_i, y_i. It holds over $\mathbb{Q}$: there $z^{\wedge n/2} = (n/2)! \operatorname{Pf}(\Omega_z) \, e_1 \wedge \cdots \wedge e_n$, the 2-forms $x_i \wedge y_i$ commute and have zero square, so $z^{\wedge n/2} = (n/2)! \sum_{|I| = n/2} \bigwedge_{i \in I} (x_i \wedge y_i)$, and $\bigwedge_{i \in I} (x_i \wedge y_i) = \det(x_{i_1}, y_{i_1}, \dots) \, e_1 \wedge \cdots \wedge e_n$. Hence it holds over $\mathbb{Z}$, and after reduction mod 2 over every commutative ring of characteristic 2.

Step 2 (bound). Put $\beta_i = \|x_i\| \, \|y_i\|$. A determinant is a sum of products containing one entry from each column, so by the ultrametric inequality each determinant in (N.6) has absolute value at most $\prod_{i \in I} \beta_i$. By the inequality of arithmetic and geometric means, $\prod_{i \in I} \beta_i \leq \big(\frac{2}{n} \sum_{i \in I} \beta_i\big)^{n/2} \leq \big(\frac{2}{n} \sum_{i \in I_0} \beta_i\big)^{n/2}$. By the ultrametric inequality again, $\sum_{i \in I_0} \beta_i \geq \frac{n}{2} |\operatorname{Pf} \Omega_z|^{2/n}$. Taking the infimum over the representations of z gives (1) on $\Lambda^2_\pi c_0(n)$. By (N.2) both sides are continuous for the coordinate topology, in which $\Lambda^2 K^n$ is dense in the completion, so (1) holds on the completion.

Step 3 (item (2)). For $\mathsf{Z} = \sum_{i \in I_0} x_i y_i^{\mathsf{T}}$, the Cauchy–Binet formula gives

$$\det \mathsf{Z} = \sum_{I \subseteq I_0, \ |I| = n} \det(x_i : i \in I) \, \det(y_i : i \in I), \tag{N.7}$$

and Step 2 applies to (N.7) with $n/2$ replaced by n; the parity of n plays no role. □

Lemma N.16 (contractions)**.** *Let $n' \geq k \geq 1$, and let $l \colon K^{n'} \to K$ be linear with $|l(x)| \leq \|x\|_\infty$ for all x. The contraction $l \lrcorner \colon \Lambda^k K^{n'} \to \Lambda^{k-1} K^{n'}$, $l \lrcorner (x_1 \wedge \cdots \wedge x_k) = \sum_j (-1)^{j-1} l(x_j) \, x_1 \wedge \cdots \wedge \widehat{x_j} \wedge \cdots \wedge x_k$, satisfies $\|l \lrcorner z\|_\pi \leq \|z\|_\pi$ for $z \in \Lambda^k_\pi c_0(n')$, and so does its extension to the completions.*

Proof. It suffices to show $\|l \lrcorner \omega\|_\pi \leq \prod_j \|x_j\|$ for $\omega = x_1 \wedge \cdots \wedge x_k$ with all $x_j \neq 0$. If $l(x_j) = 0$ for all j, then $l \lrcorner \omega = 0$. Otherwise choose j_0 with $|l(x_{j_0})| / \|x_{j_0}\|$ maximal, so that $l(x_{j_0}) \neq 0$, and put $x'_j = x_j - \big(l(x_j)/l(x_{j_0})\big) x_{j_0}$ for $j \neq j_0$. Then $l(x'_j) = 0$, and by the ultrametric inequality and the choice of j_0, $\|x'_j\| \leq \max\big(\|x_j\|, \, |l(x_j)| \, \|x_{j_0}\| / |l(x_{j_0})|\big) = \|x_j\|$. Replacing x_j by x'_j for $j \neq j_0$ does not change ω, so $l \lrcorner \omega = \pm l(x_{j_0}) \bigwedge_{j \neq j_0} x'_j$, whose projective norm is at most $|l(x_{j_0})| \prod_{j \neq j_0} \|x_j\| \leq \prod_j \|x_j\|$. □

Proof of Theorem 7.20. Step 1 (item (1)). The first assertion is Proposition N.14, with $\det = \operatorname{Pf}^2$. The assertions on σ and τ are Propositions N.12 and N.13. The map $(x, y) \mapsto \mathcal{P}_<(xy^{\mathsf{T}} + yx^{\mathsf{T}})$

is bilinear and vanishes for $x = y$, so it induces $\sigma^<$. The matrix $\mathsf{Z} = \mathcal{P}_< \Omega_{x\wedge y}$ is strictly upper triangular, of rank at most $n-1$, and $\Omega_{\wedge \mathsf{Z}} = \mathsf{Z} + \mathsf{Z}^\mathsf{T} = \Omega_{x\wedge y}$, as $\Omega_{x\wedge y}$ has zero diagonal.

Step 2 (item (2): a multiplier lift on K^n). Let $\iota_n \colon K^n \to E$ be the inclusion of the first n coordinates, for $n' \geq 1$ let $\mathrm{res}_{n'} \colon E \to c_0(n')$ be the restriction to the first n' coordinates, and for $n \leq n'$ let $\mathrm{res}_n^{n'} \colon c_0(n') \to c_0(n)$ be the restriction. The restrictions are contractive since $\|x\|_\infty \leq \|x\|_E$, so their exterior powers (Proposition 3.1) and tensor powers are contractive; they extend to the completions, and the extensions are written with the same symbols. The 0/1 vectors 1_S, $S \subseteq [n]$, have $\|1_S\|_E \leq \bar\varphi_E(n)$ and $\|1_S\|_\infty \leq 1$.

Let $n' = n + k - 2$, let $1_{>n}$ be the indicator of $\{n+1, \dots, n'\}$ and $e_{>n} = e_{n+1} \wedge \cdots \wedge e_{n'}$, the empty product for $k = 2$. Let $\eta_{>n} \colon \Lambda^k K^{n'} \to \Lambda^2 K^{n'}$ be the composite of the contractions by the coordinate functionals of $n+1, \dots, n'$, in this order, and the identity for $k = 2$; then $\eta_{>n}(w \wedge e_{>n}) = w$ for $w \in \Lambda^2 K^n$. Put $\theta(w) = (\Lambda^2_\pi \iota_n w) \wedge e_{>n} \in \Lambda^k_\pi E$ for $w \in \Lambda^2 K^n$. As $\Lambda^k_\pi \mathrm{res}_{n'}(\theta w) = w \wedge e_{>n}$ in $\Lambda^k K^{n'}$,

$$\Lambda^2_\pi \mathrm{res}_n^{n'} \circ \eta_{>n} \circ \Lambda^k_\pi \mathrm{res}_{n'} \circ \theta = \mathrm{id} \qquad \text{on } \Lambda^2 K^n. \tag{N.8}$$

For $a, b \in K^n$, regarded in $c_{00}(\mathbb{N}, K)$, let $Q_2(a,b)$ be the sum, over the pairs of slots $j < j'$, of Q evaluated with a in slot j, b in slot j' and $1_{>n}$ in the other slots, and put

$$\Psi(a,b) = \Lambda^2_\pi \mathrm{res}_n^{n'} \circ \eta_{>n} \circ \Lambda^k_\pi \mathrm{res}_{n'} \circ Q_2(a,b) \circ \theta. \tag{N.9}$$

This is a K-linear map $\Lambda^2 K^n \to \Lambda^2 \widehat K^n$, K-bilinear in (a,b); extend it $\widehat K$-linearly in w and $\widehat K$-bilinearly in (a,b). For $\lambda \in K$, multilinearity gives $Q(a+\lambda 1_{>n}, \dots, a+\lambda 1_{>n}) = \sum_{j=0}^k \lambda^{k-j} q_j(a)$, where $q_j(a)$ is the sum, over the j-element sets of slots, of Q with a in these slots and $1_{>n}$ elsewhere; so $q_2(a) = Q_2(a,a)$. As $\mu_{a+\lambda 1_{>n}}$ multiplies e_t by a_t for $t \leq n$ and by λ for $n < t \leq n'$, $\Lambda^k_\pi \mu_{a+\lambda 1_{>n}}(\theta w) = \lambda^{k-2} \theta(\Lambda^2 \mu_a w)$. As K is infinite, comparing the coefficients of λ^{k-2} gives $Q_2(a,a)(\theta w) = \theta(\Lambda^2 \mu_a w)$, and (N.8) gives $\Psi(a,a) = \Lambda^2 \mu_a$ for $a \in K^n$. Taking $a = e_t$, $a = e_{t'}$ and $a = e_t + e_{t'}$, $t \neq t'$, (7.4) gives $\Psi(e_t, e_{t'}) + \Psi(e_{t'}, e_t) = \Lambda^2 \mu_{e_t + e_{t'}} - \Lambda^2 \mu_{e_t} - \Lambda^2 \mu_{e_{t'}} = \pi_{tt'}$, so Ψ is a multiplier lift on $\widehat K^n$.

Step 3 (item (2): the bound). Let $a, b, x, y \in \{0,1\}^n$. Each of the $k(k-1)/2$ terms of $Q_2(a,b)$ has norm at most $\|Q\|$; $\Lambda^2_\pi \mathrm{res}_n^{n'}$ and $\Lambda^k_\pi \mathrm{res}_{n'}$ are contractive; $\eta_{>n}$ is contractive by Lemma N.16; and $\theta(x \wedge y)$ is a single decomposable vector, so $\|\theta(x\wedge y)\|_\pi \leq \|x\|_E \|y\|_E \prod_{t=n+1}^{n'} \|e_t\|_E \leq \bar\varphi_E(n)^2$. Hence

$$\|\Psi(a,b)(x \wedge y)\|_\pi \leq \tfrac{k(k-1)}{2} \, \|Q\| \, \bar\varphi_E(n)^2, \tag{N.10}$$

the norm being that of the completion of $\Lambda^2_\pi c_0(n)$. By (N.3) over $\widehat K$, a finite sum of the values $\mathrm{Pf}\, \mathsf{M}_\Psi(a,b,x,y)$ at 0/1 inputs is 1. As $\widehat K$ is nonarchimedean, one of them has absolute value at least 1, and Lemma N.15(1) gives $\|\Psi(a,b)(x\wedge y)\|_\pi \geq n/2$ for that input. With (N.10) this is (7.7).

Step 4 (item (3): sections). Let $\sigma = (\mathrm{res}_n \otimes \mathrm{res}_n) \circ s \circ \Lambda^2_\pi \iota_n$, extended $\widehat K$-linearly to $\Lambda^2 \widehat K^n \to \widehat K^{n\times n}$. By naturality of $\wedge$, $\wedge \circ \sigma = \Lambda^2_\pi \mathrm{res}_n \circ \widehat\wedge{}^2_E \circ s \circ \Lambda^2_\pi \iota_n = \mathrm{id}$ on $\Lambda^2 K^n$, hence on $\Lambda^2 \widehat K^n$ by linearity. Proposition N.12 over $\widehat K$ gives $\sum_{S,T\subseteq[n]} \det \mathsf{N}(1_S, 1_T) = 1$ for $\mathsf{N}(x,y) = \sigma(x\wedge y) + xy^\mathsf{T}$, so some $|\det \mathsf{N}(1_S,1_T)| \geq 1$, and Lemma N.15(2) gives $n \leq \|\mathsf{N}(1_S,1_T)\|_\pi \leq \|s\| \, \bar\varphi_E(n)^2 + 1$, so $\|s\| \geq (n-1)/\bar\varphi_E(n)^2$. For $c_0(n)$ itself take ι_n and res_n to be the identity here and in Steps 5 and 6.

Step 5 (item (3): retractions). Let n be even; for odd n apply the even case to $n-1$, as $\bar\varphi_E$ is nondecreasing. Let $\tau = \Lambda^2_\pi \mathrm{res}_n \circ \rho \circ (\iota_n \otimes \iota_n)$, extended $\widehat K$-linearly to $\widehat K^{n\times n} \to \Lambda^2 \widehat K^n$. For $w \in \Lambda^2 K^n$, $\mathrm{alt}^2 w = \Omega_w$, so $\tau(\Omega_w) = \Lambda^2_\pi \mathrm{res}_n \circ \rho \circ \mathrm{alt}^2_E(\Lambda^2_\pi \iota_n w) = w$, and by linearity $\tau(\Omega_w) = w$

for $w \in \Lambda^2 \widehat{K}^n$. Proposition N.13 over $\widehat{K}$ gives $\sum_{S\subseteq[n]} \operatorname{Pf} \Omega_{\tau(1_S 1_S^\mathsf{T})} = 1$, so some $|\operatorname{Pf} \Omega_{\tau(1_S 1_S^\mathsf{T})}| \geq 1$, and Lemma N.15(1) gives $n/2 \leq \|\tau(1_S 1_S^\mathsf{T})\|_\pi \leq \|\rho\|\, \bar\varphi_E(n)^2$.

Step 6 (item (3): projections). Let $\mathsf{N}(x,y) = (\operatorname{res}_n \otimes \operatorname{res}_n) R_2(\iota_n x \otimes \iota_n y)$, extended $\widehat{K}$-bilinearly. Its values are symmetric matrices, since $\widehat{\Gamma}^2_\pi E$ is the closed span of the pure squares and $\operatorname{res}_n \otimes \operatorname{res}_n$ maps pure squares to symmetric matrices, which form a closed subspace. As R_2 fixes pure squares, $\mathsf{N}(e_t, e_t) = e_t e_t^\mathsf{T}$ and, polarizing $(e_t + e_{t'})^{\otimes 2}$, $\mathsf{N}(e_t, e_{t'}) + \mathsf{N}(e_{t'}, e_t) = e_t e_{t'}^\mathsf{T} + e_{t'} e_t^\mathsf{T}$; so $\mathsf{N}(x,x) = xx^\mathsf{T}$ for $x \in \widehat{K}^n$ by bilinear expansion. So N is an $[n]$-polarization over $\widehat{K}$ (Definition N.7), and Lemmas N.9, N.10 and N.11 give $\sum_{S,T} \det \mathsf{N}(1_S, 1_T) = 1$. As in Step 4, $n \leq \|\mathsf{N}(1_S, 1_T)\|_\pi \leq \|R_2\|\, \bar\varphi_E(n)^2$.

Step 7 (item (4): goodness). Each listed E is a sequence space with $\|e_t\|_E = 1$ and $\|\mu_a\|_{E\to E} \leq \|a\|_\infty$ for finitely supported a, and $\sup_n n/\bar\varphi_E(n)^2 = \infty$. For a symmetric sequence space, Definition 7.10 gives a norm that is monotone in $|x|$ and equal to 1 at the unit vectors; so $\bar\varphi_E = \varphi_E$, $\|x\|_\infty \leq \|x\|_E$, and $|ax| \leq |\lambda x|$ for a coordinate λ of a with $|\lambda| = \|a\|_\infty$ gives $\|\mu_a\| \leq \|a\|_\infty$. As $\bar\varphi_E$ is nondecreasing, $(n-1)/\bar\varphi_E(n-1)^2 \geq (n-1)/\bar\varphi_E(n)^2$, so $n/\bar\varphi_E(n)^2$ is also unbounded along even n. If (E,E) were good for Banach values in degree k, Lemma 3.4 would give a bounded k-linear $\widetilde{Q}\colon \mathcal{L}(E,E)^k \to \mathcal{L}(\Lambda^k_\pi E, \widehat{\Lambda}^k_\pi E)$ with $\widetilde{Q}(u,\dots,u) = \Lambda^k_\pi u$, and $Q(a_1,\dots,a_k) = \widetilde{Q}(\mu_{a_1},\dots,\mu_{a_k})$ would satisfy the hypotheses of item (2) with $\|Q\| \leq \|\widetilde{Q}\|$, contradicting (7.7) for large even n. Lemma 3.4, Theorem 3.2(1) and the dichotomy of Proposition 2.5 then give the assertion on $A^k_{E,E;F}$, and Proposition 3.5(1),(2) the failure of (S^k_E) and (R^k_E). Item (3) gives the failure of (Γ^2_E).

Step 8 (item (4): nonarchimedean norms). Let $h \geq 1$, $n = 2h$, $n' = n + k - 2$, and $z_h = \sum_{i=1}^h e_{2i-1} \wedge e_{2i} \wedge e_{>n} \in \Lambda^k_\pi E$, a sum of h decomposable vectors of norm at most 1. The contractive maps $\Lambda^k_\pi \operatorname{res}_{n'}$, $\eta_{>n}$ and $\Lambda^2_\pi \operatorname{res}^{n'}_n$ send it to $\sum_{i\leq h} e_{2i-1} \wedge e_{2i}$, whose Pfaffian is 1, so $\|z_h\|_\pi \geq h$ by Lemma N.15(1). If $\|\cdot\|'$ were a nonarchimedean norm on $\widehat{\Lambda}^k_\pi E$ with $c\|\cdot\|_\pi \leq \|\cdot\|' \leq C\|\cdot\|_\pi$, then $ch \leq \|z_h\|' \leq C$ for all h, which is impossible.

Step 9 (item (4): pairs). Let $2 < r_1, r_2 \leq \infty$, write ℓ^{r_i} for $\ell^{r_i}(\mathbb{N}, K)$, or for $c_0(\mathbb{N}, K)$ if $r_i = \infty$, and put $1/\infty = 0$. If (ℓ^{r_1}, ℓ^{r_2}) were good for Banach values in degree k, Lemma 3.4 would give a bounded k-linear $\widetilde{Q}\colon \mathcal{L}(\ell^{r_1}, \ell^{r_2})^k \to \mathcal{L}(\Lambda^k_\pi \ell^{r_1}, \widehat{\Lambda}^k_\pi \ell^{r_2})$ with $\widetilde{Q}(u,\dots,u) = \Lambda^k_\pi u$. Run Steps 2 and 3 with $Q(a_1,\dots,a_k) = \widetilde{Q}(\mu_{a_1},\dots,\mu_{a_k})$, where μ_a is now a map $\ell^{r_1} \to \ell^{r_2}$, with $\theta_1(w) = (\Lambda^2_\pi \iota_n w) \wedge e_{>n}$ in $\Lambda^k_\pi \ell^{r_1}$ in place of θ on the right of (N.9) and the same expression $\theta_2(w)$ in $\Lambda^k_\pi \ell^{r_2}$ in the identity $\Lambda^k_\pi \mu_{a+\lambda 1_{>n}} \circ \theta_1 = \lambda^{k-2} \theta_2 \circ \Lambda^2 \mu_a$, and with $\operatorname{res}_{n'}$ defined on ℓ^{r_2}. Put $\delta = \max(0, 1/r_2 - 1/r_1)$. Hölder's inequality gives $\|\mu_a\| \leq |\operatorname{supp} a|^\delta$ for 0/1 vectors a, and $\|1_S\|_{r_1} \leq n^{1/r_1}$, so $n/2 \leq \frac{k(k-1)}{2} \|\widetilde{Q}\| (k-2)^{(k-2)\delta} n^{2\delta + 2/r_1}$, where $(k-2)^{(k-2)\delta} = 1$ for $k = 2$. As $2\delta + 2/r_1 = 2/\min(r_1, r_2) < 1$, this fails for large even n. □

Appendix O. Proofs for familywise analyticity

This appendix proves the results of Section 11 not proved there, in its notation. We write $h^{\times n} = (h,\dots,h) \in P^n$, h_I for the tuple of the h_i with $i \in I$, and $\operatorname{Mult}^0(P;Y) = Y$.

O.1. Power series and mixed exterior powers.

Lemma O.1. (1) *If Y is a Banach space and $b_n \in \operatorname{Mult}^n(P;Y)$, $\|b_n\| \leq CR^{-n}$, then $x \mapsto \sum_n b_n(x^{\times n})$ is analytic on the open ball of radius R about 0.*

(2) *Let $\Phi\colon Y_1 \times \cdots \times Y_m \to Y$ be bounded m-linear, the spaces normed, and $f_a\colon U \to Y_a$ with $f_a(x_0 + h) = \sum_n b^a_n(h^{\times n})$ for $\|h\| < R$, $\|b^a_n\| \leq CR^{-n}$. Then $\Phi(f_1(x_0+h),\dots,f_m(x_0+h)) = \sum_N \Phi_N(h^{\times N})$ for $\|h\| < R$, where $\Phi_N(h_1,\dots,h_N) = \sum_{n_1+\cdots+n_m=N} \Phi(b^1_{n_1}(h_{I_1}),$*

$\dots, b^m_{n_m}(h_{I_m})$), I_a consecutive of sizes n_a, has norm at most $\|\Phi\|\binom{N+m-1}{m-1}C^m R^{-N}$. So the composite is analytic at x_0, and on U if the f_a are.

Proof. (1) For $\|x_1\| < R$, the sum b'_j over $n \geq j$ and j-element $S \subseteq \{1,\dots,n\}$ of b_n with $h_1,\dots,h_j$ in the slots of S and x_1 elsewhere converges in $\mathrm{Mult}^j(P;Y)$ with a geometric bound, as $\sum_{n\geq j}\binom{n}{j}R^{-n}\|x_1\|^{n-j} = R(R-\|x_1\|)^{-j-1}$; for $\|h\| < R - \|x_1\|$ the majorant $\sum_n \|b_n\|(\|x_1\|+\|h\|)^n$ of the multilinear expansion gives $\sum_n b_n((x_1+h)^{\times n}) = \sum_j b'_j(h^{\times j})$. (2) The bound counts compositions of N. For $\|h\| < R$ the terms $\Phi(b^1_{n_1}(h^{\times n_1}),\dots,b^m_{n_m}(h^{\times n_m}))$ have the majorant $\|\Phi\|\prod_a \sum_n C(\|h\|/R)^n$; their sums over $\max_a n_a \leq L$ are values of Φ at partial sums of the f_a, and differ from their sums over $n_1+\cdots+n_m \leq L$ by terms with $\max_a n_a > L/m$, whose majorant tends to 0. □

Lemma O.2. *Let G be a Banach space, $v_1,\dots,v_m \in \mathcal{L}(E,G)$, and $\mathbf{i} \in \mathbb{N}_0^m$ a* type, *$i_1+\cdots+i_m = k \geq 1$; a map $\sigma\colon \{1,\dots,k\} \to \{1,\dots,m\}$ has type $\mathbf{i}$ if $|\sigma^{-1}(l)| = i_l$ for all l, and there are $\binom{k}{\mathbf{i}} = k!/(i_1!\cdots i_m!)$ of them. Entries $i_l = 0$ use $\Lambda^0_\pi v = \mathrm{id}_K$ and the conventions stated with (6.3).*

1. *There is a unique $\Lambda^{\mathbf{i}}_\pi(v_1,\dots,v_m) \in \mathcal{L}(\Lambda^k_\pi E, \widehat{\Lambda}^k_\pi G)$ sending $x_1\wedge\cdots\wedge x_k$ to $\sum_\sigma v_{\sigma(1)}x_1 \wedge \cdots \wedge v_{\sigma(k)}x_k$, over the σ of type $\mathbf{i}$; its norm is at most $\binom{k}{\mathbf{i}}\prod_l \|v_l\|^{i_l}$.*
2. *The iterated shuffle $\mathrm{sh}^{\mathbf{i}}_E = (\mathrm{id}\otimes \mathrm{sh}^{(i_2,\dots,i_m)}_E)\circ \mathrm{sh}^{i_1,k-i_1}_E$, $\mathrm{sh}^{(k)}_E = \mathrm{id}$, into $\Lambda^{i_1}_\pi E \otimes_\pi \cdots \otimes_\pi \Lambda^{i_m}_\pi E$ has norm at most $\binom{k}{\mathbf{i}}$, the iterated wedge product $\wedge^{\mathbf{i}}_G$ from $\widehat{\Lambda}^{i_1}_\pi G \mathbin{\widehat{\otimes}}_\pi \cdots \mathbin{\widehat{\otimes}}_\pi \widehat{\Lambda}^{i_m}_\pi G$ to $\widehat{\Lambda}^k_\pi G$ is contractive, and $\Lambda^{\mathbf{i}}_\pi(v_1,\dots,v_m) = \wedge^{\mathbf{i}}_G \circ (\Lambda^{i_1}_\pi v_1 \otimes \cdots \otimes \Lambda^{i_m}_\pi v_m) \circ \mathrm{sh}^{\mathbf{i}}_E$.*

Proof. Lemma 6.1, iterated and extended to completions, and $\|\mathrm{id}\otimes T\| \leq \|T\|$ give the norm bounds of (2). By induction on m and (6.1), $\mathrm{sh}^{\mathbf{i}}_E(x_1\wedge\cdots\wedge x_k)$ is the sum of the $\epsilon(S_1,\dots,S_m)\, x_{S_1}\otimes\cdots\otimes x_{S_m}$ over the ordered partitions with $|S_l| = i_l$, where $\epsilon(S_1,\dots,S_m)$, the sign of the listing of $S_1,\dots,S_m$ in turn, each increasingly, is $\epsilon(S_1)$ times the sign of the listing of $S_2,\dots,S_m$ relative to the increasing listing of the complement of S_1. With $\sigma(c) = l$ for $c \in S_l$ and $y_c = v_{\sigma(c)}x_c$, the composite in (2) sends $x_1\wedge\cdots\wedge x_k$ to the sum of the $\epsilon(S_1,\dots,S_m)\, y_{S_1}\wedge\cdots\wedge y_{S_m} = y_1\wedge\cdots\wedge y_k$, and ordered partitions correspond bijectively to maps of type $\mathbf{i}$. So the composite is the operator of (1), unique as pure wedges span, and $\|\Lambda^{i_l}_\pi v_l\| \leq \|v_l\|^{i_l}$ gives its bound. □

Lemma O.3. *Let G be a Banach space and $a,b \in \mathcal{L}(E,G)$. Then $\Lambda^k_\pi(a+\lambda b) = \sum_{j=0}^k \lambda^j \Lambda^{(k-j,j)}_\pi(a,b)$ for $\lambda \in K$, where $\Lambda^{(k-j,j)}_\pi(a,b) = \wedge^{k-j,j}_G \circ (\Lambda^{k-j}_\pi a \otimes \Lambda^j_\pi b)\circ \mathrm{sh}^{k-j,j}_E$ has norm at most $\binom{k}{j}\|a\|^{k-j}\|b\|^j$, and $\Lambda^{(0,k)}_\pi(a,b) = \Lambda^k_\pi b$. If $\lambda_0,\dots,\lambda_k \in K$ are distinct and (χ_{ji}) is the inverse of $(\lambda_i^j)_{0\leq i,j\leq k}$, then $\Lambda^k_\pi b = \sum_i \chi_{ki}\Lambda^k_\pi(a+\lambda_i b)$.*

Proof. Equation (6.3) with $w_1 = a$, $w_2 = \lambda b$ and $\Lambda^j_\pi(\lambda b) = \lambda^j \Lambda^j_\pi b$ gives the expansion, and Lemma 6.1 the bound. The Vandermonde matrix being invertible, the identities $\Lambda^k_\pi(a+\lambda_i b) = \sum_j \lambda_i^j \Lambda^{(k-j,j)}_\pi(a,b)$ give $\Lambda^{(k-j,j)}_\pi(a,b) = \sum_i \chi_{ji}\Lambda^k_\pi(a+\lambda_i b)$; take $j = k$. □

Lemma O.4. *Let H be a Banach space, $U' \subseteq P$ open and $w\colon U' \to \mathcal{L}(H)$ analytic with $\sup_{U'}\|w\| < 1$. Then $1+w$ is an invertible family on U'.*

Proof. Fix $x_1 \in U'$ and let $g_1 = 1+w(x_1)$, so $\|g_1^{-1}\| \leq (1-\sup_{U'}\|w\|)^{-1}$ by the Neumann series. Write $w(x_1+h) - w(x_1) = \delta(h) = \sum_{n\geq 1} b_n(h^{\times n})$, $\|b_n\| \leq CR^{-n}$. With $C' = \|g_1^{-1}\|C$ and $t = \|h\|/R < 1/(1+C')$, $\|g_1^{-1}\delta(h)\| \leq C't/(1-t) < 1$, so $(1+w(x_1+h))^{-1} = \sum_l (-g_1^{-1}\delta(h))^l g_1^{-1}$, whose multilinear expansion has the majorant $\|g_1^{-1}\|\sum_l (C't/(1-t))^l$; its degree-$N$ coefficient,

the sum over the compositions $(n_1,\dots,n_l)$ of N of $(-1)^l g_1^{-1}b_{n_1}(h_{I_1})g_1^{-1}\cdots b_{n_l}(h_{I_l})g_1^{-1}$, has norm at most $\|g_1^{-1}\|(1+C')^N R^{-N}$. □

O.2. **Reductions and what bundles need.**

Proof of Proposition 11.2. Step 1 (item (1)). Let $u\colon U\to\mathcal{L}(H)$ be analytic and $x_0\in U$. Choose an open $U_0\subseteq U$ containing x_0 with $\|u\|\le C=\|u(x_0)\|+1$ on U_0, an integer $m\ge 0$ with $|\vartheta|^m C<1$, and the distinct $\lambda_i=\vartheta^{m+i}$, $0\le i\le k$. By Lemma O.4 every $1+\lambda_i u$ is an invertible family on U_0, and Lemma O.3 with $G=H$, $a=1$, $b=u(x)$ gives $\Lambda^k_\pi u=\sum_i \chi_{ki}\Lambda^k_\pi(1+\lambda_i u)$ on U_0.

Step 2 (item (2)). Let $\alpha_1\colon E_0\to E$, $\alpha_2\colon D_0\to D$, $\psi\colon P_0\to P$ have bounded linear left inverses α_1', α_2', ψ'. For analytic $u_0\colon U'\to\mathcal{L}(E_0,\widehat{D}_0)$, $U'\subseteq P_0$ open, $u=\hat{\alpha}_2 u_0(\psi'\cdot)\alpha_1'$ is analytic on $\psi'^{-1}(U')$, hats denoting extensions to completions; hence so is $\Lambda^k_\pi u$, and $\Lambda^k_\pi u_0(y)=\widehat{\Lambda}^k_\pi\hat{\alpha}_2'\circ\Lambda^k_\pi u(\psi y)\circ\Lambda^k_\pi\alpha_1$ (Proposition 3.1). For $E\oplus E'$, "only if" is the retract case; conversely, for analytic $u\colon U\to\mathcal{L}(E\oplus E',\widehat{D})$, the injections ι_l and projections π_l of $E\oplus E'$, $u_l=u\iota_l$ and $k'\le k$, $u=u_1\pi_1+u_2\pi_2$, (6.3) and Proposition 3.1 give

$$\Lambda^{k'}_\pi u=\sum_{j=0}^{k'}\wedge^{j,k'-j}_{\widehat{D}}\circ\big((\Lambda^j_\pi u_1\circ\Lambda^j_\pi\pi_1)\otimes(\Lambda^{k'-j}_\pi u_2\circ\Lambda^{k'-j}_\pi\pi_2)\big)\circ\mathrm{sh}^{j,k'-j}_{E\oplus E'},$$

bounded bilinear (linear for $j\in\{0,k'\}$) in analytic maps; apply Lemma O.1(2). The target slot is the same, with $u=\sum_l \iota_l'\pi_l' u$ for the injections ι_l' and projections π_l' of $\widehat{D}\oplus\widehat{D'}$ and $\Lambda^j_\pi(\iota_l'\pi_l' u)=\widehat{\Lambda}^j_\pi\iota_l'\circ\Lambda^j_\pi(\pi_l' u)$.

Step 3 (item (3)). Let g be as in Proposition 10.1, with $H=D\oplus E$, π_D the projection onto D and i_E the injection of E. For analytic $u\colon U\to\mathcal{L}(E,D)$, $x\mapsto g(u(x))$ is an invertible family, with inverse $g(-u(x))$, and $\pi_D g(u(x))i_E=u(x)$ and Proposition 3.1 give

$$\Lambda^k_\pi u(x)=\widehat{\Lambda}^k_\pi\pi_D\circ\Lambda^k_\pi g(u(x))\circ\Lambda^k_\pi i_E. \tag{O.1}$$

In degrees $j\le k$, analyticity along invertible families into $\mathcal{L}(H)$ gives goodness of $(P;H)$ by (1), hence of the four triples by (2); conversely, goodness of the four triples gives that of $(P;H)$ by (2) in both slots, and invertible families are analytic. For $\mathcal{C}$, goodness of $(P;G,G)$ for $G\in\mathcal{C}$ covers invertible families into $\mathcal{L}(G)$; conversely, for $E,D\in\mathcal{C}$, $D\oplus E\in\mathcal{C}$ and (O.1) applies. □

Proof of Corollary 11.3. Step 1 ((2) implies the last assertion and (1)). Let $E,D\in\mathcal{C}$, let F,F' be Banach spaces, and let $u\colon U\to\mathcal{L}(E,D)$ and $v\colon U\to\mathcal{L}(F,F')$ be analytic on an open $U\subseteq P$. By (2) and the paragraph after Definition 11.1, $x\mapsto A^k_{E,D;F}(u(x))$ is analytic. The joint action $B(u(x),v(x))$, which is $A^k_{E,D;F}(u(x))$ followed by postcomposition with $v(x)$, is bounded bilinear in the two families, hence analytic (Lemma O.1(2)). The last assertion of Theorem 2.1, for restricted categories of fibers as in the paragraph after it, gives the last assertion; (1) is its case of a trivial bundle with fiber F.

Step 2 ((1) implies (2)). Let $E,D\in\mathcal{C}$, let $u\colon U\to\mathcal{L}(E,D)$ be analytic on an open $U\subseteq P$, and let $H=D\oplus E\in\mathcal{C}$ and g be as in Proposition 10.1. The trivial bundle $U\times H$ with the trivializations id and $(x,z)\mapsto(x,g(u(x))z)$ is analytic, since its transitions $g(\pm u(x))$ are affine in $u(x)$. By (1), the induced transition $x\mapsto g(u(x))^*$ on $\mathrm{Alt}^k(H;F)$ is analytic for every Banach space F. The compression (10.1) makes $x\mapsto A^k_{E,D;F}(u(x))$ analytic, so $\Lambda^k_\pi u$ is analytic on U by Theorem 3.2(1) with $Y=\widehat{\Lambda}^k_\pi D$; as D is complete, this is (2). □

O.3. **Homogeneous families.** A *content* is a finitely supported $\mathbf{m}\colon \mathbb{N}_0 \to \mathbb{N}_0$ with $\sum_n \mathbf{m}(n) = k$; its degree is $N(\mathbf{m}) = \sum_n n\mathbf{m}(n)$, and its type is $(\mathbf{m}(n_1), \dots, \mathbf{m}(n_m))$ for its support $n_1 < \cdots < n_m$. The content of $(n_1', \dots, n_k') \in \mathbb{N}_0^k$ counts repeated values, so at most $\binom{N+k-1}{k-1}$ contents have degree N.

Lemma O.5. *Let $n \geq 0$, $j \geq 1$ and $\beta \in \mathrm{Mult}^n(P; \mathcal{L}(E, \widehat{D}))$.*

1. *$c_j(\beta)$ depends only on u_β, and $c_j(\lambda\beta) = |\lambda|^j c_j(\beta)$ for $\lambda \in K^\times$.*
2. *$c_1(\beta) \leq \|\beta\|$, and $c_j(\beta) \leq (j!/|j!|)\|\beta\|^j$ if $j! \neq 0$ in K, the integer $j!$ in the numerator.*
3. *For $n \geq 1$, $h \mapsto \Lambda_\pi^j u_\beta(h)$ is analytic at some point of P if and only if $c_j(\beta) < \infty$, and it is then a continuous polynomial.*
4. *$c_j(b\beta a) \leq \|b\|^j \|a\|^j c_j(\beta)$ for bounded linear $a\colon E' \to E$, $b\colon \widehat{D} \to \widehat{D'}$, with $(b\beta a)(h_1, \dots, h_n) = b\,\beta(h_1, \dots, h_n)\,a$.*

Proof. (1) The definition involves only $\Lambda_\pi^j u_\beta$, and $\lambda^j q$ lifts $u_{\lambda\beta}$ if q lifts u_β; apply this also to λ^{-1}. (2) β lifts u_β in degree 1. If $j! \neq 0$, the section $s = \mathrm{alt}_E^j / j!$ of $\wedge_E^j$ has norm at most $j!/|j!|$ (Proposition 4.1(1)), and $q(h_1, \dots, h_{jn}) = \widehat{\wedge}_D^j \circ (\beta(h_{I_1}) \otimes \cdots \otimes \beta(h_{I_j})) \circ s$, blocks of size n, has diagonal $\Lambda_\pi^j u_\beta(h) \wedge_E^j s = \Lambda_\pi^j u_\beta(h)$ by naturality of the wedge, and norm at most $(j!/|j!|)\|\beta\|^j$. (3) $T \mapsto T \circ \wedge_E^j$ is an isometry of $\mathcal{L}(\Lambda_\pi^j E, \widehat{\Lambda}_\pi^j D)$ onto the operators in $\mathcal{L}(\otimes_\pi^j E, \widehat{\Lambda}_\pi^j D)$ vanishing on $\ker \wedge_E^j$ (Lemma 3.7), a closed subspace, and $(h_1, \dots, h_{jn}) \mapsto \widehat{\wedge}_D^j \circ (\beta(h_{I_1}) \otimes \cdots \otimes \beta(h_{I_j}))$ is an ambient representative of degree $jn \geq 1$; apply Proposition 2.5. (4) If q lifts u_β, then $\widehat{\Lambda}_\pi^j b \circ q \circ \Lambda_\pi^j a$ lifts $u_{b\beta a}$ (Proposition 3.1). □

Lemma O.6. *Let $u(x_0 + h) = \sum_{n \geq 0} \beta_n(h^{\times n})$ for $\|h\| < R$, $\beta_n \in \mathrm{Mult}^n(P; \mathcal{L}(E, \widehat{D}))$, $\|\beta_n\| \leq CR^{-n}$, $a = \beta_0$, and let $C_0, C_1 \geq 1$ and lifts $q_{n,j}$ of u_{β_n} in degree j with $\|q_{n,j}\| \leq C_0 C_1^n$ be given for $j \leq k$, $n \geq 1$. Then $\Lambda_\pi^k u$ is analytic at x_0, with Nth coefficient of norm at most $k!\binom{N+k-1}{k-1} \max(1, \|a\|)^k C_0^k C_1^N$.*

Proof. Step 1 (one content). For a content $\mathbf{m}$ of support $n_1 < \cdots < n_m$ and type $\mathbf{i}$ let $Q_l = q_{n_l, i_l}$ if $n_l \geq 1$, $Q_l = \Lambda_\pi^{i_l} a$ if $n_l = 0$, and, with consecutive blocks I_l of sizes $n_l i_l$, $q_{\mathbf{m}}(h_1, \dots, h_{N(\mathbf{m})}) = \wedge_{\widehat{D}}^{\mathbf{i}} \circ (Q_1(h_{I_1}) \otimes \cdots \otimes Q_m(h_{I_m})) \circ \mathrm{sh}_E^{\mathbf{i}}$. By Lemma O.2(2), $q_{\mathbf{m}}(h^{\times N(\mathbf{m})}) = \Lambda_\pi^{\mathbf{i}}(u_{\beta_{n_1}}(h), \dots, u_{\beta_{n_m}}(h))$ and $\|q_{\mathbf{m}}\| \leq \binom{k}{\mathbf{i}} \prod_l \|Q_l\| \leq k! \max(1, \|a\|)^k C_0^k C_1^{N(\mathbf{m})}$, as $m \leq k$ and $C_0, C_1 \geq 1$. The sum q_N of the $q_{\mathbf{m}}$ of degree N obeys the stated bound, and $\sum_N \|q_N\| \|h\|^N < \infty$ for $\|h\| < 1/(2C_1)$.

Step 2 (the expansion). For $\|h\| < R$ and $y = x_1 \wedge \cdots \wedge x_k$, as in Lemma O.1(2), $\Lambda_\pi^k u(x_0 + h)y = \omega_{\widehat{D}}(\sum_n u_{\beta_n}(h)x_1, \dots, \sum_n u_{\beta_n}(h)x_k)$ is the absolutely convergent sum of the $u_{\beta_{\sigma(1)}}(h)x_1 \wedge \cdots \wedge u_{\beta_{\sigma(k)}}(h)x_k$, $\sigma\colon \{1, \dots, k\} \to \mathbb{N}_0$. Grouped by the content of $(\sigma(1), \dots, \sigma(k))$ and then by degree, they give $\sum_N q_N(h^{\times N})y$ (Lemma O.2(1)); as pure wedges span, $\Lambda_\pi^k u(x_0 + h) = \sum_N q_N(h^{\times N})$ for $\|h\| < \min(R, 1/(2C_1))$. □

Lemma O.7. (1) *If $(P; E, D)$ is good in degrees $a, b \geq 1$ and $\binom{a+b}{a} \neq 0$ in K, it is good in degree $a + b$.*

(2) *Let $k \geq 2$, with maximum norms on $E \oplus K$, $D \oplus K$. If $(P; E \oplus K, D \oplus K)$ is good in degree k, then $(P; E, D)$ is good in degree $k - 1$.*

Proof. (1) Let $n = a + b$. For $v \in \mathcal{L}(E, \widehat{D})$, naturality of the wedge product and (6.2), extended to completions, give $\Lambda_\pi^n v = \binom{n}{a}^{-1} \wedge_{\widehat{D}}^{a,b} \circ (\Lambda_\pi^a v \otimes \Lambda_\pi^b v) \circ \mathrm{sh}_E^{a,b}$, bounded bilinear in $(\Lambda_\pi^a v, \Lambda_\pi^b v)$; apply Lemma O.1(2). (2) Let $u\colon U \to \mathcal{L}(E, \widehat{D})$ be analytic and F a Banach

space. As $u \oplus 1$ is analytic into $\mathcal{L}(E \oplus K, \widehat{D} \oplus \widehat{K})$ and $\widehat{D} \oplus \widehat{K}$ completes $D \oplus K$, goodness and the prose after Definition 11.1 make $x \mapsto A^k_{E\oplus K, \widehat{D}\oplus\widehat{K};F}(u(x) \oplus 1)$ analytic, hence also $x \mapsto A^k_{E\oplus K, \widehat{D}\oplus K;F}(u(x) \oplus 1)$, restriction $\mathrm{Alt}^k(\widehat{D} \oplus \widehat{K}; F) \to \mathrm{Alt}^k(\widehat{D} \oplus K; F)$ being an isometric isomorphism by density. The proof of Theorem 6.4, with $\widehat{D}$ for D, gives fixed bounded linear R_h, J_h with $A^{k-1}_{E,\widehat{D};F}(v) = R_h \circ A^k_{E\oplus K, \widehat{D}\oplus K;F}(v \oplus 1) \circ J_h$; so $x \mapsto A^{k-1}_{E,\widehat{D};F}(u(x))$ is analytic for every Banach F, and the paragraph after Definition 11.1 applies. □

Proof of Theorem 11.5. Step 1 (item (1)). Let (β_n) be as in (11.1). Each u_{β_n} is analytic (Proposition 2.5), so $c_j(\beta_n) < \infty$ by goodness and Lemma O.5(3). If $\limsup_n c_j(\beta_n)^{1/n} = \infty$, choose $n_1 < n_2 < \dots$ with $n_{i+1} > jn_i$ and $c_j(\beta_{n_i})^{1/n_i} \geq i$. By Lemma O.1(1) and goodness, $u(h) = \sum_i \beta_{n_i}(h^{\times n_i})$ is analytic on the open unit ball and $\Lambda^j_\pi u(h) = \sum_N q_N(h^{\times N})$ for $\|h\| < R'$, $\|q_N\| \leq C'R'^{-N}$. As in Step 2 of the proof of Lemma O.6, $\Lambda^j_\pi u(h)(x_1 \wedge \dots \wedge x_j) = \sum_N s_N(h)$, with $s_N(h)$ the sum of the $u_{\beta_{\sigma(1)}}(h)x_1 \wedge \dots \wedge u_{\beta_{\sigma(j)}}(h)x_j$ over the $\sigma\colon \{1, \dots, j\} \to \{n_1, n_2, \dots\}$ with $\sum_a \sigma(a) = N$, and Lemma A.1 on the lines $t \mapsto th$ gives $s_N(h) = q_N(h^{\times N})(x_1 \wedge \dots \wedge x_j)$. For $N = jn_i$ only $\sigma \equiv n_i$ qualifies, as $n_{i+1} > jn_i$ and $jn_{i-1} < n_i$; so q_{jn_i} lifts $u_{\beta_{n_i}}$, and $c_j(\beta_{n_i})^{1/n_i} \leq C'^{1/n_i}R'^{-j}$ stays bounded, a contradiction.

Step 2 (item (2)). Let $u\colon U \to \mathcal{L}(E, \widehat{D})$ be analytic, $u(x_0 + h) = \sum_n \beta_n(h^{\times n})$, $\|\beta_n\| \leq CR^{-n}$. For integers $m, m' \geq 0$ with $|\vartheta|^{m'} < R$ and $|\vartheta|^{-m} \geq \max(C, 1)$ and for $\lambda = \vartheta^{m'}$, $\|\vartheta^m\lambda^n\beta_n\| \leq 1$, so (11.1) gives $C'_j, C''_j \geq 1$ with $c_j(\vartheta^m\lambda^n\beta_n) < C'_jC''^n_j$ for $n \geq 1$, $j \leq k$, and Lemma O.5(1) gives lifts of u_{β_n} in degree j of norm at most $|\vartheta|^{-mj}C'_j(C''_j|\lambda|^{-j})^n$; apply Lemma O.6 with $C_0 = \max_j |\vartheta|^{-mj}C'_j$ and $C_1 = \max_j C''_j|\lambda|^{-j}$.

Step 3 (item (3)). The first assertion is Lemma O.5(2). In the second, "only if" is (1); conversely, we show goodness in degree $n \leq k$ by induction. If $n < p$, apply (2) and the first assertion. If $n > p$ is not a power of p and p^l is the largest power of p below n, then $n = p^l + (n - p^l)$ is carry-free, so $\binom{n}{p^l} \neq 0$ in K by Lucas's theorem (proof of Corollary 6.3); apply Lemma O.7(1). If $n = p^i$, (G_j) holds for $j < n$ by induction and (1), and for $j = n$ by hypothesis; apply (2).

Step 4 (item (4)). The linearization $\tau_\beta\colon Y_n \to \widehat{D}$ of $(h_1, \dots, h_n, y) \mapsto \beta(h_1, \dots, h_n)y$ has norm $\|\beta\|$ and $\beta = \tau_\beta \circ \beta^{\mathrm{u}}_n$; apply Lemma O.5(4). If $(P; E, D_{\mathrm{u}})$ is good in degrees $j \leq k$, then, with J_n the nth injection of D_{u}, (G_j) for $\beta_n = J_n \circ \beta^{\mathrm{u}}_n$, of norm at most 1, and $c_j(\beta^{\mathrm{u}}_n) \leq c_j(\beta_n)$, by Lemma O.5(4) with the nth projection, make every $c_j(\beta^{\mathrm{u}}_n)$ finite with geometric growth; with the inequality, this gives (G_j) for every normed D, hence goodness in degrees $j \leq k$ by (2). In characteristic p the degrees $p^i \leq k$ give (G_{p^i}) for every D, and (3) applies. □

Proof of Corollary 11.6. (1) Degrees below p are good by Theorem 11.5(2),(3), carry-free sums by Lemma O.7(1) and Lucas's theorem; a least missing N that is not a power of p would be the carry-free sum $p^l + (N - p^l)$ of smaller elements, p^l the largest power of p below N. (2) By Proposition 11.2(2), $(P; E, D)$ and $(P; E \oplus K, D \oplus K)$ are good in the same degrees, so Lemma O.7(2) makes $\mathfrak{D}(P; E, D)$ closed under $k \mapsto k - 1$, $k \geq 2$; apply (1). (3) is Proposition 11.11. (4) Lemma O.7(2), for all normed E, D at once, gives closure under $k \mapsto k - 1$; by (1) the set contains $1, \dots, p - 1$ and is closed under carry-free sums, so its least missing element is a power of p. □

O.4. **The familywise class.**

Lemma O.8. *Let $u(x_0 + h) = \sum_n \beta_n(h^{\times n})$ for $\|h\| < R$, $\beta_n \in \mathrm{Mult}^n(P; \mathcal{L}(E, \widehat{D}))$, $\|\beta_n\| \leq CR^{-n}$, and $\lambda \in K$, $0 < |\lambda| < R$. Let $\beta^{\otimes}_n\colon \widehat{\otimes}^n_\pi P \to \mathcal{L}(E, \widehat{D})$ be the linearization of β_n for*

$n \ge 1$, and $\beta_0^{\otimes}(c) = c\beta_0$ for c in slot 0, $\widehat{K}$, by the $\widehat{K}$-vector space structure of $\mathcal{L}(E, \widehat{D})$. Then $\theta((w_n)_n) = \sum_n \lambda^n \beta_n^{\otimes}(w_n)$ is a linear family on $\mathcal{T}(P)$, $\|\theta\| \le C$, and $u(x_0 + \lambda h) = \theta(\mathrm{ver}(h))$ for $\|h\| < 1$. Conversely, $\theta \circ \mathrm{ver}$ is analytic on the open unit ball for every linear family θ on $\mathcal{T}(P)$.

Proof. $\|\lambda^n \beta_n^{\otimes}(w_n)\| \le C(|\lambda|/R)^n \|w_n\|$, and $\theta(\mathrm{ver}(h)) = \sum_n \beta_n((\lambda h)^{\times n})$. The converse is Lemma O.1(1), as $\|\theta \circ \mathrm{ver}_n\| \le \|\theta\|$. □

Lemma O.9. *Let Y be a Banach space, $0 < R' \le R < 1$, and $C_{k,R} = \sup_N \binom{N+k-1}{k-1} R^N < \infty$. For $\Phi \in \mathrm{Mult}^k(\mathcal{T}(P); Y)$, $\|\mathrm{ver}^\sharp \Phi\|_R \le C_{k,R}\|\Phi\|$ and $\Phi(\mathrm{ver}(h), \dots, \mathrm{ver}(h)) = \sum_N (\mathrm{ver}^\sharp \Phi)_N(h^{\times N})$ for $\|h\| < 1$; if Φ has zero diagonal, $\mathrm{ver}^\sharp \Phi \in \mathrm{Mult}^\omega_{R,0}(P; Y)$; and $\iota_{R'R} \circ \mathrm{ver}^\sharp$ is $\mathrm{ver}^\sharp$ with values in $\mathrm{Mult}^\omega_{R'}(P; Y)$. Hence vanishing after shrinking does not depend on R.*

Proof. There are at most $\binom{N+k-1}{k-1}$ compositions of N into k parts, and $\|\mathrm{ver}_n\| \le 1$; the identity is Lemma O.1(2) with radius 1. For zero diagonal, $\sum_N t^N (\mathrm{ver}^\sharp \Phi)_N(h^{\times N}) = 0$ for $|t| < 1/\|h\|$; apply Lemma A.1. The inclusion keeps components, so $(\iota_{R'R})_* \mathrm{ob}^\omega_k(\theta; R) = \mathrm{ob}^\omega_k(\theta; R')$ by (11.2) and Corollary N.3(b), and pushouts of 0 vanish. □

Proof of Theorem 11.7. Step 1 (item (1)). The second and third conditions are equivalent: bounded linear maps into $\mathrm{Mult}^\omega_{R',0}(P; \widehat{\Lambda}^k_\pi D)$ are sequences of maps into the $\mathrm{Mult}^N_0(P; \widehat{\Lambda}^k_\pi D)$ with $\sup_N R'^N \|\cdot\| < \infty$, $\iota_{R'R}$ keeps components, and a pushout vanishes if and only if the map extends (Lemma 3.6(3)). Each $q_N \in \mathrm{Mult}^N(P; \mathcal{L}(\Lambda^k_\pi E, \widehat{\Lambda}^k_\pi D))$ gives $q_N^\sharp \colon \widehat{\Lambda}^k_\pi E \to \mathrm{Mult}^N(P; \widehat{\Lambda}^k_\pi D)$, $q_N^\sharp(y)(h_1, \dots, h_N) = q_N(h_1, \dots, h_N)y$, of the same norm, and conversely. If $\Lambda^k_\pi \theta(\mathrm{ver}(h)) = \sum_N q_N(h^{\times N})$ for $\|h\| < R_0$, $\|q_N\| \le C' R_0^{-N}$, then for $z \in \widehat{\otimes}^k_\pi E$ the Nth terms of $\mathrm{ver}^\sharp \mathrm{ev}_\theta(z)$ and $q_N^\sharp \widehat{\wedge}^k_E z$ have the same diagonal, both giving the t^N-coefficient of $\mathrm{ev}_\theta(z)(\mathrm{ver}(th), \dots, \mathrm{ver}(th)) = \widehat{\Lambda}^k_\pi \theta(\mathrm{ver}(th))(\widehat{\wedge}^k_E z)$ (Lemmas O.9 and A.1). So $\xi_N = \mathrm{pr}_N \mathrm{ver}^\sharp \mathrm{ev}_\theta - q_N^\sharp \widehat{\wedge}^k_E$ takes values in Mult^N_0, extends $(\mathrm{pr}_N \mathrm{ver}^\sharp \mathrm{ev}_\theta)|_{\mathcal{N}^k_E}$ and has norm at most $C_{k,R}\|\theta\|^k R^{-N} + C' R_0^{-N}$. Conversely, extensions ξ_N of norm at most CR'^{-N} make $\mathrm{pr}_N \mathrm{ver}^\sharp \mathrm{ev}_\theta - \xi_N$ vanish on $\mathcal{N}^k_E$, so it is $q_N^\sharp \widehat{\wedge}^k_E$ with $\|q_N^\sharp\| \le (C_{k,R}\|\theta\|^k + C) \min(R, R')^{-N}$ (Lemma 3.7), and $\sum_N q_N(h^{\times N}) = \Lambda^k_\pi \theta(\mathrm{ver}(h))$ for $\|h\| < \min(R, R')$ by Lemma O.9.

Step 2 (item (2)). For analytic u and x_0, Lemma O.8 gives λ, θ with $u(x_0 + \lambda h) = \theta(\mathrm{ver}(h))$, and (1), with $h = \lambda^{-1}(x - x_0)$, gives analyticity of $\Lambda^k_\pi u$ at x_0. Conversely, $\theta \circ \mathrm{ver}$ is analytic on the open unit ball (Lemma O.8); apply goodness and (1).

Step 3 (item (3)). As $\theta(w) = \tau_\theta \circ \theta_{\mathrm{u}}(w)$, $\mathrm{ev}_\theta = (\widehat{\Lambda}^k_\pi \tau_\theta)_\flat \mathrm{ev}_{\theta_{\mathrm{u}}}$ on pure tensors. The map $(\widehat{\Lambda}^k_\pi \tau_\theta)_\flat$ commutes with $\mathrm{ver}^\sharp$ and $\iota_{R'R}$, preserves zero diagonals and has norm at most $\|\tau_\theta\|^k$; so Corollary N.3(b) gives the identity, vanishing after shrinking passes from θ_{u} to every θ, and (2) gives goodness for every D. Conversely, (2) for the Banach space $D = \mathcal{T}(P) \widehat{\otimes}_\pi E$ gives vanishing for θ_{u}. Finally $\theta_{\mathrm{u}} \circ \mathrm{ver} = \mathcal{U}$; apply (1). □

O.5. **Polarizable base models.**

Lemma O.10. *Let $n, i \ge 1$ and R_i a bounded projection of $\widehat{\otimes}^i_\pi P$ onto $\widehat{\Gamma}^i_\pi P$. Identify $\widehat{\otimes}^{ni}_\pi P$ with $(\widehat{\otimes}^i_\pi P)^{\widehat{\otimes} n}$ by grouping the positions $(a, 1), \dots, (a, i)$ into the ath factor, and let reg be the isometry of $\widehat{\otimes}^{ni}_\pi P$ moving position (a, b) to (b, a), with target $(\widehat{\otimes}^n_\pi P)^{\widehat{\otimes} i}$. Then $q_{n,i}(h_1, \dots, h_{ni}) = \mathrm{reg} \circ R_i^{\otimes n}(h_1 \otimes \cdots \otimes h_{ni})$ is bounded ni-linear, $\|q_{n,i}\| \le \|R_i\|^n$, with values in $\widehat{\Gamma}^i_\pi(\widehat{\otimes}^n_\pi P)$, and $q_{n,i}(h^{\times ni}) = (h^{\otimes n})^{\otimes i}$.*

Proof. $R_i^{\otimes n}$ has norm at most $\|R_i\|^n$, takes values in the closed span of the $y_1^{\otimes i}\otimes\cdots\otimes y_n^{\otimes i}$, which reg sends to $(y_1\otimes\cdots\otimes y_n)^{\otimes i}$, and fixes $(h^{\otimes i})^{\otimes n}$, which reg sends to $(h^{\otimes n})^{\otimes i}$. □

Lemma O.11. *Let G be a Banach space and $\mathbf{m}$ a content of type $\mathbf{i}=(i_1,\dots,i_m)$, let $\Omega_{\mathbf{m}}$ represent the left cosets in S_k of the subgroup preserving the consecutive blocks of sizes $i_1,\dots,i_m$, acting on $\widehat{\otimes}_\pi^k G$ by permuting factors, and $\mathrm{per}_{\mathbf{m}}=\sum_{\sigma\in\Omega_{\mathbf{m}}}\sigma$. Then $\|\mathrm{per}_{\mathbf{m}}\|\le\binom{k}{\mathbf{i}}$; on tensors invariant under that subgroup, $\mathrm{per}_{\mathbf{m}}$ does not depend on $\Omega_{\mathbf{m}}$; for $y_1,\dots,y_m\in G$, $\mathrm{per}_{\mathbf{m}}(y_1^{\otimes i_1}\otimes\cdots\otimes y_m^{\otimes i_m})$ is the coefficient of $t_1^{i_1}\cdots t_m^{i_m}$ in $(\sum_l t_l y_l)^{\otimes k}$ and lies in the span of the powers $y^{\otimes k}$, $y\in G$; and $\mathrm{per}_{\mathbf{m}}$ maps the closed span of such tensors into $\widehat{\Gamma}_\pi^k G$.*

Proof. The bound is the triangle inequality; on invariant tensors σ and $\sigma\sigma'$, σ' in the subgroup, agree. The cosets correspond bijectively to the words in which l occurs i_l times, whose sum is the coefficient. A linear functional on the span of the coefficients of $(\sum_l t_l y_l)^{\otimes k}$ vanishing on these powers vanishes on every coefficient, by Lemma B.6, as in Step 1 of the proof of Proposition 5.13. Continuity gives the last claim. □

Proof of Theorem 11.9. Step 1 (affine maps suffice). Affine maps are analytic. Conversely, fix $n\le k$, let $G=\widehat{K}\oplus\widehat{P}$ with the maximum norm, $\gamma(x)=(1,x)$, $Z=\widehat{\otimes}_\pi^n P=\widehat{\otimes}_\pi^n\widehat{P}$ and $W=\widehat{\Gamma}_\pi^n P=\widehat{\Gamma}_\pi^n\widehat{P}$ (density; paragraph before Theorem 3.16). Scalar multiplication on Z extends from K to $\widehat{K}$ by continuity and preserves the closed subspace W. The k-linear map $\Phi((\lambda_1,x_1),\dots,(\lambda_k,x_k))=\lambda_{n+1}\cdots\lambda_k\,x_1\otimes\cdots\otimes x_n$ has $\|\Phi\|\le 1$ and $\Phi((\lambda,x),\dots,(\lambda,x))=\lambda^{k-n}x^{\otimes n}\in W$, so $x\mapsto x^{\otimes n}=\Phi(\gamma(x),\dots,\gamma(x))$ is analytic at 0 into W; Proposition 2.5, with the canonical map $P^n\to Z$, and Lemma 3.11 give polarizability in degree n.

Step 2 (the ambient sum). Let G,Z,W,Φ,U,γ be as in Definition 11.8, $x_0\in U$ and $\gamma(x_0+h)=\sum_n\gamma_n(h^{\times n})$ for $\|h\|<R$, $\|\gamma_n\|\le CR^{-n}$. Let $\gamma_n^{\otimes}\colon\widehat{\otimes}_\pi^n P\to G$ and $\Phi^{\otimes}\colon\widehat{\otimes}_\pi^k G\to Z$ be the linearizations, with $\widehat{\otimes}_\pi^0 P=\widehat{K}$ and $\gamma_0^{\otimes}(c)=c\gamma_0$; as W is closed, $\Phi^{\otimes}(\widehat{\Gamma}_\pi^k G)\subseteq W$. By Lemma O.1(2), for $\|h\|<R$, $\Phi(\gamma(x_0+h),\dots,\gamma(x_0+h))$ is the absolutely convergent sum of the $\Phi(\gamma_{n_1'}(h^{\times n_1'}),\dots,\gamma_{n_k'}(h^{\times n_k'}))$, $(n_1',\dots,n_k')\in\mathbb{N}_0^k$, and by Lemma O.11 the terms of content $\mathbf{m}$, of support $n_1<\cdots<n_m$ and type $\mathbf{i}$, sum to $\Phi^{\otimes}\,\mathrm{per}_{\mathbf{m}}(\gamma_{n_1}(h^{\times n_1})^{\otimes i_1}\otimes\cdots\otimes\gamma_{n_m}(h^{\times n_m})^{\otimes i_m})$.

Step 3 (lifting each content). Fix $\bar g$ as in (2) and projections R_i onto $\widehat{\Gamma}_\pi^i P$ with $\|R_i\|\le\bar g$, $i\le k$. Let $q_l=q_{n_l,i_l}$ (Lemma O.10) if $n_l\ge 1$, $q_l=1\in\widehat{K}$ if $n_l=0$, and, with consecutive blocks I_l of sizes $n_l i_l$, $Q_{\mathbf{m}}(h_1,\dots,h_{N(\mathbf{m})})=\Phi^{\otimes}\,\mathrm{per}_{\mathbf{m}}(\bigotimes_l(\gamma_{n_l}^{\otimes})^{\otimes i_l}q_l(h_{I_l}))$. Then $\|Q_{\mathbf{m}}\|\le\|\Phi\|\binom{k}{\mathbf{i}}\prod_l\|\gamma_{n_l}\|^{i_l}\bar g^{n_l}\le\|\Phi\|k!C^k(\bar g/R)^{N(\mathbf{m})}$, as $\bar g\ge 1$. Its values lie in W, as q_l takes values in $\widehat{\Gamma}_\pi^{i_l}(\widehat{\otimes}_\pi^{n_l}P)$, which $(\gamma_{n_l}^{\otimes})^{\otimes i_l}$ maps into $\widehat{\Gamma}_\pi^{i_l}G$, and Lemma O.11 applies. By Lemma O.10, $Q_{\mathbf{m}}(h^{\times N(\mathbf{m})})$ is the content-$\mathbf{m}$ sum of Step 2.

Step 4 (item (1)). The sum q_N of the $Q_{\mathbf{m}}$ of degree N is bounded N-linear into W, with

$$\|q_N\|\le\|\Phi\|\,k!\,C^k\tbinom{N+k-1}{k-1}(\bar g/R)^N, \tag{O.2}$$

and for $\|h\|<R/\bar g$, $\sum_N q_N(h^{\times N})$ converges in the Banach space W to $\Phi(\gamma(x_0+h),\dots,\gamma(x_0+h))$, by Step 2.

Step 5 (items (2)–(4)). (2) For analytic $u\colon U\to\mathcal{L}(E,\widehat{D})$ take $G=\mathcal{L}(E,\widehat{D})$, $Z=\mathrm{Mult}^k(E;\widehat{\Lambda}_\pi^k D)$, W the image of $\mathcal{L}(\Lambda_\pi^k E,\widehat{\Lambda}_\pi^k D)$ under the closed isometry $T\mapsto T\circ\omega_E$ (Proposition 3.1), and $\Phi(u_1,\dots,u_k)(x_1,\dots,x_k)=u_1x_1\wedge\cdots\wedge u_kx_k$, of norm at most 1, with $\Phi(u,\dots,u)=\Lambda_\pi^k u\circ\omega_E$; apply (1) and (O.2). For bundles, $x\mapsto A^k_{E,\widehat{D};F}(u(x))$ is analytic for Banach F (prose after Definition 11.1), and the last assertion of Theorem 2.1 applies as in the proof of Corollary 11.3. (3) If $n!\ne 0$ in K, P is polarizable in degree n (Proposition 4.1(2)); so (2) applies if $\mathrm{char}\,K=0$ or $k<p$. Polarizability in degree p gives it in every degree

(Theorem 6.10). A lower r-estimate, $r < 2$, bounds $\mathcal{P}_<$ (Theorem 5.6(1)), and Proposition 5.7(3) gives polarizability in every degree. (4) Projections with $\|R_n\| \le 2g_n(P)$ and Theorem G.3 give the first equivalence, and (1) the second. For nonarchimedean K, $g_n(\ell^1(\mathbb{N}, K)) = n!$ (Propositions 5.13(1) and G.4) is finite, while $(n!)^{1/n}$ is unbounded; apply (1) and (2). □

Proof of Theorem 11.10. Step 1 (item (1)). As the statement depends only on K as a normed field, Lemma 7.1 lets us assume K dense in $\kappa((X))$, as in the proof of Theorem 7.13. By Lemma 7.11(1) for P, $\|ay\|_P \le \|y\|_\infty \|a\|_P$ for $a \in P$ and $y \in E_0$, so θ_0 is a linear family, and $\|\theta_0(a)\| = \|\mu_a\|_{E_0 \to P} = \|a\|_P$ for $a \in c_{00}(\mathbb{N}, K)$ (Lemma 7.11(2),(4)). The fifth condition gives the first, the first the second. If $\Lambda^k_\pi \theta_0$ is analytic at a point, θ_0 has a lift Q in degree k (prose after Definition 11.1). Its restriction to $c_{00}(\mathbb{N}, K)^k$ is a multiplier lift for (E_0, P) (Section K.4) satisfying (K.6) with $C_Q = \|Q\|$, and E_0, P are symmetric sequence spaces; so Steps 1–5 of the proof of Theorem 7.13 apply to (E_0, P), as in Step 2 of the proof of Proposition 7.16, and $x = 1_{\operatorname{supp} a}$, $y = 1_{\operatorname{supp} b}$ in (7.2) bound $\mathcal{P}_<$ on $P \otimes_\pi P$. This bound, item (3) of Theorem 8.1, gives polarizability in degree p (Proposition 5.7(3)), hence the fifth condition (Theorem 11.9(3)). The examples follow from Corollary 8.2 and Theorem 11.9(3), every $k \ge p$ being admissible in (L).

Step 2 (item (2)). Apply Theorem 7.18 with $E = c_{00}(\mathbb{N}, K)$, $E' = \ell^\infty(\mathbb{N}, \widehat{K})$, $\psi = \mathrm{id}$, $\theta = \mathrm{id}$, $u_a = \mu_a = \mu_{w(i(a))}$, $F = \widehat{\Lambda}^k_\pi E'$, $\nu = \omega_{E'}$, $\theta_F = \mathrm{id}$: no bounded k-linear Q on $c_{00}(\mathbb{N}, K)^k$ has $Q(a^{\times k}) = \omega_{E'} \circ (\mu_a, \dots, \mu_a)$. Analyticity of $x \mapsto \Lambda^k_\pi \mu_{w(x)}$ at a point would give a lift Q_P of the linear family $x \mapsto \mu_{w(x)}$ (prose after Definition 11.1), and $(a_1, \dots, a_k) \mapsto Q_P(ia_1, \dots, ia_k) \circ \omega_E$ would be such a Q. For $c_0(\Sigma, K)$ and $\ell^\infty(\Sigma, K)$ take $\mathbb{N} \subseteq \Sigma$, i extension by zero and w restriction; for P with an equivalent nonarchimedean norm and $\dim_{\widehat{K}} \widehat{P} = \infty$, take $i(\xi) = \sum_n \xi_n e_n$ and $w = \theta_P|_P$ with e_n, θ_P from Lemma 7.17. Then (U^k_P) fails, and with it polarizability in degree p (Theorem 11.9(3)). If $\dim_{\widehat{K}} \widehat{P} = d < \infty$, a bounded $\widehat{K}$-linear isomorphism $T\colon \widehat{P} \to \ell^1(\{1, \dots, d\}, \widehat{K})$ exists as $\widehat{K}$ is complete, and $(T^{-1})^{\otimes n} R_n T^{\otimes n}$ transfers projections (Step 3 of the proof of Proposition 5.13); as passing to a dense subspace changes neither $\widehat{\otimes}^n_\pi$ nor $\widehat{\Gamma}^n_\pi$, Proposition 5.13(1) for $\ell^1(\{1, \dots, d\}, K)$ makes P polarizable in every degree; apply Theorem 11.9(2). □

O.6. **Fibers, parameter-relative ordering and the staircase.**

Proof of Proposition 11.11. (1) For an analytic $u\colon U \to \mathcal{L}(E, \widehat{D})$, $\Lambda^k_\pi u(x) = Q(u(x), \dots, u(x))$ is analytic on U by Lemma O.1(2). For β as stated, $q(h_1, \dots, h_{kn}) = Q(\beta(h_{I_1}), \dots, \beta(h_{I_k}))$, with $I_1, \dots, I_k$ the consecutive blocks of size n, is a lift of u_β in degree k of norm at most $\|Q\|\|\beta\|^k$. (2) The "if" direction is (1). Conversely, the identity of the open set $\mathcal{L}(E, \widehat{D})$ is an analytic family, so $u \mapsto \Lambda^k_\pi u$ is analytic at some point of $\mathcal{L}(E, \widehat{D})$; since $\widehat{\Lambda}^k_\pi \widehat{D} = \widehat{\Lambda}^k_\pi D$, Lemma 3.4 applies to $(E, \widehat{D})$. The second assertion follows from (1) and the first. □

Proof of Theorem 11.13. Step 1 (the map Π). Let G_r and δ_t be as in Section J.4; by density $\widehat{\otimes}^k_\pi G_r = \widehat{\otimes}^k_\pi \ell^r(\Sigma, \widehat{K})$ and $\widehat{\Gamma}^k_\pi G_r = \widehat{\Gamma}^k_\pi \ell^r(\Sigma, \widehat{K})$. For a composition $\mathbf{l} = (l_1, \dots, l_m)$ of k let $I_1, \dots, I_m$ be the consecutive blocks of sizes l_i and $\Omega_\mathbf{l}$ represent the left cosets of $S_{l_1} \times \cdots \times S_{l_m}$ in S_k, acting by permuting factors. For $a_1, \dots, a_k \in c_{00}(\Sigma, K)$ put

$$\Pi_\mathbf{l}(a_1, \dots, a_k) = \sum_{t_1 < \cdots < t_m} \prod_{i=1}^m \prod_{j \in I_i} a_j(t_i)\, \delta_{t_1}^{\otimes l_1} \otimes \cdots \otimes \delta_{t_m}^{\otimes l_m}, \qquad \Pi = \sum_\mathbf{l} \sum_{\sigma \in \Omega_\mathbf{l}} \sigma \circ \Pi_\mathbf{l}.$$

As $\sum_{\sigma \in \Omega_\mathbf{l}} \sigma \delta_\mathbf{t} = O_\mathbf{t}$ for nondecreasing $\mathbf{t}$ with constant blocks of sizes $\mathbf{l}$, Π is R_k of Proposition 5.7(3), for (δ_t), applied to $a_1 \otimes \cdots \otimes a_k$. So $\Pi(a, \dots, a) = a^{\otimes k}$, and Π takes values in the

span of the $O_{\mathbf{t}}$, which lie in the span of the powers by Step 4 of the proof of that proposition in Appendix J.

Step 2 (one composition). Let $\Delta = \{i : l_i \geq 2\}$, j_i the element of I_i for $i \notin \Delta$, and $b_i = \prod_{j\in I_i} a_j$ for $i \in \Delta$, so $\|b_i\|_1 \leq \prod_{j\in I_i}\|a_j\|_2$ by Hölder's inequality and $\|\cdot\|_{l_i} \leq \|\cdot\|_2$. For fixed strictly increasing $(t_i)_{i\in\Delta}$, the $i \notin \Delta$ form maximal runs of consecutive indices, and for a run the admissible t_i are the strictly increasing tuples in the open interval J of Σ between the neighboring fixed values. So $\Pi_{\mathbf{l}}(a_1,\dots,a_k)$ is the sum over $(t_i)_{i\in\Delta}$ of $\prod_{i\in\Delta} b_i(t_i)$ times the tensor product, in block order, of the $\delta_{t_i}^{\otimes l_i}$, $i \in \Delta$, of norm 1, and, for each run of length n, of $\mathcal{P}^{(n)}(\bigotimes_i a_{j_i} 1_J)$, of norm at most $C'_n \prod_i \|a_{j_i}\|_2$ with $C'_n = \max(1, C_{n,r}|\vartheta|^{-n})$ (Theorem J.5). Hence $\|\Pi_{\mathbf{l}}(a_1,\dots,a_k)\| \leq C_{\mathbf{l}} \prod_j \|a_j\|_2$, with $C_{\mathbf{l}}$ the product of the C'_n of the runs.

Step 3 (item (1)). So Π extends from the dense subspace $c_{00}(\Sigma, K)$ of $\ell^2(\Sigma, \widehat{K})$, with values in the closed subspace $\widehat{\Gamma}^k_\pi G_r$, and $\Pi(a,\dots,a) = a^{\otimes k}$ persists by continuity.

Step 4 (item (2), linear maps). Let $T\colon P \to G$ be bounded linear, and extend Φ to $\widehat{G}^k$; its diagonal stays in W. The coordinate map $V \to \ell^2(\Sigma, \widehat{K})$ extends to $w\colon P \to \ell^2(\Sigma, \widehat{K})$, $\|w\| \leq C_2$; write $h(t) = (wh)(t)$ and $\iota\colon \ell^2 \to \ell^r$ for the inclusion. As in Step 1 of the proof of Theorem 5.9, with P, G, T for E, D, u, the map $\xi \mapsto \sum_t \xi(t)T(e_t)$ extends from $c_{00}(\Sigma, K)$ to $\widetilde{T}\colon \ell^r(\Sigma, \widehat{K}) \to \widehat{G}$ with $T = \widetilde{T} \circ \iota \circ w$. The linearization $\Phi_r^{\otimes}$ of $(\xi_1,\dots,\xi_k) \mapsto \Phi(\widetilde{T}\xi_1,\dots,\widetilde{T}\xi_k)$, extended to $\widehat{\otimes}^k_\pi G_r$, maps $\widehat{\Gamma}^k_\pi G_r$ into W, and $Q(a_1,\dots,a_k) = \Phi_r^{\otimes}\Pi(wa_1,\dots,wa_k)$ satisfies $Q(a,\dots,a) = \Phi_r^{\otimes}((\iota wa)^{\otimes k}) = \Phi(Ta,\dots,Ta)$, by continuity from $c_{00}(\Sigma, K)$.

Step 5 (item (2), analytic maps). Let $\gamma(x_0 + h) = \sum_n \gamma_n(h^{\times n})$ for $\|h\| < R$, $\|\gamma_n\| \leq CR^{-n}$, let Σ_* be the disjoint union of the Σ^n, $n \geq 0$, totally ordered in any way, and $\lambda \in K$ with $0 < |\lambda| \leq R/C_e$, $C_e = \sup_t \|e_t\|$. For $\|h\| < |\lambda|/C_2$ the vector $\xi(h) \in \ell^2(\Sigma_*, \widehat{K})$ with entries $\xi(h)_{(n,\mathbf{t})} = \lambda^{-n} h(t_1)\cdots h(t_n)$ has nth block of norm $(\|wh\|_2/|\lambda|)^n$, and ξ is analytic at 0 with nth coefficient of norm at most $(C_2/|\lambda|)^n$. As $\|\lambda^n \gamma_n(e_{t_1},\dots,e_{t_n})\| \leq C$, (5.2) for $\widehat{G}$ (Proposition 5.8) extends $\delta_{(n,\mathbf{t})} \mapsto \lambda^n \gamma_n(e_{t_1},\dots,e_{t_n})$ to $T_*\colon \ell^r(\Sigma_*, \widehat{K}) \to \widehat{G}$, and $T_*\iota\xi(h) = \sum_n \gamma_n(h^{\times n}) = \gamma(x_0 + h)$ for $\|h\| < \min(R, |\lambda|/C_2)$: first for $h \in V$, where the blocks of $\xi(h)$ are finitely supported, then by continuity. Step 4 for $\ell^2(\Sigma_*, \widehat{K})$ with its unit vectors, $\widehat{G}$ and $T_* \circ \iota$ gives a bounded k-linear Q_{T_*} into W with $Q_{T_*}(\eta,\dots,\eta) = \Phi(T_*\iota\eta,\dots,T_*\iota\eta)$; so $\Phi(\gamma(x_0 + h),\dots,\gamma(x_0 + h)) = Q_{T_*}(\xi(h),\dots,\xi(h))$ near 0, analytic by Lemma O.1(2).

Step 6 (item (3)). The space $\widehat{D}$ satisfies (5.2): by Proposition 5.8 and continuity if D has an equivalent r-norm, and, if D has an equivalent nonarchimedean norm, as $\max_t|\xi_t| \leq (\sum_t |\xi_t|^r)^{1/r}$, as in the proof of Corollary 5.10. Hence so does $G = \mathcal{L}(E, \widehat{D})$, as $\|\sum_t \xi_t u_t\| = \sup_{\|x\|\leq 1}\|\sum_t \xi_t u_t x\|$, and G has an equivalent r-norm (Proposition 5.8). Let $Z = \mathcal{L}(\widehat{\otimes}^k_\pi E, \widehat{\Lambda}^k_\pi D)$, W its operators vanishing on $\mathcal{N}^k_E$, and $\Phi(u_1,\dots,u_k) = \widehat{\wedge}^k_D \circ (u_1 \otimes \cdots \otimes u_k)$, so $\Phi(u,\dots,u) = \widehat{\Lambda}^k_\pi u \circ \widehat{\wedge}^k_E$. For analytic $u\colon U \to G$, (2) makes $x \mapsto \widehat{\Lambda}^k_\pi u(x) \circ \widehat{\wedge}^k_E$ analytic into W; the inverse of the isometry $S \mapsto S \circ \widehat{\wedge}^k_E$ onto W (Lemma 3.7) and restriction to $\Lambda^k_\pi E$ give analyticity of $\Lambda^k_\pi u$. □

Proof of Theorem 11.14. Step 1 (the input from Theorem 7.7). Suppose such a Q exists. Step 1 of the proof of Theorem 7.7 in Appendix K uses only that Q is k-linear on $c_{00}(\mathbb{N}, K)^k$ with $Q(a,\dots,a) = \Lambda^k_\pi \mu_a$, and that $\|y\|_\infty \leq \|y\|_E$, so that the determinant array of Section K.1 is defined on $\widehat{\Lambda}^k_\pi E$; it gives τ, slots $\gamma_1 \neq \gamma_2$ and $\delta \neq 0$, independent of m.

Step 2 (the norm bound for ζ). Fix $m \geq 1$, and let $a_1,\dots,a_k$ and $x_1,\dots,x_k$ be the inputs of Step 2 of that proof: their entries lie in $\{0, \pm 1\}$, a_{γ_1} and a_{γ_2} have $2m$ nonzero entries and the

other a_γ two, and x_{γ_1}, x_{γ_2} have $4m$ and the other x_γ four. So $\zeta = Q(a_1,\dots,a_k)(x_1\wedge\cdots\wedge x_k)$ has $\|\zeta\| \le \|Q\|\varphi_P^\pm(2)^{k-2}\varphi_E^\pm(4)^{k-2}\varphi_P^\pm(2m)^2\varphi_E^\pm(4m)^2$.

Step 3 (the rank bound). The slice of Lemma K.3 for the slots γ_1,γ_2 and $c_\gamma = C_\gamma(1)$, followed by restriction to the coordinates $A_n(1)$ and $B_{n'}(1)$, contractive as $\|y\|_\infty \le \|y\|_E$, gives z with $\operatorname{coeff}_0 z = \mathsf{Z}$ and $\|z\| \le k(k-1)\|\zeta\|$; so Lemma K.2(2) and $\operatorname{rank}\mathsf{Z} \ge m-1$ give $m-1 \le M_\varrho k(k-1)\|\zeta\|$ for every m, contradicting the hypothesis by Step 2.

Step 4 (the linear family and symmetric spaces). If $\Lambda^k_\pi T$ were analytic at a point, T would have a lift (prose after Definition 11.1), whose restriction to $c_{00}(\mathbb{N},K)^k$ Steps 1–3 exclude. For a symmetric sequence space G and $y \in c_{00}(\mathbb{N},K)$ with entries in $\{0,\pm1\}$, $\|y\|_G = \|1_{\operatorname{supp} y}\|_G = \varphi_G(|\operatorname{supp} y|)$ by permutation invariance, and φ_G is nondecreasing, so $\varphi_G^\pm = \varphi_G$. □

Proof of Corollary 11.15. Step 1 (item (1)). For $r_1 < 2$, P has a lower r_1-estimate (example after Theorem 5.6); apply Theorem 11.9(3). For $E = \ell^{r_2}(\mathbb{N},K)$, $r_2 < 2$, $(E,\widehat{E})$ is good for Banach values (Theorem 5.6(3)); apply Proposition 11.11. For $r_1 = 2$ apply Theorem 11.13(3) with the unit vectors of $\ell^2(\mathbb{N},K)$, and with $r = r_2$ and the r_2-norm $\|\cdot\|_{r_2}$ of $\ell^{r_2}(\mathbb{N},K)$ over the nonarchimedean K (paragraph after Corollary 5.10), or with any $r \in (2,\infty)$ and an equivalent nonarchimedean norm of E.

Step 2 (item (2)). For $E = \ell^{r_2}(\mathbb{N},K)$ or $c_0(\mathbb{N},K)$, $\|\mu_a\|_{E\to E} \le \|a\|_\infty \le \|a\|_P$, so $a \mapsto \mu_a$ extends from $c_{00}(\mathbb{N},K)$ to a linear family $T\colon P \to \mathcal{L}(E,\widehat{E})$, and analyticity of $\Lambda^k_\pi T$ at a point would give a lift (prose after Definition 11.1). For $P = c_0(\mathbb{N},K)$ and E equal to $\ell^{r_2}(\mathbb{N},K)$, $r_2 \ge 2$, or $c_0(\mathbb{N},K)$, its restriction to $c_{00}(\mathbb{N},K)^k$ is bounded for the supremum norm, which Theorem 7.7 excludes. For $2 < r_1 < \infty$ and E equal to $c_0(\mathbb{N},K)$ or $\ell^{r_2}(\mathbb{N},K)$ with $1/r_1 + 1/r_2 < 1/2$, Theorem 11.14 applies with $\varphi_{\ell^{r_1}}(n) = n^{1/r_1}$ and $\varphi_E(n) = 1$ or n^{1/r_2}, as $(2m)^{2/r_1}(4m)^{2/r_2}/m \to 0$. If K is complete and E has an equivalent nonarchimedean norm, Lemma 7.17 gives bounded linear $\widehat{\psi}_E\colon c_0(\mathbb{N},K) \to E$ and $\theta_E\colon E \to c_0(\mathbb{N},K)$ with $\theta_E\widehat{\psi}_E = \mathrm{id}$, and $T_E(a) = \widehat{\psi}_E\mu_a\theta_E$ is a linear family on P; a lift Q_{T_E} of it gives the lift $\widehat{\Lambda}^k_\pi\theta_E \circ Q_{T_E}(a_1,\dots,a_k)\circ\Lambda^k_\pi\widehat{\psi}_E$ of $a \mapsto \Lambda^k_\pi\mu_a$ on $c_0(\mathbb{N},K)$, which the case $E = c_0(\mathbb{N},K)$ excludes.

Step 3 (item (3)). As $\|\mu_a\|_{E\to E} = \|a\|_\infty$ for $a \in c_{00}(\mathbb{N},K)$ (Lemma 7.11(1),(3)), the argument of Step 1 of the proof of Theorem 11.10, with (E,E) for (E_0,P) and with $a = 1_{\operatorname{supp} x}$, $b = 1_{\operatorname{supp} y}$ in (7.2), shows that a lift in a degree $k \ge p$ bounds $\mathcal{P}_<$ on $E\otimes_\pi E$; Theorem 8.1, (3)⇒(7), then makes (E,E) and $(E,\widehat{E})$ good for Banach values in every degree. Conversely, if (E,E) is good for Banach values in a degree $k \ge p$, (1)⇒(7) of that theorem gives the same; then Proposition 11.11(1) makes $(c_0(\mathbb{N},K);E)$ good in every degree, and the prose after Definition 11.1 makes the multiplier family liftable in every degree. □

O.7. **Tangent bundles and structure groups.**

Proof of Theorem 11.17. Step 1 (item (1), the shear chart). The factors P_1, D, E of P_0 are Banach spaces. As $(x,e) \mapsto u(x)e$ composes $(x,e) \mapsto (u(x),e)$ with the bounded bilinear evaluation, ψ and $(x,d,e) \mapsto (x,d-u(x)e,e)$ are analytic (Lemma O.1(2)); they map $U\times D\times E$ to itself, are mutually inverse, and have analytic derivatives (paragraph after Definition 11.16). By the product rule $D\psi(x,d,e)(h,b,c) = (h, b+(Du(x)h)e+u(x)c, c)$, so $\pi_D D\psi(x,d,e)i_E = u(x)$, and Proposition 3.1 gives the identity. Analyticity of $\Lambda^k_\pi D\psi$ at (x_0,d_0,e_0), composed with $x \mapsto (x,d_0,e_0)$ and $T \mapsto \widehat{\Lambda}^k_\pi\pi_D\circ T\circ\Lambda^k_\pi i_E$, would make $\Lambda^k_\pi u$ analytic at x_0; for ψ^{-1} use $\pi_D D(\psi^{-1})i_E = -u$.

Step 2 (item (1), the manifold M). The chart change from id_M to $J\circ\psi\circ J^{-1}$ is $J\circ\psi\circ J^{-1}$, with derivative $J\circ D\psi(J^{-1}y)\circ J^{-1}$ at $y\in M$; so the transitions of $\widehat{\Lambda}^k_\pi TM$ are $y\mapsto \widehat{\Lambda}^k_\pi J\circ\Lambda^k_\pi D\psi(J^{-1}y)\circ \Lambda^k_\pi J^{-1}$ and the same expression with ψ^{-1} (Proposition 3.1). They are analytic exactly where $\Lambda^k_\pi D\psi$ and $\Lambda^k_\pi D(\psi^{-1})$ are, so not at the points $J(x_0,d,e)$, by Step 1. For $\mathrm{Alt}^k(TM;\widehat{\Lambda}^k_\pi D)$ use $A^k_{E,D;F}(u)=A^k(i_E)\circ A^k(D\psi)\circ A^k(\pi_D)$, $A^k(J\circ D\psi\circ J^{-1})=A^k(J^{-1})\circ A^k(D\psi)\circ A^k(J)$ and the single target $F=\widehat{\Lambda}^k_\pi D$ of Theorem 3.2(1).

Step 3 (item (2)). If $(P_0;P_0)$ is good, then for f as in Definition 11.16, Df is an analytic $\mathcal{L}(P_0)$-valued family, so P_0 is tangent-good; tangent goodness gives the bundle statement (sentence after Definition 11.16); if the bundles are analytic, (1) with $P_1=D=E=P_0$ and $P_0\cong P_0^3$ makes $\Lambda^k_\pi u$ analytic for every analytic u into $\mathcal{L}(P_0)$.

Step 4 (item (3)). For $E=c_0(\mathbb{N},K)$ and $E=\ell^r(\mathbb{N},K)$, $4<r<\infty$, $\|a\cdot e\|_E\le\|a\|_\infty\|e\|_E$, so $(a,e)\mapsto a\cdot e$ is bounded bilinear, $D\psi(a,d,e)(h,b,c)=(h,b+h\cdot e+a\cdot c,c)$ and, with $i_3(c)=(0,0,c)$ and $\pi_2(h,b,c)=b$, $\pi_2D(\psi^{\pm1})(a,d,e)i_3=\pm\mu_a$. As in Step 1, analyticity at a point would make $a\mapsto\Lambda^k_\pi\mu_a$ analytic at a point of E, which Theorem 11.14 with $P=E$ excludes, as $\varphi_E(n)$ is 1 or $n^{1/r}$ and $(2m)^{2/r}(4m)^{2/r}/m\to0$; interleaving coordinates gives $E\cong E^3$. If K is complete and P_0 is an infinite-dimensional Banach space with an equivalent nonarchimedean norm, Lemma 7.17 gives $\widehat{\psi}_{P_0}$, θ_{P_0} with $\theta_{P_0}\widehat{\psi}_{P_0}=\mathrm{id}$, so $P_0\cong c_0\oplus P_2$ with $c_0=c_0(\mathbb{N},K)$ and $P_2=\ker\theta_{P_0}$, and $P_0\cong c_0\times c_0\times(c_0\oplus P_2)$; the chart $(a,d,e,z)\mapsto(a,d+a\cdot e,e,z)$ again has $\pi_2D\psi i_3=\mu_a$. □

Proof of Theorem 11.19. Step 1 (item (1)). Let g be an H-family and $x_0\in U$. Left multiplication by $g(x_0)$ is a homeomorphism of H, so $g(U')\subseteq g(x_0)\mathsf{H}_0$ for an open $U'\ni x_0$; then $x\mapsto g(x_0)^{-1}g(x)$ is a V_0-family on U', and $\Lambda^k_\pi g(x)=\widehat{\Lambda}^k_\pi g(x_0)\circ\Lambda^k_\pi(g(x_0)^{-1}g(x))$ (Proposition 3.1). Conversely, for a V_0-family u and $x_0\in U$ choose an open $U'\ni x_0$ and $N\ge0$ with $|\vartheta|^N\|u\|<\varepsilon$ on U', and $\lambda_i=\vartheta^{N+i}$, $0\le i\le k$. Each $1+\lambda_iu$ is an H-family on U', with values in $1+\mathsf{V}_0(\varepsilon)$, and Lemma O.3 with $G=\widehat{E}$, a the inclusion of E and $b=u(x)$ gives $\Lambda^k_\pi u=\sum_i\chi_{ki}\Lambda^k_\pi(1+\lambda_iu)$. For Banach E take $\mathsf{H}=\mathsf{H}_0=\mathrm{GL}(E)$, $\mathsf{V}_0=\mathcal{L}(E)$ and $\varepsilon=1$ (Neumann series).

Step 2 (item (2)). For $\|w\|<1$, $\|(1+w)x\|=\|x\|$ by the nonarchimedean inequality and $1+w$ is invertible; apply (1) with $\mathsf{H}=\mathsf{H}_0=\mathrm{Isom}(E)$, $\mathsf{V}_0=\mathcal{L}(E)$, $\varepsilon=1$.

Step 3 (item (3)). As $\sigma_\sharp\mu_a\sigma'_\sharp\mu_b=(\sigma\sigma')_\sharp\mu_{(a\circ\sigma')b}$ and $(\sigma_\sharp\mu_a)^{-1}=\mu_{1/a}\sigma_\sharp^{-1}$, H_m is a group. If $\sigma(t)\ne t$, the tth coordinate of $(\sigma_\sharp\mu_a-1)e_t$ is -1, so $\|\sigma_\sharp\mu_a-1\|\ge1$; hence $\{h\in\mathsf{H}_\mathrm{m}:\|h-1\|<1\}\subseteq\mathsf{H}_\mathrm{d}$, and its translates show that H_d is open in H_m. As $\|\mu_a\|=\|a\|_\infty$, the nonarchimedean inequality gives $1+\mathsf{V}_\mathrm{d}(1)\subseteq\mathsf{H}_\mathrm{d}$; apply (1) with $\mathsf{H}_0=\mathsf{H}_\mathrm{d}$, $\mathsf{V}_0=\mathsf{V}_\mathrm{d}$, $\varepsilon=1$, and H equal to H_m or H_d.

Step 4 (item (4)). By (3) it suffices that $\Lambda^k_\pi\mu_x$ is analytic at no point for the V_d-family $x\mapsto\mu_x$ on $c_0(\mathbb{N},K)$, by Proposition 2.5 and Theorem 7.7 with $r=\infty$. □

Proof of Proposition 11.20. Step 1 (H is a discrete group). As $g(u)g(u')=g(u+u')$, H is a group. For $u\ne u'$ in U some $e_i^*((u-u')e_j)$ is a nonzero element of κ', so $\|g(u)-g(u')\|\ge\|u-u'\|\ge|e_i^*((u-u')e_j)|=1$; hence H is discrete, and H-families and their exterior powers are locally constant.

Step 2 ($g(\mu_x)$ lies in the closed span). For $x\in c_0(\mathbb{N},K)$, the expansions of the coordinates give $x=\sum_{n\ge n_0}\vartheta^nx_{[n]}$, $|\vartheta|^{n_0}\ge\|x\|_\infty$, with $x_{[n]}\in\kappa'^{\mathbb{N}}$ finitely supported as $x(t)\to0$; so $\mu_{x_{[n]}}\in\mathsf{U}$, $\|\vartheta^n\mu_{x_{[n]}}\|\le|\vartheta|^n$, and $g(\mu_x)-1=\sum_n\vartheta^n(g(\mu_{x_{[n]}})-1)$ lies in the closed span of H.

Step 3 (no point of analyticity). By (O.1), $\Lambda^k_\pi\mu_x=\widehat{\Lambda}^k_\pi\pi_D\circ\Lambda^k_\pi g(\mu_x)\circ\Lambda^k_\pi i_E$, so analyticity of $x\mapsto\Lambda^k_\pi g(\mu_x)$ at a point is excluded by Step 4 of the proof of Theorem 11.19. □

Proof of Proposition 11.21. Step 1 (item (1)). Suppose that Q is as in Theorem 11.14, with $\|Q(a_1,\dots,a_k)\| \le \|Q\| \prod_i \|a_i\|_P$. Steps 2 and 3 of the proof of Theorem 7.20 apply to Q with $\|a_i\|_P$ in place of $\|a_i\|_\infty$. For 0/1 vectors a,b,x,y supported in $\{1,\dots,n\}$, $\|a\|_P, \|b\|_P \le \varphi_P^\pm(n)$, $\|1_{>n}\|_P \le \varphi_P^\pm(k-2)$, $\|x\|_E, \|y\|_E \le \varphi_E^\pm(n)$ and $\|e_t\|_E \le \varphi_E^\pm(1)$. So for every even $n \ge 2$

$$n \le k(k-1)\, c_k\, \|Q\|\, \varphi_P^\pm(n)^2 \varphi_E^\pm(n)^2, \qquad c_k = \varphi_P^\pm(k-2)^{k-2} \varphi_E^\pm(1)^{k-2},$$

with $c_2 = 1$. As $\varphi_P^\pm$ and $\varphi_E^\pm$ are nondecreasing, $n/(\varphi_P^\pm(n)^2 \varphi_E^\pm(n)^2)$ is unbounded along even n, which is a contradiction. The assertion on T follows as in Step 4 of the proof of Theorem 11.14.

Step 2 (items (2)–(4)). For $n \ge 1$, $\varphi^\pm_{c_0(\mathbb{N},K)}(n) = 1$ and $\varphi^\pm_{\ell^r(\mathbb{N},K)}(n) = n^{1/r}$. Item (2) follows from item (1) as in Step 2 of the proof of Corollary 11.15, since $n^{1-2/r_1-2/r_2} \to \infty$ when $1/r_1 + 1/r_2 < 1/2$, with $1/r_1 = 0$ for $P = c_0(\mathbb{N},K)$ and $1/r_2 = 0$ for $E = c_0(\mathbb{N},K)$. Item (3) follows as in Step 4 of the proof of Theorem 11.17, with item (1) for $P = E$ in place of Theorem 11.14, since $n^{1-4/r} \to \infty$ for $r > 4$. Item (4) follows as in Step 4 of the proof of Theorem 11.19, with item (1) for $P = E = c_0(\mathbb{N},K)$ in place of Theorem 7.7. □

Appendix P. Deferred proofs

This appendix contains the proofs to which Sections 3–10 refer, in the order of their statements; the notation is that of the corresponding section.

P.1. **Proofs for Section 3.** The notation is that of Section 3.

Proof of Proposition 3.5. The maps Q are bounded and k-linear, and naturality gives $Q(u,\dots,u) = \Lambda^k_\pi u \circ \widehat{\wedge}^k_E \circ s = \Lambda^k_\pi u$ in (1) and $Q(u,\dots,u) = \rho \circ \mathrm{alt}^k_D \circ \Lambda^k_\pi u = \Lambda^k_\pi u$ in (2); Lemma 3.4 proves (1)–(3). For (4), given s, send $m \in \mathrm{Mult}^k(E;F)$ to the alternating map $m^\otimes \circ s \circ \omega_E$. If $m = m' \circ \omega_E$ with $m' \in \mathcal{L}(\Lambda^k_\pi E, F)$, then $m^\otimes = m' \circ \wedge^k_E$ and $m^\otimes \circ s = m'$; so this is a retraction, and it commutes with postcomposition since $(g \circ m)^\otimes = g \circ m^\otimes$. Conversely, the retraction for $F = \widehat{\otimes}^k_\pi E$ sends the canonical k-linear map to $s \circ \omega_E$ for some $s \in \mathcal{L}(\Lambda^k_\pi E, \widehat{\otimes}^k_\pi E)$. Postcomposition by $\widehat{\wedge}^k_E$ turns the canonical map into the alternating map ω_E, so $\widehat{\wedge}^k_E \circ s \circ \omega_E = \omega_E$ and $\widehat{\wedge}^k_E \circ s = \mathrm{id}$. Without completion use $F = \otimes^k_\pi E$. □

Proof of Lemma 3.6. (1) Given $\tilde\alpha_1$, $\alpha_2 - i'\tilde\alpha_1$ vanishes on $\ker q$, so $\tilde\alpha_3(qy) = (\alpha_2 - i'\tilde\alpha_1)y$ defines a linear map, bounded because q is open, with $q'\tilde\alpha_3 q = q'\alpha_2 = \alpha_3 q$. Given $\tilde\alpha_3$, $q'(\alpha_2 - \tilde\alpha_3 q) = 0$, so $\alpha_2 - \tilde\alpha_3 q = i'\tilde\alpha_1$ for a linear $\tilde\alpha_1$, bounded because i' is bounded below, and $i'\tilde\alpha_1 i = \alpha_2 i = i'\alpha_1$. (2) Apply (1) to $\mathcal{Y}' = \mathcal{Y}$ and the identities. (3) The space Y_α is a Banach space, and q_α is well defined, surjective and open, with kernel $\{[(y', iy)]\} = \{[(y' + \alpha y, 0)]\} = i_\alpha(Y')$; and $\|y'\| \le \|y' + \alpha y\| + C\|\alpha\|\|iy\|$ for all y shows that i_α is injective and bounded below. An extension $\tilde\alpha$ gives the left inverse $[(y',y)] \mapsto y' + \tilde\alpha y$ of i_α, and a left inverse λ gives the extension $y \mapsto \lambda[(0,y)]$, since $[(0,iy)] = [(\alpha y, 0)]$. Now apply (2). □

Proof of Lemma 3.7. Since $\wedge^k_E = \omega^\otimes_E$ is contractive, it suffices to find for y and $\delta > 0$ a preimage z with $\|z\| < \|y\| + 3\delta$. For $y \in \Lambda^k_\pi E$ take $z = \sum_j x_{j1} \otimes \cdots \otimes x_{jk}$ for a representation $y = \sum_j x_{j1} \wedge \cdots \wedge x_{jk}$ with $\sum_j \prod_a \|x_{ja}\| < \|y\| + \delta$. For $y \in \widehat{\Lambda}^k_\pi E$ write $y = \sum_{l \ge 1} y_l$ with $y_l \in \Lambda^k_\pi E$, $\|y_1\| < \|y\| + \delta$ and $\sum_{l \ge 2} \|y_l\| < \delta$, and choose preimages z_l with $\|z_l\| < \|y_l\| + 2^{-l}\delta$; the series $z = \sum_l z_l$ converges, $\widehat{\wedge}^k_E z = y$ and $\|z\| < \|y\| + 3\delta$. □

Proof of Lemma 3.10. (1)⇒(2): a retraction ρ, extended to $\widehat{\otimes}^k_\pi D$, satisfies $\rho\,\mathrm{alt}^k_D = \mathrm{id}$ on $\widehat{\Lambda}^k_\pi D$; so $\|y\| \le \|\rho\|\|\mathrm{alt}^k_D\, y\|$, ρ is a left inverse of the first map of $\mathcal{R}^k_D$, and Lemma 3.6(2) applies. (2)⇒(1): restrict a bounded left inverse to $\otimes^k_\pi D$. (1)⇔(3): $\mathrm{alt}^k_D\, \omega_D$ is the antisymmetrization

of $(y_1,\dots,y_k)\mapsto y_1\otimes\cdots\otimes y_k$. So $\eta(y_1,\dots,y_k)=\rho(y_1\otimes\cdots\otimes y_k)$ gives $\mathcal{A}_D\eta=\rho\,\mathrm{alt}^k_D\,\omega_D=\omega_D$; conversely $\rho=\eta^{\otimes}$ satisfies $\rho\,\mathrm{alt}^k_D\,\omega_D=\mathcal{A}_D\eta=\omega_D$, and the wedges span $\Lambda^k_\pi D$. □

Proof of Lemma 3.11. (1)⇒(3): choose a projection R_n onto $\widehat{\Gamma}^n_\pi P$ with $\|R_n\|\le g_n(P)+\delta$. The linearization $\Phi^{\otimes}\colon\widehat{\otimes}^n_\pi P\to Z$ of Φ, of norm $\|\Phi\|$, maps each $h^{\otimes n}$, hence $\widehat{\Gamma}^n_\pi P$, into W. So $Q(h_1,\dots,h_n)=\Phi^{\otimes}R_n(h_1\otimes\cdots\otimes h_n)$ takes values in W and has diagonal $\Phi^{\otimes}(h^{\otimes n})$. (3)⇒(2): take $W=\widehat{\Gamma}^n_\pi P$, $Z=\widehat{\otimes}^n_\pi P$ and $\Phi(h_1,\dots,h_n)=h_1\otimes\cdots\otimes h_n$. (2)⇒(1): the linearization of the n-linear map maps $\widehat{\otimes}^n_\pi P$ into $\widehat{\Gamma}^n_\pi P$ and fixes every $h^{\otimes n}$, hence every point of $\widehat{\Gamma}^n_\pi P$. For the last assertion, for $\Phi\in\mathrm{Mult}^n(P;Y)$ the n-linear map $(h_1,\dots,h_n)\mapsto\Phi^{\otimes}R_n(h_1\otimes\cdots\otimes h_n)$ has the diagonal of Φ, and it is 0 when Φ has zero diagonal, since $\Phi^{\otimes}$ then vanishes on $\widehat{\Gamma}^n_\pi P$. So it defines a right inverse of the quotient map, of norm at most $\|R_n\|$, that commutes with postcomposition. □

Proof of Proposition 3.13. Naturality of $\wedge^k$, extended to completions, gives $\widehat{\wedge}^k_G w^{\otimes k}=\widehat{\Lambda}^k_\pi w\,\widehat{\wedge}^k_E$. So $w^{\otimes k}(\mathcal{N}^k_E)\subseteq\mathcal{N}^k_G$, and $w^{\otimes k}|_{\mathcal{N}^k_E}$, $w^{\otimes k}$, $\widehat{\Lambda}^k_\pi w$ satisfy the hypotheses of Lemma 3.6(1) for $\mathcal{S}^k_E$ and $\mathcal{S}^k_G$. Now (a)⇔(b) because sections extend by continuity and restrict, (b)⇔(c) is Lemma 3.6(1), and (c)⇔(d) is Lemma 3.6(3). In (2), naturality of $\wedge^k$ and alt^k gives $\widehat{\wedge}^k_{G_1}v^{\otimes k}s\,\Lambda^k_\pi v'=\widehat{\Lambda}^k_\pi v\,\widehat{\wedge}^k_G s\,\Lambda^k_\pi v'$ and $\widehat{\Lambda}^k_\pi v\,\rho\,v'^{\otimes k}\,\mathrm{alt}^k_{G_0}=\widehat{\Lambda}^k_\pi v\,\rho\,\mathrm{alt}^k_E\,\Lambda^k_\pi v'$; in either case the result is $\widehat{\Lambda}^k_\pi v\,\Lambda^k_\pi w\,\Lambda^k_\pi v'=\Lambda^k_\pi(vwv')$. □

From here on, q and $\overline{\mathrm{ev}}$ are the maps defined after Lemma 3.14.

Proof of Theorem 3.16. In each case we give an extension as in Theorem 3.15(2), or a lift.

Step 1 (source). By Proposition 3.13(1), $w^{\otimes k}|_{\mathcal{N}^k_E}$ has a bounded linear extension $\xi_w\colon\widehat{\otimes}^k_\pi E\to\mathcal{N}^k_G$, and $\tau\xi_w$ extends $\mathrm{ev}\,|_{\mathcal{N}^k_E}$. For (S^k_E) take $w=\mathrm{id}_E$ and $\tau=\mathrm{ev}\,|_{\mathcal{N}^k_E}$. For the data (G,w,ε), the formula for τ in part (1) defines a bounded linear map on $\widehat{\otimes}^k_\pi G$ with values in $\widehat{\Lambda}^k_\pi\widehat{D}=\widehat{\Lambda}^k_\pi D$ (the paragraph before Theorem 3.16) and $\|\tau\|\le\|\varepsilon\|^k$. Naturality of $\wedge^k$ gives $\tau(\zeta)(u,\dots,u)=\widehat{\Lambda}^k_\pi\varepsilon(u)\,\widehat{\wedge}^k_G\zeta$, which vanishes for $\zeta\in\mathcal{N}^k_G$. On pure tensors, $\tau(w^{\otimes k}(x_1\otimes\cdots\otimes x_k))(u_1,\dots,u_k)=\varepsilon(u_1)wx_1\wedge\cdots\wedge\varepsilon(u_k)wx_k=u_1x_1\wedge\cdots\wedge u_kx_k$, so $\tau w^{\otimes k}=\mathrm{ev}$ by (3.2) and continuity.

Step 2 (parameter). Let γ be a bounded linear right inverse of q. Then $\xi=\mathrm{ev}-\gamma\overline{\mathrm{ev}}\,\widehat{\wedge}^k_E$ equals ev on $\mathcal{N}^k_E$, and $q\xi=q\,\mathrm{ev}-\overline{\mathrm{ev}}\,\widehat{\wedge}^k_E=0$. If $(\Gamma^k_{\mathcal{L}(E,D)})$ holds, $\mathcal{M}^k(\mathcal{L}(E,D);\widehat{\Lambda}^k_\pi D)$ splits by Lemma 3.11. In the general form, the operators vanishing on $\mathcal{N}^k_E$ form a closed subspace W of $Z=\mathcal{L}(\widehat{\otimes}^k_\pi E,\widehat{\Lambda}^k_\pi D)$, and $T\mapsto T\widehat{\wedge}^k_E$ is an isometry of $\mathcal{L}(\widehat{\Lambda}^k_\pi E,\widehat{\Lambda}^k_\pi D)$ onto W, by Lemma 3.7. Lemma 3.11(3) gives a bounded k-linear $Q'\colon P^k\to\mathcal{L}(\widehat{\Lambda}^k_\pi E,\widehat{\Lambda}^k_\pi D)$ with $Q'(h,\dots,h)\widehat{\wedge}^k_E=\Phi(h,\dots,h)$. Then $Q(u_1,\dots,u_k)=Q'(\psi u_1,\dots,\psi u_k)|_{\Lambda^k_\pi E}$ is bounded and k-linear with $Q(u,\dots,u)=\Lambda^k_\pi u$, and Lemma 3.4 applies.

Step 3 (target). The formula $\xi'(z)(u_1,\dots,u_k)=\mathrm{alt}^k_D\big(\mathrm{ev}(z)(u_1,\dots,u_k)\big)-(u_1\otimes\cdots\otimes u_k)\,\mathrm{alt}^k_E(\widehat{\wedge}^k_E z)$ defines a bounded linear $\xi'\colon\widehat{\otimes}^k_\pi E\to\mathrm{Mult}^k(\mathcal{L}(E,D);\widehat{\otimes}^k_\pi D)$ that equals $\mathrm{alt}_*\,\mathrm{ev}$ on $\mathcal{N}^k_E$. Its values have zero diagonal, since $\xi'(z)(u,\dots,u)=\mathrm{alt}^k_D\,\widehat{\Lambda}^k_\pi u\,\widehat{\wedge}^k_E z-u^{\otimes k}\,\mathrm{alt}^k_E\,\widehat{\wedge}^k_E z=0$ by naturality of alt^k. Lemma 3.6(3) gives the first assertion. If ρ is a retraction of alt^k_D, extended to $\widehat{\otimes}^k_\pi D$, then $\rho\circ\xi'$ extends $\rho\,\mathrm{alt}^k_D\,\mathrm{ev}\,|_{\mathcal{N}^k_E}=\mathrm{ev}\,|_{\mathcal{N}^k_E}$.

Step 4 (relative target). Let $\rho'\colon\otimes^k_\pi G\to\widehat{\Lambda}^k_\pi\widehat{D}=\widehat{\Lambda}^k_\pi D$ satisfy $\rho'\,\mathrm{alt}^k_G=\Lambda^k_\pi w'$, and put

$$\xi(z)(u_1,\dots,u_k)=\mathrm{ev}(z)(u_1,\dots,u_k)-\rho'\big(\varepsilon(u_1)\otimes\cdots\otimes\varepsilon(u_k)\big)\,\mathrm{alt}^k_E(\widehat{\wedge}^k_E z).$$

This is a bounded linear map, of norm at most $1 + k!\|\rho'\|\|\varepsilon\|^k$, that equals ev on $\mathcal{N}_E^k$. By naturality of alt^k for $\varepsilon(u)\colon E \to G$, $\rho'\varepsilon(u)^{\otimes k}\,\mathrm{alt}_E^k = \rho'\,\mathrm{alt}_G^k\,\widehat{\Lambda}_\pi^k\varepsilon(u) = \widehat{\Lambda}_\pi^k w'\,\widehat{\Lambda}_\pi^k\varepsilon(u) = \widehat{\Lambda}_\pi^k u$, so $\xi(z)(u,\dots,u) = 0$. □

Proof of Theorem 3.18. Step 1 (part (1)). By (3.2) and (3.4), $\mathrm{ev}(j_\otimes e_t^\otimes) = Q_<^\sharp(e_t^\wedge)$ for $t \in \Sigma_<^k$, which is the first identity on a basis. Hence $q\circ Q_<^\sharp = \overline{\mathrm{ev}}\circ\widehat{\wedge}_E^k\circ j_\otimes\circ s_< = \overline{\mathrm{ev}}\circ j_\Lambda\circ\wedge_V\circ s_< = \overline{\mathrm{ev}}\circ j_\Lambda$.

Step 2 (part (2)). Since Q is a lift, $q\circ Q^\sharp = \overline{\mathrm{ev}}$, so $q\circ L = \overline{\mathrm{ev}}\circ j_\Lambda - q\circ Q_<^\sharp = 0$ by part (1). Thus L takes values in $\ker q = \mathrm{Mult}_0^k$, and Φ is k-linear with zero diagonal.

Step 3 (part (3)). Put $T = Q_<^\sharp + L$. Since $\|Ty\| \le \|T\|\|j_\Lambda y\|$, $j_\Lambda(\Lambda_K^k V)$ is dense and Mult^k is complete, $T = \widehat{T}\circ j_\Lambda$ for a bounded linear $\widehat{T}\colon \widehat{\Lambda}_\pi^k E \to \mathrm{Mult}^k$ with $\|\widehat{T}\| = \|T\|$. By part (1) and $q\circ L = 0$, $q\circ\widehat{T} = \overline{\mathrm{ev}}$ on a dense subspace, hence everywhere. So $\widehat{T} = Q^\sharp$ for a lift Q, with $\|Q\| = \|T\|$, and Q is unique by density. Conversely, a lift Q gives the correction L of part (2), with $\|Q_<^\sharp + L\| = \|Q^\sharp\circ j_\Lambda\| = \|Q\|$ by density. This proves the formula for $c_k(E,D)$ and the last assertion.

Step 4 (part (4)). Since $q\circ\mathrm{ev} = \overline{\mathrm{ev}}\circ\widehat{\wedge}_E^k$, part (1) gives $q(\mathrm{ev}(x_1\otimes\cdots\otimes x_k)) = \overline{\mathrm{ev}}(j_\Lambda(x_1\wedge\cdots\wedge x_k)) = q(Q_<^\sharp(x_1\wedge\cdots\wedge x_k))$, so $\Omega_<$ takes values in Mult_0^k. For $x \in V^k$,

$$\Omega_<(x) - L(x_1\wedge\cdots\wedge x_k) = \mathrm{ev}(x_1\otimes\cdots\otimes x_k) - (Q_<^\sharp + L)(x_1\wedge\cdots\wedge x_k);$$

since $\|\mathrm{ev}\| \le 1$ and $\|x_1\wedge\cdots\wedge x_k\|_\pi \le \prod_a\|x_a\|$, this gives the second inequality. For the first, let $\delta > 0$ and write $y \in \Lambda_K^k V$ as $\sum_i x_{i1}\wedge\cdots\wedge x_{ik}$ with $x_{ia} \in V$ and $\sum_i\prod_a\|x_{ia}\| < \|y\|_\pi + \delta$. Summing the identity for $(x_{i1},\dots,x_{ik})$ over i gives $\|(Q_<^\sharp + L)y\| \le (1 + \|\Omega_< - L\circ\wedge\|)(\|y\|_\pi + \delta)$; let $\delta \to 0$. The alternating k-linear maps $V^k \to \mathrm{Mult}_0^k$ are the maps $L\circ\wedge$, by the universal property of $\Lambda_K^k V$. So part (3) and the two inequalities give the remaining assertions. □

Proof of Lemma 3.19. Step 1 (algebra). Let $I_E \subseteq \otimes_K^k E$ be the span of the pure tensors with $x_a = x_{a+1}$ for some $a < k$. Modulo I_E, exchanging two adjacent entries changes the sign of a pure tensor, since $x\otimes y + y\otimes x = (x+y)\otimes(x+y) - x\otimes x - y\otimes y$. So I_E contains every pure tensor with two equal entries, and it is the kernel of the quotient map $\otimes_K^k E \to \Lambda_K^k E$, by the universal property of $\Lambda_K^k E$. The same holds for V.

Step 2 (closure). Let I be the image of I_E in $\otimes_\pi^k E$, let $z \in \otimes_\pi^k E$ be the class of $\tilde{z} \in \otimes_K^k E$, and let $\delta > 0$. Write the image of $\tilde{z}$ in $\Lambda_K^k E$ as $\sum_i x_{i1}\wedge\cdots\wedge x_{ik}$ with $\sum_i\prod_a\|x_{ia}\| < \|\wedge_E^k z\| + \delta$; by Step 1, $\tilde{z} - \sum_i x_{i1}\otimes\cdots\otimes x_{ik} \in I_E$, so the distance from z to I is at most $\|\wedge_E^k z\|$. If $z \in \mathcal{N}_E^k$ and $z_l \in \otimes_\pi^k E$ converge to z, then $\|\wedge_E^k z_l\| = \|\widehat{\wedge}_E^k(z_l - z)\| \to 0$, so z lies in the closure of I. As $I \subseteq \mathcal{N}_E^k$ and $\mathcal{N}_E^k$ is closed, $\mathcal{N}_E^k$ is the closure of I; with Step 1 this proves the first assertion.

Step 3 (entries in V). A pure tensor $x_1\otimes\cdots\otimes x_k$ with $x_a = x_{a+1}$ is a limit of pure tensors with entries in V and equal entries in the slots a and $a+1$, since V is dense and $(x_1,\dots,x_k) \mapsto x_1\otimes\cdots\otimes x_k$ is continuous. With Step 2 this proves the last assertion. □

Proof of Corollary 3.20. By linearization and continuity, the bounded k-linear maps $\Psi\colon E^k \to \mathrm{Mult}_0^k$ correspond to the bounded linear maps $\xi\colon \widehat{\otimes}_\pi^k E \to \mathrm{Mult}_0^k$, by $\xi(x_1\otimes\cdots\otimes x_k) = \Psi(x_1,\dots,x_k)$, with $\|\xi\| = \|\Psi\|$. By (3.2) the displayed identity says $\xi(x_1\otimes\cdots\otimes x_k) = \mathrm{ev}(x_1\otimes\cdots\otimes x_k)$. If it holds whenever $x_a = x_{a+1}$ for some a, then $\xi = \mathrm{ev}$ on $\mathcal{N}_E^k$ by Lemma 3.19 and continuity, so it holds whenever two entries coincide; conversely, an extension of $\mathrm{ev}\big|_{\mathcal{N}_E^k}$ gives the identity on these tuples, whose pure tensors lie in $\mathcal{N}_E^k$. So the maps Ψ correspond to the extensions ξ of Theorem 3.15(2), and that theorem with its constants gives the corollary. □

Proof of Corollary 3.21. By Theorem 3.18(1), $q\circ\mathrm{ev}\circ\nu\circ j_\Lambda=\overline{\mathrm{ev}}\circ\widehat{\wedge}^k_E\circ\nu\circ j_\Lambda=\overline{\mathrm{ev}}\circ j_\Lambda=q\circ Q^\sharp_<$, so $\Omega^\nu_<$ takes values in Mult^k_0; it is homogeneous because ν is homogeneous and ev, j_Λ, $Q^\sharp_<$ are linear. Since $Q^\sharp_<$ is linear, $\Omega^\nu_<(\sum_i y_i)-\sum_i\Omega^\nu_<(y_i)=\mathrm{ev}\big(\nu(\sum_i j_\Lambda y_i)-\sum_i\nu(j_\Lambda y_i)\big)$; the argument of ev lies in $\mathcal{N}^k_E$ and has norm at most $2(1+\delta)\sum_i\|y_i\|_\pi$ by the bound on ν, and $\|\mathrm{ev}\|\le 1$. For a linear $L\colon\Lambda^k_K V\to\mathrm{Mult}^k_0$, $\Omega^\nu_<-L=\mathrm{ev}\circ\nu\circ j_\Lambda-(Q^\sharp_<+L)$ with $\|\mathrm{ev}(\nu(j_\Lambda y))\|\le(1+\delta)\|y\|_\pi$. So the least constant for $\Omega^\nu_<-L$ and $\|Q^\sharp_<+L\|$ each exceed the other by at most $1+\delta$, and Theorems 3.18(3) and 3.15 give the remaining assertions. □

Proof of Theorem 3.22. Put $C_\alpha=2(c_+/c_-)\|\alpha\|(C_cC_e)^k$. For $z\in\otimes^k_K V$, $j_\otimes(z-s_<(\wedge_V z))$ lies in $\mathcal{N}^k_E$, since $\widehat{\wedge}^k_E\circ j_\otimes=j_\Lambda\circ\wedge_V$ and $\wedge_V\circ s_<=\mathrm{id}$; let $\alpha_0(z)$ be its image under α.

Step 1 (expansion). For $t\in\Sigma^k$ with a repeated entry put $\zeta_t=e^\otimes_t$. For $t\in\Sigma^k$ with distinct entries let $\sigma\in S_k$ be the permutation with $t\circ\sigma=(t_{\sigma(1)},\dots,t_{\sigma(k)})\in\Sigma^k_<$, and put $\zeta_t=e^\otimes_t-\mathrm{sgn}(\sigma)\,e^\otimes_{t\circ\sigma}$. Since $e_{t_1}\wedge\cdots\wedge e_{t_k}$ is $\mathrm{sgn}(\sigma)\,e^\wedge_{t\circ\sigma}$ for distinct entries and 0 otherwise, $\wedge_V\zeta_t=0$; and $\|j_\otimes\zeta_t\|\le 2C^k_e$. For $x_1,\dots,x_k\in V$, expanding $x_l=\sum_t x_l(t)e_t$ gives the finite sum

$$x_1\otimes\cdots\otimes x_k-s_<(x_1\wedge\cdots\wedge x_k)=\sum_{t\in\Sigma^k}\prod_{l=1}^k x_l(t_l)\,\zeta_t.$$

Step 2 (bound). Since $|||\cdot|||_Y$ is a nonarchimedean norm, Step 1 gives

$$|||\alpha_0(x_1\otimes\cdots\otimes x_k)|||_Y\le\max_t\prod_{l=1}^k|x_l(t_l)|\,|||\alpha(j_\otimes\zeta_t)|||_Y\le C^k_c\prod_{l=1}^k\|x_l\|\cdot\frac{2C^k_e\|\alpha\|}{c_-},$$

the maximum over the finitely many t with $\prod_l x_l(t_l)\neq 0$. As $\|\cdot\|\le c_+|||\cdot|||_Y$, this gives $\|\alpha_0(x_1\otimes\cdots\otimes x_k)\|\le C_\alpha\prod_l\|x_l\|$, and so $\|\alpha_0(z)\|\le C_\alpha\sum_i\prod_l\|x_{il}\|$ for every representation $z=\sum_i x_{i1}\otimes\cdots\otimes x_{ik}$ of $z\in\otimes^k_K V$ with entries in V. By the paragraph before Theorem 3.16, $\|\alpha_0(z)\|\le C_\alpha\|j_\otimes z\|$. Since $j_\otimes(\otimes^k_K V)$ is dense and Y is complete, there is a bounded linear $\tilde\alpha\colon\widehat{\otimes}^k_\pi E\to Y$ with $\tilde\alpha\circ j_\otimes=\alpha_0$ and $\|\tilde\alpha\|\le C_\alpha$.

Step 3 (agreement). If $z\in\otimes^k_K V$ and $\wedge_V z=0$, then $\tilde\alpha(j_\otimes z)=\alpha_0(z)=\alpha(j_\otimes z)$. The pure tensors with entries in V and two equal adjacent entries lie in $\ker\wedge_V$, and their images span a dense subspace of $\mathcal{N}^k_E$ by Lemma 3.19. So $\tilde\alpha=\alpha$ on $\mathcal{N}^k_E$ by continuity, and $\alpha_*[\mathcal{S}^k_E]=0$ by Lemma 3.6(3).

Step 4 (the class). Let $|||\cdot|||_D$ be a nonarchimedean norm on $\widehat{\Lambda}^k_\pi D$ with $c_-|||\cdot|||_D\le\|\cdot\|\le c_+|||\cdot|||_D$. Take $Y=\mathrm{Mult}^k_0$ and, for $\Phi\in Y$, let $|||\Phi|||_Y$ be the least C with $|||\Phi(u_1,\dots,u_k)|||_D\le C\prod_l\|u_l\|$ for all $u_1,\dots,u_k\in\mathcal{L}(E,D)$. Then $|||\cdot|||_Y$ is a nonarchimedean norm on the Banach space Y, since $|||\cdot|||_D$ is one, and $c_-|||\cdot|||_Y\le\|\cdot\|\le c_+|||\cdot|||_Y$, since $\|\Phi\|$ is the least C with $\|\Phi(u_1,\dots,u_k)\|\le C\prod_l\|u_l\|$. The first part, with this Y and $\alpha=\mathrm{ev}\,|_{\mathcal{N}^k_E}$, gives $\mathrm{ob}_k(E,D)=0$. □

P.2. **Proofs for Section 4.** The notation is that of Section 4.

Proof of Theorem 4.2. Since $k!=0$ in K, $p=\mathrm{char}\,K$ satisfies $2\le p\le k$. Let $\nu\in S_k$ be a cycle of length p and g the permutation matrix with $ge_i=e_{\nu(i)}$, which is monomial. Corollary 4.4(2) excludes (1)–(3) already for this g. For the algebraic powers, let H be the group generated by ν, acting on $I=[k]$, so that $V=K^k$. As in the proof of Corollary 4.4(2), a map equivariant for g is H-equivariant, and $\bar{\mathsf{H}}_{[k]}=\bar{\mathsf{H}}_{\mathbf{m}}$ has order p for the k-multiset $\mathbf{m}$ with value 1 on $[k]$; so Theorem 4.3(1)–(3) excludes such maps. □

Proof of Corollary 4.4. (1) We treat (1) and (2) of Theorem 4.3; for (3) replace T by $\mathrm{supp}\,\mathbf{m}$ and the stabilizers of T by those of $\mathbf{m}$. The first statement implies the second by restriction. If

$h \in \mathsf{H}$ maps a k-subset T onto itself and induces on it a permutation of order $p = \operatorname{char} K$, then p divides $|\bar{C}_T|$ for the group C generated by h, and Theorem 4.3 excludes a C-equivariant map; so the second statement implies the third. If the first fails, Theorem 4.3 gives a k-subset T such that $p = \operatorname{char} K > 0$ divides $|\bar{\mathsf{H}}_T|$, and by Cauchy's theorem some element of H_T induces on T a permutation of order p; so the third fails. If $\operatorname{char} K = 0$, all $|\bar{\mathsf{H}}_T|$ and $|\bar{\mathsf{H}}_{\mathbf{m}}|$ are nonzero in K.

(2) Let H be the group generated by ν, acting on $I = [k]$ and on E_0 by $he_i = e_{h(i)}$; these operators are bounded since the e_i^* are, and satisfy (4.3) since $e_i^*(he_j) = \delta_{i,h(j)} = e^*_{h^{-1}(i)}(e_j)$. For $T = I$ and the k-multiset $\mathbf{m}$ with value 1 on I, $\mathsf{H}_T = \mathsf{H}_{\mathbf{m}} = \mathsf{H}$, and $\bar{\mathsf{H}}_T = \bar{\mathsf{H}}_{\mathbf{m}}$ has order p, which is 0 in K. An equivariance identity for g holds for its powers, since $(g^j)^{\otimes k} = (g^{\otimes k})^j$ and $\Lambda^k_\pi(g^j) = (\Lambda^k_\pi g)^j$, hence for all of H, which Theorem 4.3(4) excludes. □

Proof of Proposition 4.6. (1) This is Proposition B.8(1) applied to Q. With $u_S := \sum_{i\in S} u_i$, every k-linear Φ satisfies $\sum_{S\subseteq[k]}(-1)^{k-|S|}\Phi(u_S,\dots,u_S) = \sum_\sigma \Phi(u_{\sigma(1)},\dots,u_{\sigma(k)})$, as in (4.2). Apply this to $Q(\cdot)\circ\omega_E$ and to $(u_1,\dots,u_k) \mapsto \big((x_b) \mapsto u_1x_1 \wedge \cdots \wedge u_kx_k\big)$, which agree on the diagonal since $Q(u,\dots,u) = \Lambda^k_\pi u$: the right sides give $\sum_\sigma Q(u_{\sigma(1)},\dots,u_{\sigma(k)})\circ\omega_E = \Lambda_{\mathrm{sym}}(u_1,\dots,u_k)\circ\omega_E$, and pure wedges span $\Lambda^k_\pi E$. (2) A symmetric Q would give $\Lambda_{\mathrm{sym}} = k!\,Q = 0$ by (1). Let $f_1,\dots,f_k$ be the functionals, choose $x_1,\dots,x_k \in E$ with $f_i(x_j) = \delta_{ij}$ and, since $\Lambda^k_\pi D$ is dense in $\widehat{\Lambda}^k_\pi D \neq 0$, $d_1,\dots,d_k \in D$ with $d_1 \wedge \cdots \wedge d_k \neq 0$. For $u_i = f_i(\cdot)\,d_i$ only $\sigma = \mathrm{id}$ contributes to (4.4), so $\Lambda_{\mathrm{sym}}(u_1,\dots,u_k)(x_1 \wedge \cdots \wedge x_k) = d_1 \wedge \cdots \wedge d_k \neq 0$. (3) By (4.1), $\wedge^k_D = \rho \circ \mathrm{alt}^k_D \circ \wedge^k_D = \sum_\sigma \operatorname{sgn}(\sigma)\rho\circ\tau_\sigma = k!\,\rho = 0$, so $\Lambda^k_\pi D = \wedge^k_D(\otimes^k_\pi D) = 0$. (4) For $w \in \Lambda^k_\pi E$, (4.1) gives $\mathrm{alt}^k_E(w) = \mathrm{alt}^k_E \widehat{\wedge}^k_E s(w) = \sum_\sigma \operatorname{sgn}(\sigma)\tau_\sigma s(w) = k!\,s(w) = 0$. With f_i and x_j as in (2), the bounded functional on $\widehat{\otimes}^k_\pi E$ linearizing $(y_a) \mapsto \prod_a f_a(y_a)$ takes the value $\sum_\sigma \operatorname{sgn}(\sigma)\prod_a f_a(x_{\sigma(a)}) = 1$ at $\mathrm{alt}^k_E(x_1 \wedge \cdots \wedge x_k)$, a contradiction. □

P.3. **Proofs for Section 5.**

Proof of Theorem 5.2. Step 1 (simultaneous extension). Let Y be a nonarchimedean seminormed space, G a nonarchimedean spherically complete space, $T_\alpha\colon Y \to G$ a nonempty family of bounded operators and $r_\alpha \geq 0$ radii with $\|T_\alpha - T_\beta\| \leq \max(r_\alpha, r_\beta)$, and T_0 a linear map on a subspace Y_0 with $\|T_0 d - T_\alpha d\| \leq r_\alpha\|d\|$ for all $d \in Y_0$ and all α. We extend T_0 to Y with all these bounds. For $x \notin Y_0$, the balls

$$\overline{B}\big(T_\alpha(d+x) - T_0 d,\ r_\alpha\|d+x\|\big), \qquad \alpha,\ d \in Y_0,$$

meet pairwise: for $r_\alpha \leq r_\beta$, the centers for (α,d) and (β,e) differ by at most $\max(r_\alpha\|d+x\|, r_\beta\|e+x\|)$, by the hypotheses and the nonarchimedean inequality. A common point, given by spherical completeness, is a value at x extending T_0 to $Y_0 + Kx$ with all bounds, and Zorn's lemma extends T_0 to Y. With the single operator 0 and radius one, contractions extend to contractions: this is the extension mechanism of Ingleton's theorem [11].

Step 2 ($\mathrm{Alt}^k(E;F)$ is spherically complete). Let Y be the free vector space on the symbols $[x_1,\dots,x_k]$, $x \in E^k$, with the seminorm $\|\sum_x c_x[x]\| = \max_x |c_x| \prod_i \|x_i\|$, and let Y_0 be the span of the multilinearity and repeated-input relations. Alternating maps $E^k \to F$ correspond isometrically to bounded linear maps $Y \to F$ vanishing on Y_0: one direction uses the nonarchimedean inequality in F, the other evaluates on generators. Pairwise-meeting balls $\overline{B}(m_\alpha, r_\alpha)$ in $\mathrm{Alt}^k(E;F)$ satisfy $\|m_\alpha - m_\beta\| \leq \max(r_\alpha, r_\beta)$, since this operator norm is nonarchimedean. So the maps T_α corresponding to the m_α, and $T_0 = 0$ on Y_0, satisfy the hypotheses of Step 1 with $G = F$; the extension of T_0 vanishes on Y_0, and its alternating map lies in every ball.

Step 3 (the retraction). The operator norm of $\mathrm{Mult}^k(E;F)$ is nonarchimedean. By Step 1 with $G = \mathrm{Alt}^k(E;F)$, spherically complete by Step 2, the inverse of the isometric inclusion

$j_{E,F}$ extends to a contraction of $\mathrm{Mult}^k(E;F)$ onto $\mathrm{Alt}^k(E;F)$, which is the required retraction. Proposition 5.1 finishes the proof. In degree zero the alternating space is F. □

Proof of Theorem 5.4. Step 1 (ultrametric envelopes). As in the proof of Proposition I.1, with the nonarchimedean inequality of F in place of that of K, every bounded k-linear map m into F satisfies $\|m(x_1,\dots,x_k)\| \le \|m\| \prod_a q_Y(x_a)$. Hence $m \mapsto m \circ j_Y^{\times k}$ is an isometry of $\mathrm{Alt}^k(Y^u;F)$ onto $\mathrm{Alt}^k(Y;F)$, every $u \in \mathcal{L}(E,D)$ induces $u^u \in \mathcal{L}(E^u,D^u)$ with $\|u^u\| \le \|u\|$, linearly in u, and $A^k_{E,D;F}(u)$ corresponds to $A^k_{E^u,D^u;F}(u^u)$. A lift for (E^u,D^u) therefore gives one for (E,D) of no larger norm. Since $(Y^u)^u = Y^u$, we may assume $E = E^u$ and $D = D^u$.

Step 2 (the expansion on $c_0(I,K)$). Order I, and let (e_i) and (e_i^*) be the unit vectors and coordinate functionals of $c_0(I,K)$. For $m \in \mathrm{Alt}^k(c_0(I,K);F)$ and $x_1,\dots,x_k \in c_0(I,K)$, the family $\det(e_{i_a}^*(x_b))_{a,b}\, m(e_{i_1},\dots,e_{i_k})$, $i_1 < \cdots < i_k$, is null: if a determinant has absolute value at least $\delta > 0$, every i_a lies in the finite set where some $|e_i^*(x_b)| \ge \delta/\max_b\|x_b\|^{k-1}$. Its sum is

$$m(x_1,\dots,x_k) = \sum_{i_1<\cdots<i_k} \det\bigl(e_{i_a}^*(x_b)\bigr)_{a,b}\, m(e_{i_1},\dots,e_{i_k}), \tag{P.1}$$

by sorting the multilinear expansion for finitely supported x_b, and by density in general, both sides being bounded and multilinear.

Step 3 (lifts). The maps

$$Q(u_1,\dots,u_k)(m)(x) = \sum_{i_1<\cdots<i_k} \det\bigl(e_{i_a}^*(x_b)\bigr)_{a,b}\, m(u_1e_{i_1},\dots,u_ke_{i_k}),$$

for $u_a \in \mathcal{L}(c_0(I,K),D)$, and

$$Q'(u_1,\dots,u_k)(m)(x) = \sum_{i_1<\cdots<i_k} \det\bigl(e_{i_a}^*(u_ax_b)\bigr)_{a,b}\, m(e_{i_1},\dots,e_{i_k}),$$

for $u_a \in \mathcal{L}(E,c_0(I,K))$, are well defined by Step 2, with u_ax_b in place of x_b in row a for Q'. They are alternating in x, of norm at most one, and have diagonal $A^k(u)(m)$ by (P.1). If Q_0 lifts $A^k_{c_0(I,K),D;F}$ and $\iota'\iota = \mathrm{id}_E$, then $Q_0(u_1\iota',\dots,u_k\iota')(m)\circ\iota^{\times k}$ lifts $A^k_{E,D;F}$; if Q_0 lifts $A^k_{E,c_0(I,K);F}$ and $\iota'\iota = \mathrm{id}_D$, then $Q_0(\iota u_1,\dots,\iota u_k)(m\circ\iota'^{\times k})$ does. Both have norm at most $(\|\iota\|\|\iota'\|)^k\|Q_0\|$.

Step 4 (countable type). If E is of countable type, so is E^u, which then has a t-orthogonal basis for every $0 < t < 1$ [19, Proposition 1.4]: nonzero vectors b_i with dense span and $\|\sum_i \lambda_i b_i\| \ge t\max_i|\lambda_i|\|b_i\|$ for finitely many scalars λ_i. The formulas for Q and (P.1), with this basis and its coordinate functionals in place of (e_i) and (e_i^*), give a lift of norm at most t^{-k}, as in the proof of Proposition I.1; similarly for D with Q'. □

Proof of Proposition 5.5. (1) The map $\Phi\colon \delta_i \mapsto e_{i_1}\wedge\cdots\wedge e_{i_k}$ is contractive on finitely supported vectors. The alternating map $\Omega(x_1,\dots,x_k) = (\det(x_b(i_a))_{a,b})_{i\in J^k_<}$ satisfies $\|\Omega(x_1,\dots,x_k)\|_1 \le \sum_{i\in J^k_<}\sum_{\sigma\in S_k}\prod_a|x_{\sigma(a)}(i_a)| \le \prod_b\|x_b\|_1$, since $(i,\sigma)\mapsto i\circ\sigma^{-1}$ is injective into J^k, tuples being maps $\{1,\dots,k\}\to J$. By Proposition 3.1, Ω induces a contraction $\widetilde{\Omega}\colon \Lambda^k_\pi\ell^1(J,K)\to\ell^1(J^k_<,\widehat{K})$ with $\widetilde{\Omega}\Phi\delta_i = \delta_i$, because $(e_{i_b}(i'_a))_{a,b}$ is the identity for $i' = i$ and has a zero row for every other $i' \in J^k_<$. So Φ is isometric. Its range is dense, since wedges of finitely supported vectors span a dense subspace and expand into the $e_{i_1}\wedge\cdots\wedge e_{i_k}$, $i \in J^k_<$. Completion gives (1).

(2) By Lemma 3.7, $\widehat{\wedge}^k_E\colon \widehat{\otimes}^k_\pi E \to \widehat{\Lambda}^k_\pi E$ is a bounded linear surjection of Banach spaces. Projectivity gives a bounded linear s' with $\widehat{\wedge}^k_E\circ s' = \mathrm{id}$, whose restriction to $\Lambda^k_\pi E$ satisfies (S^k_E).

(3) Order J, and let $\iota\colon E\to\ell^1(J,K)$ and $\iota'\colon\ell^1(J,K)\to E$ be bounded linear with $\iota'\iota = \mathrm{id}_E$. The inverse of the isomorphism of (1), followed by $\delta_i \mapsto e_{i_1}\otimes\cdots\otimes e_{i_n}$, restricts to a contraction

$s_0\colon \Lambda_\pi^n \ell^1(J,K) \to \widehat{\otimes}_\pi^n \ell^1(J,K)$ with $\widehat{\wedge}^n \circ s_0 = \mathrm{id}$, by the density shown in (1). By naturality of $\wedge^n$, the map $s = \iota'^{\otimes n} \circ s_0 \circ \Lambda_\pi^n \iota$ satisfies $\widehat{\wedge}_E^n \circ s = \widehat{\Lambda}_\pi^n \iota' \circ \Lambda_\pi^n \iota = \mathrm{id}$, and $\|s\| \le (\|\iota\|\|\iota'\|)^n$. □

Proof of Corollary 5.10. Let ν be a nonarchimedean norm on D with $a\nu(y) \le \|y\| \le b\nu(y)$. Then $\|\sum_t \xi_t y_t\| \le b \max_t |\xi_t| \nu(y_t) \le (b/a)(\sum_t |\xi_t|^3)^{1/3} \max_t \|y_t\|$, so (5.2) holds with $r = 3$, D has an equivalent 3-norm by Proposition 5.8, and Theorem 5.9(2) gives the first assertion. The extension of the coordinate map is the map w of Theorem 5.9(1) for $r = 3$ followed by the contractive inclusion $\ell^3(\Sigma, \widehat{K}) \to c_0(\Sigma, \widehat{K})$, hence S^n-split by the ideal property in Proposition 3.13. □

Proof of Theorem 5.12. Step 1 (the space c_0). Order I and group the coordinate coefficients of each ambient b_n into nondecreasing words. Finite-coordinate uniqueness, as in the proof of Theorem 5.11, puts the grouped coefficients $c_{i_1,\dots,i_n}$ in W, and the nonarchimedean inequality bounds their norms by $\|b_n\|$. The formula

$$q_n(x_1,\dots,x_n) = \sum_{i_1 \le \cdots \le i_n} c_{i_1,\dots,i_n} (x_1)_{i_1} \cdots (x_n)_{i_n}$$

converges, since outside finite sets of words some coordinate factor is arbitrarily small while the others are bounded; completeness of W supplies the sum, of norm at most $\|b_n\| \prod_a \|x_a\|$. The diagonals of q_n and b_n agree on finite supports, hence by density. Theorem 2.3 gives (1).

Step 2 (sorting in ℓ^1). For every set J, the sorting of Step 1 in degree d gives each bounded d-linear $B'\colon \ell^1(J,K)^d \to Z$ whose diagonal lies in W a lift into W with the same diagonal and of norm at most $d!\|B'\|$: W is complete, being closed in Z, each grouped coefficient has at most $d!$ terms, and the resulting series is absolutely summable, bounded by $d!\|B'\| \prod_a \|x_a\|_1$.

Step 3 (factorization of γ). Fix $x_0 \in U$, write $\gamma(x_0 + x) = \sum_n a_n(x,\dots,x)$, and choose $\lambda \in K$ with $0 < |\lambda|$ within the radius of this expansion. Set $J = \coprod_{n \ge 0} I^n$ and

$$T(e_{n,i_1,\dots,i_n}) = \lambda^n a_n(e_{i_1},\dots,e_{i_n}), \qquad g(x)_{n,i_1,\dots,i_n} = \lambda^{-n} x_{i_1} \cdots x_{i_n}.$$

The values of T on unit vectors are uniformly bounded, so T extends by absolute summation to a bounded linear map $\ell^1(J,K) \to \widehat{H}$. The degree-$n$ multilinear coefficient of g has entries $\lambda^{-n}(x_1)_{i_1} \cdots (x_n)_{i_n}$ in its nth block and zero elsewhere, so its norm is at most $|\lambda|^{-n}$; its diagonal block has sum norm $(\|x\|_1/|\lambda|)^n$. So g has a convergent multilinear expansion for $\|x\|_1 < |\lambda|$, without completeness of K, and $T(g(x)) = \gamma(x_0 + x)$ there.

Step 4 (conclusion of (2)). Extend B to $\widehat{H}^d$ by completeness of Z; its diagonal still lies in W, by closedness. By Step 2, the composite of this extension with $T^{\times d}$ has a bounded d-linear lift q into W with the same diagonal. Then $ja(\gamma(x_0 + x)) = jq(g(x),\dots,g(x))$ for $\|x\|_1 < |\lambda|$, so near x_0 the map $a\gamma$ is the diagonal of q, a bounded polynomial, composed with the analytic map g. Hence $a\gamma$ is analytic at every point x_0.

Step 5 (retracts). If $ri = \mathrm{id}_P$ for bounded linear $i\colon P \to V$ and $r\colon V \to P$, apply (1) or (2) to the family composed with r on $r^{-1}(U)$, and restrict along i; in (1) this multiplies the norms of the lifted coefficients by at most $(\|r\|\|i\|)^n$. □

Proof of Proposition 5.13. Step 1 (orbit sums). Let $L = \ell^1(J,K)$ with its unit vectors e_i, and order J. For nondecreasing $t \in J^n$, let O_t be the sum of the tensors $e_{\tilde{t}_1} \otimes \cdots \otimes e_{\tilde{t}_n}$ over the distinct rearrangements $\tilde{t}$ of t, let $b_1,\dots,b_m$ be the distinct entries of t, and for $\alpha \in \mathbb{N}^m$ with $\alpha_1 + \cdots + \alpha_m = n$ let $t(\alpha)$ be the nondecreasing tuple with α_i entries b_i. For $\lambda \in K^m$, expanding the tensor power gives $(\sum_i \lambda_i e_{b_i})^{\otimes n} = \sum_\alpha \lambda^\alpha O_{t(\alpha)}$. A linear functional φ on the span of the $O_{t(\alpha)}$ that vanishes on these pure powers makes the polynomial function $\lambda \mapsto \sum_\alpha \lambda^\alpha \varphi(O_{t(\alpha)})$ vanish on K^m, so $\varphi(O_{t(\alpha)}) = 0$ for all α, since K is infinite. Hence O_t lies in the span of the

pure powers, and in $\widehat{\Gamma}^n_\pi L$. This argument is algebraic: it holds for arbitrary vectors $e_{b_1}, \dots, e_{b_m}$ of a vector space over K in place of the unit vectors.

Step 2 (the projection). As in the proof of Proposition 5.5(1), with the coordinate map $(x_1, \dots, x_n) \mapsto (x_1(t_1) \cdots x_n(t_n))_{t \in J^n}$ in place of Ω, the map $e_{t_1} \otimes \cdots \otimes e_{t_n} \mapsto \delta_t$ onto the unit vectors δ_t of $\ell^1(J^n, \widehat{K})$ extends to an isometric isomorphism $\widehat{\otimes}^n_\pi L \cong \ell^1(J^n, \widehat{K})$. Let $R_n \delta_t = O_t$ for nondecreasing t and $R_n \delta_t = 0$ otherwise. Then $\|R_n\| \le n!$, since O_t is a sum of at most $n!$ unit vectors, and R_n maps into the closed subspace $\widehat{\Gamma}^n_\pi L$ by Step 1. For finitely supported $h \in L$, the coefficients of $h^{\otimes n} = \sum_t h(t_1) \cdots h(t_n) \delta_t$ are constant on rearrangements of t, so $R_n h^{\otimes n} = h^{\otimes n}$. By continuity R_n fixes every pure power, hence $\widehat{\Gamma}^n_\pi L$. This proves (1).

Step 3 (finite coordinates). A normed space P with finite continuous coordinates has a bounded linear isomorphism T onto $\ell^1(\{1, \dots, d\}, K)$. The extension of $T^{\otimes n}$ to the completions is an isomorphism mapping $\widehat{\Gamma}^n_\pi P$ onto $\widehat{\Gamma}^n_\pi \ell^1(\{1, \dots, d\}, K)$, since $T^{\otimes n} h^{\otimes n} = (Th)^{\otimes n}$. So $(T^{-1})^{\otimes n} \circ R_n \circ T^{\otimes n}$, with R_n from Step 2 for $J = \{1, \dots, d\}$, is a projection onto $\widehat{\Gamma}^n_\pi P$. □

Proof of Proposition 5.14. Alternation is vanishing on repeated arguments, so $\Lambda^2_K P$ is the quotient of $P \otimes_K P$ by the span of the tensors $h \otimes h$. Hence $\widehat{\wedge}^2_P (h \otimes h) = 0$, and $\widehat{\Gamma}^2_\pi P \subseteq \mathcal{N}^2_P$ because $\mathcal{N}^2_P$ is closed. Conversely, let $z \in \otimes^2_\pi P$ with $\wedge^2_P z = 0$, represented by $\tilde{z} \in P \otimes_K P$. The image of $\tilde{z}$ in $\Lambda^2_K P$ has seminorm zero, so for $\epsilon > 0$ it equals $\sum_j x_j \wedge y_j$ with $\sum_j \|x_j\| \|y_j\| < \epsilon$; then $\tilde{z} - \sum_j x_j \otimes y_j$ lies in the span of the tensors $h \otimes h$, so z lies within ϵ of that span, and $z \in \widehat{\Gamma}^2_\pi P$. For $z \in \mathcal{N}^2_P$ choose $z_m \in \otimes^2_\pi P$ with $z_m \to z$. Then $\wedge^2_P z_m \to 0$, Lemma 3.7 gives $w_m \in \otimes^2_\pi P$ with $\wedge^2_P w_m = \wedge^2_P z_m$ and $\|w_m\| \le 2 \|\wedge^2_P z_m\|$, and $z_m - w_m \in \widehat{\Gamma}^2_\pi P$ converges to z. The norm bounds are those of Lemma 3.9. □

P.4. **Proofs for Section 6.** The notation and the standing hypotheses are those of Section 6.

Proof of Lemma 6.1. Concatenating representations gives $\|z \wedge w\| \le \|z\| \|w\|$. The right side of (6.1) is n-linear in $(x_1, \dots, x_n)$, of norm at most $\binom{n}{a}$. It is alternating: if $x_i = x_j$ with $i \ne j$, the terms with i and j in one block vanish, and the others cancel in pairs by exchanging i and j between the blocks. Proposition 3.1 gives $\mathrm{sh}^{a,b}_E$; naturality is checked on pure wedges. Each of the $\binom{n}{a}$ terms of $\wedge^{a,b}_E \mathrm{sh}^{a,b}_E (x_1 \wedge \cdots \wedge x_n)$ is $\epsilon(S) x_S \wedge x_{S^c} = x_1 \wedge \cdots \wedge x_n$. The factorization of $\wedge^n_E$ holds on pure tensors; that of alt^n_E holds because every permutation of $\{1, \dots, n\}$ is uniquely a shuffle of two blocks followed by permutations within the blocks, and its sign is the product of their signs. □

Proof of Theorem 6.2. Identify $\mathrm{Alt}^j(Y; F')$ with $\mathcal{L}(\Lambda^j_\pi Y, F')$ by Proposition 3.1. For $T \in \mathrm{Alt}^n(D; F)$ define $T^\flat \in \mathrm{Alt}^a(D; G)$ by $T^\flat(z)(w) = T(z \wedge w)$; then $\|T^\flat\| \le \|T\|$. Let Q_a and Q_b be bounded lifts of $A^a_{E,D;G}$ and $A^b_{E,D;F}$ (Proposition 2.5). For $u_1, \dots, u_n \in \mathcal{L}(E, D)$ the bilinear map on $\Lambda^a_\pi E \times \Lambda^b_\pi E$ given by

$$(z, w) \longmapsto \big[Q_b(u_{a+1}, \dots, u_n)\big(Q_a(u_1, \dots, u_a)(T^\flat)(z)\big)\big](w)$$

is bounded. Composing its linearization with $\binom{n}{a}^{-1} \mathrm{sh}^{a,b}_E$ gives $Q_n(u_1, \dots, u_n)(T) \in \mathrm{Alt}^n(E; F)$, bounded and n-linear in $(u_1, \dots, u_n)$, with values in $\mathcal{L}(\mathrm{Alt}^n(D; F), \mathrm{Alt}^n(E; F))$. On the diagonal the bilinear map is $(z, w) \mapsto T(\Lambda^a_\pi u\, z \wedge \Lambda^b_\pi u\, w)$, so naturality and (6.2) give $Q_n(u, \dots, u)(T) = T \circ \Lambda^n_\pi u = A^n_{E,D;F}(u)(T)$. For the last assertion take $a = 1$: $A^1_{E,D;G}(u)(T) = T \circ u$ is linear in u. □

Proof of Corollary 6.3. By Lucas's theorem, $\binom{a+b}{a} \ne 0$ in K exactly when the base-p addition of a and b has no carry: over $\mathbb{F}_p$, $(1 + X)^{a+b} = \prod_i (1 + X^{p^i})^{c_i}$, where the c_i are the base-p digits of $a + b$.

(1) Degrees below p belong to $\mathfrak{D}$ by Proposition 5.1(1). For a carry-free sum apply Theorem 6.2; if F is normed or Banach, so is $G = \mathrm{Alt}^b(D;F)$. For the scalar set, Theorem 3.2(2) shows that $A^j_{E,D;K}$ is analytic exactly when $A^j_{E,D;Y^*}$ is analytic for every normed Y; apply Theorem 6.2 with $F = Y^*$, for which $G = \mathrm{Alt}^b(D;Y^*)$ is the dual of $\Lambda^b_\pi D \otimes_\pi Y$.

(2) Let N be the least missing element; $N \geq p$ by (1). If N is not a power of p and p^j is the largest power of p below N, then $N = p^j + (N - p^j)$ is a carry-free sum with $0 < N - p^j < N$, so $N \in \mathfrak{D}$, a contradiction. The second assertion follows.

(3) The pair (E,D) is not good in degree n_F; by (2), the least degree in which it is not good is p^i for some $i \geq 1$, and $n_F \geq p^i$. If n_F had a nonzero digit in a position $j < i$, then (E,D) would be good in degree p^j, $A^{n_F - p^j}_{E,D;F}$ would be analytic, and $n_F = p^j + (n_F - p^j)$ would be carry-free, so Theorem 6.2 would make $A^{n_F}_{E,D;F}$ analytic. □

Proof of Theorem 6.4. Let $y = (0,1) \in E \oplus K$, and let pr_D and pr_K be the coordinate projections of $D \oplus K$ onto D and K. Define bounded linear maps $R\colon \mathrm{Alt}^k(E \oplus K;F) \to \mathrm{Alt}^{k-1}(E;F)$, of norm at most one, and $J\colon \mathrm{Alt}^{k-1}(D;F) \to \mathrm{Alt}^k(D \oplus K;F)$, of norm at most k, by $R(m)(x_1,\dots,x_{k-1}) = m((x_1,0),\dots,(x_{k-1},0),y)$ and

$$J(m)(z_1,\dots,z_k) = \sum_{i=1}^{k} (-1)^{k-i}\,\mathrm{pr}_K(z_i)\,m(\mathrm{pr}_D\,z_1,\dots,\widehat{\mathrm{pr}_D\,z_i},\dots,\mathrm{pr}_D\,z_k).$$

The form $J(m)$ is alternating in every characteristic: if $z_a = z_b$ with $a < b$, only the terms $i = a$ and $i = b$ can be nonzero, their argument lists differ by moving one entry across $b - a - 1$ others, and $(-1)^{k-a} + (-1)^{k-b}(-1)^{b-a-1} = 0$. For $u \in \mathcal{L}(E,D)$, $(u \oplus 1)(x_i,0) = (ux_i,0)$ and $(u \oplus 1)y = (0,1)$, so only the term $i = k$ survives and

$$A^{k-1}_{E,D;F}(u) = R \circ A^k_{E\oplus K, D\oplus K;F}(u \oplus 1) \circ J.$$

The right side is analytic in u, since $u \mapsto u \oplus 1$ is affine and bounded. □

Proof of Theorem 6.13. Step 1 (pieces). Let $n \leq k$. For a piece of source type and a section s_n for w in degree n, the map $Q_n(u_1,\dots,u_n) = \widehat{\wedge}^n_{\widehat{D}} \circ (w'\varepsilon(u_1) \otimes \cdots \otimes w'\varepsilon(u_n)) \circ s_n$ has diagonal $\Lambda^n_\pi(w'\varepsilon(u)) \circ \Lambda^n_\pi w$, as in the proof of Proposition 6.11(1). For a piece of target type and a retraction ρ_n for w' in degree n, the map $Q_n(u_1,\dots,u_n) = \rho_n \circ (\varepsilon(u_1)w \otimes \cdots \otimes \varepsilon(u_n)w) \circ \mathrm{alt}^n_E$ has diagonal $\Lambda^n_\pi w' \circ \Lambda^n_\pi(\varepsilon(u)w)$, as in the proof of Proposition 6.11(2). Both diagonals equal $\Lambda^n_\pi(w'\varepsilon(u)w)$, which proves (1).

Step 2 (two maps). Let $\theta, \tilde\theta\colon \mathcal{L}(E,D) \to \mathcal{L}(E,\widehat{D})$ be liftable in every degree $a \leq k$, with lifts Q_a and $\tilde Q_a$, and put $Q_0 = \tilde Q_0 = \mathrm{id}_K$. For $n \leq k$ the map

$$Q^+_n(u_1,\dots,u_n) = \sum_{a=0}^{n} \wedge^{a,n-a}_{\widehat{D}} \circ \big(Q_a(u_1,\dots,u_a) \otimes \tilde Q_{n-a}(u_{a+1},\dots,u_n)\big) \circ \mathrm{sh}^{a,n-a}_E,$$

with $\wedge^{a,n-a}_{\widehat{D}}$ extended to the completions, is bounded and n-linear with diagonal $\Lambda^n_\pi(\theta(u) + \tilde\theta(u))$, by (6.3) for $G = \widehat{D}$.

Step 3 (conclusion). By Step 2 and induction on m, the map $u \mapsto \sum_i \theta_i(u) = u$ is liftable in degree k, and Lemma 3.4 gives (2); under the hypotheses of its last sentence, the formula of Step 2 keeps values in $\Lambda^n_\pi D$. Item (3) follows from (1) and (2) for decompositions into pieces of source and target type, with Step 1 for decompositions without completion, and from Proposition 6.11 for relative splittings. □

Proof of Proposition 6.15. Step 1 (the two bilinear maps). Given a decomposition (6.4) into pieces of source and target type, let b_S and b_R be the sums of the bounded bilinear maps

$b_i(u,x) = w_i'\varepsilon_i(u)w_i x$ over the pieces of source type and of target type; then $b_S(u,x)+b_R(u,x) = ux$. For a piece of source type, the map $\xi_i\colon G_i \to \mathcal{L}(\mathcal{L}(E,D),\widehat{D})$, $\xi_i(g)(u) = w_i'\varepsilon_i(u)g$, is bounded linear and $w_{b_i} = \xi_i w_i$. For a piece of target type, the linear map $\lambda_i\colon \mathcal{L}(E,D)\otimes_\pi E \to G_i'$ with $\lambda_i(u\otimes x) = \varepsilon_i(u)w_i x$ is bounded and $w_{b_i}' = w_i'\lambda_i$. By the paragraph on split operators in Section 6.1, $w_{b_S} = \sum w_{b_i}$ is $\mathrm{S}^{\le k}$-split and $w_{b_R}' = \sum w_{b_i}'$ is $\mathrm{R}^{\le k}$-split.

Step 2 (the two pieces). Conversely, given b_S and b_R, the piece $(\mathcal{L}(\mathcal{L}(E,D),\widehat{D}),\widehat{D},w_{b_S},\mathrm{id}_{\widehat{D}},\varepsilon_S)$ with $\varepsilon_S(u)(T) = T(u)$ is of source type, the piece $(E,\mathcal{L}(E,D)\otimes_\pi E,\mathrm{id}_E,w_{b_R}',\varepsilon_R)$ with $\varepsilon_R(u)(x) = u\otimes x$ is of target type, and $\varepsilon_S(u)w_{b_S}x + w_{b_R}'\varepsilon_R(u)x = b_S(u,x)+b_R(u,x) = ux$. □

Proof of Lemma 6.16. (1) By (6.4), $v = \Pi\Phi(v)J = \sum_i \Pi w_i'\,\varepsilon_i(\Phi v)\,w_i J$, and by the ideal property (Proposition 3.13(2)) the pieces $(G_i,G_i',w_iJ,\Pi w_i',\varepsilon_i\circ\Phi)$ have the types of the given pieces. A source-side relative splitting (G,w,ε) gives $(G,wJ,v\mapsto\Pi\varepsilon(\Phi v))$, and a target-side one (G,w',ε) gives $(G,\Pi w',v\mapsto\varepsilon(\Phi v)J)$. (2) Apply Proposition 6.14 with $G=E$, $G'=D$, $w=J$, $w'=\Pi|_D$ and $\varepsilon=\Phi$. □

Proof of Proposition 6.18. Step 1 (items (1) and (3)). Item (1) is Proposition 5.13. For (3), let $J\colon P_0\to P$ and $\Pi'\colon P\to P_0$ be bounded linear with $\Pi'J = \mathrm{id}_{P_0}$, and let R_n be a projection of $\widehat{\otimes}_\pi^n P$ onto $\widehat{\Gamma}_\pi^n P$. Then $\Pi'^{\otimes n}R_nJ^{\otimes n}$ takes values in $\widehat{\Gamma}_\pi^n P_0$, since $\Pi'^{\otimes n}$ maps pure powers to pure powers, and it fixes $h^{\otimes n} = \Pi'^{\otimes n}(Jh)^{\otimes n}$ for $h\in P_0$.

Step 2 (item (2)). Let $n\le k$, let J_1,J_2 and Π_1,Π_2 be the inclusions and projections of $P = P_1\oplus P_2$, and let $R_a^{(1)}$, $R_a^{(2)}$ be projections onto $\widehat{\Gamma}_\pi^a P_1$, $\widehat{\Gamma}_\pi^a P_2$ for $a\le n$, with $R_0^{(1)} = R_0^{(2)} = \mathrm{id}_K$. For $0\le a\le n$ let $\mathrm{sh}_a\colon \widehat{\otimes}_\pi^a P_1\,\widehat{\otimes}_\pi\,\widehat{\otimes}_\pi^{n-a}P_2 \to \widehat{\otimes}_\pi^n P$ send $y\otimes z$ to the sum of the $\binom{n}{a}$ tensors obtained by distributing the factors of $J_1^{\otimes a}y$ and of $J_2^{\otimes(n-a)}z$ over the n slots, keeping their order, and put

$$R = \sum_{a=0}^{n} \mathrm{sh}_a\circ\big(R_a^{(1)}\otimes R_{n-a}^{(2)}\big)\circ\big(\Pi_1^{\otimes a}\otimes\Pi_2^{\otimes(n-a)}\big).$$

For $h = J_1h_1+J_2h_2$, $(\Pi_1^{\otimes a}\otimes\Pi_2^{\otimes(n-a)})(h^{\otimes n}) = h_1^{\otimes a}\otimes h_2^{\otimes(n-a)}$ and $h^{\otimes n} = \sum_a \mathrm{sh}_a(h_1^{\otimes a}\otimes h_2^{\otimes(n-a)})$, so R fixes $h^{\otimes n}$, hence $\widehat{\Gamma}_\pi^n P$. The tensor $\mathrm{sh}_a(y^{\otimes a}\otimes z^{\otimes(n-a)})$ is the coefficient of λ^a in the polynomial $\lambda\mapsto(\lambda J_1y + J_2z)^{\otimes n}$, hence a linear combination of its values, K being infinite. Since $R_a^{(1)}\otimes R_{n-a}^{(2)}$ takes values in the closed span of the tensors $y^{\otimes a}\otimes z^{\otimes(n-a)}$, the map R takes values in $\widehat{\Gamma}_\pi^n P$.

Step 3 (item (4)). Let $n\le k$ and let R_n be a projection onto $\widehat{\Gamma}_\pi^n P$. The operators that vanish on $\mathcal{N}_E^n$ form a closed subspace W of $Z = \mathcal{L}(\widehat{\otimes}_\pi^n E,\widehat{\Lambda}_\pi^n D)$, and $T\mapsto T\circ\widehat{\wedge}_E^n$ is an isometry of $\mathcal{L}(\widehat{\Lambda}_\pi^n E,\widehat{\Lambda}_\pi^n D)$ onto W (Lemma 3.7). Let $\Theta\colon \widehat{\otimes}_\pi^n P\to Z$ be the linearization of the bounded n-linear map $(y_1,\dots,y_n)\mapsto \widehat{\wedge}_{\widehat{D}}^n\circ\big(w'\varepsilon'(y_1)w\otimes\cdots\otimes w'\varepsilon'(y_n)w\big)$. By naturality of $\wedge^n$, $\Theta(y^{\otimes n}) = \widehat{\Lambda}_\pi^n(w'\varepsilon'(y)w)\circ\widehat{\wedge}_E^n$ lies in W, so Θ maps $\widehat{\Gamma}_\pi^n P$ into W. Hence $Q_n(u_1,\dots,u_n) = \Theta\big(R_n(\psi u_1\otimes\cdots\otimes\psi u_n)\big)$, read in $\mathcal{L}(\widehat{\Lambda}_\pi^n E,\widehat{\Lambda}_\pi^n D)$ and restricted to $\Lambda_\pi^n E$, is bounded and n-linear, and its diagonal is $\Lambda_\pi^n(w'\varepsilon(u)w)$, since R_n fixes $(\psi u)^{\otimes n}$. The last assertion follows from Theorem 6.13(1)–(3). □

Proof of Corollary 6.24. (1) Extend w' to $\widehat{G'}$. Then $\theta_{E,D}(u) = w'\circ\theta_{G,G'}(\varepsilon(u))\circ w$ for $u\in\mathcal{L}(E,D)$. Proposition 6.23(b), (c) and (a), applied in this order to w, w' and $\psi=\varepsilon$, give $\mathrm{ob}_k(E,D) = (\widehat{\Lambda}_\pi^k w)^*((\widehat{\Lambda}_\pi^k w')_\flat)_*(\varepsilon^\sharp)_*\,\mathrm{ob}_k(G,G')$, and pullbacks commute with pushouts (Corollary N.3). (2) The family ε on $\mathcal{L}(E,D)$ satisfies $\theta_{E,D} = \varepsilon_w$, so Proposition 6.23(b) gives the formula. If w is S^k-split, then $\mathrm{sp}_k(w) = 0$ by Proposition 3.13(1), and the pushout of a split sequence splits by Lemma 3.6(2) and (3). □

Proof of Proposition 6.27. Step 1 (the alternating map). Fix $u^1,\dots,u^k\in\mathcal{U}$, and let $E_{\mathrm{fin}}\subseteq E$ be the subspace of vectors with finitely many nonzero blocks. For $\nu=(\nu_1,\dots,\nu_k)\in I^k$ let $i_1<\cdots<i_m$ be the distinct entries of ν, let $S_l=\{a:\nu_a=i_l\}$ and $n_l=|S_l|$, and let $\epsilon(S_1,\dots,S_m)$ be the sign of the permutation of $\{1,\dots,k\}$ that lists S_1, then S_2, and so on, each in increasing order. For $x_1,\dots,x_k\in E_{\mathrm{fin}}$ put

$$\Phi(x_1,\dots,x_k)=\sum_{\nu\in I^k}\epsilon(S_1,\dots,S_m)\bigwedge_{l=1}^{m}\widehat{\Lambda}^{n_l}_{\pi}J'_{i_l}\,Q^{i_l}_{n_l}\big(u^{(l)}_{i_l}\big)\Big(\bigwedge_{a\in S_l}\Pi_{i_l}x_a\Big),$$

a finite sum in $\widehat{\Lambda}^k_\pi D$. Here $u^{(l)}_{i_l}$ is the tuple of the $u^j_{i_l}$ with $n_1+\cdots+n_{l-1}<j\le n_1+\cdots+n_l$, in increasing order of j; the wedges over S_l are taken in increasing order; and the outer wedge is the iterated wedge product, extended to completions. Each u^j enters one slot of one map $Q^{i_l}_{n_l}$, determined by the sizes n_l, so Φ is k-linear in $(x_1,\dots,x_k)$ and in $(u^1,\dots,u^k)$. It is alternating: let $x_a=x_b$ with $a\ne b$. The terms with $\nu_a=\nu_b$ vanish, since one block wedge contains $\Pi_{\nu_a}x_a$ twice. The term of $\nu\circ(a\ b)$ is minus the term of ν with x_a and x_b exchanged: the two tuples have the same entries and block sizes, the permutations defining their signs differ by the transposition $(a\ b)$ followed by permutations inside the blocks, and each block factor is alternating in the vectors of its block; so the remaining terms cancel in pairs.

Step 2 (the bound). The wedge product is contractive and $\|\widehat{\Lambda}^n_\pi J'_i\|\le 1$, so the term of ν has norm at most $C^m\prod_j(c\|u^j\|)\prod_a\|\Pi_{\nu_a}x_a\|\le C^kc^k\prod_j\|u^j\|\prod_a\|\Pi_{\nu_a}x_a\|$. Summing over ν gives $\|\Phi(x_1,\dots,x_k)\|\le C^kc^k\prod_j\|u^j\|\prod_a\|x_a\|$, since $\sum_\nu\prod_a\|\Pi_{\nu_a}x_a\|=\prod_a\|x_a\|$. Hence Φ extends by continuity to a bounded k-linear map on E^k, alternating because a tuple in E^k with two equal entries is a limit of such tuples in E^k_{fin}, and Proposition 3.1 gives $Q(u^1,\dots,u^k)\in\mathcal{L}(\Lambda^k_\pi E,\widehat{\Lambda}^k_\pi D)$, k-linear in $(u^1,\dots,u^k)$, with $\|Q\|\le C^kc^k$.

Step 3 (the diagonal). Let $u^j=u$ for every j. The l-th block factor of the term of ν is then $\bigwedge_{a\in S_l}J'_{i_l}u_{i_l}\Pi_{i_l}x_a$, so by the choice of the sign the term is $\bigwedge_{a=1}^k J'_{\nu_a}u_{\nu_a}\Pi_{\nu_a}x_a$. Summing over ν gives $\bigwedge_a\big(\sum_i J'_iu_i\Pi_ix_a\big)=ux_1\wedge\cdots\wedge ux_k$ for $x_1,\dots,x_k\in E_{\mathrm{fin}}$. Wedges of vectors of E_{fin} span a dense subspace of $\Lambda^k_\pi E$, so $Q(u,\dots,u)=\Lambda^k_\pi u$. The last assertion is Lemma 3.4. □

Proof of Proposition 6.28. Step 1 (retracts and sums). Let $J\colon Y_0\to Y$ and $\Pi'\colon Y\to Y_0$ be bounded linear with $\Pi'J=\mathrm{id}_{Y_0}$. Since $\mathrm{id}_{Y_0}=\Pi'\,\mathrm{id}_Y\,J$, the conditions (S^k) and (R^k) pass to Y_0 by Proposition 3.13(2), and polarizability by Proposition 6.18(3). For pairs, apply Lemma 6.16(2) in the situation of its second sentence. In (2), the conditions pass from the sum to Y_1 and Y_2 by (1). Conversely, $\mathrm{id}_{Y_1\oplus Y_2}$ is the sum of the inclusion-projection composites of the two summands, each $\mathrm{S}^{\le k}$-split, respectively $\mathrm{R}^{\le k}$-split, by the ideal property, so their sum is split in the same degrees by the paragraph on split operators in Section 6.1; polarizability is Proposition 6.18(2). For the pairs, let J_a,Π_a be the inclusion and projection of E_a in $E_1\oplus E_2$, and J'_b,Π'_b those of D_b in $D_1\oplus D_2$. Every $u\in\mathcal{L}(E_1\oplus E_2,D_1\oplus D_2)$ is $\sum_{a,b}J'_b(\Pi'_buJ_a)\Pi_a$. For each (a,b) the map $u\mapsto J'_b(\Pi'_buJ_a)\Pi_a$ is liftable in every degree $n\le k$, by the formula of Proposition 6.14 for the pair (E_a,D_b), with $w=\Pi_a$, $w'=J'_b$ and $\varepsilon(u)=\Pi'_buJ_a$. Theorem 6.13(2), applied in each degree $n\le k$, concludes.

Step 2 (completion of a space). By the paragraph before Theorem 3.16, applied to the dense subspace Y of $\widehat{Y}$, the inclusion induces isometries $\otimes^k_\pi Y\to\otimes^k_\pi\widehat{Y}$ and $\Lambda^k_\pi Y\to\Lambda^k_\pi\widehat{Y}$ with dense images. So the completed powers of Y and $\widehat{Y}$ agree, as do the maps $\widehat{\wedge}^k$ and alt^k on them, and $\widehat{\Gamma}^k_\pi Y=\widehat{\Gamma}^k_\pi\widehat{Y}$, since $h\mapsto h^{\otimes k}$ is continuous. Sections and retractions for id_Y extend by continuity to sections and retractions for $\mathrm{id}_{\widehat{Y}}$, since both sides of the defining identities are continuous, and those for $\mathrm{id}_{\widehat{Y}}$ restrict to ones for id_Y. The projections onto $\widehat{\Gamma}^k_\pi Y=\widehat{\Gamma}^k_\pi\widehat{Y}$ are the same.

Step 3 (completion of a pair). Let D be complete. Restriction is an isometry of $\mathcal{L}(\widehat{E},D)$ onto $\mathcal{L}(E,D)$; write $\widehat{u}$ for the extension of $u\in\mathcal{L}(E,D)$. By Step 2, $\Lambda^k_\pi E$ is a dense subspace of $\Lambda^k_\pi\widehat{E}$ with the induced norm. A lift $\widehat{Q}$ for $(\widehat{E},D)$ gives the lift $Q(u_1,\dots,u_k)=\widehat{Q}(\widehat{u}_1,\dots,\widehat{u}_k)|_{\Lambda^k_\pi E}$ for (E,D). A lift Q for (E,D) gives the lift for $(\widehat{E},D)$ whose value at $(v_1,\dots,v_k)$ is the extension by continuity of $Q(v_1|_E,\dots,v_k|_E)$; its diagonal is $\Lambda^k_\pi v$, since $\Lambda^k_\pi v$ extends $\Lambda^k_\pi(v|_E)$. Finally $\mathcal{L}(E,D)\subseteq\mathcal{L}(E,\widehat{D})$ isometrically for any normed D, and $\widehat{\Lambda}^k_\pi\widehat{D}=\widehat{\Lambda}^k_\pi D$, so a lift for $(E,\widehat{D})$ restricts to a lift for (E,D). Lemma 3.4 concludes. □

P.5. **Proofs for Section 7.** The notation is that of Section 7.

Proof of Lemma 7.1. Step 1 (value group and residue field). The field $\widehat{K}$ is closed in the complete field $\kappa((X))$, hence complete. Put $\mathcal{O}=\{a\in\widehat{K}:|a|\le 1\}$ and $\mathfrak{m}=\{a\in\widehat{K}:|a|<1\}$. For $a\in\widehat{K}^\times$ there is $a'\in K$ with $|a-a'|<|a|$, hence $|a'|=|a|$; so $|\widehat{K}^\times|=|K^\times|=|\vartheta|^{\mathbb{Z}}$, and $|a|\le|\vartheta|$ for every $a\in\mathfrak{m}$. Since $\mathcal{O}=\widehat{K}\cap\kappa[[X]]$ and $\mathfrak{m}=\widehat{K}\cap X\kappa[[X]]$, the map coeff_0 restricts to a ring homomorphism $\mathcal{O}\to\kappa$ with kernel $\mathfrak{m}$. So $\mathcal{O}/\mathfrak{m}$ is isomorphic to a subfield κ_1 of κ, and $|\kappa_1|$ is a power of p.

Step 2 (Teichmüller representatives). Since $\widehat{K}$ has characteristic p, $\mathrm{Fr}(a)=a^{|\kappa_1|}$ is a continuous ring endomorphism of $\mathcal{O}$; let κ' be its subring of fixed points. A nonzero $b\in\kappa'$ satisfies $b^{|\kappa_1|-1}=1$, since $\mathcal{O}$ is a domain; so κ' is a field, $|b|=1$, and the reduction map $\kappa'\to\mathcal{O}/\mathfrak{m}$ is injective. For surjectivity let $a\in\mathcal{O}$ and $a_n=\mathrm{Fr}^n(a)$. Since $y^{|\kappa_1|}=y$ on κ_1, $w=\mathrm{Fr}(a)-a$ lies in $\mathfrak{m}$, and $|a_{n+1}-a_n|=|w^{|\kappa_1|^n}|\le|\vartheta|^{|\kappa_1|^n}$. Hence $a_n\to b\in\mathcal{O}$ with $\mathrm{Fr}(b)=b$ and $|b-a|<1$. So $\kappa'\cong\kappa_1$ is a finite field of characteristic p.

Step 3 (expansions). Let $a\in\mathcal{O}$. By Step 2 there is $b_0\in\kappa'$ with $a-b_0\in\mathfrak{m}$, and $(a-b_0)/\vartheta\in\mathcal{O}$ by Step 1. Repeating this gives $b_0,b_1,\dots\in\kappa'$ with $a-\sum_{i<n}b_i\vartheta^i\in\vartheta^n\mathcal{O}$ for every n, so $a=\sum_{i\ge0}b_i\vartheta^i$. If all b_i lie in κ' and $b_n\ne0$, then $|\sum_{i\ge n}b_i\vartheta^i|=|\vartheta|^n$, since $|b_n\vartheta^n|=|\vartheta|^n$ and the later terms have absolute value at most $|\vartheta|^{n+1}$. Applied to the difference of two expansions, this shows that the expansion is unique.

Step 4 (the isomorphism). For $g=\sum_{i\ge m}b_iX^i$ in $\kappa'((X))$ put $\sigma(g)=\sum_{i\ge m}b_i\vartheta^i$, which converges since $|b_i\vartheta^i|\le|\vartheta|^i$. The map σ is additive, and it is multiplicative since in the complete nonarchimedean field $\widehat{K}$ the product of two convergent series is the sum of their Cauchy product, whose coefficients $\sum_{i+j=n}b_ib'_j$ lie in κ'. If $g\ne0$ has order m, Step 3 applied to $\vartheta^{-m}\sigma(g)$ gives $|\sigma(g)|=|\vartheta|^m$; so σ is isometric, hence injective. For $a\in\widehat{K}^\times$ choose n with $|a|=|\vartheta|^n$ (Step 1); then $\vartheta^{-n}a=\sum_{i\ge0}b_i\vartheta^i$ by Step 3, and $a=\sigma(\sum_i b_iX^{i+n})$. So σ is an isometric field isomorphism with $\sigma(X)=\vartheta$. It restricts to an isometric isomorphism of the dense subfield $\sigma^{-1}(K)$ of $\kappa'((X))$ onto K, and every statement that depends only on K as a normed field transfers along this isomorphism. □

Proof of Lemma 7.11. (1) For $x\in c_{00}(\mathbb{N},K)$ and $t\in\mathbb{N}$, $|x(t)|e_t\le|x|$ pointwise, so $|x(t)|\le\|x\|_E$ by monotonicity and $\|e_t\|_E=1$. For $x\in E$ and $t\in[n]$ this gives $|x(t)|\le\|x1_{[n]}\|_E\le\|x\|_E$. If b is finitely supported with $|b|\le1$ and $[n]$ contains its support, then $bx=b(x1_{[n]})$ and $|bx|\le|x1_{[n]}|$, so $\|bx\|_E\le\|x1_{[n]}\|_E\le\|x\|_E$.

(2) For $x\in E$, ax is finitely supported and $\|ax\|_D\le\sum_{t\in\operatorname{supp}a}|a(t)|\|e_t\|_D\|x\|_E$ by (1); so μ_a is bounded. Moreover $ax=a(x1_{\operatorname{supp}a})$, where $x1_{\operatorname{supp}a}$ is supported in $\operatorname{supp}a$ and has norm at most $\|x\|_E$ by (1). This gives the formula.

(3) The map $a\mapsto\mu_a$ is linear, and $\|\mu_ae_t\|_D=|a(t)|$ gives $\|\mu_a\|_{E\to D}\ge\|a\|_\infty$; so $a\mapsto\|\mu_a\|_{E\to D}$ is a norm. Since $\mu_{e_t}x=x(t)e_t$, (1) gives $\|\mu_{e_t}\|_{E\to D}=1$. For an injective map $\gamma\colon\mathbb{N}\to\mathbb{N}$ and finitely supported y, let $y^\gamma(\gamma(t))=y(t)$ and $y^\gamma=0$ off $\gamma(\mathbb{N})$. Then $\|y^\gamma\|=\|y\|$ in E and

in D, since γ restricted to $\operatorname{supp} y$ extends to a permutation of $\mathbb{N}$. The vectors supported in $\operatorname{supp} a^\gamma$ are the x^γ with $\operatorname{supp} x \subseteq \operatorname{supp} a$, and $a^\gamma x^\gamma = (ax)^\gamma$; so (2) gives $\|\mu_{a^\gamma}\|_{E\to D} = \|\mu_a\|_{E\to D}$. Multiplication of a by a sequence of absolute value 1 does not change $|ax|$. If $|a| \le |b|$, let $c = a/b$ on $\operatorname{supp} b$ and $c = 0$ elsewhere; then $|c| \le 1$, $ax = b(cx)$ and $\|cx\|_E \le \|x\|_E$ by (1).

(4) For $x \in c_{00}(\mathbb{N}, K)$, $|ax| \le \|a\|_\infty |x|$ pointwise, so $\|ax\|_D \le C\|ax\|_E \le C\|a\|_\infty \|x\|_E$. If D is symmetric and $\|x\|_\infty \le 1$, then $|ax| \le |a|$, so $\|ax\|_D \le \|a\|_D$, with equality at $x = 1_{\operatorname{supp} a}$. Finally $\|ax\|_\infty \le \|a\|_\infty \|x\|_\infty \le \|a\|_\infty \|x\|_E$ by (1), and (3) gives the reverse inequality. □

P.6. **Proofs for Section 9.**

Proof of Proposition 9.1. Step 1 (the lift). Each $u \in \mathcal{L}(E, D)$ is λu_0 for a unique $\lambda \in K$, and $|\lambda| = \|u\|/\|u_0\|$. Hence Q is a well-defined k-linear map $\mathcal{L}(E, D)^k \to \mathcal{L}(\Lambda^k_\pi E, \Lambda^k_\pi D)$ of norm at most $\|\Lambda^k_\pi u_0\|/\|u_0\|^k \le 1$. Its diagonal is $\lambda^k \Lambda^k_\pi u_0 = \Lambda^k_\pi(\lambda u_0)$, so Lemma 3.4 gives (1).

Step 2 (sufficiency in (2)). Put $\varepsilon(\lambda u_0) = \lambda \operatorname{id}_D$ and $\varepsilon'(\lambda u_0) = \lambda \operatorname{id}_E$. Then $\varepsilon(u) \circ u_0 = u = u_0 \circ \varepsilon'(u)$ for every $u \in \mathcal{L}(E, D)$. If u_0 is S^k-split, (D, u_0, ε) is a source-side relative splitting; if u_0 is R^k-split, (E, u_0, ε') is a target-side relative splitting. Either is a splitting decomposition of degree k (Definition 6.12).

Step 3 (necessity in (2), one piece). A splitting decomposition with one piece writes u_0 as a composite of bounded operators one of which is S^k-split or R^k-split (Definition 6.12). By Proposition 3.13(2), u_0 is S^k-split or R^k-split.

Step 4 (necessity in (2), several pieces). For a decomposition into at least two pieces of source and target type, Proposition 6.15 gives, in its notation, bounded bilinear maps b_S, b_R with $b_S(u, x) + b_R(u, x) = ux$, w_{b_S} $\mathrm{S}^{\le k}$-split and w'_{b_R} $\mathrm{R}^{\le k}$-split; their values lie in D, since D is complete. Put $u_S = b_S(u_0, \cdot)$ and $u_R = b_R(u_0, \cdot)$. Since u_S is w_{b_S} followed by evaluation at u_0, and u_R is $x \mapsto u_0 \otimes x$ followed by w'_{b_R}, Proposition 3.13(2) makes u_S S^k-split and u_R R^k-split. Both lie in $K\, u_0$ and $u_S + u_R = u_0$, so there is $a \in K$ with $u_S = a\, u_0$ and $u_R = (1 - a)\, u_0$. If $a \ne 0$, then $u_0 = a^{-1} u_S$ is S^k-split; otherwise $u_0 = u_R$ is R^k-split.

Step 5 (the dual). If f is a nonzero bounded linear functional on E, the operators $x \mapsto f(x)\, d$ with $d \in D$ form a subspace of $\mathcal{L}(E, D)$ of dimension $\dim D \ge 2$, contrary to $\mathcal{L}(E, D) = K\, u_0$. □

P.7. **Proofs for Section 10.**

Proof of Proposition 10.3. Choose E, D from Theorem 7.2(2), so that $A = A^k_{E,D;K}$ is analytic at no point.

Step 1 (assertion (1)). Put $P = \mathcal{L}(E, D) \times D \times E$ and use the charts id_P and $\psi(u, d, e) = (u, d + ue, e)$. Operator evaluation is bounded bilinear, so ψ and $\psi^{-1}(u, d, e) = (u, d - ue, e)$ are polynomial analytic, with $D\psi(u, d, e)(v, a, b) = (v, a + ve + ub, b)$. Let J extend forms from D along the projection $\pi_D \colon P \to D$, and let R restrict forms on P along $i_E(b) = (0, 0, b)$. These bounded linear maps give

$$R \circ A^k_{P,P;K}(D\psi(u, d, e)) \circ J = A(u).$$

If the transition were analytic at (u_0, d_0, e_0), restriction to the affine slice $u \mapsto (u, d_0, e_0)$ would make A analytic at u_0. The reverse transition is treated in the same way with $-u$.

Step 2 (assertion (2)). Let $U = \operatorname{Alt}^k(D; K)$, $Y = U \times D$ and $N = \mathcal{L}(E, D) \times U \times E$. Define the bounded linear map $\omega_0(m, d) = m \circ \pi_D^{\times k}$, with $\pi_D \colon Y \to D$ the projection, and the polynomial map $h(u, m, e) = (m, ue)$. Since $Dh(u, m, e)(v, a, b) = (a, ve + ub)$, restricting the values of $h^*\omega_0$ to the E-directions gives $f(u, m) = A(u)(m)$. If $h^*\omega_0$ were analytic as an $\operatorname{Alt}^k(N; K)$-valued map at (u_0, m_0, e_0), fixing $e = e_0$ would make f analytic near (u_0, m_0); since Banach-valued power series differentiate termwise on a smaller ball, $\partial_m f(u, m) = A(u)$ would

be analytic near u_0, a contradiction. In $\mathrm{Mult}^k(N;K)$ the pullback is a bounded multilinear expression in Dh and $\omega_0 \circ h$, hence analytic. □

Proof of Proposition 10.4. Throughout, for Banach spaces E and D' put $P = \mathcal{L}(E, D') \times D' \times E$, with the maximum norm, and use the charts id_P and $\psi(u, d, e) = (u, d + ue, e)$ of Step 1 of the proof of Proposition 10.3. Both charts are polynomial analytic, and $g = D\psi(u, d, e)$ is $g(v, a, b) = (v, a + ve + ub, b)$, with $g^{-1}(v, a, b) = (v, a - ve - ub, b)$.

Step 1 (values in F). For (1), take E, F from Theorem 7.2(1) and $D' = E$. The maps J, R of Step 1 of the proof of Proposition 10.3, with values in F, give

$$R \circ A^k_{P,P;F}(D\psi(w, d, e)) \circ J = A^k_{E,E;F}(w) \qquad (w \in \mathcal{L}(E, E)),$$

which is analytic at no point; restriction to affine slices, as in that step, shows that neither transition is analytic at any point. Only bounded linear compression and affine slices are used, so K need not be complete, and F has no equivalent nonarchimedean norm by Theorem 7.2(1). For $K = \mathbb{F}_q((X))$ with its X-adic absolute value, Remark E.2 allows $E = \ell^\infty(\mathbb{N}, K)$ and $F = F_1$, which has no equivalent nonarchimedean norm by Proposition E.1. Then P is nonarchimedean, since operator norms into a nonarchimedean space and maximum norms of products are nonarchimedean.

Step 2 (tangent values). Suppose that $A^k_{E,D;F}$ is analytic at no point, and put $D' = D \times F$. As functions of the point (u, d, e), the induced transitions on tangent-valued forms are $m \mapsto g \circ m \circ (g^{-1}, \dots, g^{-1})$ and $m \mapsto g^{-1} \circ m \circ (g, \dots, g)$. Let $\pi_D \colon P \to D$ and $\pi_F \colon P \to F$ be the components of the middle factor, and put $i_F(y) = (0, (0, y), 0)$ and $i_E(b) = (0, 0, b)$. The bounded linear map

$$\Phi(T)(m) = \pi_F \circ T\big(i_F \circ m \circ \pi_D^{\times k}\big) \circ i_E^{\times k}$$

sends operators T on $\mathrm{Alt}^k(P;P)$ to operators $\mathrm{Alt}^k(D;F) \to \mathrm{Alt}^k(E;F)$, and it sends $m \mapsto h \circ m \circ (f, \dots, f)$, for $f, h \in \mathcal{L}(P, P)$, to $m \mapsto (\pi_F h i_F) \circ m \circ (\pi_D f i_E, \dots, \pi_D f i_E)$. Since $\pi_F g^{\pm 1} i_F = \mathrm{id}_F$ and $\pi_D g^{\pm 1} i_E = \pm u_D$, where $u_D \in \mathcal{L}(E, D)$ is the D-component of u, Φ sends the two transitions to $A^k_{E,D;F}(\mp u_D) = (\mp 1)^k A^k_{E,D;F}(u_D)$. If either transition were analytic at (u_0, d_0, e_0), restriction to the affine slice $w \mapsto (u_0 + (w, 0), d_0, e_0)$, $w \in \mathcal{L}(E, D)$, would make $A^k_{E,D;F}$ analytic at the D-component of u_0.

Step 3 (the models). For (2), apply Step 2 with E, F from Theorem 7.2(1) and $D = E$; then P contains the isometric copy $i_F(F)$ of F, so it admits no equivalent nonarchimedean norm. For (3), apply Step 2 with E, D from Theorem 7.2(2) and $F = K$; then P is a nonarchimedean Banach space, as in Step 1. □

Proof of Theorem 10.7. Step 1 (sheaf and pullback). Locality of analytic coefficient maps gives the sheaf property. Addition and multiplication by an analytic function are bounded multilinear operations on the coefficient spaces. For an analytic map h, the formula

$$h^*\eta = (\eta \circ h) \circ (Dh, \dots, Dh)$$

is analytic in the ambient space and has alternating values. The chain rule gives identity and composition laws for pullback.

Step 2 (the product). The shuffle formula (10.2) has $\binom{k+l}{k}$ terms, each of norm at most $\|\eta\|\|\zeta\|$, so it is a bounded bilinear operation on multilinear coefficient spaces, of norm at most $\binom{k+l}{k}$. For alternating inputs it is alternating: terms with two repeated arguments in one block vanish, and the remaining terms cancel in pairs by exchanging those arguments between blocks; this argument includes characteristic two. Formula (10.2) commutes with simultaneous precomposition, hence glues in charts and commutes with pullback. Complementing S proves

graded commutativity. Expanding either bracketing of a triple product yields the same signed sum over ordered three-block partitions, proving associativity.

Step 3 ($d\eta$ is a form). On an open subset of a model space P, the operation (10.3) applied to $D\eta$ is bounded linear with norm at most $k+1$, so $d\eta$ is ambient analytic. Each $D\eta(y)(v_i)$ is alternating: difference quotients have alternating values, and that subspace is closed. If $v_a = v_b$ with $a < b$, the summands other than $i = a, b$ vanish; alternation moves the remaining repeated argument across $b - a - 1$ positions, making those two terms cancel. Hence $d\eta$ is a form.

Step 4 ($d^2 = 0$). The second derivative of an analytic map is symmetric in every characteristic: if c_2 is the bilinear coefficient of degree two of a power series, the second derivative at its center is $(v, w) \mapsto c_2(v, w) + c_2(w, v)$. In the expansion of $d(d\eta)$, pair the terms differentiated in directions (v_a, v_b) and (v_b, v_a) for $a < b$. Their values agree and their signs are $(-1)^{a+b-1}$ and $(-1)^{a+b}$, so $d^2\eta = 0$.

Step 5 (naturality and gluing of d). Differentiate the pullback formula. Terms differentiating η give $h^*(d\eta)$ after the signed sum (10.3). The other terms insert $D^2h(v_a, v_b)$ into one argument of η, and symmetry of D^2h and alternation of η cancel them in pairs. Therefore $d(h^*\eta) = h^*(d\eta)$; applied to chart changes, this identity shows that the local derivatives glue to $d\colon \Omega^k_{\mathrm{amb}}(M) \to \Omega^{k+1}_{\mathrm{amb}}(M)$.

Step 6 (Leibniz rule). Differentiating the shuffle formula (10.2) gives the Leibniz rule: placing the derivative index in the second block moves it past the first k arguments and contributes $(-1)^k$. Every identity in Steps 1–6 uses finite sums and cancellation, with no division by a factorial. □

Proof of Proposition 10.9. Clearly (2) implies (1). If (1) holds, put $T = D\eta(y)$ and $\beta(v_0, \dots, v_{p-1}) = T(v_0)(v_1, \dots, v_{p-1})$. Grouping the permutations by the image of 0 and using alternation of each $T(v_i)$,

$$\mathcal{A}_Y\beta(v_0, \dots, v_{p-1}) = (p-1)! \sum_{i=0}^{p-1} (-1)^i T(v_i)(v_0, \dots, \widehat{v_i}, \dots, v_{p-1}) = (p-1)!\,\omega(v_0, \dots, v_{p-1}),$$

and $(p-1)! = -1$ in K by Wilson's theorem, so $\omega = \mathcal{A}_Y(-\beta)$. If (3) holds, the bounded linear map $\eta(y)(v_1, \dots, v_{p-1}) = \sum_\sigma \operatorname{sgn}(\sigma)\, \beta(y, v_{\sigma(1)}, \dots, v_{\sigma(p-1)})$, summed over the permutations σ of $\{1, \dots, p-1\}$, satisfies $D\eta(y) = \eta$, and the same grouping gives $d\eta = \mathcal{A}_Y\beta = \omega$. For $F = \widehat{\Lambda}^p_\pi Y$ the equivalence with (R^p_Y) is Lemma 3.10, whose proof also gives the uncompleted case, and Proposition 3.5 gives goodness. □

Appendix Q. Proofs for Section 14

This appendix uses the notation of Section 14. Subsection Q.2 holds over every nontrivially normed field. Elsewhere K is nonarchimedean and ϖ is as there.

Q.1. **Laws.**

Lemma Q.1 (linearization)**.** *Let $n \geq 0$.*

1. *A bounded polynomial law $a\colon P \to W$ of degree n has a unique linear map $L_a\colon \Gamma^n(P) \to W$ with $L_a(h^{\otimes n}) = a(h)$ for $h \in P$, and $\|L_a(t)\| \leq \|a\|\,\|t\|_\Gamma$.*
2. *If $\mathsf{Lin}\colon \Gamma^n(P) \to W$ is linear with $\|\mathsf{Lin}(t)\| \leq C\|t\|_\Gamma$, then $h \mapsto \mathsf{Lin}(h^{\otimes n})$ is a bounded polynomial law of degree n and norm at most C.*
3. *If a map $a\colon P \to W$ satisfies $ja(h) = \beta(h, \dots, h)$ for a linear isometry $j\colon W \to Z$ and a bounded n-linear $\beta\colon P^n \to Z$, then a is a bounded polynomial law of degree n, with $\|a\| \leq \|\beta\|$ and $\operatorname{amb}(a) \leq \|\beta\|$.*
4. *If $\operatorname{amb}(a) < \infty$, then a is ambient of degree n, with an n-linear map of norm $\operatorname{amb}(a)$.*

Without norms, and since K is an infinite field, the maps of Definition 14.8(1) are the maps induced by homogeneous polynomial laws in the sense of Roby, and each determines its law [21, Th. I.1, Props. I.1, I.8, IV.6]; items (1) and (2) are then the linearization [21, Th. IV.1, Prop. IV.5].

Proof. Step 1 (the linear map, item (1)). Let $\sum_i \lambda_i h_i^{\otimes n} = 0$. Choose a basis $e_1, \dots, e_{m'}$ of the span of the h_i and write $h_i = \sum_c \theta_{i,c} e_c$, and put $\theta_i^\alpha = \prod_c \theta_{i,c}^{\alpha_c}$. By (14.1), $\sum_\alpha (\sum_i \lambda_i \theta_i^\alpha)$ $O_\alpha(e_1, \dots, e_{m'}) = 0$. The tensors $O_\alpha(e_1, \dots, e_{m'})$ are sums of disjoint sets of basis tensors, so they are linearly independent and every $\sum_i \lambda_i \theta_i^\alpha$ vanishes. Hence $\sum_i \lambda_i a(h_i) = \sum_\alpha (\sum_i \lambda_i \theta_i^\alpha) a^{[\alpha]}(e_1, \dots, e_{m'}) = 0$. So $L_a(\sum_i \lambda_i h_i^{\otimes n}) = \sum_i \lambda_i a(h_i)$ is well defined on the span of the powers, which is $\Gamma^n(P)$, and this span gives uniqueness. Also $\|L_a(t)\| \le \|a\| \sum_i |\lambda_i| \|h_i\|^n$; take the infimum over representations of t.

Step 2 (the converse, item (2)). By (14.1), $\mathsf{Lin}((\sum_i \lambda_i h_i)^{\otimes n}) = \sum_\alpha \lambda^\alpha \mathsf{Lin}(O_\alpha(h_1, \dots, h_r))$ is a homogeneous polynomial of degree n with coefficients in W, and $\|\mathsf{Lin}(h^{\otimes n})\| \le C \|h\|^n$.

Step 3 (ambient diagonals are laws, item (3)). For $h_1, \dots, h_r \in P$, the map $\lambda \mapsto \beta(\sum_c \lambda_c h_c, \dots, \sum_c \lambda_c h_c)$ is a homogeneous polynomial of degree n with coefficients in Z. As K is infinite, interpolation in each variable writes its coefficients as combinations of its values, which lie in $j(W)$. So a is a law, and $\|a(h)\| = \|\beta(h, \dots, h)\| \le \|\beta\| \|h\|^n$. Let $\beta_\otimes \colon P^{\otimes n} \to Z$ linearize β. Then jL_a and $\beta_\otimes$ agree on powers, hence on $\Gamma^n(P)$, and $\|L_a(t)\| = \|\beta_\otimes(t)\| \le \|\beta\| \, \pi(t)$.

Step 4 (the pushout of one law, item (4)). Put $C = \mathrm{amb}(a)$. On $W \oplus P^{\otimes n}$ let $\|(v, y)\|_Z = \inf\{\|v + L_a(t)\| + C\pi(y - t) : t \in \Gamma^n(P)\}$. This is the quotient seminorm of $\|v\| + C\pi(y)$ modulo the pairs $(L_a(t), -t)$. Let Z be the quotient by its kernel, with classes $[v, y]$, and put $j(v) = [v, 0]$. Taking $t = 0$ gives $\|j(v)\| \le \|v\|$, and $\|v + L_a(t)\| + C\pi(t) \ge \|v\| - \|L_a(t)\| + C\pi(t) \ge \|v\|$; so j is an isometry. The map $\beta(h_1, \dots, h_n) = [0, h_1 \otimes \cdots \otimes h_n]$ has norm at most C, and taking $t = h^{\otimes n}$ shows $\beta(h, \dots, h) = [a(h), 0] = ja(h)$. Item (3) gives $\|\beta\| \ge C$. □

Lemma Q.2 (mixed terms)**.** *Let $a \colon W \to Y$ be a bounded polynomial law of degree m.*

(1) *For $w_1, \dots, w_r \in W$ and $\alpha \in \mathbb{N}^r$ with $|\alpha| = m$,*

$$\|a^{[\alpha]}(w_1, \dots, w_r)\| \le (8/|\varpi|)^m \|a\| \, (\|w_1\| + \cdots + \|w_r\|)^m. \tag{Q.1}$$

(2) *For $x, y \in W$ and $0 \le i \le m$,*

$$\|a^{[(m-i,i)]}(x, y)\| \le (16/|\varpi|^2)^m \|a\| \, \|x\|^{m-i} \|y\|^i. \tag{Q.2}$$

So $a(x+y) = \sum_{i \le m} a^{[(m-i,i)]}(x, y)$ is a finite expansion at x by laws in y: a is continuous and DP on W. It is Lipschitz on bounded sets: for $m \ge 1$, $R > 0$ and $\|x\|, \|y\| \le R$,

$$\|a(x + y) - a(x)\| \le m(16/|\varpi|^2)^m \|a\| \, R^{m-1} \|y\|. \tag{Q.3}$$

(3) *With $C_0 = 8\mathrm{e}/|\varpi|^2$, for w_i and α as in (1),*

$$\|a^{[\alpha]}(w_1, \dots, w_r)\| \le C_0^m \|a\| \, \frac{m!}{\alpha_1! \cdots \alpha_r!} \prod_i \|w_i\|^{\alpha_i}. \tag{Q.4}$$

(4) *Let $K' \supseteq K$ be a nonarchimedean field whose absolute value extends that of K, let W be $K'^{m'}$ with the maximum norm, written $c_0([m'], K')$, with unit vectors $e_1, \dots, e_{m'}$, and let Y be an ultrametric normed K'-space and a a law over K'. For $\alpha \in \mathbb{N}^{m'}$ with $|\alpha| = m$,*

$$\|a^{[\alpha]}(e_1, \dots, e_{m'})\| \le |\varpi|^{-m} \|a\|. \tag{Q.5}$$

(5) *If P is ultrametric and $x_1,\dots,x_m\in P$, then*

$$\|O_{(1,\dots,1)}(x_1,\dots,x_m)\|_\Gamma \le (8/|\varpi|^2)^m \prod_i \|x_i\|. \tag{Q.6}$$

Proof. Step 1 (nodes). For an integer $\ell\ge 0$ with binary digits ℓ_u, so that $\ell=\sum_u \ell_u 2^u$, put $\xi_\ell=\sum_u \ell_u\varpi^u\in K$. Then $|\xi_\ell|\le 1$. For $\ell\ne\ell'$ the least u with $\ell_u\ne\ell'_u$ is $\mathrm{ord}_2(\ell-\ell')$, so $|\xi_\ell-\xi_{\ell'}|=|\varpi|^{\mathrm{ord}_2(\ell-\ell')}$. For $0\le\ell\le i$ the numbers $|\ell-\ell'|$, with $\ell'\le i$ and $\ell'\ne\ell$, are $1,\dots,\ell$ and $1,\dots,i-\ell$, and $\mathrm{ord}_2(\ell!)\le\ell$, $\mathrm{ord}_2((i-\ell)!)\le i-\ell$. So $\prod_{\ell'\le i,\ell'\ne\ell}|\xi_\ell-\xi_{\ell'}|^{-1}\le|\varpi|^{-i}$. Hence the divided difference $\sum_{\ell\le i}\tau(\xi_\ell)\prod_{\ell'\le i,\ell'\ne\ell}(\xi_\ell-\xi_{\ell'})^{-1}$ of order i of a map τ into a normed space is a combination of its $i+1$ values at these nodes with coefficients of absolute value at most $|\varpi|^{-i}$.

Step 2 (interpolation on the nodes, item (1)). If $m=0$, a is constant. Arguments with $\alpha_c=0$ do not change $a^{[\alpha]}(w)$, so assume all $\alpha_c\ge 1$; then $r\le m$. If $r=1$, $a^{[\alpha]}(w_1)=a(w_1)$. Let $r\ge 2$ and $N=r-1$. The map $\Pi(\lambda)=a(\lambda_1w_1+\dots+\lambda_Nw_N+w_r)$ on K^N is a polynomial of total degree at most m with coefficients in Y, and $a^{[\alpha]}(w)$ is its coefficient of $\lambda_1^{\alpha_1}\cdots\lambda_N^{\alpha_N}$. Newton's interpolation formula on $\xi_0,\dots,\xi_m$ in each variable gives

$$\Pi(\lambda)=\sum_{\mathbf{i}\in\{0,\dots,m\}^N}\mathsf{c}_\mathbf{i}\prod_{c=1}^N \mathsf{N}_{i_c}(\lambda_c),\qquad \mathsf{N}_i(\lambda_c)=\prod_{\ell<i}(\lambda_c-\xi_\ell),$$

where $\mathsf{c}_\mathbf{i}\in Y$ is the divided difference of Π of order i_c in λ_c on $\xi_0,\dots,\xi_{i_c}$, for each c. A divided difference of order i kills polynomials of degree below i, so $\mathsf{c}_\mathbf{i}=0$ unless $|\mathbf{i}|\le m$. By Step 1, $\mathsf{c}_\mathbf{i}$ is a combination of at most $\prod_c(i_c+1)\le 2^{|\mathbf{i}|}$ values of Π at points with coordinates ξ_ℓ, with coefficients of absolute value at most $|\varpi|^{-|\mathbf{i}|}$. Each value has norm at most $\|a\|(\sum_c\|w_c\|)^m$. The N_i have coefficients of absolute value at most 1. So the coefficient of $\lambda_1^{\alpha_1}\cdots\lambda_N^{\alpha_N}$ is a sum of at most $\binom{m+N}{N}\le 4^m$ terms, each of norm at most $(2/|\varpi|)^m\|a\|(\sum_c\|w_c\|)^m$.

Step 3 (two arguments, item (2)). If $x,y\ne 0$, choose $\lambda_1,\lambda_2\in K$ with $|\varpi|<|\lambda_1|\|x\|\le 1$ and $|\varpi|<|\lambda_2|\|y\|\le 1$. As $a^{[(m-i,i)]}(\lambda_1x,\lambda_2y)=\lambda_1^{m-i}\lambda_2^i a^{[(m-i,i)]}(x,y)$, (Q.1) gives (Q.2), which is trivial if $x=0$ or $y=0$. For fixed x, $y\mapsto a^{[(m-i,i)]}(x,y)$ is a coefficient of a polynomial on the span of finitely many vectors, so by (Q.2) it is a bounded law of degree i. Summing (Q.2) over $1\le i\le m$ gives (Q.3), since $a^{[(m,0)]}(x,y)=a(x)$.

Step 4 (scaling each argument, item (3)). If $m=0$, a is constant. Otherwise drop the w_i with $\alpha_i=0$; if some remaining $w_i=0$, both sides vanish. Choose $\lambda_i\in K$ with $|\varpi|\alpha_i/m<|\lambda_i|\|w_i\|\le\alpha_i/m$. Then $\sum_i|\lambda_i|\|w_i\|\le 1$, so $\|a^{[\alpha]}(\lambda_1w_1,\dots,\lambda_rw_r)\|\le(8/|\varpi|)^m\|a\|$ by (Q.1). Undoing the scaling gives

$$\|a^{[\alpha]}(w_1,\dots,w_r)\|\le(8/|\varpi|^2)^m\|a\|\,m^m\prod_i\alpha_i^{-\alpha_i}\prod_i\|w_i\|^{\alpha_i},$$

and $m^m\le \mathrm{e}^m m!$, $\alpha_i^{\alpha_i}\ge\alpha_i!$.

Step 5 (ultrametric cases, items (4) and (5)). For (4), follow Step 2 with the unit vectors e_c, $\alpha_c\ge 1$, in place of the w_c. Every node point has maximum norm 1, so every value of Π has norm at most $\|a\|$. In the ultrametric Y this gives $\|\mathsf{c}_\mathbf{i}\|\le|\varpi|^{-|\mathbf{i}|}\|a\|$ with no count of terms, and the coefficient is an ultrametric sum over $|\mathbf{i}|\le m$. For (5), let W_m be the quotient of $\Gamma^m(P)$ by the kernel of $\|\cdot\|_\Gamma$. By Lemma Q.1(2), $h\mapsto[h^{\otimes m}]$ is a law into W_m of norm at most 1, and by (14.1) its coefficient of index $(1,\dots,1)$ is the class of $O_{(1,\dots,1)}$. In Step 2 the node points have norm at most $\max_i\|x_i\|$, as P is ultrametric, and the rest is unchanged; so $\|O_{(1,\dots,1)}(x_1,\dots,x_m)\|_\Gamma\le(8/|\varpi|)^m\max_i\|x_i\|^m$. Both sides of (Q.6) vanish if some $x_i=0$; otherwise scale each x_i to $|\varpi|<|\lambda_i|\|x_i\|\le 1$. □

Proof of Proposition 14.9. Step 1 (coefficients). Let f be DP at x_0 with radius R, bound $\mathfrak{b}$ and laws a_n. For $h \neq 0$, $f(x_0+\lambda h) = \sum_n \lambda^n a_n(h)$ for $|\lambda| < R/\|h\|$, with $\|a_n(h)\|(R/\|h\|)^n \leq \mathfrak{b}$. By Lemma A.1, f determines the a_n. If f also has an ambient expansion (14.2), the same lemma in Z gives $ja_n(h) = b_n(h,\dots,h)$; the case $h=0$ is clear.

Step 2 (ambient analytic implies DP). Let f have an ambient expansion with radius R and bound $\mathfrak{b}$, and put $a_n(h) = j^{-1}(b_n(h,\dots,h))$. By Lemma Q.1(3), a_n is a law with $\|a_n\|$ and $\mathrm{amb}(a_n)$ at most $\|b_n\| \leq \mathfrak{b}R^{-n}$. The partial sums $\sum_{n\leq N} a_n(h)$ lie in W, and their images under the isometry j converge to $jf(x_0+h)$; so they converge to $f(x_0+h)$.

Step 3 (the pushout). On $\bigoplus_n P^{\otimes n}$, with ι_n the inclusion of the n-th summand, put $\|y\|_{\pi,R} = \sum_n R^{-n}\pi(y_n)$. Let f be DP at x_0 with radius R and $\mathrm{amb}(a_n) \leq \mathfrak{b}R^{-n}$ for all n. Then $L_\oplus(t) = \sum_n L_{a_n}(t_n)$ on $\bigoplus_n \Gamma^n(P)$ satisfies $\|L_\oplus(t)\| \leq \mathfrak{b}\|t\|_{\pi,R}$. On $W \oplus \bigoplus_n P^{\otimes n}$ let

$$\|(v,y)\|_Z = \inf\Big\{\|v + L_\oplus(t)\| + \mathfrak{b}\|y-t\|_{\pi,R} :\ t \in \bigoplus_n \Gamma^n(P)\Big\}, \tag{Q.7}$$

the quotient seminorm of $\|v\| + \mathfrak{b}\|y\|_{\pi,R}$ modulo the pairs $(L_\oplus(t), -t)$. Let Z be the quotient by its kernel, with classes $[v,y]$, and put $j(v) = [v,0]$. As in Step 4 of the proof of Lemma Q.1, j is an isometry. The maps $b_n(h_1,\dots,h_n) = [0, \iota_n(h_1\otimes\cdots\otimes h_n)]$ have norm at most $\mathfrak{b}R^{-n}$, and taking $t = \iota_n h^{\otimes n}$ shows $b_n(h,\dots,h) = [a_n(h),0] = ja_n(h) \in j(W)$. So $jf(x_0+h) = \sum_n b_n(h,\dots,h)$ for $\|h\| < R$. Completing Z keeps all of this. With Steps 1 and 2 this proves (1).

Step 4 (one point suffices, item (2)). Let a be a bounded law of degree n with $ja(h_0+h) = \sum_i b_i(h,\dots,h)$ near $h=0$. The polynomial $\lambda \mapsto ja(h_0+\lambda h)$ has coefficient $ja(h)$ at λ^n. Lemma A.1 on the line gives $b_n(h,\dots,h) = ja(h)$, so a is ambient of degree n. □

Q.2. **Fixed envelopes.** Let β be a bounded d-linear map and let $\varsigma_1,\dots,\varsigma_d$ be maps analytic at a point. Then $\beta(\varsigma_1,\dots,\varsigma_d)$ is analytic at that point: for small increments the family of all values of β at terms of the expansions is absolutely summable in the completion of the target, and grouping it by total degree gives finite sums in the target, whose partial sums converge there, with multilinear coefficients of geometric bound after the radius is shrunk.

Proof of Proposition 14.3. Step 1 (charts, module). Let $s_{\mathsf{a}}, s_{\mathsf{b}}$ be local expressions of a section of $\mathcal{B}$ in charts of the analytic atlas of $\widetilde{\mathcal{B}}$, so that $s_{\mathsf{a}}(x) = \rho(\Theta_{\mathsf{ab}}(x))s_{\mathsf{b}}(x)$. If $s_{\mathsf{b}} = \mathrm{qt}\circ\varsigma$ with ς analytic into $\tilde{V}$ and values in V_1, then $s_{\mathsf{a}} = \mathrm{qt}\circ(\Theta_{\mathsf{ab}}\varsigma)$, as $\rho(\Theta)\,\mathrm{qt} = \mathrm{qt}\,\Theta$ on V_1. The map $x \mapsto \Theta_{\mathsf{ab}}(x)\varsigma(x)$ is analytic with values in V_1. Sums and analytic scalar multiples lift in the same way.

Step 2 (operations). Let o be an operation on ambient sections $s_1,\dots,s_d$, with lifts $\varsigma_1,\dots,\varsigma_d$, with local expression $\rho_d(\Theta(x))$, and with quotient map qt' of its target. The identities $\rho_d(\Theta)\circ(\rho(\Theta_1)\times\cdots\times\rho(\Theta_d)) = \rho_d(\Theta\circ(\Theta_1\times\cdots\times\Theta_d))$ and $\rho(\Theta')\circ\rho_d(\Theta) = \rho_d(\Theta'\circ\Theta)$ show that this form of the local expression does not depend on the charts. The local expression of $\mathsf{o}(s_1,\dots,s_d)$ is qt' applied to $x \mapsto \Theta(x)(\varsigma_1(x),\dots,\varsigma_d(x))$, which is analytic with values in V_1'.

Step 3 (the split case). Let l be a bounded linear section of qt and r a bounded linear retraction of $\tilde{V}$ onto V_1. For T preserving the filtration, $\rho(T) = \mathrm{qt}\circ\mathsf{r}\circ T\circ\mathsf{l}$, since $T\mathsf{l}v \in V_1$ is fixed by r. So $\rho(\Theta_{\mathsf{ab}})$ is a bounded linear image of Θ_{ab}, hence analytic, with inverse family $\rho(\Theta_{\mathsf{ba}})$. An analytic local expression s lifts to the analytic map $\mathsf{l}\circ s$. Conversely, if $s = \mathrm{qt}\circ\varsigma$, then $\varsigma = \mathsf{r}\circ\varsigma$ is analytic into V_1 and s is analytic. □

Proof of Theorem 14.4. Step 1 (the subfunctor, item (1)). By Proposition 2.2 and its proof, $\mathfrak{m}_k$ is the joint action B, the diagonal of a bounded $(k+1)$-linear map of norm at most 1. The subspace $\mathrm{Alt}^k(E;F)$ is closed in $\mathrm{Mult}^k(E;F)$, and two equal arguments stay equal under $\mathbf{f}_1$; so $\mathfrak{m}_k(\mathbf{f}_1,\mathbf{f}_0)$ preserves alternating maps, and its restriction is $\alpha_k(\mathbf{f}_1,\mathbf{f}_0)$.

Step 2 (the filtered bundle, item (2)). Over a common trivializing cover, $\Theta_{\mathsf{ab}}(x) = \mathfrak{m}_k(\mathbf{g}^1_{\mathsf{ba}}(x), \mathbf{g}^0_{\mathsf{ab}}(x))$ is jointly analytic, as a bounded multilinear evaluation of analytic operator families. Functoriality gives $\Theta_{\mathsf{ab}}\Theta_{\mathsf{bc}} = \Theta_{\mathsf{ac}}$ and the inverse Θ_{ba}, so these transitions glue the analytic bundle $\mathcal{L}(M_1, \dots, M_1; M_0)$, as in Step 1 of the proof of Theorem 2.1. By Step 1 they preserve $\mathrm{Alt}^k(E; F)$, and so does the inverse: this is a filtered analytic bundle of sub type. Their restrictions $\alpha_k(\mathbf{g}^1_{\mathsf{ba}}, \mathbf{g}^0_{\mathsf{ab}})$ are the transitions of the C^∞ bundle of Theorem 2.1, which applies by Corollary 7.3. The union of two analytic atlases of M_1, and of two of M_0, is again an analytic atlas, and the argument applies to it. So all the charts induced by analytic vector charts of M_1, M_0 form one such atlas, and by Proposition 14.3 the bundle and its ambient sections do not depend on the cover.

Step 3 (evaluation, n° 7.8.1). The local expression of an ambient ω is analytic into $\mathrm{Mult}^k(E; F)$ and those of the s_i are analytic into E. Evaluation is bounded $(k+1)$-linear, so $\omega(\mathsf{s}_1, \dots, \mathsf{s}_k)$ is analytic.

Step 4 (exterior products, n° 7.8.2). Let $\mathcal{Q}, \mathcal{Q}', \mathcal{Q}''$ have fibers $F_{\mathcal{Q}}, F_{\mathcal{Q}'}, F_{\mathcal{Q}''}$, so that $\varphi_x \colon F_{\mathcal{Q}} \times F_{\mathcal{Q}'} \to F_{\mathcal{Q}''}$ is the local expression of the pairing φ. On $\mathrm{Mult}^k(E; F_{\mathcal{Q}}) \times \mathrm{Mult}^{k'}(E; F_{\mathcal{Q}'})$ let sh_{φ_x} be Bourbaki's formula (1), the formula (10.2) with φ_x in place of the product and with degrees k, k'. In degree 0 the fiber $F_{\mathcal{Q}}$ is the filtered space $(F_{\mathcal{Q}}; F_{\mathcal{Q}}, 0)$. The map $\varphi_x \mapsto \mathrm{sh}_{\varphi_x}$ is bounded linear, of norm at most $\binom{k+k'}{k}$, so $x \mapsto \mathrm{sh}_{\varphi_x}$ is analytic. Let two arguments agree and let the inputs be alternating. Terms with both in one block vanish. Shuffles placing them in opposite blocks cancel in pairs with opposite signs, as in Step 2 of the proof of Theorem 10.7; this uses only bilinearity of φ_x, in every characteristic. So $\wedge_\varphi$ is a bilinear operation, and Proposition 14.3 applies.

Step 5 (the graded algebra, n° 7.8.3). The multiplication of A is an analytic pairing, so Step 4 gives the product on $\Omega^*(U)$, with $\mathrm{Alt}^0(M_1; \mathsf{A}) = \mathsf{A}$. The operations of Steps 3, 4 and 6 act on Mult, but the identities below are claimed only for alternating maps, at each point of U; this suffices, as ambient sections have alternating values. Associativity and anticommutativity follow as in Step 2 of the proof of Theorem 10.7, from associativity and commutativity of the fiber A_x. Formula (2) is the Laplace expansion, by induction on the number of factors, from (1).

Step 6 (interior products, n° 7.8.4). Let $\mathrm{ins}(v, \mathsf{m}) = \mathsf{m}(v, \cdot, \dots, \cdot)$. For a transition $\mathbf{f}_1$ of M_1 and a transition $\mathbf{f}_0$ of A,

$$\mathrm{ins}\big(\mathbf{f}_1 v,\ \mathbf{f}_0 \circ \mathsf{m} \circ (\mathbf{f}_1^{-1})^{\times k}\big) = \mathbf{f}_0 \circ \mathrm{ins}(v, \mathsf{m}) \circ (\mathbf{f}_1^{-1})^{\times (k-1)},$$

so the local expression of ins is the same constant in every induced chart. It is bounded bilinear of norm at most 1 and maps $E \times \mathrm{Alt}^k(E; F)$ into $\mathrm{Alt}^{k-1}(E; F)$, for the fiber F of A. So it is a bilinear operation, with E read as $(E; E, 0)$. Formulas (3) and (5) are its definition and (4) is alternation, with $i(\mathsf{s}) = 0$ on degree 0 as in [3, n° 7.8.4]. For (6), split the sum in (1) by the block containing the inserted argument; in the second block it moves past k arguments. Formula (7) follows from (5) and (6) by induction.

Step 7 (inverse images and forms, items (3)(e) and (4)). For an analytic map $\gamma \colon M' \to M$ the transitions of the pulled-back bundles are the composites with γ, so those of $\mathrm{Alt}^k(\gamma^* M_1; \gamma^* M_0)$ inside its multilinear bundle are $\Theta_{\mathsf{ab}} \circ \gamma$. This is γ^* of the filtered bundle, and lifts compose with γ. For $M_1 = TM$, $M_0 = M \times K$ and K complete the transitions are given by derivatives of analytic chart changes, and being ambient in the induced charts is the definition of $\Omega^k_{\mathrm{amb}}(M)$ in Section 10.3. □

Proof of Theorem 14.5. Step 1 (incompatible charts, item (1)). Take E, F from Theorem 7.2(1) and apply Corollary 10.2 with $D = E$. Propositions 10.4(1) and 10.3(1) give the same with $M_1 = TM$.

Step 2 (no retraction, item (2)). For these E, F, a bounded retraction of $j_{E,F}$ would give a bounded k-linear lift of $A^k_{E,E;F}$ by Proposition 5.1. Then $A^k_{E,E;F}$ would be a continuous polynomial, against Theorem 7.2(1). For $M_1 = M \times E$ and $M_0 = M \times F$ every fiber of $\mathrm{Alt}^k(M_1; M_0)$ is then not a direct factor of the fiber of $\mathcal{L}(M_1, \dots, M_1; M_0)$.

Step 3 (ambient, not analytic, item (3)). This is Proposition 10.3(2), with the Banach space named N there as the manifold M in its one chart, $M_1 = TM$ and $M_0 = M \times K$.

Step 4 (no compatible structure, item (4)). In the induced product chart, used on both sides, the local expression of the map induced by the shear $(u, z) \mapsto (u, g(u)z)$ is $u \mapsto g(u)^*$. This is not analytic at u_0 by Proposition 10.1, while an analytic morphism has analytic local expressions in all charts. For the other claim, automorphisms are taken over open subsets of M, and $A^k_{E,D;F}$ is analytic nowhere, by Proposition 2.5. Let $\mathcal{S}$ be such a structure containing the chart X over an open U induced by an analytic vector chart of $M \times H$ and the product chart of $M \times F$. After a constant isomorphism of model spaces, the frame of this vector chart is an analytic map $\mathbf{k}$ from U to the invertible operators of H, with analytic inverse family, and $(u, z) \mapsto (u, \mathbf{k}(u)z)$ is an analytic automorphism of $M \times H$ over U. Let $\mathbf{k}_*$ be the map it induces, chosen with $\mathsf{X} \circ \mathbf{k}_*^{-1}$ the induced product chart over U. By hypothesis $\mathbf{k}_*$ and $\mathbf{k}_*^{-1}$ are analytic morphisms of $\mathcal{S}$, and the transitions between X and $\mathsf{X} \circ \mathbf{k}_*^{-1}$ are local expressions of these maps; so $\mathcal{S}$ contains the induced product chart over U. The previous sentences, over U and at any $u_0 \in U$, give a contradiction.

Step 5 (when the fibers behave, item (5)). If α_k is analytic on the hom spaces of the fibers, the transitions $\alpha_k(\mathbf{g}^1_{\mathsf{ba}}, \mathbf{g}^0_{\mathsf{ab}})$ are analytic and glue an analytic bundle, by Step 1 of the proof of Theorem 2.1. Composing an analytic section with $j_{E,F}$ makes it ambient. If $j_{E,F}$ has a bounded retraction, the split clause of Proposition 14.3 applies, with $\mathrm{qt} = \mathrm{id}$ and $\mathsf{l} = \mathrm{id}$. The instances named in the statement are Corollaries 7.3 and 10.5 for analyticity, and normalized antisymmetrization and Theorem 5.2 for the retraction. □

Q.3. **Enlargements.**

Proof of Theorem 14.11. Step 1 (data). Let the germs be $f_i \colon U_i \to W$, $i \in I$, with U_i open in P_i, at x_i, with radius R_i and bound $\mathfrak{b}_i$. Let a^i_n be their coefficient laws and $L_{\oplus,i} = \sum_n L_{a^i_n}$, so that $\|L_{\oplus,i}(t)\| \le \mathfrak{b}_i \|t\|_{\pi,R_i}$ on $\bigoplus_n \Gamma^n(P_i)$ by Proposition 14.9.

Step 2 (amalgamated pushout). On $W \oplus \bigoplus_{i\in I} \bigoplus_n P_i^{\otimes n}$ put

$$\|(v, (y_i)_i)\|_Z = \inf\Big\{ \Big\| v + \sum_i L_{\oplus,i}(t_i) \Big\| + \sum_i \mathfrak{b}_i \|y_i - t_i\|_{\pi,R_i} \Big\},$$

over finitely supported families $t_i \in \bigoplus_n \Gamma^n(P_i)$. This is a quotient seminorm, as in (Q.7). Let Z be the completion of the quotient by its kernel, $j(v) = [v, 0]$, and let $b^i_n(h_1, \dots, h_n)$ be the class of $h_1 \otimes \cdots \otimes h_n$ in the n-th summand of the i-th family.

Step 3 (j is an isometry). All $t_i = 0$ gives $\|j(v)\| \le \|v\|$. For every family, $\|v + \sum_i L_{\oplus,i}(t_i)\| + \sum_i \mathfrak{b}_i \|t_i\|_{\pi,R_i} \ge \|v\| - \sum_i \|L_{\oplus,i}(t_i)\| + \sum_i \mathfrak{b}_i \|t_i\|_{\pi,R_i} \ge \|v\|$.

Step 4 (expansions). We have $\|b^i_n\| \le \mathfrak{b}_i R_i^{-n}$, and taking $t_i = \iota_n h^{\otimes n}$, the other terms zero, shows $b^i_n(h, \dots, h) = j a^i_n(h) \in j(W)$. So $j f_i(x_i + h) = \sum_n b^i_n(h, \dots, h)$ for $\|h\| < R_i$. For the last assertion take $I = U$. □

Proof of Proposition 14.12. Let f have radius R, bound $\mathfrak{b}$ and laws a_n.

Step 1 (target descent). Let $f_Y \colon U \to Y$ be DP at x_0 with laws a^Y_n, and let $j_Y f_Y$ be ambient analytic at x_0 with radius R' and bound $\mathfrak{b}'$, for a linear isometry $j_Y \colon Y \to Z_Y$. By Proposition 14.9 and Step 1 of its proof, the coefficient laws of $j_Y f_Y$ are $j_Y a^Y_n$, and $\mathrm{amb}(a^Y_n) = \mathrm{amb}(j_Y a^Y_n) \le \mathfrak{b}' R'^{-n}$, as $L_{j_Y a^Y_n} = j_Y L_{a^Y_n}$; also $\|a^Y_n\| \le \mathrm{amb}(a^Y_n)$. For $\|h\| < R'$ the

partial sums $\sum_{n\le N} a_n^Y(h)$ lie in Y, and their images under j_Y converge to $j_Y f_Y(x_0+h)$; so they converge to $f_Y(x_0+h)$, and f_Y is DP with radius R'. Proposition 14.9 applies to f_Y.

Step 2 (postcomposition, pairs, germs). For bounded linear $\Upsilon\colon W\to Y$, Υf has laws Υa_n with $L_{\Upsilon a_n}=\Upsilon L_{a_n}$, so $\operatorname{amb}(\Upsilon a_n)\le\|\Upsilon\|\mathfrak{b}R^{-n}$. For $f'\colon U\to W'$ ambient analytic at x_0 with radius $R_{W'}$ and bound $\mathfrak{b}_{W'}$ and with laws a'_n, (f,f') has laws (a_n,a'_n) into $W\oplus W'$ with the maximum norm, and $\operatorname{amb}(a_n,a'_n)\le\max(\operatorname{amb}(a_n),\operatorname{amb}(a'_n))$, so (f,f') has radius $\min(R,R_{W'})$ and bound $\max(\mathfrak{b},\mathfrak{b}_{W'})$. Sums follow with the addition map. Apply Proposition 14.9. Dependence only on the germ is clear after shrinking the radius.

Step 3 (scalar factors). Take an ambient expansion $(j,(b_n))$ of f, and let $\zeta(x_0+h)=\sum_i\zeta_i(h,\dots,h)$ with $\zeta_i\colon P^i\to K$, $\|\zeta_i\|\le\mathfrak{b}_\zeta R_\zeta^{-i}$. Put

$$\beta_N(h_1,\dots,h_N)=\sum_{i+n=N}\zeta_i(h_1,\dots,h_i)\,b_n(h_{i+1},\dots,h_N),$$

of norm at most $(N+1)\mathfrak{b}_\zeta\mathfrak{b}\min(R,R_\zeta)^{-N}$. Its diagonal $\sum_{i+n=N}\zeta_i(h,\dots,h)ja_n(h)$ lies in $j(W)$. For $R_3<\min(R,R_\zeta)$ this norm is at most $\mathfrak{b}_3R_3^{-N}$ for a suitable $\mathfrak{b}_3$. For $\|h\|<R_3$ the double series converges absolutely in the completion of Z, with sum $j(\zeta f)(x_0+h)\in Z$, and its partial sums in N lie in Z. So ζf is ambient analytic with radius R_3.

Step 4 (precomposition). Let γ map an open subset of P' into U, with $\gamma(x'_0)=x_0$ and $\gamma(x'_0+h)=x_0+\sum_{l\ge1}\gamma_l(h,\dots,h)$, $\|\gamma_l\|\le\mathfrak{b}_\gamma R_\gamma^{-l}$. Take an ambient expansion $(j,(b_m))$ of f. Put $\beta_0=b_0$ and, for $n\ge1$,

$$\beta_n(h_1,\dots,h_n)=\sum_{1\le m\le n}\ \sum_{l_1+\cdots+l_m=n,\ l_c\ge1} b_m\big(\gamma_{l_1}(h_1,\dots,h_{l_1}),\dots,\gamma_{l_m}(h_{n-l_m+1},\dots,h_n)\big).$$

With $R_1=R_\gamma/(1+\mathfrak{b}_\gamma/R)$, counting compositions of n gives

$$\|\beta_n\|\le\sum_{1\le m\le n}\binom{n-1}{m-1}\mathfrak{b}R^{-m}\mathfrak{b}_\gamma^m R_\gamma^{-n}\le\mathfrak{b}R_1^{-n}.$$

For $\|h\|<R_1$ we have $\sum_l\|\gamma_l(h,\dots,h)\|<R$, and the family, indexed by m and $(l_1,\dots,l_m)$, of the values of b_m at the vectors $\gamma_{l_c}(h,\dots,h)$ is absolutely summable in the completion of Z. Grouped by m it has sum $jf(\gamma(x'_0+h))\in Z$, and grouped by n it has the finite groups $\beta_n(h,\dots,h)\in Z$. So $jf\gamma(x'_0+h)=\sum_n\beta_n(h,\dots,h)$ in Z. For the diagonals, fix h and $1\le m\le n$, and put $\gamma_{\le n}(\lambda)=\sum_{l=1}^n\lambda^l\gamma_l(h,\dots,h)$. Then $\lambda\mapsto b_m(\gamma_{\le n}(\lambda),\dots,\gamma_{\le n}(\lambda))$ is a polynomial with coefficients in Z and values in $j(W)$, so Lagrange interpolation puts every coefficient in $j(W)$. Its coefficient at λ^n is the inner sum of $\beta_n(h,\dots,h)$. Hence $\beta_n(h,\dots,h)\in j(W)$.

Step 5 (re-expansion). Let W be Banach. Take an ambient expansion with Z Banach and $j(W)$ closed, and put $h_1=x_1-x_0$. For $\|h\|<R-\varrho$ expand $b_n(h_1+h,\dots,h_1+h)$ into 2^n terms indexed by the sets of slots holding h. The family of all terms is absolutely summable, since $\sum_n\mathfrak{b}R^{-n}(\varrho+\|h\|)^n<\infty$. Grouping by the number i of slots holding h gives bounded i-linear maps b'_i with

$$\|b'_i\|\le\sum_{n\ge i}\binom{n}{i}\mathfrak{b}R^{-n}\varrho^{n-i}=\frac{\mathfrak{b}R}{R-\varrho}(R-\varrho)^{-i},$$

and $jf(x_1+h)=\sum_i b'_i(h,\dots,h)$. Step 1 of the proof of Theorem 2.3 puts the diagonals in $j(W)$.

Step 6 (derivatives). Let W be Banach and f ambient analytic on U; fix $x_1\in U$ and an expansion at x_1 as in Step 5, with radius R_2 and bound $\mathfrak{b}_2$. Step 5 at x_1+h, $\|h\|<R_2$, shows that jf is differentiable there, the terms with $i\ge2$ being of order $\|v\|^2$ in the

increment v, with derivative $\sum_{n\geq 1} b''_{n-1}(h,\dots,h)$, where $b''_{n-1}(v_1,\dots,v_{n-1}) \in \mathcal{L}(P,Z)$ is $u \mapsto \sum_{l=1}^{n} b_n(v_1,\dots,v_{l-1},u,v_l,\dots,v_{n-1})$, of norm at most $n\mathfrak{b}_2 R_2^{-n}$. So $D(jf)$ is analytic at x_1 with every radius smaller than R_2. Postcomposition by j is a linear isometry $\mathcal{L}(P,W) \to \mathcal{L}(P,Z)$ with closed range, and by Lemma 2.4 and its proof f is differentiable with $jDf = D(jf)$. So Df is ambient analytic at x_1 with values in the Banach space $\mathcal{L}(P,W)$. Induct, using $\mathcal{L}(P,\mathrm{Mult}^{\ell}(P;W)) = \mathrm{Mult}^{\ell+1}(P;W)$. Each $D^{\ell}f$ is continuous, so f is C^{∞}. □

Q.4. **Composition.** In this subsection P, W, Y are normed spaces. The algebra of composition follows from Roby [21, VI §8, Prop. VI.6 and its Cor.; Prop. I.2; II §1]; only the estimate is added here. For a sequence of bounded laws $(a_n)_{n\geq 1}$ on P and bounded laws $(d_m)_{m\geq 0}$ on W put $\bar{a}_{\geq 1}(\theta) = \sum_{n\geq 1}\|a_n\|\theta^n$ and $\bar{d}^{\sharp}(\theta) = \sum_{m\geq 0} C_0^m\|d_m\|\theta^m$, power series in a real variable θ with nonnegative coefficients. For a finitely supported sequence $\alpha = (\alpha_n)_{n\geq 1}$ of nonnegative integers put $|\alpha| = \sum_n \alpha_n$ and $\deg\alpha = \sum_n n\alpha_n$.

Lemma Q.3 (composition of DP germs)**.** *Let f map an open subset of P into W, DP at x_0 with radius R, bound $\mathfrak{b} > 0$ and coefficient laws a_n. Let g map an open subset of W into Y, DP at $w_0 = f(x_0)$ with radius R', bound $\mathfrak{b}'$ and coefficient laws d_m. Put*

$$\Xi_{(\alpha)}(h) = d^{[\alpha]}_{|\alpha|}(a_1(h), a_2(h), \dots),$$

computed with $a_1(h),\dots,a_{\overline{n}}(h)$ for any $\overline{n}$ at least the largest index in the support of α, and $\Xi_N = \sum_{\deg\alpha = N}\Xi_{(\alpha)}$.

1. *$\Xi_{(\alpha)}$ is a bounded polynomial law of degree $\deg\alpha$, and $\sum_{\deg\alpha=N}\|\Xi_{(\alpha)}\|$ is at most the coefficient of θ^N in $\bar{d}^{\sharp}(\bar{a}_{\geq 1}(\theta))$.*
2. *Let $R_1 = \min(R/2, RR'/(8C_0\mathfrak{b}))$. For $\|h\| < R_1$ the family $(\Xi_{(\alpha)}(h))_{\alpha}$ is absolutely summable in $\widehat{Y}$, and $g(f(x_0+h)) = \sum_N \Xi_N(h)$ in Y. Moreover $\|\Xi_N\| \leq 2\mathfrak{b}' R_1^{-N}$, so $g\circ f$ is DP at x_0 with radius R_1 and bound $2\mathfrak{b}'$.*

Proof. Step 1 ($\Xi_{(\alpha)}$ is a law). On the span of finitely many vectors of P the a_n are polynomials whose coefficients span a finite-dimensional subspace of W. On that subspace $d^{[\alpha]}_{|\alpha|}$ is a polynomial in the coordinates of its arguments, with coefficients in Y. Also $\Xi_{(\alpha)}(\lambda h) = \lambda^{\deg\alpha}\Xi_{(\alpha)}(h)$, and K is infinite, so $\Xi_{(\alpha)}$ is homogeneous of degree $\deg\alpha$ on that span.

Step 2 (the bound). By (Q.4) with $m = |\alpha|$,

$$\|\Xi_{(\alpha)}(h)\| \leq C_0^m\|d_m\|\frac{m!}{\prod_n \alpha_n!}\prod_n\big(\|a_n\|\|h\|^n\big)^{\alpha_n}.$$

By the multinomial theorem, the sum of $\frac{m!}{\prod_n \alpha_n!}\prod_n\|a_n\|^{\alpha_n}$ over the α with $|\alpha| = m$, $\deg\alpha = N$ is the coefficient of θ^N in $\bar{a}_{\geq 1}(\theta)^m$. Summing over m gives (1).

Step 3 (the sum). Let $0 \leq \theta \leq R_1$. As $\|a_n\| \leq \mathfrak{b}R^{-n}$ and $\theta/R \leq \frac{1}{2}$, $\bar{a}_{\geq 1}(\theta) \leq 2\mathfrak{b}\theta/R \leq R'/(4C_0)$. As $\|d_m\| \leq \mathfrak{b}'R'^{-m}$, this gives $\bar{d}^{\sharp}(\bar{a}_{\geq 1}(\theta)) \leq \mathfrak{b}'\sum_m 4^{-m} \leq 2\mathfrak{b}'$. By (1), $\sum_{\alpha}\|\Xi_{(\alpha)}\|\theta^{\deg\alpha} \leq 2\mathfrak{b}'$. So $\|\Xi_N\| \leq 2\mathfrak{b}'R_1^{-N}$, and the family is absolutely summable for $\|h\| < R_1$. Put $\tilde{w}_h = \sum_{n\geq 1} a_n(h) = f(x_0+h) - w_0$. Then $\|\tilde{w}_h\| \leq R'/(4C_0) < R'$, as $C_0 \geq 1$, so $g(f(x_0+h)) = \sum_m d_m(\tilde{w}_h)$. For each $\overline{n}$, $d_m(\sum_{n\leq\overline{n}} a_n(h))$ is the sum of the $\Xi_{(\alpha)}(h)$ with $|\alpha| = m$ and support in $\{1,\dots,\overline{n}\}$. As $\overline{n}\to\infty$ the left side tends to $d_m(\tilde{w}_h)$, by Lemma Q.2(2). So $d_m(\tilde{w}_h) = \sum_{|\alpha|=m}\Xi_{(\alpha)}(h)$ in $\widehat{Y}$. Summing over m, and regrouping the same family by degree, gives $g(f(x_0+h)) = \sum_N \Xi_N(h)$ in $\widehat{Y}$. The left side and the partial sums lie in Y, so the series converges in Y. Coefficient laws are unique, by Lemma A.1 on lines, so the Ξ_N are the coefficient laws of $g\circ f$. □

In what follows $t \in \Gamma^N(P)$ is read in $\Gamma^{n_1}(P) \otimes \cdots \otimes \Gamma^{n_r}(P)$, as in Definition 14.21.

Lemma Q.4 (the laws of a composite)**.** *In Lemma Q.3 let $N \geq 1$. For $t \in \Gamma^N(P)$,*

$$L_{\Xi_N}(t) = \sum_{r=1}^{N} L_{d_r}\big(\mathsf{Q}_{r,N}(t)\big), \qquad \mathsf{Q}_{r,N}(t) = \sum_{\vec{n}\in\mathcal{C}(N,r)} \big(L_{a_{n_1}} \otimes \cdots \otimes L_{a_{n_r}}\big)(t) \in \Gamma^r(W). \tag{Q.8}$$

Proof. Step 1 (mixed coefficients). Let d be a law of degree r on W. Then $d(\sum_i \lambda_i w_i) = L_d((\sum_i \lambda_i w_i)^{\otimes r}) = \sum_{|\alpha|=r} \lambda^\alpha L_d(O_\alpha(w))$ by (14.1). Comparing coefficients gives $d^{[\alpha]}(w) = L_d(O_\alpha(w))$.

Step 2 (regrouping). Fix h and r. The tensor $O_\alpha(w)$ is the sum of the tensors $w_{i_1} \otimes \cdots \otimes w_{i_r}$ over the words $(i_1, \dots, i_r)$ in which each i occurs α_i times. So the sum of $O_\alpha(a_1(h), a_2(h), \dots)$ over the α with $|\alpha| = r$, $\deg \alpha = N$ is the sum of $a_{n_1}(h) \otimes \cdots \otimes a_{n_r}(h)$ over $\vec{n} \in \mathcal{C}(N, r)$, which is $\mathsf{Q}_{r,N}(h^{\otimes N})$. It lies in $\Gamma^r(W)$, as each O_α does. By Step 1, $\Xi_N(h) = \sum_{r=1}^N L_{d_r}(\mathsf{Q}_{r,N}(h^{\otimes N}))$.

Step 3 (linearity). Both sides of (Q.8) are linear in t and agree on the powers, which span $\Gamma^N(P)$. So also $\mathsf{Q}_{r,N}(t) \in \Gamma^r(W)$ for all t. □

Lemma Q.5 (density and ℓ^1-sums)**.**

(1) *Let K be complete, and let $\mathbf{X}_1, \dots, \mathbf{X}_r$ be seminormed spaces with separated completions $\widehat{\mathbf{X}}_i$. The projective seminorm of $t \in \mathbf{X}_1 \otimes \cdots \otimes \mathbf{X}_r$ equals that of its image in $\widehat{\mathbf{X}}_1 \otimes \cdots \otimes \widehat{\mathbf{X}}_r$. Hence $\widehat{\pi}_{\vec{n}} = \pi_{\vec{n}}$ and $\widehat{\mathfrak{k}}(P) = \mathfrak{k}(P)$ for K complete.*

(2) *For $i = 1, \dots, r$ let $(B_l^{(i)})_{l\in\mathcal{J}_i}$ be Banach spaces, $\mathsf{c}_l^{(i)} > 0$ weights, and $\mathbf{X}_i$ the completed ℓ^1-sum with norm $\sum_l \mathsf{c}_l^{(i)} \|x_l\|$. For a finite sum $t = \sum_{\vec{l}} t_{\vec{l}}$ with $t_{\vec{l}} \in B_{l_1}^{(1)} \otimes \cdots \otimes B_{l_r}^{(r)}$, the projective seminorm of t in $\mathbf{X}_1 \otimes \cdots \otimes \mathbf{X}_r$ is the sum over $\vec{l}$ of $\mathsf{c}_{l_1}^{(1)} \cdots \mathsf{c}_{l_r}^{(r)}$ times the projective seminorm of $t_{\vec{l}}$ in $B_{l_1}^{(1)} \otimes \cdots \otimes B_{l_r}^{(r)}$.*

Proof. Step 1 (reduction to norms in (1)). Let $\mathbf{X}$ be seminormed, with quotient map sq onto its separated quotient. The projective seminorm of $t \in \mathbf{X} \otimes \mathbf{X}'$ equals that of $(\mathrm{sq} \otimes 1)(t)$. Representations map to representations. A representation of the image lifts, with the same cost, to one of t up to an element of $\ker \mathrm{sq} \otimes \mathbf{X}'$, whose projective seminorm is 0. So we may assume the $\mathbf{X}_i$ normed and dense in $\widehat{\mathbf{X}}_i$.

Step 2 (item (1)). A representation in the $\mathbf{X}_i$ is one in the $\widehat{\mathbf{X}}_i$. Conversely, let $t = \sum_q x_{q,1} \otimes \cdots \otimes x_{q,r}$ with $x_{q,i} \in \widehat{\mathbf{X}}_i$, and let $\delta > 0$. Fix a representation of t in the $\mathbf{X}_i$, and let $\mathsf{F}_i \subseteq \widehat{\mathbf{X}}_i$ be the span of all factors in both representations. Then $t \in (\bigotimes_i \mathsf{F}_i) \cap (\bigotimes_i \mathbf{X}_i) = \bigotimes_i (\mathsf{F}_i \cap \mathbf{X}_i)$. Write $\mathsf{F}_i = (\mathsf{F}_i \cap \mathbf{X}_i) \oplus \mathsf{F}_i'$ with a basis $\upsilon_1, \dots, \upsilon_d$ of F_i'. As K is complete, the coordinate forms $\eta_1, \dots, \eta_d$ of this decomposition are bounded by Lemma A.3, say by $\bar{c} > 0$. Choose $x_l' \in \mathbf{X}_i$ with $\|x_l' - \upsilon_l\| \leq \delta/(d\bar{c})$. For $y = x + \sum_l \eta_l(y)\upsilon_l$ with $x \in \mathsf{F}_i \cap \mathbf{X}_i$ put $\mathsf{Y}_i y = x + \sum_l \eta_l(y) x_l'$. Then $\mathsf{Y}_i \colon \mathsf{F}_i \to \mathbf{X}_i$ is linear, is the identity on $\mathsf{F}_i \cap \mathbf{X}_i$, and $\|\mathsf{Y}_i y - y\| \leq \delta \|y\|$. So $t = (\mathsf{Y}_1 \otimes \cdots \otimes \mathsf{Y}_r)(t) = \sum_q \mathsf{Y}_1 x_{q,1} \otimes \cdots \otimes \mathsf{Y}_r x_{q,r}$, of cost at most $(1+\delta)^r \sum_q \prod_i \|x_{q,i}\|$. The clause on $\widehat{\pi}_{\vec{n}}$ is the case $\mathbf{X}_i = \Gamma_\pi^{n_i}(P)$.

Step 3 (item (2)). Concatenating representations of the $t_{\vec{l}}$ gives $\leq$, as an element of $B_l^{(i)}$ has norm $\mathsf{c}_l^{(i)} \|\cdot\|$ in $\mathbf{X}_i$. Let $t = \sum_q x_{q,1} \otimes \cdots \otimes x_{q,r}$ in the $\mathbf{X}_i$, with components $x_{q,i,l} \in B_l^{(i)}$. Then

$$\sum_q \prod_i \|x_{q,i}\| = \sum_{\vec{l}} \prod_i \mathsf{c}_{l_i}^{(i)} \sum_q \prod_i \|x_{q,i,l_i}\|,$$

and $\sum_q x_{q,1,l_1} \otimes \cdots \otimes x_{q,r,l_r} = t_{\vec{l}}$, by applying the coordinate projections. This gives $\geq$. □

Recall that $\widehat{\Gamma}^n_\pi P$ is the closed span of the powers in $\widehat{\otimes}^n_\pi P$. As $\Gamma^n_\pi(P)$ carries the restriction of π, the space $\widehat{\Gamma}^n_\pi P$ is also the separated completion of $\Gamma^n_\pi(P)$. Let $\mathcal{T}_1(P)$ be the completed ℓ^1-sum of the spaces $\widehat{\otimes}^n_\pi P$, $n \geq 0$, and $\mathcal{A}_1(P)$ the closure in it of the image of $\bigoplus_n \Gamma^n(P)$, which is the ℓ^1-sum of the spaces $\widehat{\Gamma}^n_\pi P$. Let ι_n be the inclusion of the n-th summand.

Proof of Theorem 14.22. Step 1 ((iii) implies (i)). By definition $\mathfrak{k}_{\vec{n}}(P) \leq \mathfrak{k}(P)^N$ for $\vec{n} \in \mathcal{C}(N,r)$. By Proposition 14.9(1) there are radii R, R' and bounds $\mathfrak{b}, \mathfrak{b}' > 0$ with $\mathrm{amb}(a_n) \leq \mathfrak{b} R^{-n}$ and $\mathrm{amb}(d_r) \leq \mathfrak{b}' R'^{-r}$, with the laws of Lemma Q.3. For $\vec{n} \in \mathcal{C}(N,r)$ and $t' \in \Gamma^{n_1}(P) \otimes \cdots \otimes \Gamma^{n_r}(P)$, applying $L_{a_{n_1}} \otimes \cdots \otimes L_{a_{n_r}}$ to a representation of t' as in the definition of $\pi_{\vec{n}}$ gives

$$\pi_{W^{\otimes r}}\big((L_{a_{n_1}} \otimes \cdots \otimes L_{a_{n_r}})(t')\big) \leq \prod_i \mathrm{amb}(a_{n_i})\, \pi_{\vec{n}}(t'). \tag{Q.9}$$

With (Q.8), $|\mathcal{C}(N,r)| = \binom{N-1}{r-1}$, and $\|L_{d_r}(s)\| \leq \mathrm{amb}(d_r)\pi_{W^{\otimes r}}(s)$ on $\Gamma^r(W)$, this gives for $t \in \Gamma^N(P)$

$$\|L_{\Xi_N}(t)\| \leq \sum_{r=1}^{N} \mathfrak{b}' R'^{-r} \binom{N-1}{r-1} \mathfrak{b}^r R^{-N} \mathfrak{k}(P)^N \pi(t) = \mathfrak{b}' \frac{\mathfrak{b}}{R'} \Big(1 + \frac{\mathfrak{b}}{R'}\Big)^{N-1} \Big(\frac{\mathfrak{k}(P)}{R}\Big)^N \pi(t).$$

Also $\Xi_0 = d_0$ is constant, with $\mathrm{amb}(\Xi_0) \leq \mathfrak{b}'$. So $\mathrm{amb}(\Xi_N)$ is at most geometric in N. By Lemma Q.3, $g \circ f$ is DP at x_0 with these laws, and Proposition 14.9(1) gives AA at x_0, with a smaller radius if needed.

Step 2 ((i) implies (ii)). An analytic map is AA, with $j = \mathrm{id}$.

Step 3 ((ii) implies (iii′): the universal germs). Let $W = \mathcal{A}_1(P)$, a Banach space, and $f(h) = \sum_{n \geq 1} \iota_n(h^{\otimes n})$ for $\|h\| < 1$. Then f is AA at 0 with radius 1 and bound 1: the inclusion $\mathcal{A}_1(P) \subseteq \mathcal{T}_1(P)$ is an isometry, the n-linear maps $\iota_n(h_1 \otimes \cdots \otimes h_n)$ into $\mathcal{T}_1(P)$ have norm at most 1, and their diagonals lie in W. Its laws are $a_n(h) = \iota_n(h^{\otimes n})$, and L_{a_n} is the map $\Gamma^n(P) \to \widehat{\Gamma}^n_\pi P$ followed by ι_n. Let Y be the completed ℓ^1-sum of the spaces $\widehat{\otimes}^r_\pi W$, $r \geq 1$, with inclusions ι^Y_r, and $g(w) = \sum_{r \geq 1} \iota^Y_r(w^{\otimes r})$ for $\|w\| < 1$. This g is analytic at 0, with the r-linear maps $\iota^Y_r(w_1 \otimes \cdots \otimes w_r)$ of norm at most 1, and $L_{d_r} = \iota^Y_r$ on $\Gamma^r(W)$. The composite $g \circ f$ is defined for $\|h\| < \frac{1}{2}$.

Step 4 ((ii) implies (iii′): the estimate). By (ii) and Proposition 14.9(1) there are $R_\Xi, \mathfrak{b}_\Xi > 0$ with $\mathrm{amb}(\Xi_N) \leq \mathfrak{b}_\Xi R_\Xi^{-N}$. By (Q.8), $L_{\Xi_N}(t) = \sum_r \iota^Y_r \mathsf{Q}_{r,N}(t)$. Here $(L_{a_{n_1}} \otimes \cdots \otimes L_{a_{n_r}})(t)$ is the image of t in the block $\widehat{\Gamma}^{n_1}_\pi P \otimes \cdots \otimes \widehat{\Gamma}^{n_r}_\pi P$ of $W^{\otimes r}$. Lemma Q.5(2), with weights 1, gives

$$\|L_{\Xi_N}(t)\|_Y = \sum_{r=1}^{N} \sum_{\vec{n} \in \mathcal{C}(N,r)} \widehat{\pi}_{\vec{n}}(t) \;\geq\; \widehat{\pi}_{\vec{n}}(t) \qquad \text{for each } \vec{n}.$$

So $\widehat{\mathfrak{k}}_{\vec{n}}(P) \leq \mathfrak{b}_\Xi R_\Xi^{-N}$ for all $\vec{n}$, and $\widehat{\mathfrak{k}}(P) \leq \max(\mathfrak{b}_\Xi, 1)/R_\Xi$.

Step 5 (the equivalences). For K complete, (iii′) and (iii) are equivalent by Lemma Q.5(1). Over every K, let W be Banach and assume (iii′). Each L_{a_n} is bounded for π with values in a complete space, so it extends to $\widehat{\Gamma}^n_\pi P$ with the same bound. Then (Q.9) holds for $t' = t \in \Gamma^N(P)$ with $\widehat{\pi}_{\vec{n}}$ in place of $\pi_{\vec{n}}$, and Step 1 runs with $\widehat{\mathfrak{k}}$ in place of $\mathfrak{k}$. So (iii′) implies (i) with Banach W. This implies (ii) with Banach W, Y, which implies (iii′) by Steps 3 and 4.

Step 6 (one infinite constant). Let $\mathfrak{k}_{\vec{n}}(P) = \infty$ for $\vec{n} \in \mathcal{C}(N,r)$, with $W_{\vec{n}}, Y_{\vec{n}}, f_{\vec{n}}, g_{\vec{n}}$ as in the statement. Let $Z_{\vec{n}} = P^{\otimes n_1} \oplus \cdots \oplus P^{\otimes n_r}$ with the seminorms π and the maximum norm. The inclusion $W_{\vec{n}} \to Z_{\vec{n}}$ is isometric. If π is not a norm, pass to the separated quotients of $W_{\vec{n}}, Y_{\vec{n}}, Z_{\vec{n}}$: the seminorm of $W_{\vec{n}}$ is the restriction of that of $Z_{\vec{n}}$, so the map of quotients is well defined and isometric. In $Z_{\vec{n}}$, $f_{\vec{n}}(x+h)$ is a finite sum of diagonals of bounded multilinear

maps in h, and these diagonals are the coefficients of the polynomial $\lambda \mapsto f_{\vec{n}}(x+\lambda h)$ with coefficients in $W_{\vec{n}}$. So $f_{\vec{n}}$ is AA at every point. The map $g_{\vec{n}}$ is the diagonal of the r-linear map $(s^1,\dots,s^r) \mapsto s^1_1 \otimes \cdots \otimes s^r_r$, which takes the i-th component of the i-th argument and has norm at most 1. The composite $h \mapsto h^{\otimes N}$ into $Y_{\vec{n}}$ is a bounded law of degree N whose linearization is the inclusion of $\Gamma^N(P)$, followed by the quotient map if there is one. The norm of $Y_{\vec{n}}$ at the image of t is $\pi_{\vec{n}}(t)$, so, with $c/0 = \infty$, its ambient constant is $\mathfrak{k}_{\vec{n}}(P) = \infty$. By Lemma Q.1(3) it is not ambient of degree N, and by Proposition 14.9(2) it is AA at no point. □

Q.5. **The failure.** We use the following notation. For a field $\Bbbk$ and integers $m, n \geq 1$, read a tensor $t \in (\Bbbk^m)^{\otimes n}$ as the function $t(i_1,\dots,i_n)$ on $[m]^n$, where $[m] = \{1,\dots,m\}$, and for linear forms $\eta_1,\dots,\eta_n$ on $\Bbbk^m$ let $t(\eta_1,\dots,\eta_n)$ be the value of $\eta_1 \otimes \cdots \otimes \eta_n$ at t. Let $\Gamma^n(\Bbbk^m)$ be the space of symmetric tensors, also for finite $\Bbbk$, where it can be larger than the span of the powers, let $\operatorname{diag}(t) \in \Bbbk^m$ have the entries $t(i,\dots,i)$, and let the *flattening rank* $\operatorname{frk}(t)$ be the rank of the $m \times m^{n-1}$ matrix of the entries $t(i_1,\dots,i_n)$, with rows indexed by i_1; it is subadditive and does not change under extension of scalars. A *projection onto the symmetric tensors* is a linear map Ψ of $(\Bbbk^m)^{\otimes n}$ into $\Gamma^n(\Bbbk^m)$ with $\Psi(t) = t$ for $t \in \Gamma^n(\Bbbk^m)$. Symmetric tensors with entries in a commutative $\Bbbk$-algebra are combinations of symmetric tensors over $\Bbbk$, so the linear extension of Ψ to such entries maps tensors with these entries to symmetric ones and fixes the symmetric ones.

Lemma Q.6 (projections onto symmetric tensors)**.** *Let $m \geq 1$.*

1. *Let $\Bbbk$ be a field, $\mathsf{z} \geq 1$, $x = (x_1,\dots,x_m)^\top$ a column of indeterminates, and $\mathsf{M}(x)$ an $m \times m$ matrix over $\Bbbk[x_1,\dots,x_m]$ with $\mathsf{M}(x)x = (x_1^{\mathsf{z}},\dots,x_m^{\mathsf{z}})^\top$. Then $\det \mathsf{M}(x)$ has the coefficient 1 at $x_1^{\mathsf{z}-1}\cdots x_m^{\mathsf{z}-1}$.*
2. *Let $\Bbbk$ have characteristic $p > 0$, let $r_{\max} \geq 0$ be an integer with $|\Bbbk| \geq 2r_{\max}+2$, and let $\mathcal{X} \subseteq \Gamma^p(\Bbbk^m)$ be a linear space whose elements have flattening rank at most $r_{\max}$. Then $\dim \operatorname{diag}(\mathcal{X}) \leq r_{\max}$.*
3. *Let n be a function from an abelian group into $[0,\infty)$ with $\mathsf{n}(\mathsf{a}+\mathsf{a}') \leq \mathsf{n}(\mathsf{a}) + \mathsf{n}(\mathsf{a}')$, and let G be a map from $(\mathbb{F}_p^m)^p$ into this group that is additive in each argument. Then the average of $\mathsf{n} \circ \mathsf{G}$ over $(\mathbb{F}_p^m)^p$ is at least 2^{-p} times its value at every point.*
4. *Let $\Bbbk$ be a field of characteristic $p > 0$, and let Ψ be a projection of $(\Bbbk^m)^{\otimes p}$ onto $\Gamma^p(\Bbbk^m)$. Then the average of $\operatorname{frk}\Psi(y^1 \otimes \cdots \otimes y^p)$ over $(y^1,\dots,y^p) \in (\mathbb{F}_p^m)^p$ is at least $m/(2^p \mathsf{d}_m)$, where $\mathsf{d}_m = 1 + \lfloor \log_p m \rfloor$.*

Proof. Step 1 (item (1)). Let D be the diagonal matrix with the entries $x_1^{\mathsf{z}-1},\dots,x_m^{\mathsf{z}-1}$, and put $\mathsf{N} = \mathsf{M} - \mathsf{D}$, so $\mathsf{N}x = 0$. As the determinant is linear in each row, $\det \mathsf{M} = \sum_{S \subseteq [m]} \det \mathsf{M}_S$, where M_S has the rows of N in the positions in S and those of D elsewhere, and $\det \mathsf{M}_\emptyset = x_1^{\mathsf{z}-1}\cdots x_m^{\mathsf{z}-1}$. Let $S \neq \emptyset$, and let J_S be the ideal generated by the x_i^{z}, $i \notin S$. The vector $\mathsf{M}_S x$ has the entries 0 in the positions in S and x_i^{z} elsewhere, so multiplication by the adjugate of M_S gives $\det(\mathsf{M}_S)x_j \in J_S$ for every j. For $j \in S$, x_j is a nonzerodivisor modulo J_S, which is generated by monomials in the other variables; so $\det \mathsf{M}_S \in J_S$. Every monomial of an element of J_S is divisible by some x_i^{z}, so $x_1^{\mathsf{z}-1}\cdots x_m^{\mathsf{z}-1}$ does not occur in $\det \mathsf{M}_S$.

Step 2 (item (2): a common kernel). For $t \in \Gamma^p(\Bbbk^m)$ and a linear form η on $\Bbbk^m$ let $t^\flat(\eta) \in (\Bbbk^m)^{\otimes(p-1)}$ have the entries $\sum_i \eta(e_i)\, t(i,i_2,\dots,i_p)$, e_i the unit vectors; the linear map $t^\flat$ has rank $\operatorname{frk}(t)$, and $t^\flat(\eta)(\eta,\dots,\eta) = t(\eta,\dots,\eta)$. Let $t_0 \in \mathcal{X}$ have the largest flattening rank $r_0 \leq r_{\max}$. If $r_0 = 0$, then $\mathcal{X} = 0$ and (2) holds, so let $r_0 \geq 1$. Let $\mathcal{K}_0$ be the kernel of $t_0^\flat$, of codimension r_0, with a complement $\mathcal{K}_0'$, and let $\mathcal{I}_0$ be the image of $t_0^\flat$, with a complement $\mathcal{I}_0'$. For $t \in \mathcal{X}$ write $t^\flat$ in blocks $\mathsf{B}_{11}(t)\colon \mathcal{K}_0' \to \mathcal{I}_0$, $\mathsf{B}_{12}(t)\colon \mathcal{K}_0 \to \mathcal{I}_0$, $\mathsf{B}_{21}(t)\colon \mathcal{K}_0' \to \mathcal{I}_0'$ and $\mathsf{B}_{22}(t)\colon \mathcal{K}_0 \to \mathcal{I}_0'$; then $\mathsf{B}_{11}(t_0)$ is invertible and the other blocks of $t_0^\flat$ vanish. Put

$\mathsf{B}_\tau = \mathsf{B}_{11}(t_0) + \tau \mathsf{B}_{11}(t)$ for $\tau \in \Bbbk$. If $\tau \neq 0$ and $\det \mathsf{B}_\tau \neq 0$, the Schur complement of B_τ in $(t_0 + \tau t)^\flat$ vanishes, since $\operatorname{frk}(t_0 + \tau t) \leq r_0$; that is,

$$\det(\mathsf{B}_\tau)\, \mathsf{B}_{22}(t) = \tau\, \mathsf{B}_{21}(t) \operatorname{adj}(\mathsf{B}_\tau) \mathsf{B}_{12}(t). \tag{Q.10}$$

Both sides are polynomials in τ of degree at most r_0, and $\det \mathsf{B}_\tau$ has at most r_0 roots, since it is nonzero at $\tau = 0$. So (Q.10) holds for all τ outside a set of at most $r_0 + 1$ elements, hence, as $|\Bbbk| \geq 2r_0 + 2$, for at least $r_0 + 1$ values of τ, and so identically. At $\tau = 0$, (Q.10) gives $\mathsf{B}_{22}(t) = 0$: $t^\flat(\eta) \in \mathcal{I}_0$ for $\eta \in \mathcal{K}_0$.

Step 3 (item (2): diagonal forms). The elements of $\mathcal{I}_0$ vanish when one of their arguments lies in $\mathcal{K}_0$, as t_0 is symmetric. For $t \in \mathcal{X}$ put $\operatorname{dg}_t(\eta) = t(\eta, \dots, \eta)$. So $\operatorname{dg}_t(\eta) = t^\flat(\eta)(\eta, \dots, \eta)$ vanishes for $\eta \in \mathcal{K}_0$. As t is symmetric and the binomial coefficients $\binom{p}{1}, \dots, \binom{p}{p-1}$ vanish in $\Bbbk$, dg_t is additive, and $\operatorname{dg}_t(\tau\eta) = \tau^p \operatorname{dg}_t(\eta)$ for $\tau \in \Bbbk$. Let $\eta'_1, \dots, \eta'_{r_0}$ be a basis of $\mathcal{K}'_0$, and write the coordinate forms as $e_i^* = \eta_i^\circ + \sum_j \tau_{ij} \eta'_j$ with $\eta_i^\circ \in \mathcal{K}_0$. Then $\operatorname{diag}(t)$ has the entries $\operatorname{dg}_t(e_i^*) = \sum_j \tau_{ij}^p \operatorname{dg}_t(\eta'_j)$, so it is the image of the vector $(\operatorname{dg}_t(\eta'_j))_j \in \Bbbk^{r_0}$, which depends linearly on t, under a matrix that does not depend on t. Hence $\dim \operatorname{diag}(\mathcal{X}) \leq r_0 \leq r_{\max}$.

Step 4 (item (3)). For a in a finite subgroup H of the group, the average of n over H is at least $\mathsf{n}(\mathsf{a})/2$, since $\mathsf{n}(\mathsf{a}') + \mathsf{n}(\mathsf{a} - \mathsf{a}') \geq \mathsf{n}(\mathsf{a})$ and $\mathsf{a}' \mapsto \mathsf{a} - \mathsf{a}'$ permutes H. For fixed other arguments, $y^j \mapsto \mathsf{G}(y^1, \dots, y^p)$ is additive on $\mathbb{F}_p^m$, so as y^j runs through $\mathbb{F}_p^m$ its value runs equally often through each element of its image, a finite subgroup. Hence the average of $\mathsf{n} \circ \mathsf{G}$ over y^j is at least half its value at any fixed y^j. Applying this in the slots $j = 1, \dots, p$ in turn gives (3).

Step 5 (item (4): an invertible diagonal). Let $x = (x_1, \dots, x_m)$ be a vector of indeterminates, extend Ψ linearly to entries in $\Bbbk[x_1, \dots, x_m]$, and let $\mathsf{M}(x)$ be the matrix over $\Bbbk[x_1, \dots, x_m]$ of the linear map $v \mapsto \operatorname{diag} \Psi(x^{\otimes(p-1)} \otimes v)$, whose columns are $\operatorname{diag} \Psi(x^{\otimes(p-1)} \otimes e_i)$. Its value at a point $\xi \in \Bbbk^m$ is the matrix of the same map with ξ in place of x. As Ψ fixes the symmetric tensor $x^{\otimes p}$, $\mathsf{M}(x)x = \operatorname{diag}(x^{\otimes p}) = (x_1^p, \dots, x_m^p)^\top$. By (1), the form $\det \mathsf{M}(x)$ of degree $m(p-1)$ has the coefficient 1 at $x_1^{p-1} \cdots x_m^{p-1}$. The sum $\sum_{\tau \in \mathbb{F}_p} \tau^i$ is -1 if $i > 0$ and $p - 1$ divides i, and 0 otherwise, and $x_1^{p-1} \cdots x_m^{p-1}$ is the only monomial of degree $m(p-1)$ whose exponents are positive multiples of $p - 1$; so $\sum_{\xi \in \mathbb{F}_p^m} \det \mathsf{M}(\xi) = (-1)^m$, and $\mathsf{M}(\xi_0)$ is invertible for some $\xi_0 \in \mathbb{F}_p^m$.

Step 6 (item (4): a value of large rank). Let $\Bbbk'' \supseteq \Bbbk$ be a field with $|\Bbbk''| \geq 2m$, with $\Bbbk'' = \Bbbk$ if $\Bbbk$ is infinite, and let Ψ'' be the linear extension of Ψ to entries in $\Bbbk''$. The tensors $\Psi''(\xi_0^{\otimes(p-1)} \otimes u)$, $u \in \Bbbk''^m$, form a linear space whose diagonals form $\mathsf{M}(\xi_0)\Bbbk''^m = \Bbbk''^m$, so by (2) with $r_{\max} = m - 1$ one of them has flattening rank m. Hence some $m \times m$ minor $\mathsf{R}(u)$ of the matrix of entries of $\Psi''(\xi_0^{\otimes(p-1)} \otimes u)$, a polynomial of degree at most m in each coordinate of u, is not the zero polynomial. Then d_m is the least integer with $p^{\mathsf{d}_m} > m$. As $|\Bbbk''| > m$, and $|\Bbbk''|$ is a power of p if $\Bbbk''$ is finite, $\Bbbk''$ contains an $\mathbb{F}_p$-subspace $\mathcal{W}_0$ of dimension d_m, so $|\mathcal{W}_0| = p^{\mathsf{d}_m} > m$. A nonzero polynomial in q variables of degree at most m in each variable does not vanish on all of $\mathcal{W}_0^{\mathsf{q}}$: for $\mathsf{q} = 1$ it has at most m roots, and for $\mathsf{q} > 1$ write it as $\sum_{c=0}^m \varphi_c u_{\mathsf{q}}^c$ with polynomials φ_c in the other variables; some φ_c is nonzero, by induction it does not vanish at some point of $\mathcal{W}_0^{\mathsf{q}-1}$, and there the polynomial is a nonzero polynomial in u_{q} of degree at most m. With $\mathsf{q} = m$ and the minor R, there is $v \in \mathcal{W}_0^m$ with $\operatorname{frk} \Psi''(\xi_0^{\otimes(p-1)} \otimes v) = m$. Write $v = \sum_{i=1}^{\mathsf{d}_m} \mathsf{o}_i v_i$ with an $\mathbb{F}_p$-basis $\mathsf{o}_1, \dots, \mathsf{o}_{\mathsf{d}_m}$ of $\mathcal{W}_0$ and $v_i \in \mathbb{F}_p^m$. Then $\Psi''(\xi_0^{\otimes(p-1)} \otimes v) = \sum_i \mathsf{o}_i \Psi(\xi_0^{\otimes(p-1)} \otimes v_i)$, and subadditivity gives $\operatorname{frk} \Psi(\xi_0^{\otimes(p-1)} \otimes v_{i_0}) \geq m/\mathsf{d}_m$ for some i_0.

Step 7 (item (4): averaging). The flattening rank is subadditive, and $(y^1, \dots, y^p) \mapsto \Psi(y^1 \otimes \cdots \otimes y^p)$ is additive in each argument, so by (3) the average in (4) is at least 2^{-p} times the flattening rank of the value at $(\xi_0, \dots, \xi_0, v_{i_0})$, hence at least $m/(2^p \mathsf{d}_m)$. □

The next lemma uses the data of Definition 14.15. Let e_l be the unit vectors of $\ell^1(I,K)$.

Lemma Q.7 (averaged local complementation)**.** *Let $X \neq 0$ be a normed space and $n \geq 1$.*

(1) *For a family y, the tensor $t(y)$ lies in $\Gamma^n(X) \otimes G(y)$, $\pi(t(y)) \leq \sum_l \|y_l\|_{\mathrm{e}}$, and $\mu(t(y)) = \inf_{\Psi \in \mathcal{P}(y)} \sum_l \pi(\Psi y_l)$.*

(2) *If $\mathfrak{l}_n(X) < \infty$, then for every normed G and $t \in \Gamma^n(X) \otimes G$: $\mu(t) \leq |\varpi|^{-1} \mathfrak{l}_n(X) \pi(t)$.*

Proof. Step 1 ($t(y)$). Let $(\bar v_j)$ be a basis of $\bar E$ with dual basis $(\bar v_j^*)$. In $\bar E \otimes \ell^1(I,K)$, $\sum_l \bar y_l \otimes e_l = \sum_j \bar v_j \otimes (\bar v_j^*(\bar y_l))_l$, which lies in $\bar E \otimes \mathcal{N}$. So $\sum_l \bar y_l \otimes \varepsilon_l = 0$ in $\bar E \otimes G(y)$, and $t(y)$ lies in the kernel of $E_0 \otimes G(y) \to \bar E \otimes G(y)$, which is $(E_0 \cap \Gamma^n(X)) \otimes G(y)$. The bound on π holds as $\|\varepsilon_l\| \leq 1$.

Step 2 ($\mu \leq$). Let $\Psi \in \mathcal{P}(y)$. By Step 1, $(\Psi \otimes 1)t(y) = t(y)$, so $t(y) = \sum_l \Psi(y_l) \otimes \varepsilon_l$, of cost at most $\sum_l \pi(\Psi y_l)$.

Step 3 ($\mu \geq$). Let $t(y) = \sum_q s_q \otimes w_q$ with $s_q \in \Gamma^n(X)$, $w_q \in G(y)$, and let $\delta > 0$. Choose $\hat w_q \in \ell^1(I,K)$ with image w_q and $\|\hat w_q\| \leq (1+\delta)\|w_q\|$, and put $s'_l = \sum_q \hat w_q(l) s_q \in \Gamma^n(X)$. By the triangle inequality for π,

$$\sum_l \pi(s'_l) \leq \sum_q \|\hat w_q\| \pi(s_q) \leq (1+\delta) \sum_q \pi(s_q)\|w_q\|.$$

Let E_1 be the span of E_0 and the s_q. The tensors $\sum_l s'_l \otimes e_l = \sum_q s_q \otimes \hat w_q$ and $\sum_l y_l \otimes e_l$ of $E_1 \otimes \ell^1(I,K)$ have the same image in $E_1 \otimes G(y)$, so their difference lies in $E_1 \otimes \mathcal{N}$. The vectors $(\bar v_j^*(\bar y_l))_l$ form a basis of $\mathcal{N}$: they span it by definition, and they are independent because the $\bar y_l$ span $\bar E$. So there are $u_j \in E_1$ with $s'_l - y_l = \sum_j \bar v_j^*(\bar y_l) u_j$ for all l. Put $\Psi(v) = v + \sum_j \bar v_j^*(\bar v) u_j$ for $v \in E_0$. Then $\Psi(y_l) = s'_l \in \Gamma^n(X)$; the y_l span E_0, so $\Psi(E_0) \subseteq \Gamma^n(X)$; and $\Psi = \mathrm{id}$ on $E_0 \cap \Gamma^n(X)$. So $\Psi \in \mathcal{P}(y)$ and $\sum_l \pi(\Psi y_l) \leq (1+\delta) \sum_q \pi(s_q)\|w_q\|$.

Step 4 (item (2)). We may assume $t \neq 0$. Let $\delta > 0$ and let $t = \sum_q y_q \otimes w_q$ with y_q elementary, $w_q \neq 0$, of cost $\sum_q \|y_q\|_{\mathrm{e}} \|w_q\| \leq \pi(t) + \delta$. Choose $\lambda_q \in K$ with $\|w_q\| \leq |\lambda_q| < |\varpi|^{-1}\|w_q\|$. The family $y'_q = \lambda_q y_q$ has the same span E_0 as the y_q. Choose $\Psi \in \mathcal{P}(y')$ with $\sum_q \pi(\Psi y'_q) \leq (\mathfrak{l}_n(X) + \delta) \sum_q \|y'_q\|_{\mathrm{e}}$. As $t \in (E_0 \otimes G) \cap (\Gamma^n(X) \otimes G) = (E_0 \cap \Gamma^n(X)) \otimes G$, we get $t = (\Psi \otimes 1)t = \sum_q \Psi(y'_q) \otimes \lambda_q^{-1} w_q$, and $\|\lambda_q^{-1} w_q\| \leq 1$. Hence $\mu(t) \leq \sum_q \pi(\Psi y'_q) \leq (\mathfrak{l}_n(X) + \delta)|\varpi|^{-1}(\pi(t) + \delta)$. □

Now let $\operatorname{char} K = p$. Elements of $\mathbb{F}_p^\times$ have absolute value 1, so $\{x \in K : |x| < 1\}$ is an $\mathbb{F}_p$-subspace of K not containing 1. Fix an $\mathbb{F}_p$-linear map $\mathrm{red}\colon K \to \mathbb{F}_p$ with $\mathrm{red}(1) = 1$ that vanishes on it. Then red fixes $\mathbb{F}_p$, and $\mathrm{red}(x) \neq 0$ implies $|x| \geq 1$. Apply red to tensors entrywise. Write $\mathrm{rk}_{\mathbb{F}_p}$ for the tensor rank over $\mathbb{F}_p$; it dominates the flattening rank. Let $\mathsf{At}_m = \mathbb{F}_p^m \setminus \{0\}$ and $\mathrm{qm}\colon \ell^1(\mathsf{At}_m, K) \to X_m$ the quotient map, which sends each unit vector e_{x} to x.

Lemma Q.8 (reduction to the prime field)**.** *For $t \in (K^m)^{\otimes p}$, $\mathrm{rk}_{\mathbb{F}_p}(\mathrm{red}(t)) \leq \pi(t)$, with π the projective norm of $X_m^{\otimes p}$.*

Proof. Step 1 (a lift). Let $t = \sum_q x_{q,1} \otimes \cdots \otimes x_{q,p}$ and $\delta > 0$. Choose $\hat x_{q,j}$ in $\ell^1(\mathsf{At}_m, K)$ with $\mathrm{qm}(\hat x_{q,j}) = x_{q,j}$ and $\|\hat x_{q,j}\| \leq (1+\delta)\|x_{q,j}\|$. Put $\tilde t = \sum_q \hat x_{q,1} \otimes \cdots \otimes \hat x_{q,p}$, a function on At_m^p. The ℓ^1-norm on At_m^p is multiplicative on elementary tensors, so

$$\sum_{\vec{\mathsf{x}} \in \mathsf{At}_m^p} |\tilde t(\vec{\mathsf{x}})| \leq (1+\delta)^p \sum_q \prod_j \|x_{q,j}\|, \qquad \mathrm{qm}^{\otimes p}(\tilde t) = t.$$

Step 2 (reduction). Let $t' = \mathrm{red}(\tilde t)$. Each entry of $\mathrm{qm}^{\otimes p}(\tilde t)$ is a combination of the $\tilde t(\vec{\mathsf{x}})$ with coefficients in $\mathbb{F}_p$, and red is $\mathbb{F}_p$-linear, so $\mathrm{qm}^{\otimes p}(t') = \mathrm{red}(t)$. If $t'(\vec{\mathsf{x}}) \neq 0$ then $|\tilde t(\vec{\mathsf{x}})| \geq 1$, so the

support of t' has at most $(1+\delta)^p \sum_q \prod_j \|x_{q,j}\|$ elements. And $\mathrm{qm}^{\otimes p}(t') = \sum_{\vec{\mathsf{x}}} t'(\vec{\mathsf{x}})\, \mathsf{x}_1 \otimes \cdots \otimes \mathsf{x}_p$. So $\mathrm{rk}_{\mathbb{F}_p}(\mathrm{red}(t))$ is at most that number. Let $\delta \to 0$ and take the infimum over representations. □

Lemma Q.9 (the atomic spaces)**.** *Let* $\mathrm{char}\, K = p$*, and let* $y^{(m)}$ *be the family of Definition 14.16. Then* $\mathfrak{l}(y^{(m)}) \geq m/(2^p \mathsf{d}_m)$*, in* X_m *and in* X^{at}*, with* d_m *as in Lemma Q.6. So* $\mathfrak{l}_p(X_m) \geq m/(2^p \mathsf{d}_m)$ *and* $\mathfrak{l}_p(X^{\mathrm{at}}) = \infty$*.*

Proof. Step 1 (projections over K give projections over $\mathbb{F}_p$). In X_m, $E_0 = (K^m)^{\otimes p}$, since I_m contains the tuples of unit vectors. Write $y = y^{(m)}$, and let $\Psi \in \mathcal{P}(y)$. Its values are symmetric tensors over K. Define $\Psi'(t) = \mathrm{red}(\Psi(t))$ for $t \in (\mathbb{F}_p^m)^{\otimes p}$. Then Ψ' is $\mathbb{F}_p$-linear, and its values are symmetric, as red acts entrywise. If t is symmetric, then $\Psi(t) = t$ and $\mathrm{red}(t) = t$. So Ψ' is a projection onto the symmetric tensors, with $\Bbbk = \mathbb{F}_p$.

Step 2 (ranks). By Lemma Q.8, $\pi(\Psi y_l) \geq \mathrm{rk}_{\mathbb{F}_p}(\mathrm{red}\, \Psi y_l) = \mathrm{rk}_{\mathbb{F}_p}(\Psi' y_l) \geq \mathrm{frk}(\Psi' y_l)$.

Step 3 (averaging). By Lemma Q.6(4) with $\Bbbk = \mathbb{F}_p$, $\sum_l \mathrm{frk}(\Psi' y_l) \geq |I_m| m/(2^p \mathsf{d}_m)$. Vectors of $\mathbb{F}_p^m$ have norm at most 1 in X_m, so $\sum_l \|y_l\|_{\mathrm{e}} \leq |I_m|$. This gives the bound on $\mathfrak{l}(y)$ in X_m.

Step 4 (passage to X^{at}). The space X_m sits isometrically in X^{at}, with a contractive coordinate projection pr_m. If $\tilde{\Psi}$ is in $\mathcal{P}(y)$ for X^{at}, then $\mathrm{pr}_m^{\otimes p} \circ \tilde{\Psi}$ is in $\mathcal{P}(y)$ for X_m, with values of no larger π. So the bound holds in X^{at}. It tends to infinity with m. □

Proof of Proposition 14.18. Let $(y^{(i)})_i$ be the families that define G. Write $t_i = t(y^{(i)})$ and $\Sigma_i = \sum_l \|y_l^{(i)}\|_{\mathrm{e}} > 0$. Let $Z_X = (X^{\otimes n}, \pi) \oplus G$, with the maximum norm.

Step 1 (f_X). The inclusion $W_X \subseteq Z_X$ is isometric. For $(x,w) \in H_X$ and $h = (u,v)$, $f_X((x,w)+h) = ((x+u)^{\otimes n}, w+v)$, and $(x+u)^{\otimes n} = \sum_{d=0}^n \mathsf{b}_d(u,\dots,u)$, where the d-linear map $\mathsf{b}_d(u_1,\dots,u_d)$ is the sum, over the d-element sets of positions, of the tensor with $u_1,\dots,u_d$ at these positions, in order, and x elsewhere. Into $(X^{\otimes n}, \pi)$, b_d has norm at most $\binom{n}{d}\|x\|^{n-d}$. So f_X has a finite ambient expansion in Z_X at every point, with every radius, the bound depending on the radius and the point. Its diagonals lie in W_X, as $\mathsf{b}_d(u,\dots,u) = O_{(n-d,d)}(x,u) \in \Gamma^n(X)$.

Step 2 (g_X). The map β_X of Definition 14.15 has $\mu(s \otimes w') \leq \pi(s)\|w'\|$, so it has norm at most 1, and its diagonal is g_X. Hence $g_X(\mathsf{u}+\mathsf{u}') = g_X(\mathsf{u}) + \beta_X(\mathsf{u},\mathsf{u}') + \beta_X(\mathsf{u}',\mathsf{u}) + g_X(\mathsf{u}')$ is a finite expansion, and g_X is analytic everywhere.

Step 3 (the composite is DP). The value of $g_X \circ f_X$ at (x,w) is $x^{\otimes n} \otimes w$. On the span of finitely many vectors this is a homogeneous polynomial of degree $n+1$ with coefficients in $\Gamma^n(X) \otimes G$, and $\mu(x^{\otimes n} \otimes w) \leq \|x\|^n \|w\|$. So it is a bounded law of degree $n+1$, of norm at most 1. It is DP everywhere: for $(x,w) \in H_X$ and $h = (u,v)$, its value at $(x,w)+h$ is

$$\sum_{d=0}^{n} \big(O_{(n-d,d)}(x,u) \otimes w + O_{(n-d,d)}(x,u) \otimes v \big),$$

a finite sum of bounded laws in h, of degrees d and $d+1$, since $\pi(O_{(n-d,d)}(x,u)) \leq \binom{n}{d}\|x\|^{n-d}\|u\|^d$.

Step 4 (the insertion map). Let $\mathrm{Ins}\colon X^{\otimes n} \otimes G \to H_X^{\otimes(n+1)}$ be the linear map sending $x_1 \otimes \cdots \otimes x_n \otimes w$ to the sum of the $n+1$ tensors obtained by inserting w into the positions $1,\dots,n+1$ of $x_1 \otimes \cdots \otimes x_n$, with X, G read in H_X by the isometric inclusions. Each insertion is a permutation of the factors, so $\pi_{H_X}(\mathrm{Ins}(t)) \leq (n+1)\pi(t)$. For $u \in X$, $w \in G$, $\mathrm{Ins}(u^{\otimes n} \otimes w) = O_{(n,1)}(u,w)$, the coefficient of $\lambda_1^n \lambda_2$ in $(\lambda_1 u + \lambda_2 w)^{\otimes(n+1)}$. So $\mathrm{Ins}(\Gamma^n(X) \otimes G) \subseteq \Gamma^{n+1}(H_X)$.

Step 5 (the size of t_i). The space $G(y^{(i)})$ sits isometrically in G with a contractive coordinate projection, so μ and π of t_i do not depend on which of the two spaces is used. By Lemma Q.7(1), $\pi(t_i) \leq \Sigma_i$ and $\mu(t_i) \geq \mathfrak{l}(y^{(i)})\Sigma_i$.

Step 6 (AA at no point). Suppose $g_X \circ f_X$ is AA at a point, as a map into Y_X or into its completion. In both cases Proposition 14.9(2) gives a linear isometry j of Y_X and a bounded $(n+1)$-linear map β on H_X with diagonal $j \circ g_X \circ f_X$. Let $\beta_\otimes$ be the linearization of β on $H_X^{\otimes(n+1)}$. For $u \in X$, $w \in G$, the value of $g_X \circ f_X$ at $(\lambda_1 u, \lambda_2 w)$ is $\lambda_1^n \lambda_2\, u^{\otimes n} \otimes w$. Comparing coefficients of $\lambda_1^n \lambda_2$ in $\beta_\otimes((\lambda_1 u + \lambda_2 w)^{\otimes(n+1)})$ gives $\beta_\otimes(\mathrm{Ins}(u^{\otimes n} \otimes w)) = j(u^{\otimes n} \otimes w)$. So $\beta_\otimes \circ \mathrm{Ins} = j$ on $\Gamma^n(X) \otimes G$. Hence $\mu(t) \le (n+1)\|\beta\|\pi(t)$ there. With $t = t_i$ and Step 5, $\mathfrak{l}(y^{(i)}) \le (n+1)\|\beta\|$ for all i, which is false.

Step 7 (the constant). Let $\mathrm{pr}_X, \mathrm{pr}_G$ be the coordinate projections of H_X. The map $\mathrm{pr}_X^{\otimes n}$ sends $\Gamma^n(H_X)$ into $\Gamma^n(X)$ and does not increase π, so $\mathrm{pr}_X^{\otimes n} \otimes \mathrm{pr}_G$ does not increase cost from $\Gamma^n_\pi(H_X) \otimes H_X$ to $\Gamma^n_\pi(X) \otimes G$. This map sends $\mathrm{Ins}(t_i) \in \Gamma^{n+1}(H_X)$ to t_i: only the insertion with w in the last position survives. So $\pi_{(n,1)}(\mathrm{Ins}(t_i)) \ge \mu(t_i) \ge \mathfrak{l}(y^{(i)})\Sigma_i$, while $\pi_{H_X}(\mathrm{Ins}(t_i)) \le (n+1)\Sigma_i$. Hence $\mathfrak{k}_{(n,1)}(H_X) \ge \mathfrak{l}(y^{(i)})/(n+1)$ for all i, and $\mathfrak{k}_{(n,1)}(H_X) = \infty$.

Step 8 (completions). Let $\widehat{W}_X, \widehat{Y}_X$ be the completions. A bounded K-bilinear map extends to the completions of normed K-spaces, over every K. So g_X extends to the diagonal $\hat{g}_X \colon \widehat{W}_X \to \widehat{Y}_X$ of a bounded bilinear map $\hat{\beta}_X$, and $\hat{g}_X$ is analytic everywhere. By Proposition 14.12(1)(a), f_X is AA into $\widehat{W}_X$, and $\hat{g}_X \circ f_X$ is $g_X \circ f_X$ into $\widehat{Y}_X$, whose norm restricts to μ. □

Proof of Theorem 14.17. Take $n = p \ge 2$, $X = X^{\mathrm{at}}$ and $G = G^{\mathrm{at}}$. By Definition 14.16 and Lemma Q.9, π is a norm on $(X^{\mathrm{at}})^{\otimes p}$, and G^{at} is a bad space, as $\sup_m \mathfrak{l}(y^{(m)}) = \infty$. So Proposition 14.18 gives every assertion on f_X, g_X and $g_X \circ f_X$, and $\mathfrak{k}_{(p,1)}(H^{\mathrm{at}}) = \infty$. Then $\mathfrak{a}_{p+1}(H^{\mathrm{at}}) = \infty$, as $\mathfrak{k}_{\vec{n}} \le \mathfrak{a}_N$ by Definition 14.21. □

Proof of Corollary 14.19. Let $\hat{\beta}_X$ be as in Step 8 of the proof of Proposition 14.18, with $n = p$. By that step, f_X is AA everywhere as a map into $\widehat{W}^{\mathrm{at}}$, and $\hat{g}_X \circ f_X$ is $g_X \circ f_X$ into $\widehat{Y}^{\mathrm{at}}$. By Theorem 14.17 this map is AA at no point. The permanence used below is Proposition 14.12(1).

Step 1 (Δ is AA). The middle component of Δ is a projection minus f_X precomposed with a projection, so it is AA. The third component is analytic, as $\hat{g}_X$ is a bounded bilinear diagonal precomposed with a projection and added to a projection. The component z_1 is linear. Pairing gives AA at every point.

Step 2 (the derivative). Let $z = (z_1, z_2, z_3)$ and $u = (u_1, u_2, u_3)$. By Steps 1, 2 and 8 of the proof of Proposition 14.18, each component of $\Delta(z+u) - \Delta(z)$ is a finite sum of terms homogeneous in the increment u, of a degree $d \ge 1$ and of norm at most a constant times $\|u\|^d$. So Δ is differentiable at z, its derivative is the term of degree 1, and $D\Delta(z) = \mathrm{id} - \mathsf{U}_\Delta(z)$, where

$$\mathsf{U}_\Delta(z)(u_1, u_2, u_3) = \big(0,\ Df_X(z_1)u_1,\ \hat{\beta}_X(z_2, u_2) + \hat{\beta}_X(u_2, z_2)\big).$$

Here $Df_X(z_1)$ is the bounded linear term of the expansion in Step 1 of that proof. So $\mathsf{U}_\Delta(z)$ is bounded with $\mathsf{U}_\Delta(z)^3 = 0$, and $D\Delta(z)^{-1} = \mathrm{id} + \mathsf{U}_\Delta(z) + \mathsf{U}_\Delta(z)^2$.

Step 3 (the inverse). Substitution in (14.6) gives $\Delta \circ \Delta' = \Delta' \circ \Delta = \mathrm{id}$. Sums and pairings of DP maps are DP, so Δ' is DP everywhere by Lemma Q.3.

Step 4 (Δ' is AA nowhere). Suppose Δ' is AA at $z^0 = (z_1^0, z_2^0, z_3^0)$. Precompose with the affine map $z_1 \mapsto (z_1, z_2^0, z_3^0)$ and postcompose with the projection onto $\widehat{Y}^{\mathrm{at}}$. This gives a map AA at z_1^0, with the value

$$z_3^0 + \hat{g}_X(z_2^0) + \hat{\beta}_X(z_2^0, f_X(z_1)) + \hat{\beta}_X(f_X(z_1), z_2^0) + \hat{g}_X(f_X(z_1)).$$

The two middle terms are bounded linear maps applied to f_X, so AA at z_1^0, and the constant is analytic. Taking differences, $g_X \circ f_X$ is AA at z_1^0, against Theorem 14.17. This proves item (1).

Step 5 (item (2): target charts). Δ'' has the inverse $(z_2, z_3) \mapsto (z_2, z_3 - \hat{g}_X(z_2))$, and both are analytic, as $\hat{g}_X$ is a bounded bilinear diagonal. The map $(f_X, 0)$ is AA everywhere, by pairing with the constant 0. The composite $\Delta'' \circ (f_X, 0)$ is $(f_X, g_X \circ f_X)$. If it were AA at a point, then so would be its second component, against Theorem 14.17.

Step 6 (item (3): charts). Both ϕ_1 and $\phi_1^{-1}(z) = (z_1, z_2 - f_X(z_1), z_3)$ are AA everywhere, by the argument of Step 1, so id and ϕ_1 are compatible. The transitions between ϕ_1 and $\psi \circ \phi_1$ are ψ and $\psi^{-1}(z) = (z_1, z_2, z_3 - \hat{g}_X(z_2))$, both analytic. The transition from id to $\psi \circ \phi_1$ is $\psi \circ \phi_1 = \Delta'$, which is AA at no point.

Step 7 (item (4): sections). For $w \in G^{\mathrm{at}}$ the extension in item (4) is linear in w. Hence $\mathsf{S}(x) = \mathrm{id} + \mathsf{U}_{\mathsf{S}}(x)$, where $\mathsf{U}_{\mathsf{S}}(x)(s, z_3) = (0, s \otimes x_G)$ is linear and bounded in x, and $\mathsf{S}(x)^{-1} = \mathrm{id} - \mathsf{U}_{\mathsf{S}}(x)$, as $\mathsf{U}_{\mathsf{S}}(x)^2 = 0$. So S and $x \mapsto \mathsf{S}(x)^{-1}$ are affine bounded maps from H^{at} to $\mathcal{L}(\mathsf{E}^{\mathrm{at}})$, hence analytic. The section $\mathsf{v}(x) = (x_X^{\otimes p}, 0)$ is AA everywhere: $x \mapsto x_X^{\otimes p}$ is the component of f_X in the completion of $\Gamma^p_\pi(X^{\mathrm{at}})$, and we pair with 0. Finally $\mathsf{S}(x)\mathsf{v}(x) = (x_X^{\otimes p}, (g_X \circ f_X)(x))$. If it were AA at a point, then so would be its second component, against Theorem 14.17. □

Q.6. **The mechanism.**

Proof of Theorem 14.24. Step 1 (the law). Put $a_{\beta_0}(x, w) = \beta_0(x^{\otimes n}, w)$ on $X \oplus G$, with the maximum norm. It is a bounded law of degree $n+1$: on a finite span it is a homogeneous polynomial of degree $n+1$, by (14.1), and $\|a_{\beta_0}(x, w)\| \le \|\beta_0\|\|x\|^n\|w\|$.

Step 2 ((d) implies (c)). Take $C > 0$ in (d). Let Z_Y be the separated quotient of $Y \oplus (X^{\otimes n} \otimes G)$, with the seminorm $(\mathsf{y}, t') \mapsto \|\mathsf{y}\| + C\pi(t')$, by the subspace of the $(L_{\beta_0} t, -t)$, $t \in \Gamma^n(X) \otimes G$. Put $j_Y(\mathsf{y}) = [\mathsf{y}, 0]$. By (d), $\|\mathsf{y} + L_{\beta_0} t\| + C\pi(t) \ge \|\mathsf{y}\|$, so j_Y is a linear isometry. For $\tau \in X^{\otimes n}$ put $\tilde{\beta}(\tau, w) = [0, \tau \otimes w]$; then $\|\tilde{\beta}(\tau, w)\| \le C\pi(\tau)\|w\|$. The relation with $t = s \otimes w$ gives $\tilde{\beta}(s, w) = [L_{\beta_0}(s \otimes w), 0] = j_Y \beta_0(s, w)$ for $s \in \Gamma^n(X)$.

Step 3 ((c) implies (b), and (b) implies (a)). Let $\tilde{\beta}$ be the bilinear map of (c). The map $(h^1, \dots, h^{n+1}) \mapsto \tilde{\beta}(\mathrm{pr}_X h^1 \otimes \cdots \otimes \mathrm{pr}_X h^n, \mathrm{pr}_G h^{n+1})$ is bounded and $(n+1)$-linear, with diagonal $j_Y \circ a_{\beta_0}$. Expanding it at $z_0 = (x_0, w_0)$ writes $j_Y a_{\beta_0}(z_0 + h) = \sum_{n' \le n+1} b_{n'}(h, \dots, h)$ with bounded n'-linear $b_{n'}$. For $h = (u, v)$, $b_{n'}(h, \dots, h)$ is j_Y of the coefficient of $\lambda^{n'}$ in $\beta_0((x_0 + \lambda u)^{\otimes n}, w_0 + \lambda v)$. That coefficient lies in Y, as the coefficients of $(x_0 + \lambda u)^{\otimes n}$ are orbit sums in $\Gamma^n(X)$. So this is an ambient expansion at z_0. Item (b) gives (a), as $X \oplus G$ is not empty.

Step 4 ((a) implies (d)). By Proposition 14.9(2) and Lemma Q.1(3), $\mathrm{amb}(a_{\beta_0}) < \infty$. Let $\mathrm{Ins}\colon X^{\otimes n} \otimes G \to (X \oplus G)^{\otimes(n+1)}$ be defined as in Step 4 of the proof of Proposition 14.18, which uses neither hypothesis on G or π: $\pi(\mathrm{Ins}\, t) \le (n+1)\pi(t)$, and $\mathrm{Ins}(u^{\otimes n} \otimes w) \in \Gamma^{n+1}(X \oplus G)$ is the coefficient of λ in $(u + \lambda w)^{\otimes(n+1)}$. So $L_{a_{\beta_0}}$ sends it to the coefficient of λ in $a_{\beta_0}(u + \lambda w) = \lambda\beta_0(u^{\otimes n}, w)$. Hence $L_{a_{\beta_0}} \circ \mathrm{Ins} = L_{\beta_0}$ on $\Gamma^n(X) \otimes G$, and $\|L_{\beta_0} t\| \le (n+1)\,\mathrm{amb}(a_{\beta_0})\pi(t)$.

Step 5 (all β_0 and G). Let $\mathfrak{l}_n(X) < \infty$. By (14.4), $\|L_{\beta_0} t\| \le \|\beta_0\|\mu(t)$, so Lemma Q.7(2) gives (d). Let $\mathfrak{l}_n(X) = \infty$. Take a bad space G, the ℓ^1-sum of the $G(y^{(i)})$ with $\sup_i \mathfrak{l}(y^{(i)}) = \infty$, let Y be the separated quotient of $(\Gamma^n(X) \otimes G, \mu)$, and let $\beta_0(s, w)$ be the class of $s \otimes w$. Then $\|L_{\beta_0} t\| = \mu(t)$. Step 5 of the proof of Proposition 14.18 does not use that π is a norm. It gives, for $t_i = t(y^{(i)})$ and $\Sigma_i = \sum_l \|y_l^{(i)}\|_{\mathrm{e}} > 0$, that $\mu(t_i) \ge \mathfrak{l}(y^{(i)})\Sigma_i$ and $\pi(t_i) \le \Sigma_i$, computed in $G(y^{(i)})$ or in G. So no C satisfies (d). □

Proof of Proposition 14.26. Step 1 (composition). Let Φ_1 be a morphism germ at w_0, with value w_0' there, and Φ_2 a morphism germ at w_0'. Write $\Phi_1(w_0 + h) = w_0' + \sum_{l \ge 1} b_l^1(h, \dots, h)$ and $\Phi_2(w_0' + v) = \sum_{r \ge 0} b_r^2(v, \dots, v)$, with $\|b_l^1\| \le \mathfrak{b}_{\Phi_1} R_{\Phi_1}^{-l}$ and $\|b_r^2\| \le \mathfrak{b}_{\Phi_2} R_{\Phi_2}^{-r}$. Substitution gives the multilinear coefficients $\sum_r \sum_{l_1 + \cdots + l_r = n} b_r^2 \circ (b_{l_1}^1 \times \cdots \times b_{l_r}^1)$. Their norms are bounded

by the coefficients of the real series $\mathfrak{b}_{\Phi_2}\sum_r\big(R_{\Phi_2}^{-1}\mathfrak{b}_{\Phi_1}(\theta/R_{\Phi_1})/(1-\theta/R_{\Phi_1})\big)^r$, of positive radius of convergence. For small h we have $\sum_l\|b_l^1(h,\dots,h)\|<R_{\Phi_2}$, and the family of the values of b_r^2 at the vectors $b_{l_c}^1(h,\dots,h)$ is absolutely summable in the complete final target. Grouped by r it has sum $\Phi_2(\Phi_1(w_0+h))$, and grouped by n it gives the coefficients above. So $\Phi_2\circ\Phi_1$ is analytic. This argument uses completeness of the final target only. The composite preserves the subspaces near w_0, and it sends w_0 to the value of Φ_2 at w_0'. Identity germs are morphisms, and restriction preserves composition, so Fg is a functor to pointed Banach spaces and germs of maps, the spaces being Banach as closed subspaces.

Step 2 (the image). For a morphism germ Φ, $j'\circ\operatorname{Fg}\Phi=\Phi\circ j$ is analytic, with diagonals in $j'(W')$ by Theorem 2.3. So $\operatorname{Fg}\Phi$ is AA. Conversely, let f be AA at x_0 from a Banach space P into a Banach space W, through $j\colon W\to Z$ with Z Banach. Then $j(W)$ is closed, and $j\circ f$ is a morphism germ from (P,P) to $(Z,j(W))$. Identifying W with $j(W)$, its image under Fg is f.

Step 3 (composites). $\operatorname{Fg}(Z,W)=W=\operatorname{Fg}(W,W)$. Let Φ_1 land in (Z,W) and let Φ_2 start at (W,W), with $Z\neq W$. These two pairs are different objects, so $\Phi_2\circ\Phi_1$ is not defined, while $\operatorname{Fg}\Phi_2\circ\operatorname{Fg}\Phi_1$ is. That composite can leave the image. Let K be complete of characteristic p, and take $\Phi_1=j\circ f_X$ and $\Phi_2=\hat g_X$ as in the paragraph after the proposition. By Theorem 14.17, $\hat g_X\circ f_X$ is ambient analytic at no point, so by Step 2 it is not Fg of a morphism germ.

Step 4 (directedness). Let W be Banach and let $j_1\colon W\to Z_1$, $j_2\colon W\to Z_2$ be linear isometries into Banach spaces. As W is Banach, the subspace of the $(j_1w,-j_2w)$ in the ℓ^1-sum of Z_1,Z_2 is complete, hence closed, so the quotient Z_3 is Banach. Put $i_1(z)=[z,0]$ and $i_2(z)=[0,z]$. Then $i_1j_1=i_2j_2$. The inequality $\|z+j_1w\|+\|j_2w\|\geq\|z\|$ makes i_1 isometric, and i_2 likewise. So $i_1\circ\Phi_1$ is a morphism germ into $(Z_3,i_1j_1(W))$ with the same image under Fg. □

Proof of Theorem 14.27. Step 1 (the derivative). Let $w_0'=\Phi(w_0)\in W'$. An analytic germ is strictly differentiable: if $\|b_n\|\leq\mathfrak{b}R^{-n}$, then for $\|h\|,\|h'\|\leq r<R$ the term of degree n changes by at most $n\|b_n\|r^{n-1}\|h-h'\|$, and the terms with $n\geq2$ sum to at most $\big(\sum_{n\geq2}n\mathfrak{b}R^{-n}r^{n-1}\big)\|h-h'\|$, whose coefficient tends to 0 with r. By the chain rule T_1 is bijective, with bounded inverse $D\Phi^{-1}(w_0')$.

Step 2 (only if). For $u\in W$, $\mathsf{T}_1u=\lim_{\lambda\to0}\lambda^{-1}(\Phi(w_0+\lambda u)-w_0')$ lies in the closed subspace W'. If Φ^{-1} is a morphism germ, likewise $\mathsf{T}_1^{-1}(W')\subseteq W$. So $\mathsf{T}_1(W)=W'$.

Step 3 (if). Let $\mathsf{T}_1(W)=W'$. The map $z\mapsto z-\mathsf{T}_1^{-1}\Phi(z)$ has strict derivative 0 at w_0. Choose $r>0$ such that, on the closed ball of radius r about w_0 in Z, this map is $\frac12$-Lipschitz and $\Phi^{-1}\circ\Phi=\mathrm{id}$, and Φ maps the part of the ball in W into W'. Let B be that part of the ball, a complete metric space. Let $w'\in W'$ with $\|\mathsf{T}_1^{-1}(w'-w_0')\|\leq r/2$, and $\mathsf{Ct}_{w'}(w)=w+\mathsf{T}_1^{-1}(w'-\Phi(w))$. Then $\mathsf{Ct}_{w'}$ preserves W on B, as $\mathsf{T}_1^{-1}(W')\subseteq W$. It is $\frac12$-Lipschitz with $\|\mathsf{Ct}_{w'}(w_0)-w_0\|\leq r/2$, so it maps B to B. Its fixed point $w\in B$ has $\Phi(w)=w'$, so $\Phi^{-1}(w')=w\in W$. So $\Phi^{-1}(W')\subseteq W$ near w_0', and Φ^{-1} is a morphism germ.

Step 4 (the map Δ). Let K be complete of characteristic p, let $z^0\in H_1$, let j,j' be as in the theorem, and suppose that Φ is an analytic germ from Z to Z' at $j(z^0)$ with an analytic inverse germ and $\Phi\circ j=j'\circ\Delta$ near z^0. Then Φ is a morphism germ from $(Z,j(H_1))$ to $(Z',j'(H_1))$, and $\mathsf{T}_1j=j'D\Delta(z^0)$, by the chain rule, as Δ is differentiable. As $D\Delta(z^0)$ is bijective on H_1 (Corollary 14.19(1)), $\mathsf{T}_1(j(H_1))=j'(H_1)$ holds automatically. By Step 3, $\Phi^{-1}(j'(z'))=j(z)$ for z' near $\Delta(z^0)$, with z near z^0. Then $j'\Delta(z)=\Phi(jz)=j'(z')$, so $z=\Delta'(z')$. Hence $j\circ\Delta'=\Phi^{-1}\circ j'$ is analytic near $\Delta(z^0)$, so Δ' is AA there, against Corollary 14.19(1). □

Proof of Corollary 14.28. Step 1 (groupoid). Identity germs are local automorphisms. Composites are, by Step 1 of the proof of Proposition 14.26 and the chain rule. Inverses are, by Theorem 14.27, with $D\Phi^{-1}(\Phi(w_0)) = D\Phi(w_0)^{-1}$.

Step 2 (invariance). Let Φ be a local automorphism at $f(x_0)$. If $j \circ f$ is analytic at x_0, then $j \circ (\Phi|_W \circ f) = \Phi \circ (j \circ f)$ is analytic at x_0, by Step 1 of the proof of Proposition 14.26. Let $g = \operatorname{Fg} \Phi_g$, and let Φ, Φ' be local automorphisms at the value and at the base point of g. By the same Step and Theorem 14.27, $\Phi \circ \Phi_g \circ \Phi'^{-1}$ is a morphism germ, and its restriction to W is $\Phi|_W \circ g \circ (\Phi'|_W)^{-1}$.

Step 3 (no single envelope). Let K be complete of characteristic p, let j be the inclusion of $\widehat{W}^{\mathrm{at}} \oplus \widehat{Y}^{\mathrm{at}}$ in Z, let $j \circ (f_X, 0)$ be analytic at x_0, and let $\Delta'' = \operatorname{Fg} \Phi_{\Delta''}$ near $(f_X, 0)(x_0)$, for a morphism germ $\Phi_{\Delta''}$ of the pair. Then $j \circ \Delta'' \circ (f_X, 0) = \Phi_{\Delta''} \circ (j \circ (f_X, 0))$ is analytic at x_0, by Step 1 of the proof of Proposition 14.26. So $\Delta'' \circ (f_X, 0)$ is AA at x_0, against Corollary 14.19(2). □

Q.7. **Envelopes.** Throughout this subsection K is complete, as in Section 14.5. Where an envelope occurs, $(\mathfrak{E}, j)$ is a functorial envelope, with the notation of Definition 14.30. For alternating ω on E and $v \in E^k$,

$$\omega \circ T_v^{\times k} = \omega(v_1, \dots, v_k) \det . \tag{Q.11}$$

The bounded linear maps below extend to completions, and we do not repeat this.

Lemma Q.10 (reduction to the homogeneous part)**.**

1. *For every E and small $\lambda \in K$, $\mathfrak{E}(\lambda \mathrm{id}_E) = \sum_n \lambda^n \mathfrak{q}_n^E$ with bounded $\mathfrak{q}_n^E$, natural in E, with $(\mathfrak{q}_n^E)^2 = \mathfrak{q}_n^E$, and with $\mathfrak{q}_n^E \mathfrak{q}_{n'}^E = 0$ for $n \neq n'$. Put $\mathfrak{E}_k(E) = \mathfrak{q}_k^E \mathfrak{E}(E)$.*
2. *$\mathfrak{E}_k$ is a subfunctor of $\mathfrak{E}$, $\mathfrak{E}_k(\lambda T) = \lambda^k \mathfrak{E}_k(T)$ for all $\lambda \in K$, and j_E maps into $\mathfrak{E}_k(E)$.*
3. *For all E, D there is a bounded k-linear $\beta_{E,D}^{\mathrm{lift}} \colon \mathcal{L}(E, D)^k \to \mathcal{L}(\mathfrak{E}_k(D), \mathfrak{E}_k(E))$ with $\mathfrak{E}_k(T) = \beta_{E,D}^{\mathrm{lift}}(T, \dots, T)$.*
4. *A map into $\mathrm{Alt}^k(E; K)$ is analytic after j_E into the completion of $\mathfrak{E}(E)$ if and only if it is analytic after j_E into the completion of $\mathfrak{E}_k(E)$.*
5. *Let M be a manifold as in Section 10.3, modeled on E, and let ω be a section of $\mathrm{Alt}^k(TM; K)$. If $j_E \circ \omega$ is analytic at a point into the completion of $\mathfrak{E}(E)$ in one chart around the point, then it is so in every chart around the point.*

Proof. Step 1 (item 1). Restrict the expansion of the hom map at 0 to the line $\lambda \mathrm{id}_E$. From $\mathfrak{E}(\lambda \mathrm{id}_E)\mathfrak{E}(\lambda' \mathrm{id}_E) = \mathfrak{E}(\lambda\lambda' \mathrm{id}_E)$ and Lemma A.1 in each variable, $(\mathfrak{q}_n^E)^2 = \mathfrak{q}_n^E$ and $\mathfrak{q}_n^E \mathfrak{q}_{n'}^E = 0$ for $n \neq n'$. From $\mathfrak{E}(T)\mathfrak{E}(\lambda \mathrm{id}_D) = \mathfrak{E}(\lambda T) = \mathfrak{E}(\lambda \mathrm{id}_E)\mathfrak{E}(T)$, $\mathfrak{E}(T)\mathfrak{q}_n^D = \mathfrak{q}_n^E \mathfrak{E}(T)$.

Step 2 (item 2). Naturality gives the subfunctor. On $\mathfrak{E}_k(E)$, $\mathfrak{E}(\lambda \mathrm{id}_E) = \lambda^k$ for small λ. For general λ take small $\lambda' \neq 0$ with $\lambda\lambda'$ small: $\mathfrak{E}(\lambda \mathrm{id}_E)(\lambda')^k = \mathfrak{E}(\lambda\lambda' \mathrm{id}_E) = (\lambda\lambda')^k$ on $\mathfrak{E}_k(E)$. Also $\mathfrak{E}(\lambda \mathrm{id}_E) j_E \omega = \lambda^k j_E \omega$, so $\mathfrak{q}_k^E j_E \omega = j_E \omega$ and $\mathfrak{q}_n^E j_E \omega = 0$ for $n \neq k$.

Step 3 (item 3). Let $T \mapsto \mathfrak{E}_k(T) = \mathfrak{q}_k^E \mathfrak{E}(T) \mathfrak{q}_k^D$ have the expansion $\sum_n b_n(T, \dots, T)$ near 0. Comparison with $\mathfrak{E}_k(\lambda T) = \lambda^k \mathfrak{E}_k(T)$ on lines gives $b_n(T, \dots, T) = 0$ for $n \neq k$ and $\mathfrak{E}_k(T) = b_k(T, \dots, T)$ for small T, hence by homogeneity for all T. Take $\beta_{E,D}^{\mathrm{lift}} = b_k$. This lift is a choice, and we do not use naturality of it.

Step 4 (item 4). $\mathfrak{q}_k^E$ is bounded with $\mathfrak{q}_k^E j_E = j_E$, and the inclusion of $\mathfrak{E}_k(E)$ is isometric.

Step 5 (item 5). Let ω^1, ω^2 be the expressions of ω in two charts around the point, with $\omega^1(y) = \omega^2(\psi(y)) \circ D\psi(y)^{\times k}$ for the change of chart ψ, as in Section 10.3. By the definition of an envelope and items 2 and 3,

$$j_E \omega^1(y) = \mathfrak{E}(D\psi(y))\, j_E \omega^2(\psi(y)) = \beta_{E,E}^{\mathrm{lift}}(D\psi(y), \dots, D\psi(y))\, j_E \omega^2(\psi(y)).$$

Let $j_E \circ \omega^2$ be analytic at $\psi(y_0)$ into the completion of $\mathfrak{E}(E)$. By item 4 it is analytic there into the completion of $\mathfrak{E}_k(E)$, and ψ, $D\psi$ are analytic at y_0. The tuple of the maps $D\psi$ and $j_E \circ \omega^2 \circ \psi$ is analytic at y_0, by Step 1 of the proof of Proposition 14.26, which uses completeness of the final target only. A bounded $(k+1)$-linear map has a finite expansion at every point. So the same Step makes $j_E \circ \omega^1$ analytic at y_0 into the completion of $\mathfrak{E}_k(E)$, and item 4 gives the claim. □

Proof of Theorem 14.31. Step 1 (χ_E). For $\lambda \in K^k$, $T_{(\lambda_c v_c)_c} = T_v \circ \mathsf{t}_\lambda$, so $\mathsf{p}\,\mathfrak{E}_k(T_{(\lambda_c v_c)_c}) = \lambda_1 \cdots \lambda_k \mathsf{p}\,\mathfrak{E}_k(T_v)$. By Lemma Q.10(3), $v \mapsto \mathsf{p}\,\mathfrak{E}_k(T_v)\xi$ is a polynomial map on E^k. It is homogeneous of degree 1 in each v_c, hence linear in each v_c. As $\|T_v\| \le \max_c \|v_c\|$, $\|\chi_E(\xi)(v)\| \le \|\mathsf{p}\|\|\beta^{\mathrm{lift}}_{\mathsf{L}_k,E}\| \max_c \|v_c\|^k \|\xi\|$. Scaling each $v_c \ne 0$ by λ_c with $|\varpi| < |\lambda_c|\|v_c\| \le 1$ bounds the operator norm by $|\varpi|^{-k}\|\mathsf{p}\|\|\beta^{\mathrm{lift}}_{\mathsf{L}_k,E}\|\|\xi\|$. For $S \in \mathcal{L}(E', E)$, $ST_{v'} = T_{Sv'}$ gives naturality. By (Q.11), $\chi_E(j_E\omega)(v) = \mathsf{p}\, j_{\mathsf{L}_k}(\omega \circ T_v^{\times k}) = \omega(v) u_{\mathfrak{E}}$.

Step 2 (natural families). Let $\mathsf{V}_E \colon \mathfrak{E}(E) \to \mathrm{Mult}^k(E; \mathcal{U}')$ be a natural family of bounded linear maps, that is, $\mathsf{V}_E(\mathfrak{E}(T)\xi) = \mathsf{V}_D(\xi) \circ T^{\times k}$ for $T \in \mathcal{L}(E, D)$ and $\xi \in \mathfrak{E}(D)$. Put $\mathsf{h}(\xi') = \mathsf{V}_{\mathsf{L}_k}(\xi')(e_1, \dots, e_k)$ for $\xi' \in \mathcal{U}_{\mathfrak{E}}$, a bounded linear map. Naturality along $\lambda \mathrm{id}_E$ gives $\mathsf{V}_E\mathfrak{E}(\lambda \mathrm{id}_E) = \lambda^k \mathsf{V}_E$, so $\mathsf{V}_E = \mathsf{V}_E \mathsf{q}^E_k$. For $\xi \in \mathfrak{E}_k(E)$, naturality along T_v gives $\mathsf{V}_E(\xi)(v) = \mathsf{V}_{\mathsf{L}_k}(\mathfrak{E}_k(T_v)\xi)(e_1, \dots, e_k)$. For $\xi'' = \mathsf{p}_\alpha \xi''$ in $\mathfrak{E}_k(\mathsf{L}_k)$, multilinearity and naturality along t_λ give

$$\lambda_1 \cdots \lambda_k \mathsf{V}_{\mathsf{L}_k}(\xi'')(e_1, \dots, e_k) = \mathsf{V}_{\mathsf{L}_k}(\xi'')(\mathsf{t}_\lambda e_1, \dots, \mathsf{t}_\lambda e_k) = \lambda^\alpha \mathsf{V}_{\mathsf{L}_k}(\xi'')(e_1, \dots, e_k),$$

so this value is 0 unless $\alpha = (1, \dots, 1)$. As the p_α have sum id, this gives $\mathsf{V}_E(\xi)(v) = \mathsf{h}(\chi_E(\xi)(v))$, that is, $\mathsf{V}_E = \mathsf{h}_* \circ \chi_E \circ \mathsf{q}^E_k$. Conversely, for $\mathsf{h} \in \mathcal{L}(\mathcal{U}_{\mathfrak{E}}, \mathcal{U}')$ the family $\mathsf{h}_* \circ \chi_E \circ \mathsf{q}^E_k$ is natural, and on $\xi' \in \mathcal{U}_{\mathfrak{E}}$ its value at $(e_1, \dots, e_k)$ is $\mathsf{h}(\mathsf{p}\,\xi') = \mathsf{h}(\xi')$, as $T_{(e_1,\dots,e_k)} = \mathrm{id}$. So h is determined by the family.

Step 3 (sections). Let ω be $\mathfrak{E}$-ambient, in a chart. By Lemma Q.10(4), $j_E \circ \omega$ is analytic into the completion of $\mathfrak{E}_k(E)$. Applying χ_E, $\omega\, u_{\mathfrak{E}}$ is analytic into $\mathrm{Mult}^k(E; \widehat{\mathcal{U}}_{\mathfrak{E}})$. If the line is split, let η_0 be the form of Definition 14.30. It extends to $\widehat{\mathcal{U}}_{\mathfrak{E}}$, as K is complete. Applying $(\eta_0)_*$, ω is analytic into $\mathrm{Mult}^k(E; K)$. □

Let $E^{(k)}$ be the sum of k copies of E, with the sum norm in case (A) and the maximum norm in case (B), with inclusions in_c, projections pr_c, sum $= \sum_c \mathrm{pr}_c$, and scalings $\mathrm{sc}_\lambda = \bigoplus_c \lambda_c$. For $\omega' \in \mathrm{Alt}^k(E^{(k)}; K)$ write $\omega' \circ \mathrm{sc}^{\times k}_\lambda = \sum_{|\alpha|=k} \lambda^\alpha \omega'_\alpha$. By interpolation, as K is infinite, each ω'_α is alternating. The *part of weight* $(1, \dots, 1)$ of ω' is $\omega'_{(1,\dots,1)}$. For $\mathsf{m} \in \mathrm{Mult}^k(E; K)$ and $z_1, \dots, z_k \in E^{(k)}$ put

$$\mathrm{Alt}'(\mathsf{m})(z_1, \dots, z_k) = \sum_{\sigma \in S_k} \mathrm{sgn}(\sigma)\, \mathsf{m}(\mathrm{pr}_1 z_{\sigma(1)}, \dots, \mathrm{pr}_k z_{\sigma(k)}).$$

Lemma Q.11 (generation)**.** $\mathrm{Alt}'(\mathsf{m}) \in \mathrm{Alt}^k(E^{(k)}; K)$, $\|\mathrm{Alt}'(\mathsf{m})\| \le \|\mathsf{m}\|$, *and* $\mathrm{Alt}'(\mathsf{m}) \circ (\mathrm{in}_1 \times \cdots \times \mathrm{in}_k) = \mathsf{m}$. *For alternating* ω, $\mathrm{Alt}'(\omega)$ *is the part of weight* $(1, \dots, 1)$ *of* $\omega \circ \mathrm{sum}^{\times k}$.

Proof. Step 1 (alternating, norm). If $z_c = z_{c'}$ with $c \ne c'$, the terms of σ and $(c\,c') \circ \sigma$ are equal with opposite signs, so the sum vanishes in every characteristic. The norm bound holds as $\|\mathrm{pr}_c\| \le 1$ and K is ultrametric.

Step 2 (restriction). Since $\mathrm{pr}_c \mathrm{in}_{\sigma(c)} = 0$ unless $\sigma(c) = c$, only $\sigma = \mathrm{id}$ survives on $\mathrm{in}_1 x_1, \dots, \mathrm{in}_k x_k$.

Step 3 (the part of weight $(1, \dots, 1)$). The part of weight $(1, \dots, 1)$ of $\omega \circ \mathrm{sum}^{\times k}$ is $\sum_\sigma \omega(\mathrm{pr}_{\sigma^{-1}(1)} z_1, \dots, \mathrm{pr}_{\sigma^{-1}(k)} z_k)$. Reordering the arguments of ω by σ gives the sign $\mathrm{sgn}(\sigma)$ and $\mathrm{Alt}'(\omega)$. □

Proof of Theorem 14.32. Write $\beta = \beta^{\mathfrak{E}}$ and $\beta(\mathrm{in}) = \beta_{E,E^{(k)}}(\mathrm{in}_1, \dots, \mathrm{in}_k)$. By (14.7), $\mathfrak{E}(\lambda \mathrm{id}_E) = \lambda^k \mathrm{id}$, so $\mathsf{q}_k^E = \mathrm{id}$.

Step 1 (uniqueness). By Lemma Q.11, $\mathsf{m} = \mathrm{Alt}'(\mathsf{m}) \circ (\mathrm{in}_1 \times \cdots \times \mathrm{in}_k)$. Equivariance and $\mathsf{i}|_{\mathrm{Alt}^k} = j$ force $\mathsf{i}_E(\mathsf{m}) = \beta(\mathrm{in})\, j_{E^{(k)}}(\mathrm{Alt}'(\mathsf{m}))$. Take this as the definition of i_E; then $\|\mathsf{i}_E\| \le \|\beta_{E,E^{(k)}}\|$.

Step 2 (it extends j). Let ω be alternating on E. As $\mathrm{sum} \circ \mathrm{in}_c = \mathrm{id}_E$, (14.7) gives $j_E\omega = \beta(\mathrm{in})\mathfrak{E}(\mathrm{sum}) j_E\omega = \beta(\mathrm{in})\, j_{E^{(k)}}(\omega \circ \mathrm{sum}^{\times k})$. For $\omega' \in \mathrm{Alt}^k(E^{(k)}; K)$, as $\mathrm{sc}_\lambda \mathrm{in}_c = \lambda_c \mathrm{in}_c$,

$$\sum_{|\alpha|=k} \lambda^\alpha \beta(\mathrm{in})\, j_{E^{(k)}}\omega'_\alpha = \beta(\mathrm{in})\, j_{E^{(k)}}(\omega' \circ \mathrm{sc}_\lambda^{\times k}) = \beta(\mathrm{in})\mathfrak{E}(\mathrm{sc}_\lambda) j_{E^{(k)}}\omega' = \lambda_1 \cdots \lambda_k \beta(\mathrm{in})\, j_{E^{(k)}}\omega'.$$

Comparing coefficients of $\lambda_1 \cdots \lambda_k$ gives $\beta(\mathrm{in})\, j_{E^{(k)}}\omega'_{(1,\dots,1)} = \beta(\mathrm{in})\, j_{E^{(k)}}\omega'$. With $\omega' = \omega \circ \mathrm{sum}^{\times k}$ and Lemma Q.11, $j_E\omega = \mathsf{i}_E(\omega)$.

Step 3 (equivariance). Let $\mathsf{m} \in \mathrm{Mult}^k(D; K)$, $T_c \in \mathcal{L}(E, D)$, and $\widehat{T} = \bigoplus_c T_c \colon E^{(k)} \to D^{(k)}$. Then $\mathrm{Alt}'(\mathsf{m} \circ (T_1 \times \cdots \times T_k)) = \mathrm{Alt}'(\mathsf{m}) \circ \widehat{T}^{\times k}$ and $\widehat{T}\, \mathrm{in}_c^E = \mathrm{in}_c^D\, T_c$. By (14.7),

$$\begin{aligned}\mathsf{i}_E(\mathsf{m} \circ (T_1 \times \cdots \times T_k)) &= \beta(\mathrm{in}^E)\mathfrak{E}(\widehat{T})\, j_{D^{(k)}}\, \mathrm{Alt}'(\mathsf{m}) = \beta_{E,D^{(k)}}(\mathrm{in}_1^D\, T_1, \dots, \mathrm{in}_k^D\, T_k)\, j_{D^{(k)}}\, \mathrm{Alt}'(\mathsf{m}) \\ &= \beta_{E,D}(T_1, \dots, T_k)\, \mathsf{i}_D(\mathsf{m}).\end{aligned}$$

Taking $T_1 = \cdots = T_k$ shows that i is natural. So i is an equivariant morphism, and Step 1 makes it the only one.

Step 4 (classes). If ω is analytic into $\mathrm{Mult}^k(E; K)$ in a chart, then $j_E \circ \omega = \mathsf{i}_E \circ \omega$ is analytic into the completion of $\mathfrak{E}(E)$. So $\Omega_K \subseteq \Omega_{\mathfrak{E}}$. If the line is split, Theorem 14.31(3) gives equality.

Step 5 (all envelopes). Let $k \ge 2$, and let $(1\,2)$ be the exchange of the arguments in slots 1 and 2. The families $\mathsf{m} \mapsto \mathsf{m}$ and $\mathsf{m} \mapsto -\mathsf{m} \circ (1\,2)$ on $\mathrm{Mult}^k(-; K)$ are natural and bounded, and both are the identity on Alt^k. So they are morphisms of $\mathrm{Mult}^k(-; K)$ to itself. They are distinct: with e_c^* the coordinate forms of L_k, at $\mathsf{m} = e_1^* \otimes \cdots \otimes e_k^*$ their values at $(e_1, \dots, e_k)$ are 1 and 0. For $k = 1$, $\mathrm{Mult}^1 = \mathrm{Alt}^1$, and j is the unique morphism to any envelope. □

Lemma Q.12 (from scalars to forms)**.** *Let P be an object of $\mathcal{V}$, f a scalar germ on P at x_0, $E = P \oplus K^k$ (sum norm in case* (A)*, maximum norm in case* (B)*), $\mathrm{d}y_c$ the coordinate forms on the summand K^k, ∂_c its unit vectors, and $\omega_{\mathsf{f}}(x, y) = \mathsf{f}(x)\, \mathrm{d}y_1 \wedge \cdots \wedge \mathrm{d}y_k$.*

1. *For $\mathcal{U}$ with $\|u_0\| = 1$: $\omega_{\mathsf{f}}\, u_0$ is analytic into $\mathrm{Mult}^k(E; \mathcal{U})$ at (x_0, y_0) if and only if $\mathsf{f}\, u_0$ is analytic into $\mathcal{U}$ at x_0. In particular, ω_{f} is analytic into $\mathrm{Mult}^k(E; K)$ at (x_0, y_0) if and only if f is analytic at x_0.*
2. *ω_{f} is AA at (x_0, y_0) as a map into $\mathrm{Alt}^k(E; K)$ if and only if f is AA at x_0. The same holds for DP.*

Proof. Step 1 (item 1, if). If $\mathsf{f}\, u_0$ is analytic, precompose with the projection to P and postcompose with the bounded linear map $u \mapsto (\mathrm{d}y_1 \wedge \cdots \wedge \mathrm{d}y_k)\, u$.

Step 2 (item 1, only if). Postcompose with the evaluation at $(\partial_1, \dots, \partial_k)$, which gives $\mathsf{f}\, u_0$, and restrict to $y = y_0$.

Step 3 (item 2). For AA the same two maps pass between ω_{f} and f, by Proposition 14.12(1). For DP, the coefficient laws of ω_{f} are $a_n(\mathrm{pr}_P\, h)\, \mathrm{d}y_1 \wedge \cdots \wedge \mathrm{d}y_k$, with a_n those of f, and the same two maps pass between them. □

Proof of Theorem 14.33. Put $a_n(h) = \mathsf{e}_n h_1 \cdots h_n$ on $\ell^1(\mathbb{N}, K)$, and let e_i be the unit vectors of this space.

Step 1 (item 1, AA). For $t \in \ell^1(\mathbb{N}, K)^{\otimes n}$ with coordinates $t(\mathsf{w})$, $\mathsf{w} \in \mathbb{N}^n$, the triangle inequality gives $\sum_{\mathsf{w}} |t(\mathsf{w})| \le \pi(t)$. If t is symmetric, its $n!$ coordinates at the reorderings of $(1, \dots, n)$ agree,

so $n!\,|t(1,\dots,n)| \le \pi(t)$, with $n!$ the integer. The arithmetic and geometric means give $\|a_n\| \le |\mathsf{e}_n| n^{-n} < |\varpi|^{-1}$. Also $L_{a_n}(t) = \mathsf{e}_n t(1,\dots,n)$, so $|L_{a_n}(t)| \le |\varpi|^{-1}(n^n/n!)\pi(t) \le |\varpi|^{-1}\mathrm{e}^n\pi(t)$. By Proposition 14.9(1), f is AA at 0 with radius e^{-1} and bound $|\varpi|^{-1}$.

Step 2 (item 1, not analytic). Suppose $\mathsf{f}(h) = \sum_n b_n(h,\dots,h)$ near 0, with $b_n\colon \ell^1(\mathbb{N},K)^n \to K$ bounded n-linear. By Lemma A.1, $b_n(h,\dots,h) = a_n(h)$. The coefficient of $\lambda_1\cdots\lambda_n$ in $b_n(\sum_i \lambda_i e_i,\dots,\sum_i \lambda_i e_i)$ is $\sum_{\sigma\in S_n} b_n(e_{\sigma(1)},\dots,e_{\sigma(n)})$, and it is e_n. As K is ultrametric, $\|b_n\| \ge |\mathsf{e}_n| \ge n^n$, so no radius exists.

Step 3 (item 2). In (14.8), the sum formula is the relation $[\mathsf{r}_N] = 0$ and the bound on $\mathsf{u}_{N,i}$ is the weight. The norm of u_0 is the infimum over finitely supported (c_N) of $|1+\sum_N c_N| + \sum_N |c_N|$, which is at least 1 by the triangle inequality; so $\|u_0\| = 1$. If ϕ is a linear form on $\widehat{\mathfrak{U}}$ with $|\phi| \le C\|\cdot\|$, then $|\phi(u_0)| \le \max_i |\phi(\mathsf{u}_{N,i})| \le C/N$ for all N, so $\phi(u_0) = 0$. Let $\mathrm{nb}\colon S_n \to \{1,\dots,n!\}$ be a bijection and

$$b_n(h^1,\dots,h^n) = \mathsf{e}_n \sum_{\sigma\in S_n} h^1_{\sigma(1)}\cdots h^n_{\sigma(n)}\,\mathsf{u}_{n!,\mathrm{nb}(\sigma)}.$$

Then $\|b_n\| \le |\mathsf{e}_n|/n! \le |\varpi|^{-1}\mathrm{e}^n$, and $b_n(h,\dots,h) = \mathsf{e}_n h_1\cdots h_n u_0$. So $\mathsf{f}\,u_0$ is analytic at 0 into $\widehat{\mathfrak{U}}$ with radius e^{-1}. Regrouping $b_n(x_1+h,\dots,x_1+h)$ by multilinearity, as in Step 5 of the proof of Proposition 14.12, makes it analytic at every point x_1 of the ball of radius e^{-1}, with radius $\mathrm{e}^{-1} - \|x_1\|$. By Lemma Q.12(1), $\omega_{\mathsf{f}} \in \Omega_{\widehat{\mathfrak{U}}}$ on O, and $\omega_{\mathsf{f}} \notin \Omega_K$ by Step 2. The map $\mathsf{m} \mapsto \mathsf{m}\,u_0$ gives $\Omega_K \subseteq \Omega_{\widehat{\mathfrak{U}}}$.

For the envelope $\mathrm{Mult}^k(-;\widehat{\mathfrak{U}})$, a bounded linear form on the coefficient space is a sum, over the words $(\sigma(1),\dots,\sigma(k))$, of bounded linear forms ϕ_σ on $\widehat{\mathfrak{U}}$ applied to the coordinates, since $\|u\,e^*_{\sigma(1)}\otimes\cdots\otimes e^*_{\sigma(k)}\| \le \|u\|$ for $u \in \widehat{\mathfrak{U}}$, with $e^*_1,\dots,e^*_k$ the coordinate forms of L_k. The vector $u_{\mathfrak{E}} = \det u_0$ has the coordinate $\mathrm{sgn}(\sigma)u_0$ at that word, and $\phi_\sigma(u_0) = 0$. So the form vanishes at $u_{\mathfrak{E}}$, and the line $Ku_{\mathfrak{E}}$ is not split.

Step 4 (item 3). A section analytic into $\mathrm{Mult}^k(E;K)$ with values in the closed subspace $\mathrm{Alt}^k(E;K)$ is AA, so $\Omega_K \subseteq \Omega_{\mathrm{AA}}$. By Step 1 and Proposition 14.12(2), f is AA on the ball of radius e^{-1}, and Lemma Q.12(2) puts ω_{f} in Ω_{AA} on O. By Step 3 it is not in Ω_K.

Step 5 (item 4). $\det\cdot\,\mathsf{v}$ is AA by bounded linear postcomposition, as v is AA (Corollary 14.19(4)). If $\det\cdot(\mathsf{Sv})$ were AA at a point, evaluation at the unit vectors of K^k would make Sv AA there, against that Corollary. □

Proof of Corollary 14.34. Step 1 (from laws to coefficients). Let E be an ultrametric Banach space and ω DP at x_0 with radius R, bound $\mathfrak{b}$ and laws a_n. On the ultrametric space $E \oplus E^{(k)}$ (maximum norm), $a'_n(h,v_1,\dots,v_k) = a_n(h)(v_1,\dots,v_k)$ is a law of degree $n+k$ into K, as it is linear in each v_c, with $\|a'_n\| \le \|a_n\|$. Suppose that $C \ge 1$ and that each a'_n is the diagonal of an $(n+k)$-linear β_n into K with $\|\beta_n\| \le C^{n+k}\|a_n\|$. Let $\beta'_n(h_1,\dots,h_n;v_1,\dots,v_k)$ be the sum of β_n over all placements of $\mathrm{in}_1 v_1,\dots,\mathrm{in}_k v_k$ among its $n+k$ slots, with $h_1,\dots,h_n$ in order in the other slots. Then $\|\beta'_n\| \le \|\beta_n\|$, as K is ultrametric. Its diagonal in h is the coefficient of $\lambda_1\cdots\lambda_k$ in $\beta_n(z,\dots,z)$, $z = h + \sum_c \lambda_c\,\mathrm{in}_c\,v_c$, and $\beta_n(z,\dots,z) = \lambda_1\cdots\lambda_k a_n(h)(v_1,\dots,v_k)$. So β'_n is an n-linear map into $\mathrm{Mult}^k(E;K)$ with diagonal a_n, and ω is analytic at x_0 into $\mathrm{Mult}^k(E;K)$, with radius R/C.

Step 2 (the positive cases). If $\mathrm{char}\,K = p$ and K is spherically complete, Proposition 14.40(1) with $W = K$ gives β_n with $C = (|\varpi|\mathfrak{c}_K)^{-1}$. If $\mathrm{char}\,K = 0$, Theorem 14.39(3) gives β_n with $C = 8\mathfrak{d}/|\varpi|^2$.

Step 3 (a law analytic nowhere). Let $\mathrm{char}\,K = p$ and let K be complete, not spherically complete. Take the ultrametric Banach spaces $E_\flat, D_\flat$ of Theorem 7.2(2) in degree p. Let $\mathsf{Q}^\flat$ be the separated completion of $(E_\flat^{\otimes p}, \pi_{\max})$ modulo the closure of the kernel of the quotient

map $E_\flat^{\otimes p} \to \Lambda^p_K E_\flat$ of Section 3.1, an ultrametric Banach space. As K is ultrametric, a form on $E_\flat^{\otimes p}$ has the same norm for $\pi_{\max}$ as on elementary tensors. So a bounded form on $\mathsf{Q}^\flat$ is a form on $E_\flat^{\otimes p}$, bounded for $\pi_{\max}$, that kills that kernel; that is, an alternating p-linear form of the same norm. Hence the bounded forms on $\mathsf{Q}^\flat$ are $\mathrm{Alt}^p(E_\flat; K)$, isometrically. Let $X_\flat = \mathcal{L}(E_\flat, D_\flat) \oplus \mathrm{Alt}^p(D_\flat; K) \oplus \mathsf{Q}^\flat$ (maximum norm), an ultrametric Banach space, and $\mathsf{f}_\flat(T, \omega_\flat, \mathsf{k}) = \langle \omega_\flat \circ T^{\times p}, \mathsf{k}\rangle$. This is a bounded law of degree $p+2$ and norm at most 1, the coefficients in T being alternating by interpolation, so it is DP at every point by (Q.2).

Suppose it is analytic at $(T_0, \omega_0, \mathsf{k}_0)$, with coefficients b_n, $\|b_n\| \le \mathfrak{b}_\flat R_\flat^{-n}$. For $n \ge 2$ let $b_n^\flat(h^1, \dots, h^{n-2})(\omega_\flat, \mathsf{k})$ be the sum over ordered pairs of slots of b_n with $(0, \omega_\flat, 0)$ and $(0, 0, \mathsf{k})$ in that pair and $(h^i, 0, 0)$, in order, in the other slots. Then $\|b_n^\flat\| \le \|b_n\|$, as K is ultrametric. Fix $\omega_\flat, \mathsf{k}$ and $\|h\| < R_\flat$. For small λ_1, λ_2 the terms of the expansion at $(T_0 + h, \omega_0 + \lambda_1\omega_\flat, \mathsf{k}_0 + \lambda_2\mathsf{k})$ tend to 0 in K, so they regroup by powers of λ_1, λ_2. Lemma A.1, in λ_1 and then in λ_2, gives $\langle \omega_\flat \circ (T_0 + h)^{\times p}, \mathsf{k}\rangle = \sum_n b_n^\flat(h, \dots, h)(\omega_\flat, \mathsf{k})$ for $\|h\| < R_\flat$. The values $b_n^\flat(h^1, \dots, h^{n-2})$ are bounded operators from $\mathrm{Alt}^p(D_\flat; K)$ to $\mathrm{Alt}^p(E_\flat; K)$, as $\mathsf{Q}^\flat$ has dual $\mathrm{Alt}^p(E_\flat; K)$ isometrically, and the series converges in operator norm, as this operator space is complete. So $A^p_{E_\flat, D_\flat; K}$ is analytic at T_0, with coefficients $b_n^\flat$, against Theorem 7.2(2). Hence $\mathsf{f}_\flat$ is analytic at no point.

Step 4 (only if). For every $k \ge 1$, $E = X_\flat \oplus K^k$ (maximum norm) is an ultrametric Banach space. By Lemma Q.12, $\omega_{\mathsf{f}_\flat}$ is DP everywhere into $\mathrm{Alt}^k(E; K)$ and analytic nowhere into $\mathrm{Mult}^k(E; K)$. □

Q.8. **Tame pairs.** The completion $\widehat{P}$ of a normed space P is a normed space over $\widehat{K}$, the scalar multiplication extending by continuity, and it is ultrametric if P is.

Lemma Q.13 (coordinates)**.**

(1) *Let P, W be normed, and let $a\colon P \to W$ be a bounded polynomial law of degree $n \ge 1$. Then a extends uniquely to a continuous map $\hat{a}\colon \widehat{P} \to \widehat{W}$, and $\hat{a}$ is a bounded polynomial law of degree n over $\widehat{K}$ with $\|\hat{a}\| \le \|a\|$.*

(2) *Let $P_0 \ne \{0\}$ be a finite-dimensional normed space. Then $\widehat{P}_0$ has finite dimension $m' \ge 1$ over $\widehat{K}$, and $\mathfrak{c}_{\widehat{K}} = \mathfrak{c}_K$.*

(3) *Let K be complete and let P_0 be ultrametric of finite dimension $m' \ge 1$. If K is discretely valued, put $\mathfrak{o} = 1$ and $\mathfrak{o}' = \mathfrak{c}_K$; otherwise let $0 < \mathfrak{o} < 1$ and put $\mathfrak{o}' = \mathfrak{o}$. There is a linear bijection $\mathsf{C}\colon c_0([m'], K) \to P_0$ with $\|\mathsf{C}\| \le 1$ and $\|\mathsf{C}^{-1}\| \le (\mathfrak{o}\mathfrak{o}')^{-1}$.*

Proof. Step 1 (item (1): the extension). By (Q.3), a is Lipschitz on every bounded subset of P. So it extends uniquely to a continuous map $\hat{a}\colon \widehat{P} \to \widehat{W}$, and $\|\hat{a}(x)\| \le \|a\|\|x\|^n$ by continuity.

Step 2 (item (1): the law). Let $r \ge 1$ and $|\alpha| = n$. By interpolation in each variable on $n+1$ distinct elements of K, there are finitely many scalars λ_i and $\xi_{i,1}, \dots, \xi_{i,r}$ in K, depending only on r and α, with $a^{[\alpha]}(h_1, \dots, h_r) = \sum_i \lambda_i a(\xi_{i,1}h_1 + \cdots + \xi_{i,r}h_r)$ on P^r. Define $\hat{a}^{[\alpha]}$ on $\widehat{P}^r$ by the same formula with $\hat{a}$. The identity

$$\hat{a}(s_1h_1 + \cdots + s_rh_r) = \sum_{|\alpha|=n} s_1^{\alpha_1} \cdots s_r^{\alpha_r}\, \hat{a}^{[\alpha]}(h_1, \dots, h_r)$$

holds on $K^r \times P^r$. Both sides are continuous on $\widehat{K}^r \times \widehat{P}^r$, so it holds there. Hence $\hat{a}$ is a bounded polynomial law of degree n over $\widehat{K}$, and $\|\hat{a}\| \le \|a\|$ by Step 1.

Step 3 (item (2)). The $\widehat{K}$-span of a basis of P_0 in $\widehat{P}_0$ is complete by Lemma A.3, hence closed, and it contains the dense subspace P_0. So it is $\widehat{P}_0$. If $\lambda_i \to \lambda \ne 0$ in $\widehat{K}$ with $\lambda_i \in K$, then $|\lambda_i - \lambda| < |\lambda|$ for large i, so $|\lambda_i| = |\lambda|$. Hence $|\widehat{K}^\times| = |K^\times|$ and $\mathfrak{c}_{\widehat{K}} = \mathfrak{c}_K$.

Step 4 (item (3): distances). Let P_1 be a subspace of P_0 and $x \in P_0 \setminus P_1$. By Lemma A.3, P_1 is complete, hence closed, so $\operatorname{dist}(x, P_1) > 0$. Let K be discretely valued. If nonzero vectors $v_1, \dots, v_i$ of P_0 have norms in distinct cosets of $\mathfrak{c}_K^{\mathbb{Z}}$ in $(0, \infty)$, the nonzero numbers among $|\lambda_1|\|v_1\|, \dots, |\lambda_i|\|v_i\|$ are distinct. So the ultrametric inequality gives $\|\sum_q \lambda_q v_q\| = \max_q |\lambda_q|\|v_q\|$, the v_q are independent, and $i \le m'$. Thus the nonzero values of the norm of P_0 lie in at most m' cosets, and there are only finitely many values in $[\operatorname{dist}(x, P_1), \|x\|]$. Hence $\|x - x'\| = \operatorname{dist}(x, P_1)$ for some $x' \in P_1$.

Step 5 (item (3): orthogonal bases). Let $0 < \mathfrak{o}_1 \le 1$, with $\mathfrak{o}_1 < 1$ unless K is discretely valued. We show by induction on m', for all such $\mathfrak{o}_1$ and all ultrametric P_0 of dimension m', that P_0 has a basis $\mathsf{l}_1, \dots, \mathsf{l}_{m'}$ with

$$\|\lambda_1 \mathsf{l}_1 + \cdots + \lambda_{m'} \mathsf{l}_{m'}\| \ge \mathfrak{o}_1 \max_q |\lambda_q| \|\mathsf{l}_q\| \qquad (\lambda_1, \dots, \lambda_{m'} \in K). \tag{Q.12}$$

For $m' = 1$ every nonzero vector will do. Let $m' \ge 2$, let P_1 be a hyperplane of P_0, and let $\mathsf{l}_1, \dots, \mathsf{l}_{m'-1}$ be a basis of P_1 with (Q.12) for $\mathfrak{o}_1^{1/2}$. Let $x \in P_0 \setminus P_1$. By Step 4 (since $\operatorname{dist}(x, P_1) > 0$ if $\mathfrak{o}_1 < 1$, and by its last sentence if $\mathfrak{o}_1 = 1$) there is $x' \in P_1$ with $\|x - x'\| \le \mathfrak{o}_1^{-1/2} \operatorname{dist}(x, P_1)$. Put $\mathsf{l}_{m'} = x - x'$, so that $\operatorname{dist}(\mathsf{l}_{m'}, P_1) \ge \mathfrak{o}_1^{1/2} \|\mathsf{l}_{m'}\|$. Let $v = v' + \lambda_{m'} \mathsf{l}_{m'}$ with $v' = \sum_{q<m'} \lambda_q \mathsf{l}_q$. As $-\lambda_{m'}^{-1} v' \in P_1$ if $\lambda_{m'} \ne 0$, we get $\|v\| \ge \mathfrak{o}_1^{1/2} |\lambda_{m'}| \|\mathsf{l}_{m'}\|$. By the ultrametric inequality, $\|v'\| \le \max(\|v\|, |\lambda_{m'}|\|\mathsf{l}_{m'}\|) \le \mathfrak{o}_1^{-1/2} \|v\|$. Hence $\mathfrak{o}_1 \max_{q<m'} |\lambda_q| \|\mathsf{l}_q\| \le \mathfrak{o}_1^{1/2} \|v'\| \le \|v\|$, and $\mathfrak{o}_1 |\lambda_{m'}| \|\mathsf{l}_{m'}\| \le \|v\|$.

Step 6 (item (3): the bijection). Take the basis of Step 5 with $\mathfrak{o}_1 = \mathfrak{o}$. Scaling keeps (Q.12). If K is discretely valued, scale so that $\mathfrak{c}_K < \|\mathsf{l}_q\| \le 1$. Otherwise $|K^\times|$ is a subgroup of $(0, \infty)$ that contains a number other than 1 and is not of the form $\mathfrak{c}^{\mathbb{Z}}$, so it is dense in $(0, \infty)$; scale so that $\mathfrak{o} < \|\mathsf{l}_q\| \le 1$. Put $\mathsf{C}(\lambda) = \sum_q \lambda_q \mathsf{l}_q$. The ultrametric inequality in P_0 gives $\|\mathsf{C}(\lambda)\| \le \max_q |\lambda_q| \|\mathsf{l}_q\| \le \|\lambda\|$, with no count of the m' terms, and (Q.12) gives $\|\mathsf{C}(\lambda)\| \ge \mathfrak{o}\mathfrak{o}' \|\lambda\|$. □

The seminorm $\pi_{\max}$ of Section 14.6 is ultrametric, by concatenating representations, and $\pi_{\max} \le \pi$, as a maximum is at most a sum. For normed P and $n \ge 1$, let W_n be as in Section 14.6, and let $a_\Gamma(h) \in \mathsf{W}_n$ be the class of $h^{\otimes n}$. By (14.1), a_Γ is a bounded law of degree n with $\|a_\Gamma\| \le 1$, and $a_\Gamma^{[\alpha]}(h_1, \dots, h_r)$ is the class of $O_\alpha(h_1, \dots, h_r)$.

Proof of Theorem 14.39. Throughout, $a \colon P \to W$ is a bounded law of degree $n \ge 1$.

Step 1 (item (1): a finite-dimensional subspace). Let P, W be ultrametric, $t \in \Gamma^n(P)$ with $t \ne 0$ (for $t = 0$ there is nothing to prove), $\delta > 0$, and $t = \sum_l x_{l,1} \otimes \cdots \otimes x_{l,n}$ with $\max_l \prod_q \|x_{l,q}\| \le \pi_{\max}(t) + \delta$. Let P_0 be the span of the $x_{l,q}$; $P_0 \ne \{0\}$, as $t \ne 0$. Then t is a symmetric tensor in $P_0^{\otimes n}$, so $t \in \Gamma^n(P_0)$ by Section L.3, and $\pi_{\max}(t)$ computed in P_0 is at most $\pi_{\max}(t) + \delta$.

Step 2 (item (1): completion). By Lemma Q.13(1),(2), $a|_{P_0}$ extends to a bounded law $\hat{a} \colon \widehat{P}_0 \to \widehat{W}$ over $\widehat{K}$ with $\|\hat{a}\| \le \|a\|$, and $\widehat{P}_0$ has finite dimension $m' \ge 1$ over $\widehat{K}$. Let $\hat{t}$ be the image of t in $\widehat{P}_0^{\otimes_{\widehat{K}} n}$. It lies in $\Gamma^n(\widehat{P}_0)$, and if $t = \sum_i \lambda_i h_i^{\otimes n}$ with $h_i \in P_0$, then $L_a(t) = \sum_i \lambda_i \hat{a}(h_i) = L_{\hat{a}}(\hat{t})$. A representation of t in P_0 is one of $\hat{t}$, so $\pi_{\max}(\hat{t}) \le \pi_{\max}(t) + \delta$.

Step 3 (item (1): coordinates). The space $\widehat{P}_0$ is ultrametric, as P_0 is, so it has coordinates for the maximum norm: take $\mathsf{C} \colon c_0([m'], \widehat{K}) \to \widehat{P}_0$ and $\mathfrak{o}, \mathfrak{o}'$ from Lemma Q.13(3) over $\widehat{K}$, and put $t' = (\mathsf{C}^{-1})^{\otimes n} \hat{t}$. As t' is symmetric, $t' = \sum_\alpha t'_\alpha O_\alpha(e_1, \dots, e_{m'})$, with t'_α the common coordinate of t' at the words in $[m']^n$ of content α, that is, in which each q occurs α_q times, and so $\hat{t} = \sum_\alpha t'_\alpha O_\alpha(\mathsf{C}e_1, \dots, \mathsf{C}e_{m'})$. Comparing the coefficients of s^α in $L_{\hat{a}}((\sum_q s_q \mathsf{C}e_q)^{\otimes n}) =$

$\hat{a}(\sum_q s_q \mathsf{C} e_q)$, by (14.1), gives $L_{\hat{a}}(O_\alpha(\mathsf{C}e)) = (\hat{a}\circ\mathsf{C})^{[\alpha]}(e)$. By (Q.5) with $K' = \widehat{K}$, as $\widehat{W}$ is ultrametric and $\|\hat{a}\circ\mathsf{C}\| \le \|a\|$, this has norm at most $|\varpi|^{-n}\|a\|$. If $t' = \sum_l x'_{l,1}\otimes\cdots\otimes x'_{l,n}$, the coordinate of t' at a word w is $\sum_l \prod_q x'_{l,q}(\mathsf{w}_q)$, of absolute value at most $\max_l \prod_q \|x'_{l,q}\|$, since $\widehat{K}$ is nonarchimedean. So $|t'_\alpha| \le \pi_{\max}(t') \le \|\mathsf{C}^{-1}\|^n(\pi_{\max}(t)+\delta)$.

Step 4 (item (1): the estimate). The ultrametric inequality in $\widehat{W}$ is used here: the multi-indices α are not counted, and

$$\|L_a(t)\| = \Big\|\sum_\alpha t'_\alpha L_{\hat{a}}(O_\alpha(\mathsf{C}e))\Big\| \le (|\varpi|\mathfrak{o}\mathfrak{o}')^{-n}\|a\|\,(\pi_{\max}(t)+\delta).$$

Let $\delta\to 0$ and, if K is densely valued, $\mathfrak{o}\to 1$. As $\mathfrak{c}_{\widehat{K}} = \mathfrak{c}_K$, this is (14.9). As $\pi_{\max}\le\pi$, it gives $\mathfrak{a}_n(P,W) \le (|\varpi|\mathfrak{c}_K)^{-n}$.

Step 5 (item (1): germs). Let f be DP at x_0 with radius R, bound $\mathfrak{b}$, laws a_n, and put $R_{\max} = |\varpi|\mathfrak{c}_K R < R$. By (14.9) for each a_n, $\|L_{a_n}(t_n)\| \le \mathfrak{b}R_{\max}^{-n}\pi_{\max}(t_n)$ for $t_n\in\Gamma^n(P)$, also for $n=0$, with $\pi_{\max} = |\cdot|$ on $P^{\otimes 0} = K$, as $\|a_0\|\le\mathfrak{b}$. Let $L_\oplus(t) = \sum_n L_{a_n}(t_n)$ on $\bigoplus_n \Gamma^n(P)$. Replace the sums in (Q.7) by maxima: on $W\oplus\bigoplus_n P^{\otimes n}$ let

$$\|(v,y)\|_Z = \inf\Big\{\max\Big(\|v+L_\oplus(t)\|,\ \max_n \mathfrak{b}R_{\max}^{-n}\pi_{\max}(y_n - t_n)\Big) : t\in\bigoplus_n \Gamma^n(P)\Big\},$$

the quotient of an ultrametric seminorm by the subspace of the pairs $(L_\oplus(t),-t)$, hence ultrametric. Let Z be the quotient by its kernel, $[v,y]$ the class of (v,y), and $j(v) = [v,0]$. Then j is an isometry. Indeed $t=0$ gives $\|j(v)\|\le\|v\|$; and if $\|v+L_\oplus(t)\| < \|v\|$, the ultrametric inequality in W gives $\|v\| = \|L_\oplus(t)\| \le \max_n\|L_{a_n}(t_n)\| \le \max_n \mathfrak{b}R_{\max}^{-n}\pi_{\max}(t_n)$. So the maximum in the infimum is always at least $\|v\|$. Put $b_n(h_1,\dots,h_n) = [0,\iota_n(h_1\otimes\cdots\otimes h_n)]$. Then $\|b_n\|\le\mathfrak{b}R_{\max}^{-n}$, and taking $t = \iota_n h^{\otimes n}$ gives $b_n(h,\dots,h) = [a_n(h),0] = ja_n(h)$. So f is ambient analytic at x_0 with radius $R_{\max}$, bound $\mathfrak{b}$, through the ultrametric space Z. With Proposition 14.9(1), this gives DP = AA between ultrametric spaces.

Step 6 (item (2): the projective seminorm on ℓ^1). Let P be $\ell^1(J)$ or $\ell^1_{00}(J)$, and let $t\in P^{\otimes n}$ have coordinates $t(\mathsf{w})$, $\mathsf{w}\in J^n$. The coordinate forms separate the points of P, hence those of $P^{\otimes n}$, so t is determined by its coordinates. We claim $\pi(t) = \sum_{\mathsf{w}}|t(\mathsf{w})|$. As $\sum_{\mathsf{w}}|(x_1\otimes\cdots\otimes x_n)(\mathsf{w})| = \prod_q\|x_q\|$, the triangle inequality gives $\ge$. If t is finitely supported, $t = \sum_{\mathsf{w}} t(\mathsf{w})e_{\mathsf{w}_1}\otimes\cdots\otimes e_{\mathsf{w}_n}$ gives $\le$. In general let pr_{J_0} be the coordinate projection onto a finite $J_0\subseteq J$, of norm at most 1. Then $\mathrm{pr}_{J_0}^{\otimes n}t\to t$ for π as J_0 increases, by telescoping on elementary tensors, and the finitely supported case gives $\le$ again.

Step 7 (item (2): the bound). Let t be a symmetric tensor supported in J_0^n, $J_0\subseteq J$ finite. Then $t(\mathsf{w})$ depends only on the content $\alpha\in\mathbb{N}^{J_0}$ of w; call it t_α. So $t = \sum_\alpha t_\alpha O_\alpha(e)$, with e listing the unit vectors e_c, $c\in J_0$, and $\pi(t) = \sum_\alpha (n!/\alpha!)|t_\alpha|$ by Step 6, as $n!/\alpha!$ words have content α. By (Q.4) for a_Γ, $\|O_\alpha(e)\|_\Gamma \le C_0^n n!/\alpha!$. Hence $\|t\|_\Gamma \le C_0^n\pi(t)$. This settles $\ell^1_{00}(J)$. Let $t = \sum_i \lambda_i h_i^{\otimes n}\in\Gamma^n(\ell^1(J))$ and $t_0 = \mathrm{pr}_{J_0}^{\otimes n}t = \sum_i\lambda_i(\mathrm{pr}_{J_0}h_i)^{\otimes n}$. Then $\|t_0\|_\Gamma \le C_0^n\pi(t_0)\le C_0^n\pi(t)$. As J_0 increases, $\|t-t_0\|_\Gamma \le \sum_i|\lambda_i|\|a_\Gamma(h_i) - a_\Gamma(\mathrm{pr}_{J_0}h_i)\|\to 0$, by (Q.3). Hence $\|t\|_\Gamma\le C_0^n\pi(t)$.

Step 8 (item (3)). Let $\mathrm{char}\,K = 0$, P ultrametric. The absolute value is at most 1 on $\mathbb{Z}$. At most one prime p_0 has $|p_0|<1$, since 1 is an integer combination of any two distinct primes and K is nonarchimedean. If there is one, $|n!| = |p_0|^{\sum_{i\ge1}\lfloor n/p_0^i\rfloor}$ with $\sum_{i\ge1}\lfloor n/p_0^i\rfloor \le n/(p_0-1)$; otherwise $|n!| = 1$. So $|1/n!|\le\mathfrak{d}^n$. Put $\beta_a(x_1,\dots,x_n) = \frac{1}{n!}L_a(O_{(1,\dots,1)}(x_1,\dots,x_n))$, which uses characteristic 0. It is n-linear with values in W. Its diagonal is a, as $O_{(1,\dots,1)}(h,\dots,h) = n!\,h^{\otimes n}$. By Lemma Q.1(1) and (Q.6), where the ultrametric norm of P removes the factor n^n, $\|\beta_a\|\le$

$(8\mathfrak{d}/|\varpi|^2)^n\|a\|$. For $a = a_\Gamma$, the linearization of β_{a_Γ} on $P^{\otimes n}$ and the quotient map agree on powers, hence on $\Gamma^n(P)$. So $\|t\|_\Gamma \le \|\beta_{a_\Gamma}\|\pi(t)$, which is the bound on $\mathfrak{a}_n(P)$.

Step 9 (item (4): the hull). Let P be any normed space and W ultrametric. Put $\nu(x) = \inf\{\max_i\|x_i\| : x = \sum_i x_i \text{ a finite sum}\}$. It is an ultrametric seminorm, by concatenating sums, and $\nu \le \|\cdot\|$. Let $x = \sum_i x_i$. Then $a(x) = \sum_{|\alpha|=n} a^{[\alpha]}(x_1, x_2, \dots)$. For each α at most n vectors enter, so (Q.1) gives $\|a^{[\alpha]}(x_1, x_2, \dots)\| \le (8/|\varpi|)^n\|a\|(n\max_i\|x_i\|)^n$. The ultrametric inequality in W is used here: the number of α is not counted. Taking the infimum,

$$\|a(x)\| \le \mathfrak{b}_a\nu(x)^n, \qquad \mathfrak{b}_a = (8n/|\varpi|)^n\|a\|. \tag{Q.13}$$

Step 10 (item (4): descent to the quotient). Let $\nu(z) = 0$ and $x \in P$. As $\nu(x + sz) \le \nu(x)$ for every $s \in K$, the polynomial $s \mapsto a(x + sz) = \sum_i v_i s^i$ is bounded on K by (Q.13). In the ultrametric W, $\|\sum_i v_i s^i\| = \max_i\|v_i\||s|^i$ for all large $|s|$. So $v_i = 0$ for $i \ge 1$, and $a(x+z) = a(x)$. Thus a induces a bounded law $\tilde{a}$ of degree n on $P_\nu = P/\ker\nu$, normed by ν, with $\|\tilde{a}\| \le \mathfrak{b}_a$. The space P_ν is ultrametric.

Step 11 (item (4): the estimate). If $P_\nu = \{0\}$, then $a = 0$ by (Q.13), and there is nothing to prove. Otherwise let $\tilde{t}$ be the image of $t \in \Gamma^n(P)$ in $\Gamma^n(P_\nu)$. On powers, $L_{\tilde{a}}(\tilde{t}) = L_a(t)$, hence on $\Gamma^n(P)$. A representation of t maps to one of $\tilde{t}$, and ν is at most the norm, so $\pi_{\max}(\tilde{t}) \le \pi_{\max}(t)$. By (14.9) for $\tilde{a}$, $\|L_a(t)\| \le (|\varpi|\mathfrak{c}_K)^{-n}\mathfrak{b}_a\pi_{\max}(t)$, which is (14.10). As $\pi_{\max} \le \pi$, this bounds $\mathfrak{a}_n(P, W)$, and a is ambient by Lemma Q.1(4). □

Proof of Proposition 14.37. Step 1 (one degree). Let $n \ge 1$ and suppose $\mathrm{amb}(a) < \infty$ for every bounded law $a\colon P \to W$ of degree n. We show $\mathfrak{a}_n(P, W) < \infty$. The bounded laws of degree n form a complete normed space, whether or not K is complete: on the span of finitely many vectors the coefficients of a law are fixed linear combinations of its values at finitely many points, by interpolation, so a Cauchy sequence of laws converges pointwise to a law, in norm, as W is complete. For $t \in \Gamma^n(P)$ the map $a \mapsto L_a(t)$ is continuous and linear, by Lemma Q.1(1) and the uniqueness in it. So $\{a : \mathrm{amb}(a) \le N\} = \{a : \|L_a(t)\| \le N\pi(t) \text{ for all } t\}$ is closed for each integer $N \ge 1$. By hypothesis these sets cover the space, so by Baire's theorem one of them contains a ball of center a_0 and radius $\delta > 0$. For $\|a\| < \delta$, $\mathrm{amb}(a) \le \mathrm{amb}(a_0 + a) + \mathrm{amb}(a_0) \le 2N$. For $a \ne 0$ choose $\lambda \in K^\times$ with $|\varpi|\delta \le |\lambda|\|a\| < \delta$. Then $\mathrm{amb}(a) = |\lambda|^{-1}\,\mathrm{amb}(\lambda a) \le (2N/(|\varpi|\delta))\|a\|$.

Step 2 ((i) implies (ii), and (ii) implies (iii)). If (P, W) is tame, then $\mathfrak{a}_n(P, W) \le \mathfrak{q}^n$ for $n \ge 1$ with $\mathfrak{q} = \max(1, \sup_{n\ge1}\mathfrak{a}_n(P, W)^{1/n})$. So Theorem 14.38(1), with $C = 1$, whose proof below uses only Proposition 14.9, makes every DP germ from P to W ambient analytic. A germ at 0 given by a law series is a DP germ.

Step 3 ((iii) implies (i)). Suppose (P, W) is not tame. If $\mathfrak{a}_n(P, W) = \infty$ for one n, Step 1 gives a law a of degree n with $\mathrm{amb}(a) = \infty$. It is a DP germ at 0, and it is not ambient analytic there, by Proposition 14.9(1). Otherwise $\sup_n \mathfrak{a}_n(P, W)^{1/n} = \infty$ with all terms finite, and finitely many terms are bounded. So there are $n_1 < n_2 < \dots$ and laws a_{n_i} of degree n_i with $\mathrm{amb}(a_{n_i})/\|a_{n_i}\| > i^{n_i}$. Multiplying each law by a scalar, which does not change the quotient, we may assume $|\varpi| \le \|a_{n_i}\| \le 1$ and $\mathrm{amb}(a_{n_i}) \ge |\varpi|\,i^{n_i}$. The series $f = \sum_i a_{n_i}$ converges for $\|h\| < 1$, as W is complete, and is DP at 0 with radius 1. Its coefficient laws are the a_{n_i} and 0. As $\mathrm{amb}(a_{n_i})^{1/n_i}$ is unbounded, f is not ambient analytic at 0, by Proposition 14.9(1). □

Lemma Q.14 (inverse germs)**.** *Let P be a Banach space, U an open subset of P and $f\colon U \to P$ a map that is DP at x_0 with radius R, bound $\mathfrak{b}$ and coefficient laws a_n. Suppose that $\mathsf{T}_0 = a_1$, a bounded linear map, is bijective with bounded inverse. Put $\mathsf{P}_m = -\mathsf{T}_0^{-1} \circ a_m$ for $m \ge 2$,*

$\mathsf{g}_1 = \mathrm{id}_P$, *and, for* $d \geq 2$, *with* $|\alpha|$ *and* $\deg \alpha$ *as in Section Q.4,*

$$\mathsf{g}_d(z) = \sum_{m=2}^{d} \sum_{|\alpha|=m,\ \deg\alpha=d} \mathsf{P}_m^{[\alpha]}\big(\mathsf{g}_1(z), \dots, \mathsf{g}_{d-1}(z)\big). \tag{Q.14}$$

Then g_d *is a bounded law of degree* d, *the series* $\mathsf{g} = \sum_d \mathsf{g}_d$ *converges on a ball around* 0, *and* $y \mapsto x_0 + \mathsf{g}(\mathsf{T}_0^{-1}(y - f(x_0)))$ *is DP at* $f(x_0)$. *There are open sets* $U_0 \ni x_0$ *and* $U_1 \ni f(x_0)$ *such that* f *restricts to a bijection* $U_0 \to U_1$ *whose inverse is this map on* U_1.

The formal inverse is Roby's [21, Th. VI.3, Prop. VI.8]; only the estimates are added here.

Proof. Step 1 (normal form). Put $\tilde f(z) = \mathsf{T}_0^{-1}(f(x_0+z) - f(x_0))$ for $\|z\| < R$. As $a_0 = f(x_0)$, we have $\tilde f(z) = z - \mathsf{P}(z)$ with $\mathsf{P} = \sum_{m\geq 2} \mathsf{P}_m$. Each P_m is a bounded law of degree m with $\|\mathsf{P}_m\| \leq \mathfrak{b}_\mathsf{T} R^{-m}$, where $\mathfrak{b}_\mathsf{T} = \|\mathsf{T}_0^{-1}\|\mathfrak{b}$.

Step 2 (the recursion). Since $m \geq 2$, every α with $|\alpha| = m$ and $\deg\alpha = d$ is supported in $\{1, \dots, d-1\}$, so (Q.14) is a recursion. By Lemma Q.3(1), for the polynomial germs $\sum_{i<d} \mathsf{g}_i$ and P_m at 0, which are DP, each term is a bounded law of degree d.

Step 3 (majorant). Put $\varkappa = C_0/R$ and $\mathsf{K}(\theta) = \mathfrak{b}_\mathsf{T} \sum_{m\geq 2} (\varkappa\theta)^m$. Define $\mathsf{y}_1 = 1$ and $\mathsf{y}_d = [s^d]\,\mathsf{K}(\sum_{i<d} \mathsf{y}_i s^i)$ for $d \geq 2$, where $[s^d]$ denotes the coefficient of s^d in a power series in s. We show $\|\mathsf{g}_d\| \leq \mathsf{y}_d$ by induction. By (Q.4),

$$\big\|\mathsf{P}_m^{[\alpha]}(\mathsf{g}_1(z), \dots, \mathsf{g}_{d-1}(z))\big\| \leq C_0^m \|\mathsf{P}_m\| \frac{m!}{\prod_i \alpha_i!} \prod_i \|\mathsf{g}_i\|^{\alpha_i}\ \|z\|^{\deg\alpha}.$$

Summing over α with the multinomial theorem, and then over m, as $C_0^m\|\mathsf{P}_m\| \leq \mathfrak{b}_\mathsf{T}\varkappa^m$, $\|\mathsf{g}_d\| \leq [s^d]\,\mathsf{K}(\sum_{i<d}\|\mathsf{g}_i\| s^i) \leq \mathsf{y}_d$, since the coefficients of K are nonnegative.

Step 4 (a positive radius). The series $\sum_d \mathsf{y}_d s^d$ is the unique formal power series with zero constant term that solves $\mathsf{y} = s + \mathsf{K}(\mathsf{y})$, the coefficients being forced by the recursion. As $\mathsf{K}(\theta) = \mathfrak{b}_\mathsf{T}\varkappa^2\theta^2/(1-\varkappa\theta)$, this equation reads $\varkappa(1+\mathfrak{b}_\mathsf{T}\varkappa)\mathsf{y}^2 - (1+\varkappa s)\mathsf{y} + s = 0$, whose root vanishing at $s = 0$ is

$$\mathsf{y}_*(s) = \frac{(1+\varkappa s) - \sqrt{(1+\varkappa s)^2 - 4\varkappa(1+\mathfrak{b}_\mathsf{T}\varkappa)s}}{2\varkappa(1+\mathfrak{b}_\mathsf{T}\varkappa)}.$$

The polynomial under the root has value 1 at $s = 0$, so y_* is a convergent power series near 0. It satisfies the equation near 0, so its Taylor series is the formal solution. Hence there is $R_0 > 0$ with $\mathsf{y}(\theta) = \sum_d \mathsf{y}_d\theta^d < \infty$ for $0 \leq \theta < R_0$, and $\mathsf{y}(\theta) \to 0$ as $\theta \to 0$.

Step 5 (a right inverse). As P is complete, $\mathsf{g}(z) = \sum_d \mathsf{g}_d(z)$ converges absolutely for $\|z\| < R_0$, with $\|\mathsf{g}(z)\| \leq \mathsf{y}(\|z\|)$, and g is DP at 0, with radius R_3 and bound $\mathsf{y}(R_3) > 0$ for any $0 < R_3 < R_0$. Apply Lemma Q.3 with inner germ g, whose laws are 0 in degree 0 and g_d for $d \geq 1$, and outer germ $\tilde f$ at 0, whose laws are 0, id_P and $-\mathsf{P}_m$, $m \geq 2$ (radius R, bound $\max(R, \mathfrak{b}_\mathsf{T})$). The law of the composite in degree 0 is 0, in degree 1 it is $\mathsf{g}_1 = \mathrm{id}_P$, and in degree $d \geq 2$ it is g_d minus the right side of (Q.14), which is 0. So $\tilde f(\mathsf{g}(z)) = z$ for small z.

Step 6 (injectivity). Let $\|z\|, \|z'\| \leq R_\mathrm{i}$. By (Q.3) with radius $2R_\mathrm{i}$,

$$\|\mathsf{P}_m(z) - \mathsf{P}_m(z')\| \leq m(16/|\varpi|^2)^m \|\mathsf{P}_m\| (2R_\mathrm{i})^{m-1}\|z - z'\| \leq \frac{16\mathfrak{b}_\mathsf{T}}{|\varpi|^2 R}\, m\Big(\frac{32R_\mathrm{i}}{|\varpi|^2 R}\Big)^{m-1}\|z - z'\|.$$

The sum over $m \geq 2$ of these coefficients tends to 0 with R_i. Fix R_i with $32R_\mathrm{i} < |\varpi|^2 R$ for which it is at most $\frac12$. Then $\frac12\|z - z'\| \leq \|\tilde f(z) - \tilde f(z')\| \leq \frac32\|z - z'\|$ on the closed ball of radius R_i. So $\tilde f$ is injective and continuous there.

Step 7 (the local inverse). Choose $0 < R_2 < R_0$ with $\mathsf{y}(R_2) < R_\mathrm{i}$ and $\tilde f(\mathsf{g}(z)) = z$ for $\|z\| < R_2$. Let B_2 and B_i be the open balls around 0 of radii R_2 and R_i, and $B = B_\mathrm{i} \cap \tilde f^{-1}(B_2)$, an open set. Then $\mathsf{g}(B_2) \subseteq B$, and $\tilde f\colon B \to B_2$ is a bijection with inverse $\mathsf{g}|_{B_2}$. As $f(x_0+z) = f(x_0) + \mathsf{T}_0\tilde f(z)$, the sets $U_0 = x_0 + B$ and $U_1 = f(x_0) + \mathsf{T}_0(B_2)$ have the stated property. The germ at $f(x_0)$ is DP: its laws are x_0 in degree 0 and $\mathsf{g}_d \circ \mathsf{T}_0^{-1}$ in degree $d \geq 1$, of norm at most $\mathsf{y}_d\|\mathsf{T}_0^{-1}\|^d$, so for $0 < R_3 < R_0$ the radius $R_3/\|\mathsf{T}_0^{-1}\|$ and the bound $\max(\|x_0\|, \mathsf{y}(R_3))$ will do. □

Proof of Theorem 14.38. Step 1 (item (1)). For $n \geq 1$, $\mathrm{amb}(a_n) \leq C\mathfrak{q}^n\|a_n\| \leq C\mathfrak{b}(R/\mathfrak{q})^{-n}$. For $n = 0$, $\mathrm{amb}(a_0) = \|a_0\| \leq \mathfrak{b} \leq C\mathfrak{b}$. As f is DP with radius $R/\mathfrak{q} \leq R$, Proposition 14.9(1) applies. A tame pair has such constants: $C = 1$ and $\mathfrak{q} = \max(1, \sup_{n\geq 1} \mathfrak{a}_n(P,W)^{1/n})$.

Step 2 (item (2)). Ambient analytic germs are DP (Proposition 14.9(1)). DP germs compose (Lemma Q.3), with no condition on Y. Apply item (1), with constants for (P, W), to the composite, a DP germ from P to W.

Step 3 (item (3)). By Lemma Q.14, f has a local inverse germ that is DP at $f(x_0)$. By item (1) for the pair (P, P), with constants for that pair, it is ambient analytic at $f(x_0)$, with a radius that we shrink so that the ball lies in the set U_1 of Lemma Q.14. As P is Banach, Proposition 14.12(2) makes it ambient analytic at every point of that ball. □

Proof of Proposition 14.40. Step 1 (item (1): laws). Let $a\colon P \to W$ be a bounded law of degree n, and $C_a = (|\varpi|\mathfrak{c}_K)^{-n}\|a\|$. For $n = 0$ take $\beta = a$. Let $n \geq 1$. By (14.9), $\|L_a(t)\| \leq C_a\pi_{\max}(t)$ on the subspace $\Gamma^n(P)$ of $(P^{\otimes n}, \pi_{\max})$, an ultrametric seminormed space: here the ultrametric norms of P and W are used. Step 1 of the proof of Theorem 5.2, with the single operator 0 and radius C_a, extends L_a to a linear map $L'_a\colon P^{\otimes n} \to W$ with $\|L'_a(t)\| \leq C_a\pi_{\max}(t)$. This uses that W is spherically complete. Put $\beta(h_1, \dots, h_n) = L'_a(h_1 \otimes \cdots \otimes h_n)$. It is n-linear with $\|\beta\| \leq C_a$, and $\beta(h, \dots, h) = a(h)$.

Step 2 (item (1): discretely valued K). Let $|K^\times| = \mathfrak{c}_K^{\mathbb{Z}}$ and let W be an ultrametric Banach space. As in the proof of Corollary 5.3, let $\|w\|'$ be the least element of $\mathfrak{c}_K^{\mathbb{Z}}$ at least $\|w\|$, with $\|0\|' = 0$. It is an ultrametric norm with $\|\cdot\| \leq \|\cdot\|' \leq \mathfrak{c}_K^{-1}\|\cdot\|$, complete with distances in $\mathfrak{c}_K^{\mathbb{Z}} \cup \{0\}$, hence spherically complete by Lemma D.1. For this norm a has norm at most $\mathfrak{c}_K^{-1}\|a\|$. Step 1 gives β with $\|\beta\| \leq \mathfrak{c}_K^{-1}C_a$, also for the norm of W.

Step 3 (item (1): germs). If f is DP at x_0 with radius R, bound $\mathfrak{b}$, laws a_n, the maps β_n that Steps 1 and 2 give for a_n have norm at most $\mathfrak{b}(|\varpi|\mathfrak{c}_K R)^{-n}$, with one more factor $\mathfrak{c}_K^{-1}$ in Step 2. So f is analytic at x_0 with radius $|\varpi|\mathfrak{c}_K R$ and bound $\mathfrak{b}$, respectively $\mathfrak{c}_K^{-1}\mathfrak{b}$.

Step 4 (item (2)). Let f be DP at x_0 with radius R, bound $\mathfrak{b}$, laws a_n. By Theorem 14.39(3), a_n is for $n \geq 1$ the diagonal of an n-linear map of norm at most $\mathfrak{b}(|\varpi|^2 R/(8\mathfrak{d}))^{-n}$, and a_0 is a constant. So f is analytic at x_0. Analytic germs are ambient analytic, and ambient analytic germs are DP.

Step 5 (item (3)). Let K be complete, not spherically complete, of characteristic p. Let $\mathsf{f}_\flat$ on the ultrametric Banach space $X_\flat$ be as in Step 3 of the proof of Corollary 14.34. That step shows that $\mathsf{f}_\flat$ is DP at every point and analytic at no point. As $X_\flat$ and K are ultrametric, Theorem 14.39(1) makes $\mathsf{f}_\flat$ ambient analytic at every point. □

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